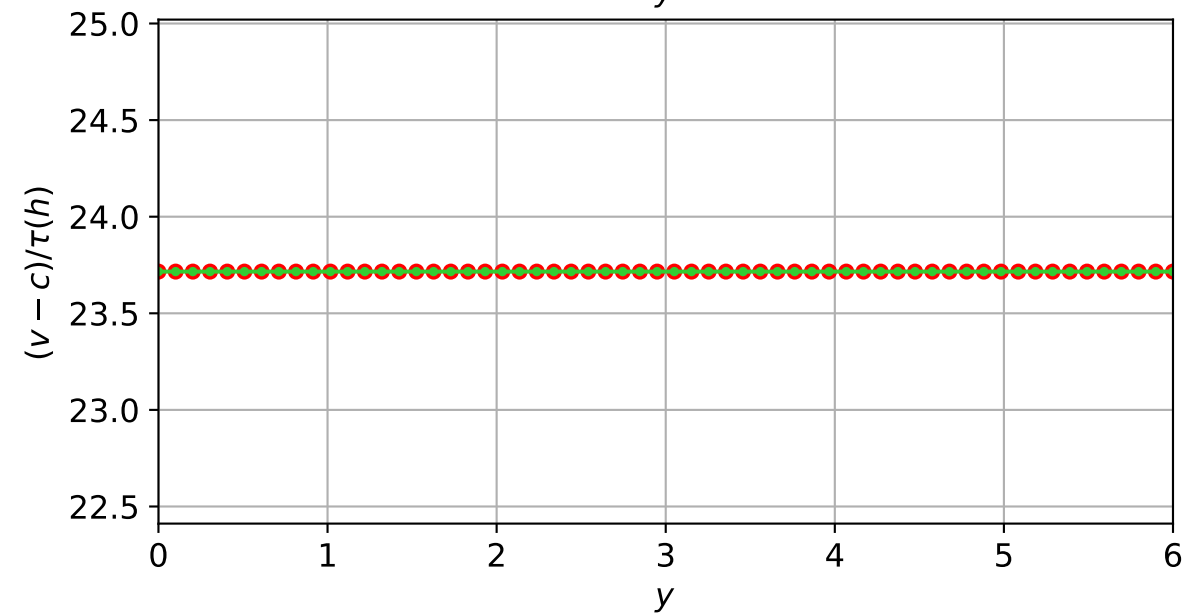
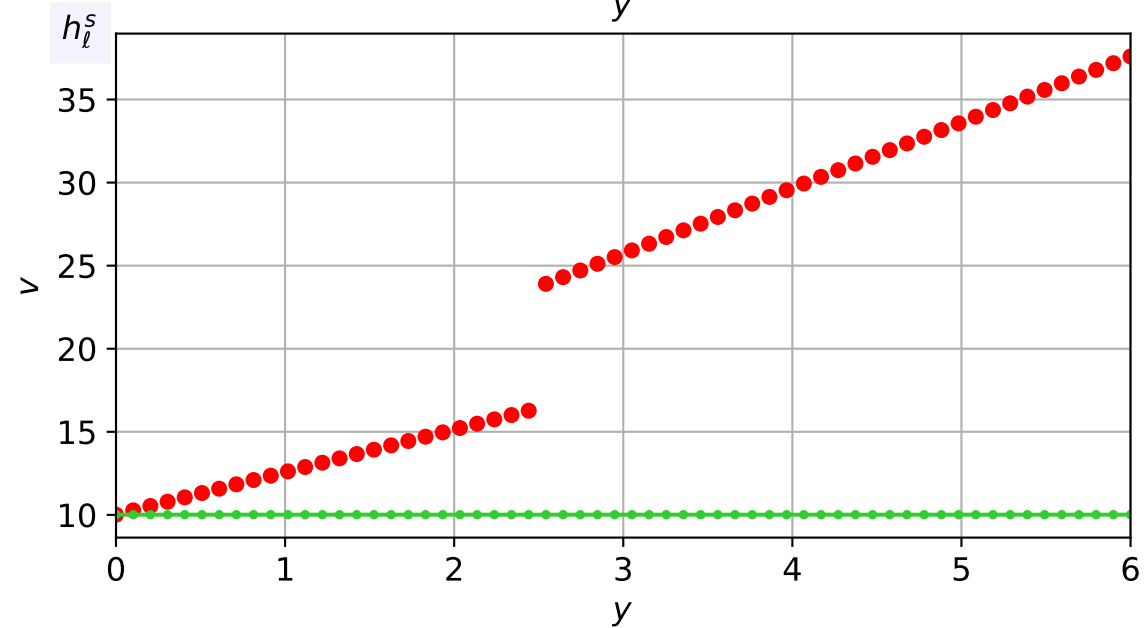
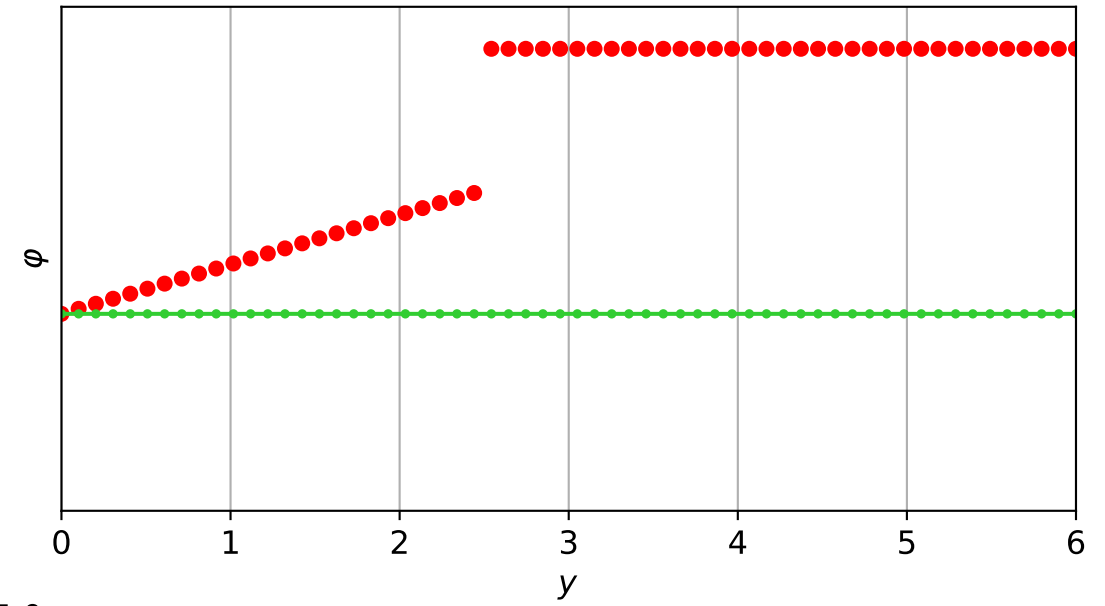
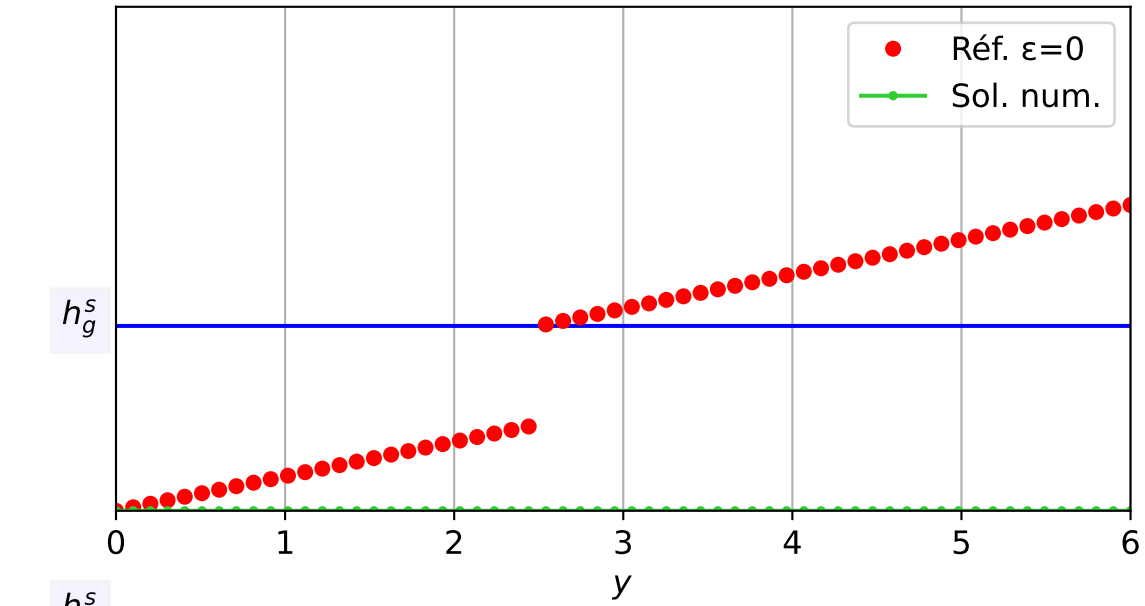
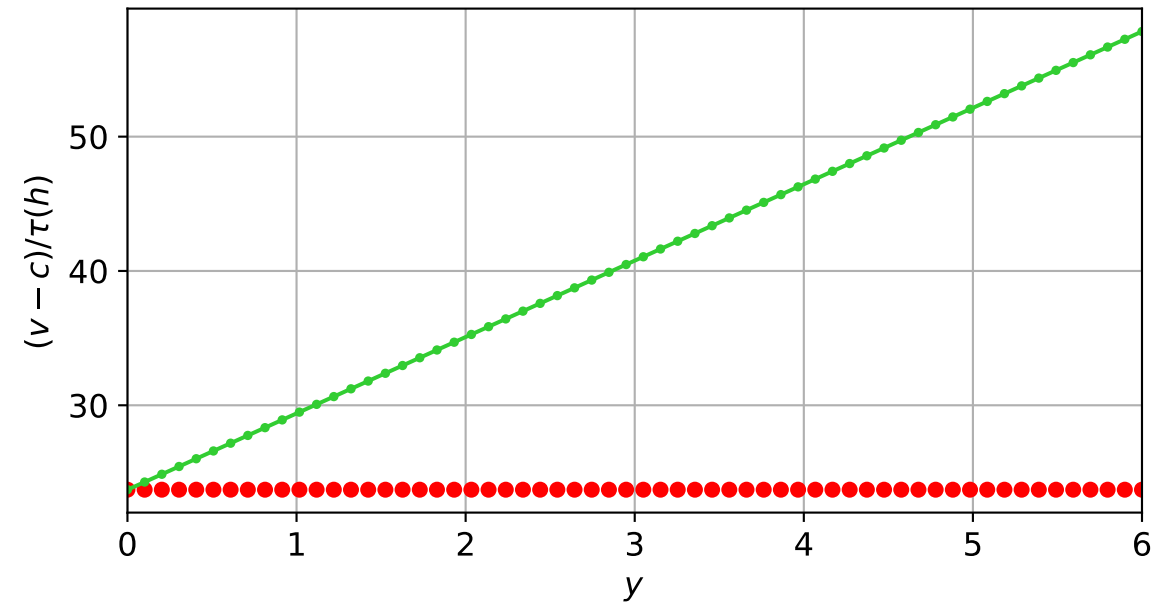
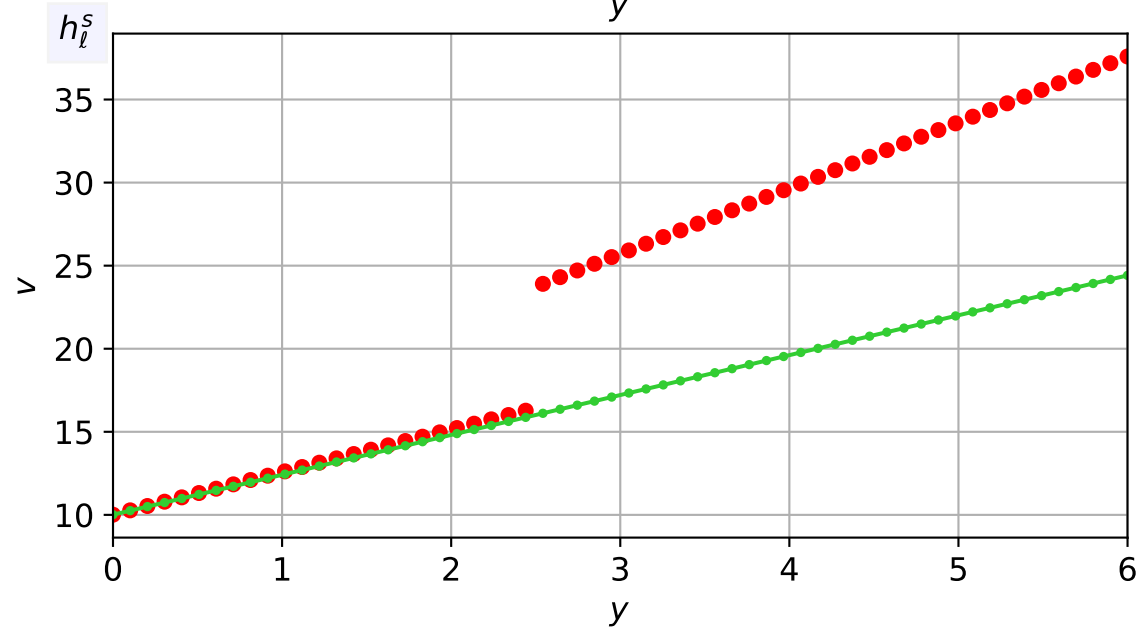
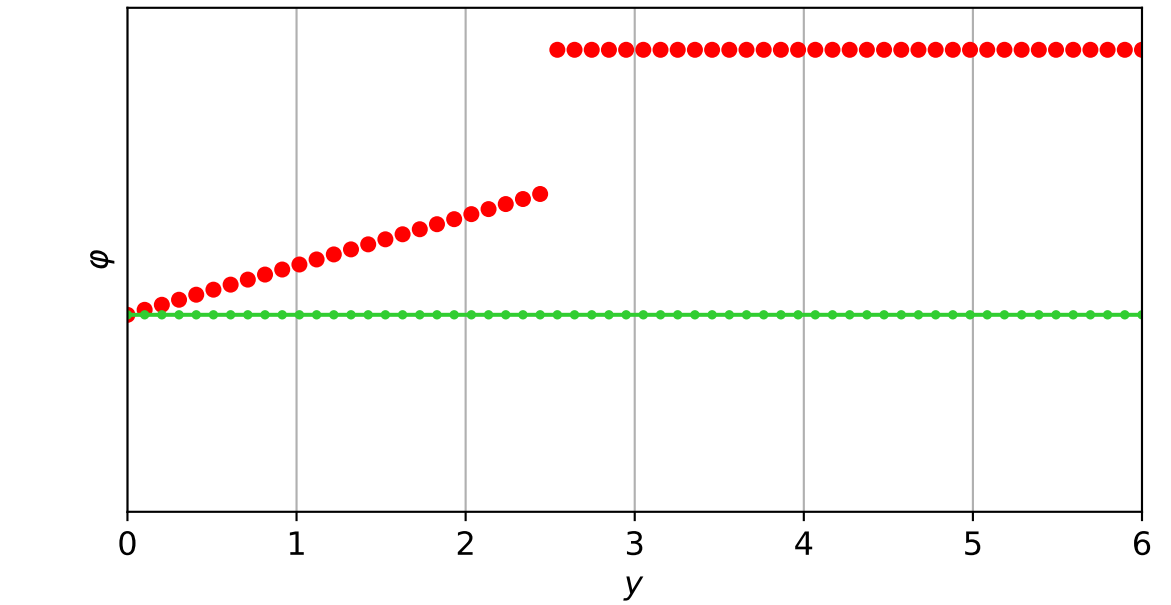
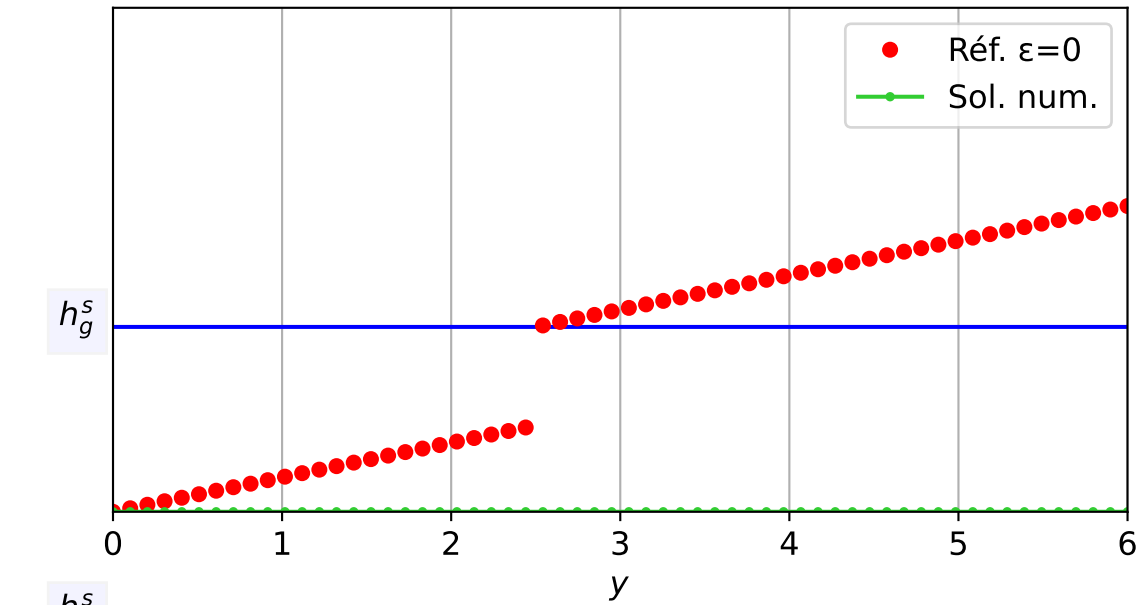


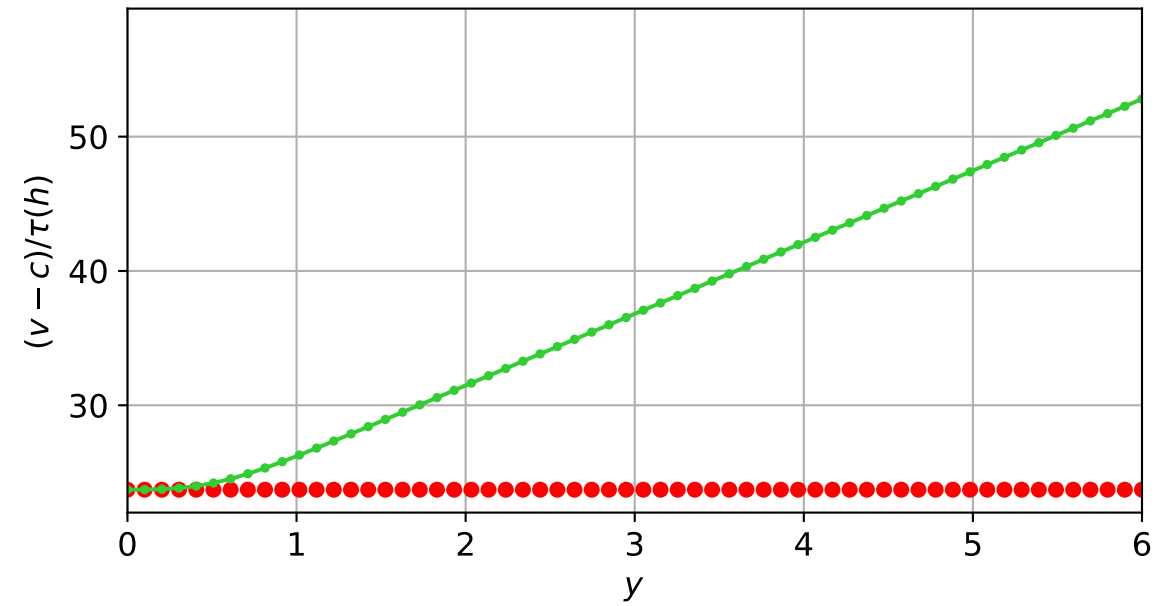
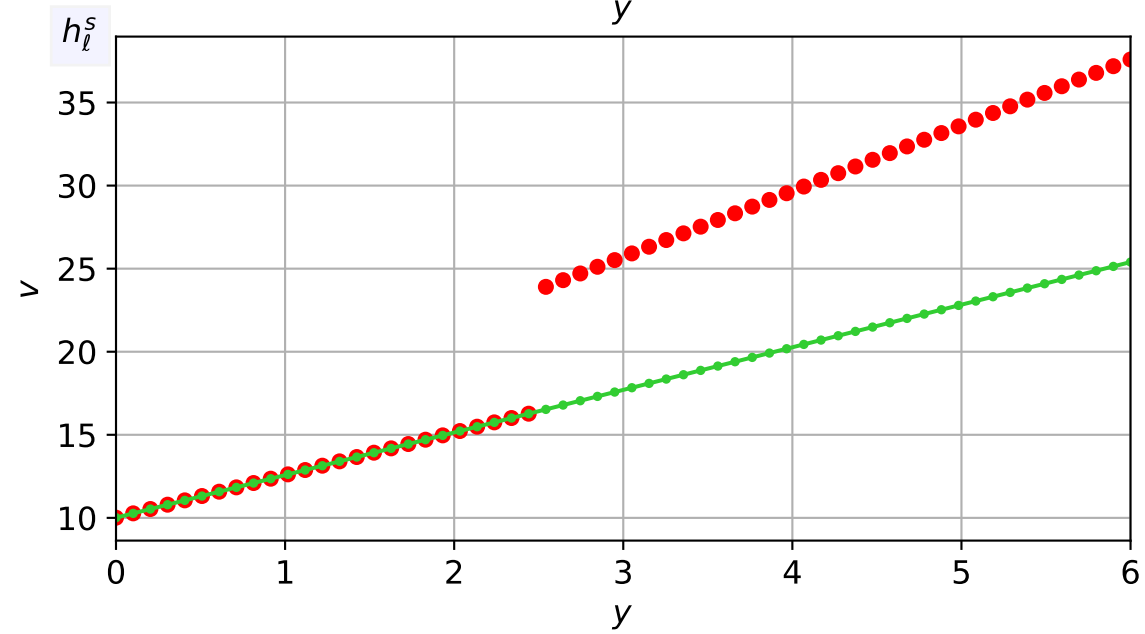
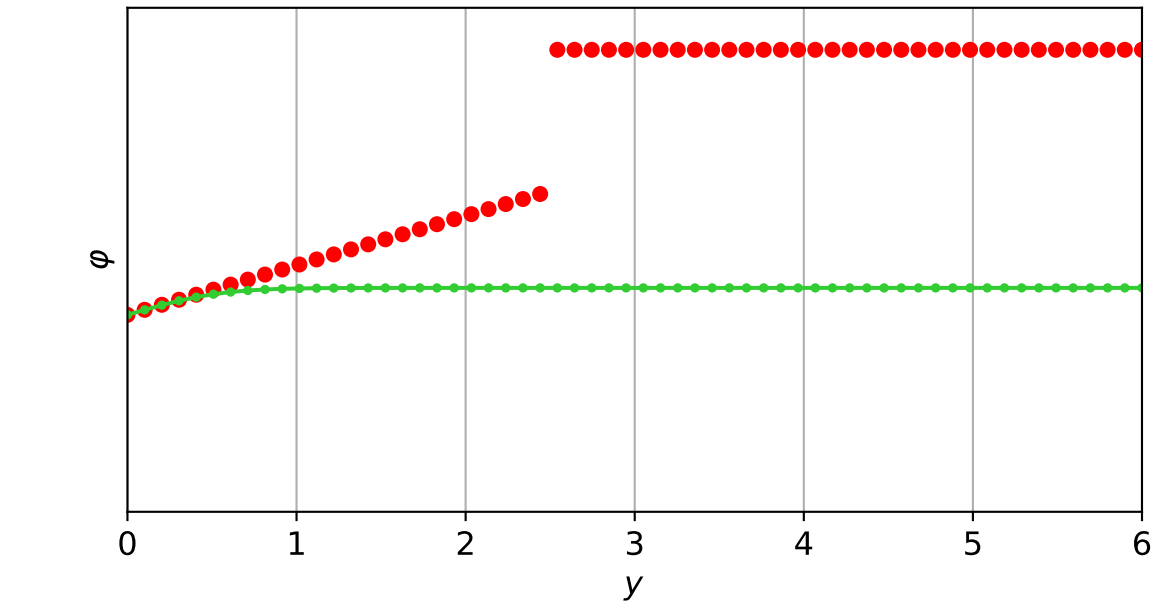
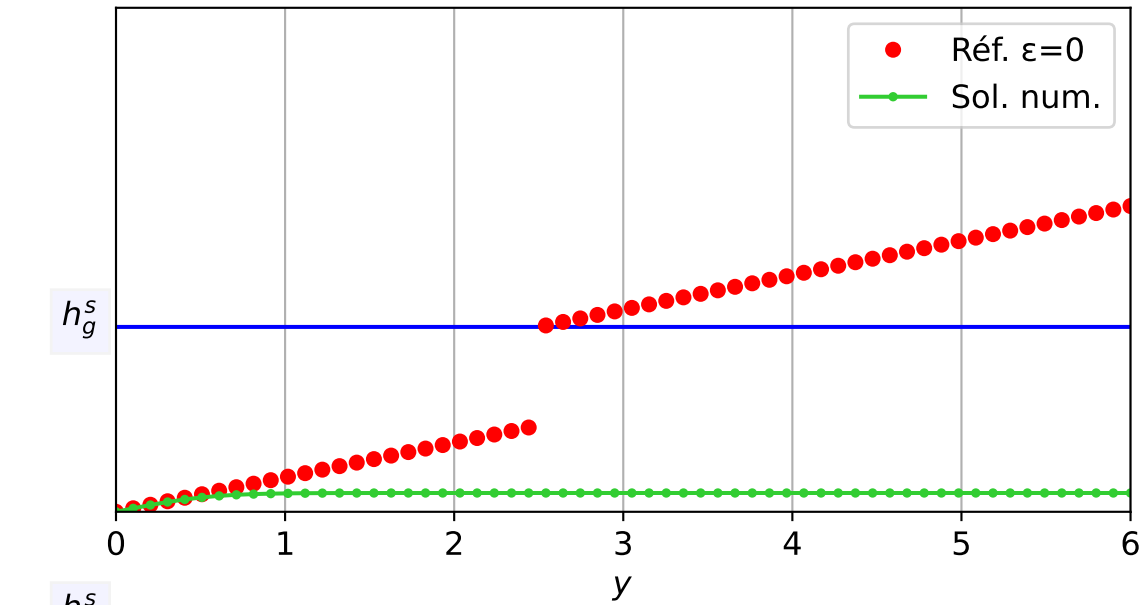
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 9.58e-04$  —  $t = 0/12$  (0 iter) — résidu eq. =  $0.00e+00$



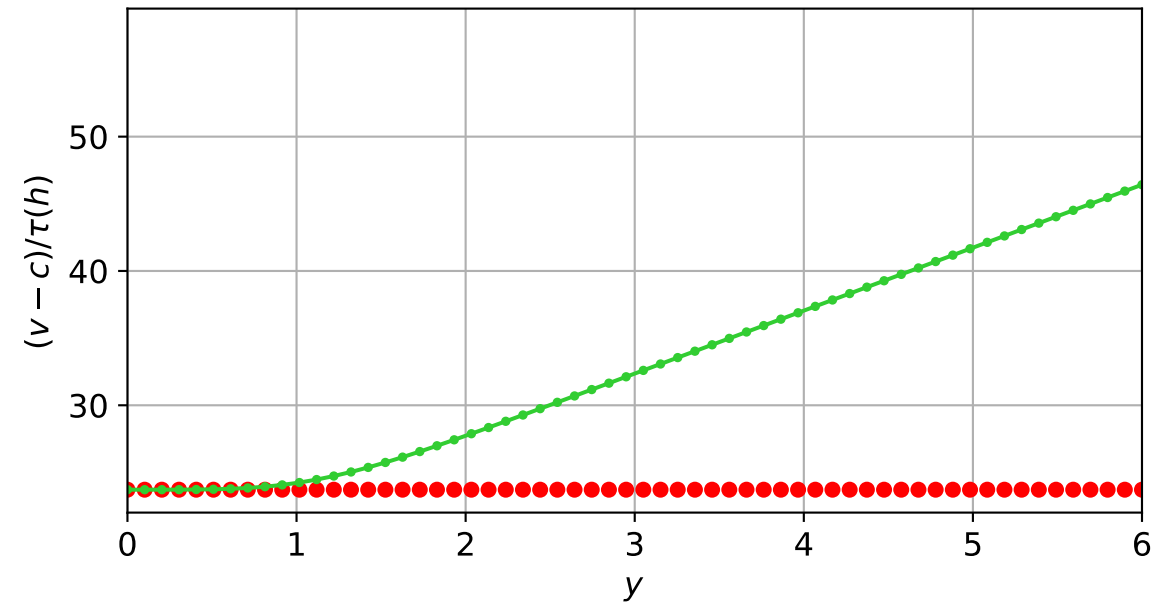
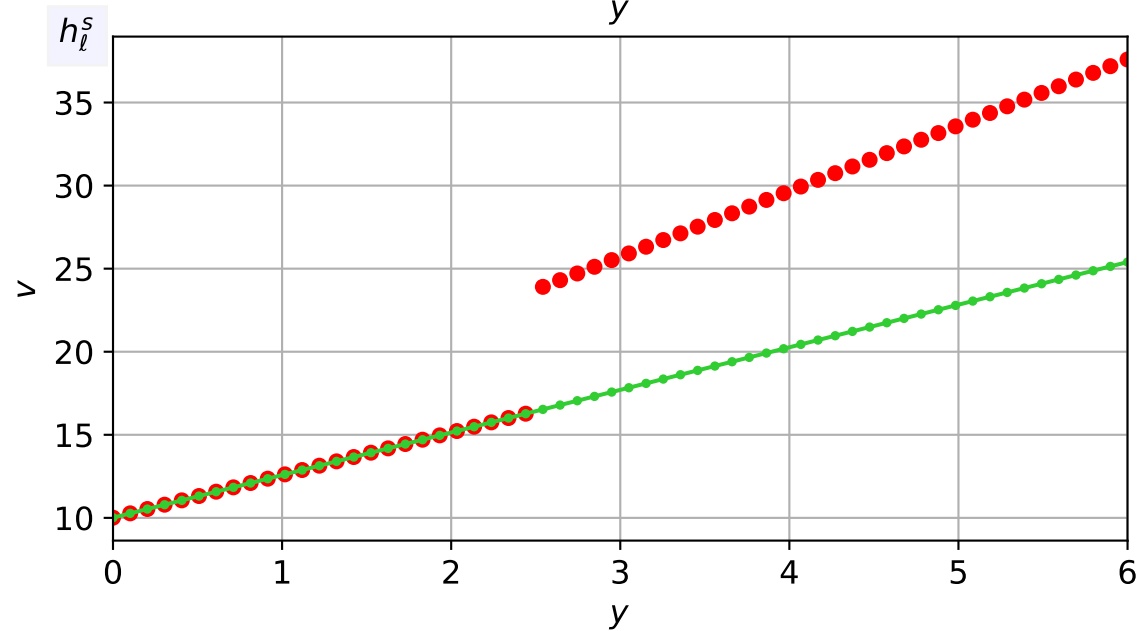
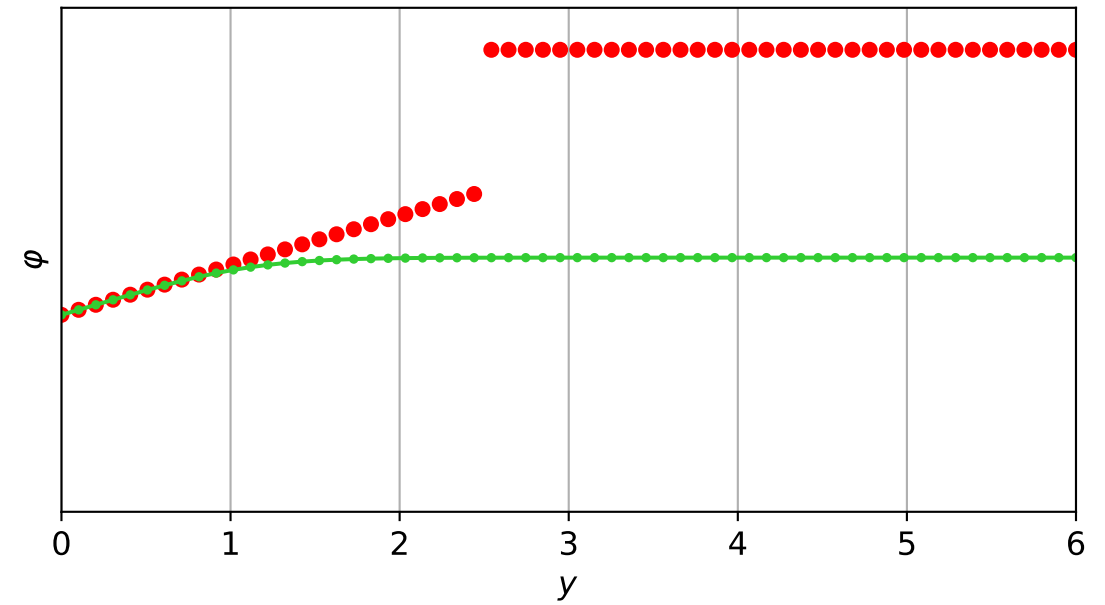
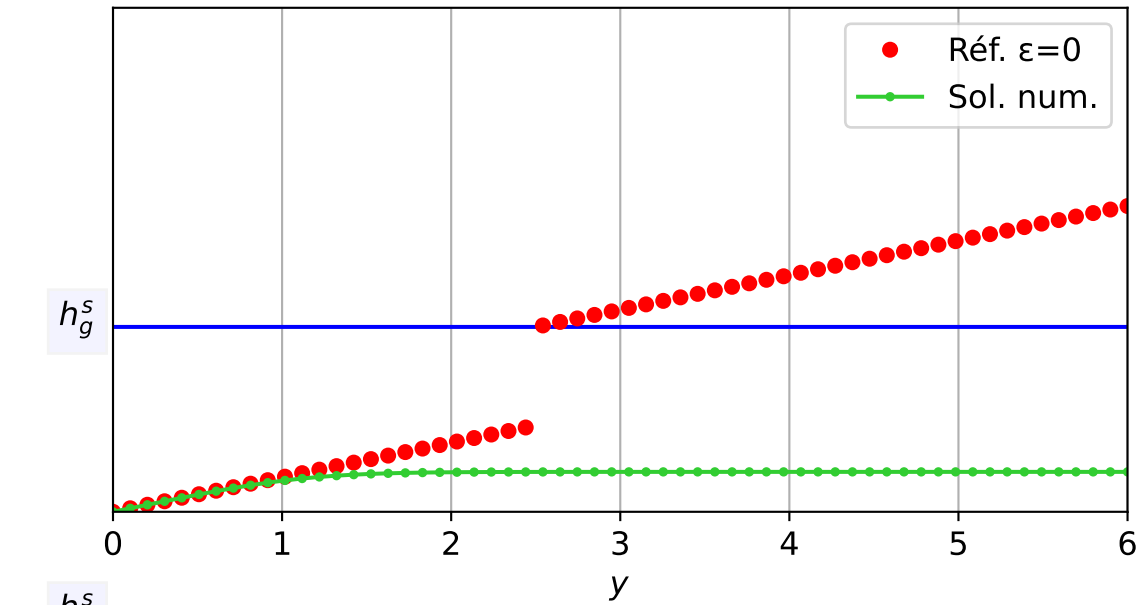
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 1.20e-04$  —  $t = 0.0001/12$  (1 iter) — résidu eq. =  $5.04e-11$



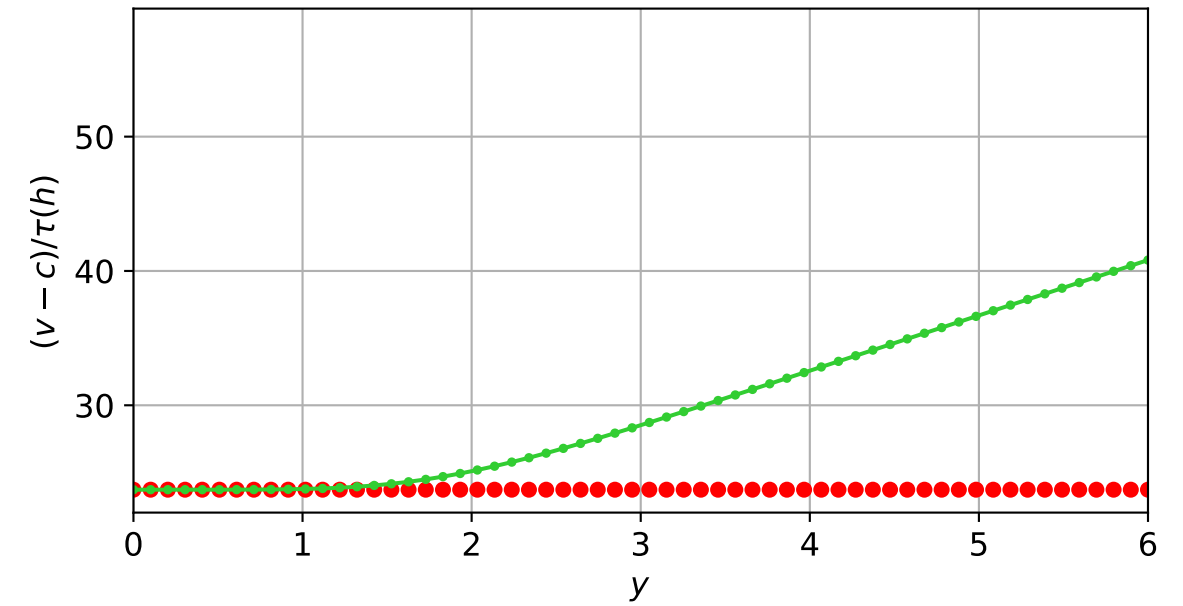
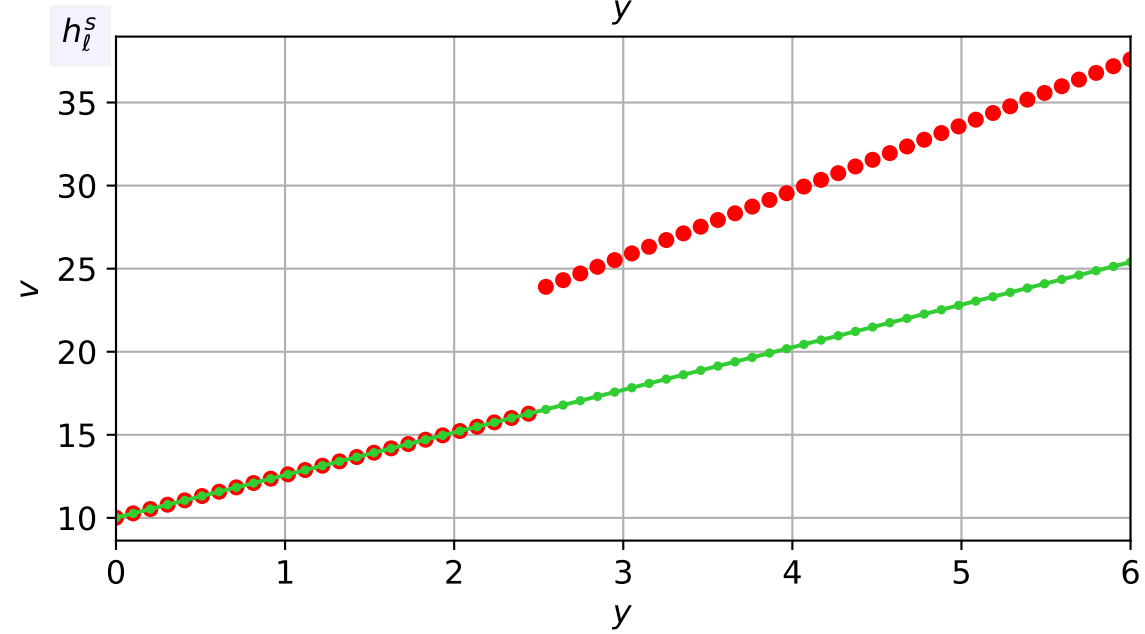
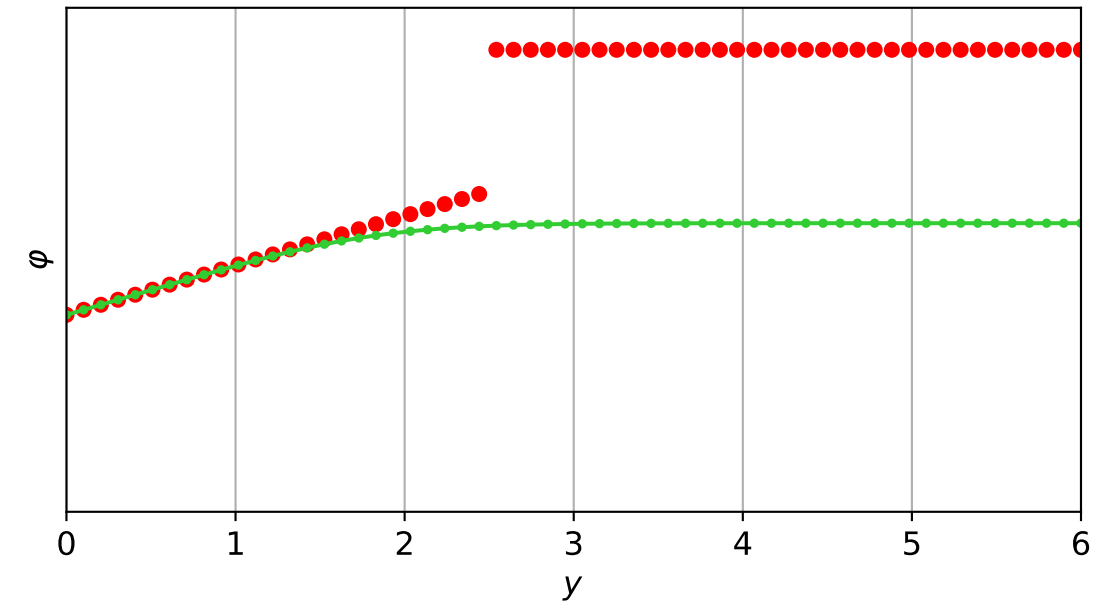
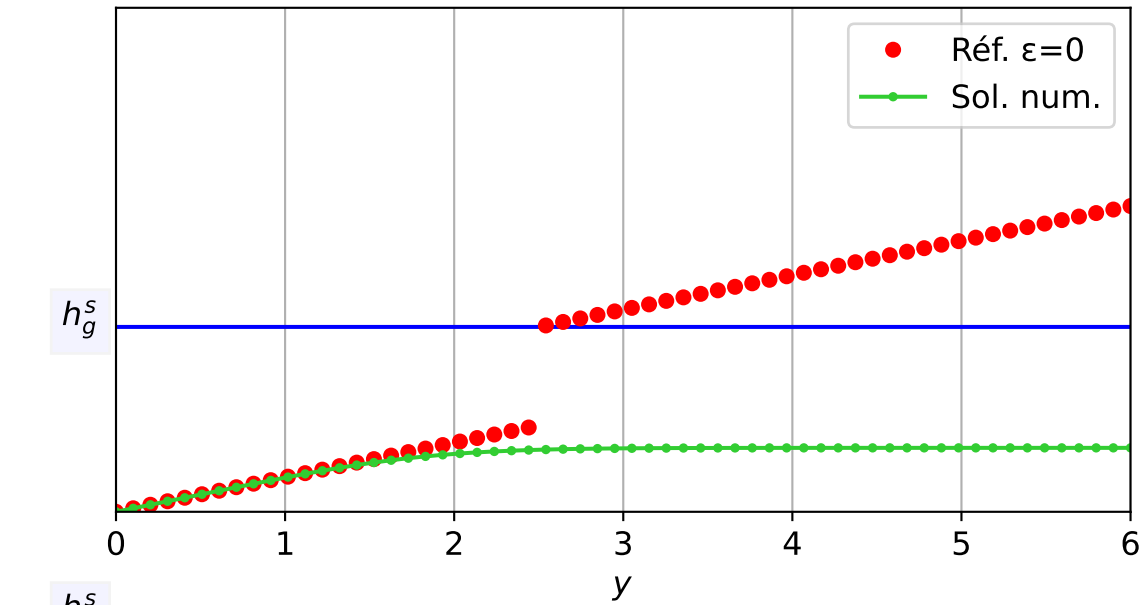
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.60e-03$  —  $t = 0.0511039/12$  (29 iter) — résidu eq. =  $1.88e-07$



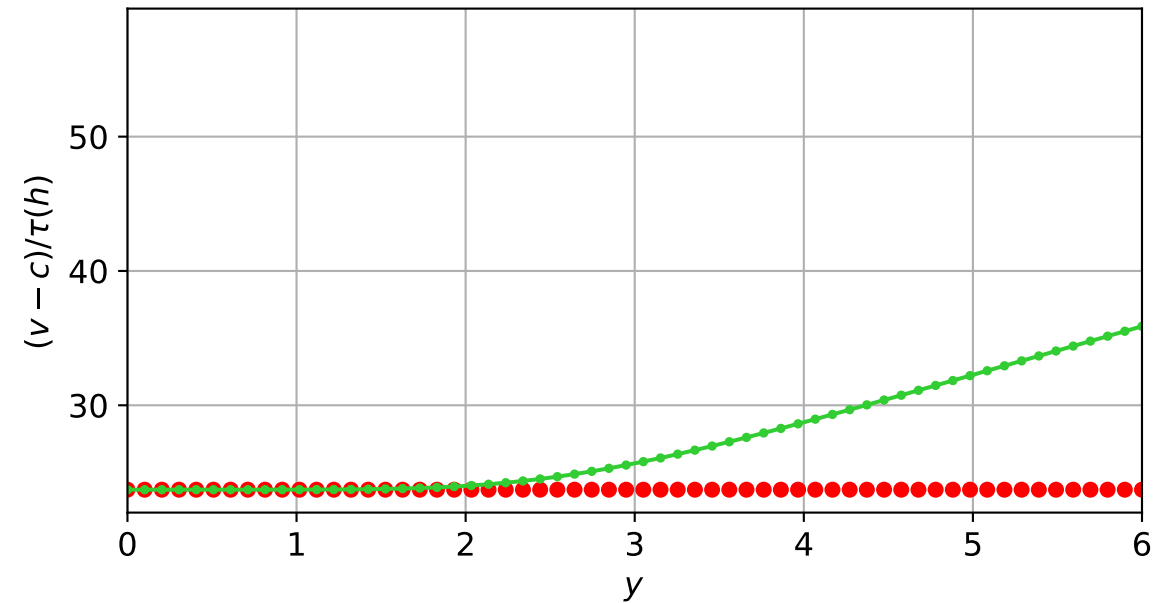
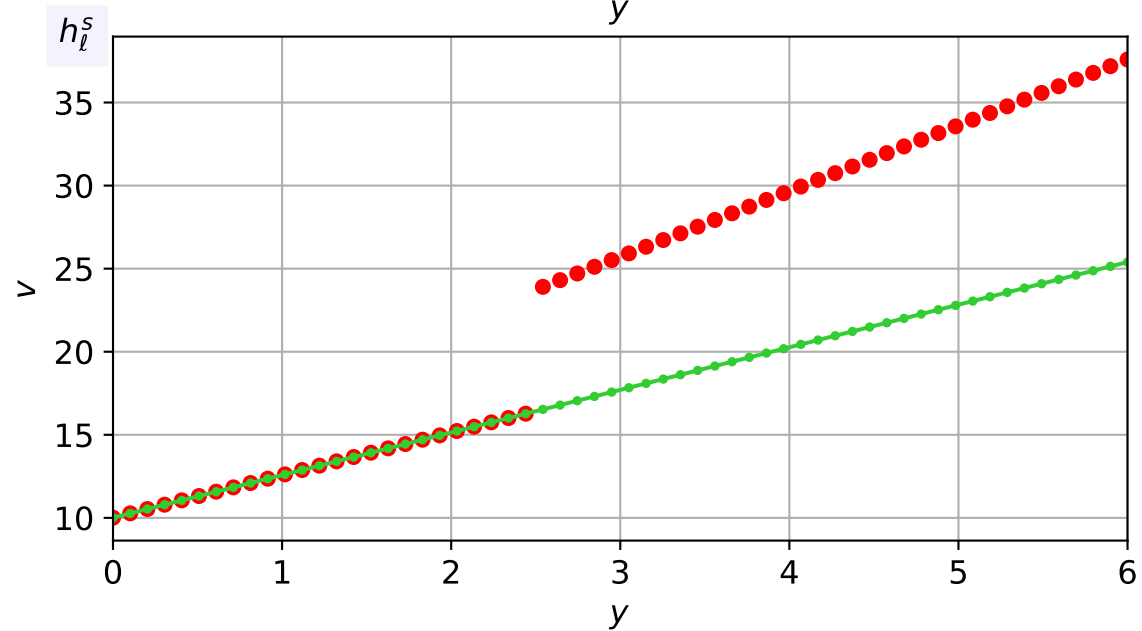
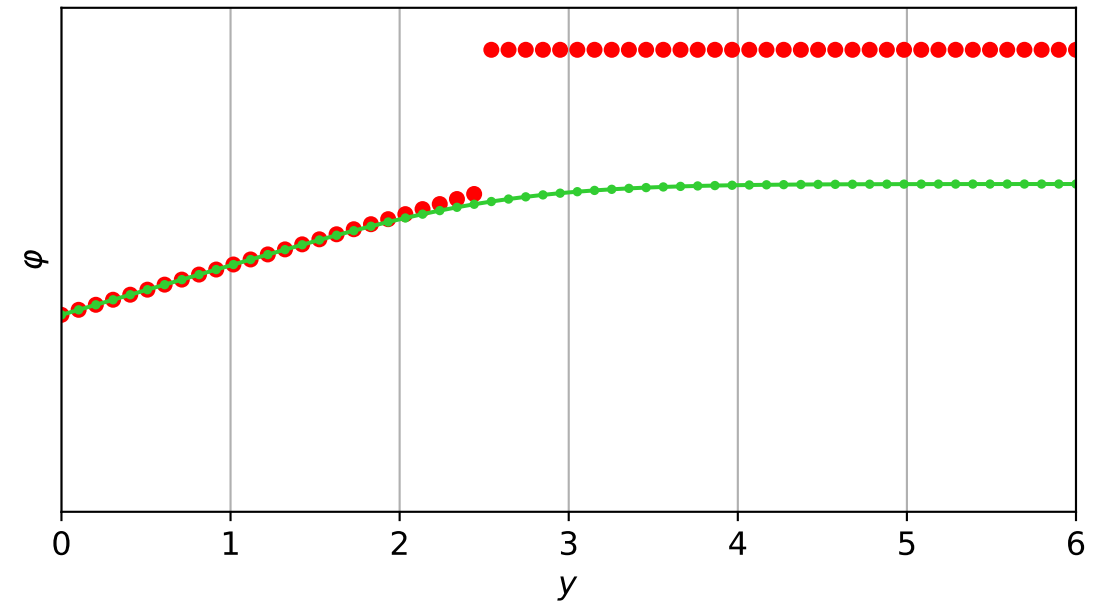
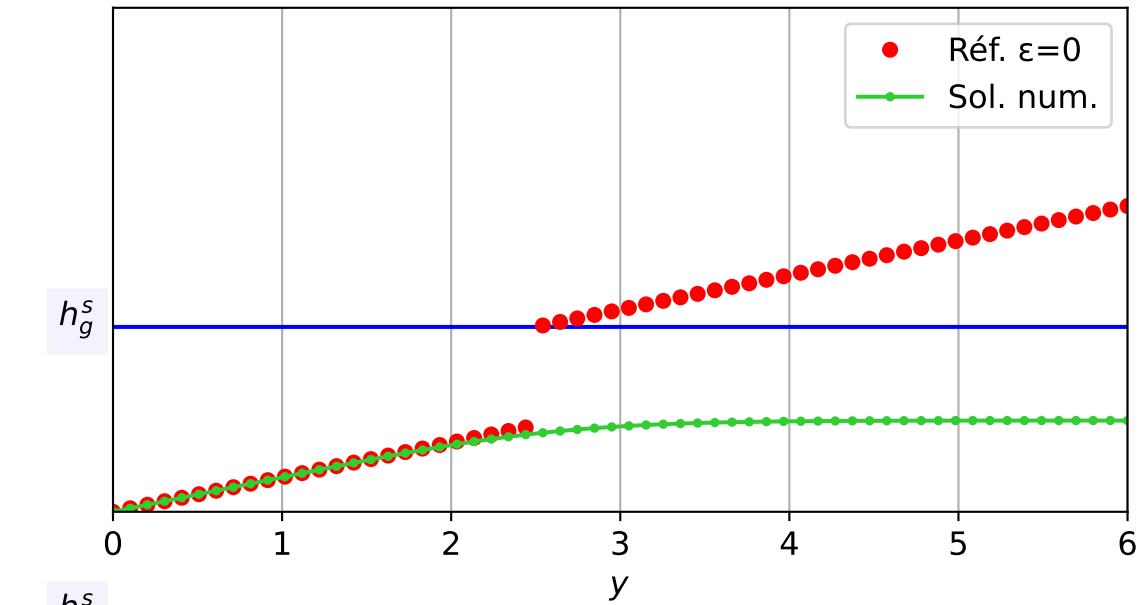
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.60e-03$  —  $t = 0.101558/12$  (43 iter) — résidu eq. =  $1.66e-07$



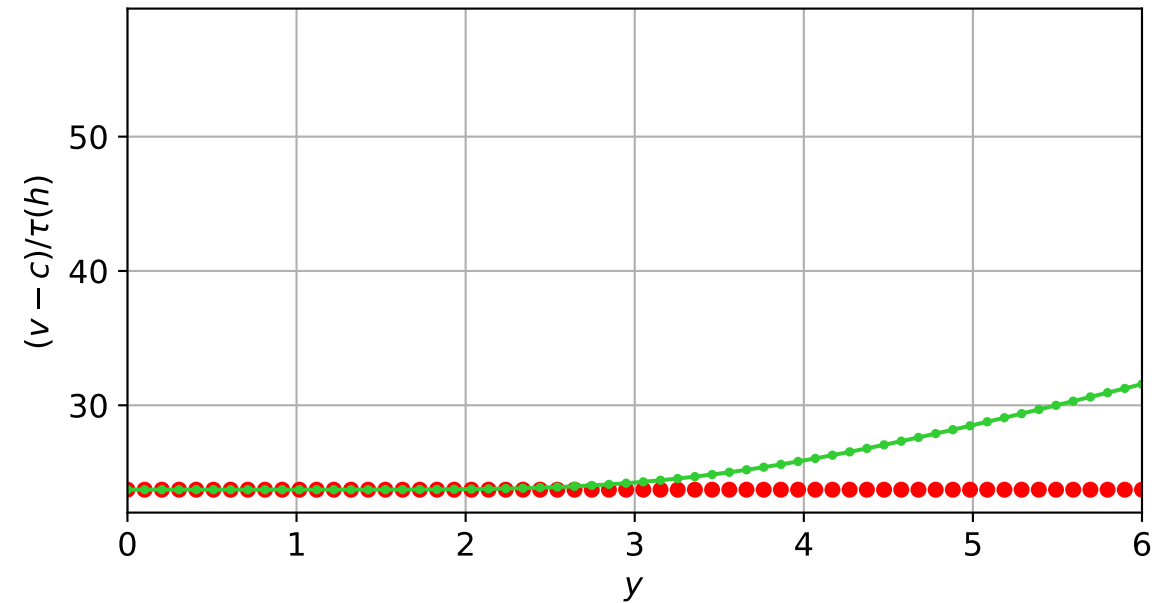
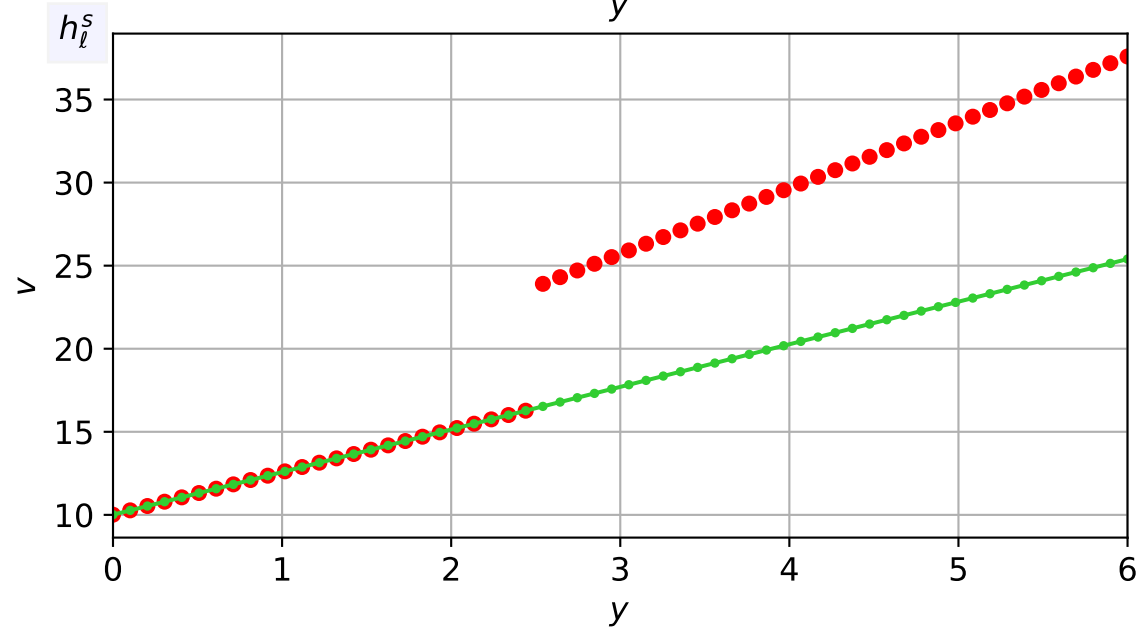
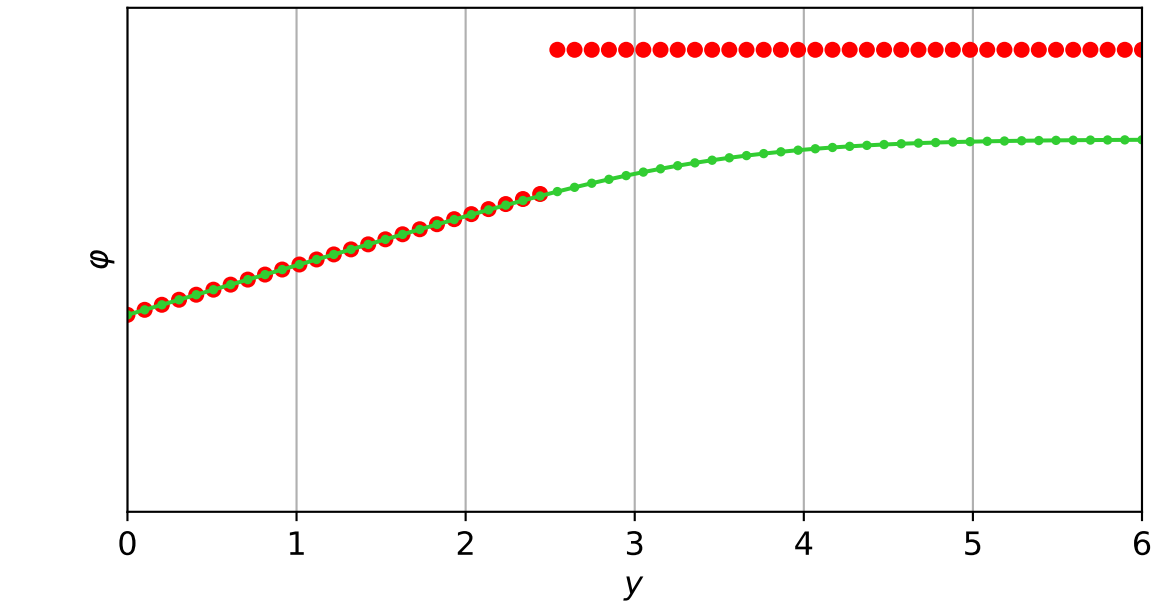
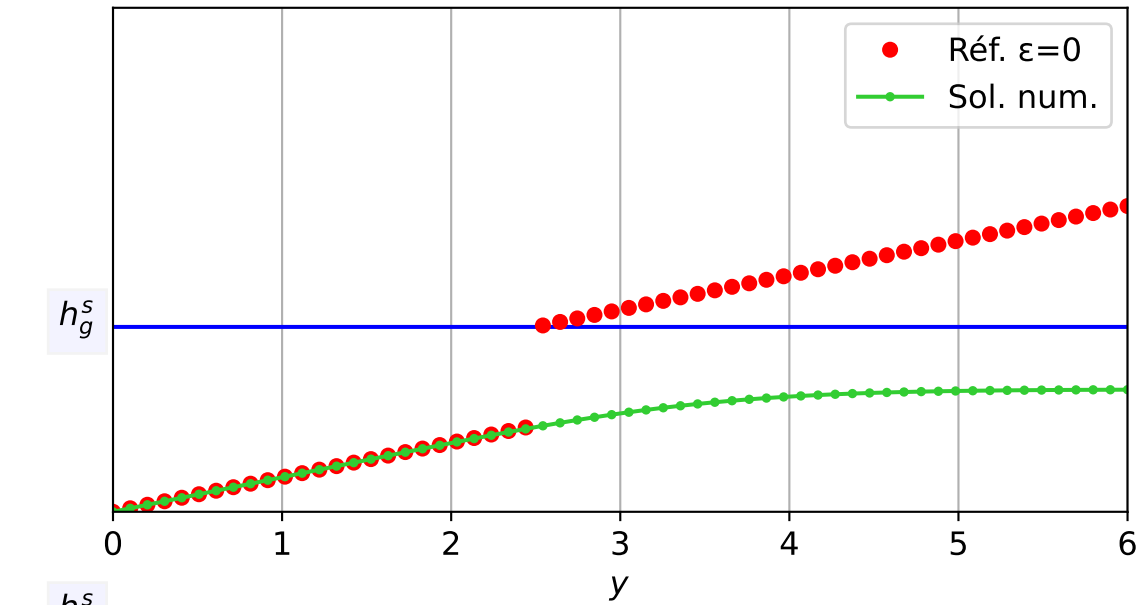
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.60e-03$  —  $t = 0.152012/12$  (57 iter) — résidu eq. =  $2.54e-07$



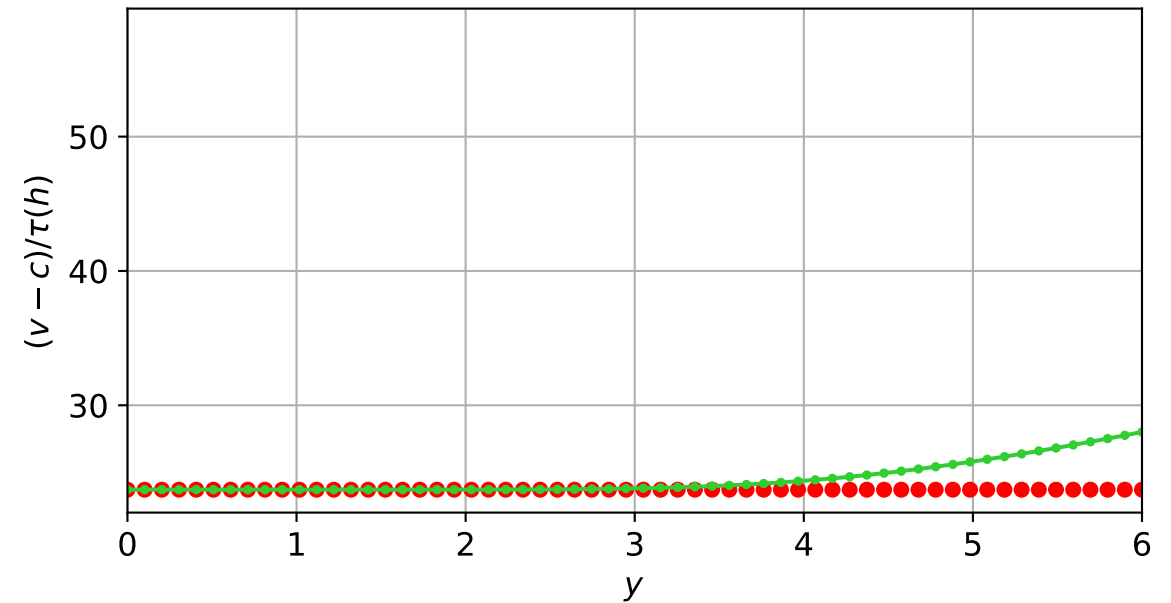
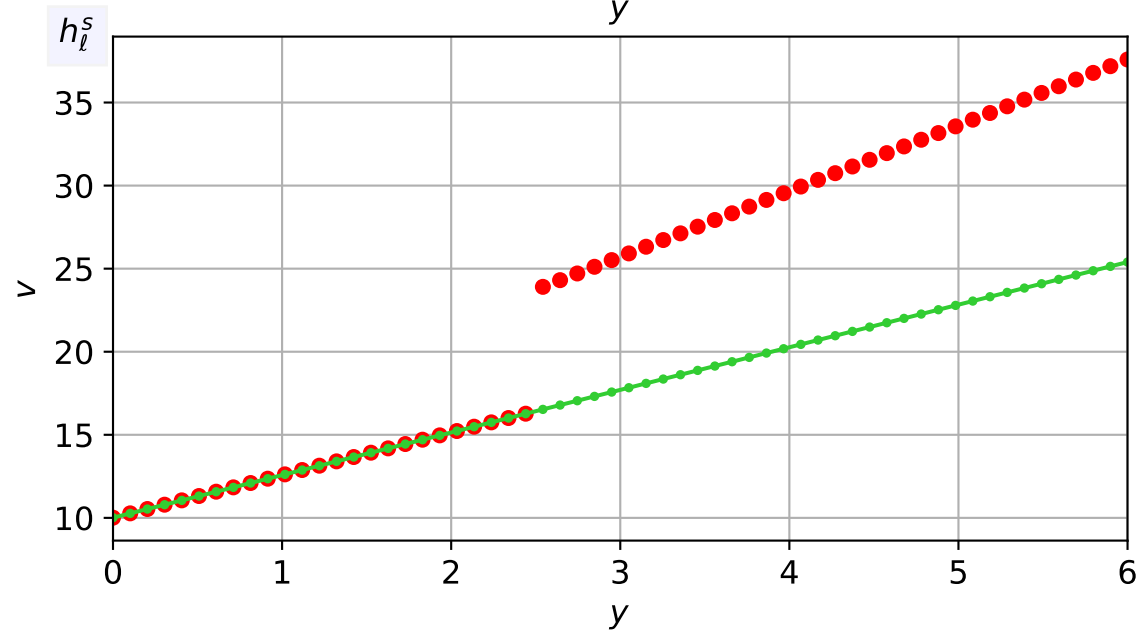
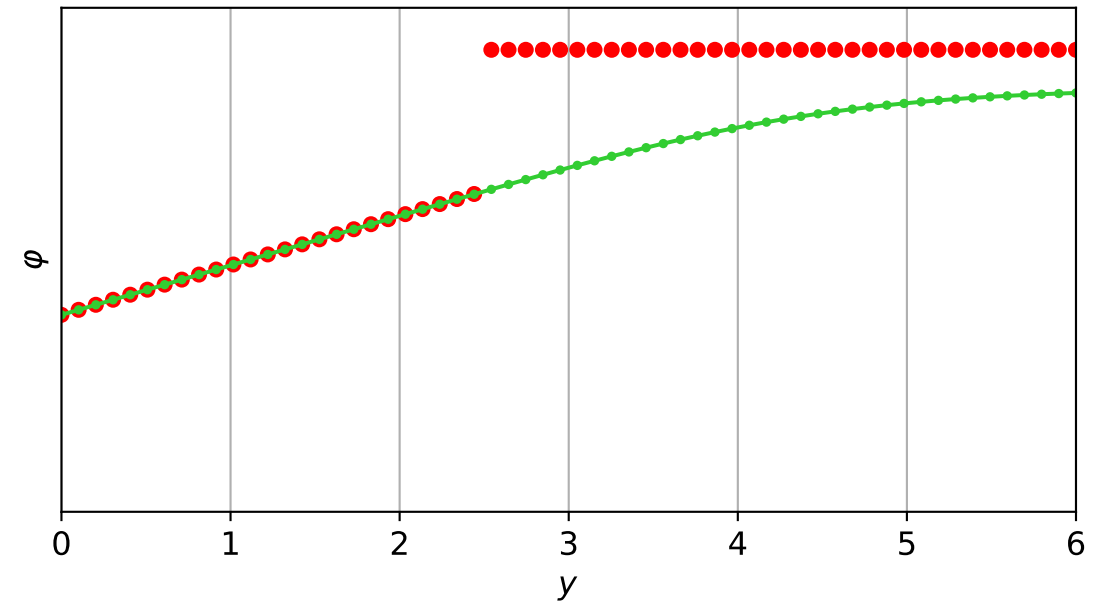
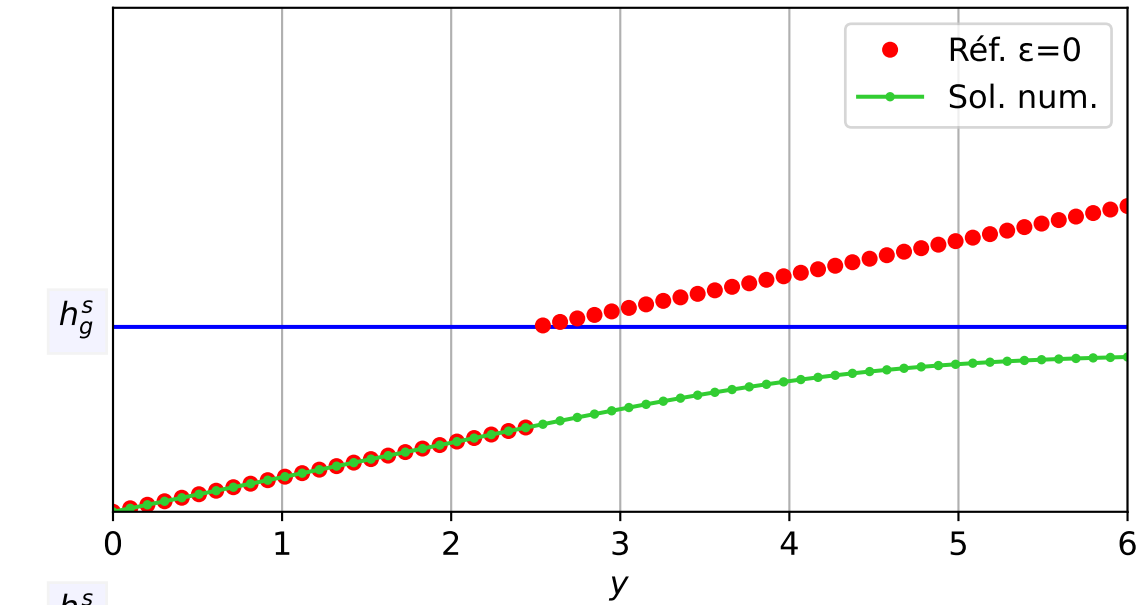
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.60e-03$  —  $t = 0.202466/12$  (71 iter) — résidu eq. =  $5.05e-07$



TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\epsilon = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.60e-03$  —  $t = 0.25292/12$  (85 iter) — résidu eq. =  $4.02e-06$

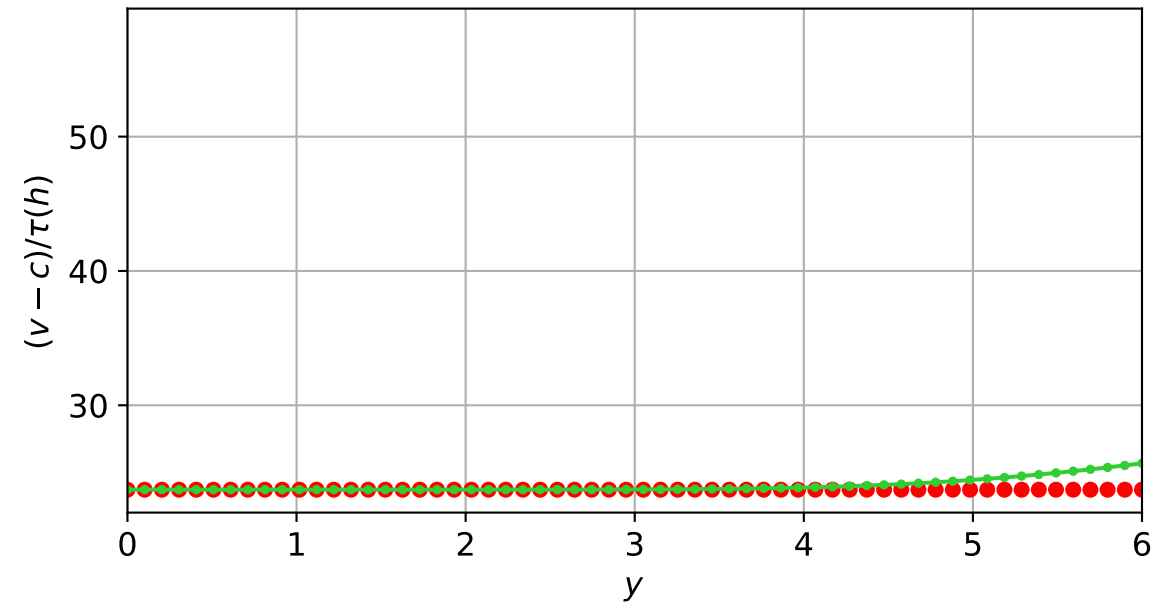
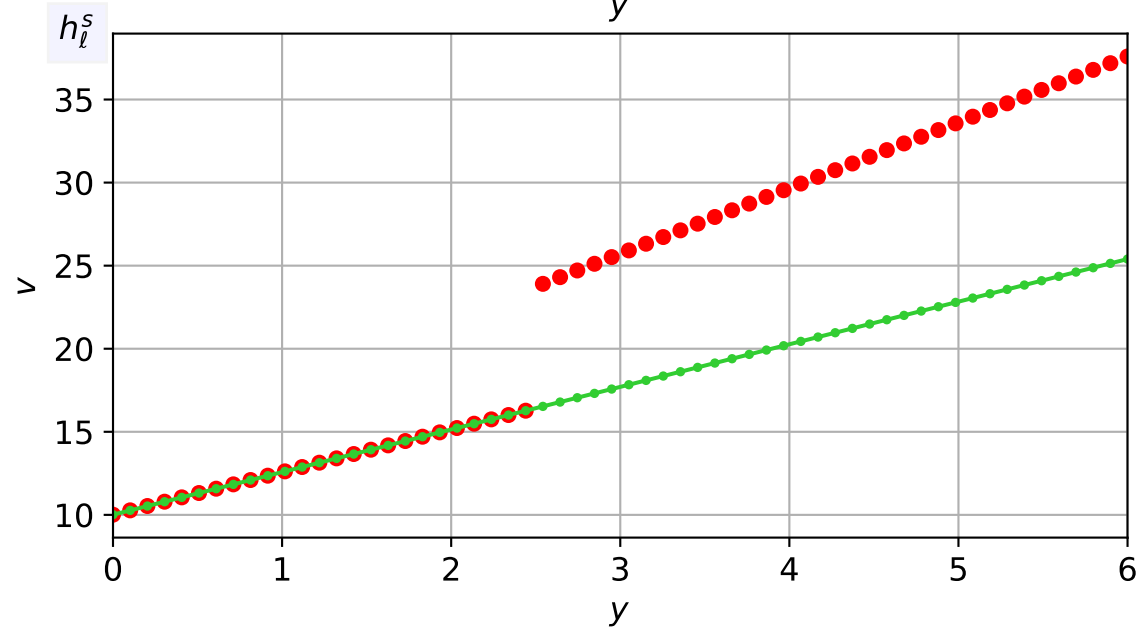
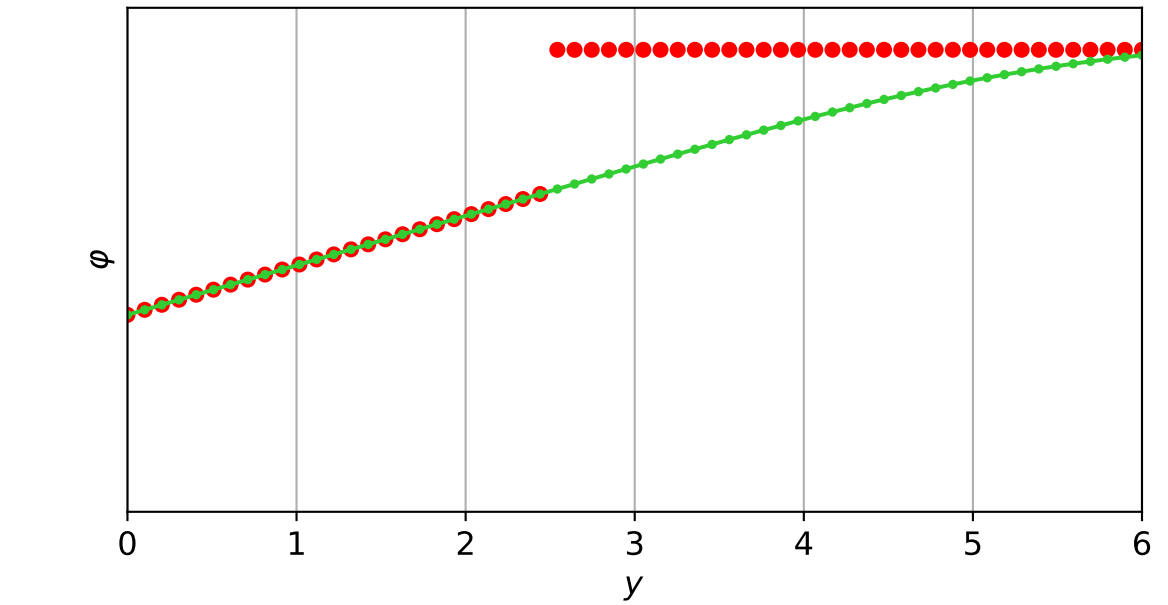
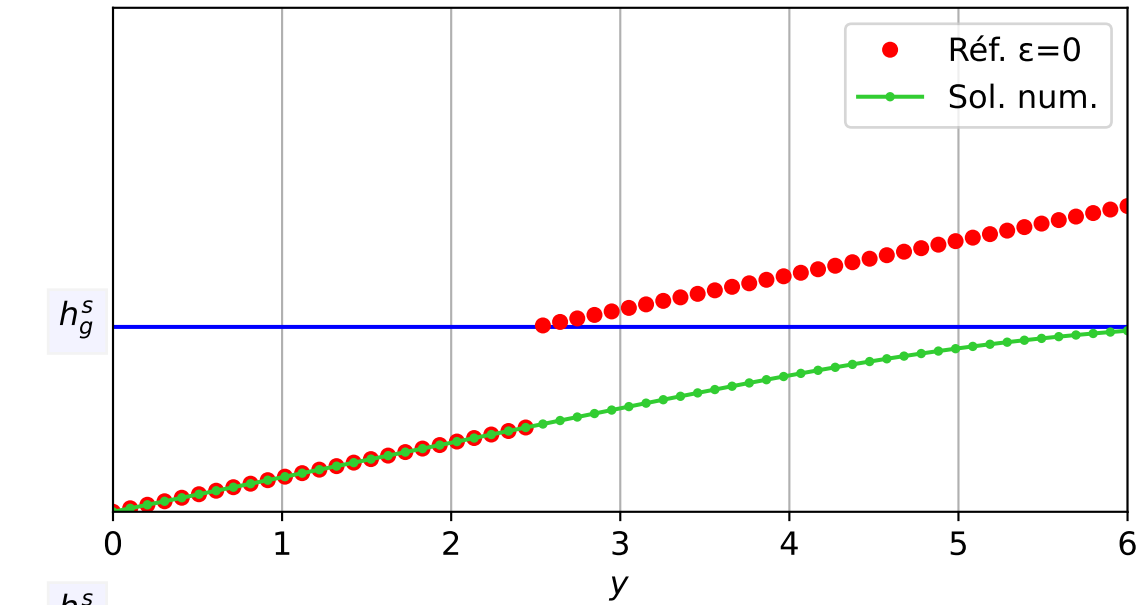


TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\epsilon = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.60e-03$  —  $t = 0.303374/12$  (99 iter) — résidu eq. =  $4.29e-07$

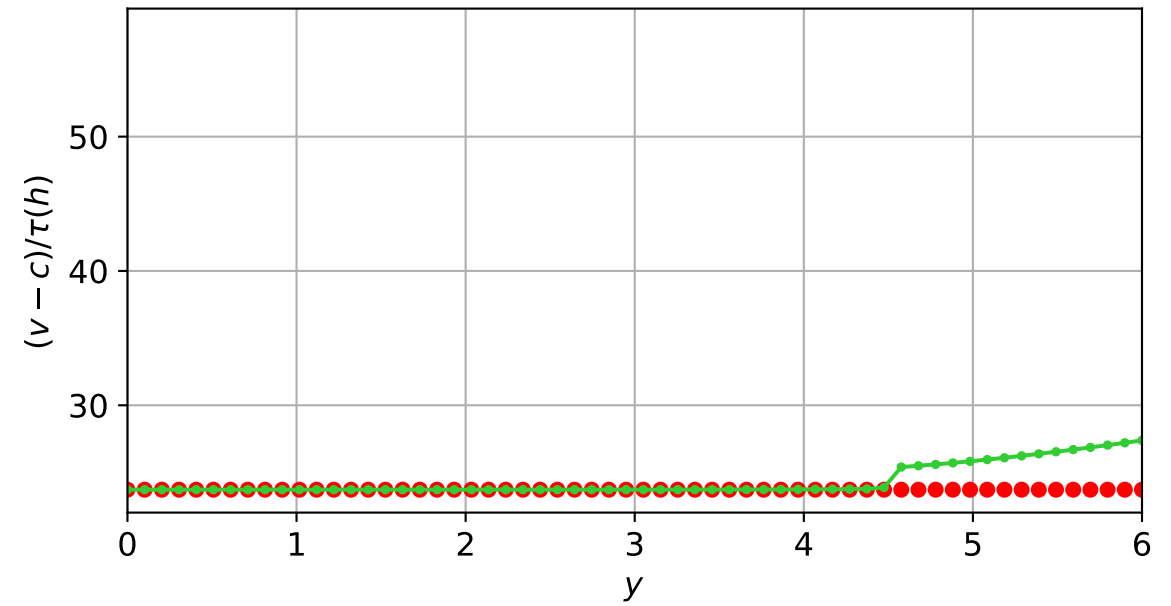
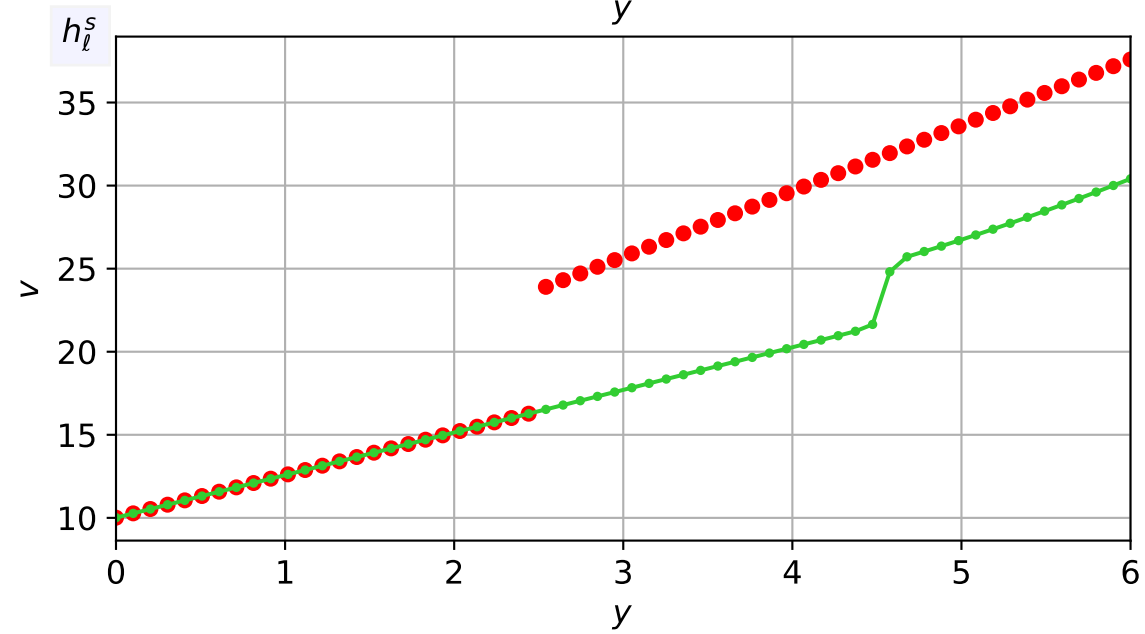
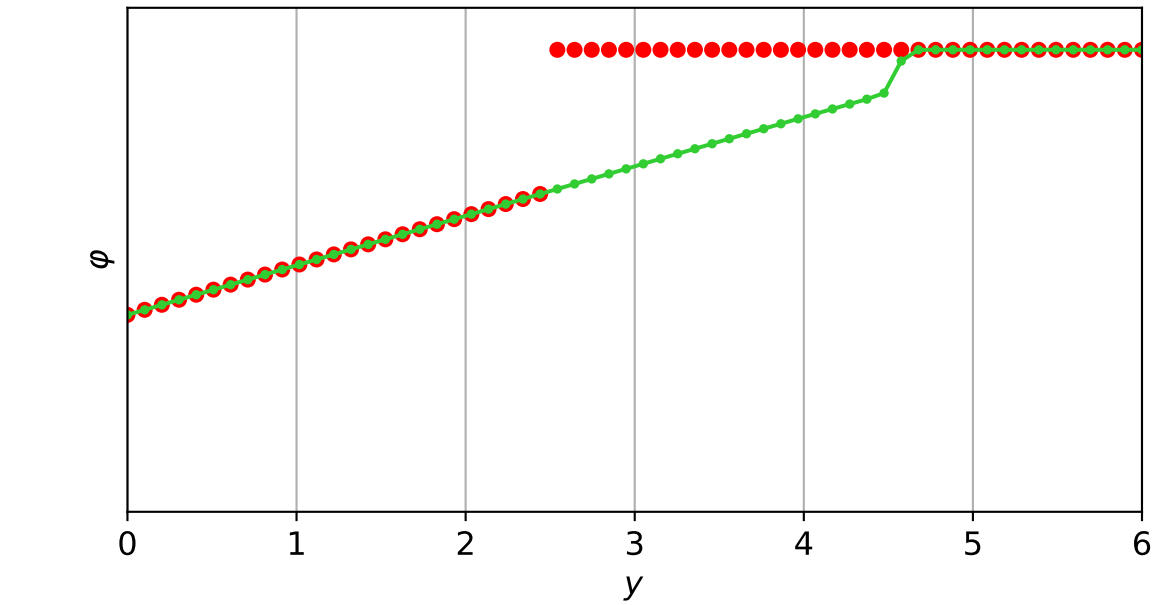
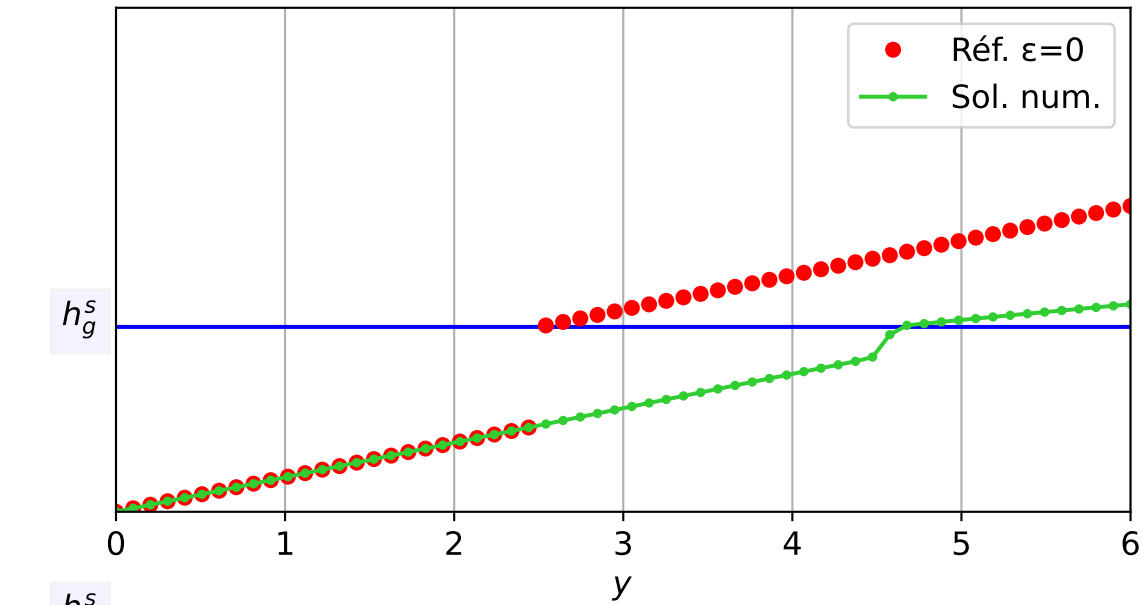




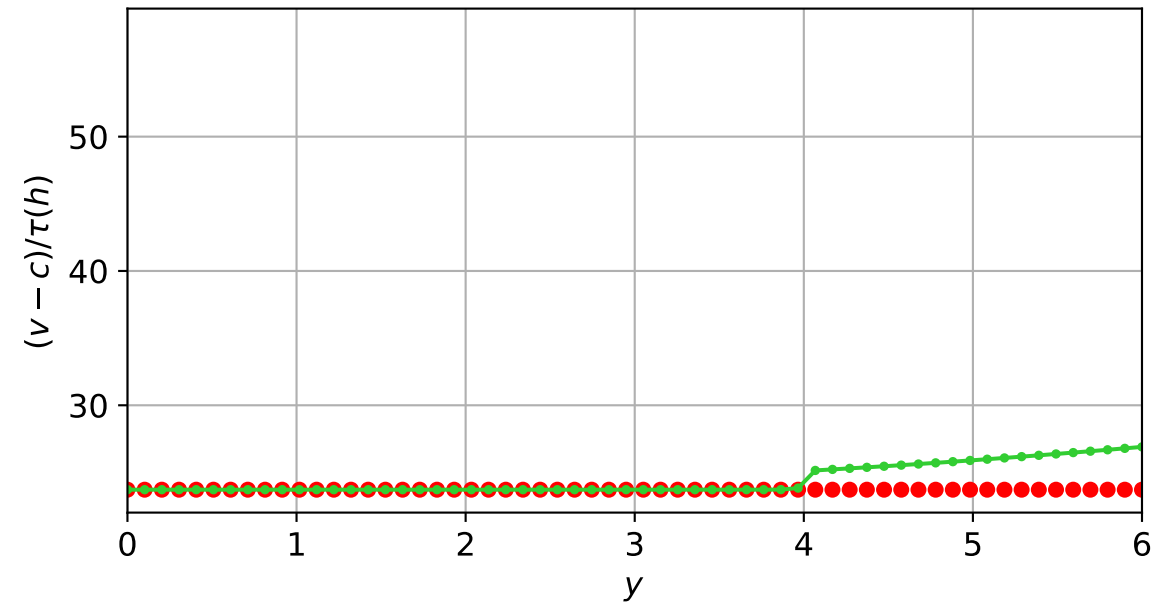
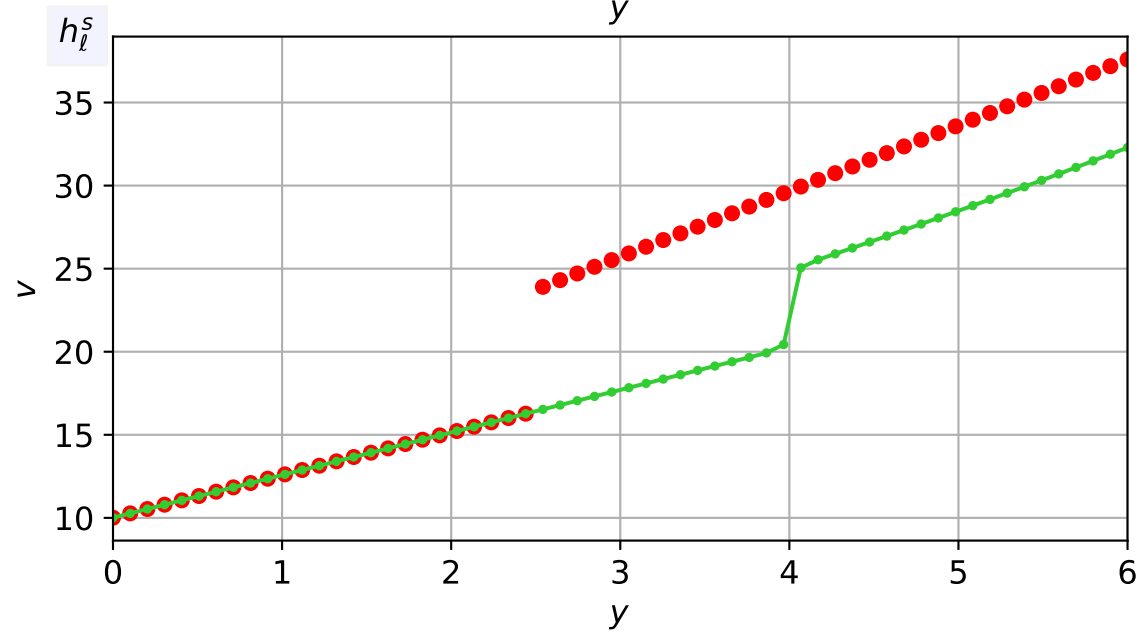
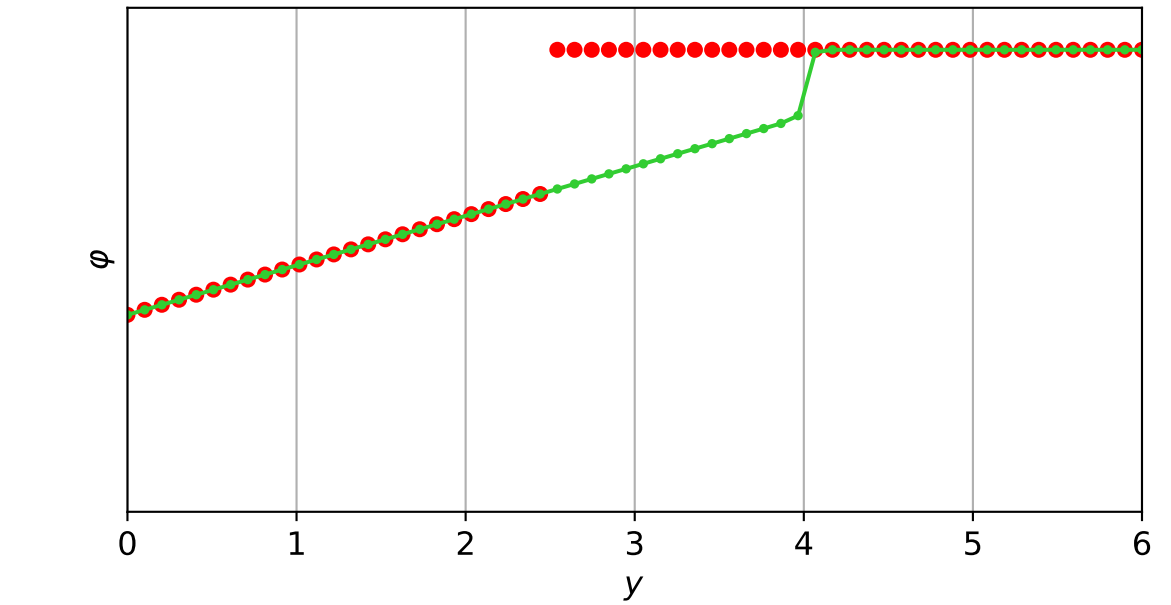
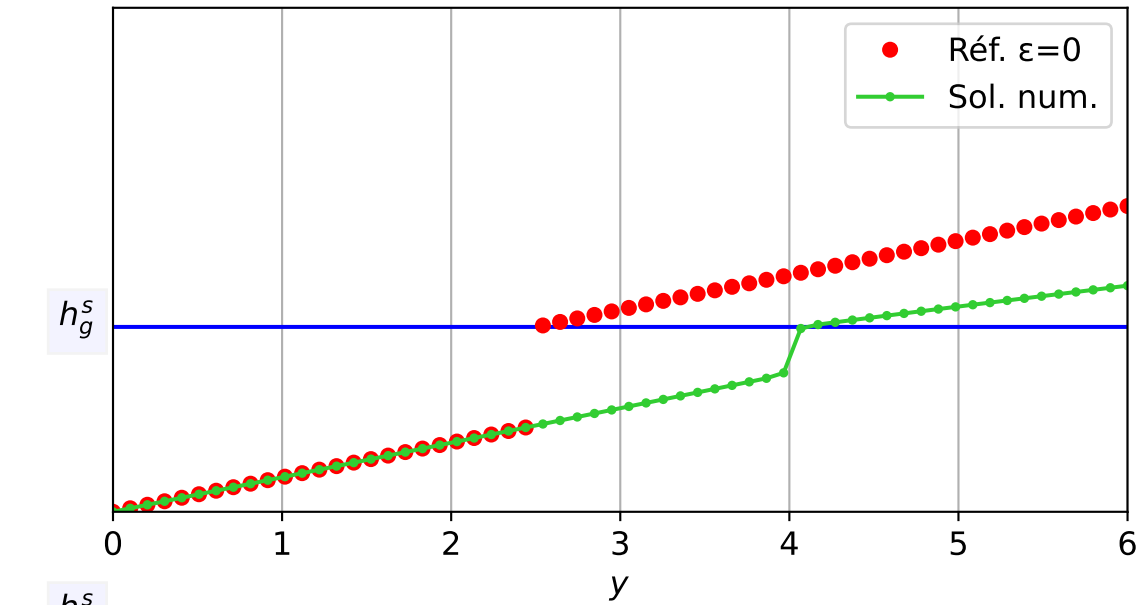
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\epsilon = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.60e-03$  —  $t = 0.350223/12$  (112 iter) — résidu eq. =  $8.08e-06$



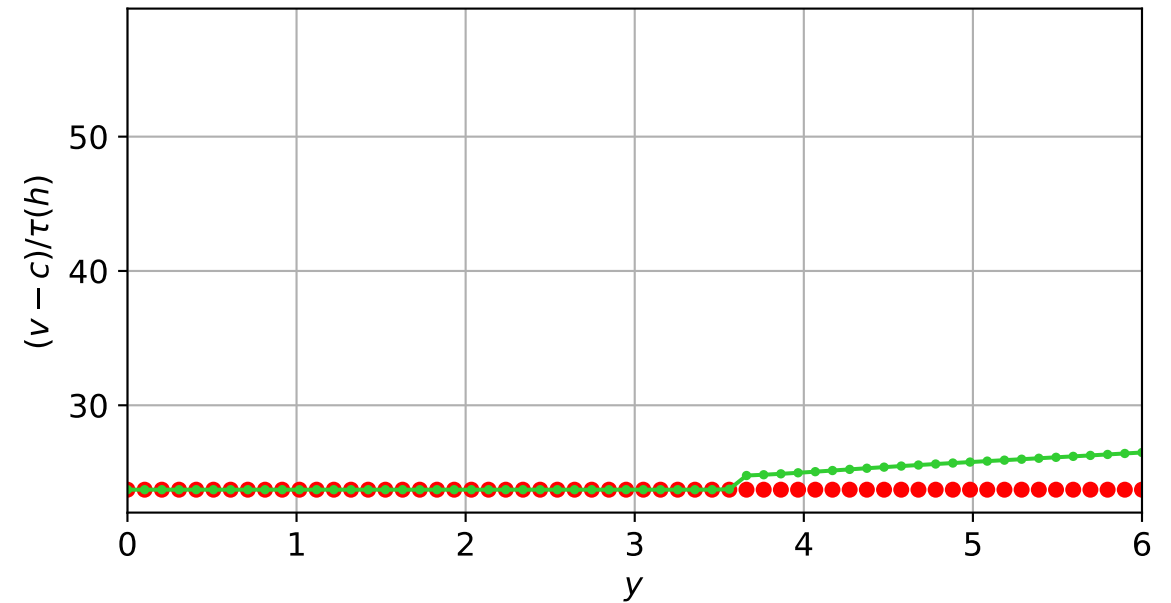
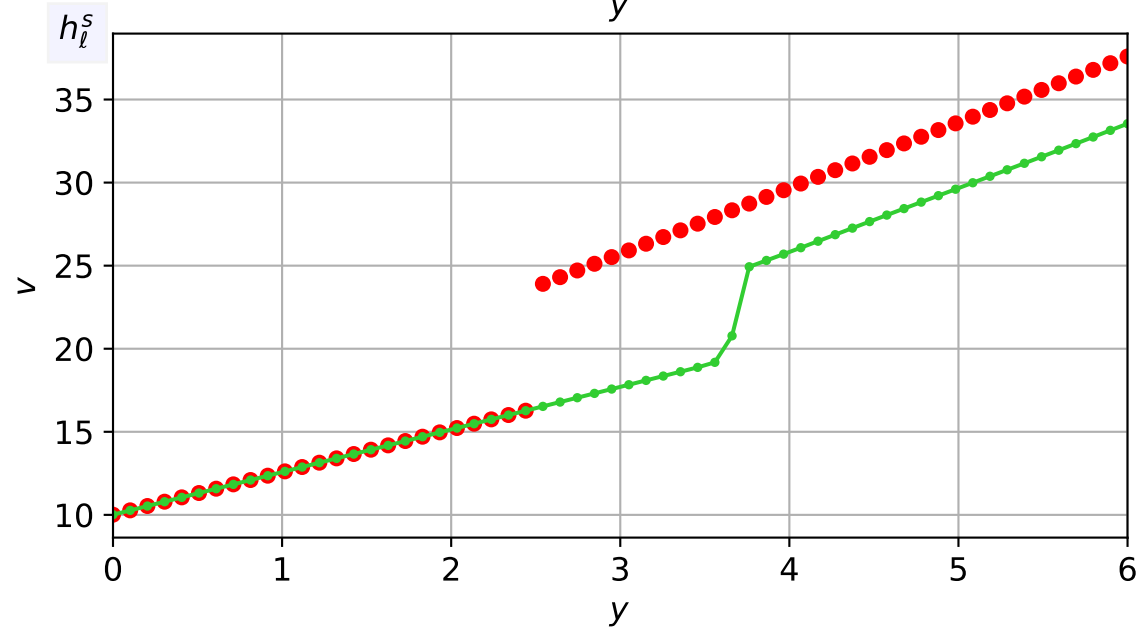
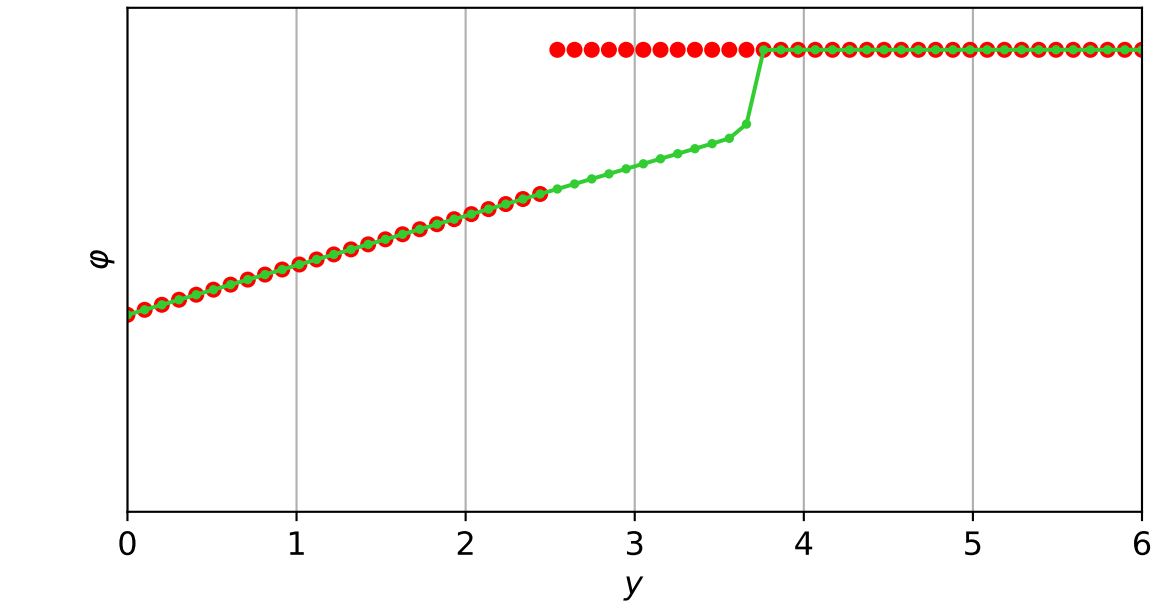
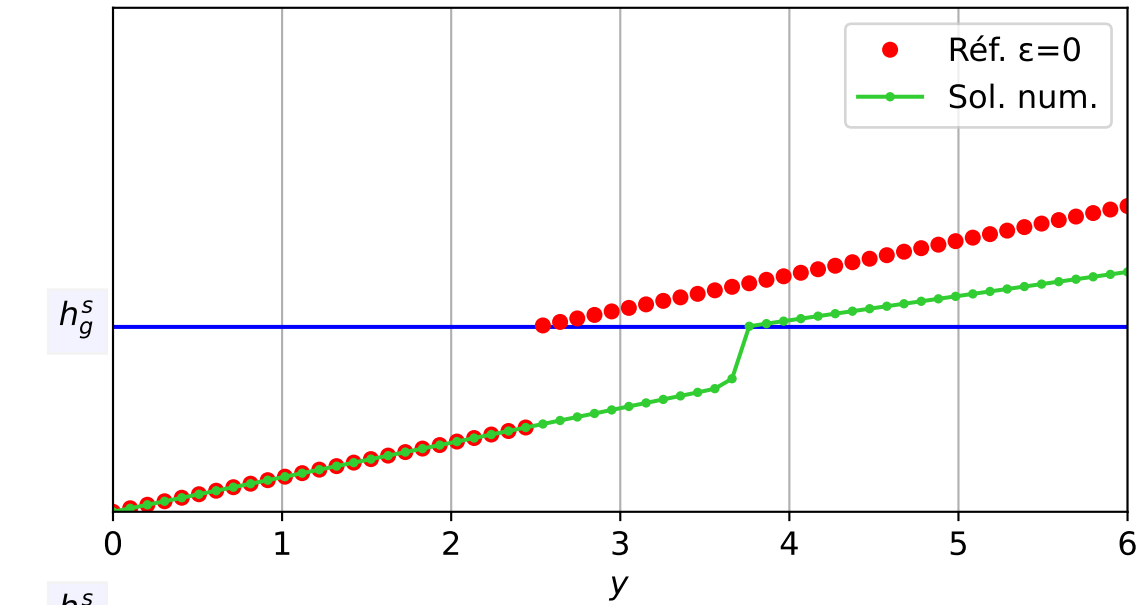
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\epsilon = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 0.400224/12$  (278 iter) — résidu eq. =  $7.47e-06$



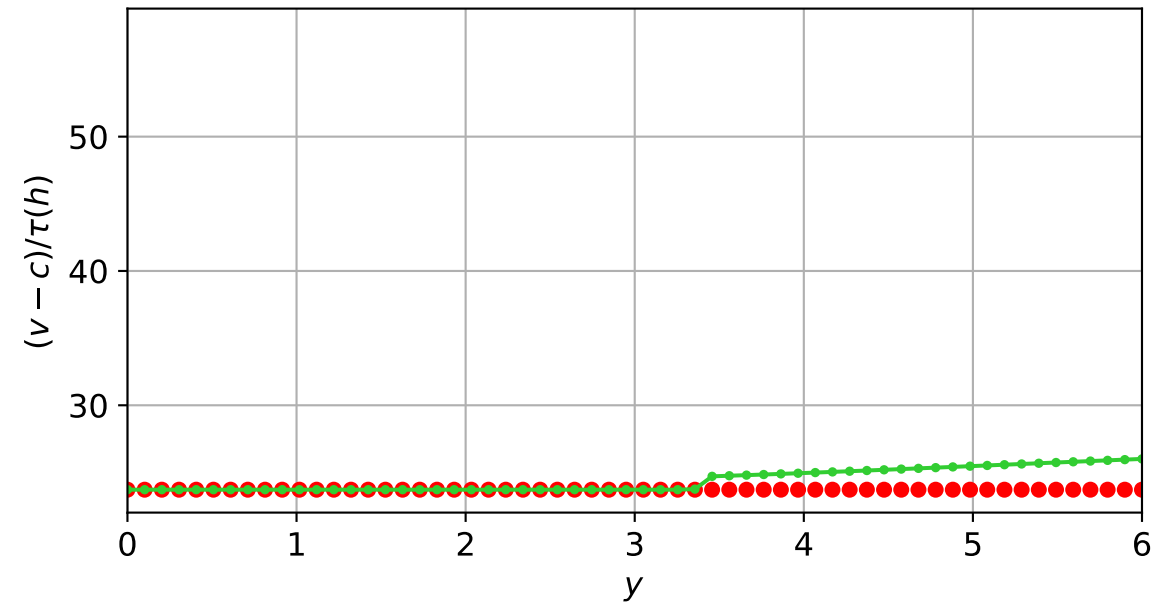
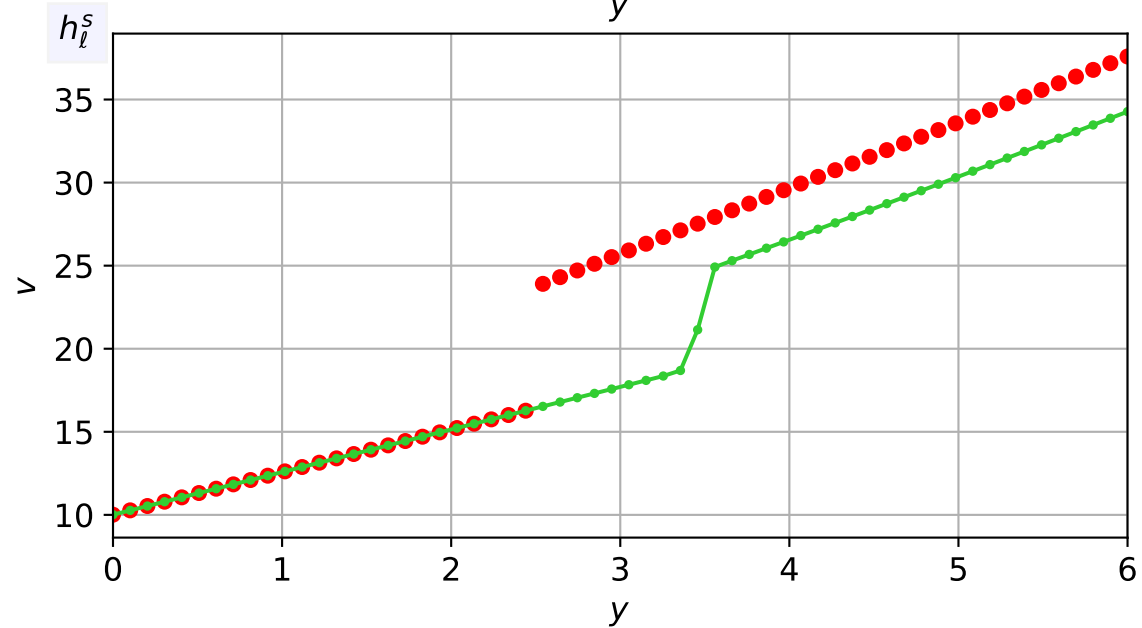
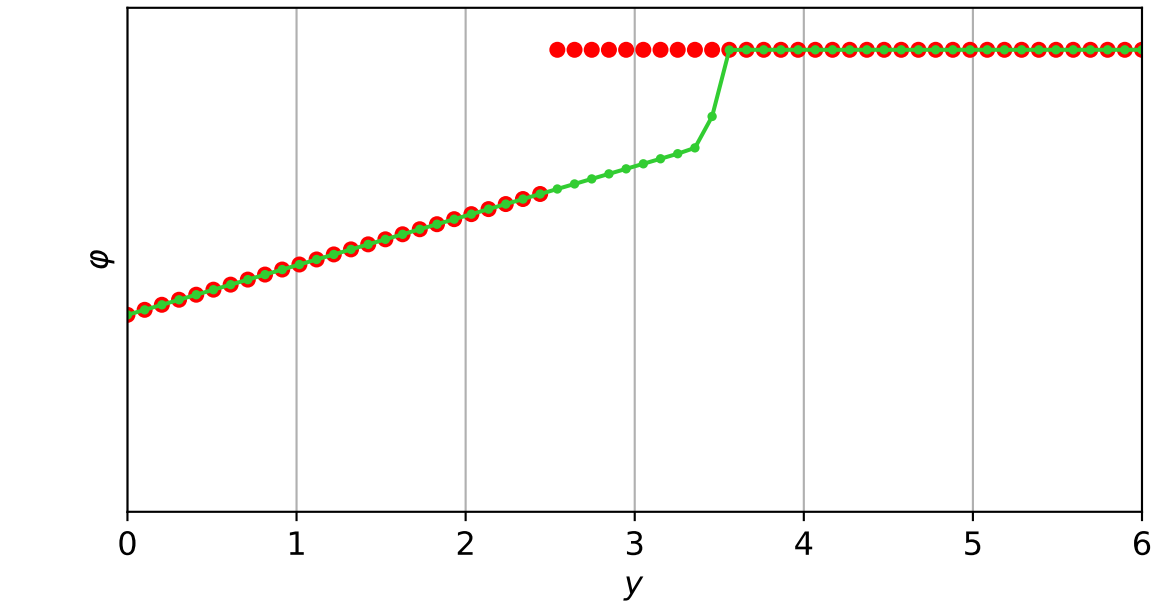
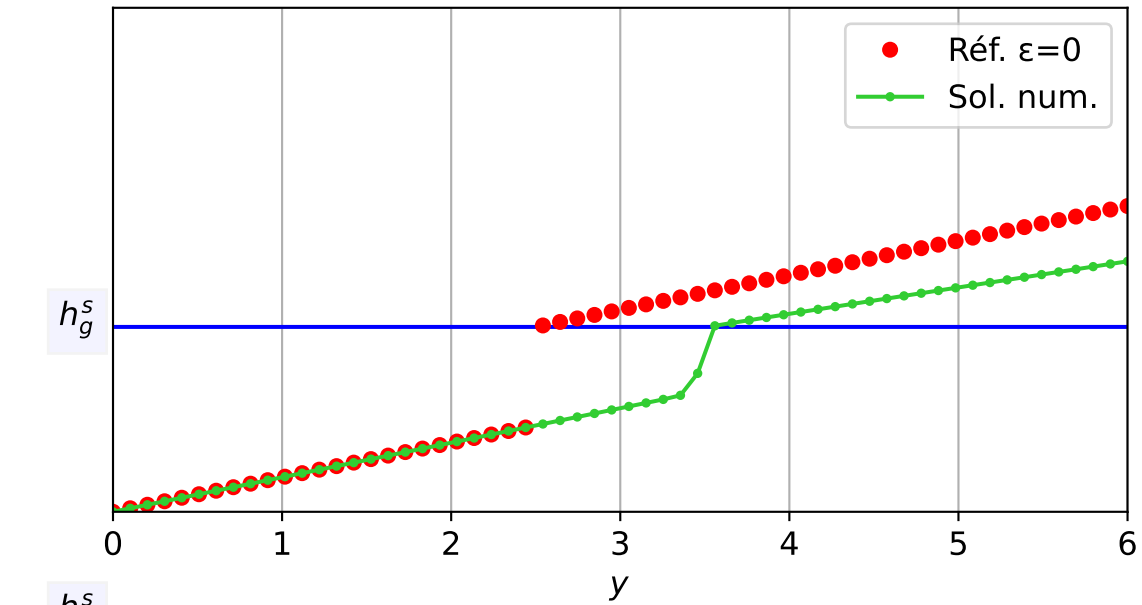
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\epsilon = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 0.450001/12$  (499 iter) — résidu eq. =  $6.08e-06$



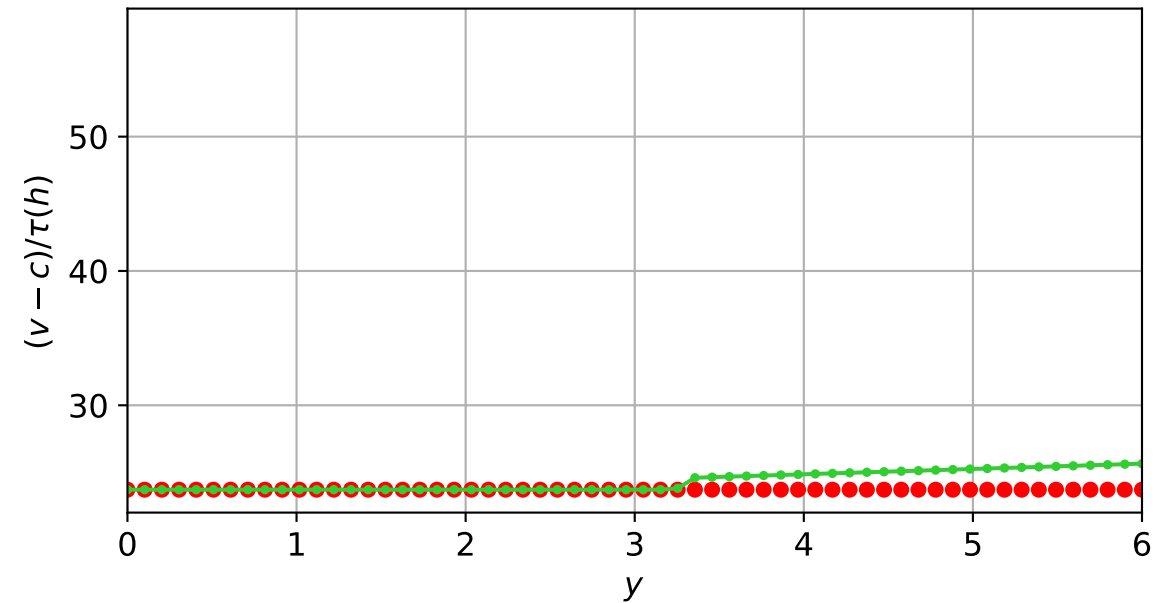
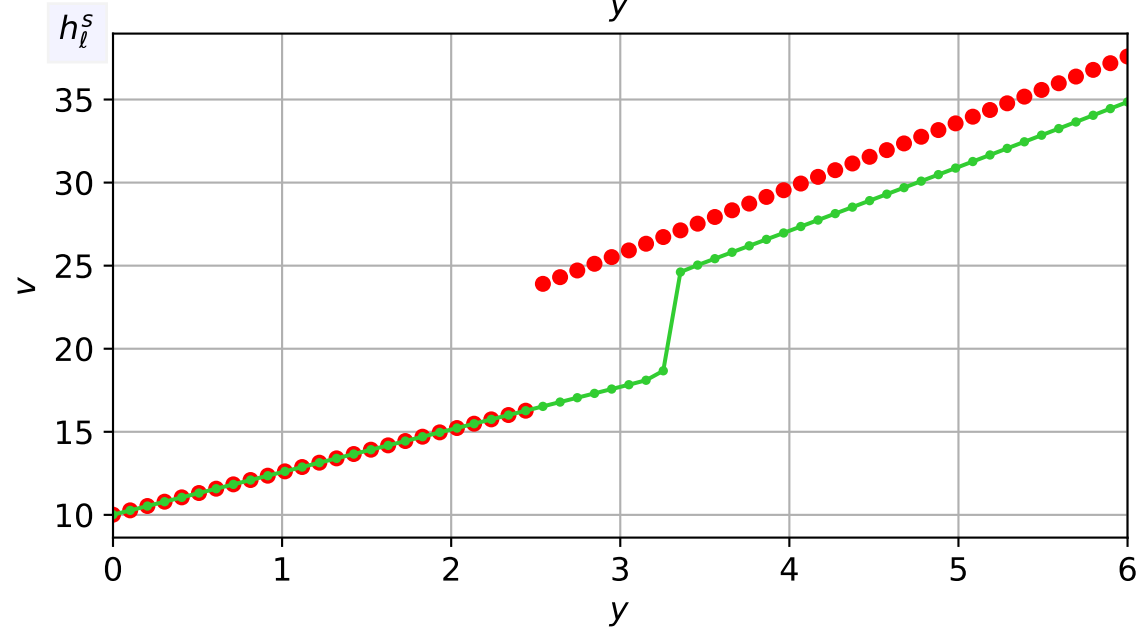
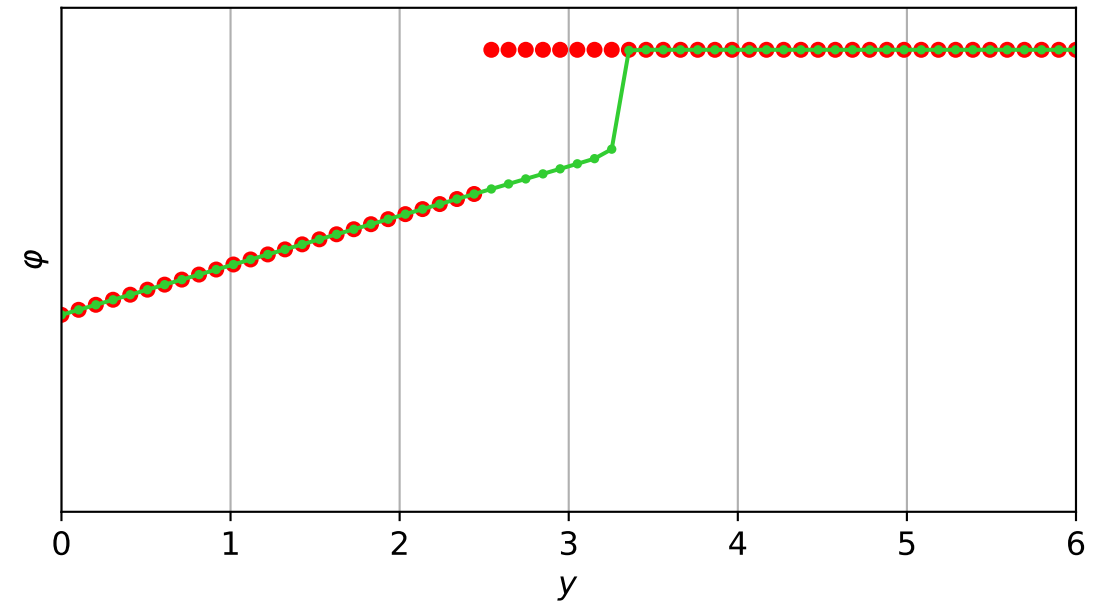
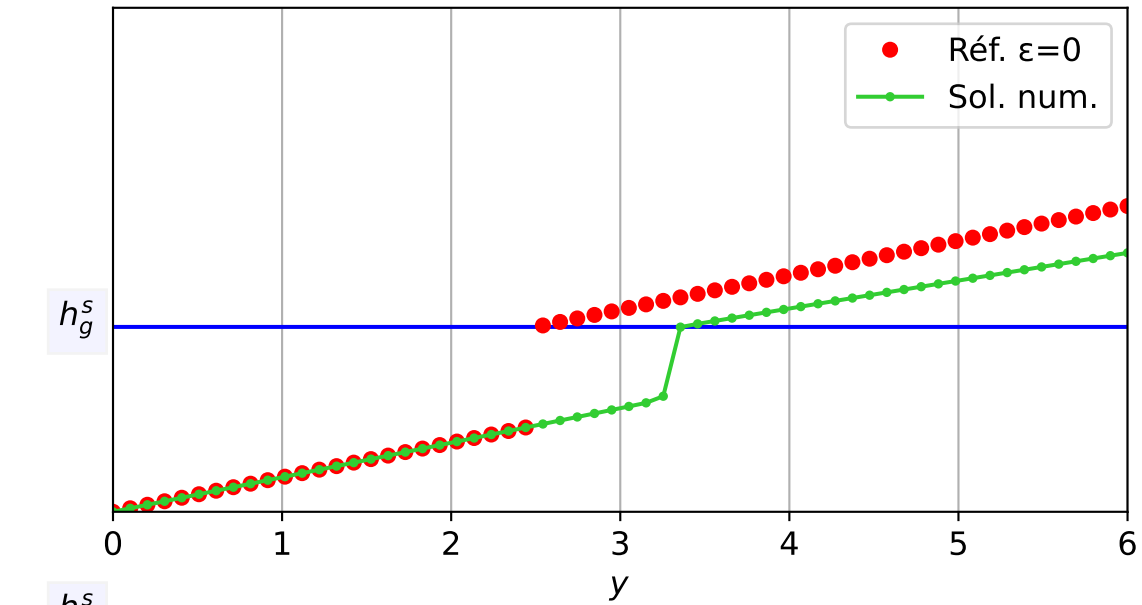
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\epsilon = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 0.500003/12$  (721 iter) — résidu eq. =  $9.09e-06$



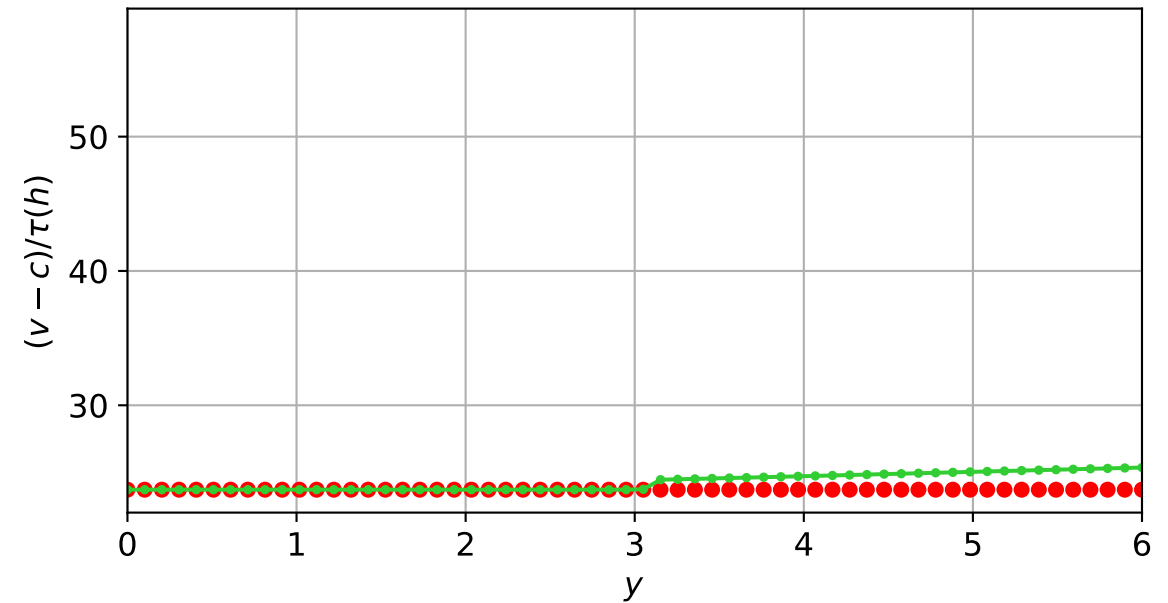
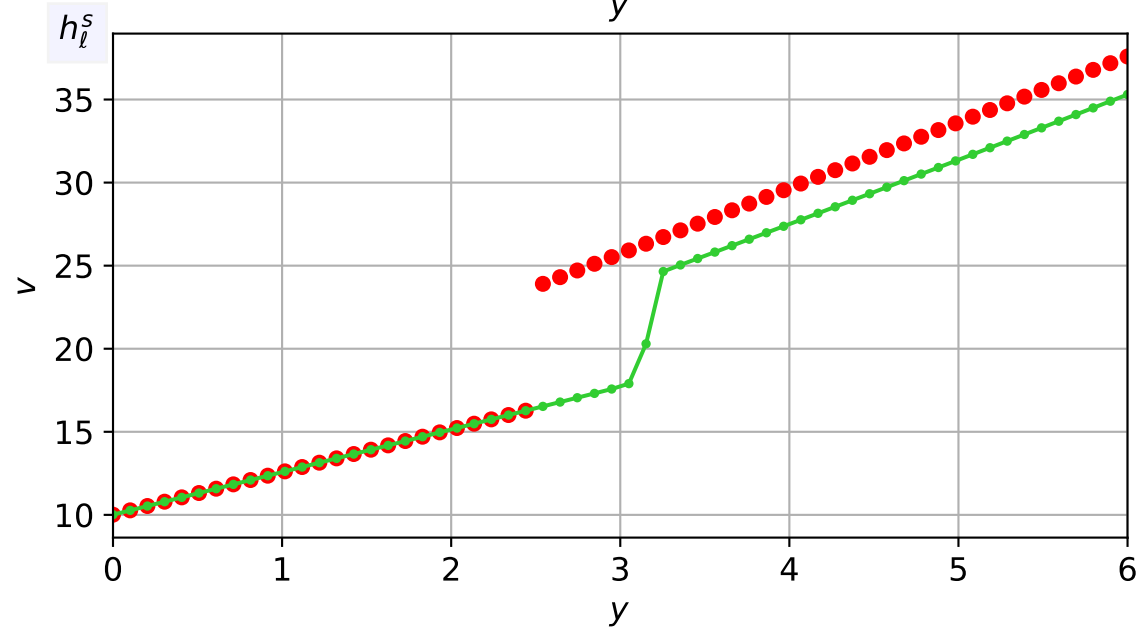
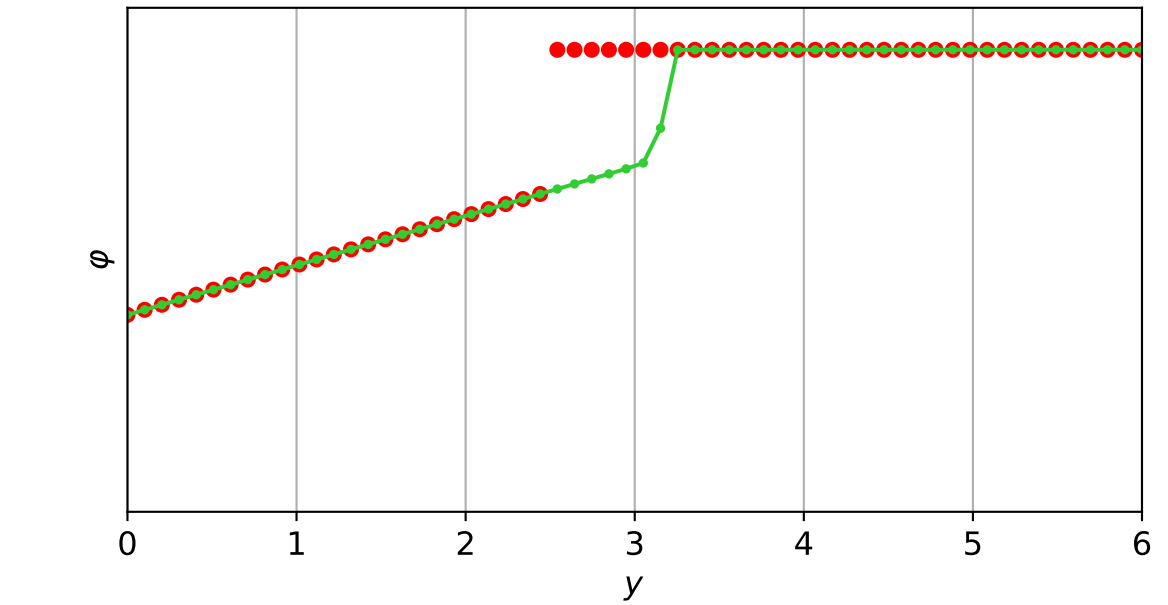
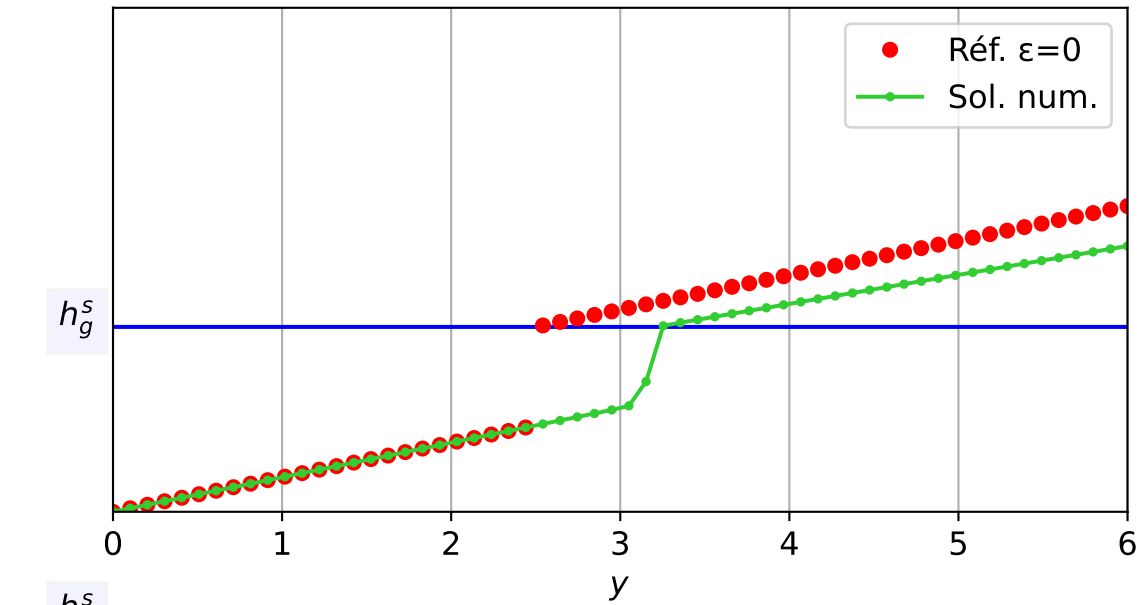
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\epsilon = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 0.550005/12$  (943 iter) — résidu eq. =  $5.27e-06$



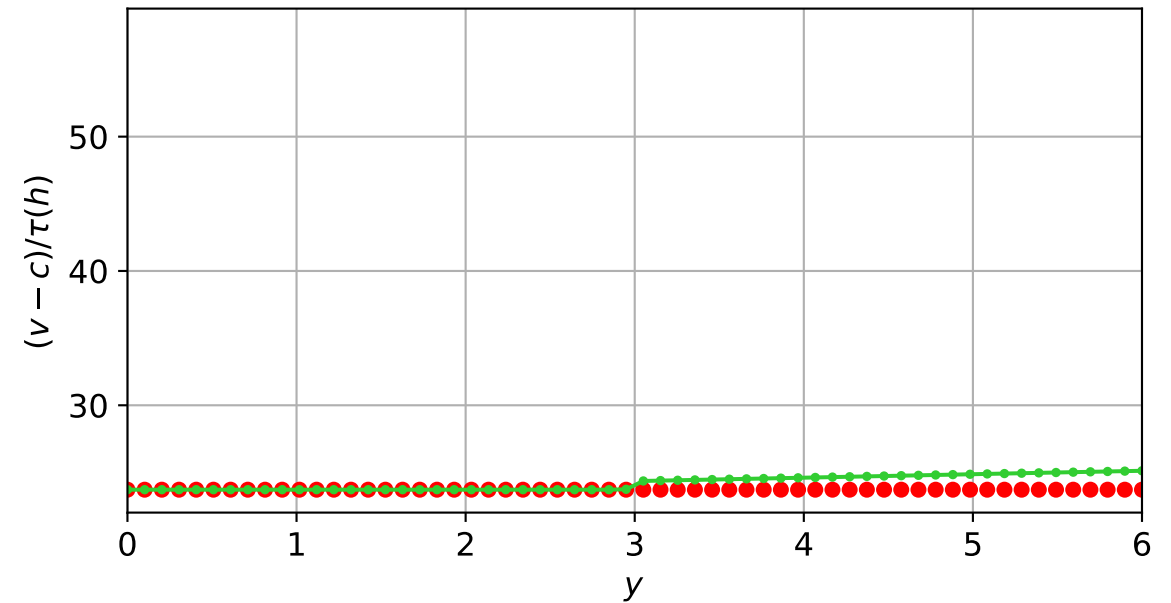
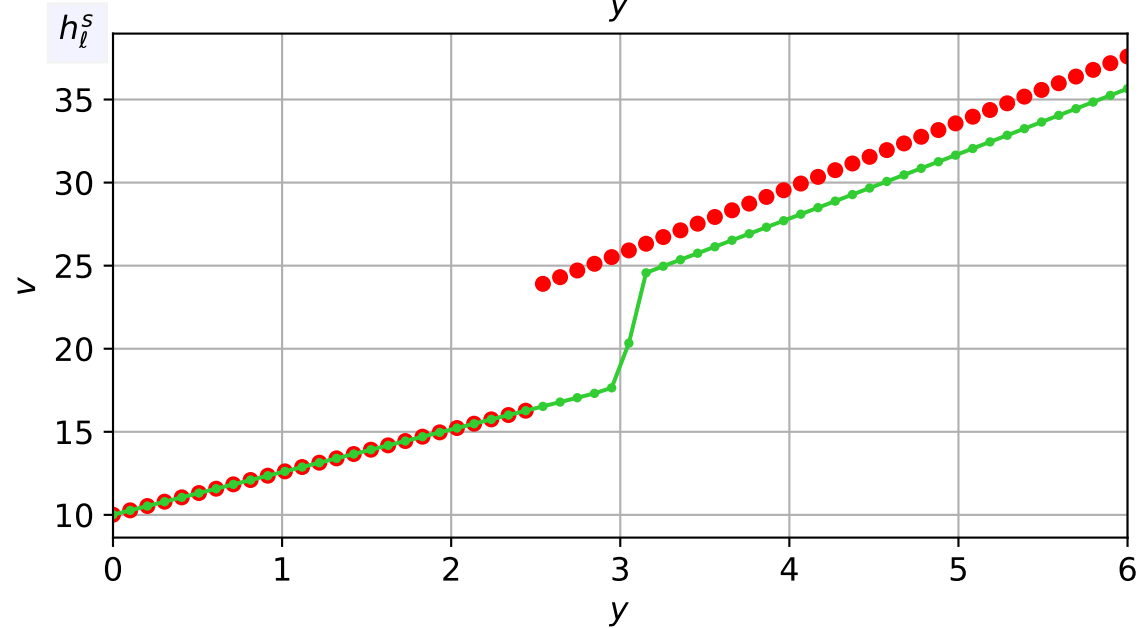
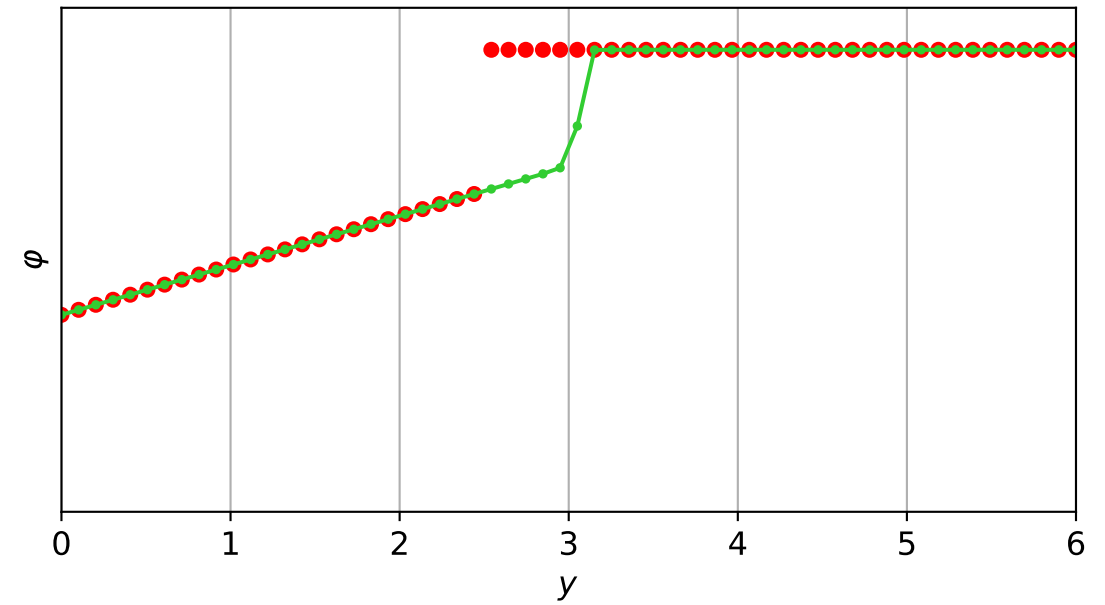
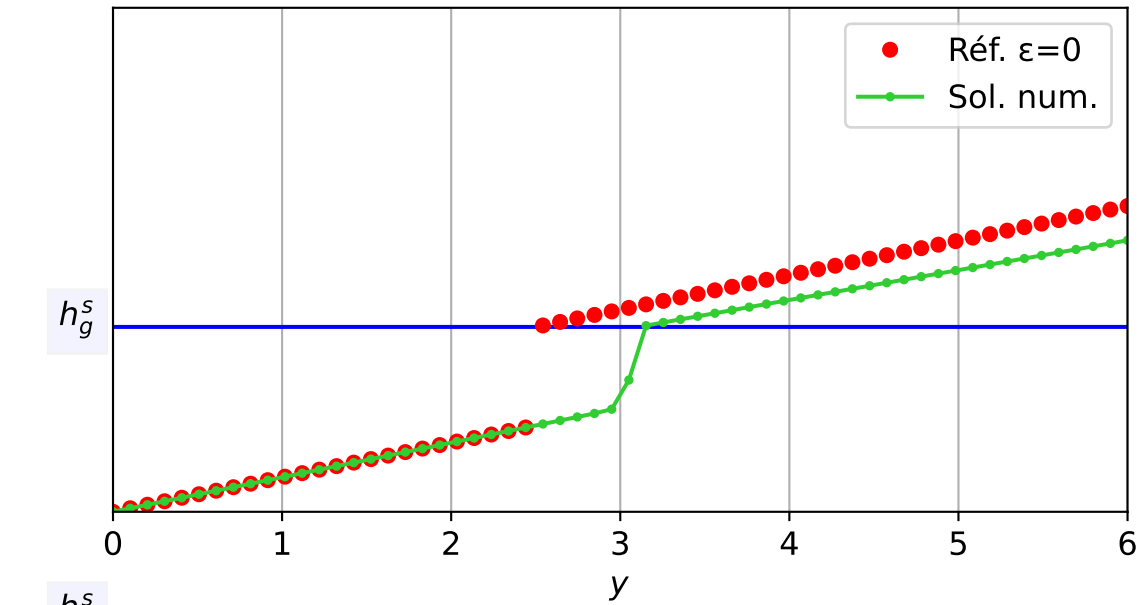
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 0.600007/12$  (1165 iter) — résidu eq. =  $5.90e-06$



TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 0.650009/12$  (1387 iter) — résidu eq. =  $7.99e-06$

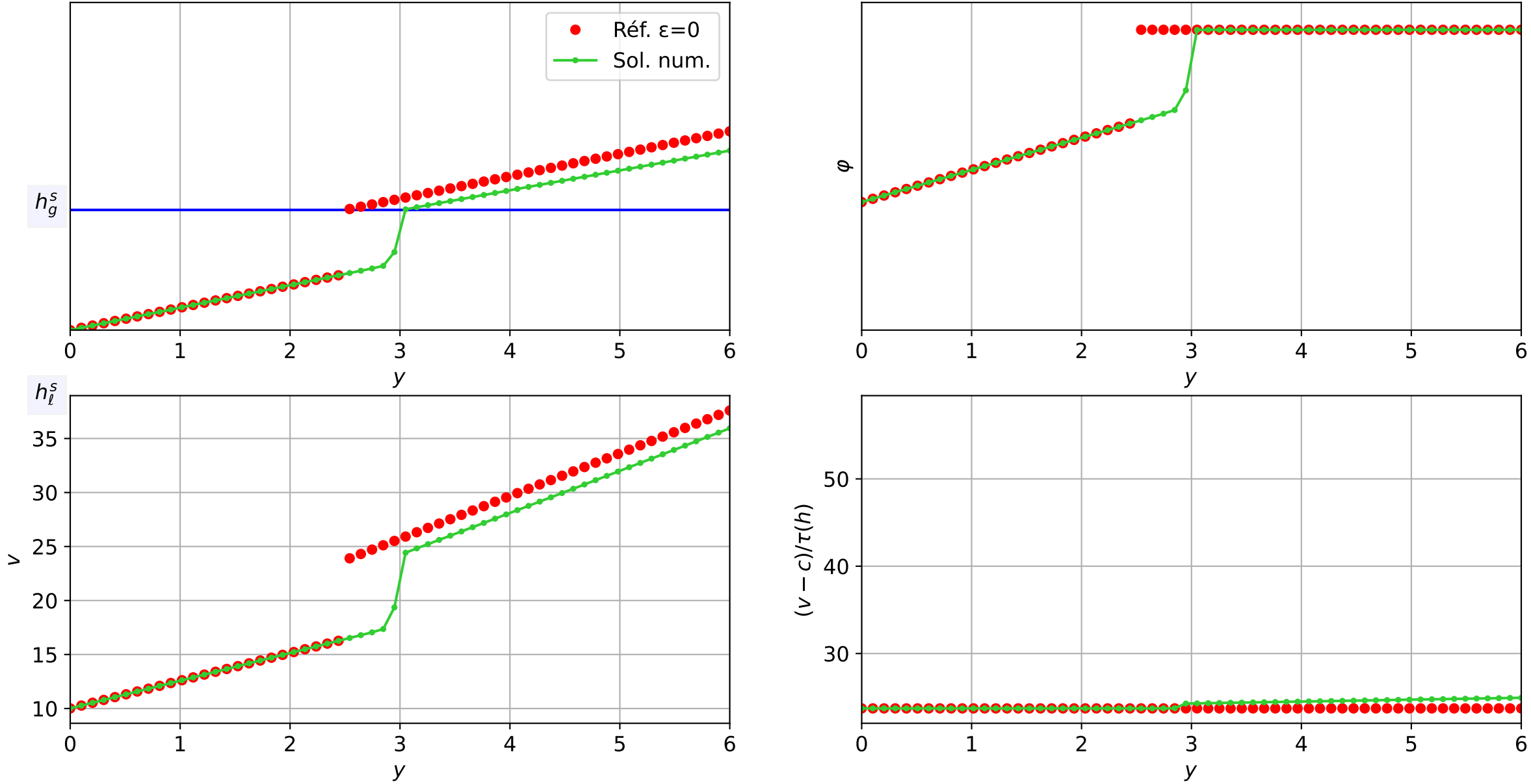


TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 0.70001/12$  (1609 iter) — résidu eq. =  $7.85e-06$

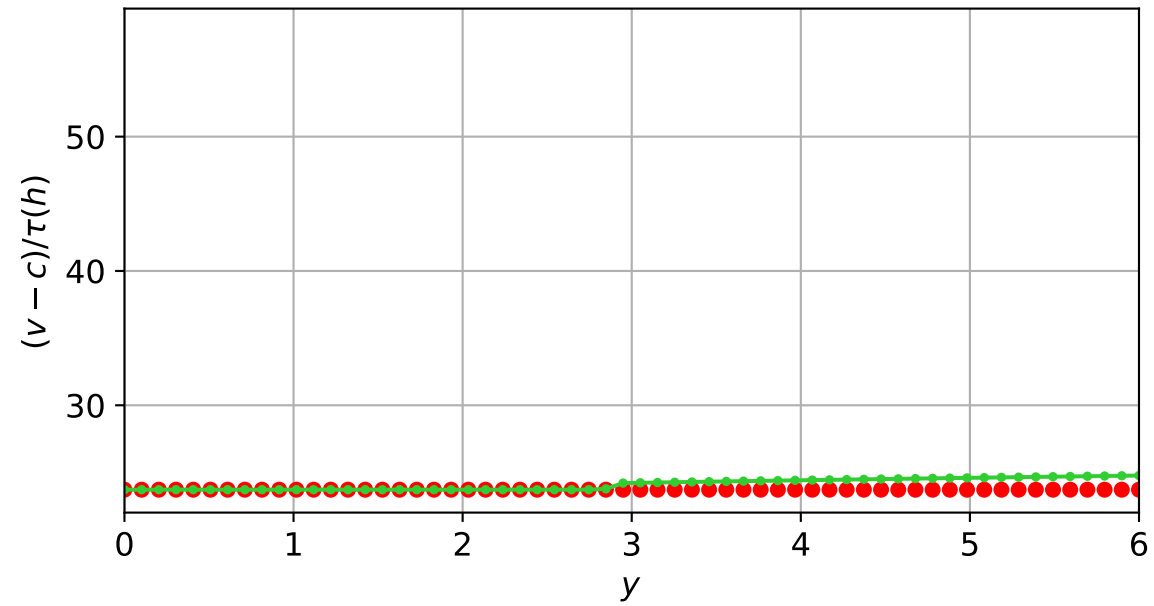
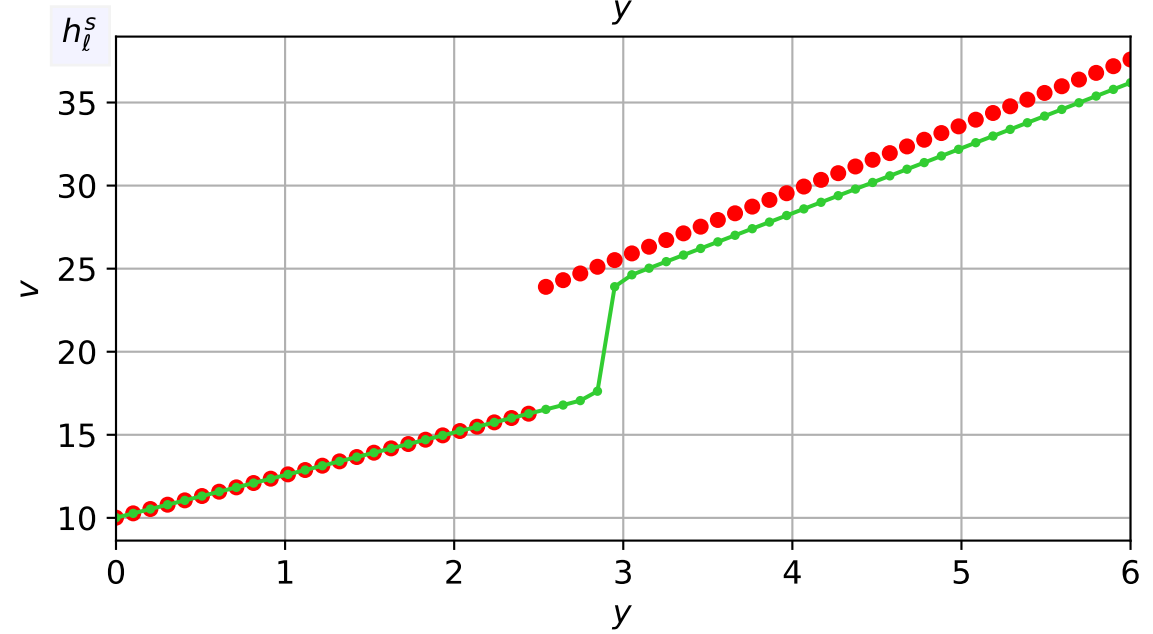
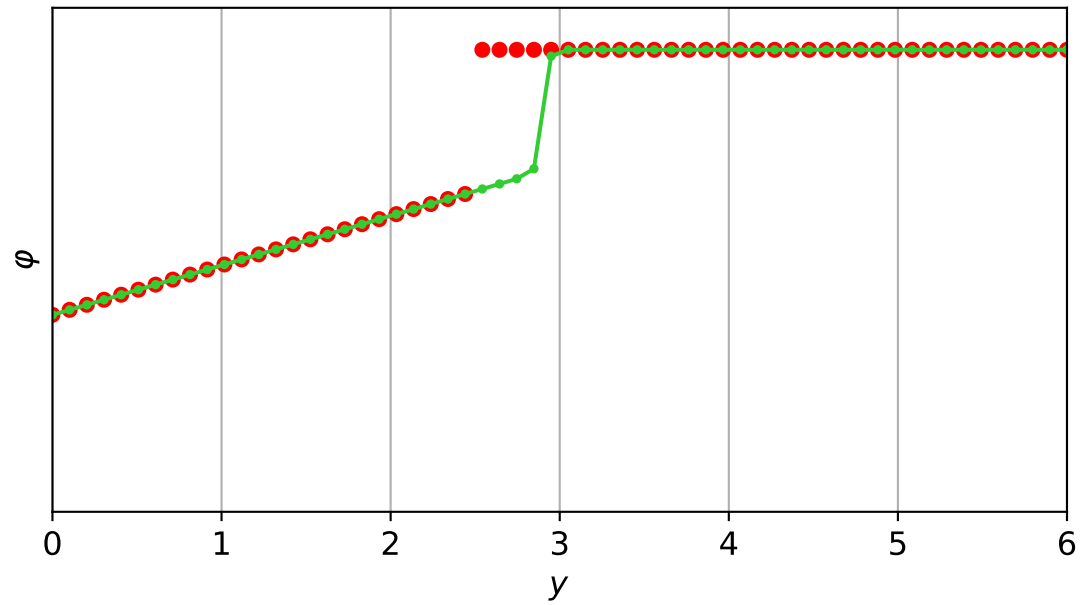
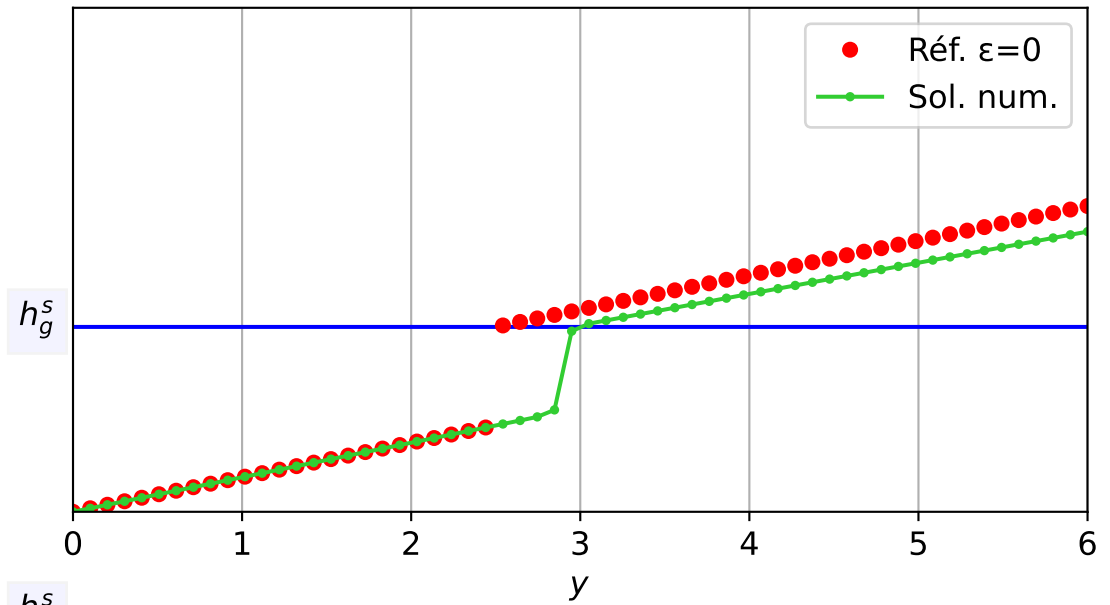




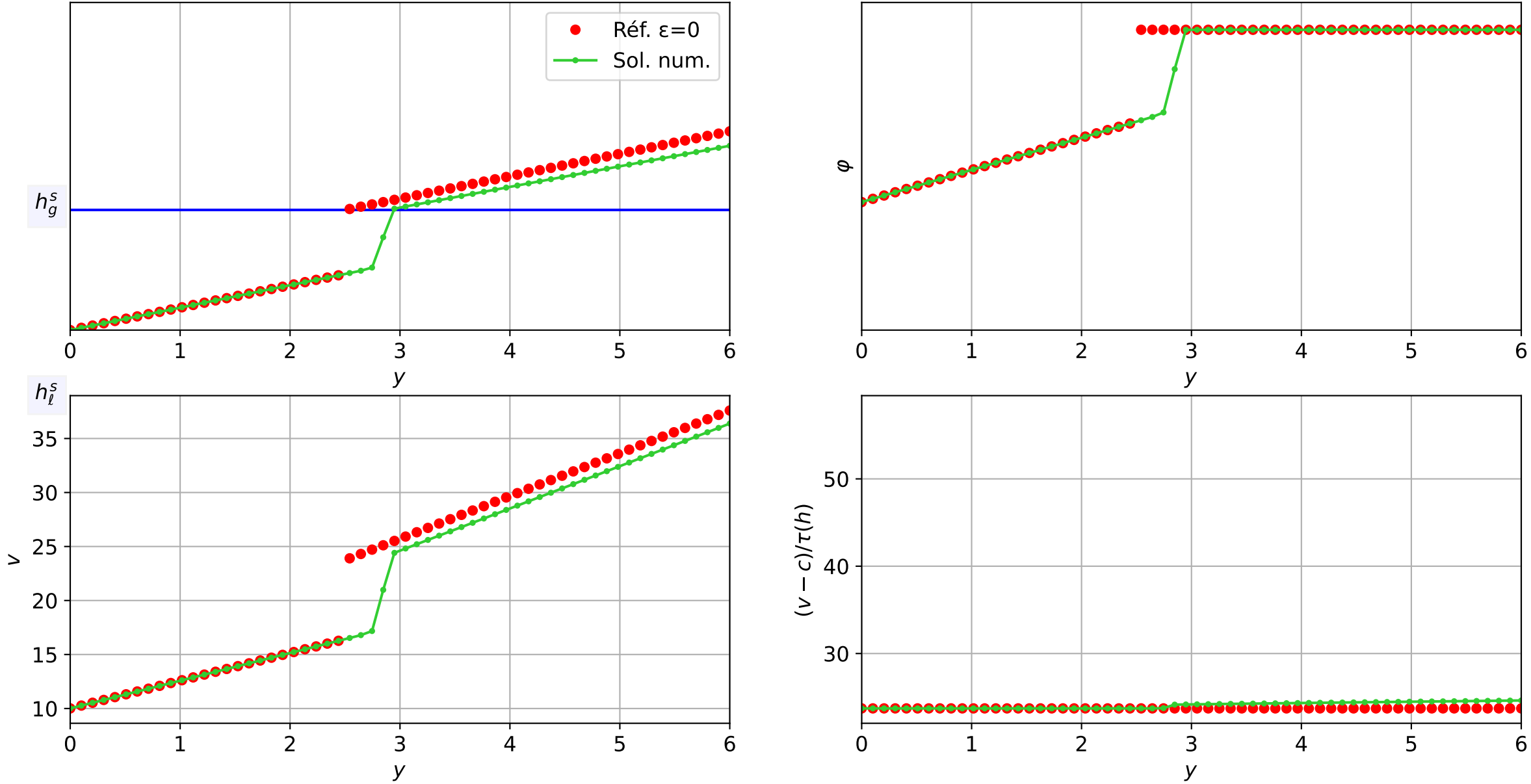
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\epsilon = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 0.750012/12$  (1831 iter) — résidu eq. =  $5.60e-06$



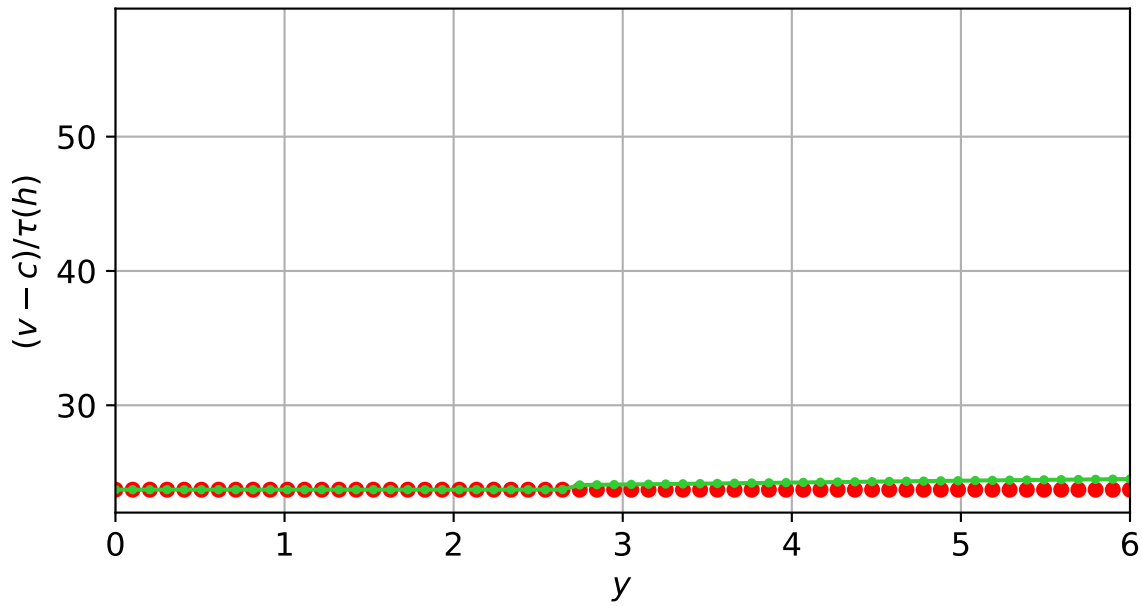
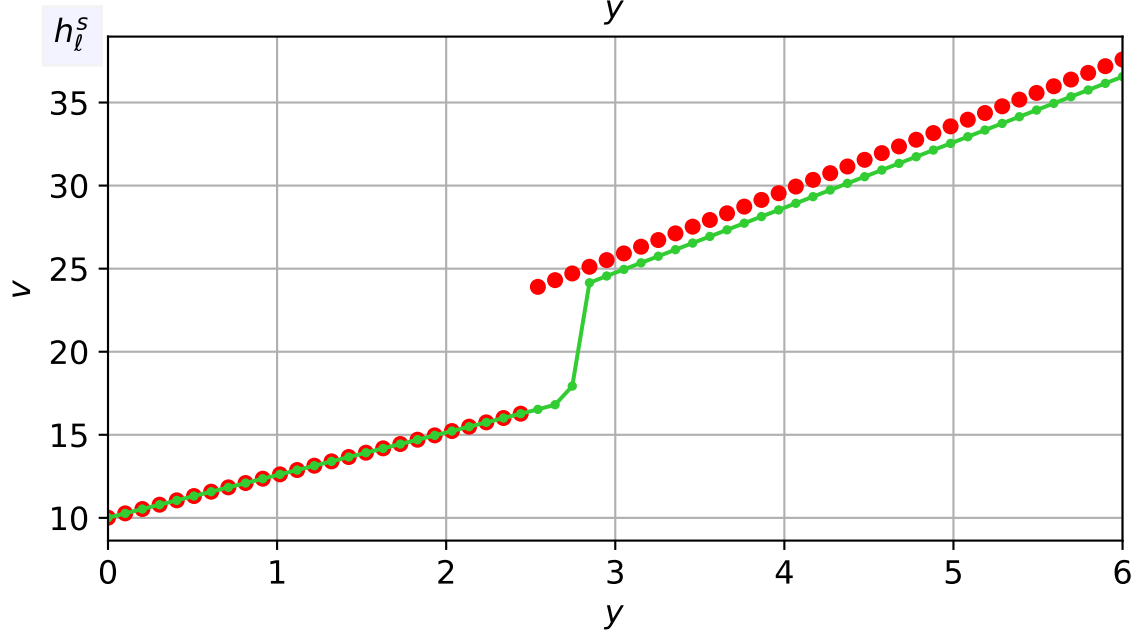
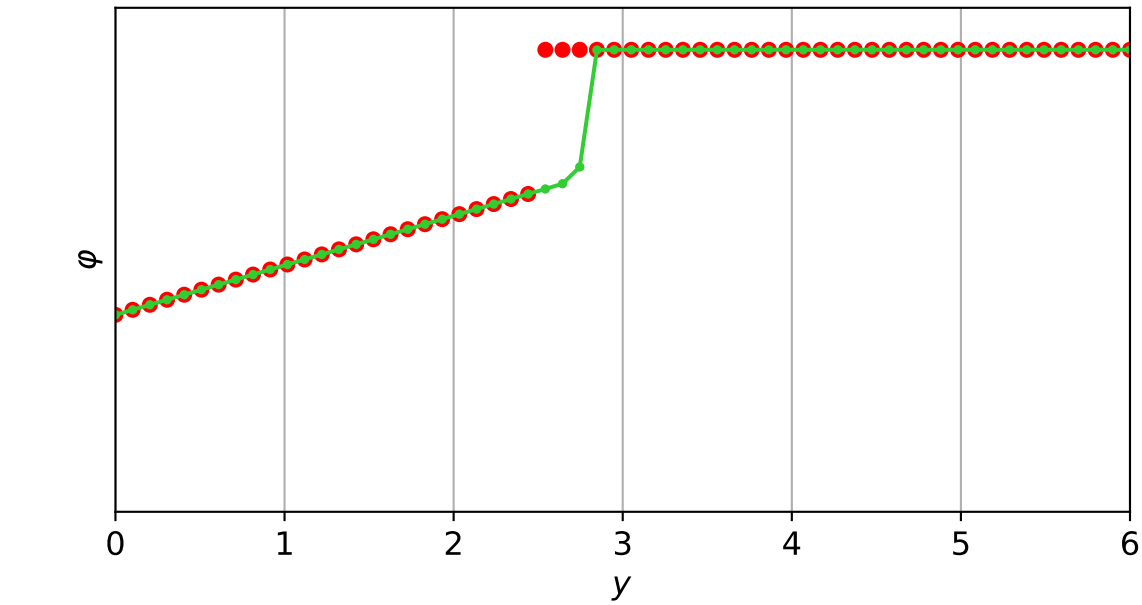
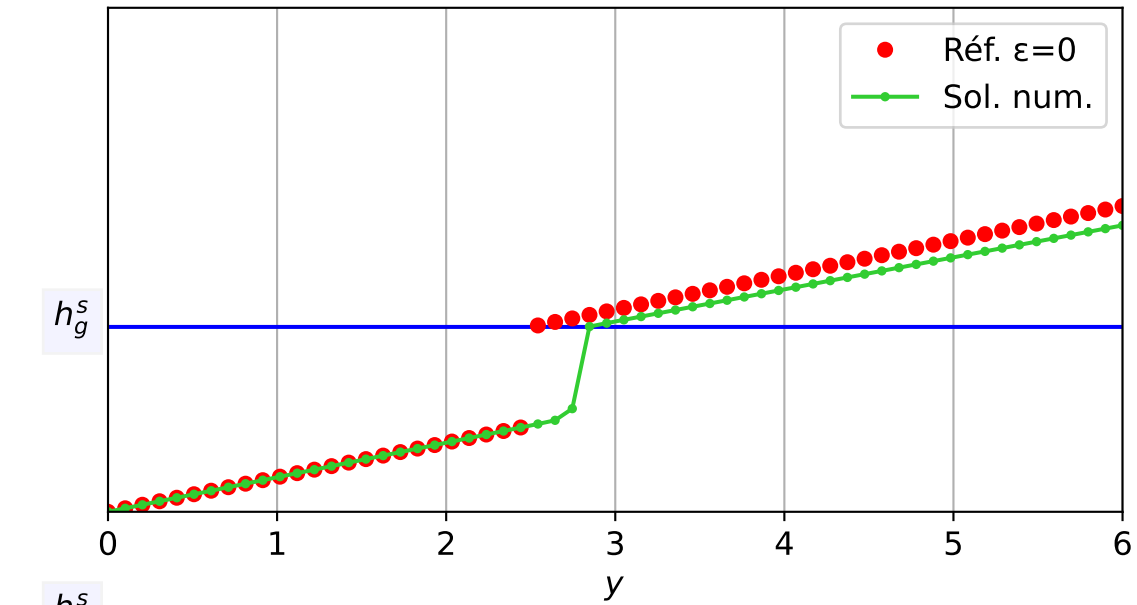
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\epsilon = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 0.800014/12$  (2053 iter) — résidu eq. =  $5.23e-06$



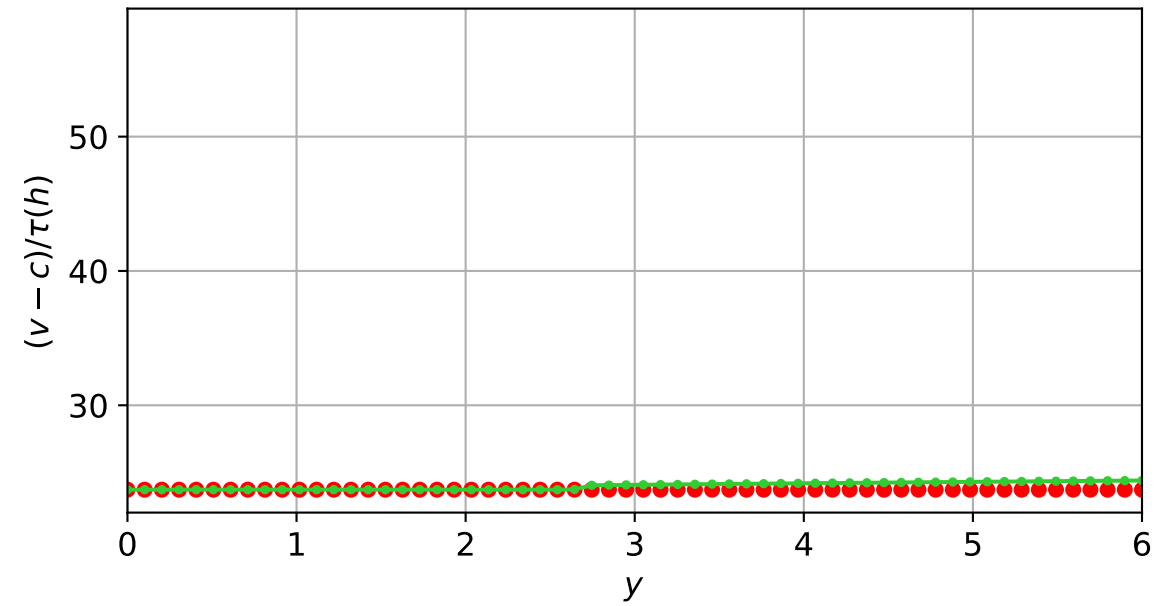
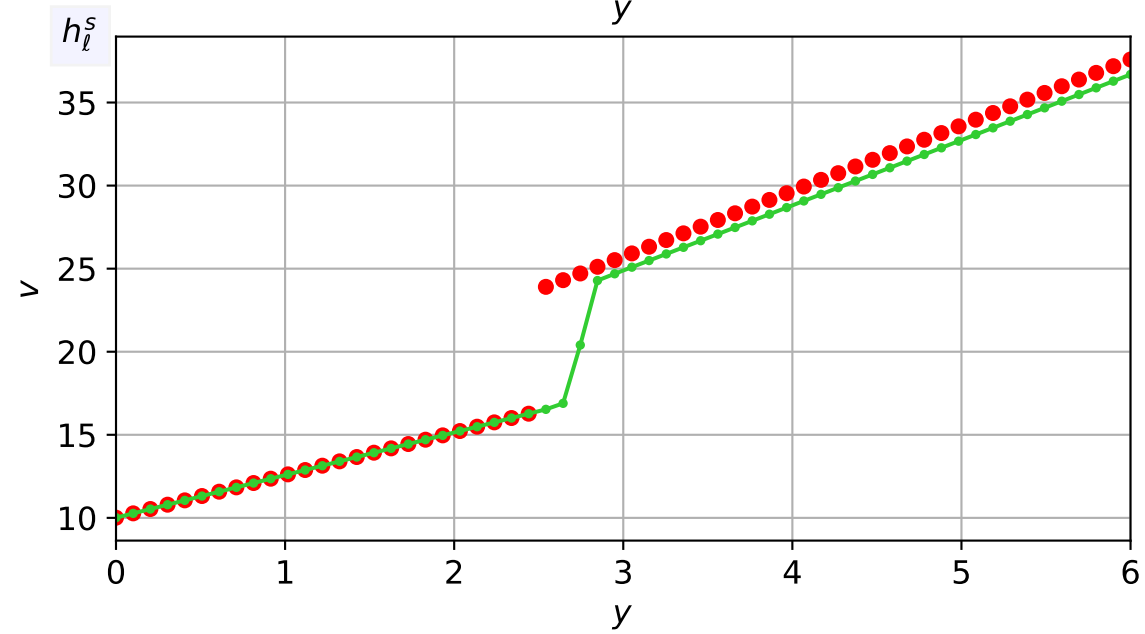
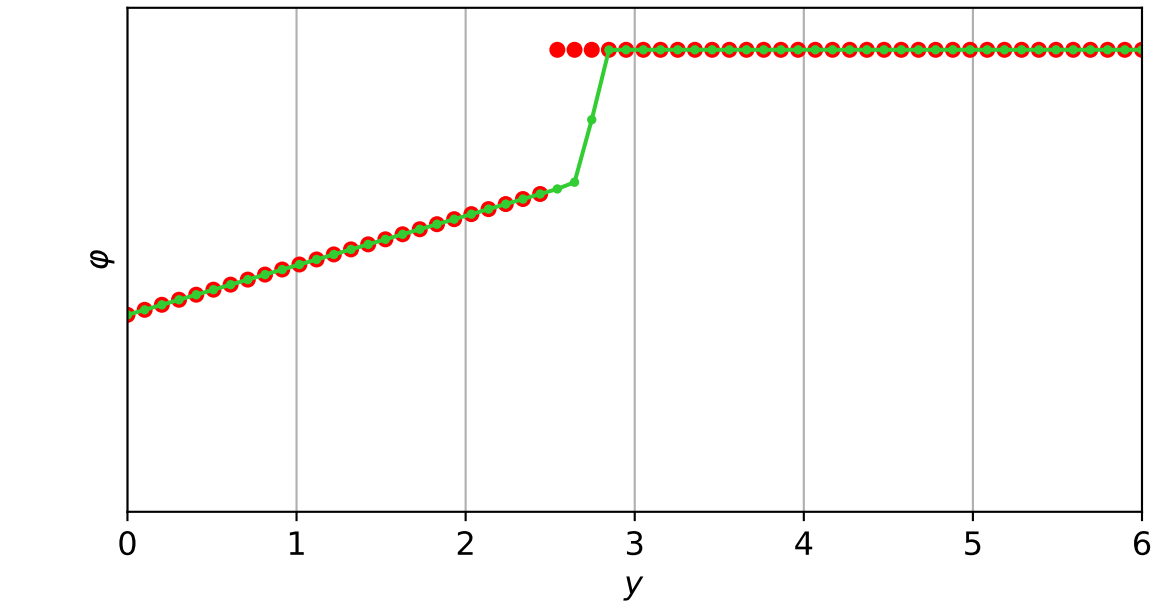
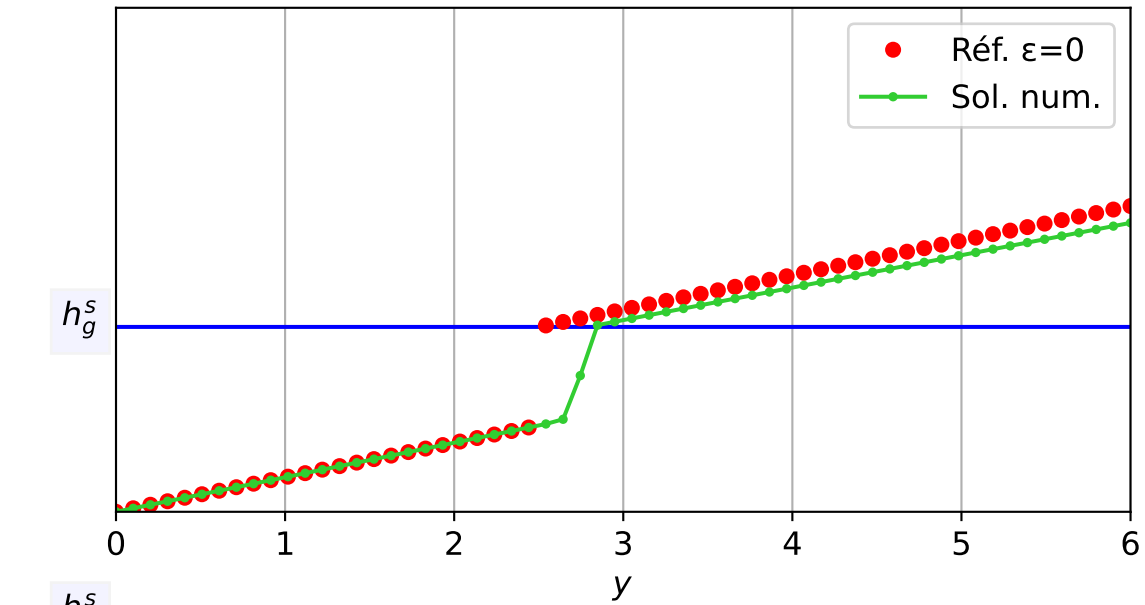
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\epsilon = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 0.850016/12$  (2275 iter) — résidu eq. =  $5.10e-06$



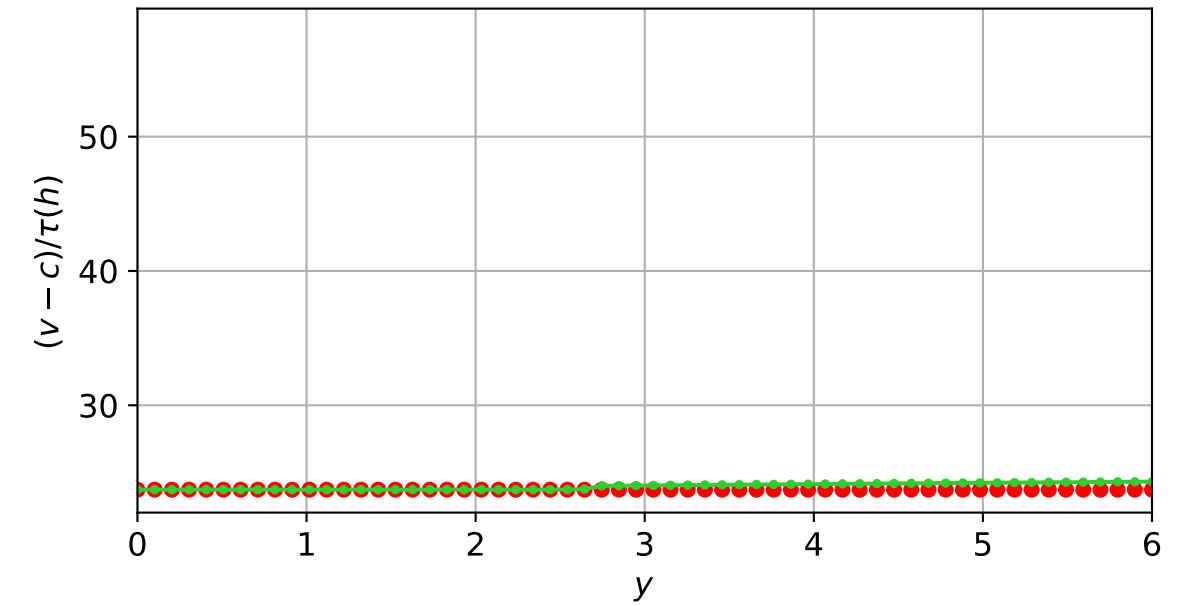
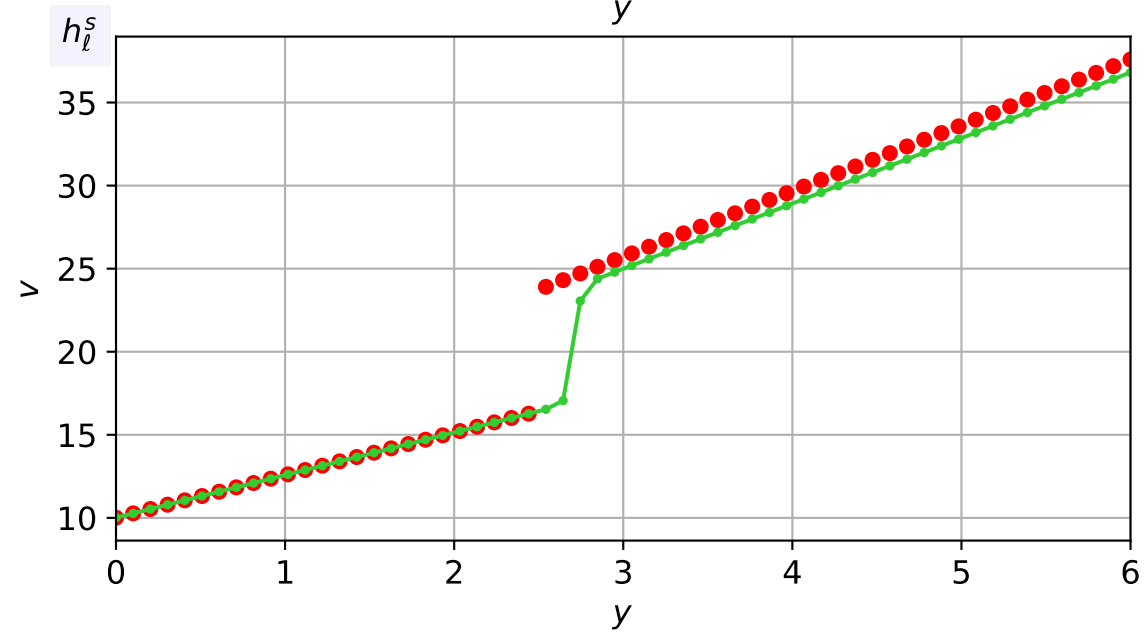
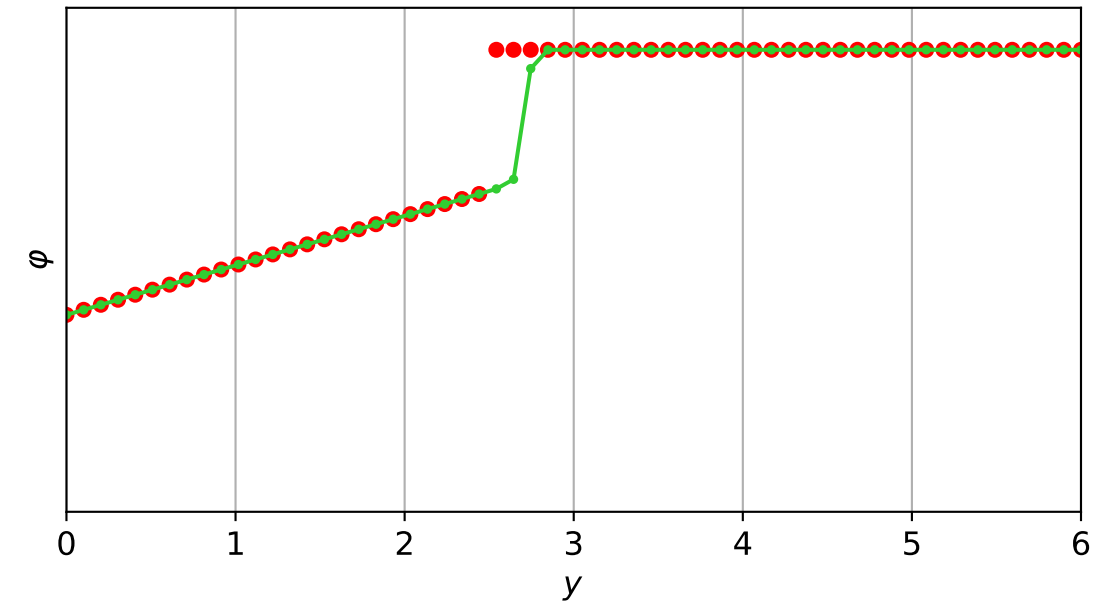
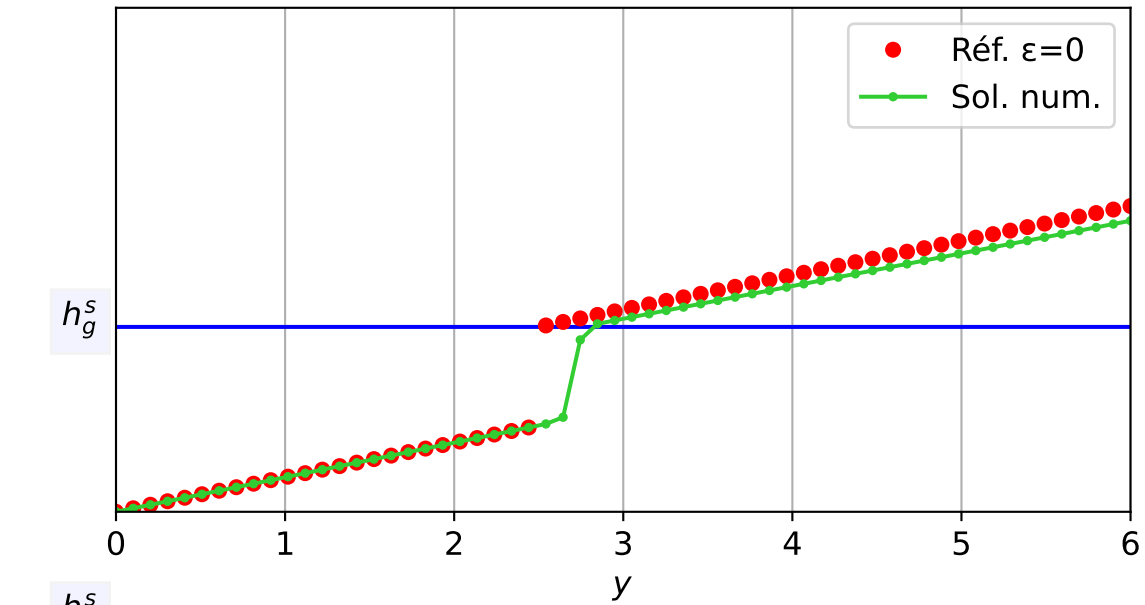
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\epsilon = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 0.900018/12$  (2497 iter) — résidu eq. =  $5.12e-06$



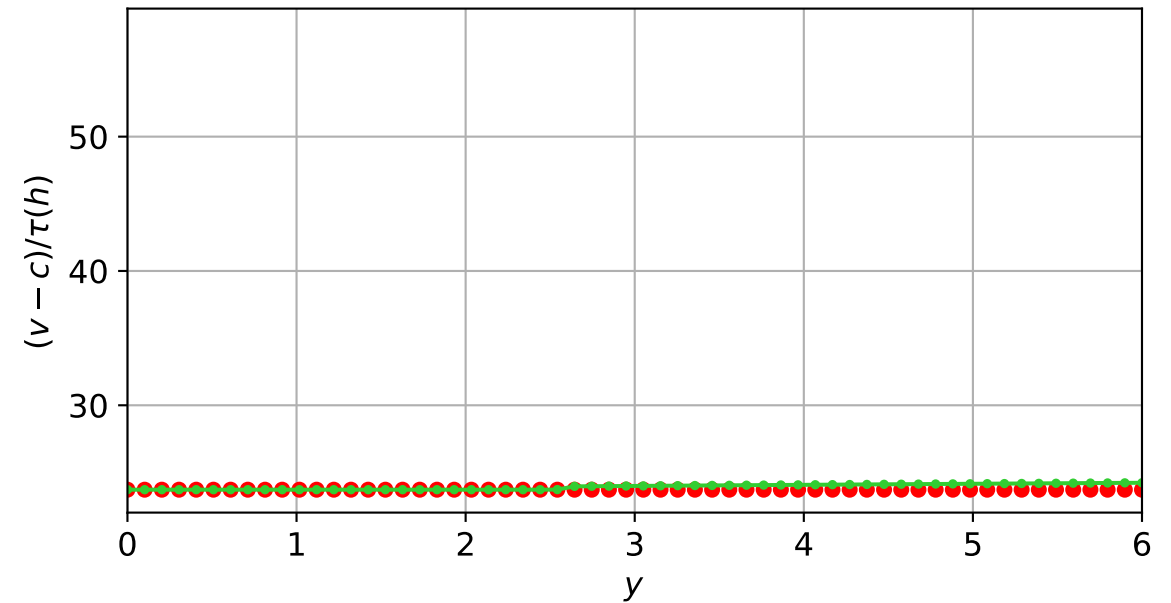
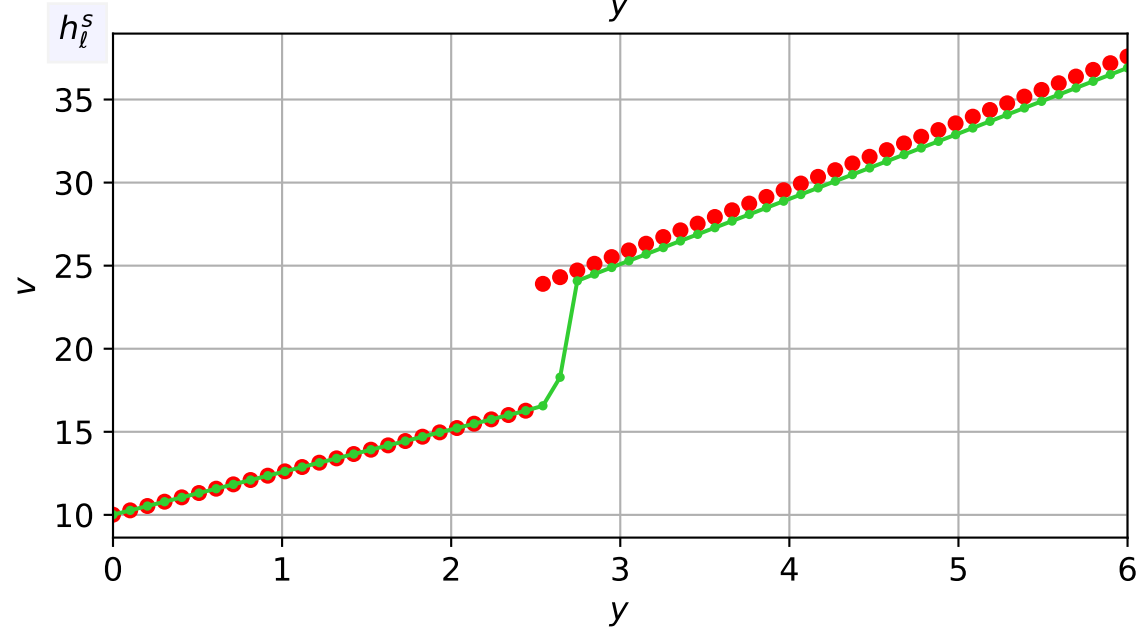
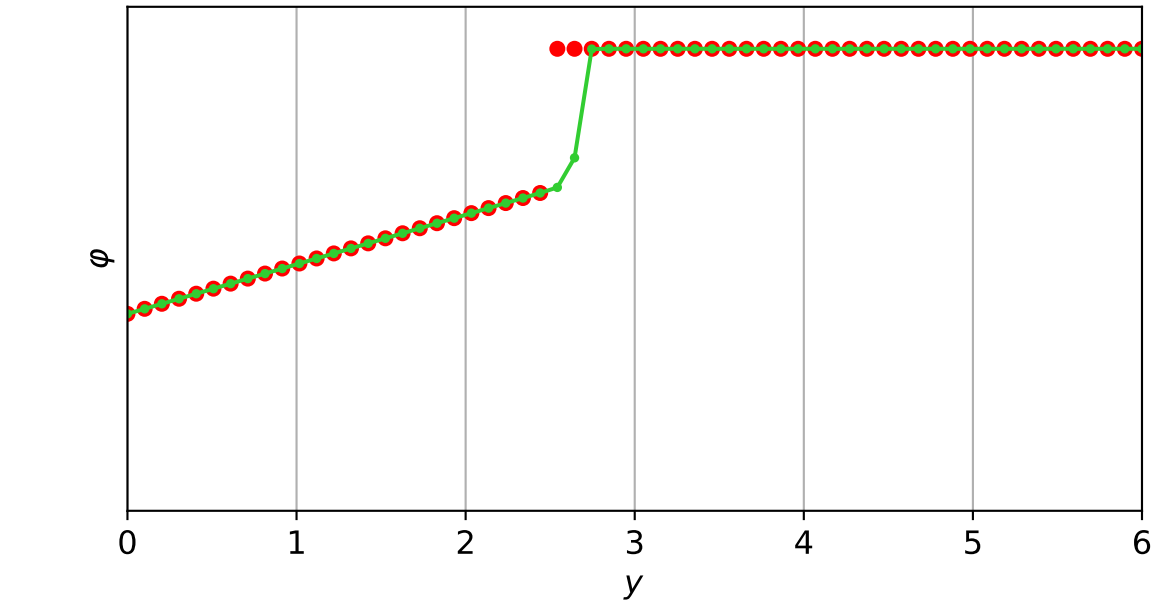
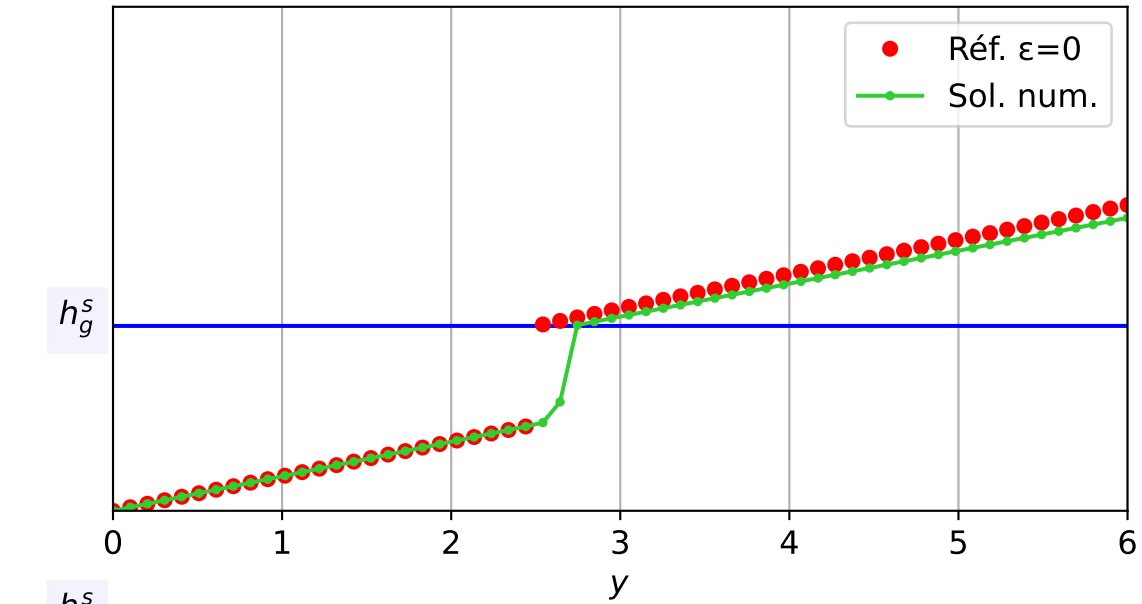
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 0.95002/12$  (2719 iter) — résidu eq. =  $7.31e-06$



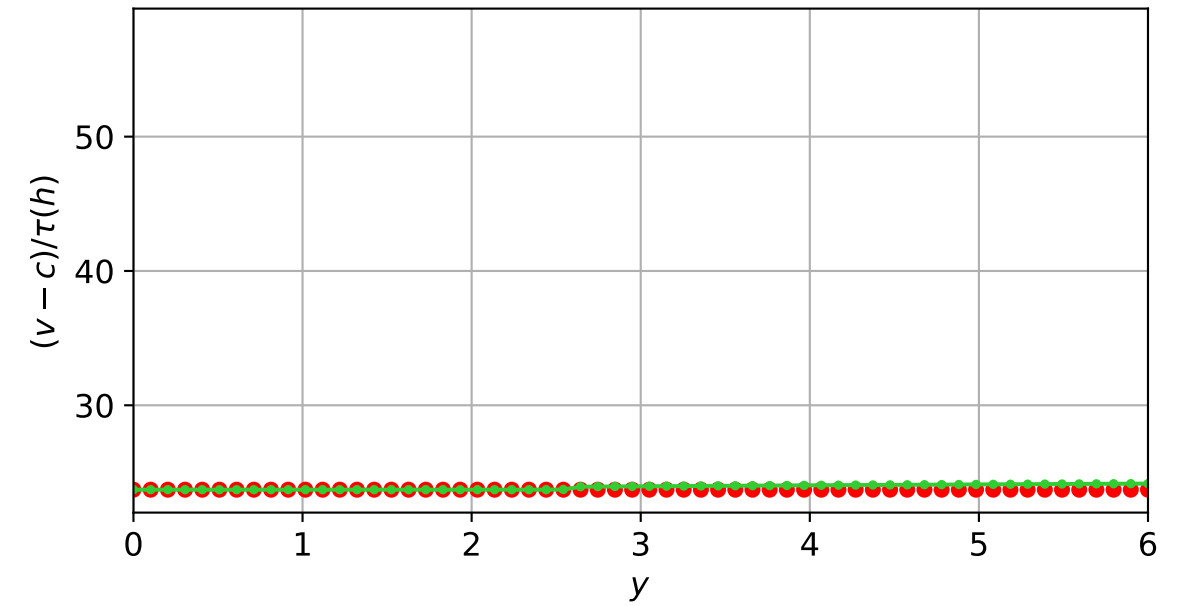
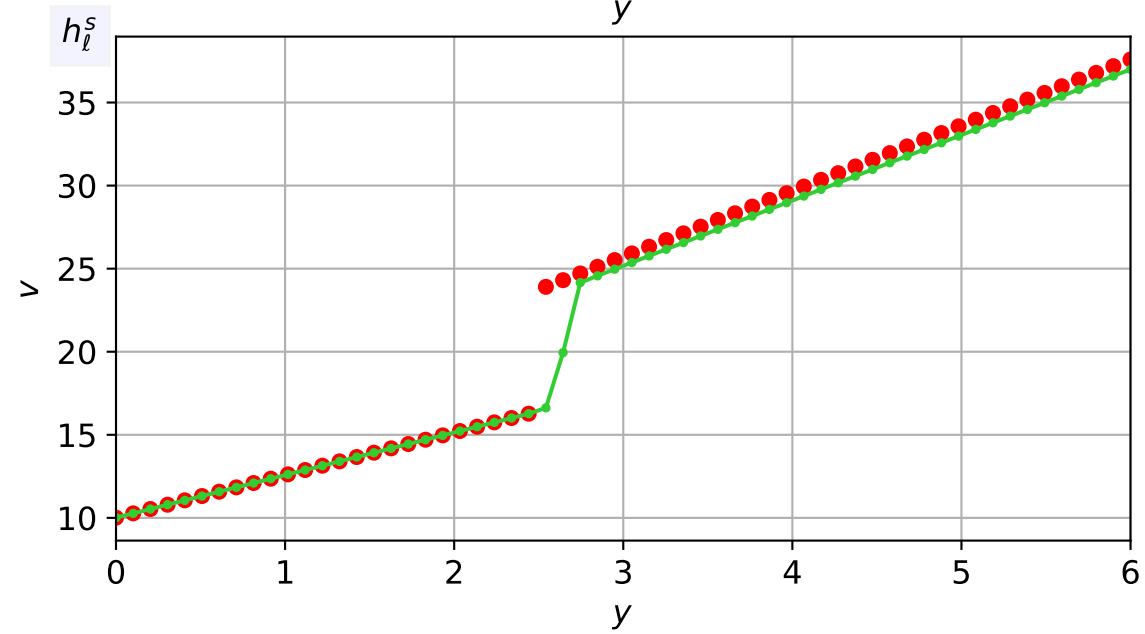
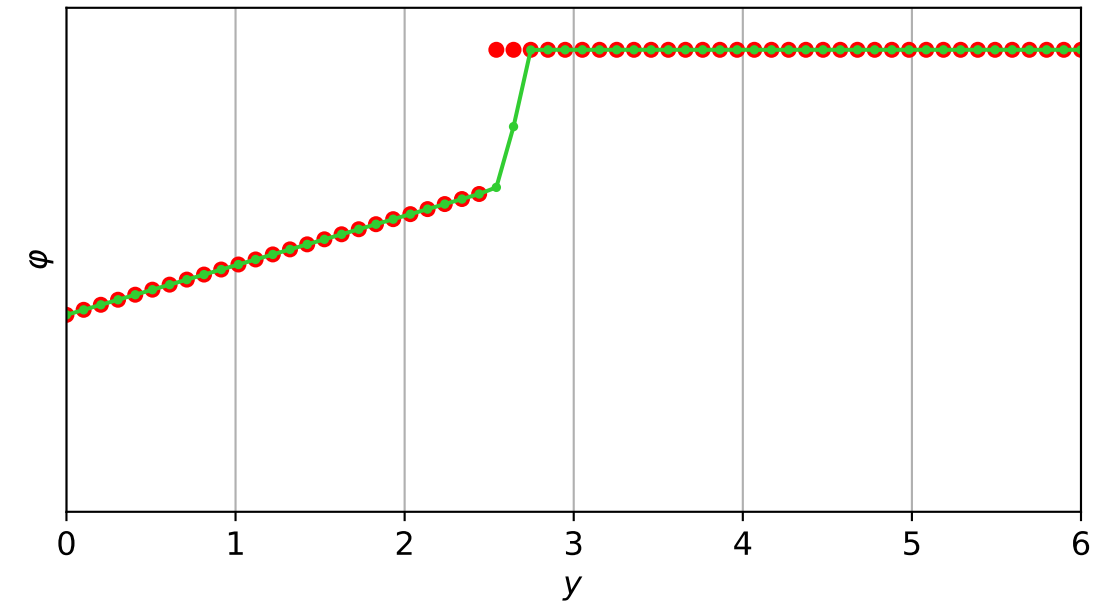
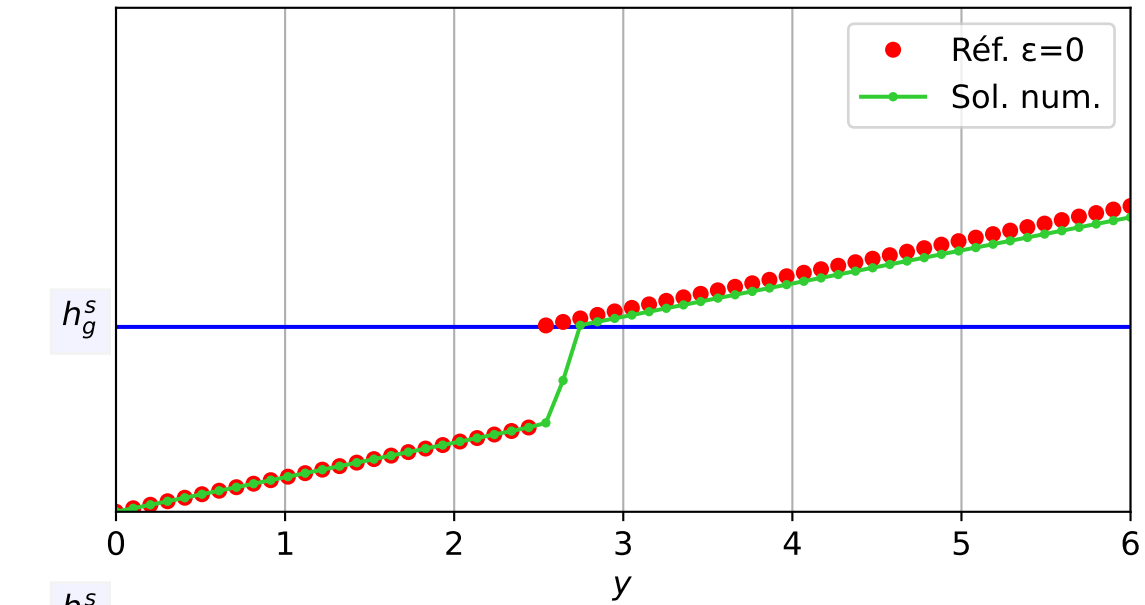
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 1.00002/12$  (2941 iter) — résidu eq. =  $5.35e-06$



TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 1.05002/12$  (3163 iter) — résidu eq. =  $5.99e-06$

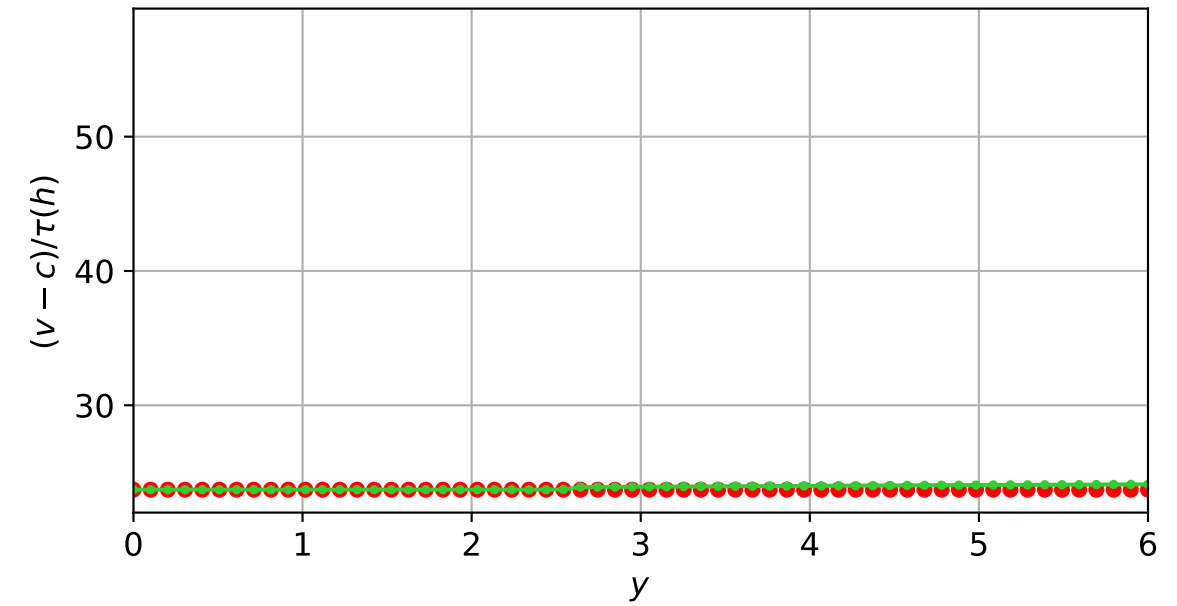
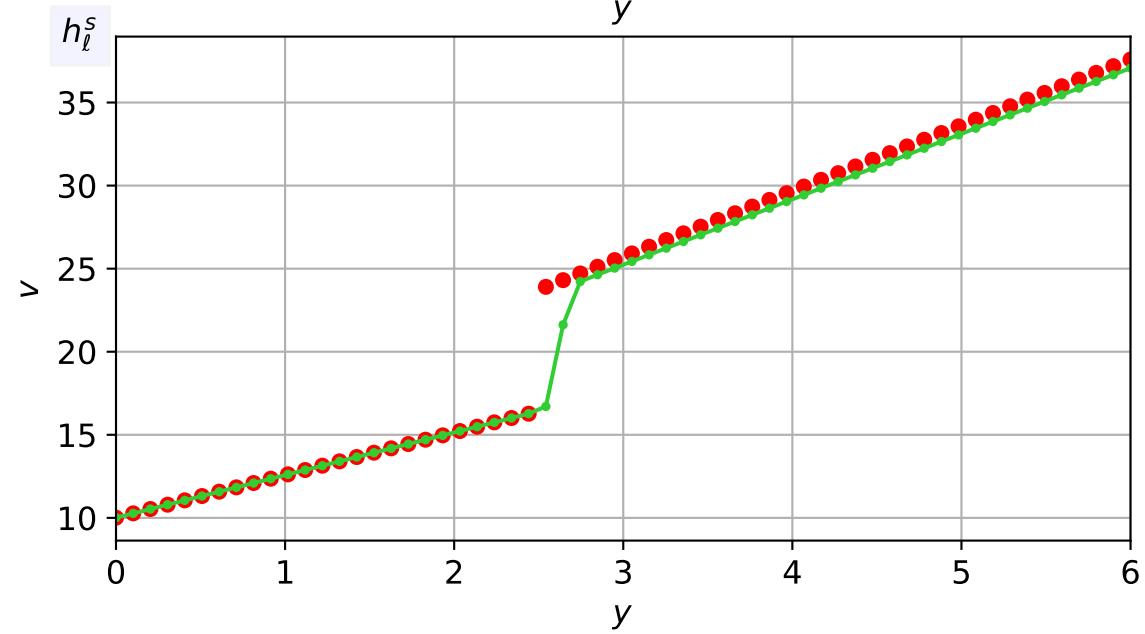
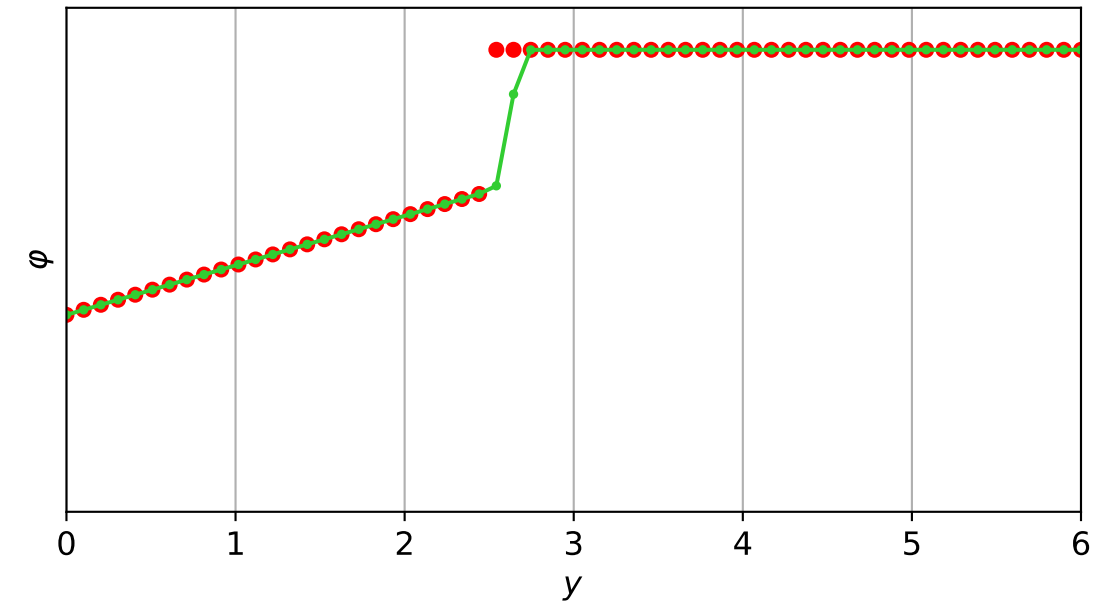
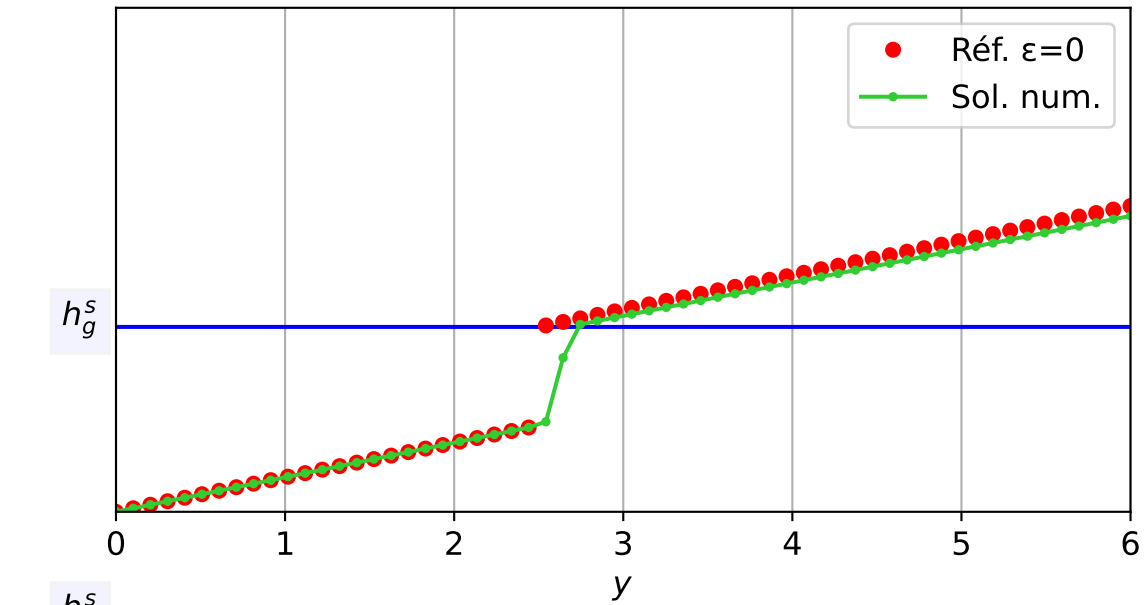


TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 1.10003/12$  (3385 iter) — résidu eq. =  $8.84e-06$

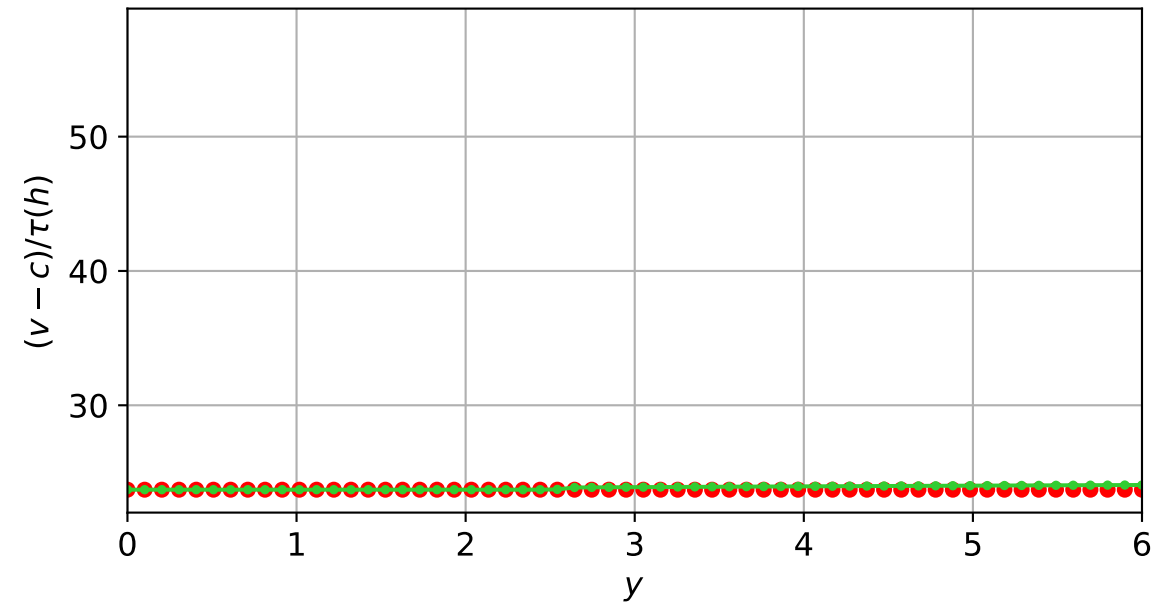
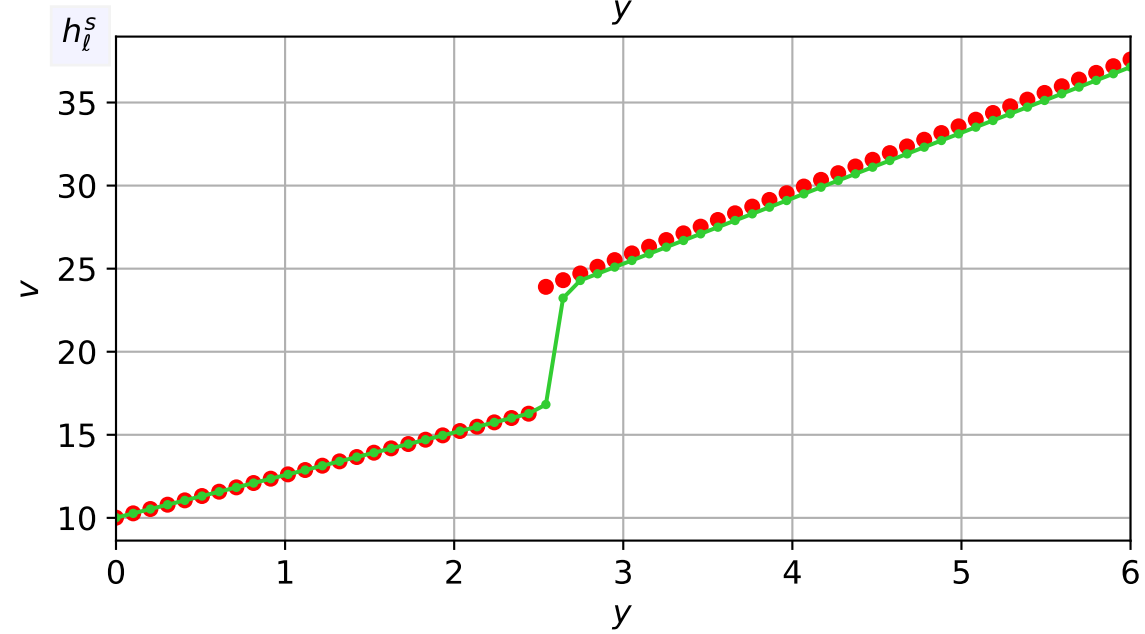
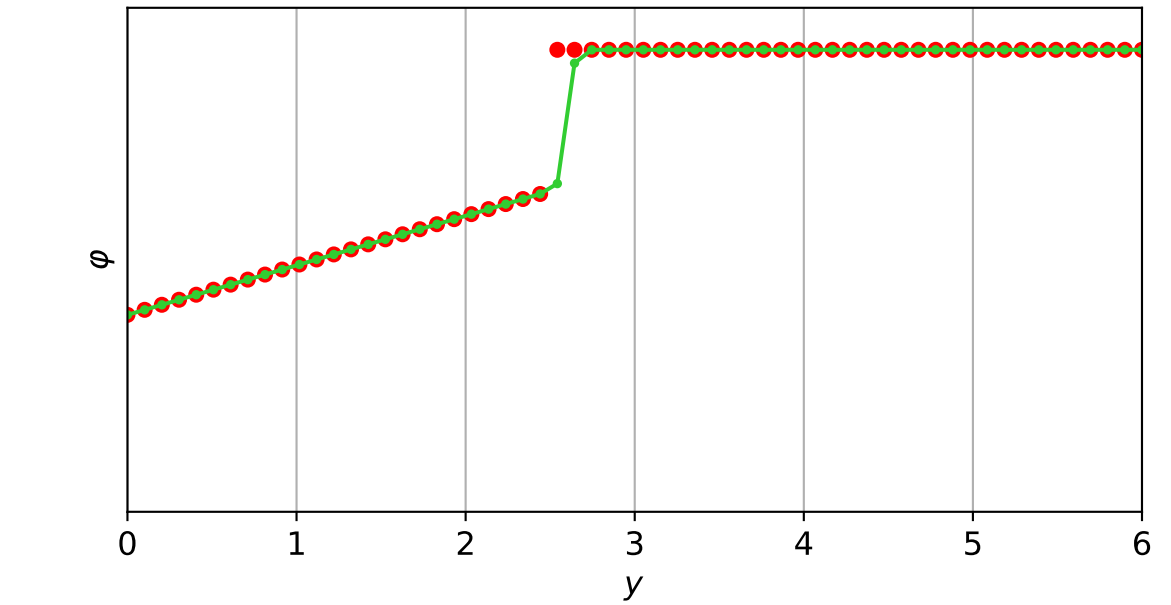
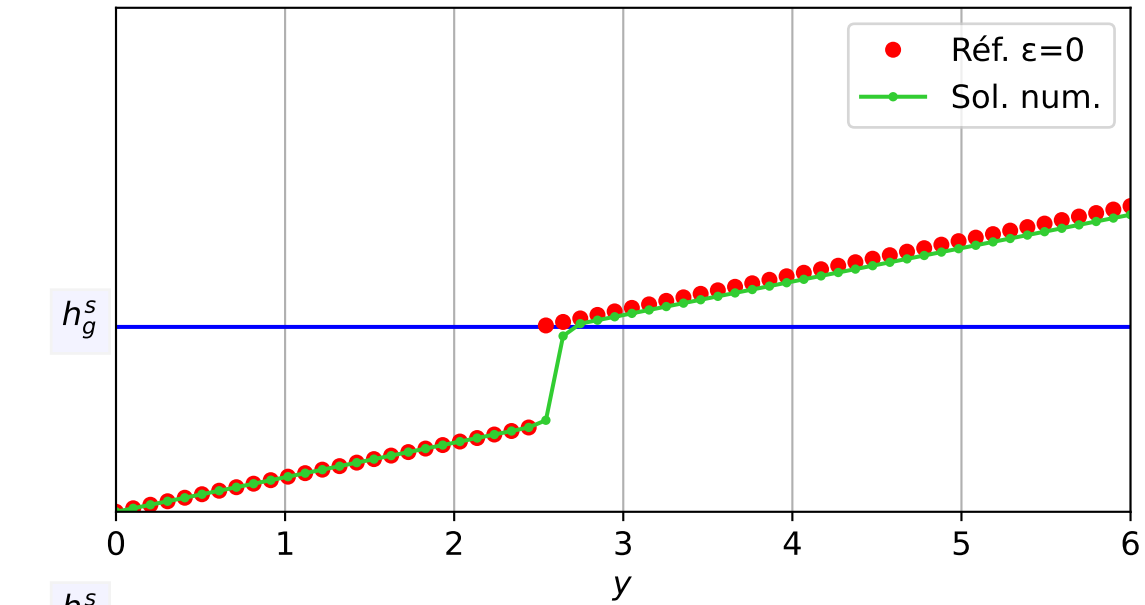




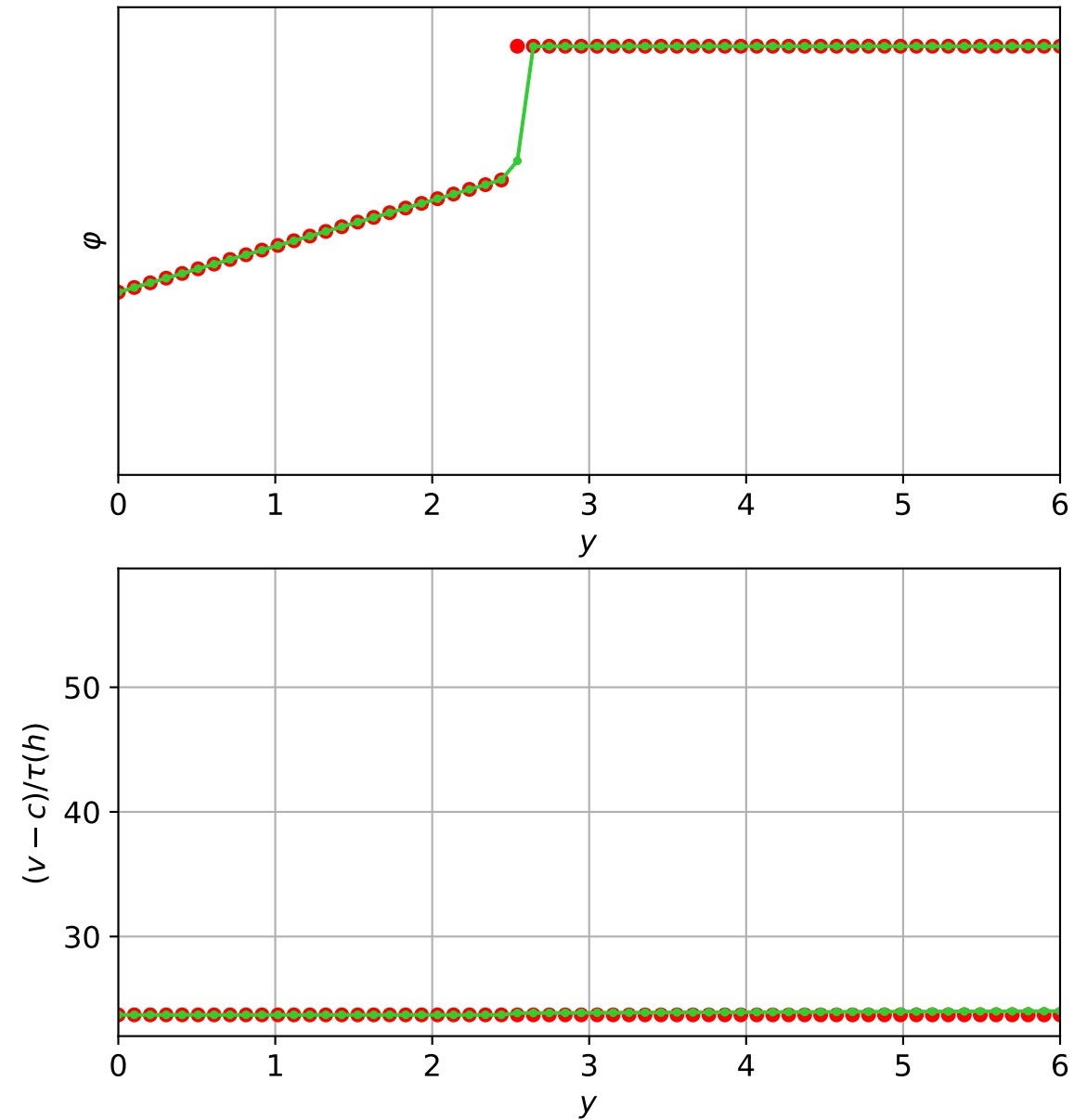
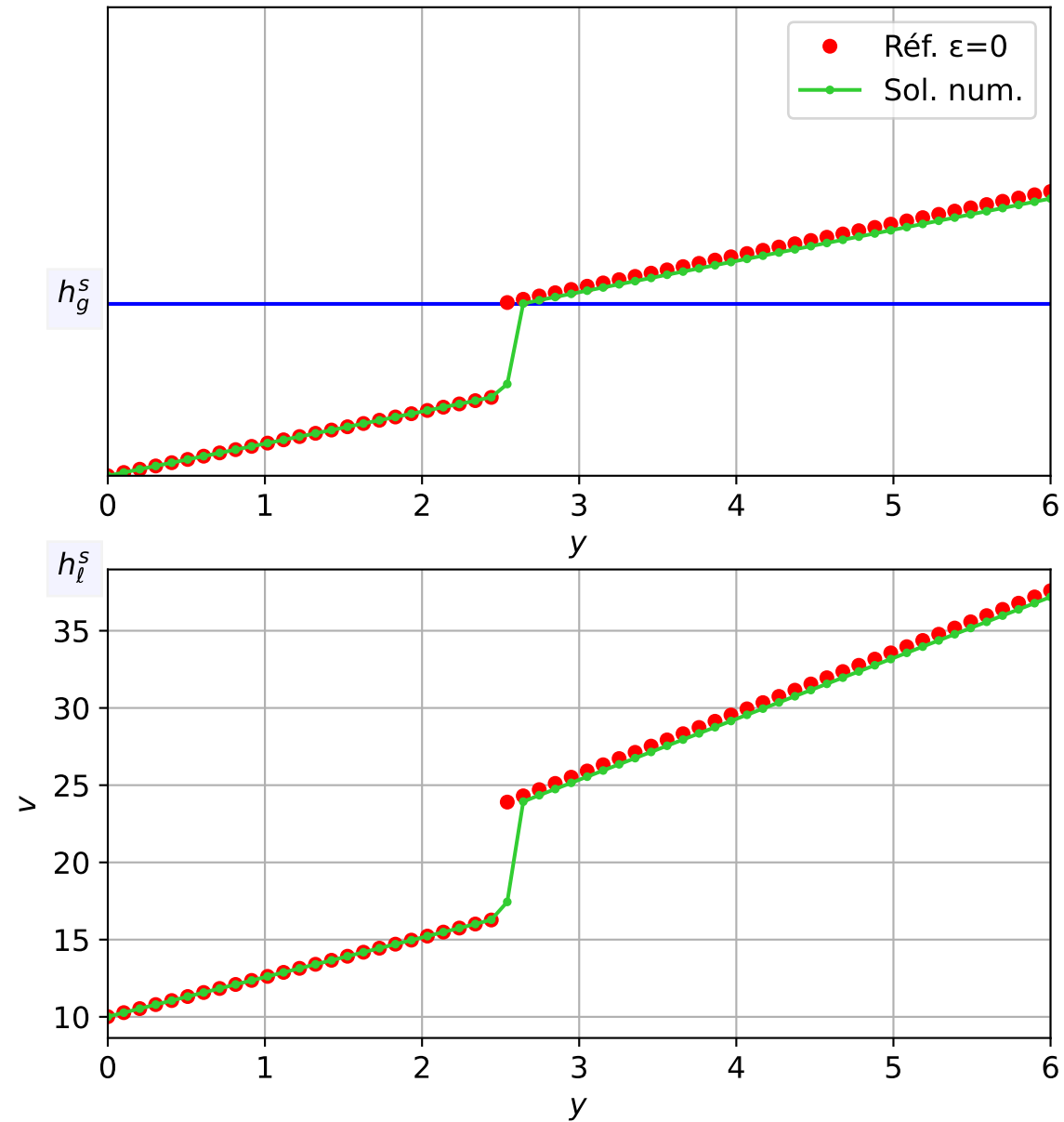
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 1.15003/12$  (3607 iter) — résidu eq. =  $6.91e-06$



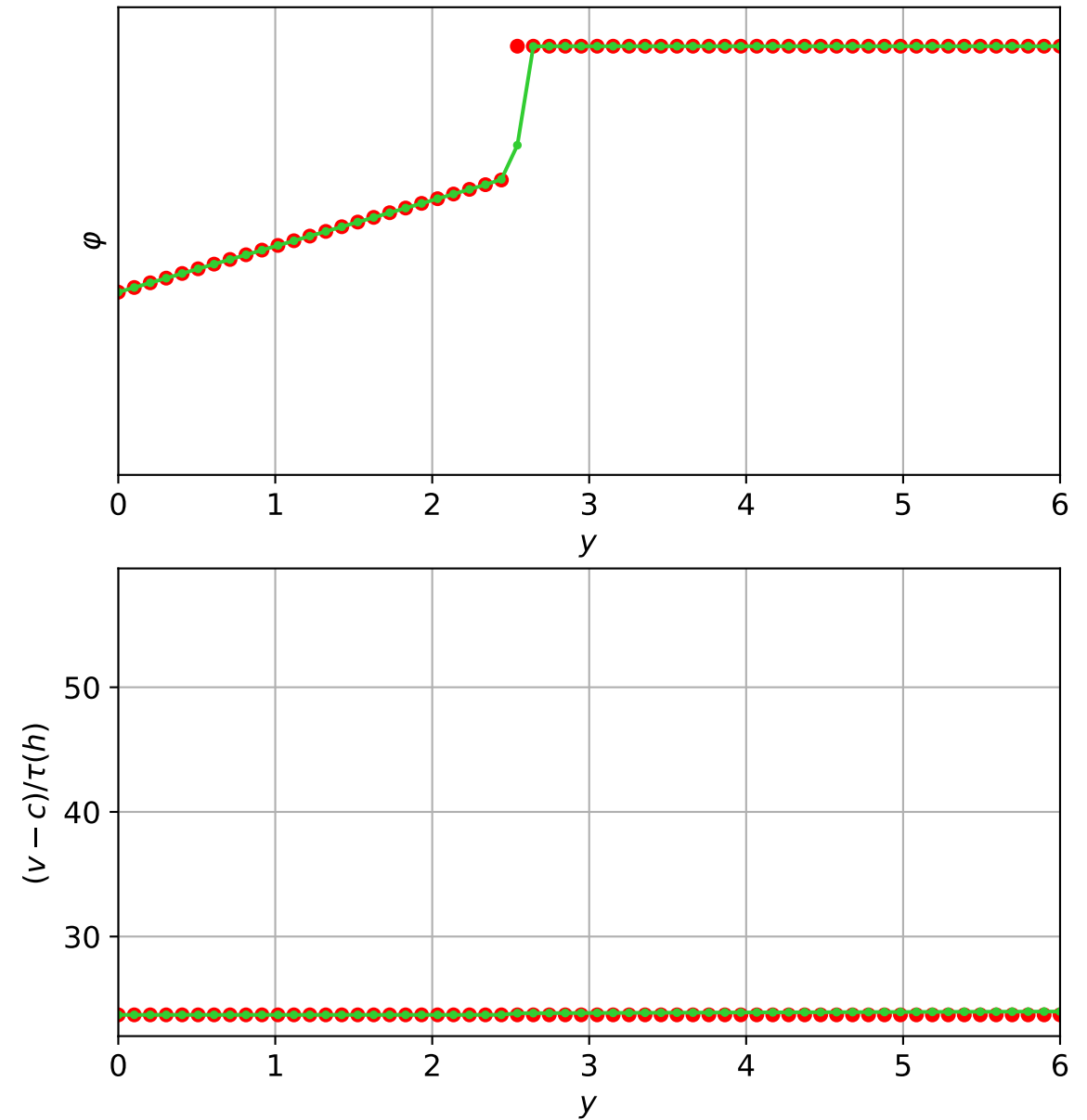
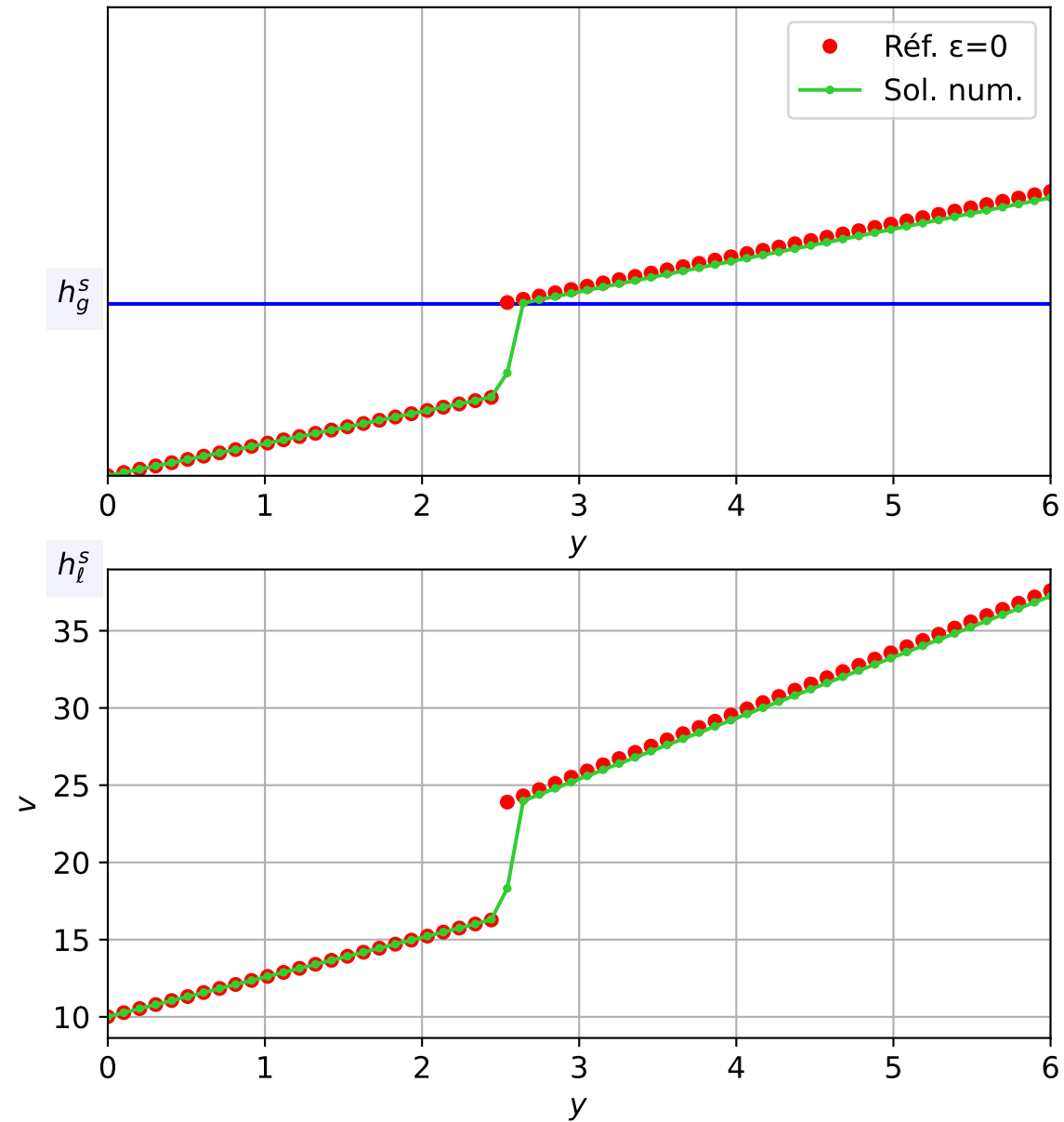
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 1.20003/12$  (3829 iter) — résidu eq. =  $3.06e-06$



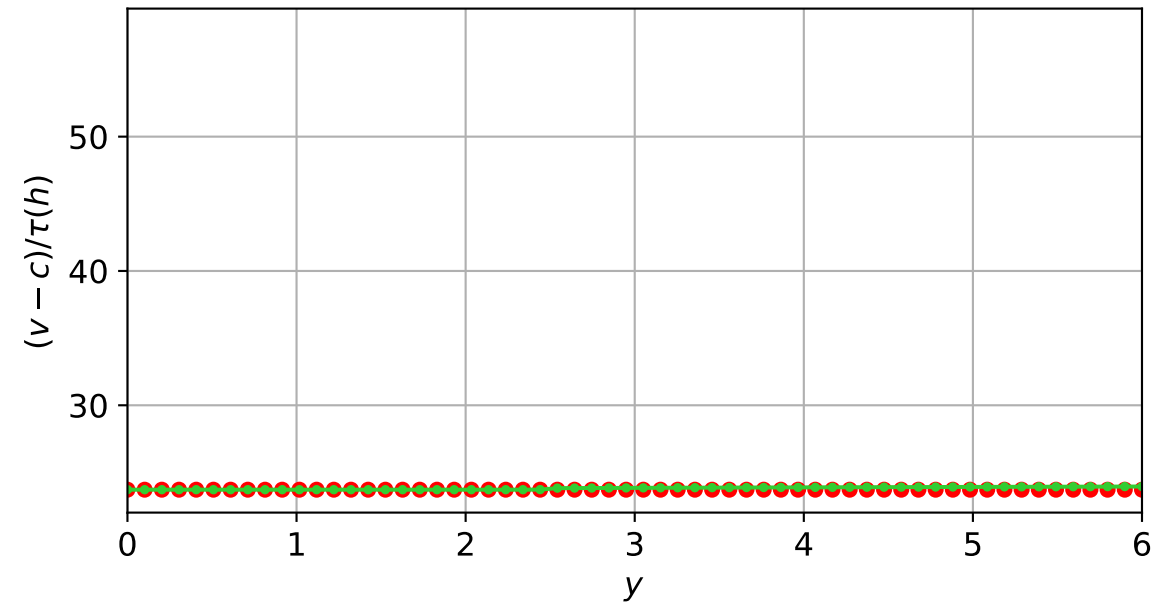
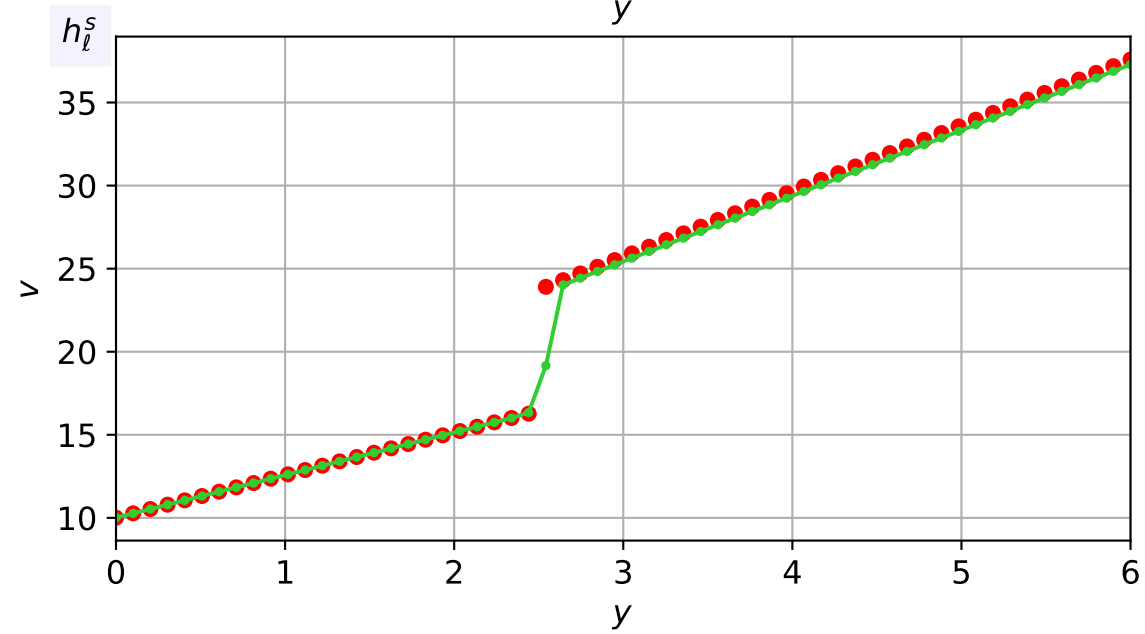
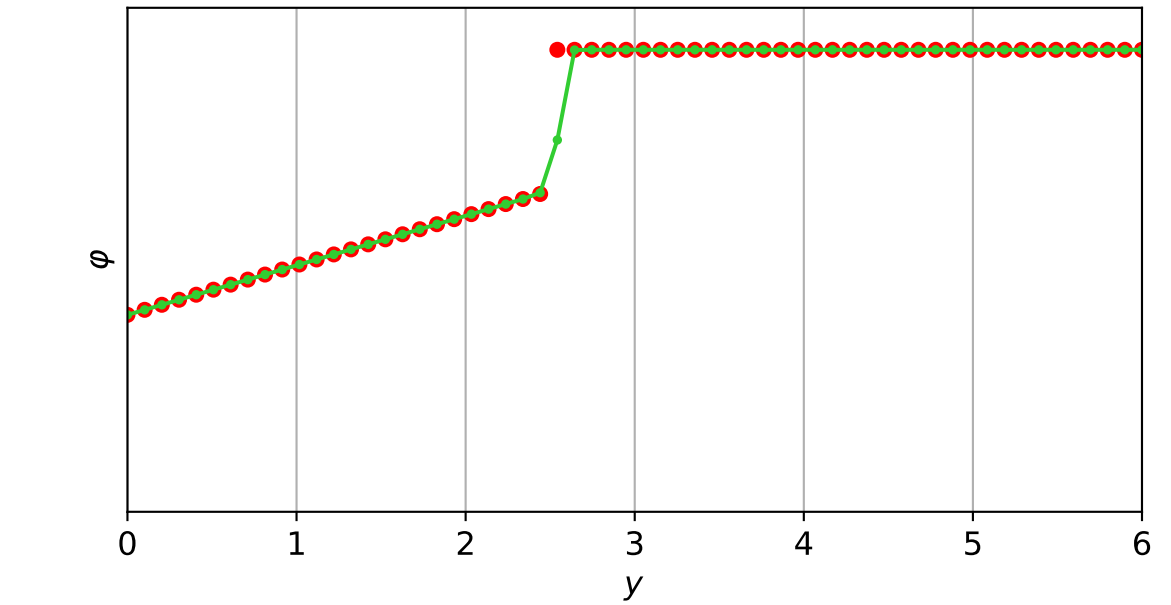
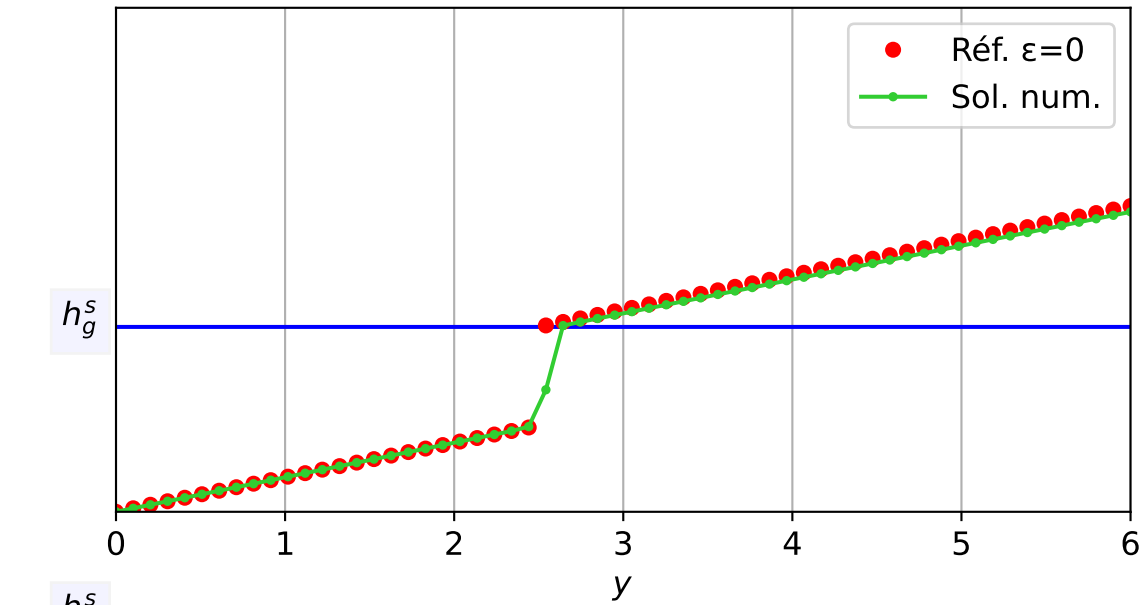
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 1.25003/12$  (4051 iter) — résidu eq. =  $6.76e-06$



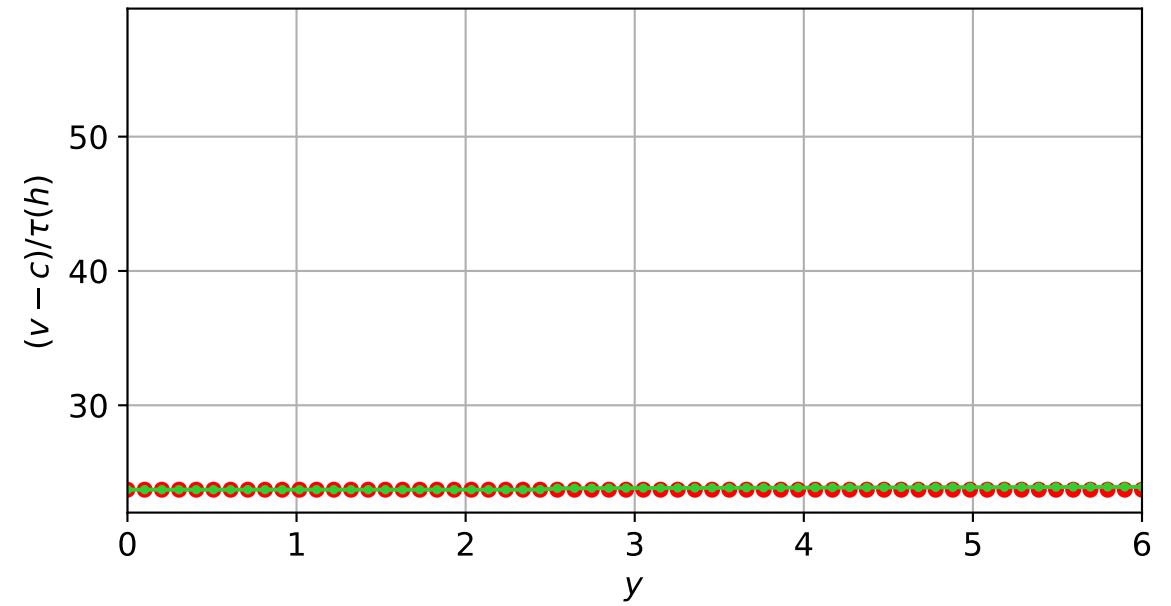
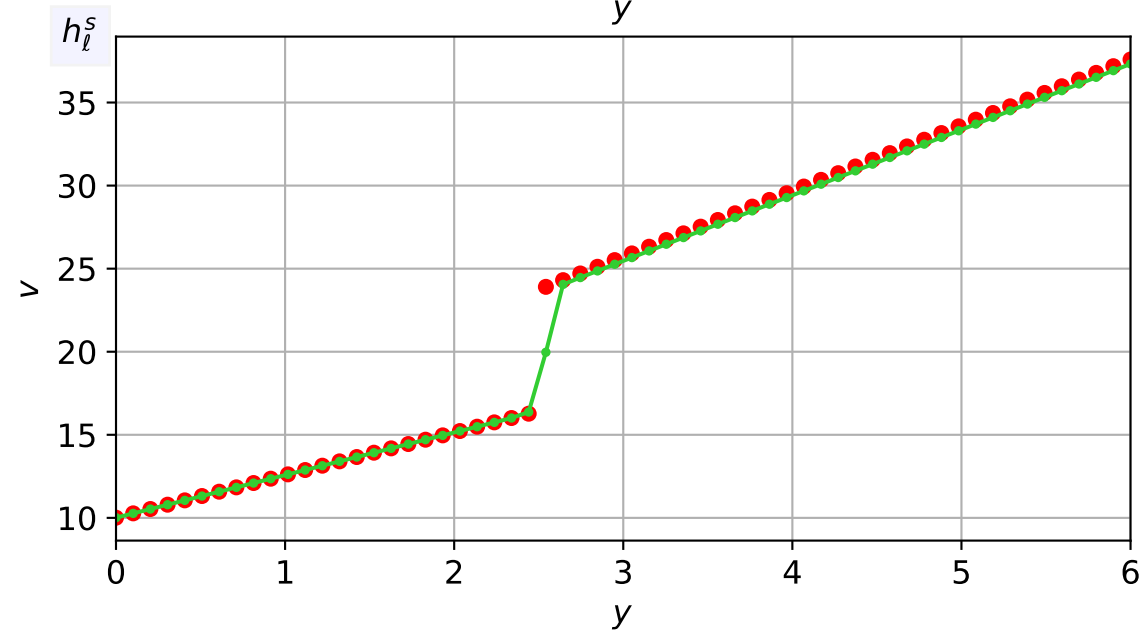
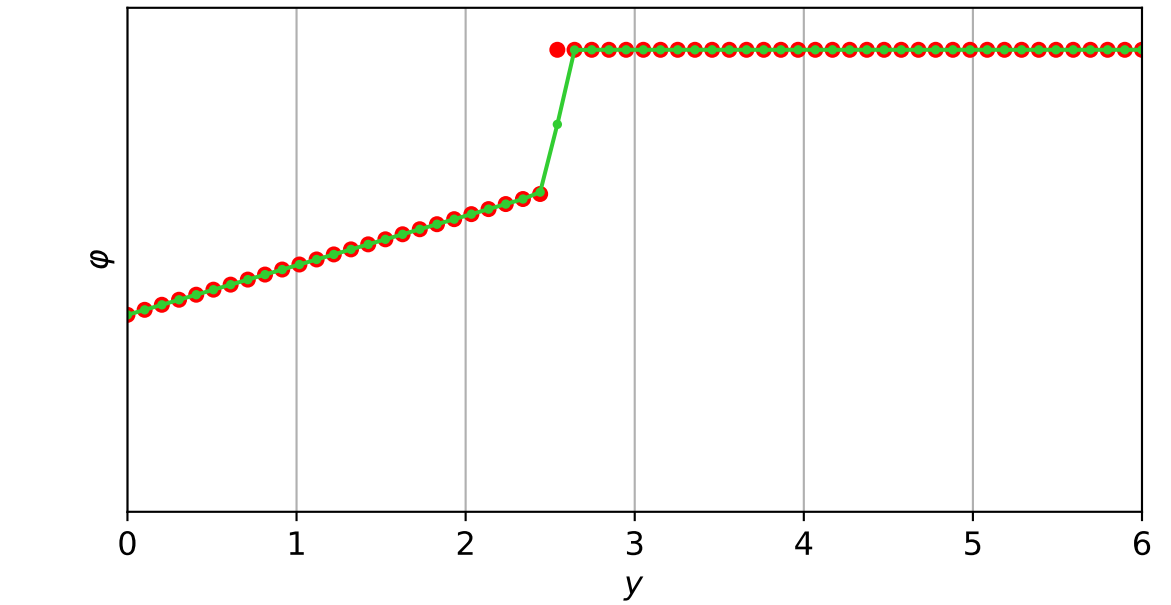
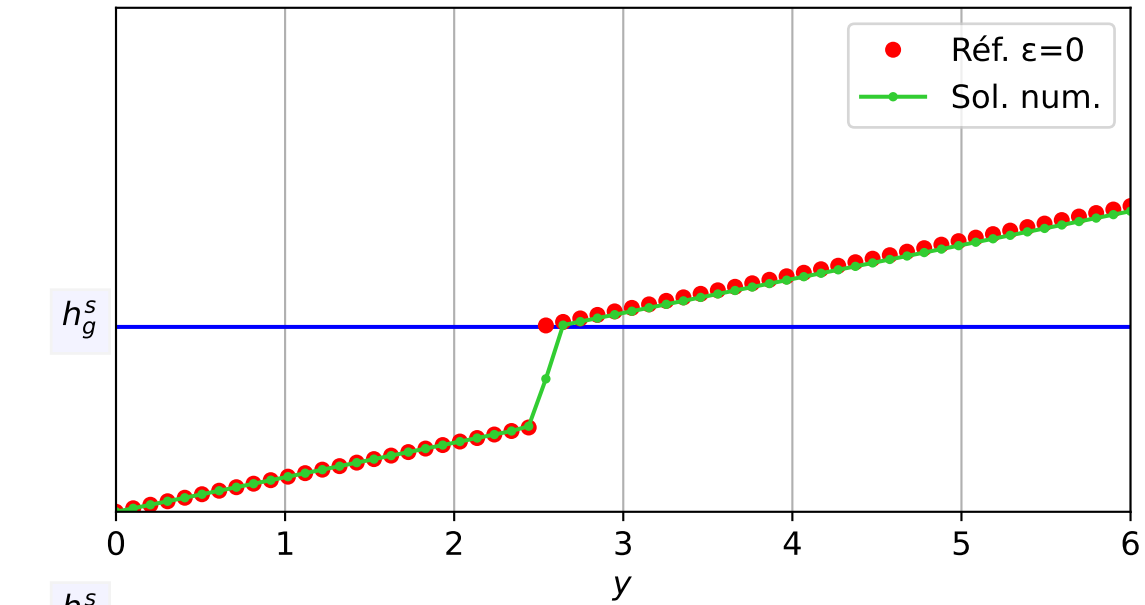
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 1.30003/12$  (4273 iter) — résidu eq. =  $5.46e-06$



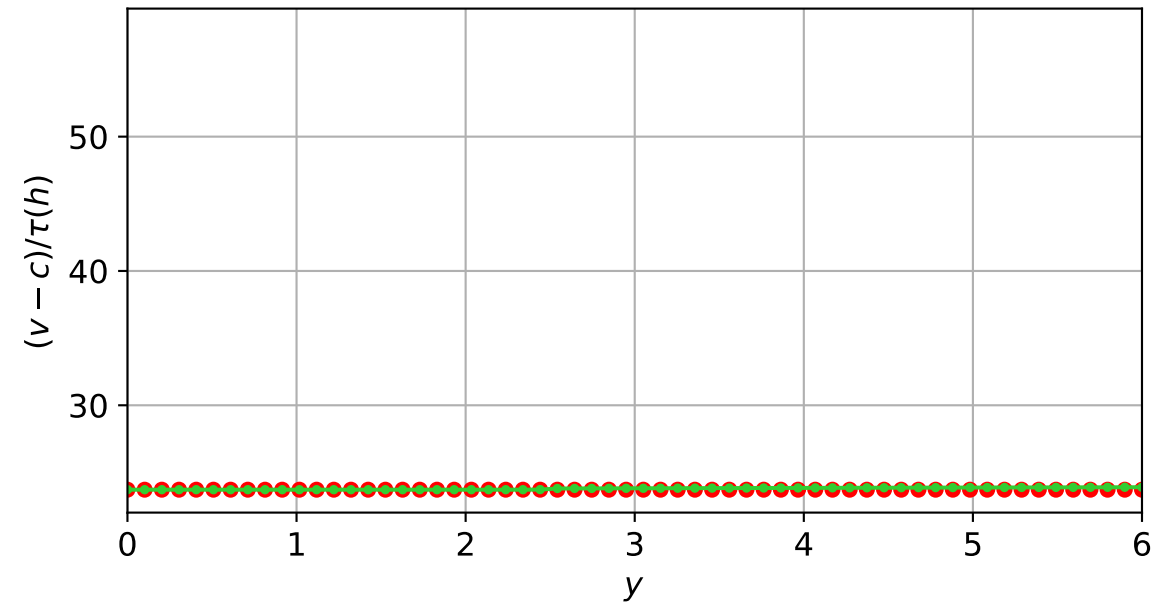
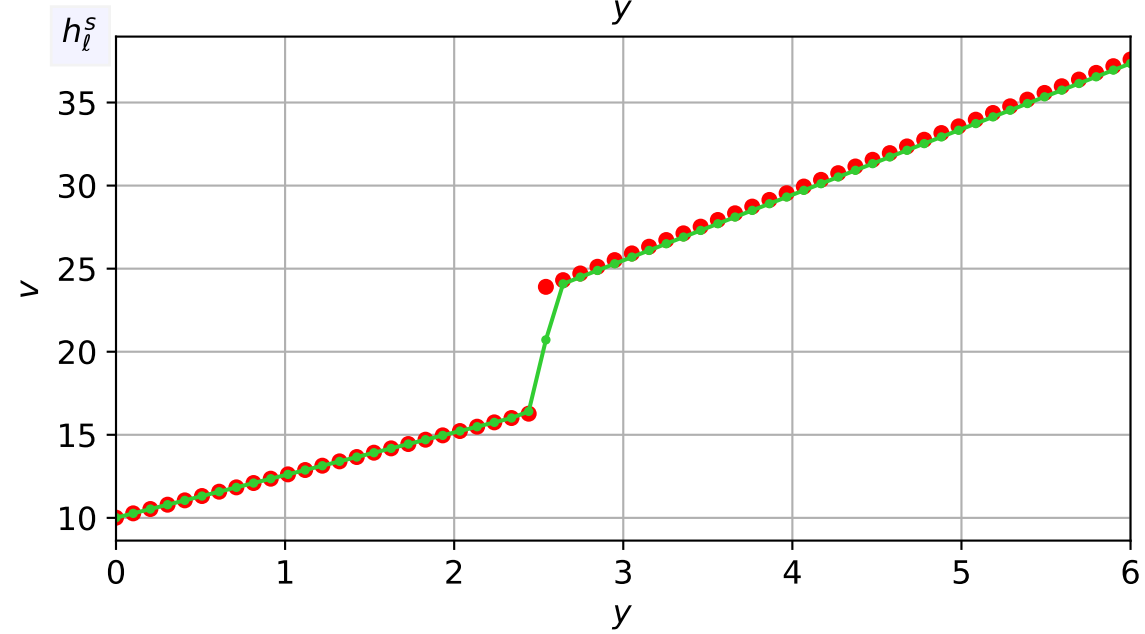
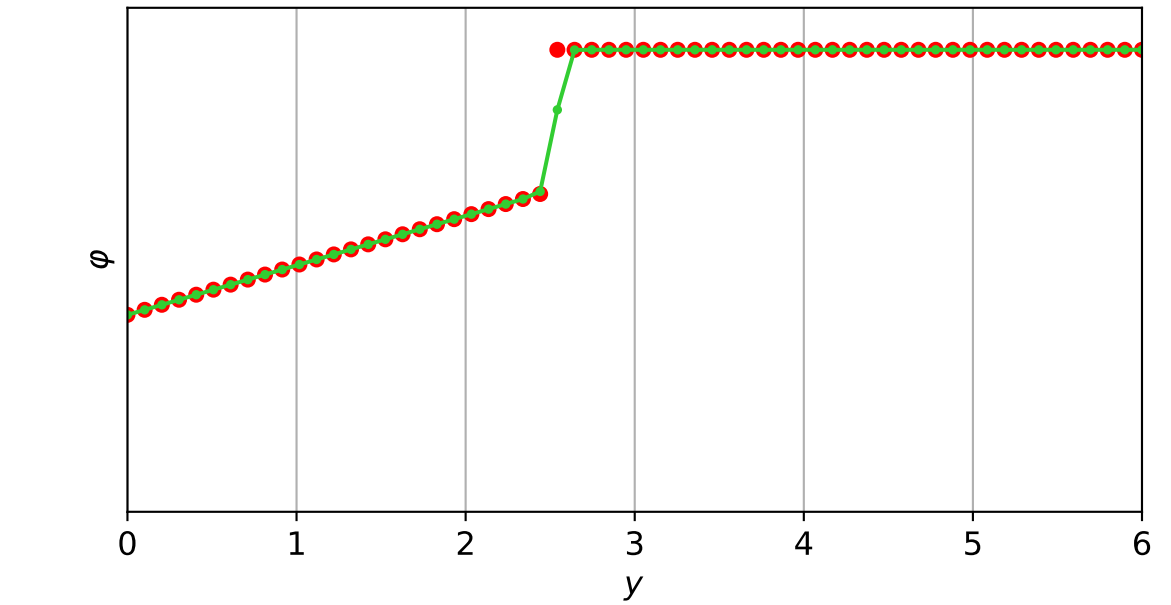
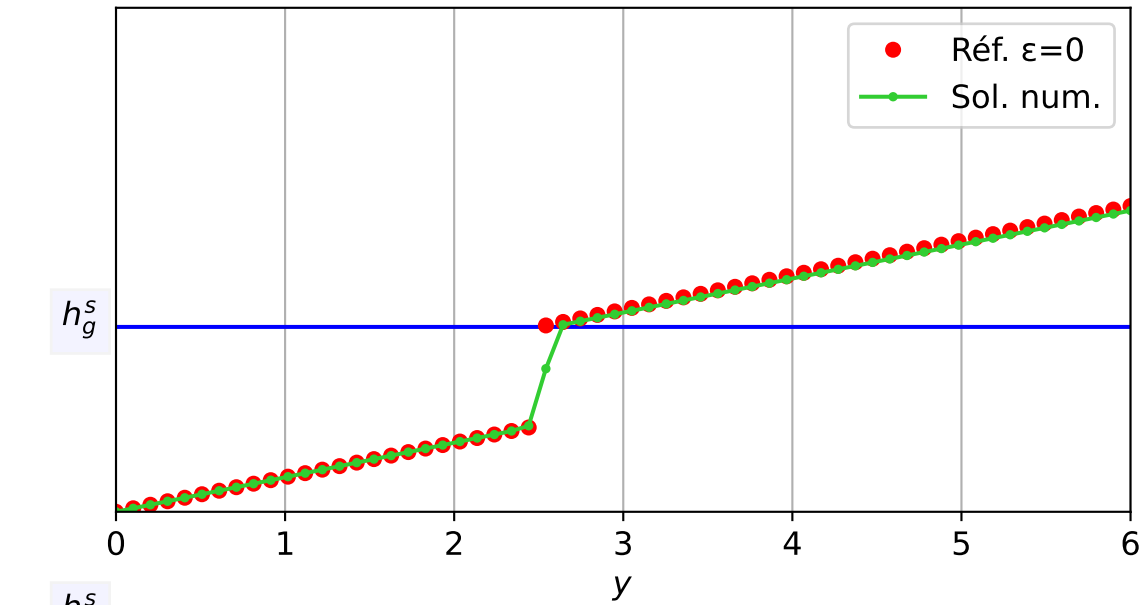
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 1.35003/12$  (4495 iter) — résidu eq. =  $8.22e-06$



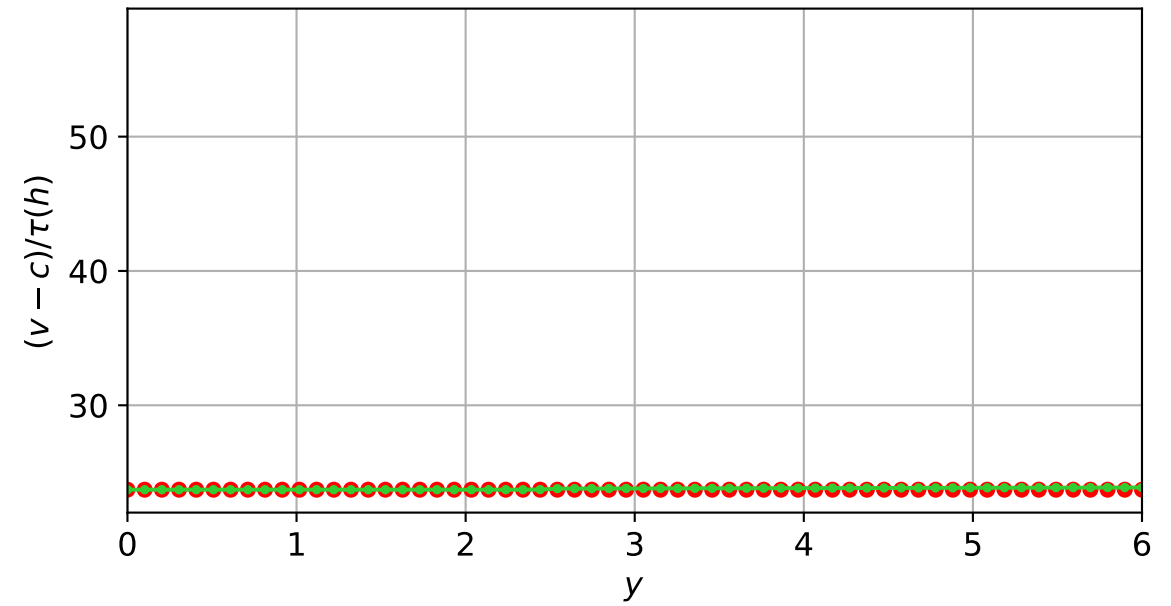
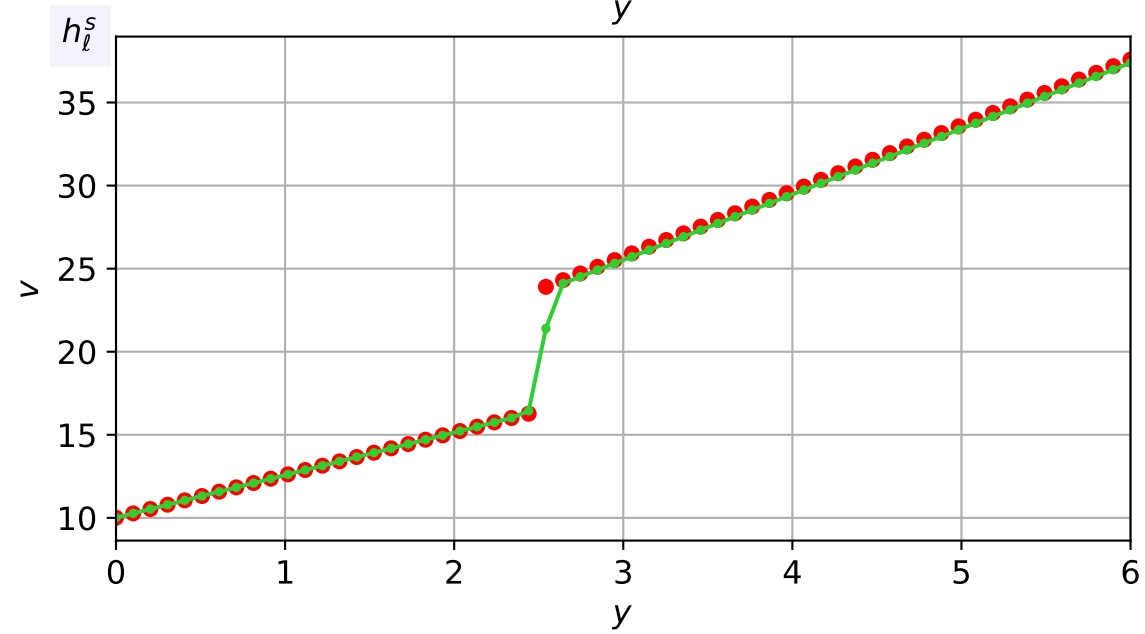
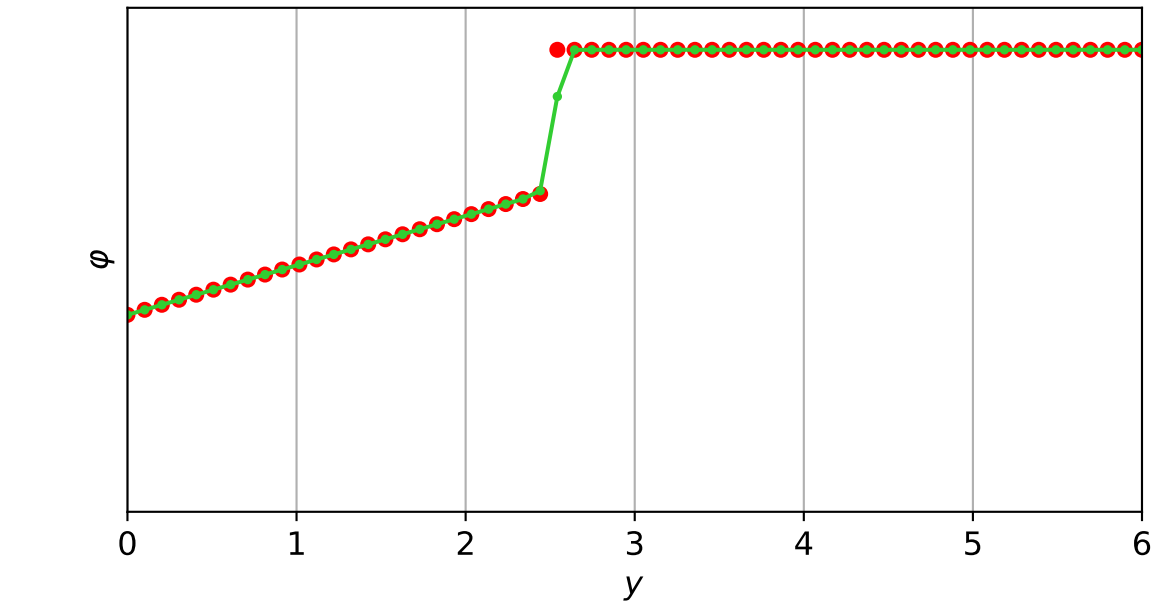
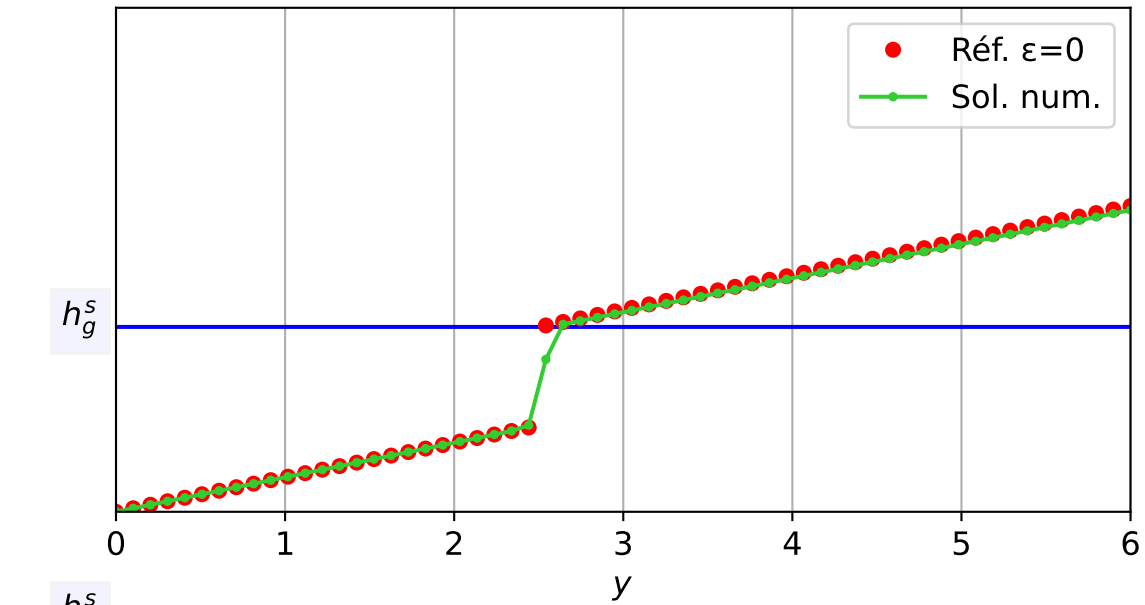
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 1.40004/12$  (4717 iter) — résidu eq. =  $6.87e-06$



TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 1.45004/12$  (4939 iter) — résidu eq. =  $6.00e-06$

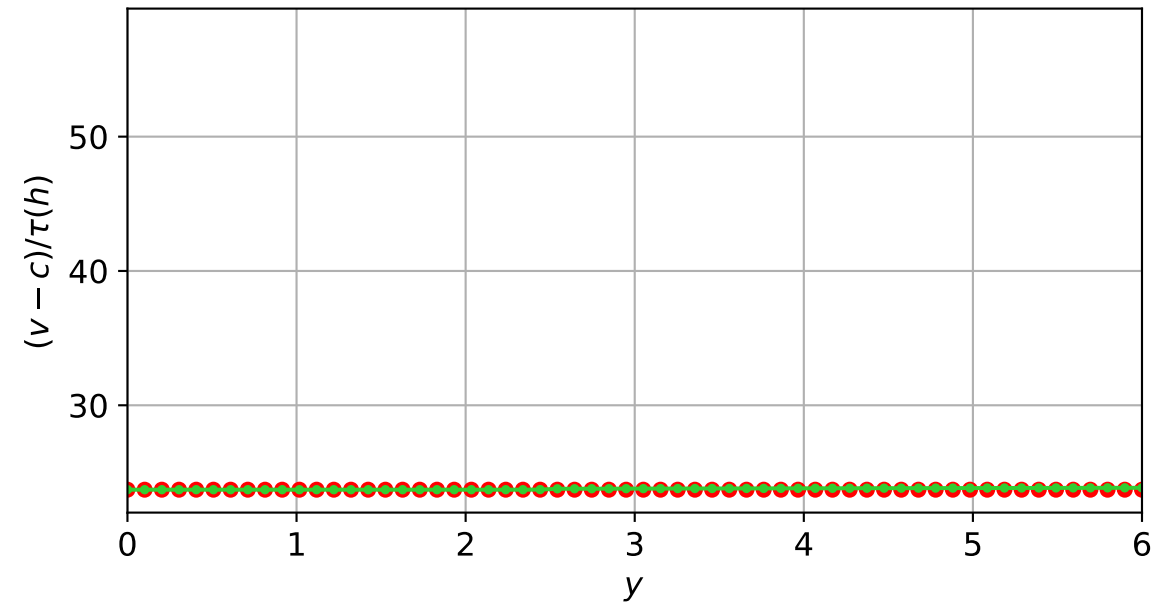
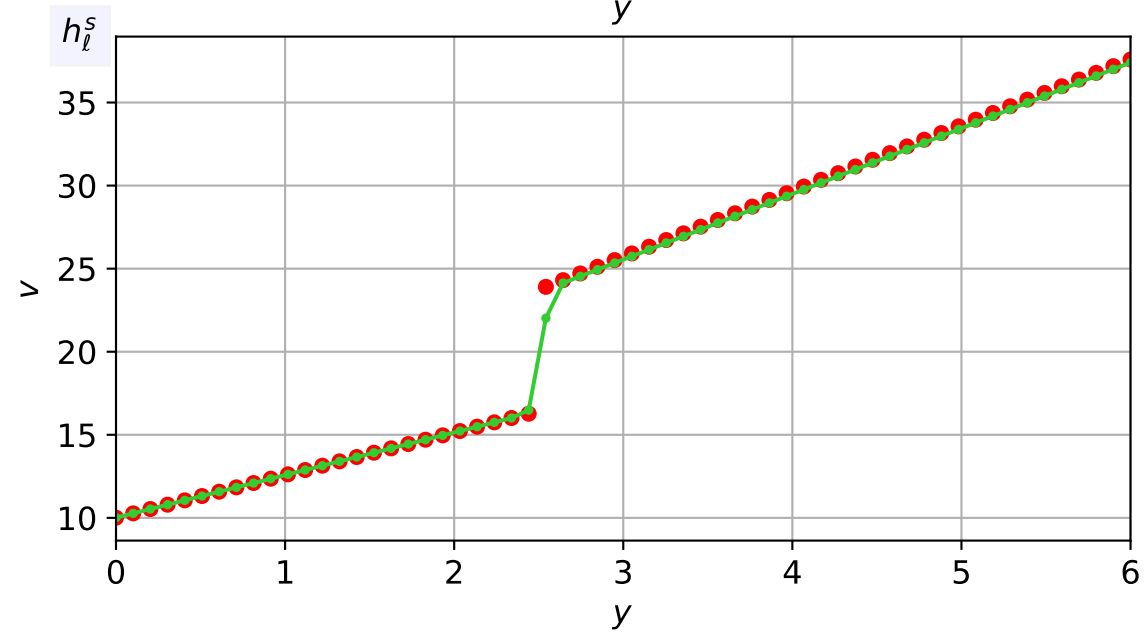
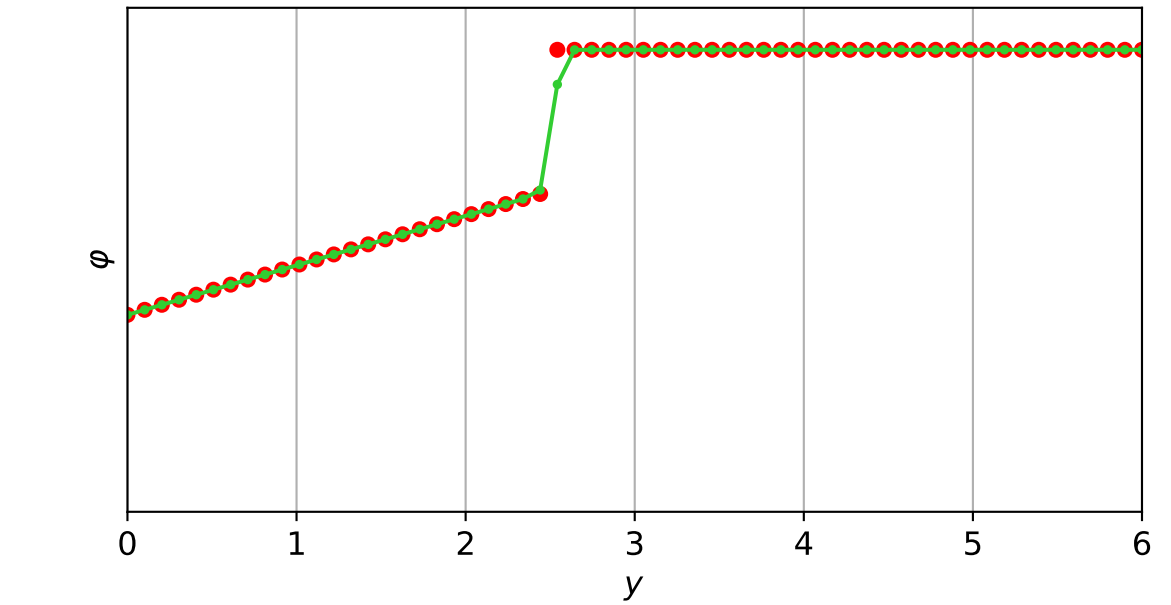
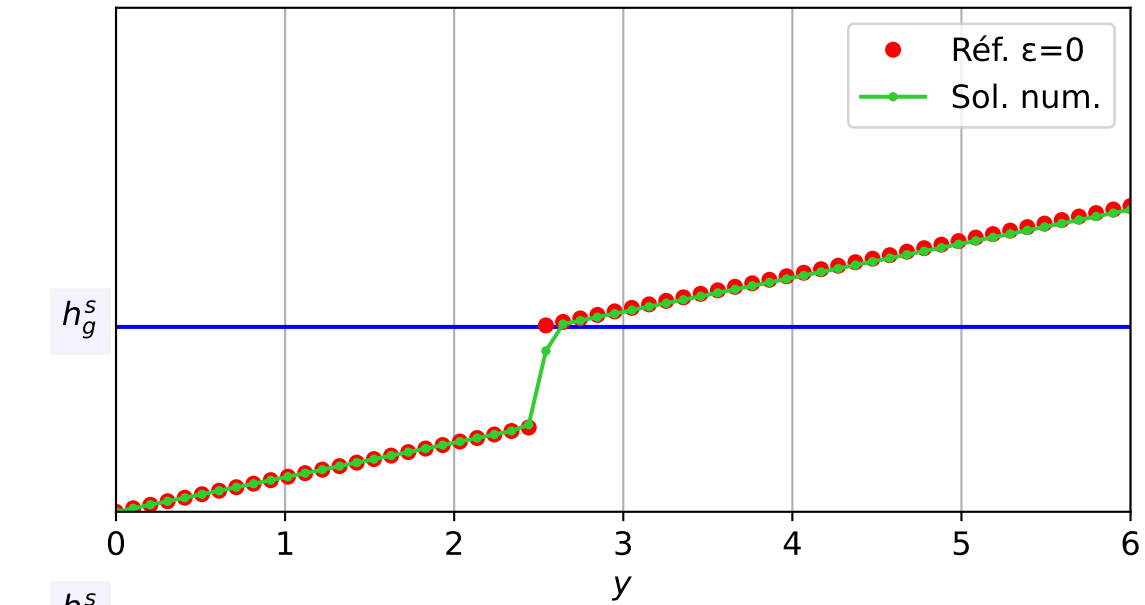


TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.25e-04$  —  $t = 1.50004/12$  (5161 iter) — résidu eq. =  $5.36e-06$

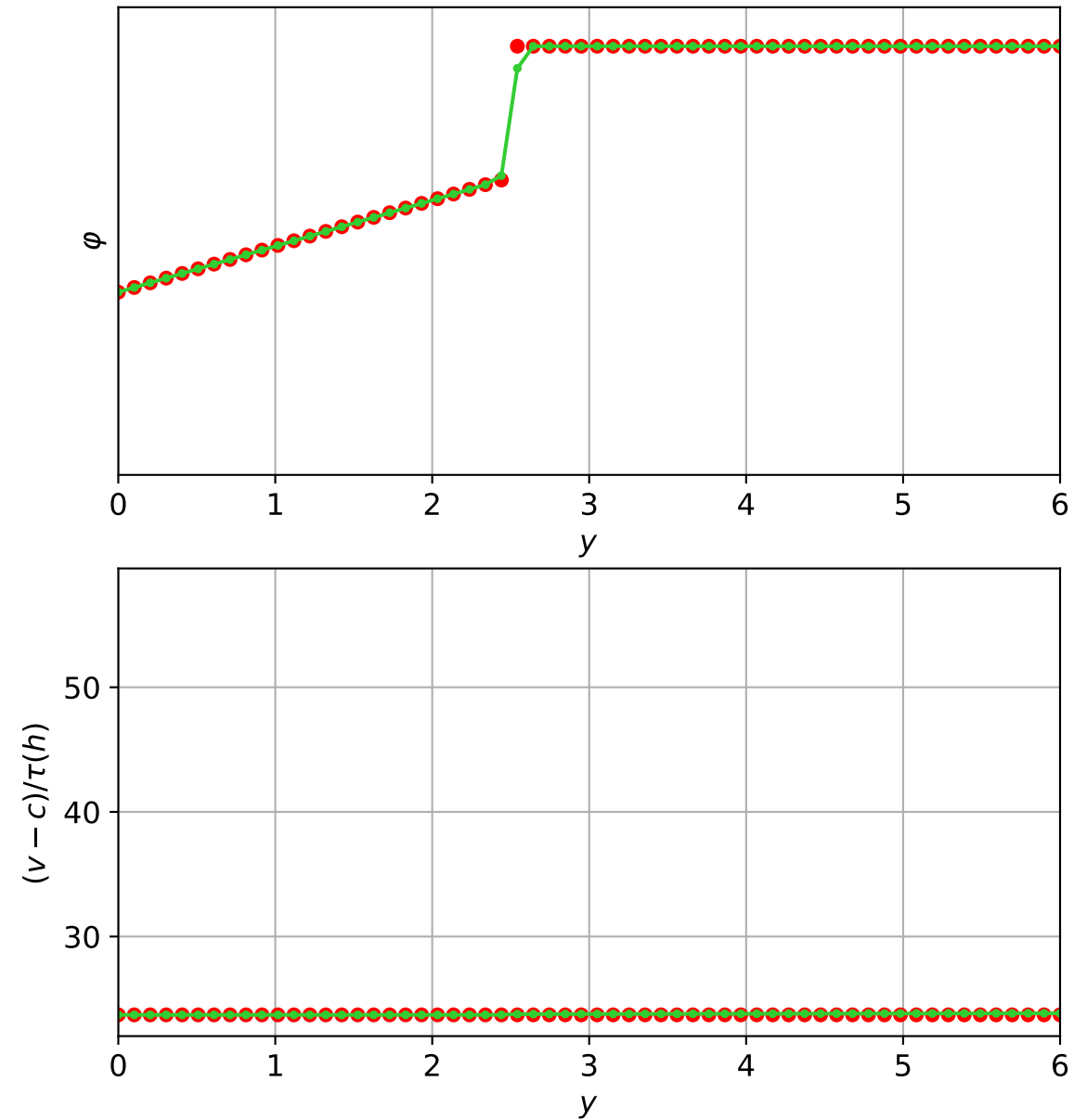
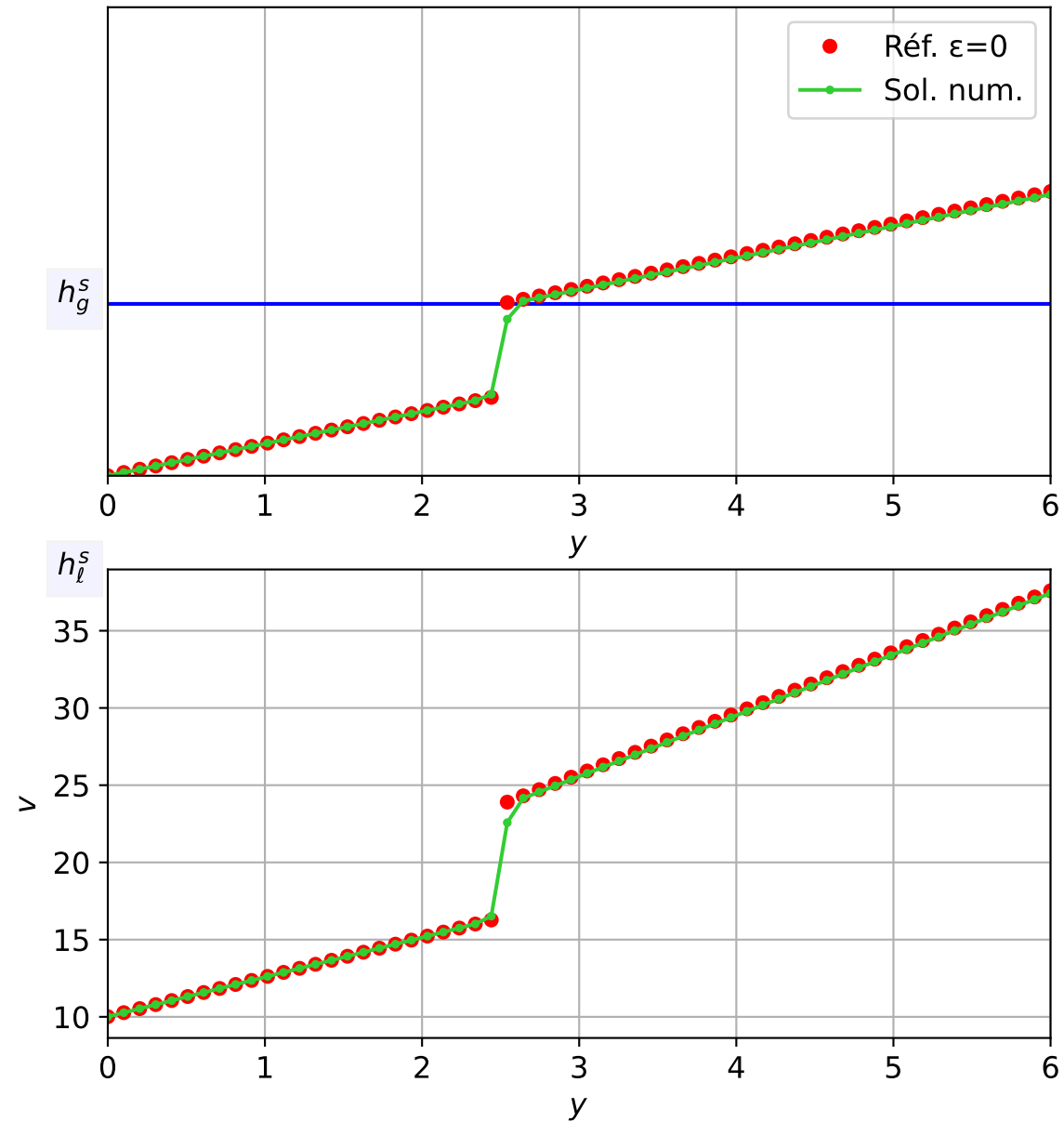




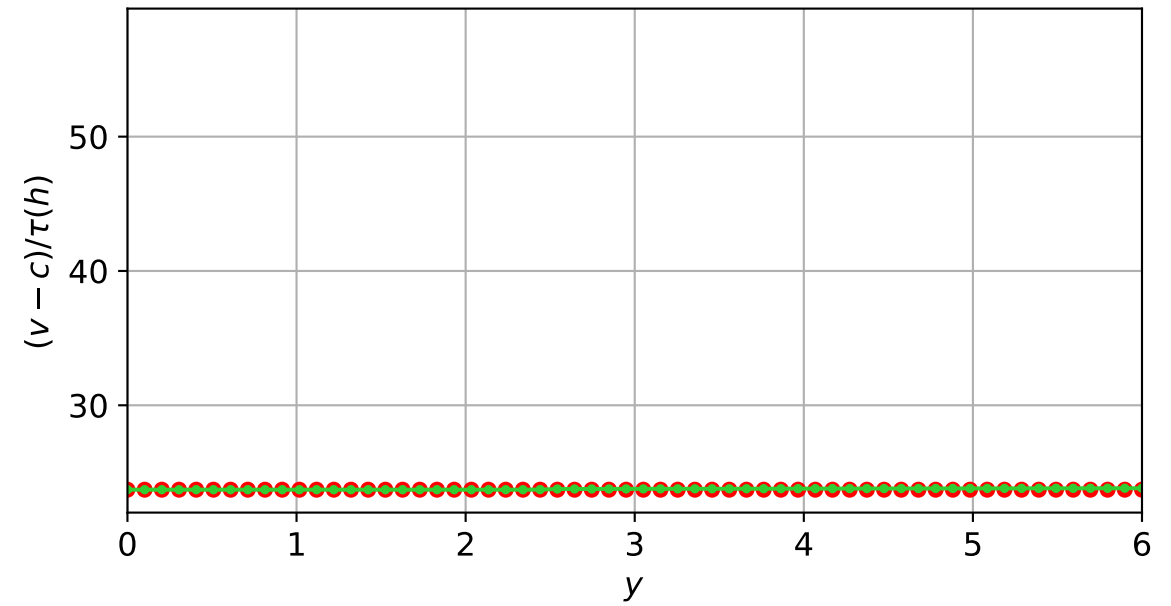
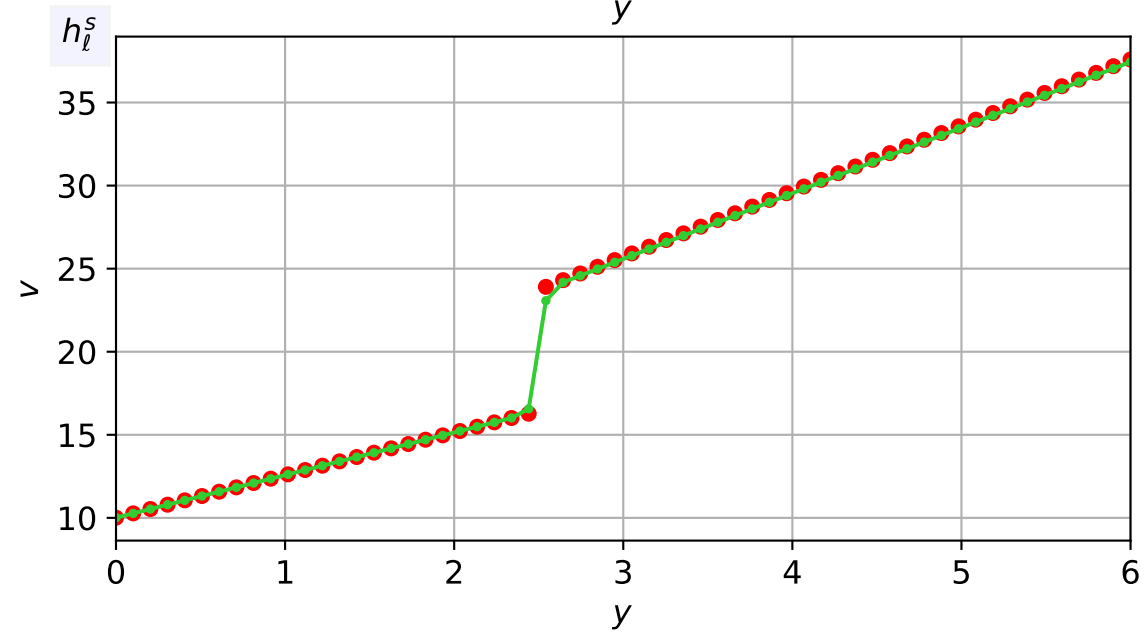
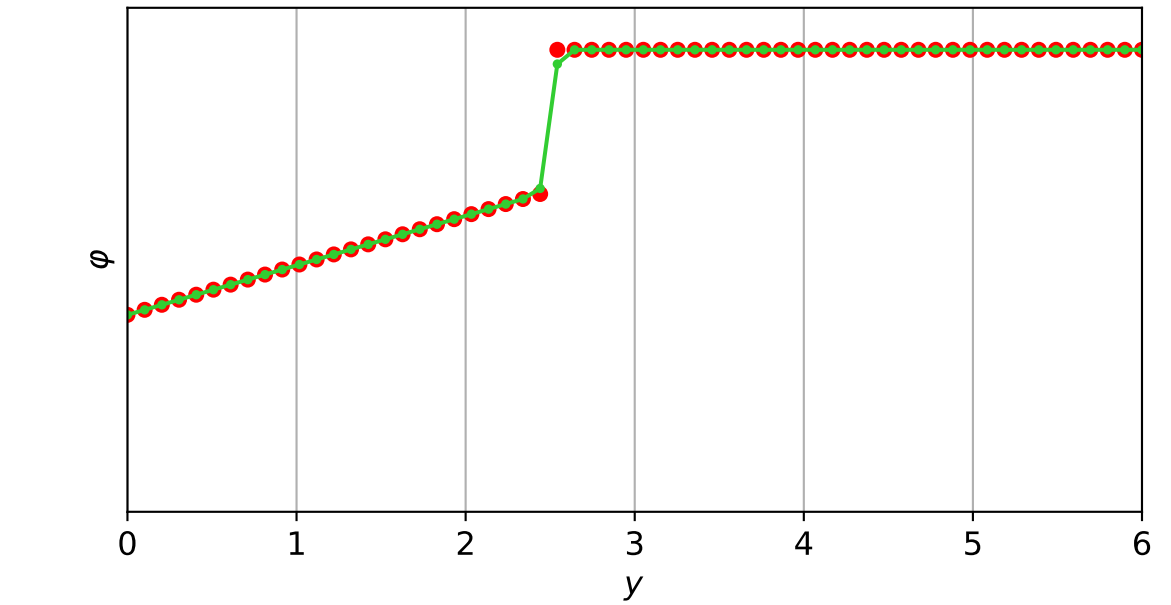
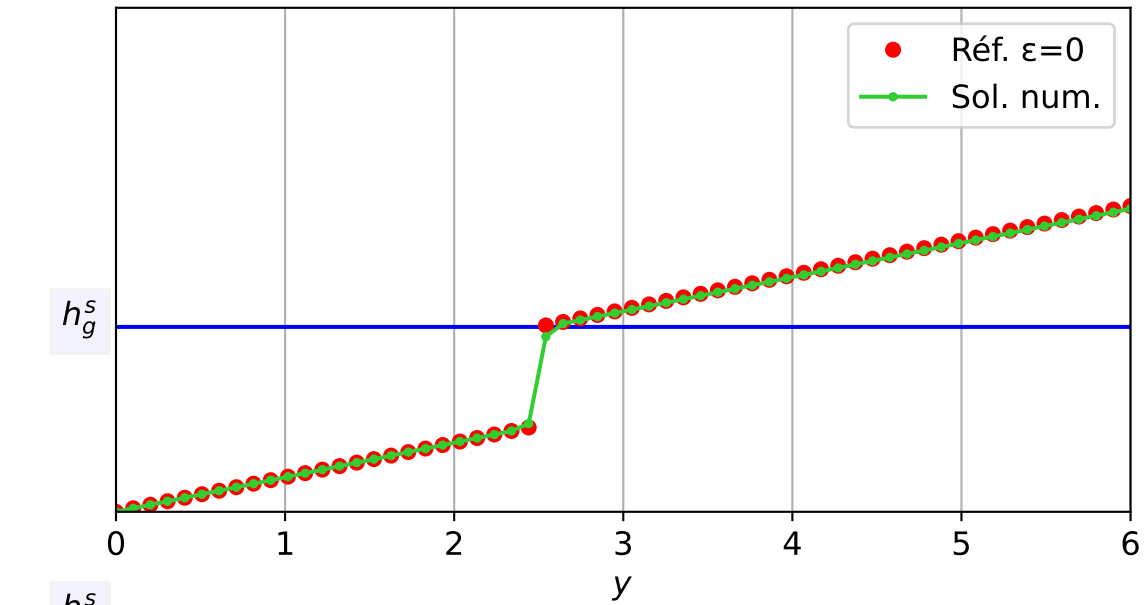
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.70e-04$  —  $t = 1.55022/12$  (5359 iter) — résidu eq. =  $7.54e-06$



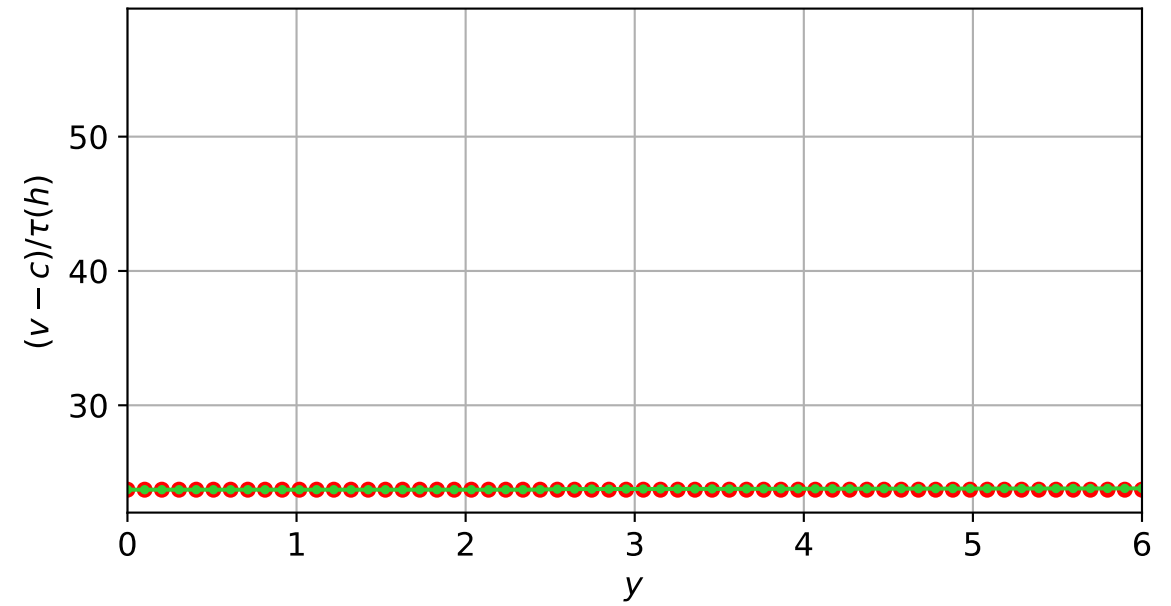
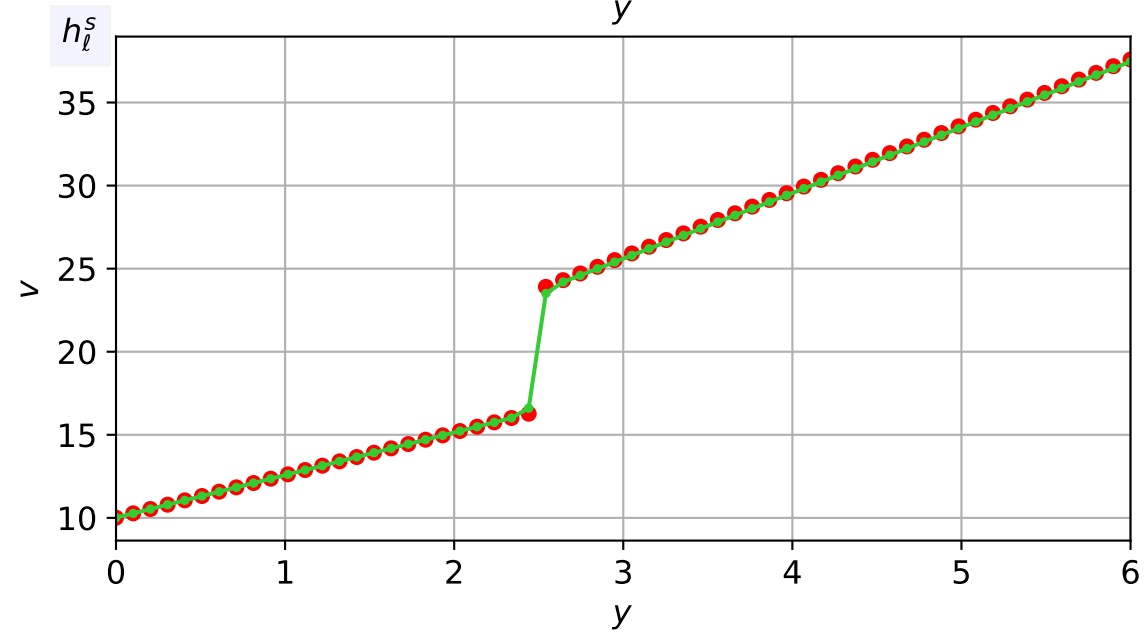
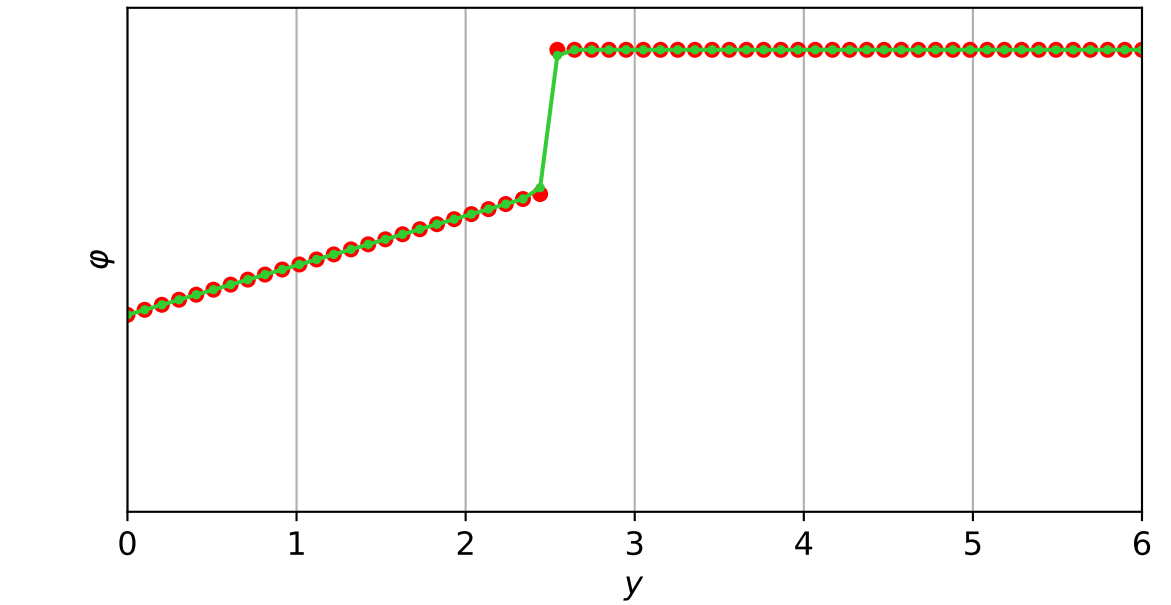
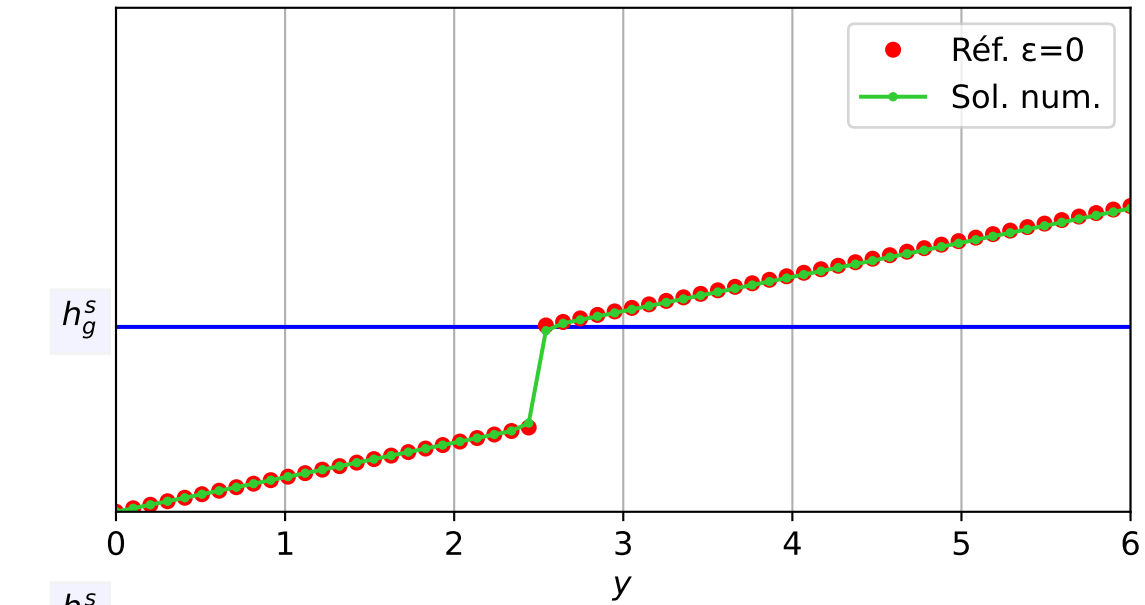
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.70e-04$  —  $t = 1.60022/12$  (5544 iter) — résidu eq. =  $6.78e-06$



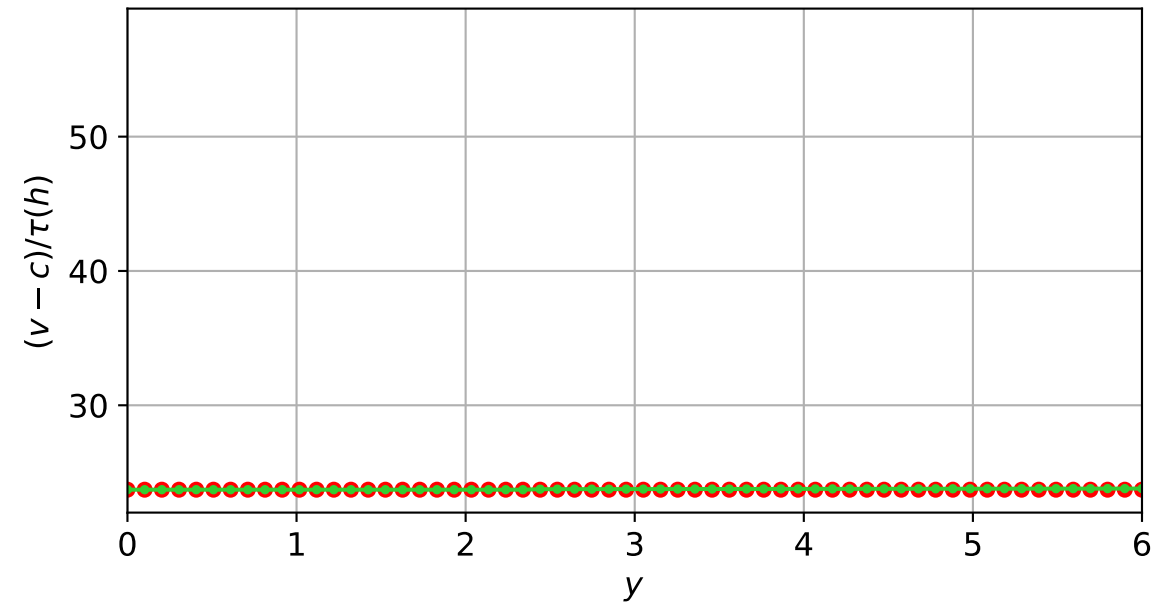
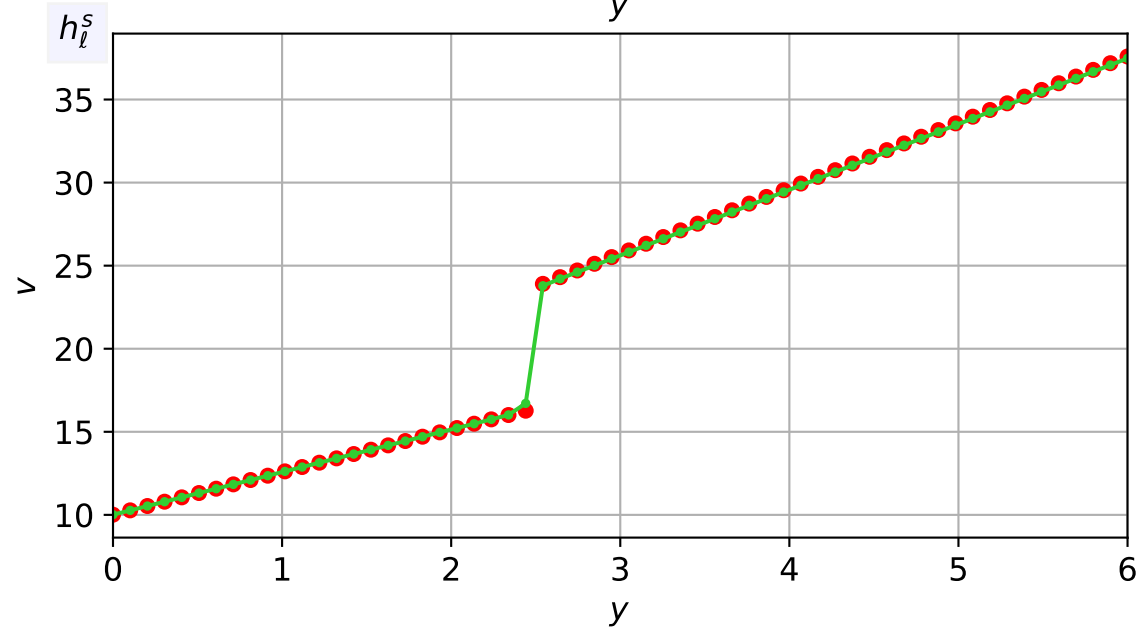
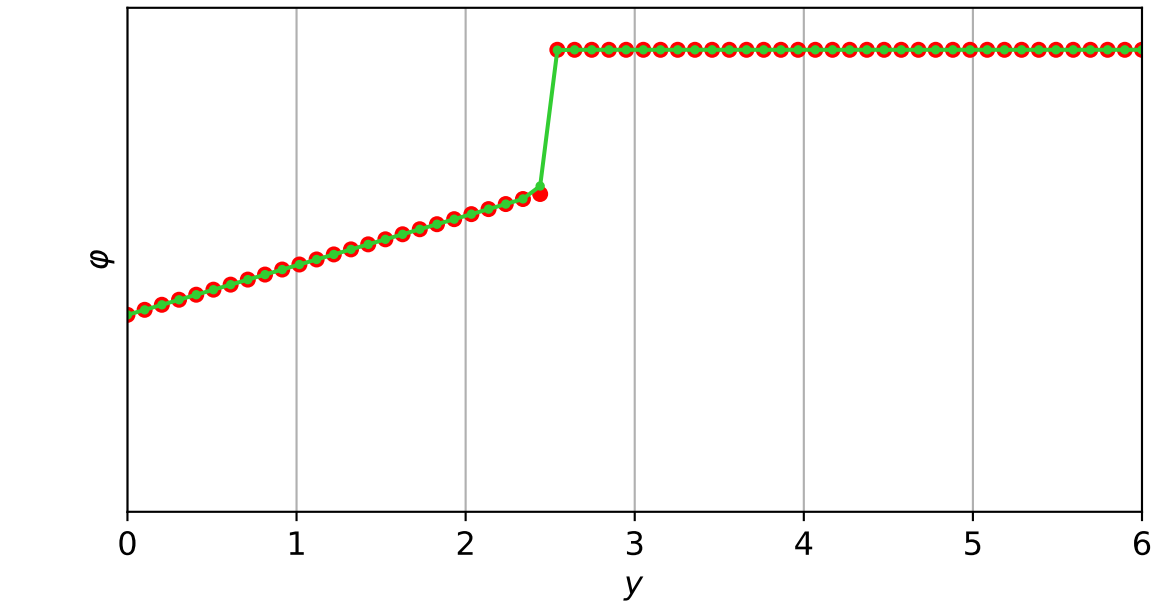
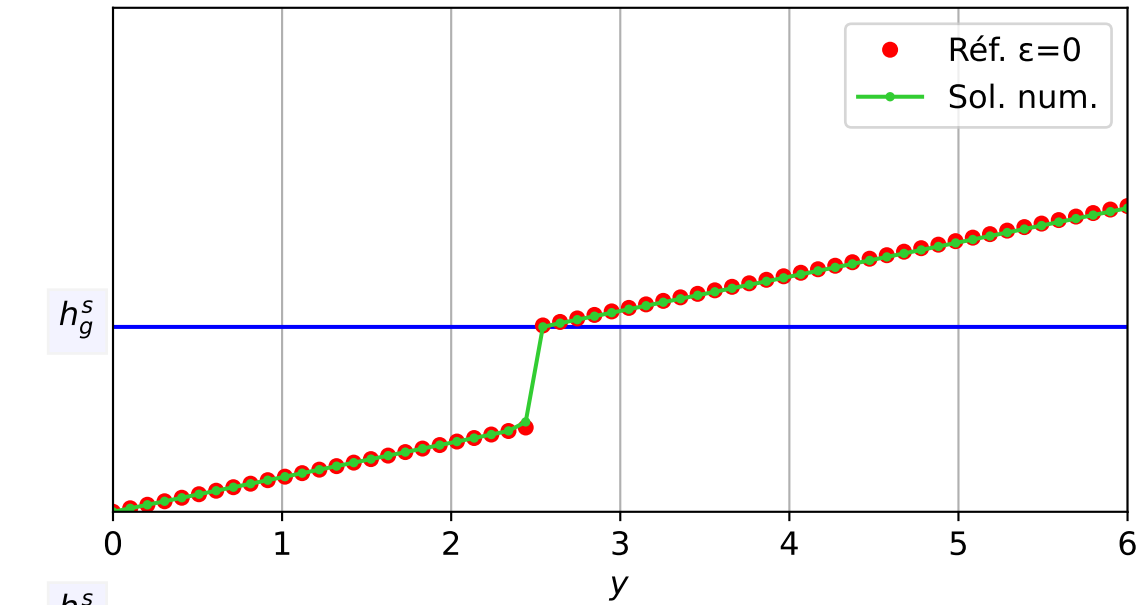
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.70e-04$  —  $t = 1.65023/12$  (5729 iter) — résidu eq. =  $8.62e-06$



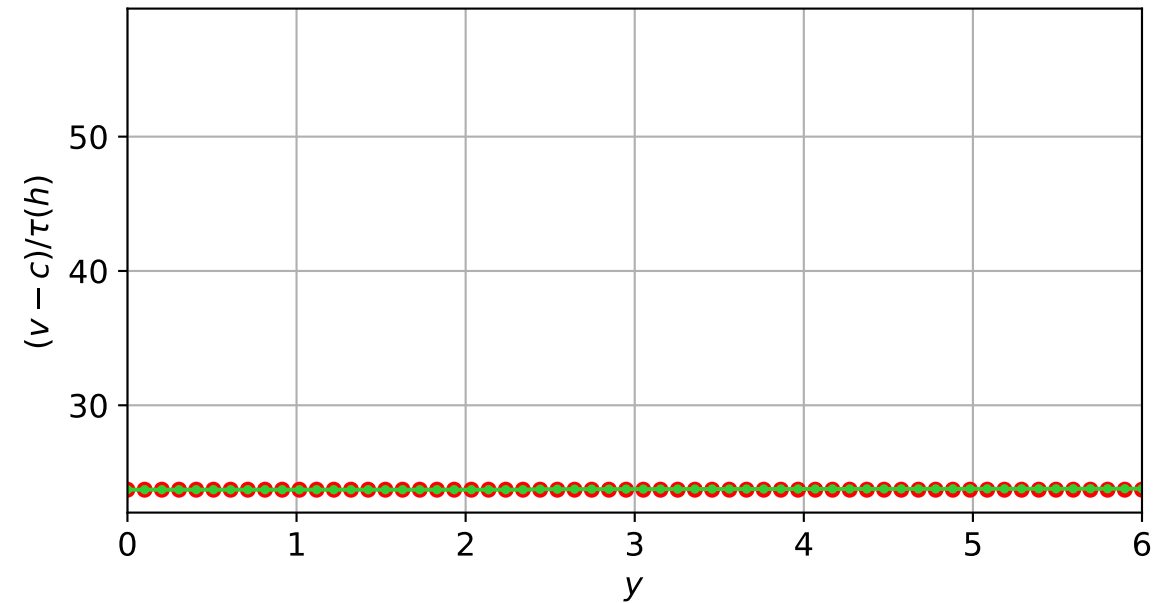
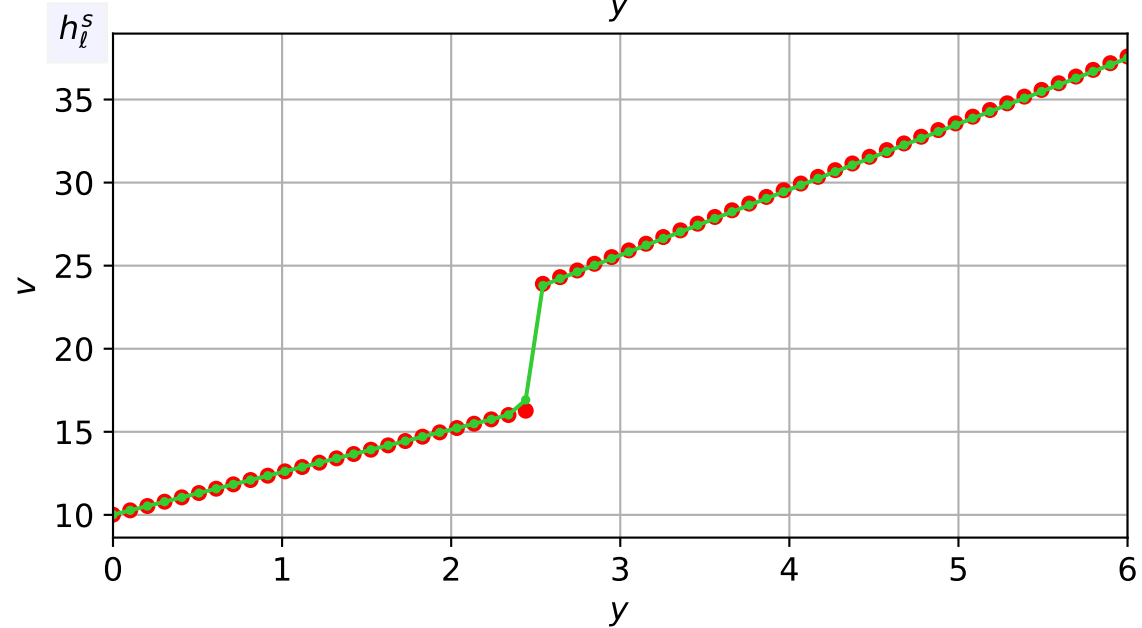
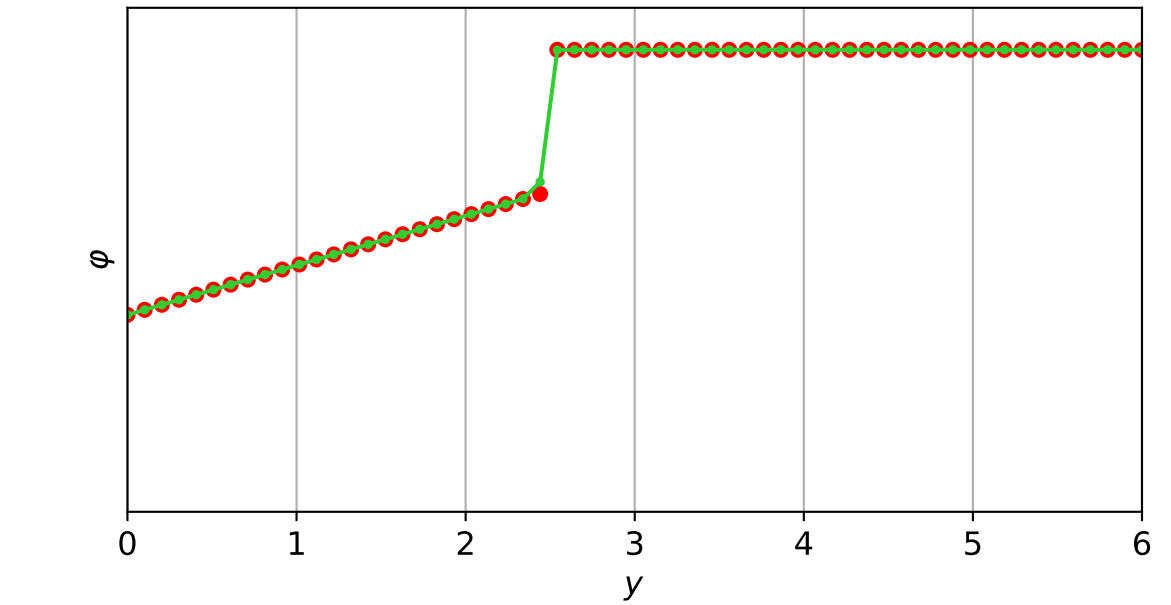
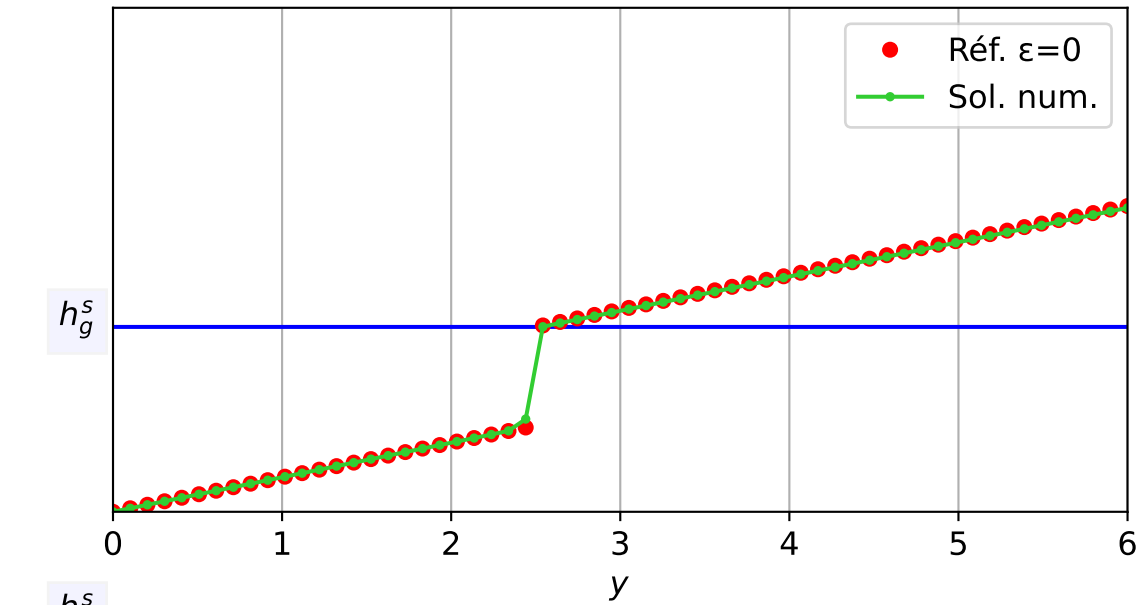
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.70e-04$  —  $t = 1.70023/12$  (5914 iter) — résidu eq. =  $6.49e-06$



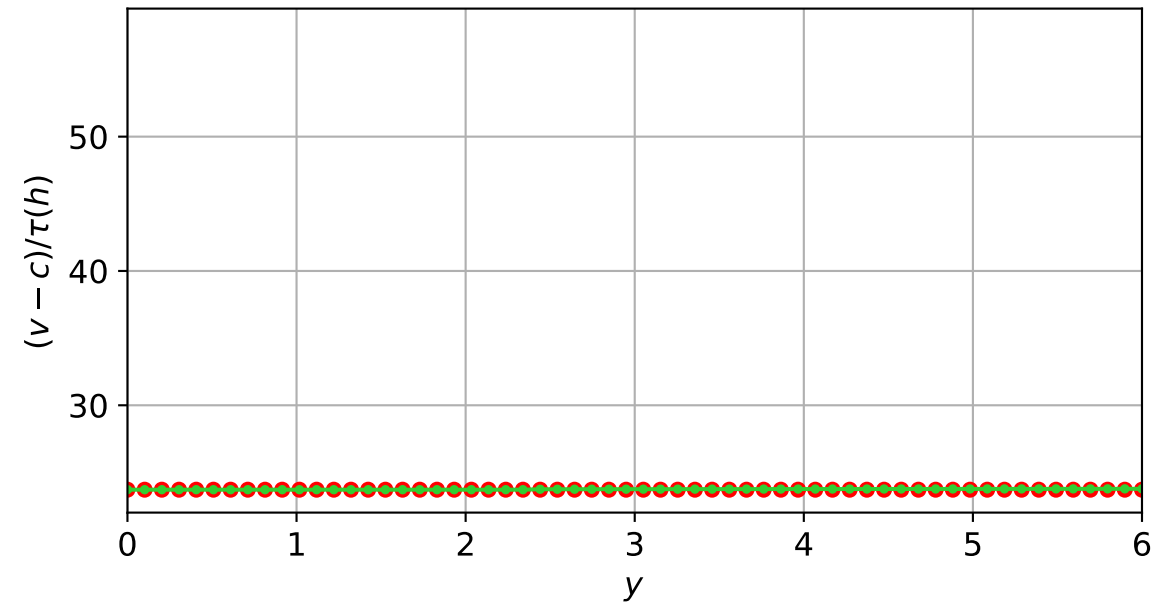
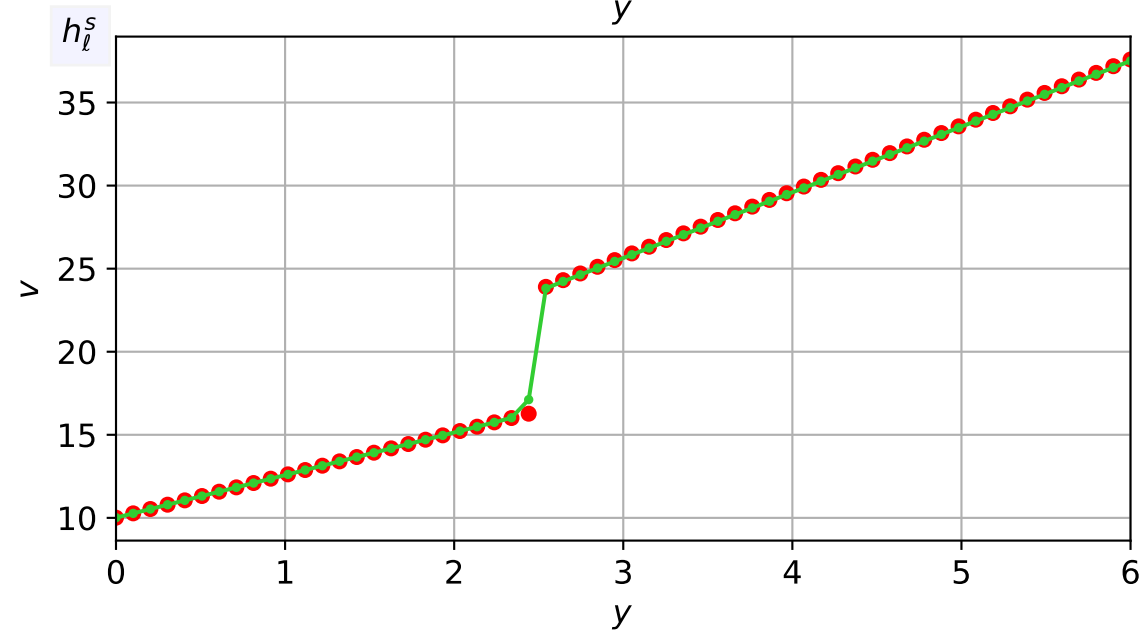
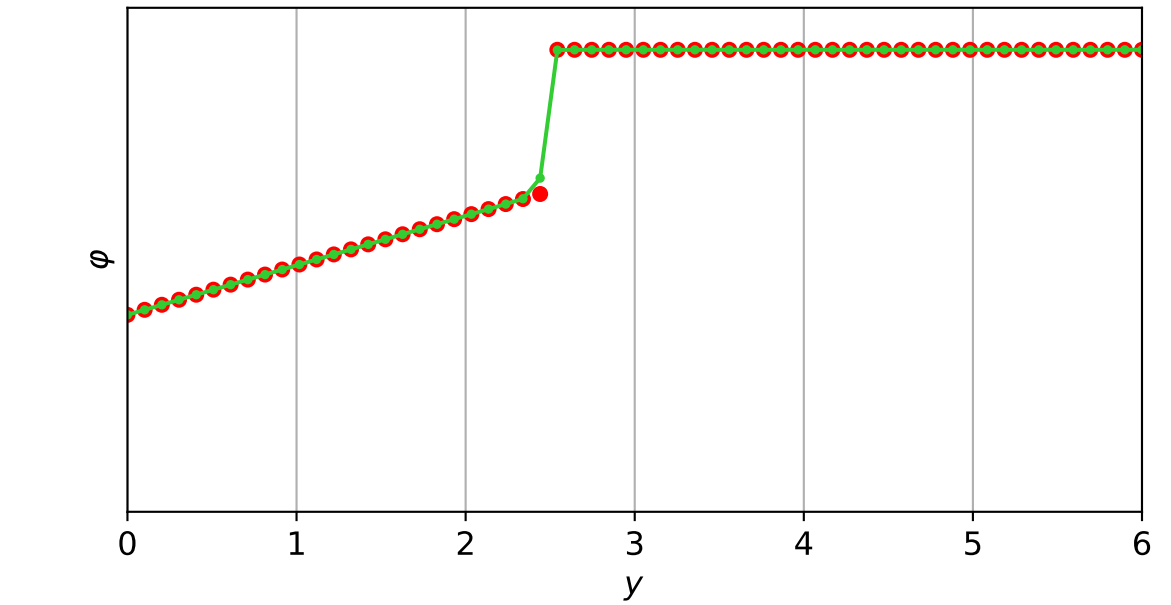
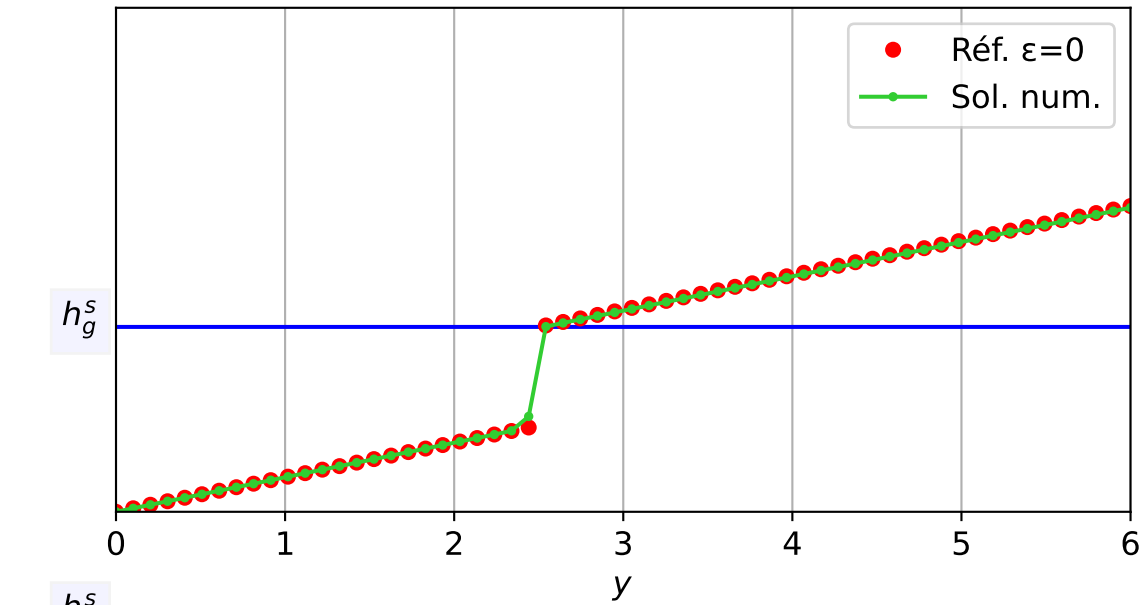
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.36e-04$  —  $t = 1.75009/12$  (6137 iter) — résidu eq. =  $6.98e-06$



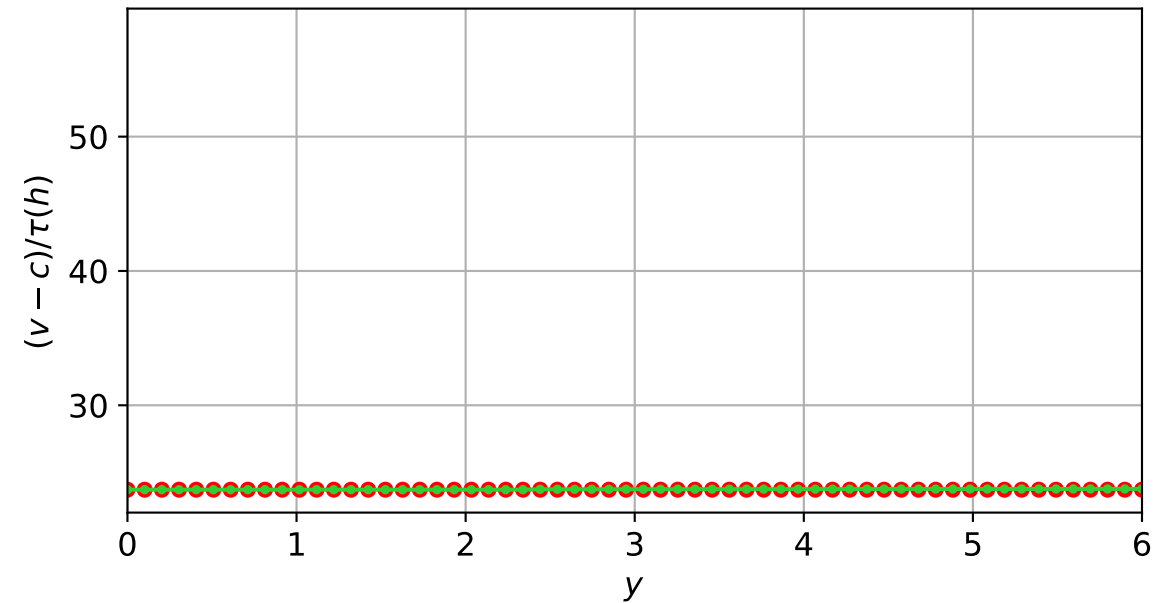
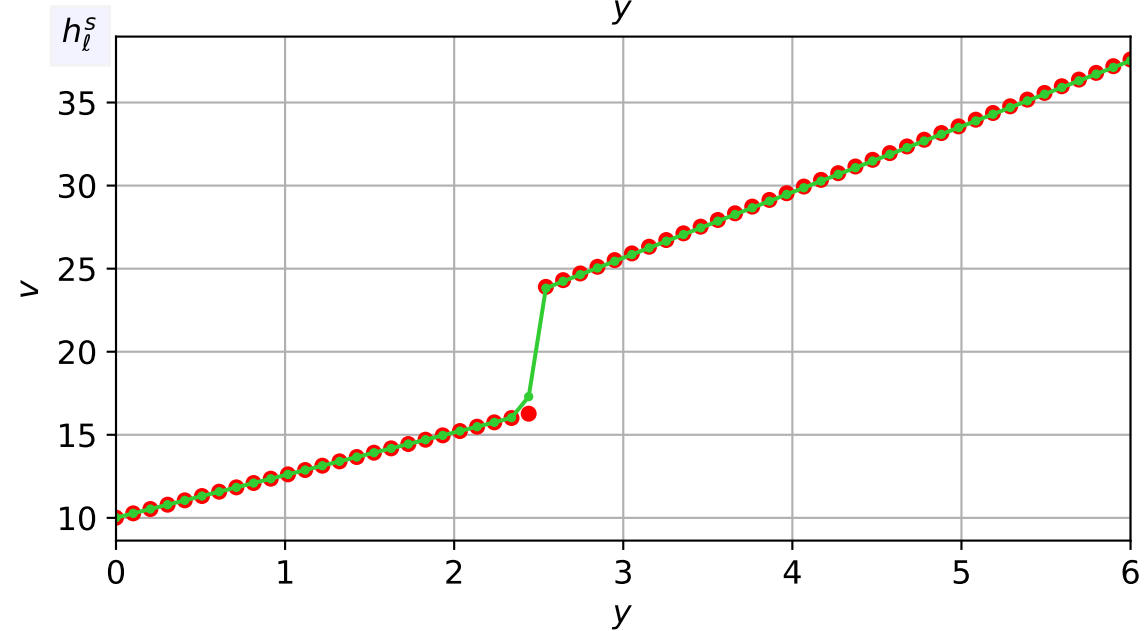
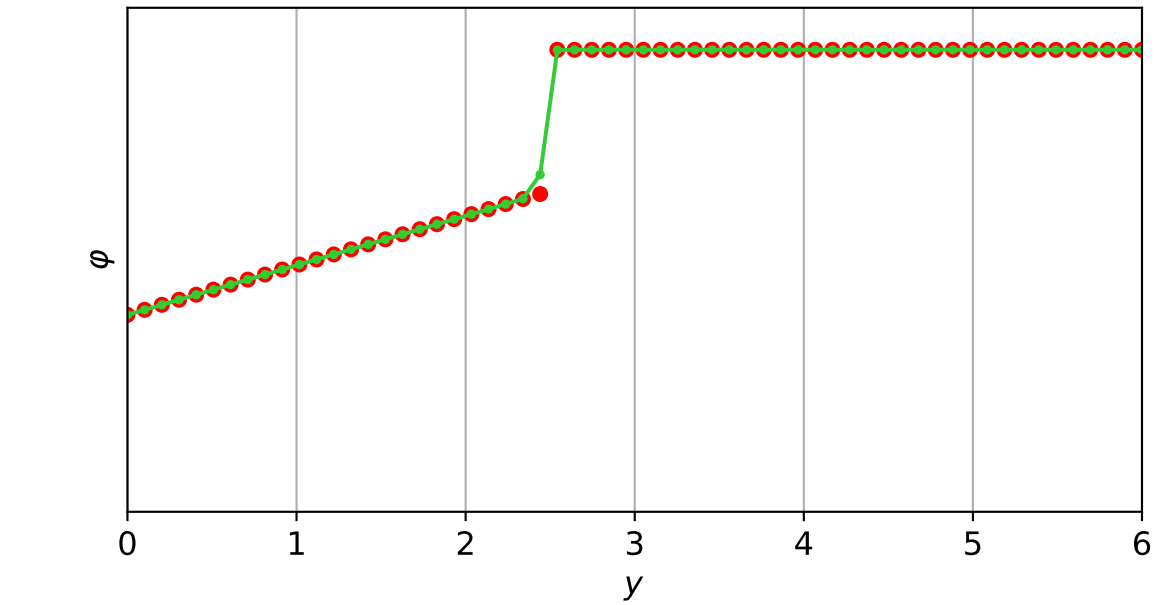
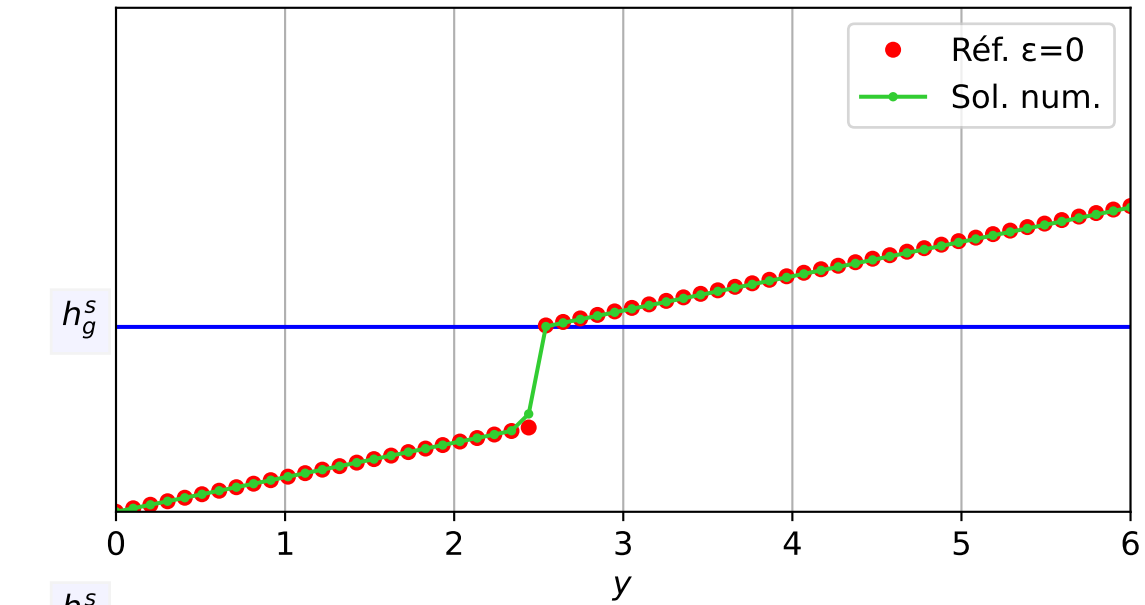
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.36e-04$  —  $t = 1.8002/12$  (6286 iter) — résidu eq. =  $8.30e-06$



TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 3.36e-04$  —  $t = 1.8503/12$  (6435 iter) — résidu eq. =  $6.19e-06$

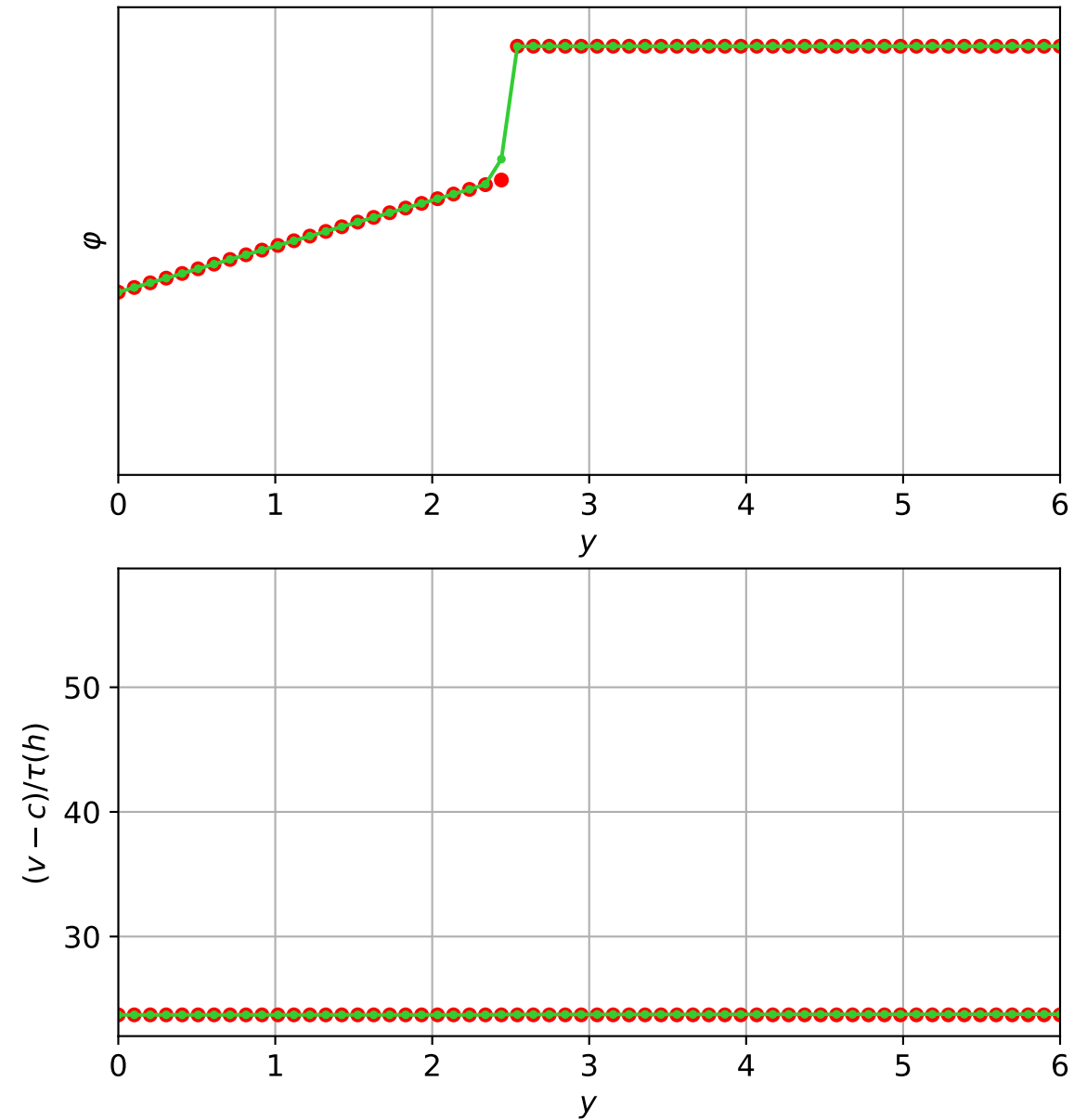
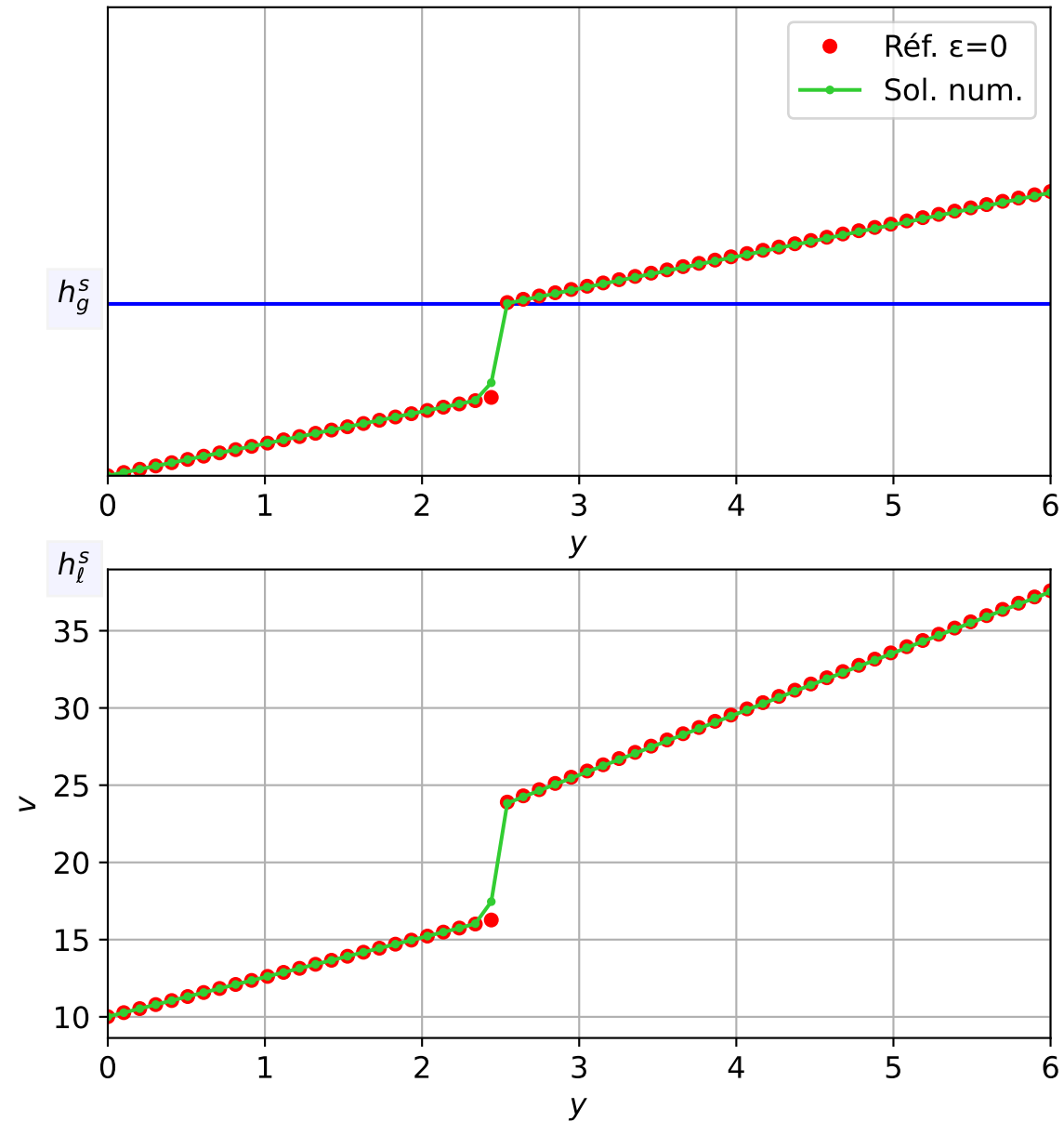


TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 4.04e-04$  —  $t = 1.90034/12$  (6566 iter) — résidu eq. =  $7.41e-06$

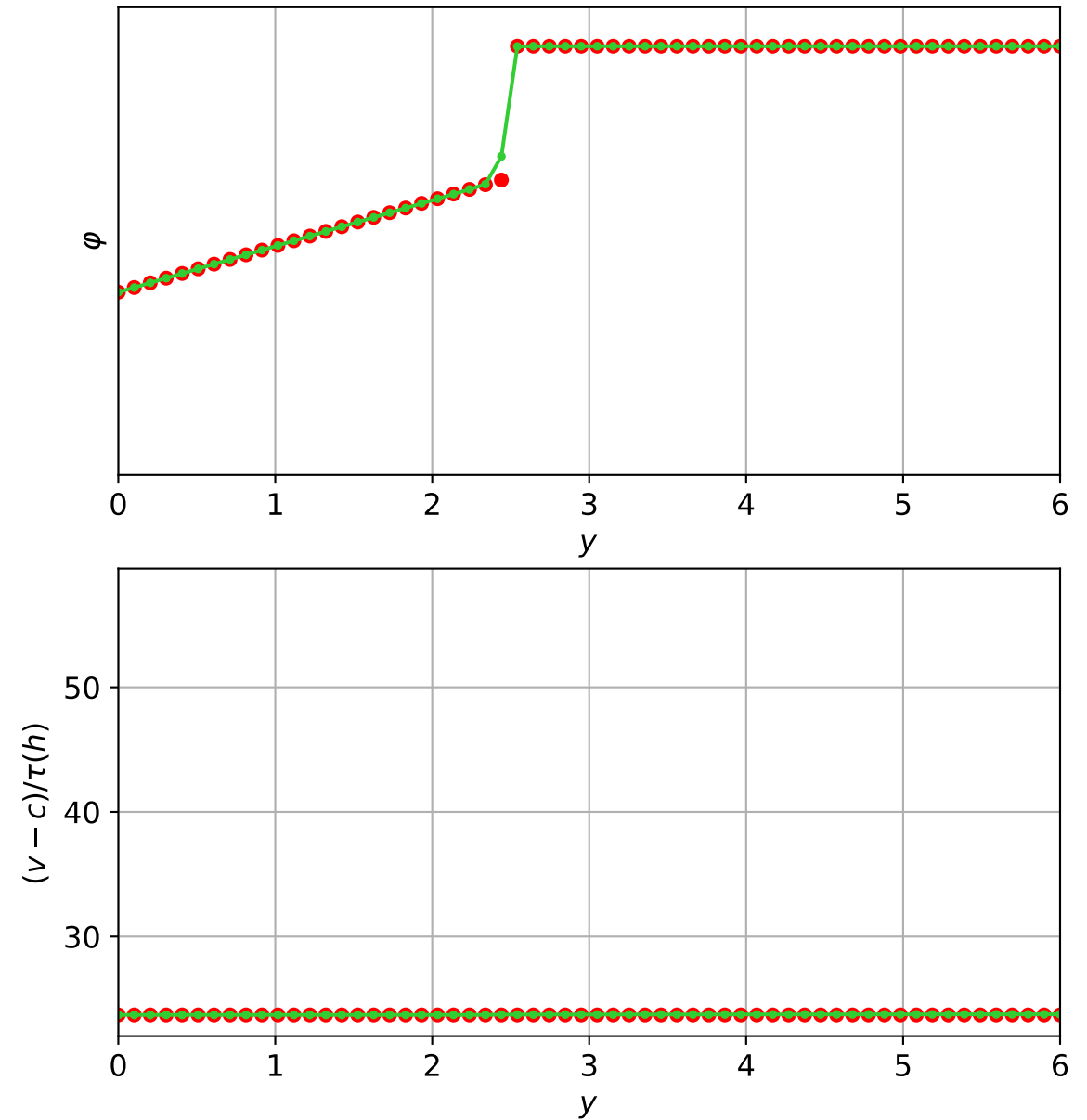
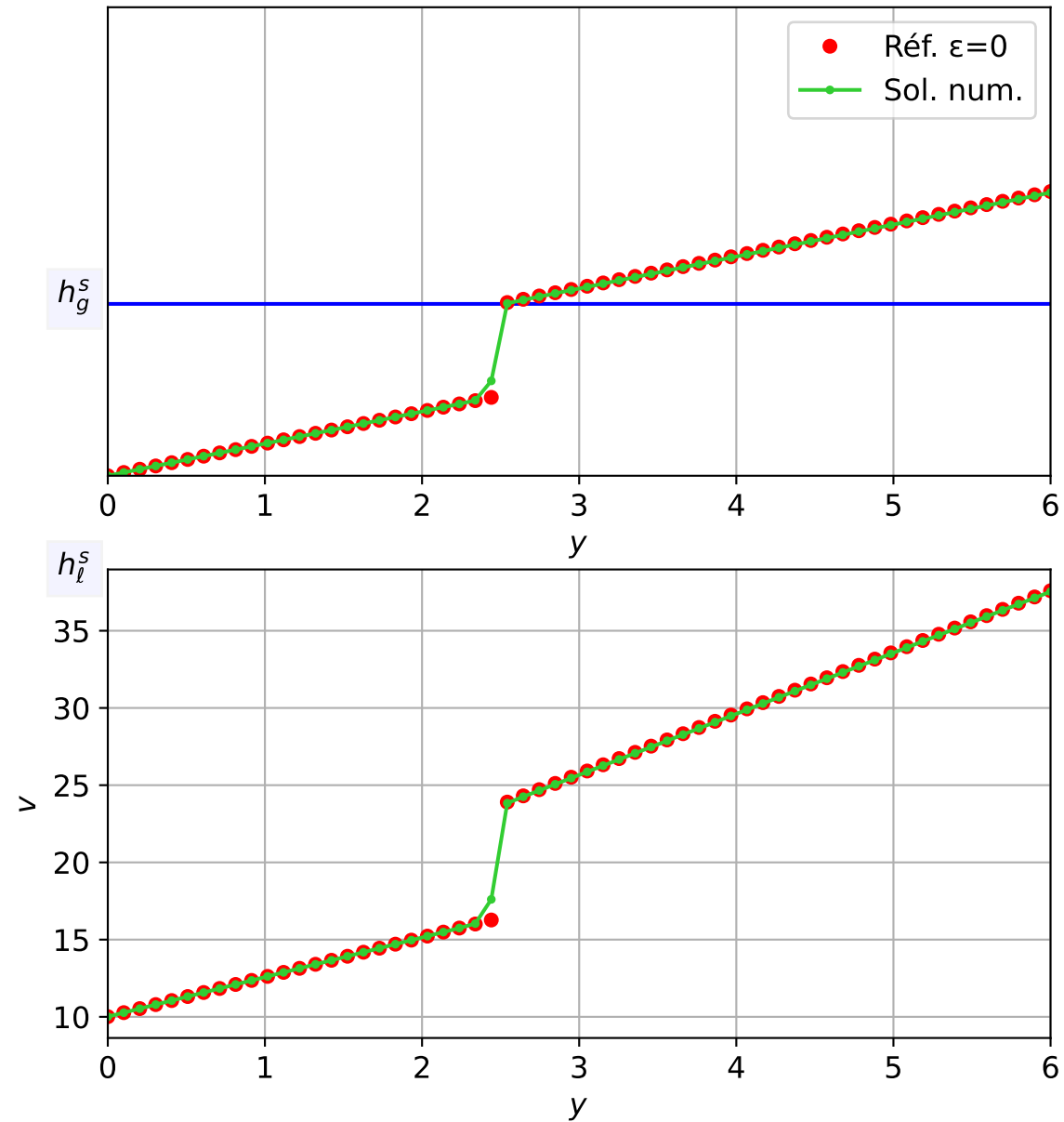




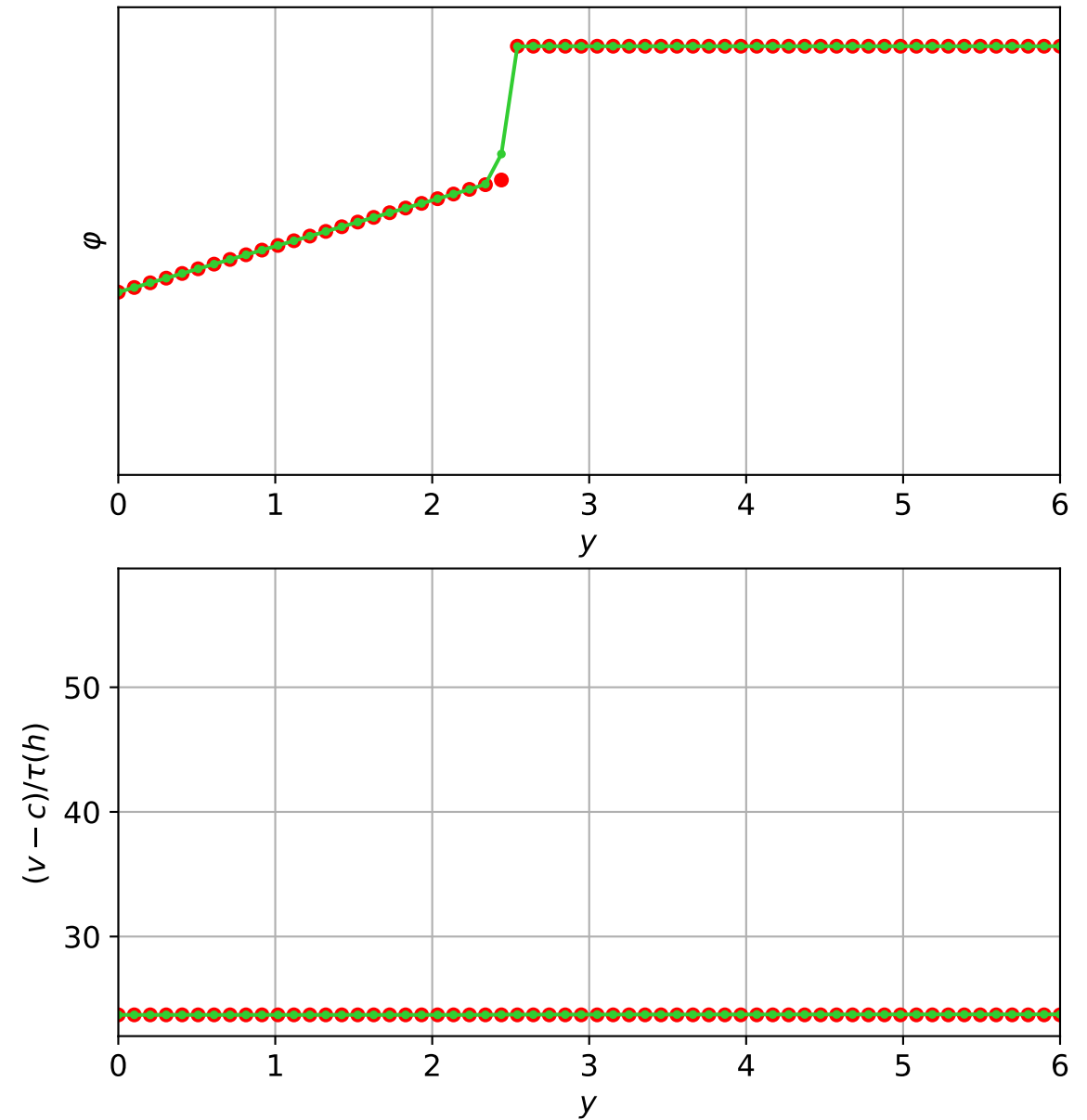
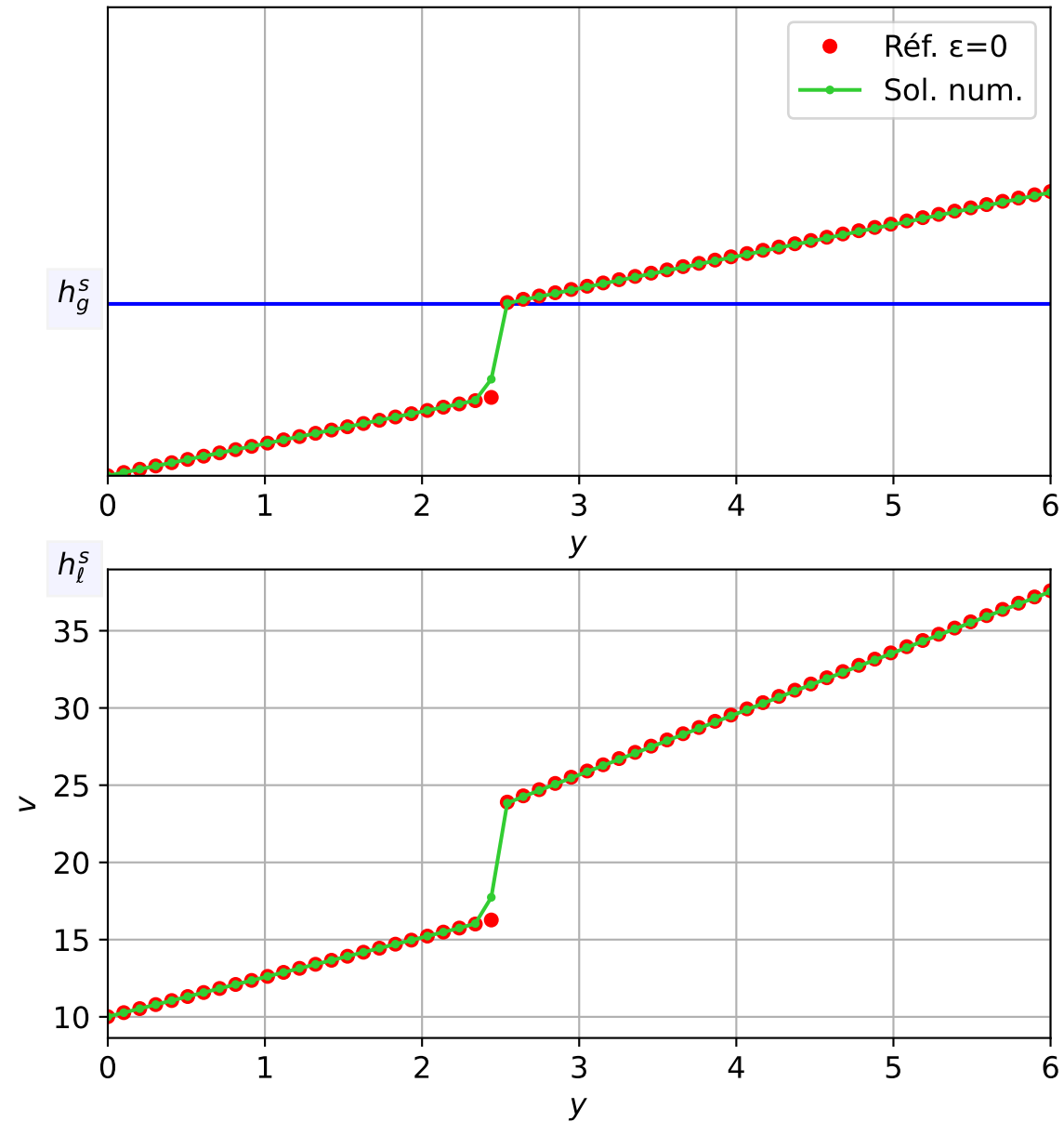
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 4.04e-04$  —  $t = 1.95038/12$  (6690 iter) — résidu eq. =  $6.28e-06$



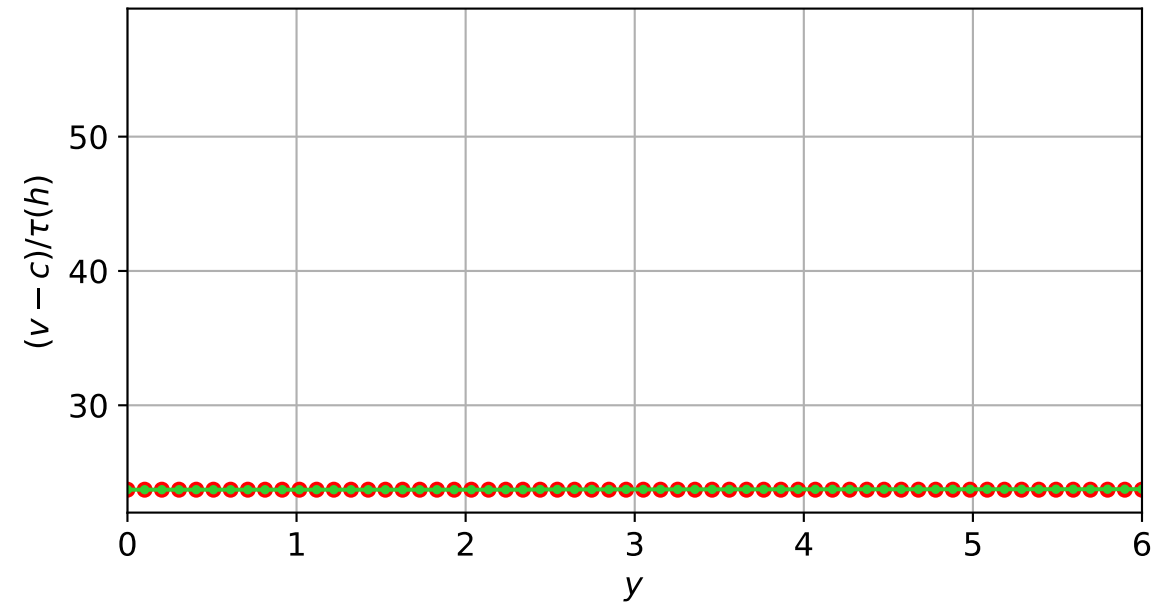
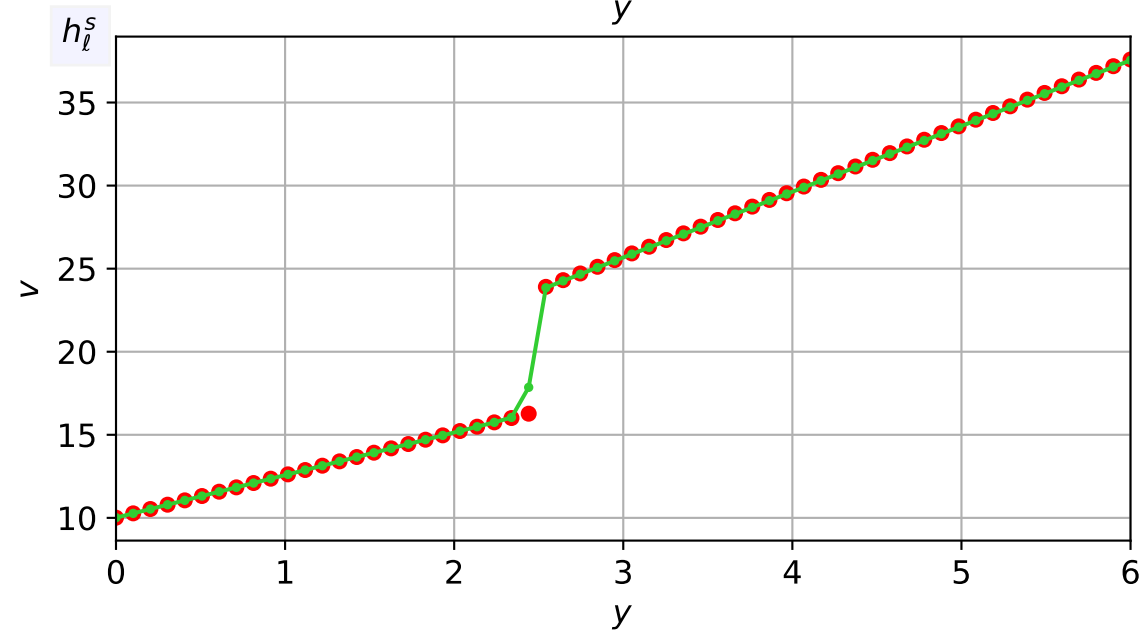
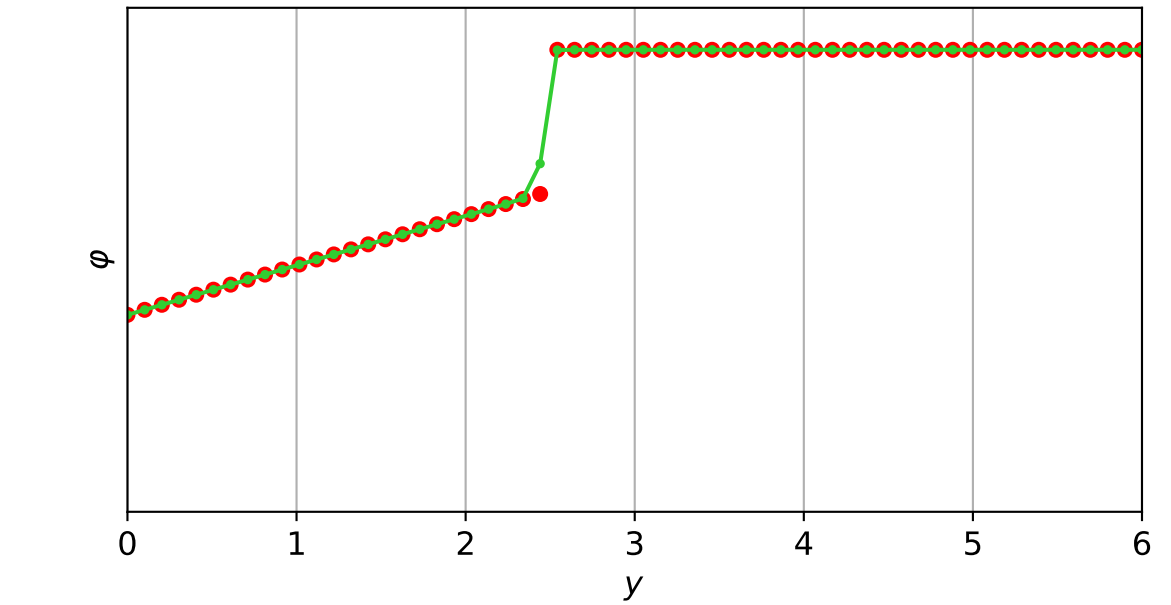
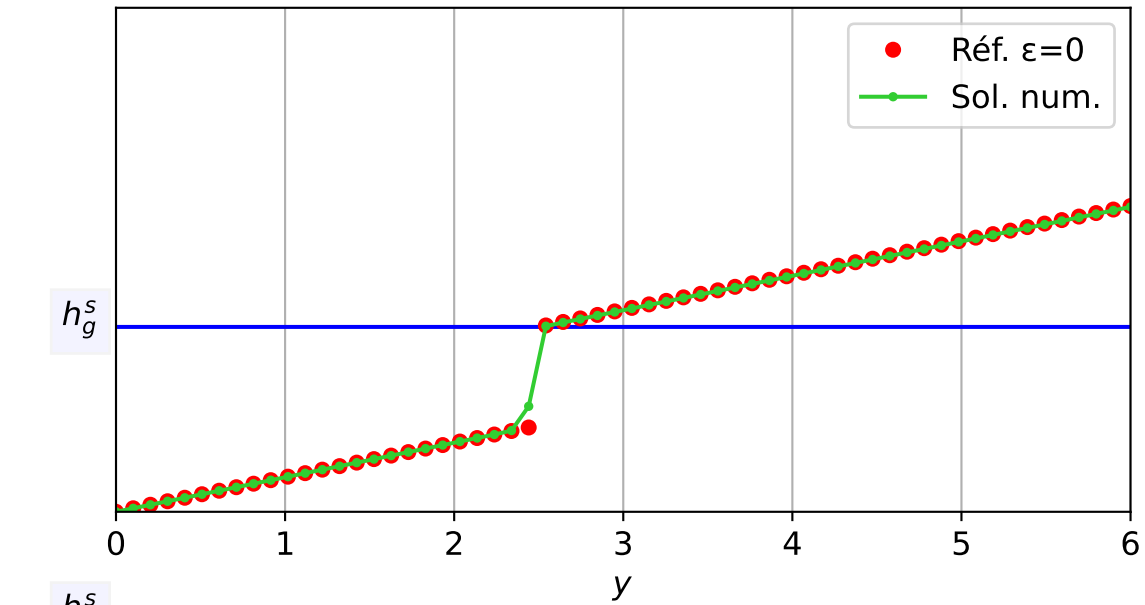
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 4.84e-04$  —  $t = 2.00025/12$  (6798 iter) — résidu eq. =  $7.94e-06$



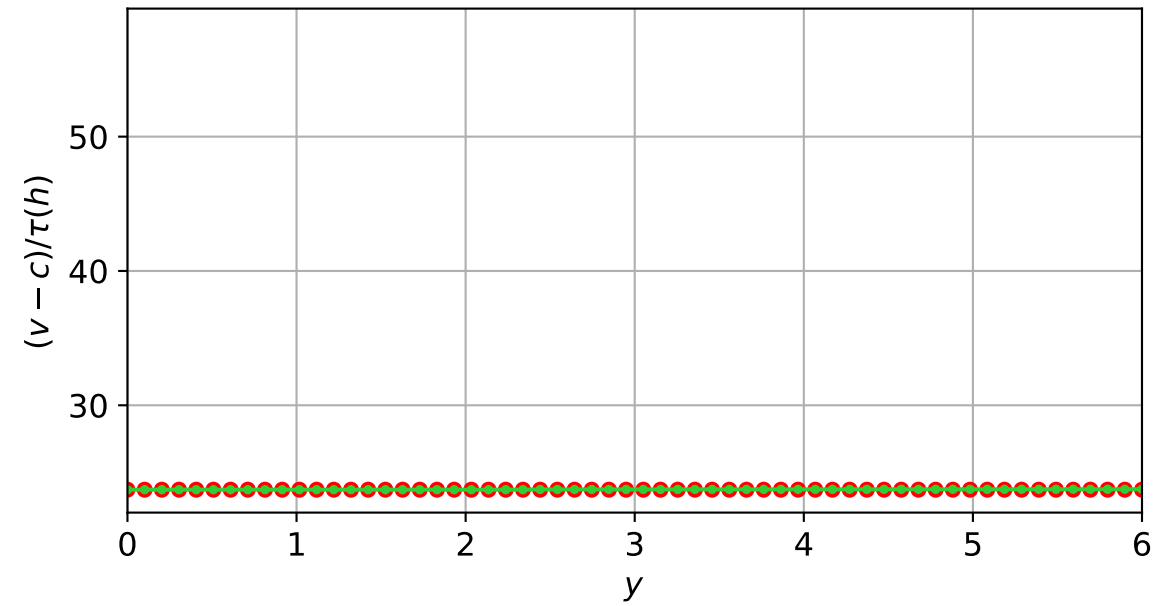
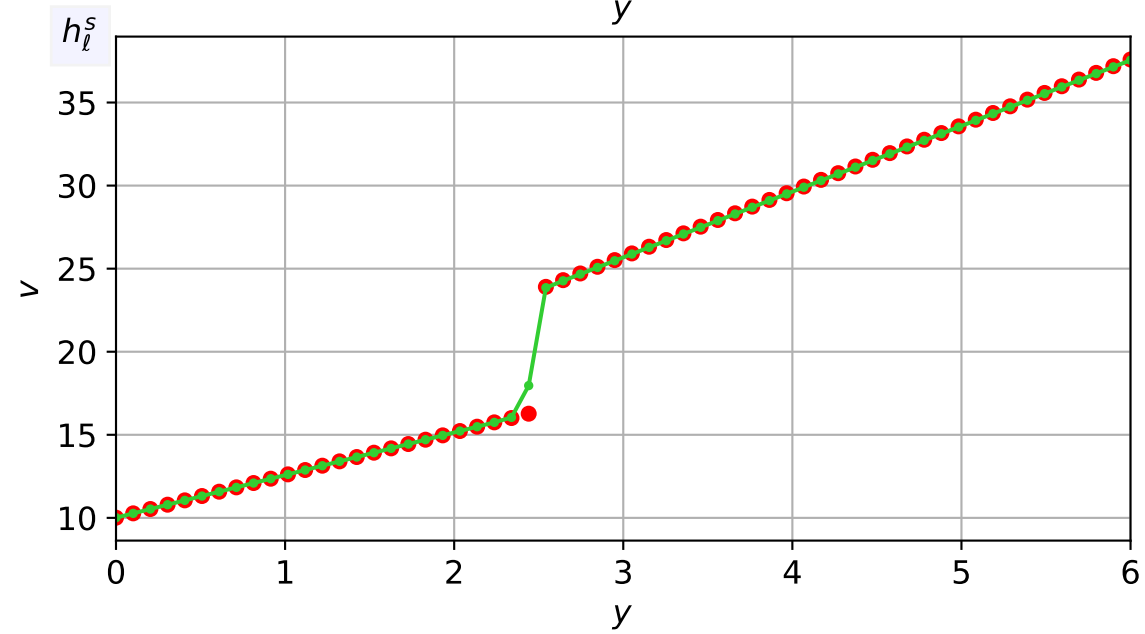
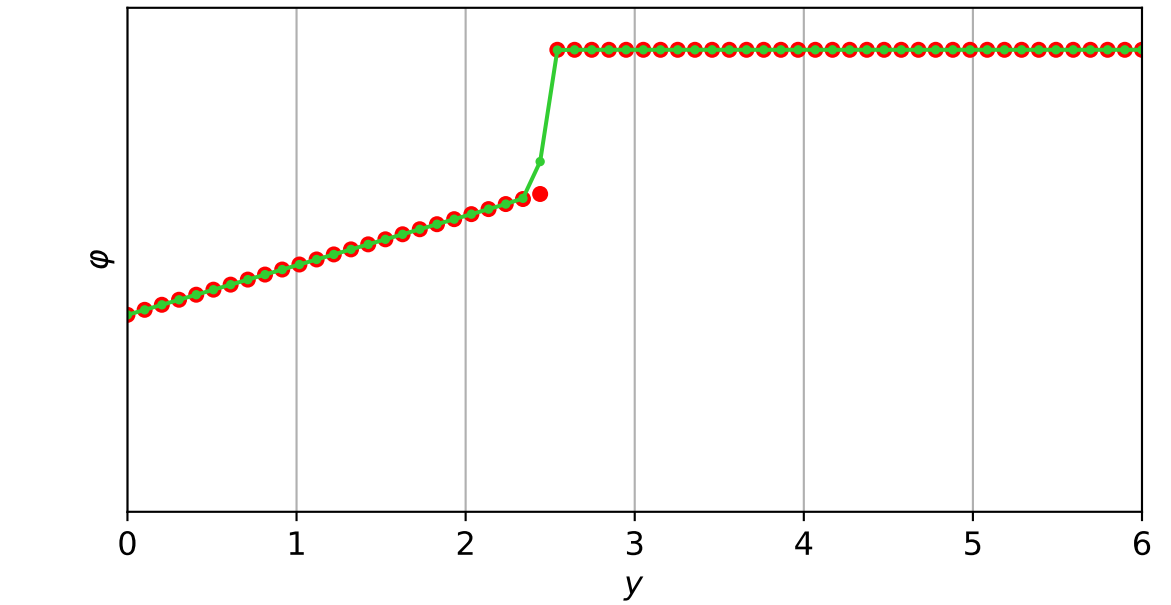
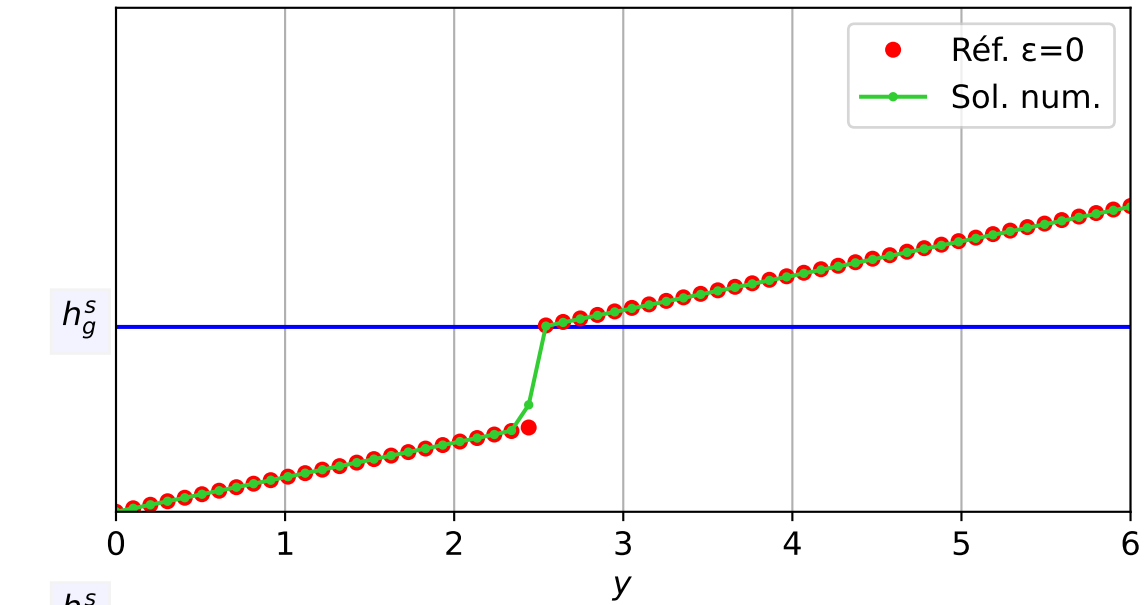
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 4.84e-04$  —  $t = 2.05013/12$  (6901 iter) — résidu eq. =  $7.00e-06$



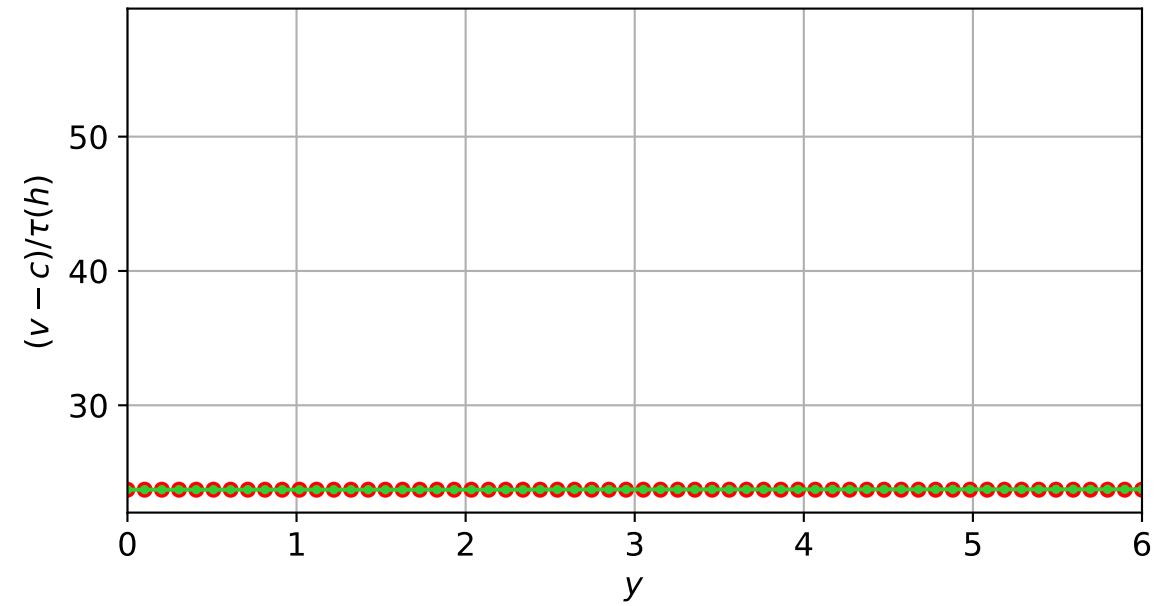
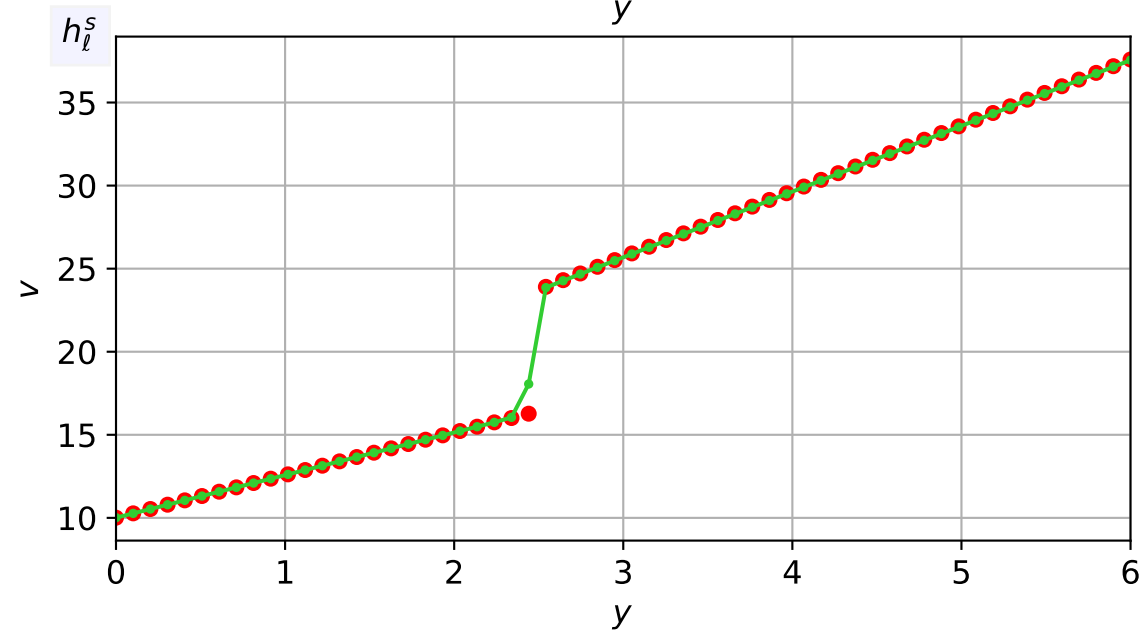
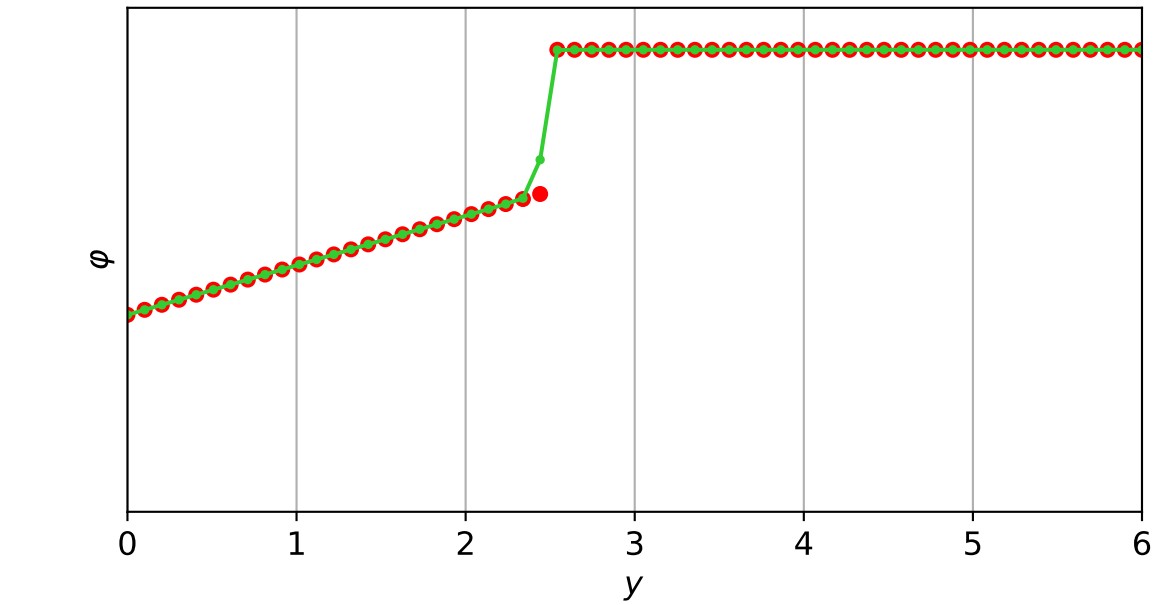
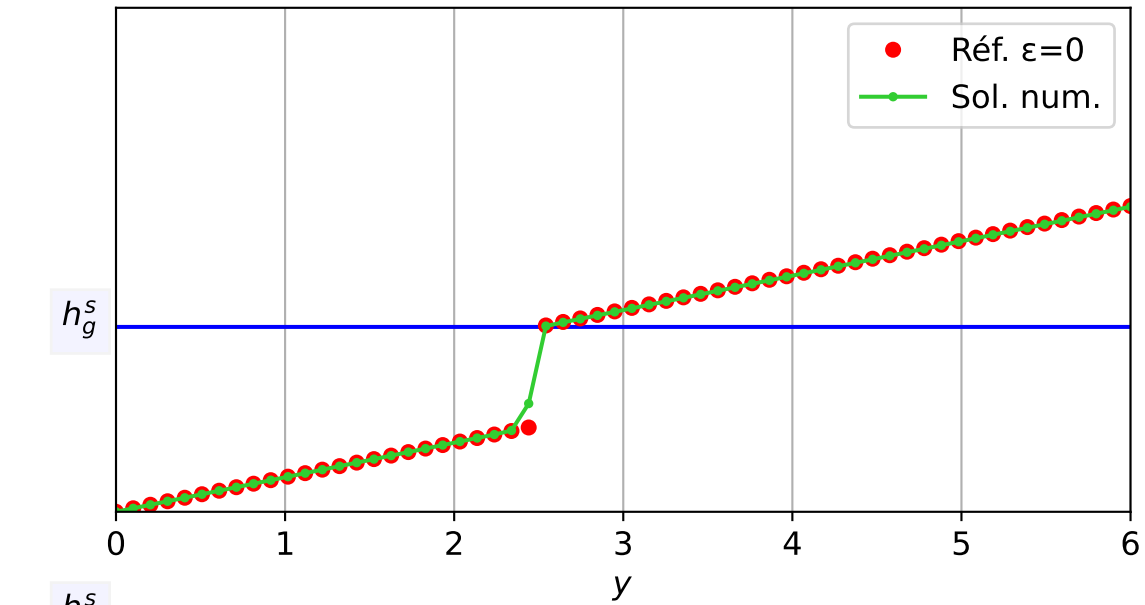
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 5.81e-04$  —  $t = 2.1002/12$  (7002 iter) — résidu eq. =  $8.82e-06$



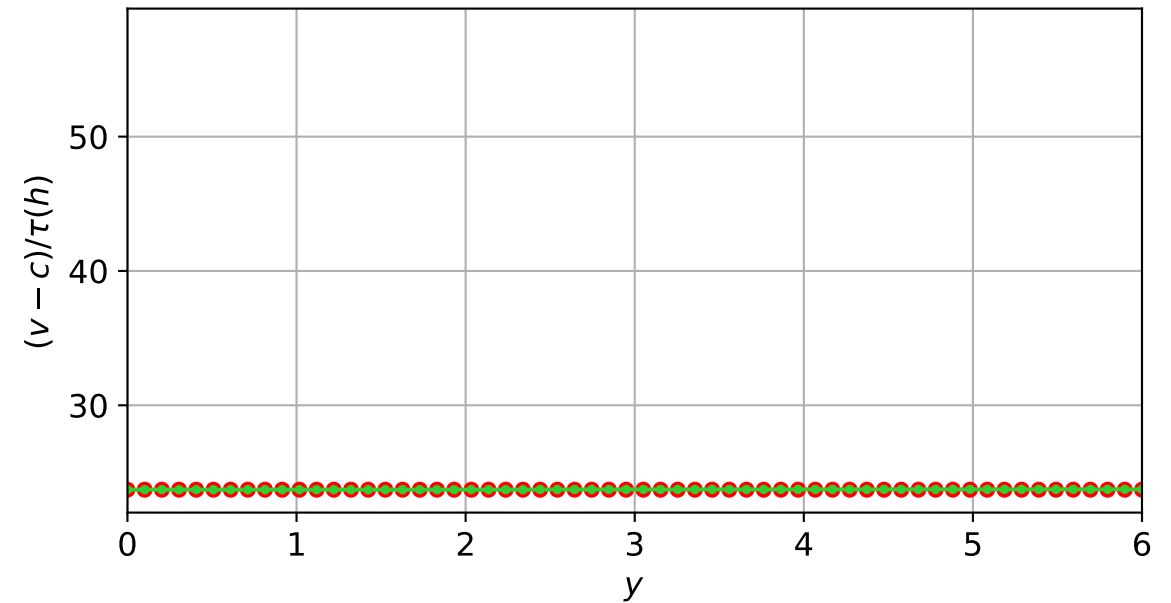
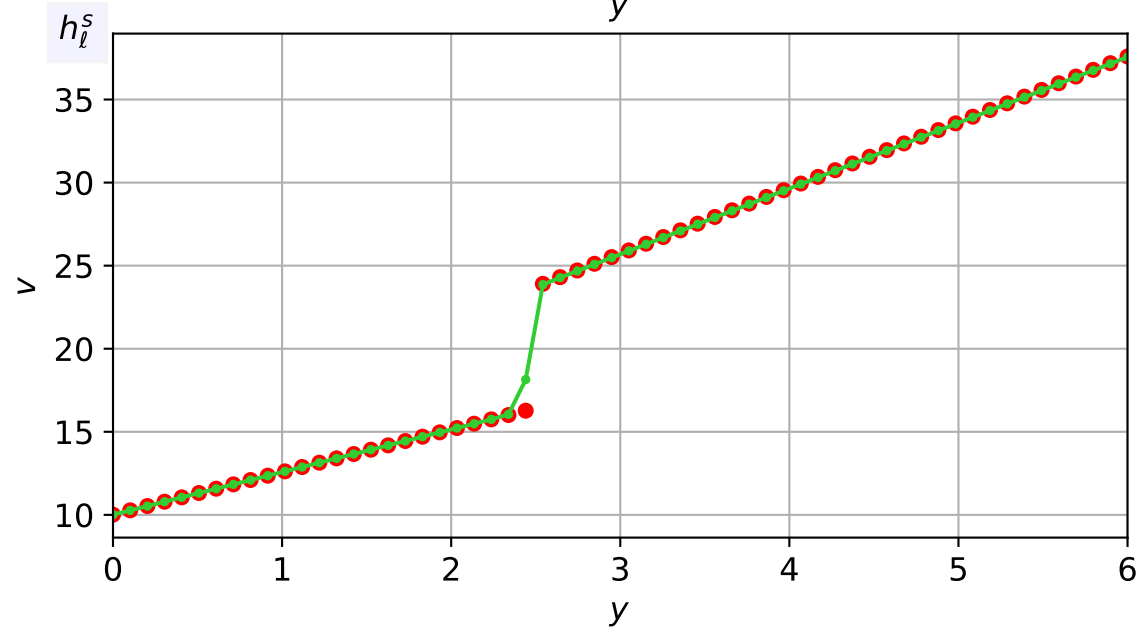
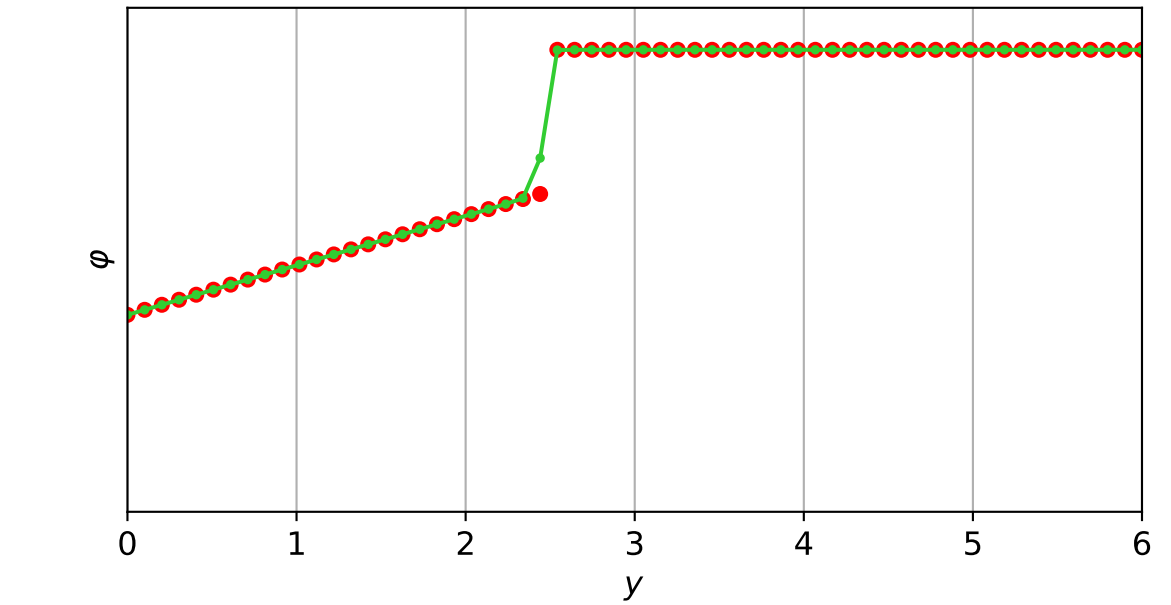
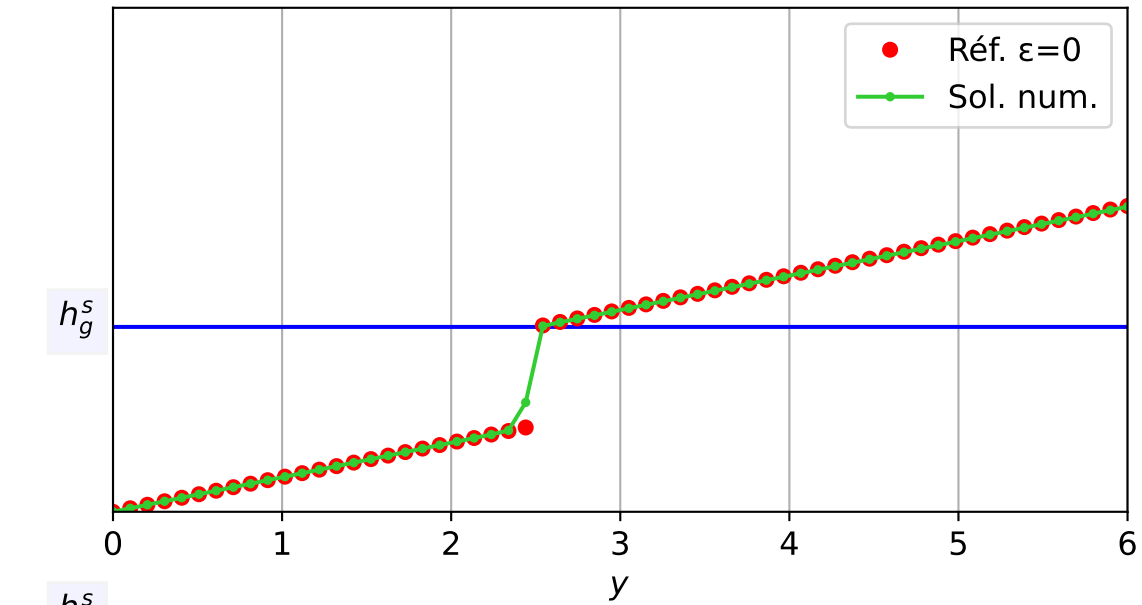
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 5.81e-04$  —  $t = 2.15017/12$  (7088 iter) — résidu eq. =  $7.86e-06$



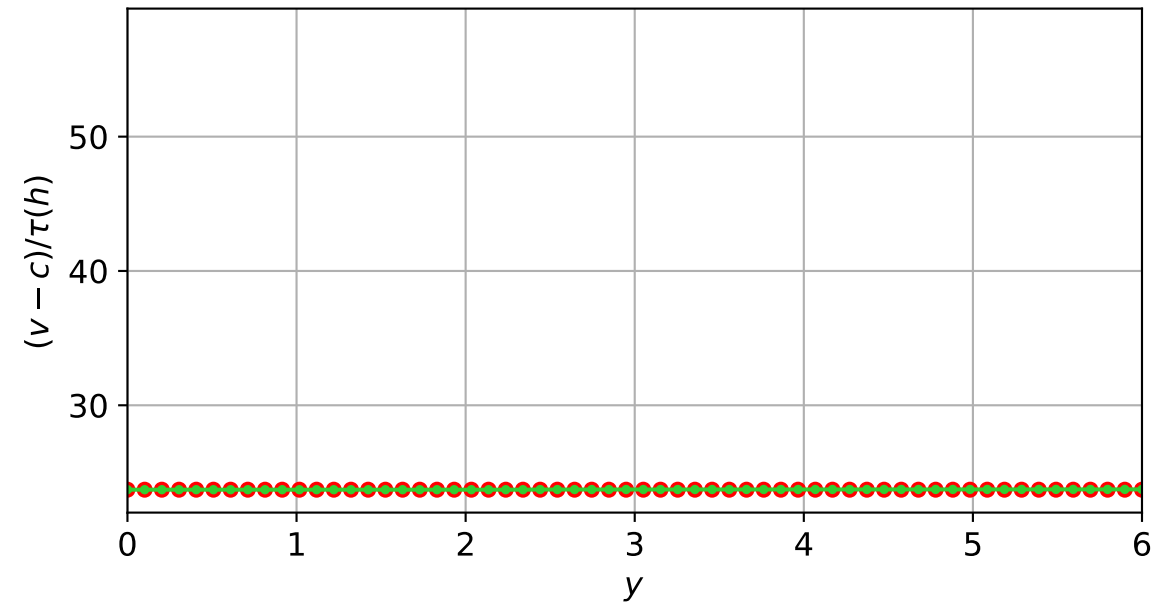
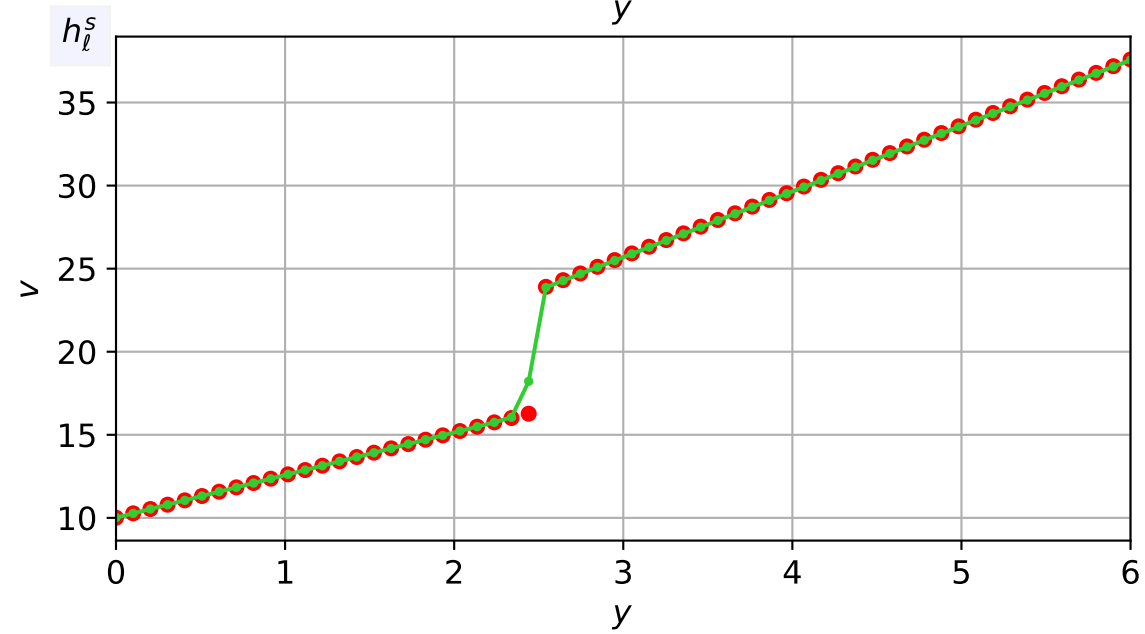
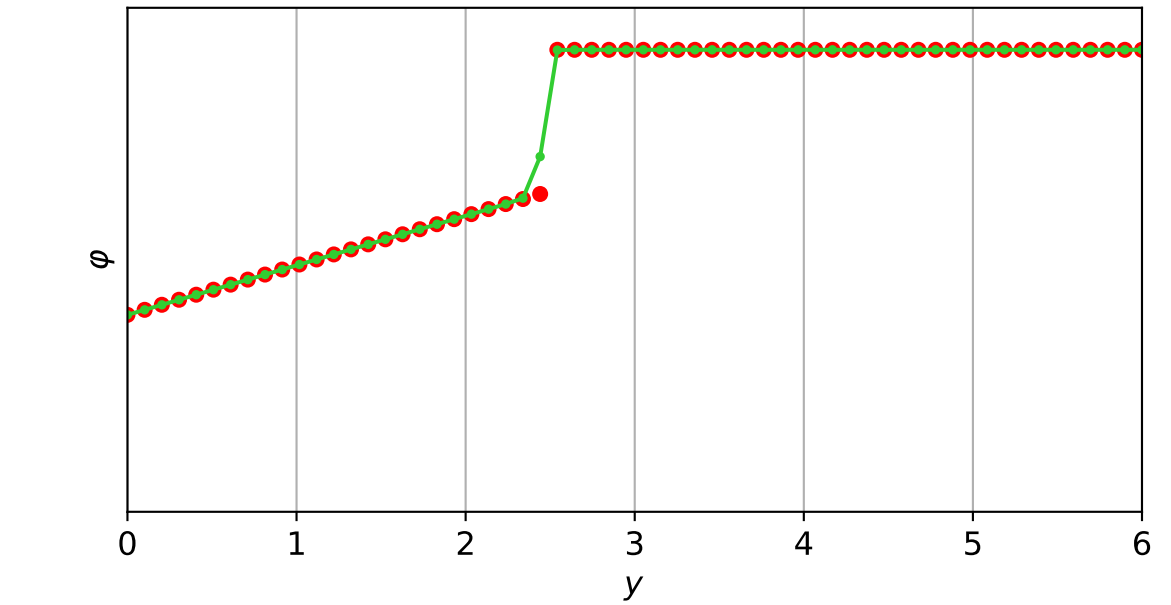
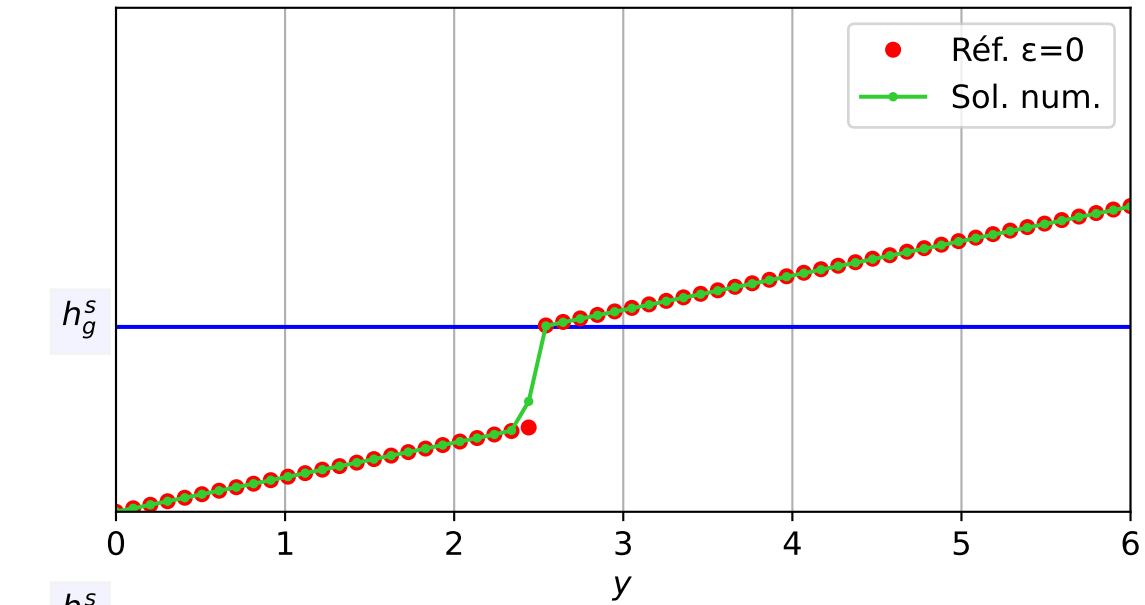
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 5.81e-04$  —  $t = 2.20014/12$  (7174 iter) — résidu eq. =  $7.00e-06$



TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 6.97e-04$  —  $t = 2.25023/12$  (7254 iter) — résidu eq. =  $8.70e-06$

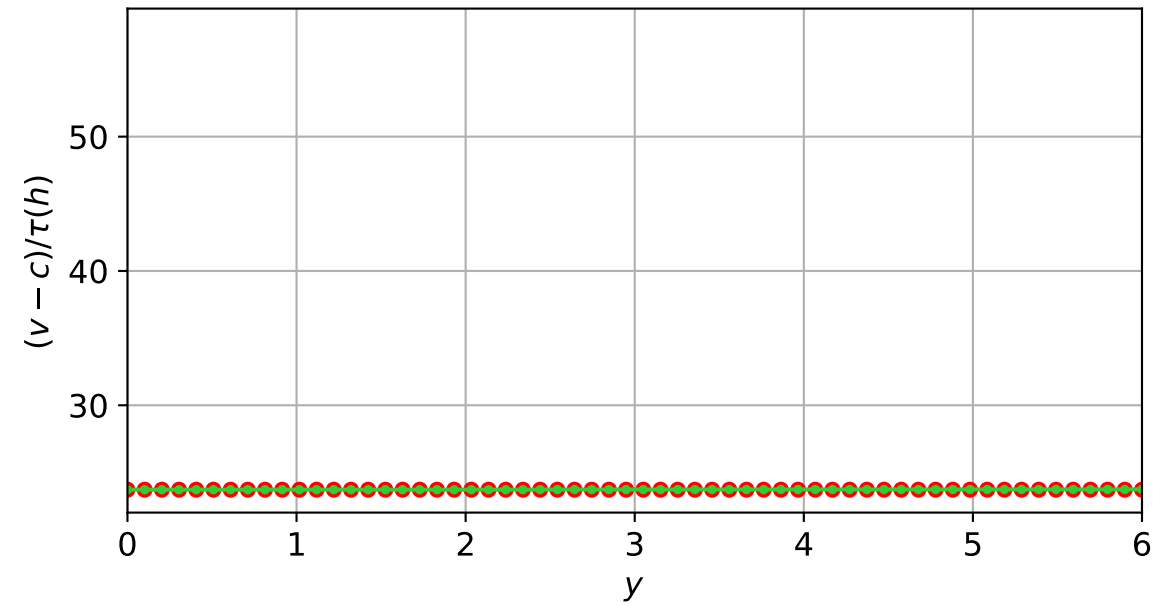
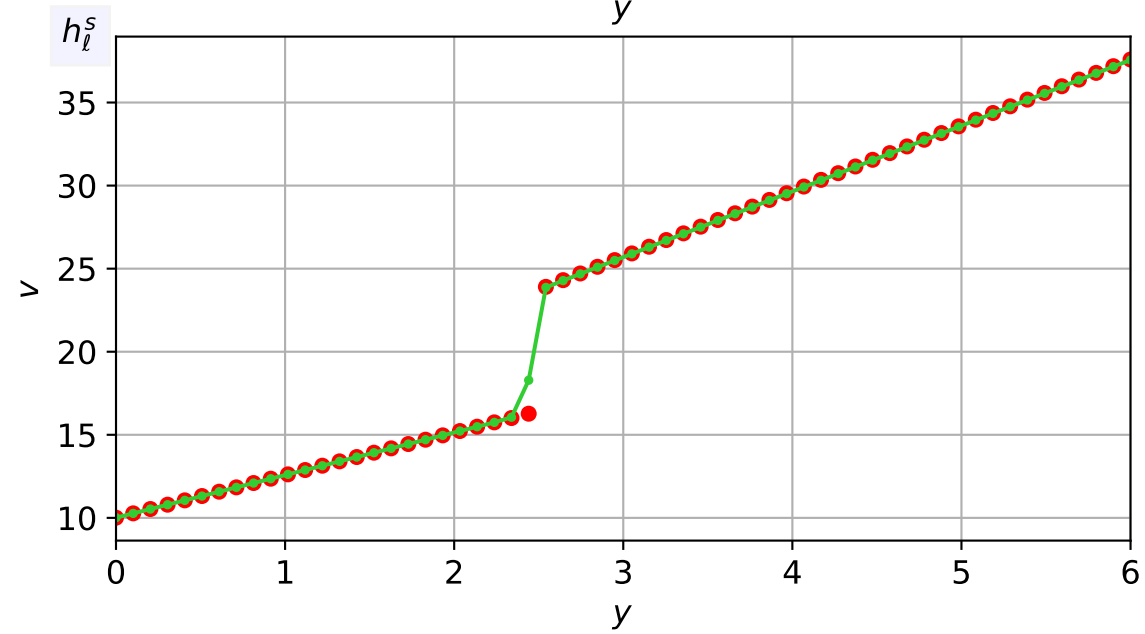
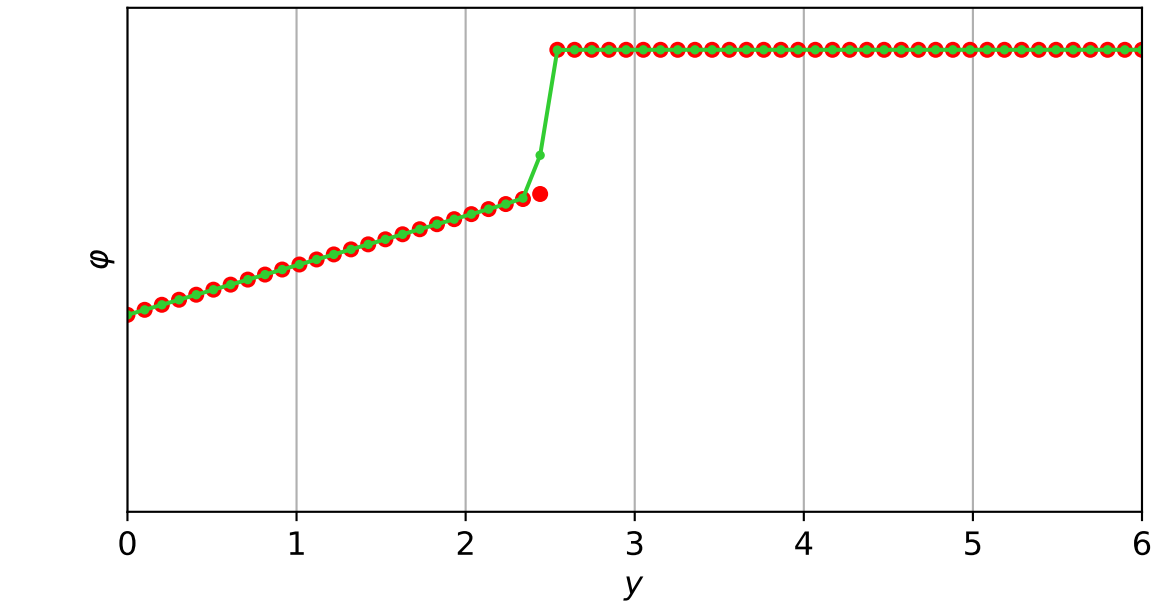
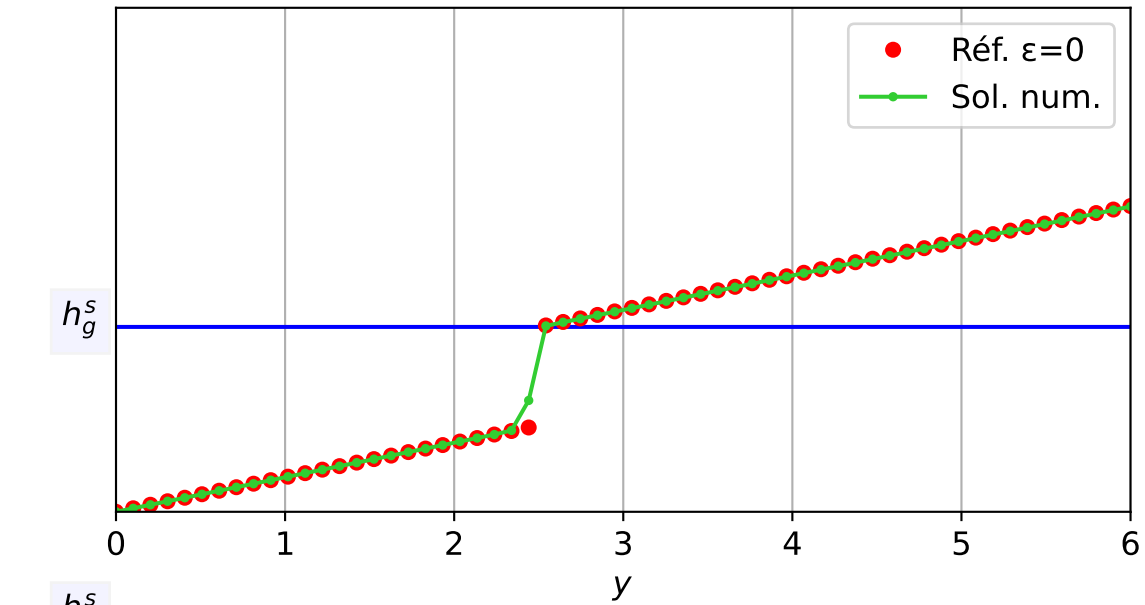


TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 6.97e-04$  —  $t = 2.30044/12$  (7326 iter) — résidu eq. =  $7.75e-06$

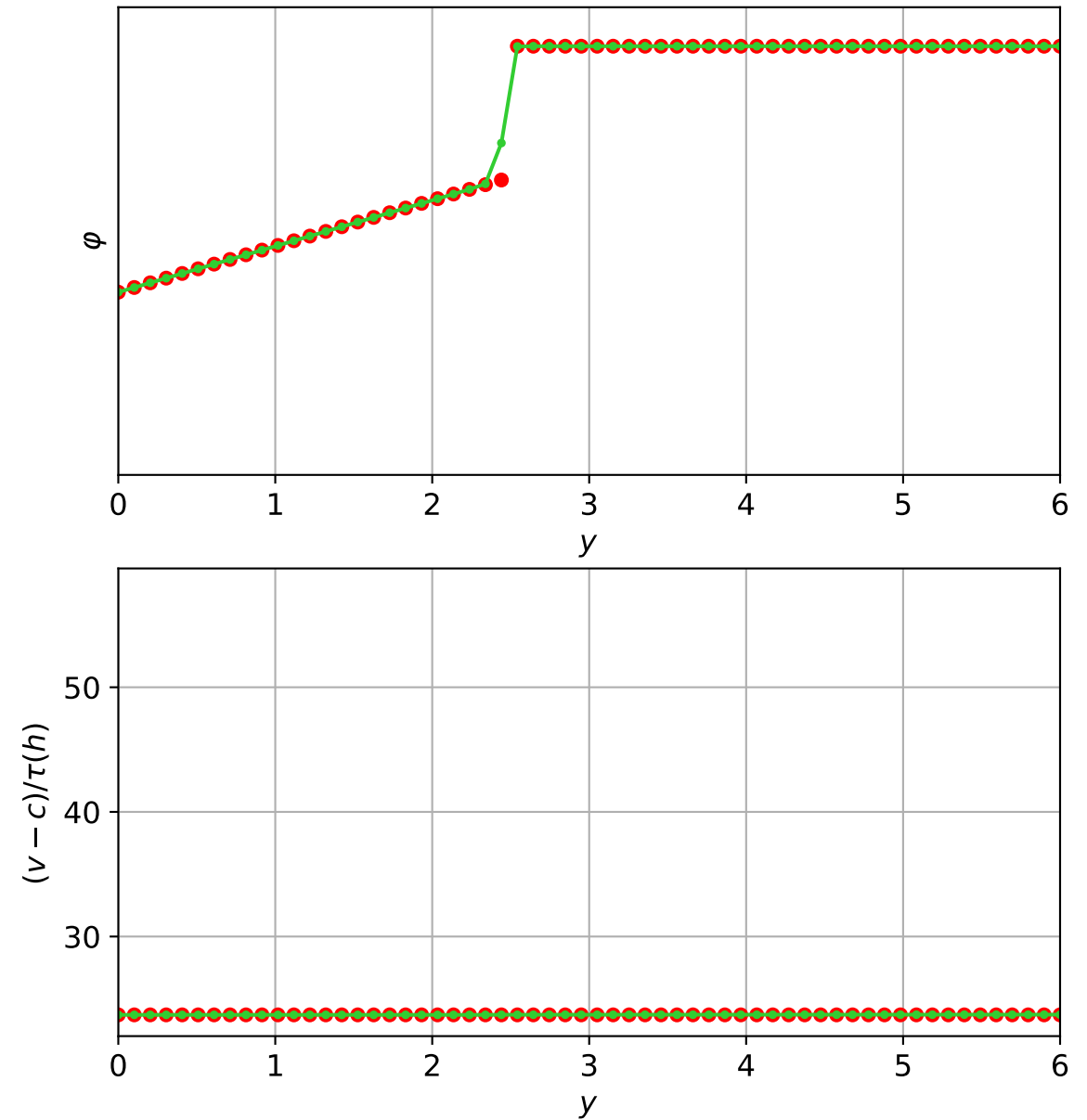
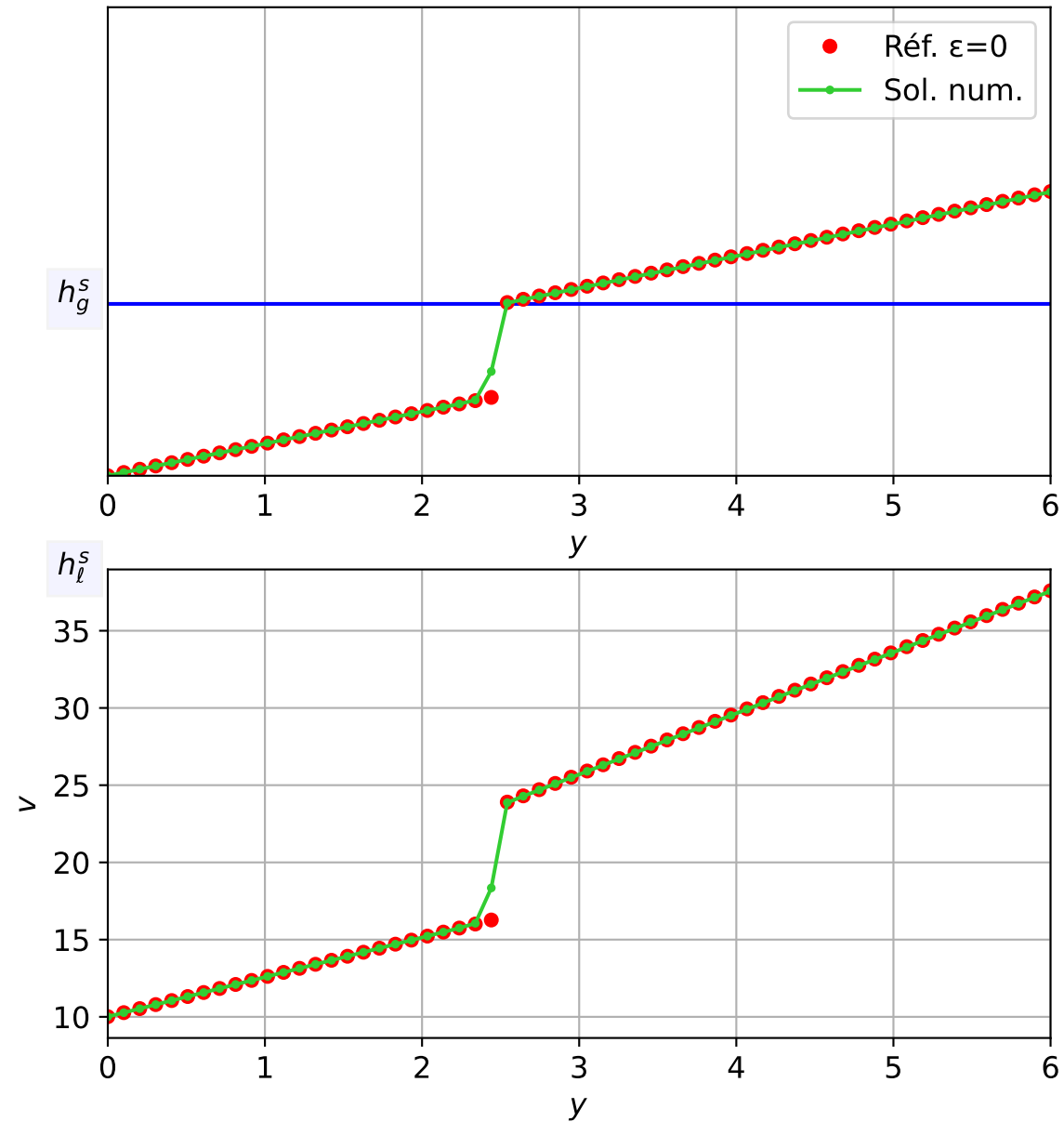




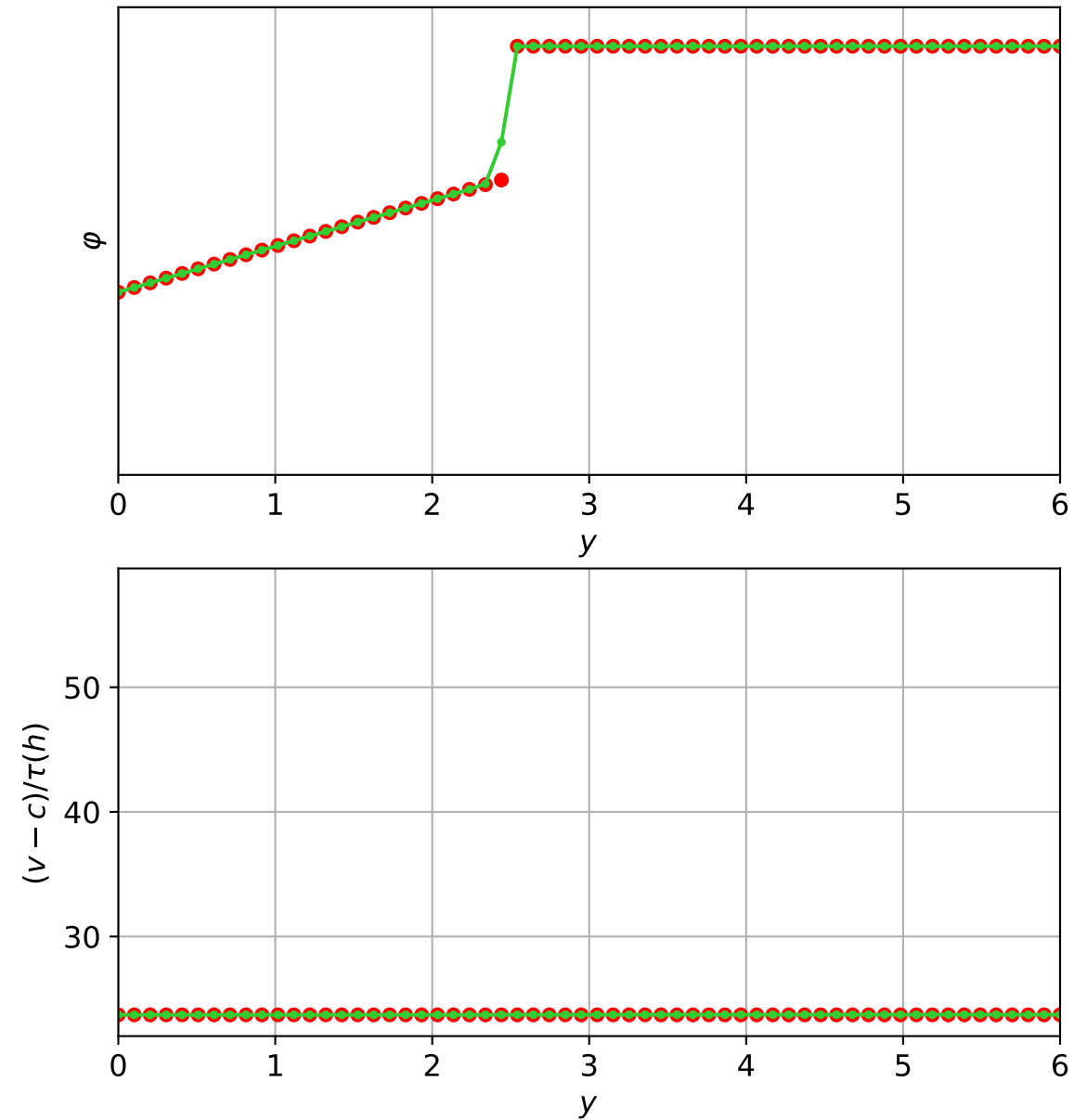
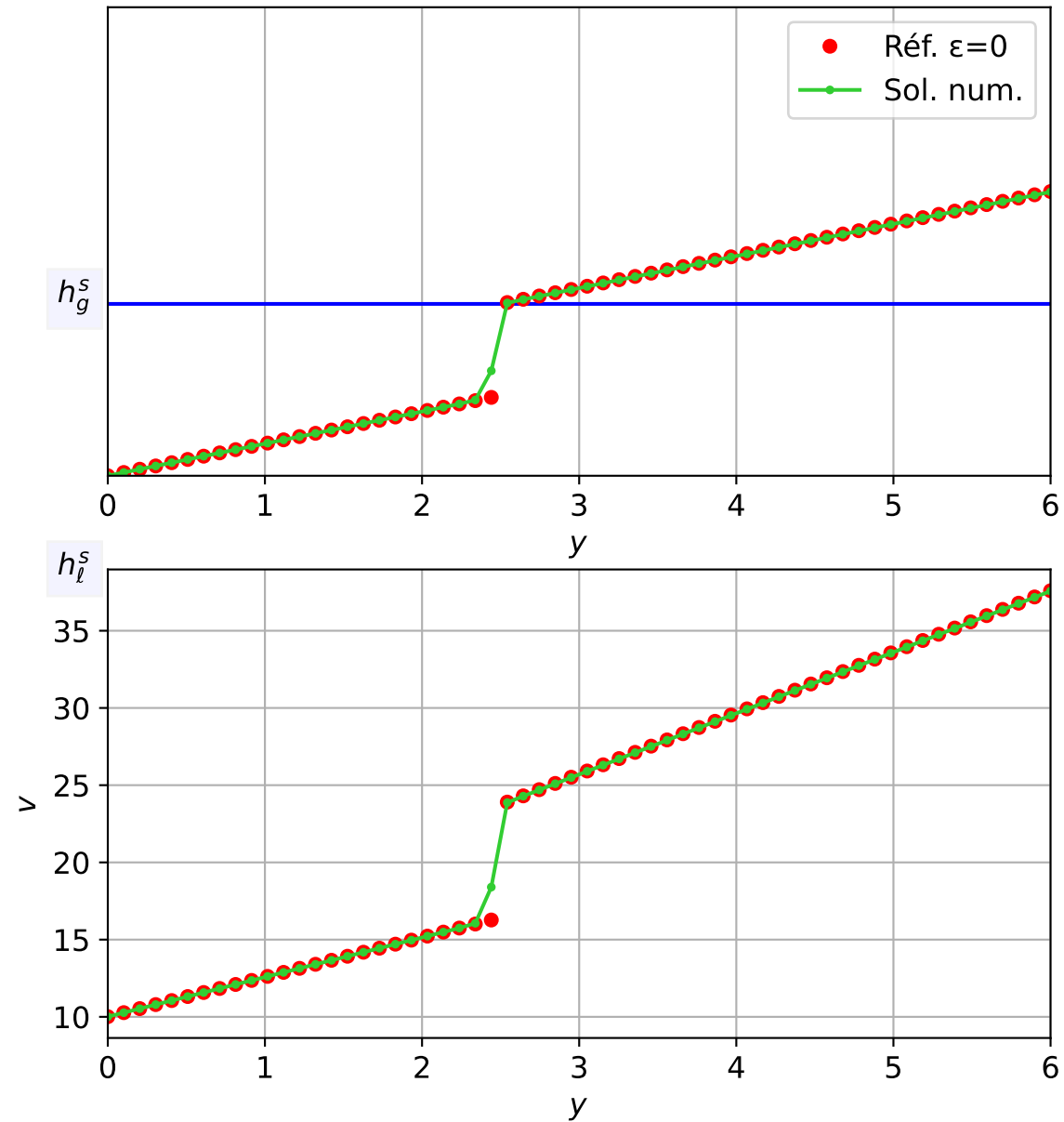
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 6.97e-04$  —  $t = 2.35064/12$  (7398 iter) — résidu eq. =  $6.89e-06$



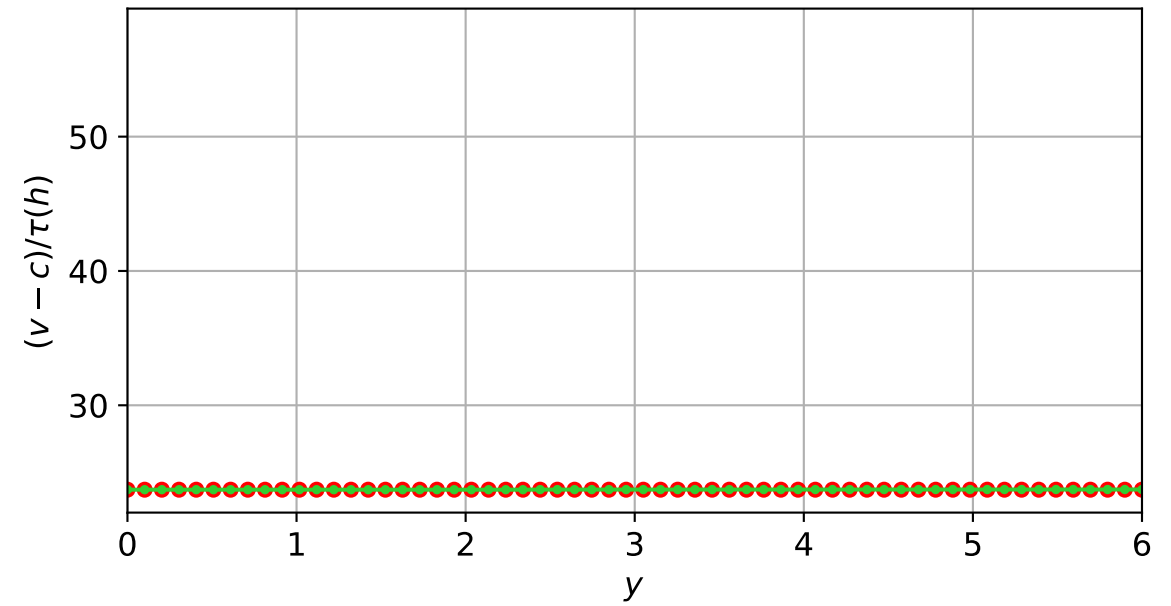
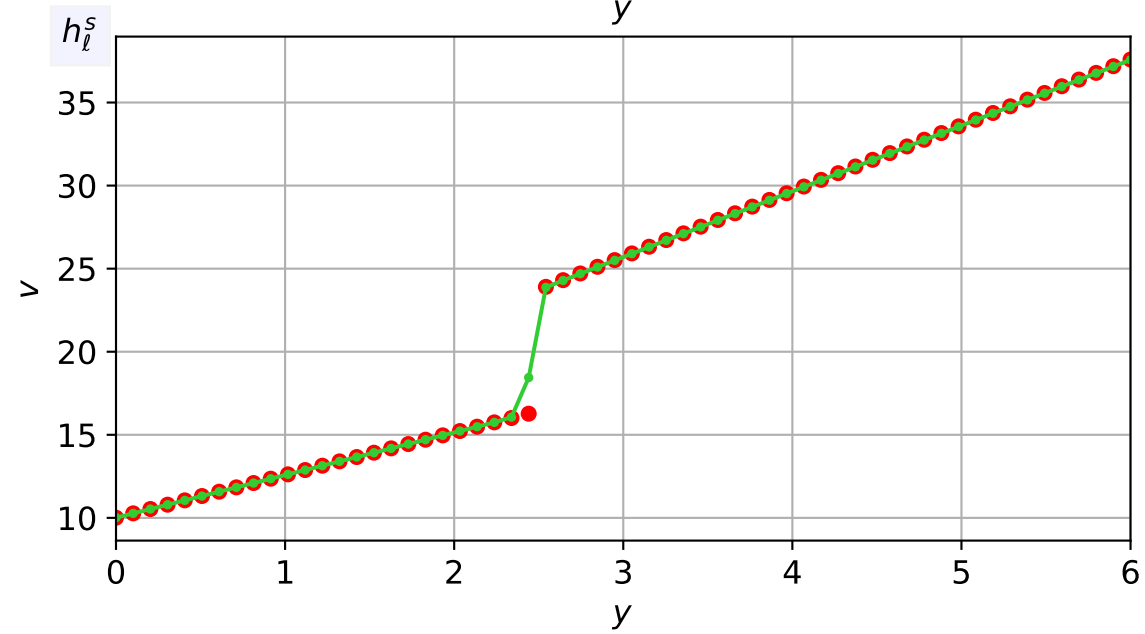
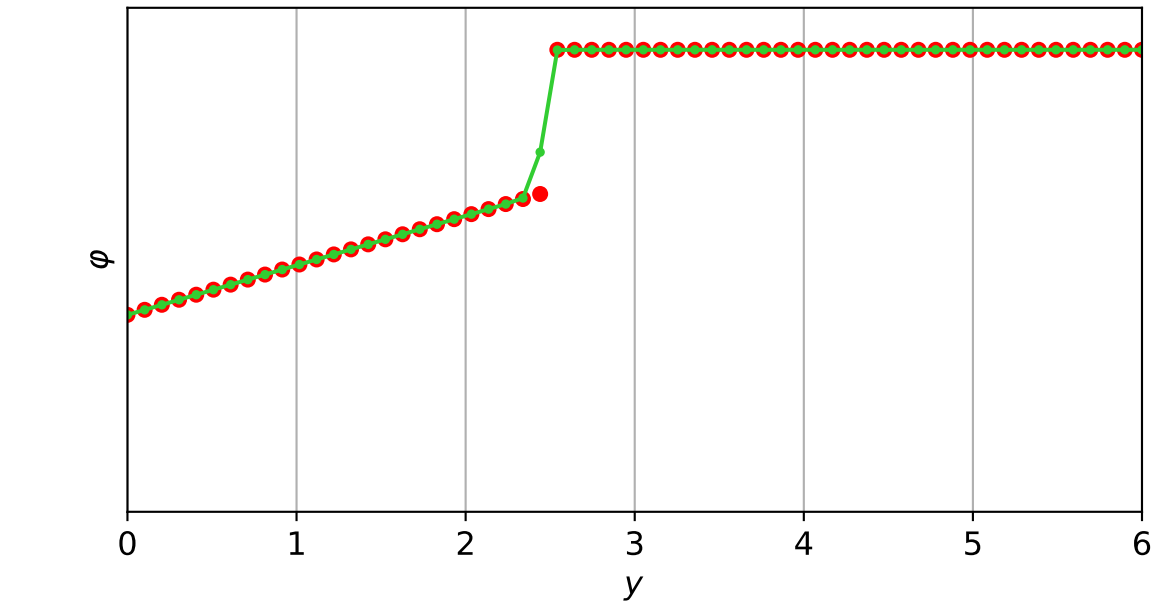
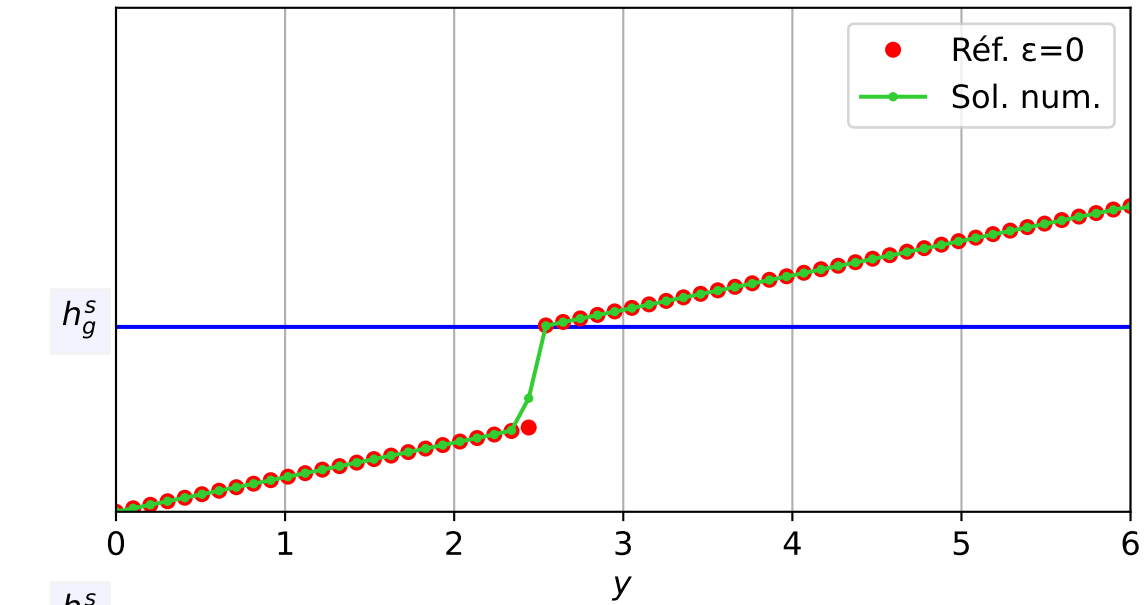
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 8.37e-04$  —  $t = 2.40043/12$  (7459 iter) — résidu eq. =  $8.42e-06$



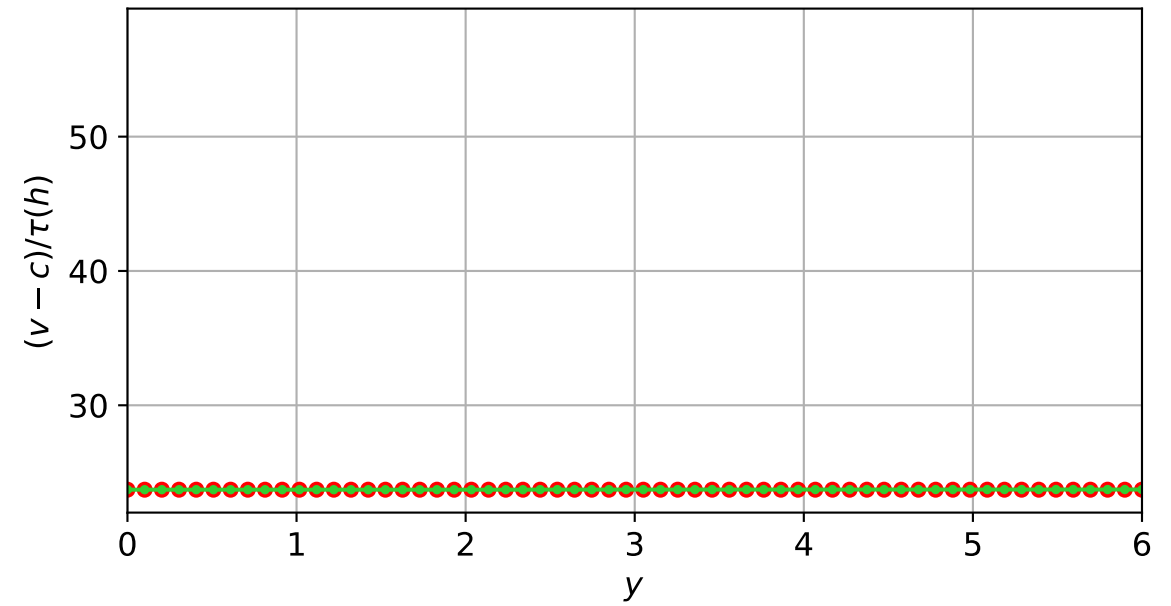
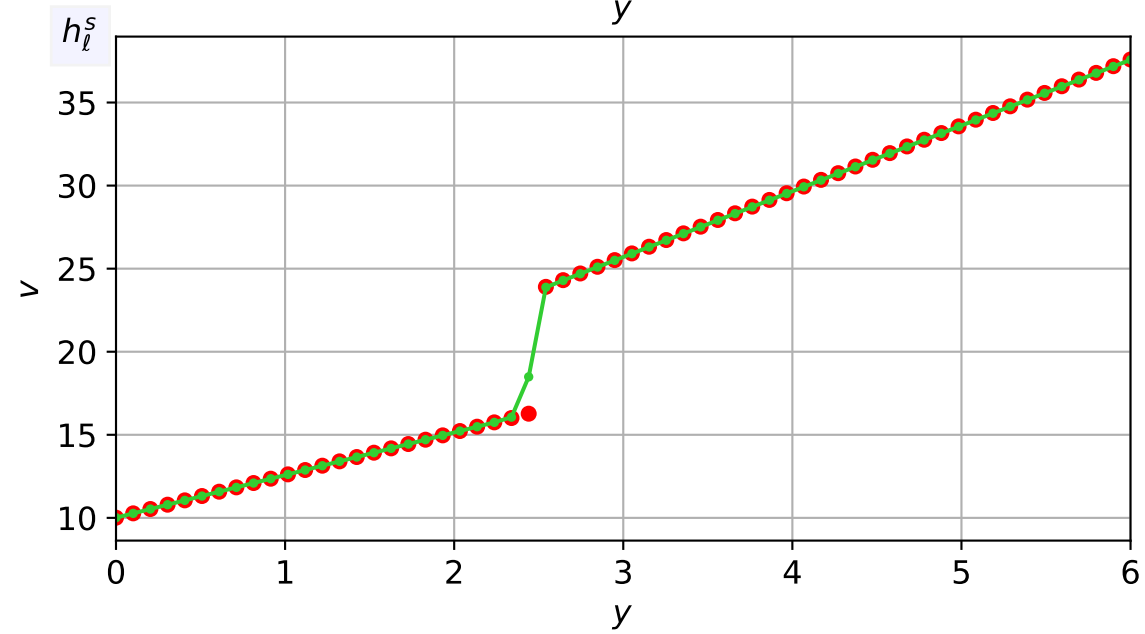
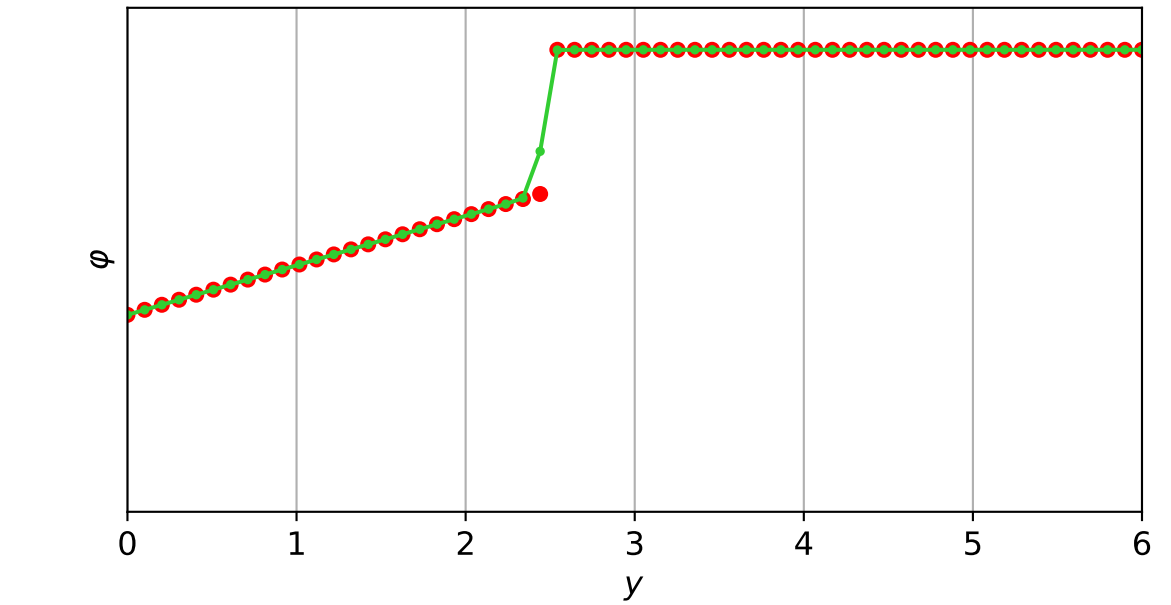
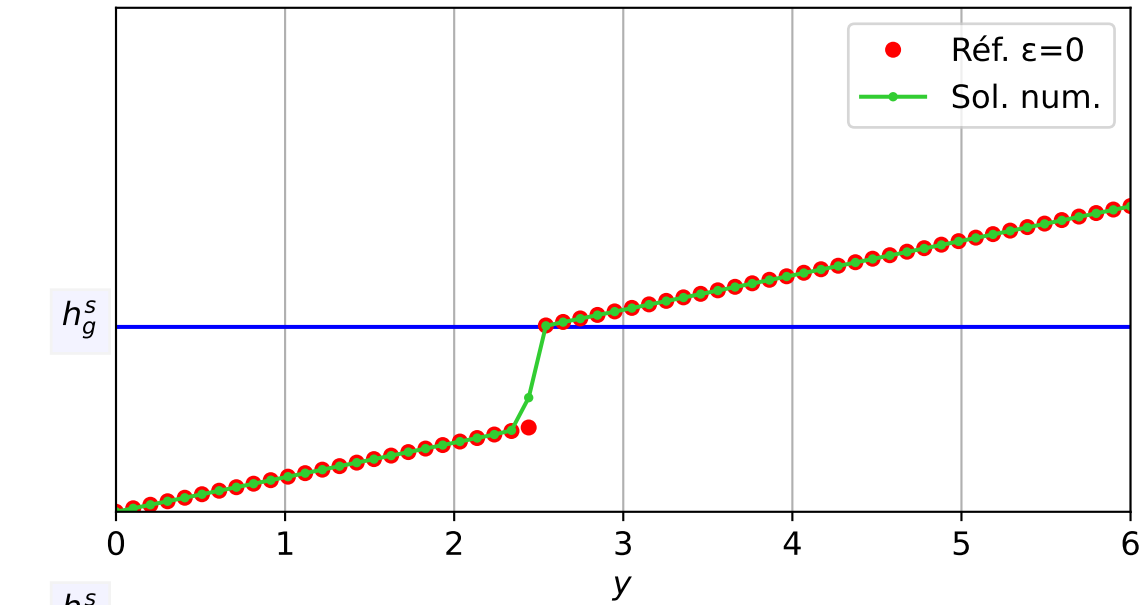
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 8.37e-04$  —  $t = 2.45063/12$  (7519 iter) — résidu eq. =  $7.49e-06$



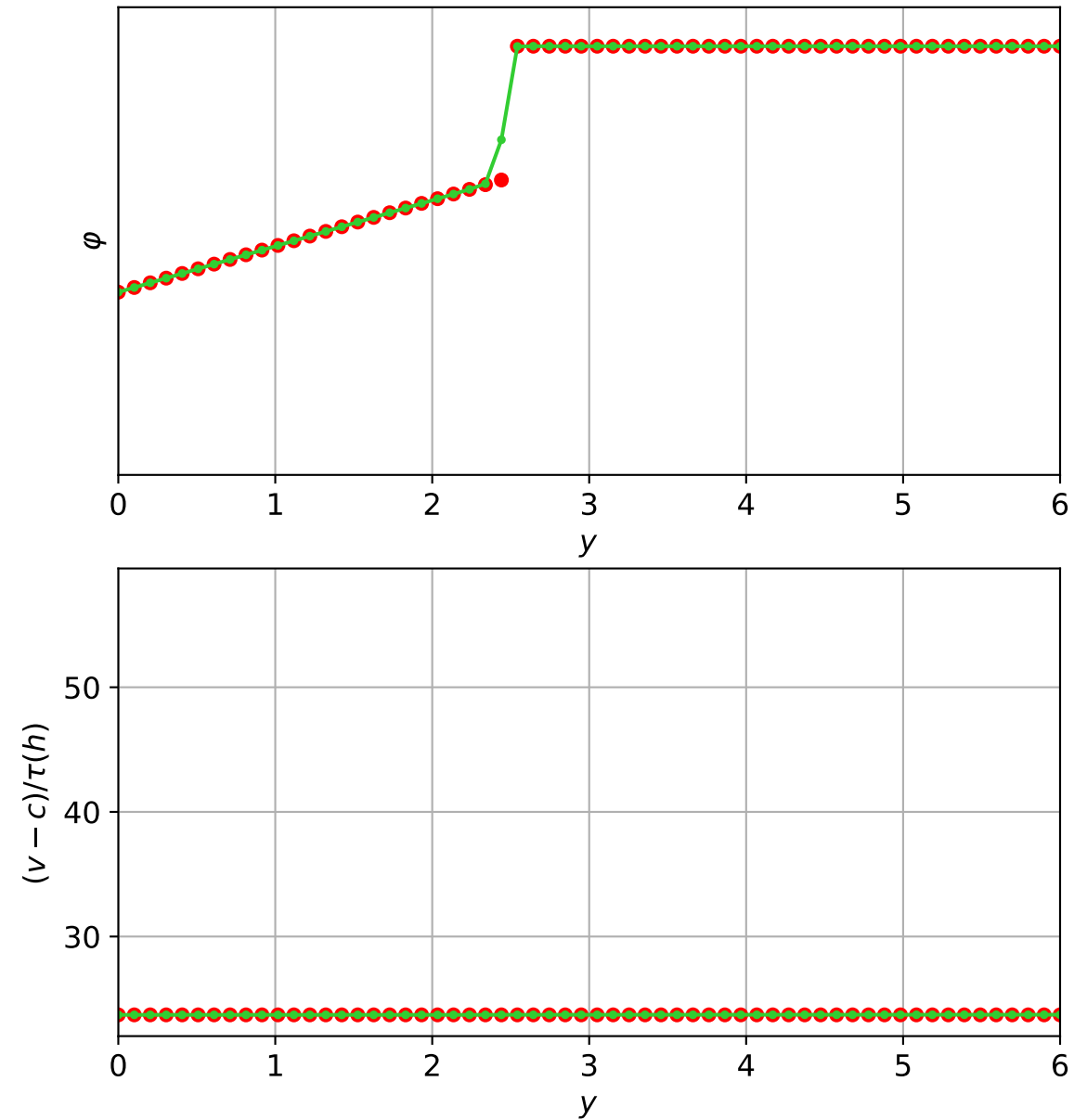
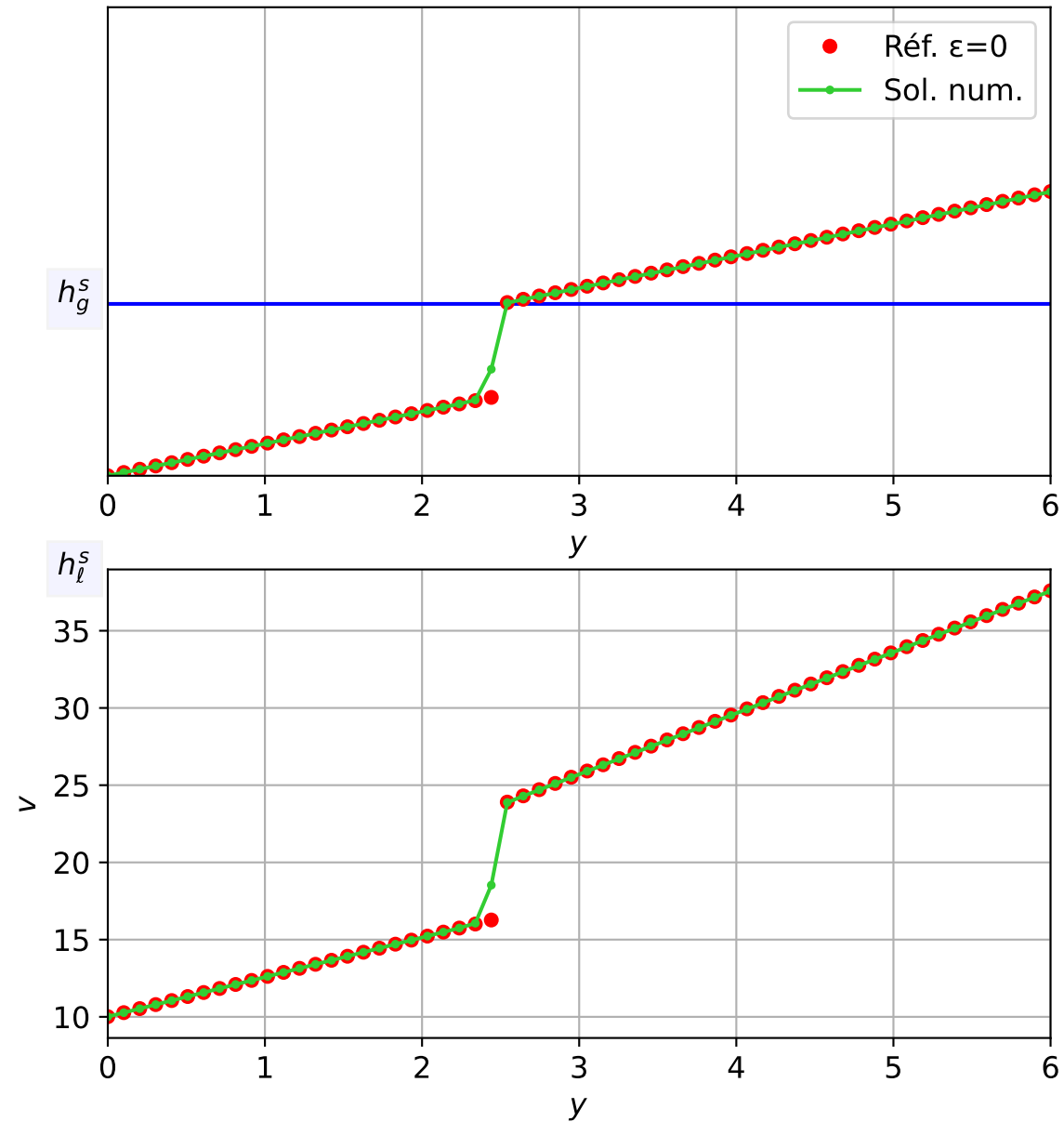
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 1.00e-03$  —  $t = 2.50017/12$  (7574 iter) — résidu eq. =  $9.01e-06$



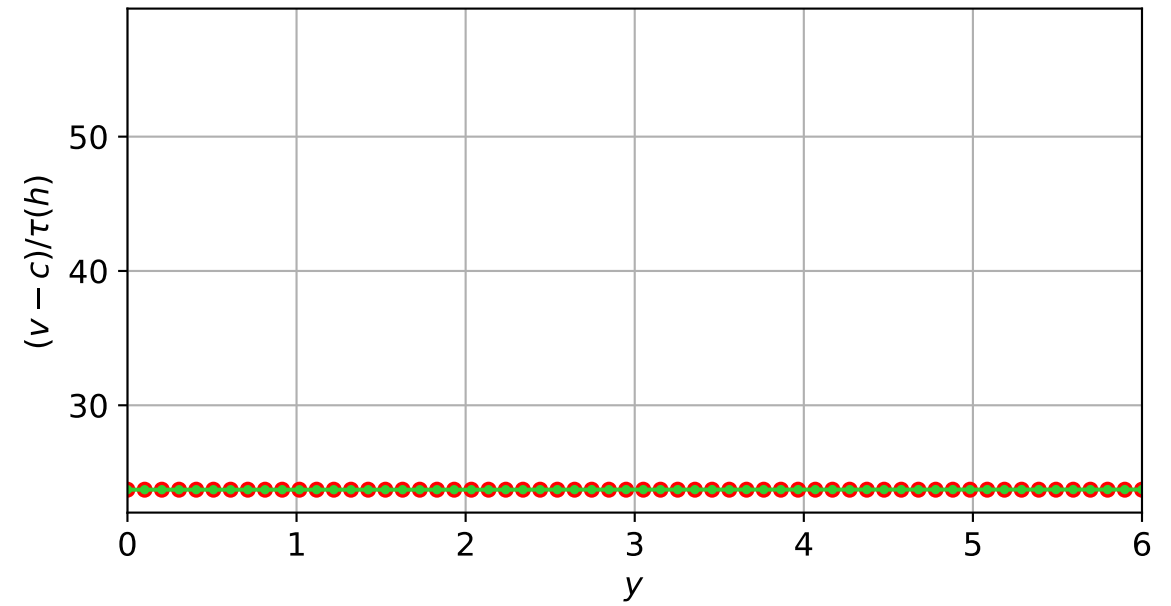
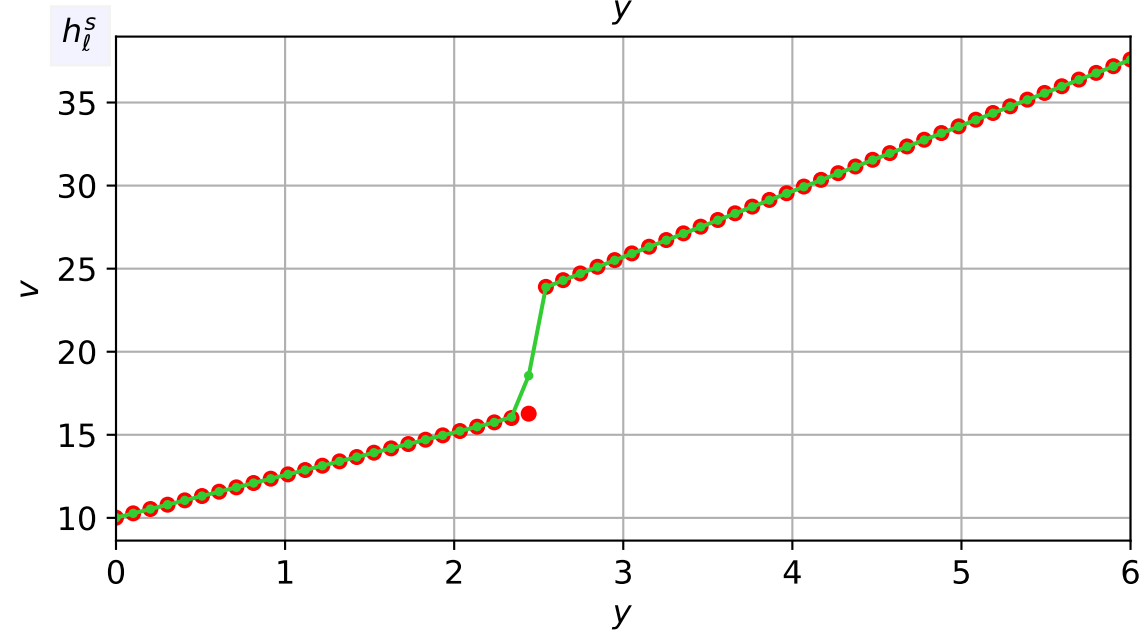
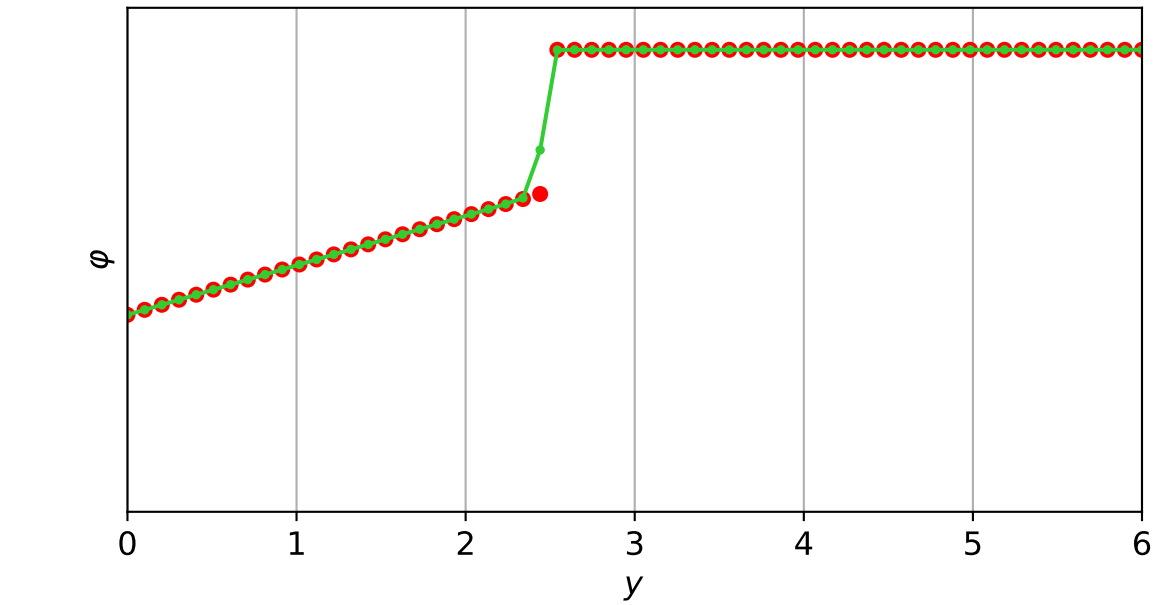
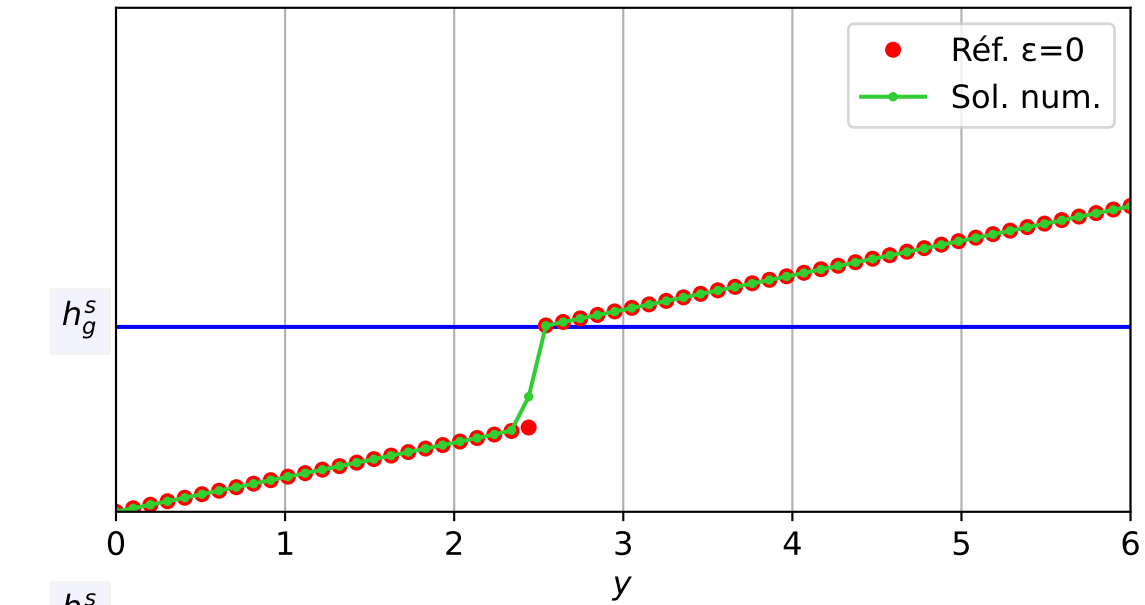
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 1.00e-03$  —  $t = 2.55037/12$  (7624 iter) — résidu eq. =  $8.01e-06$



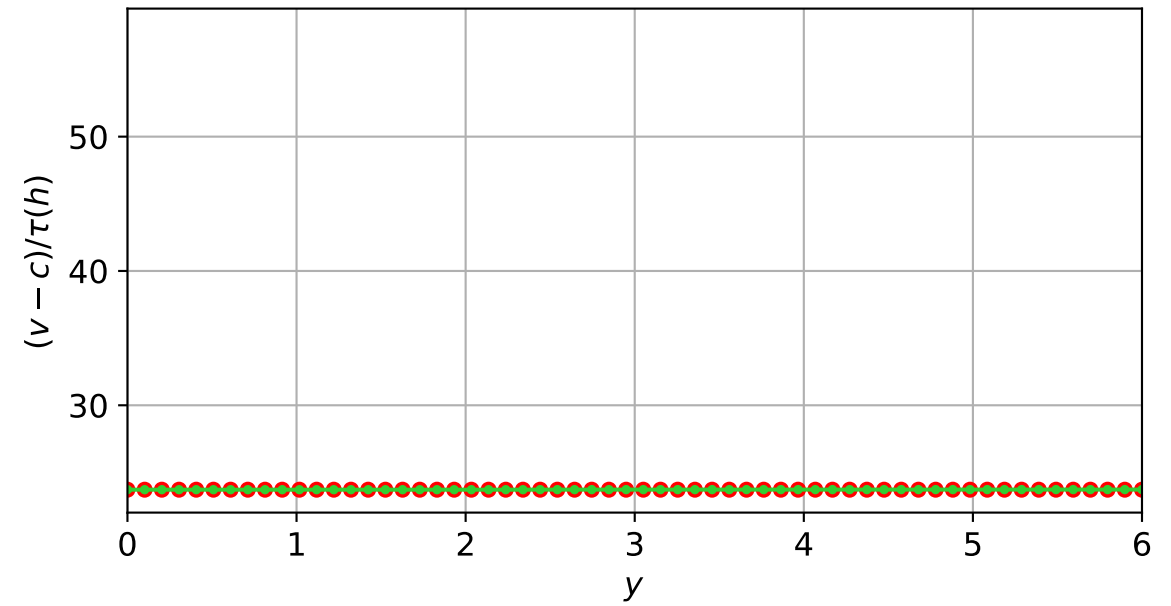
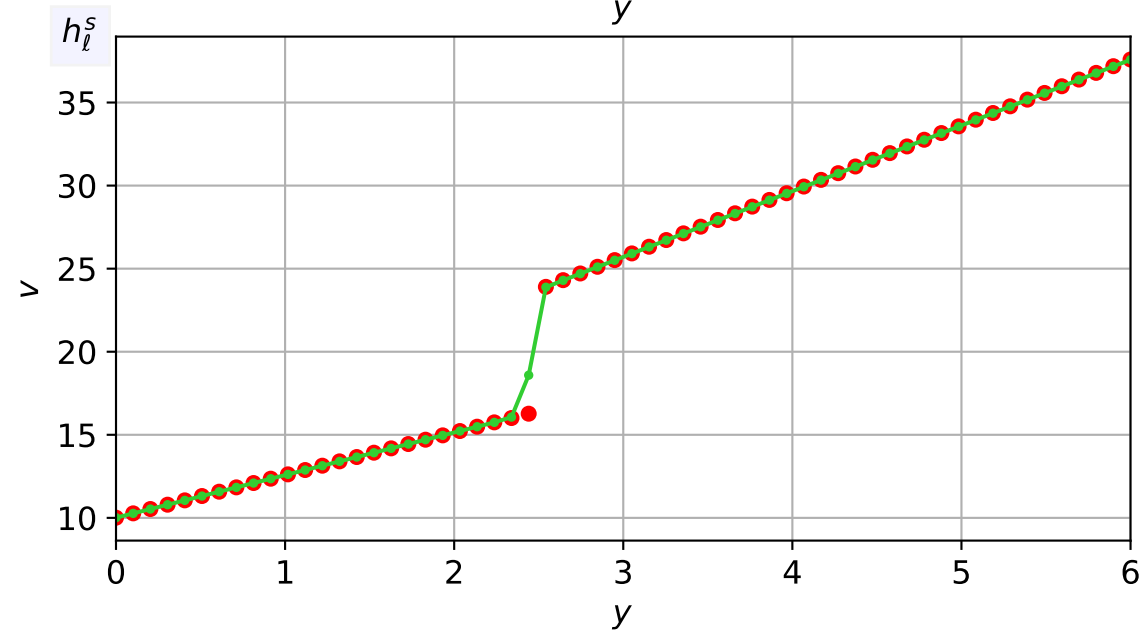
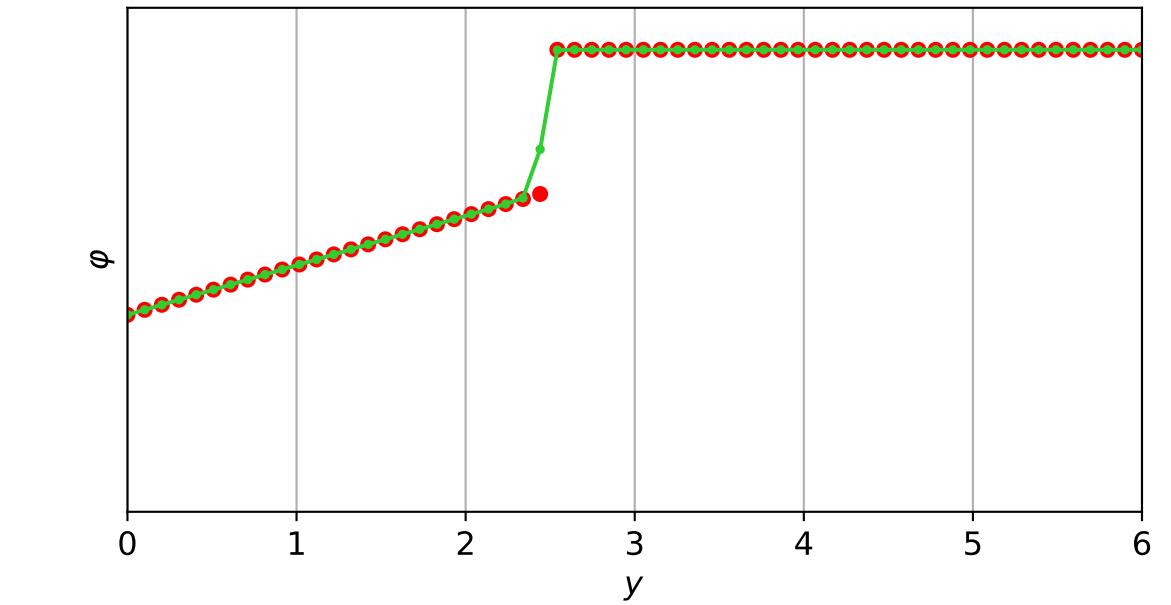
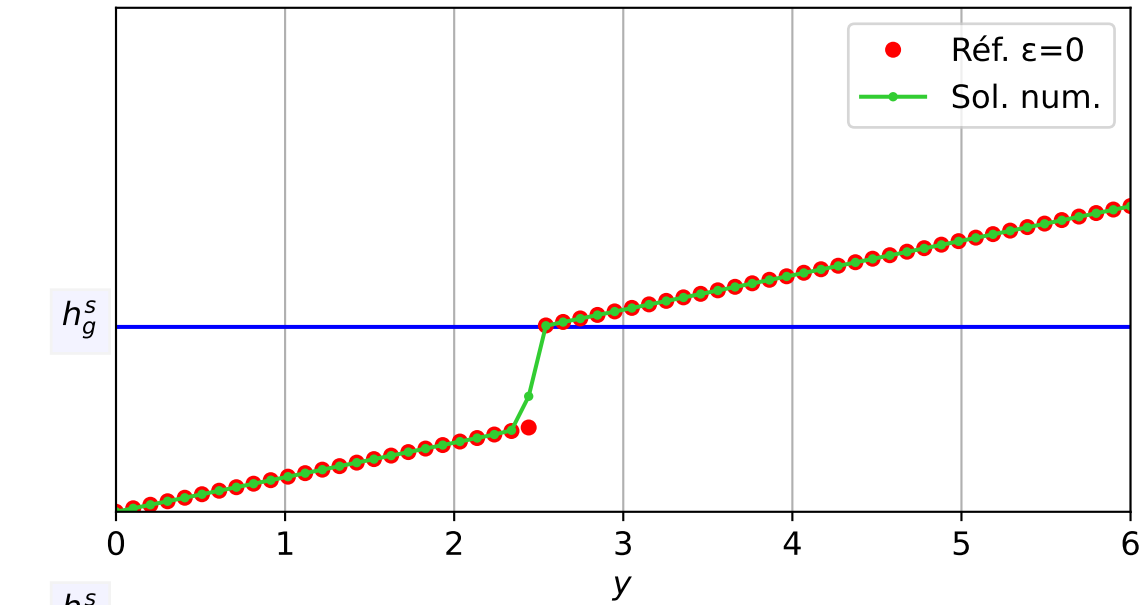
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 1.20e-03$  —  $t = 2.60058/12$  (7673 iter) — résidu eq. =  $9.46e-06$



TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 1.20e-03$  —  $t = 2.65119/12$  (7715 iter) — résidu eq. =  $8.44e-06$

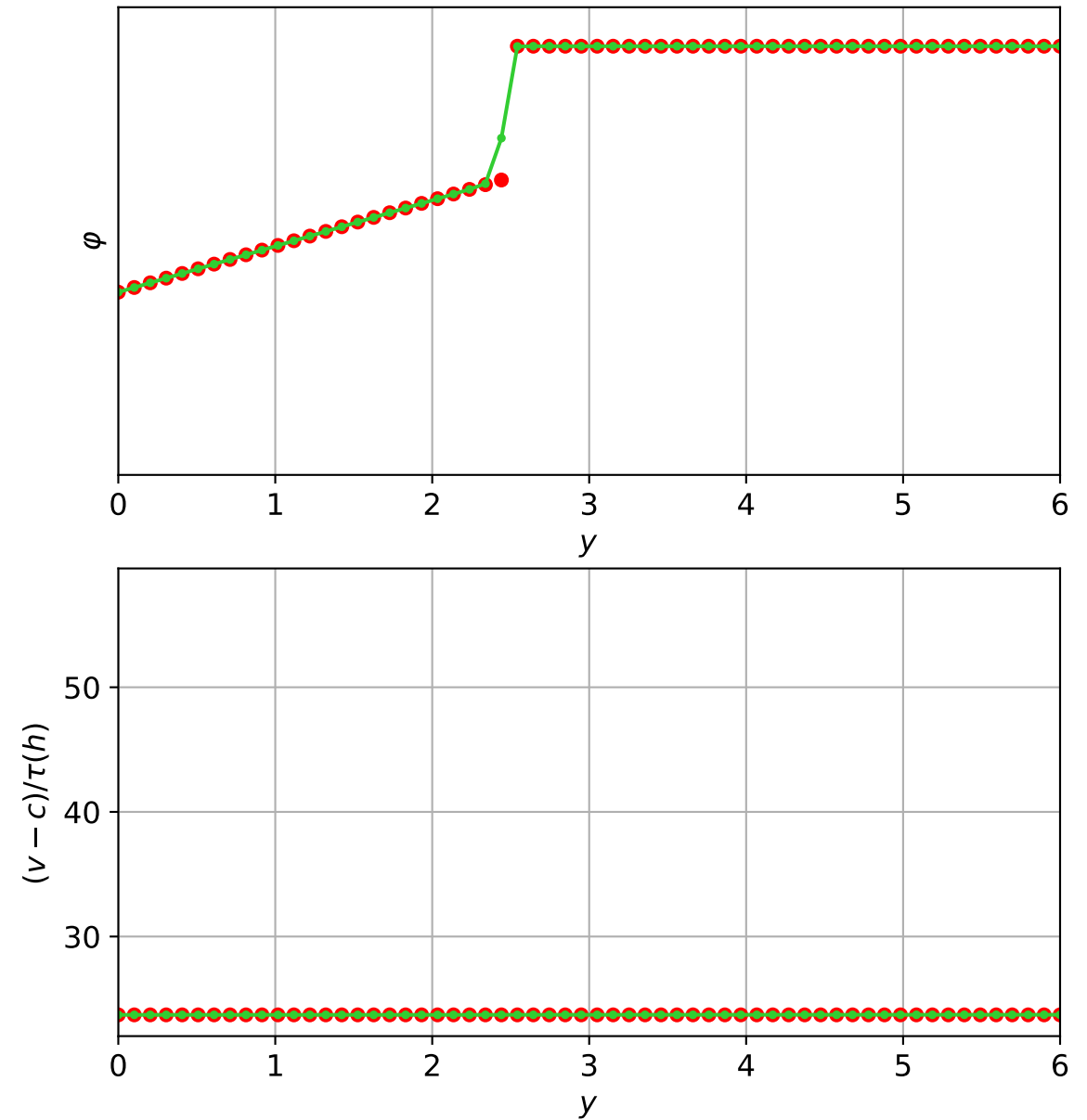
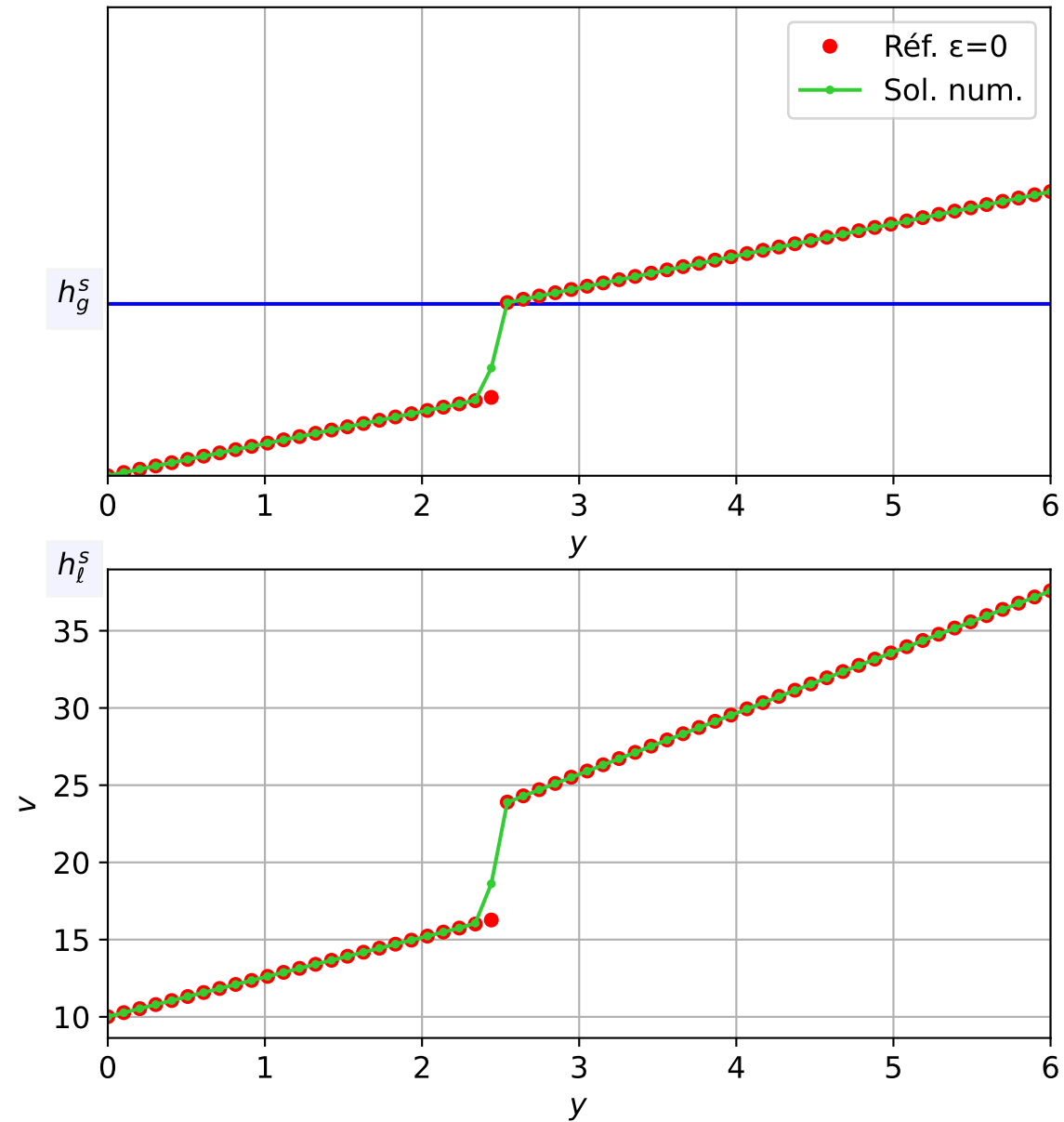


TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 1.20e-03$  —  $t = 2.70059/12$  (7756 iter) — résidu eq. =  $7.50e-06$

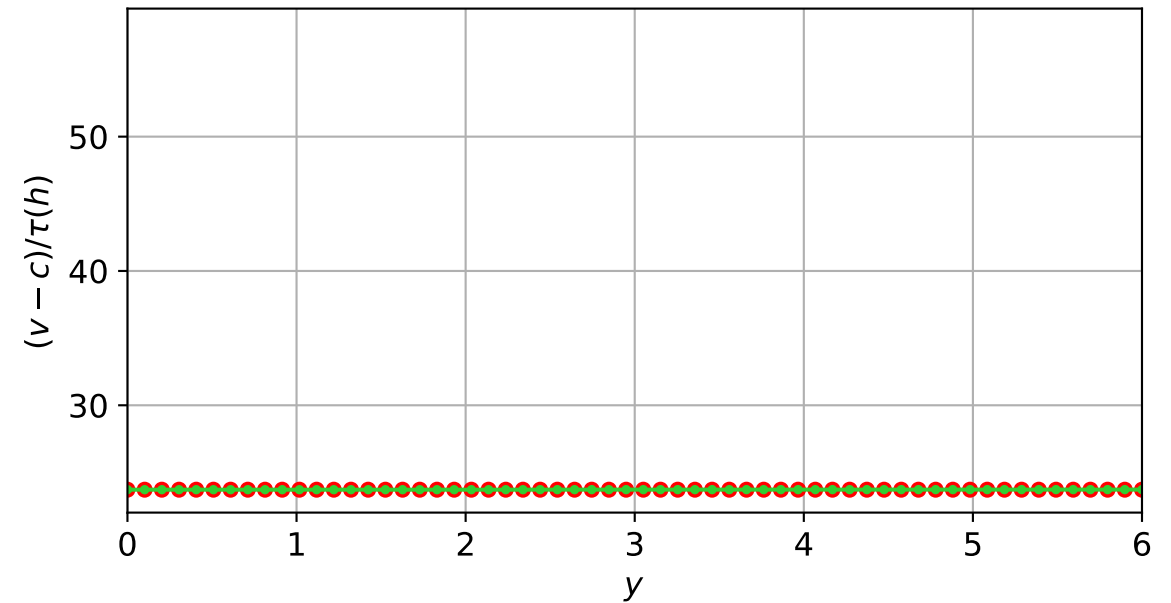
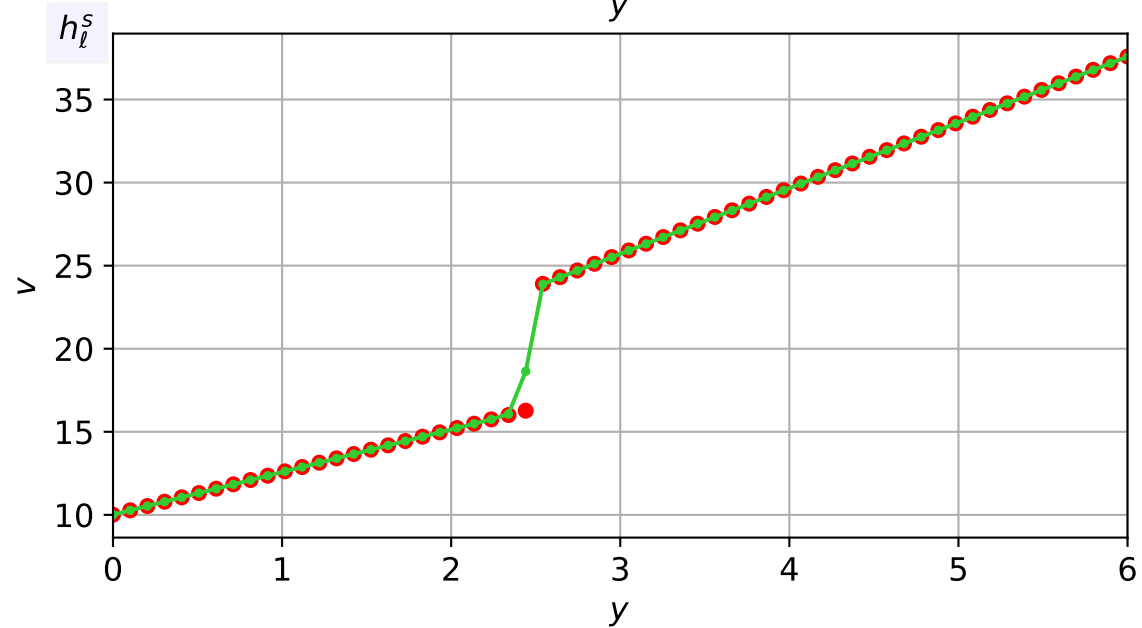
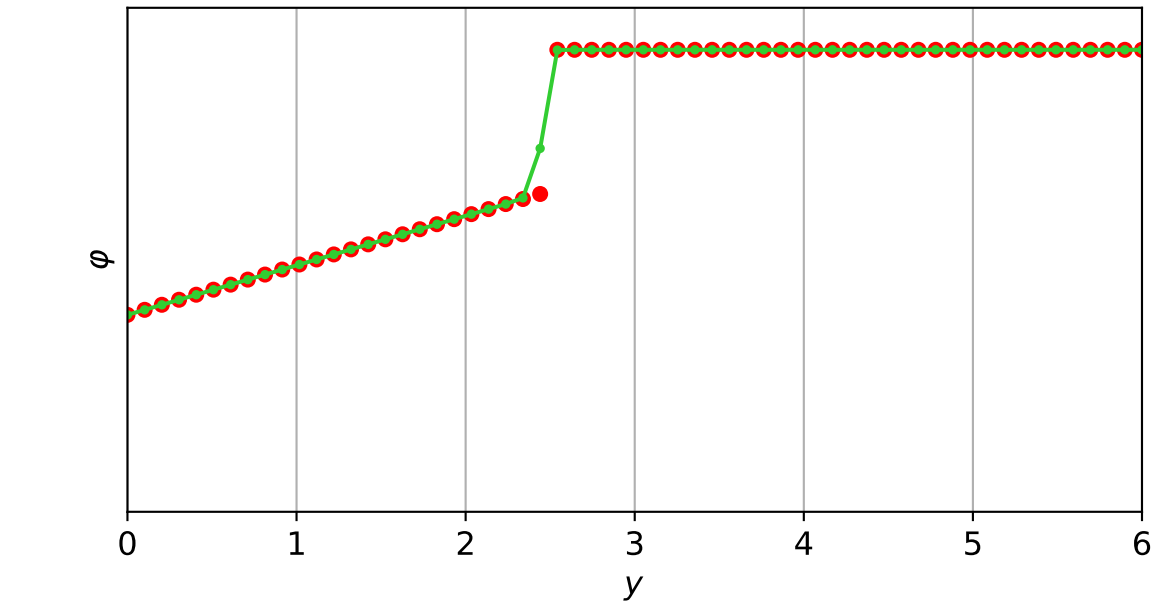
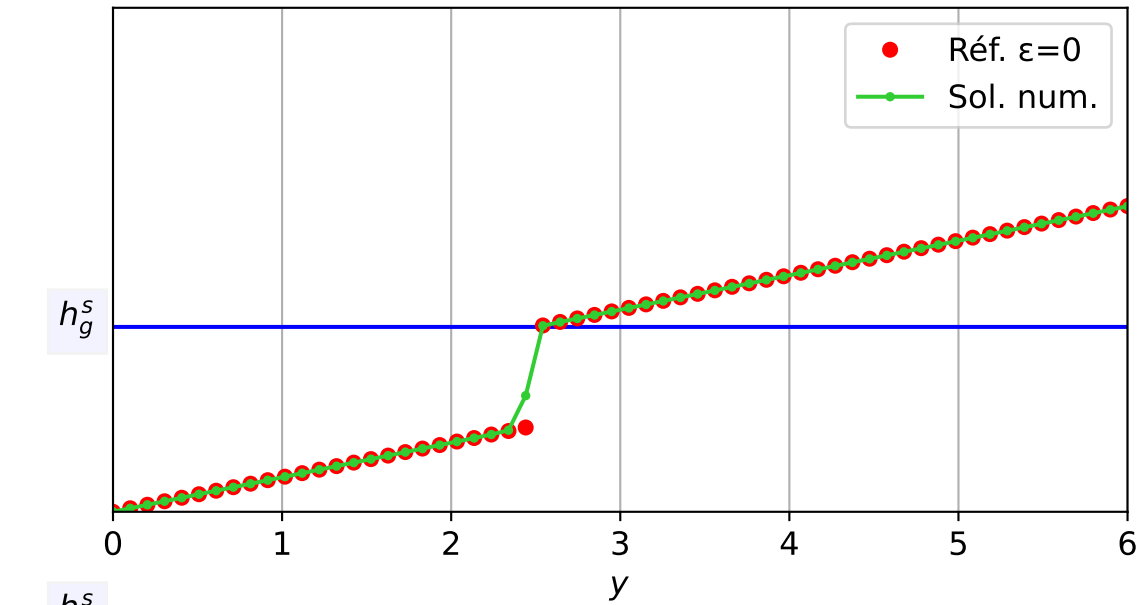




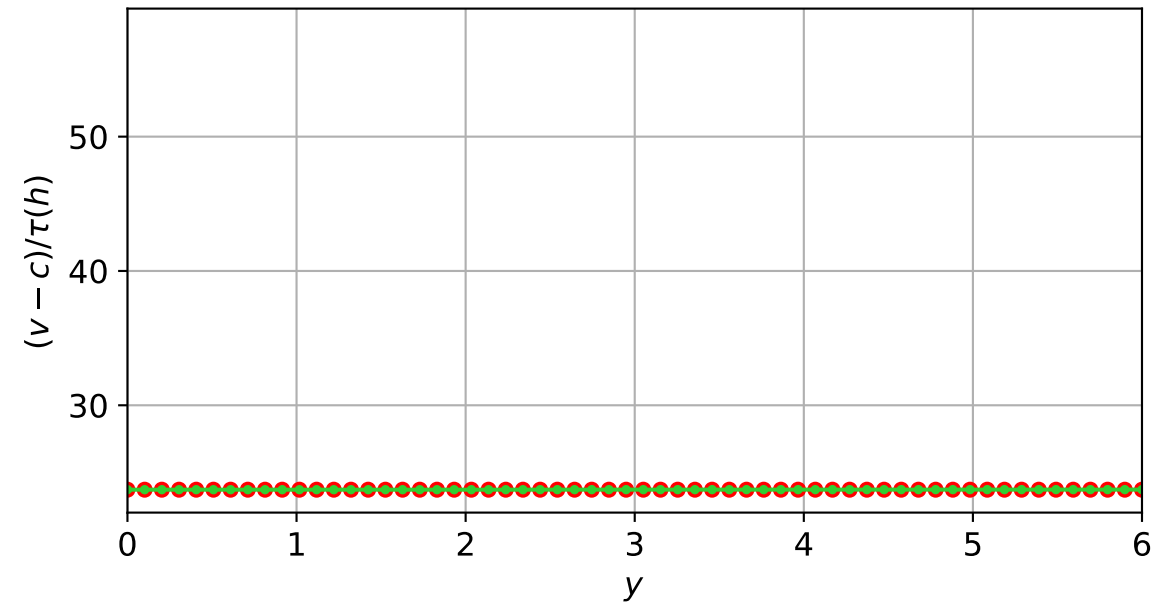
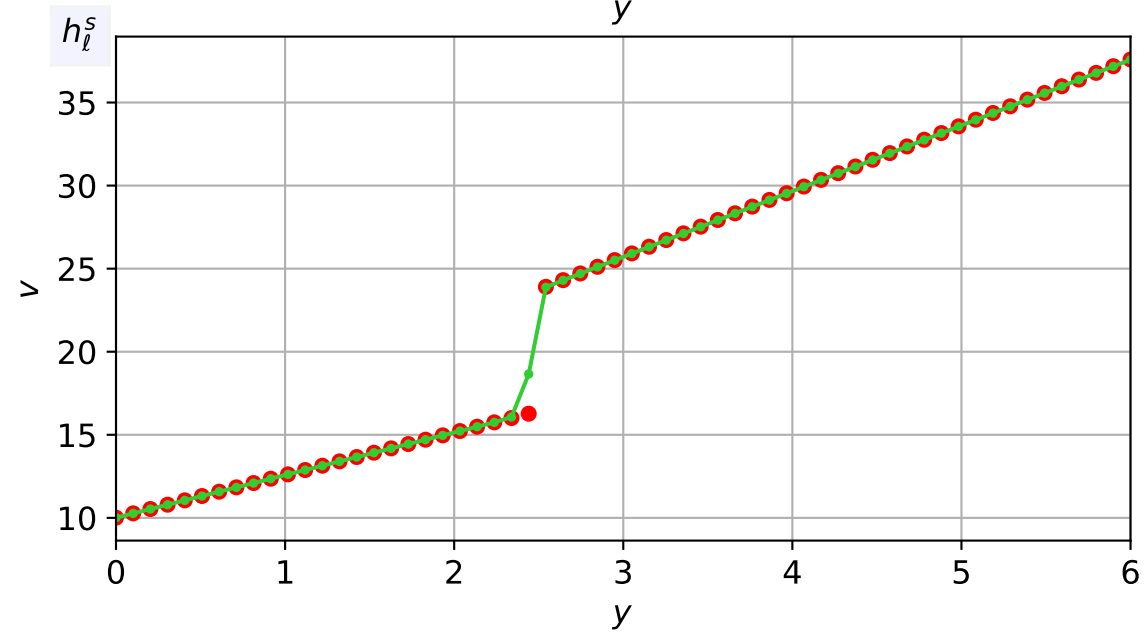
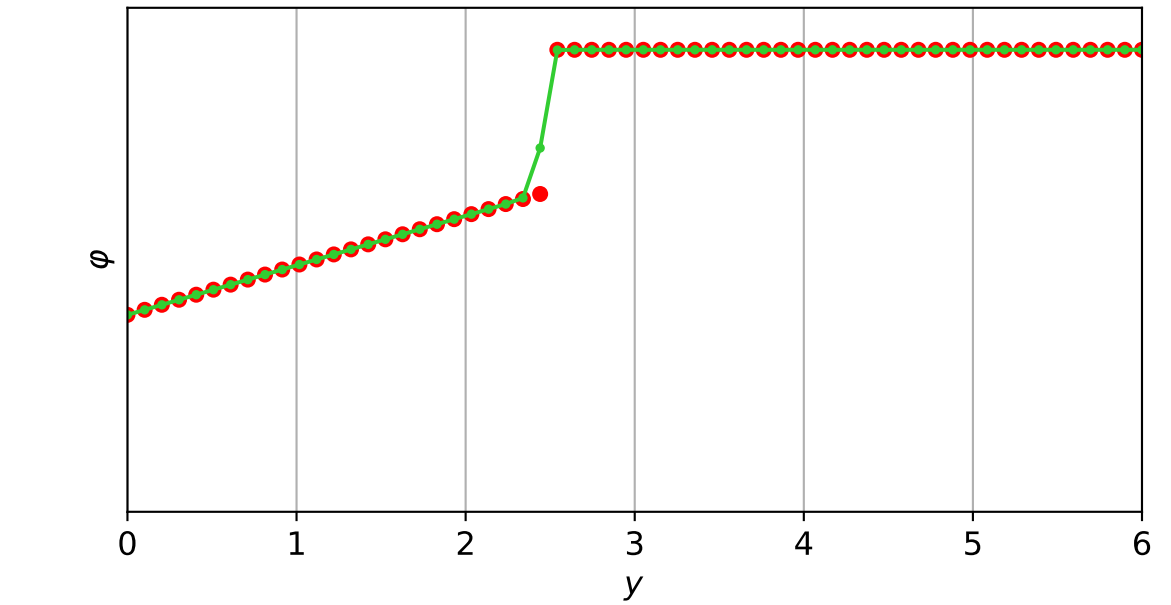
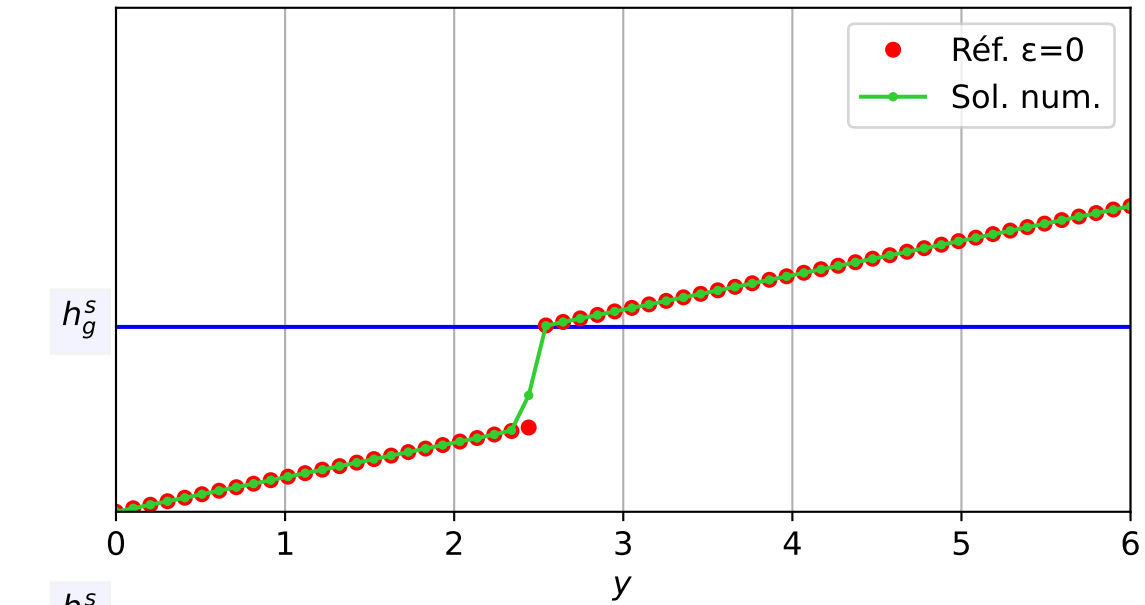
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 1.45e-03$  —  $t = 2.75144/12$  (7792 iter) — résidu eq. =  $8.79e-06$



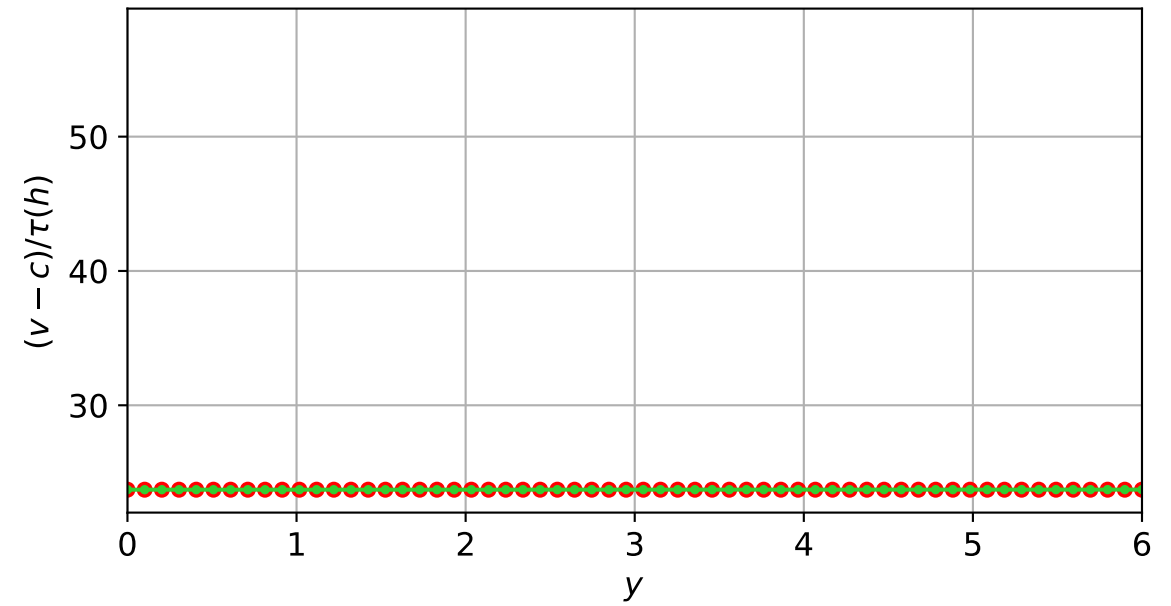
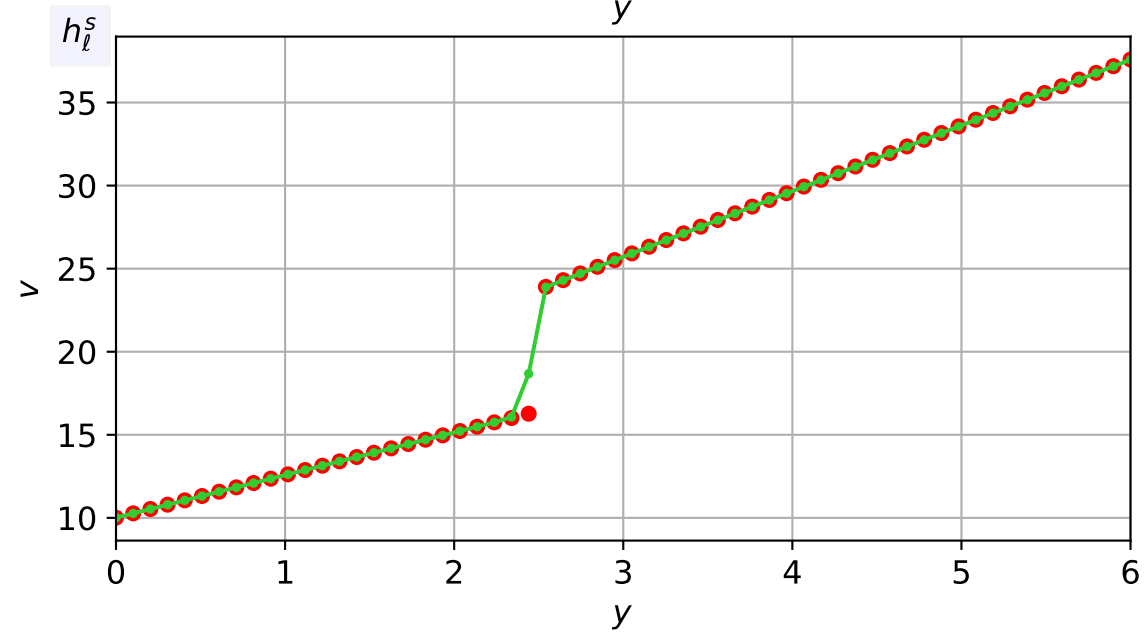
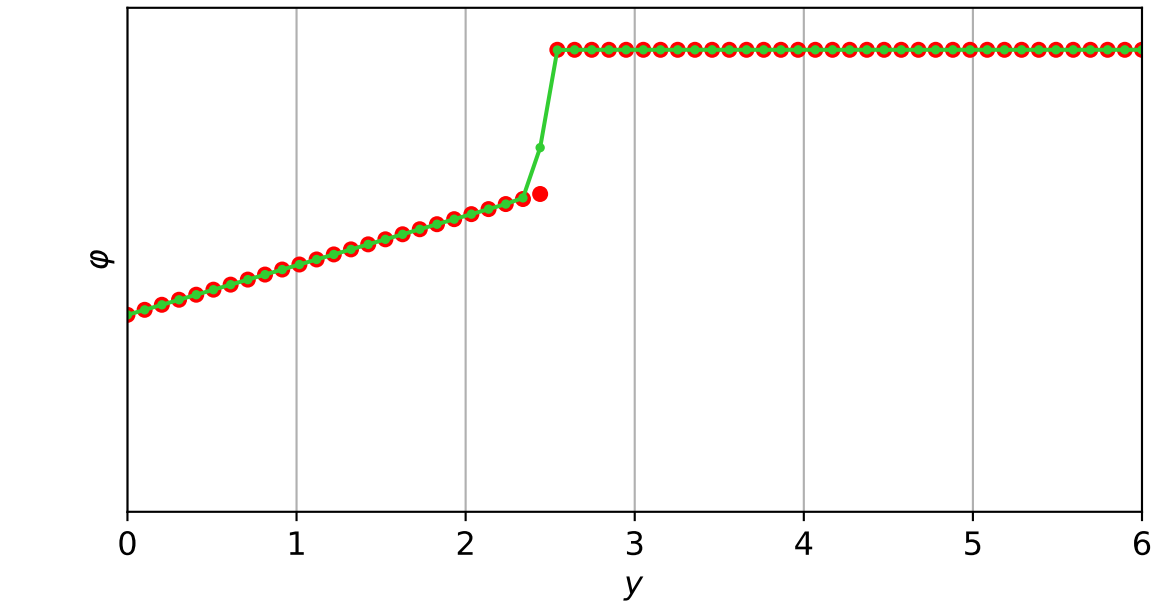
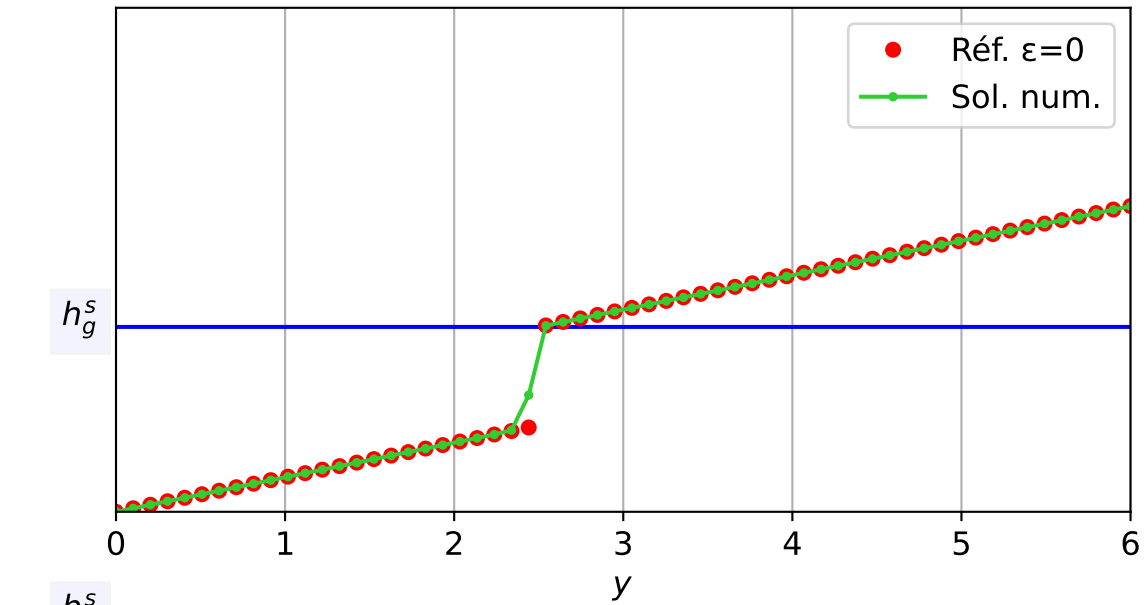
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 1.45e-03$  —  $t = 2.8006/12$  (7826 iter) — résidu eq. =  $7.82e-06$



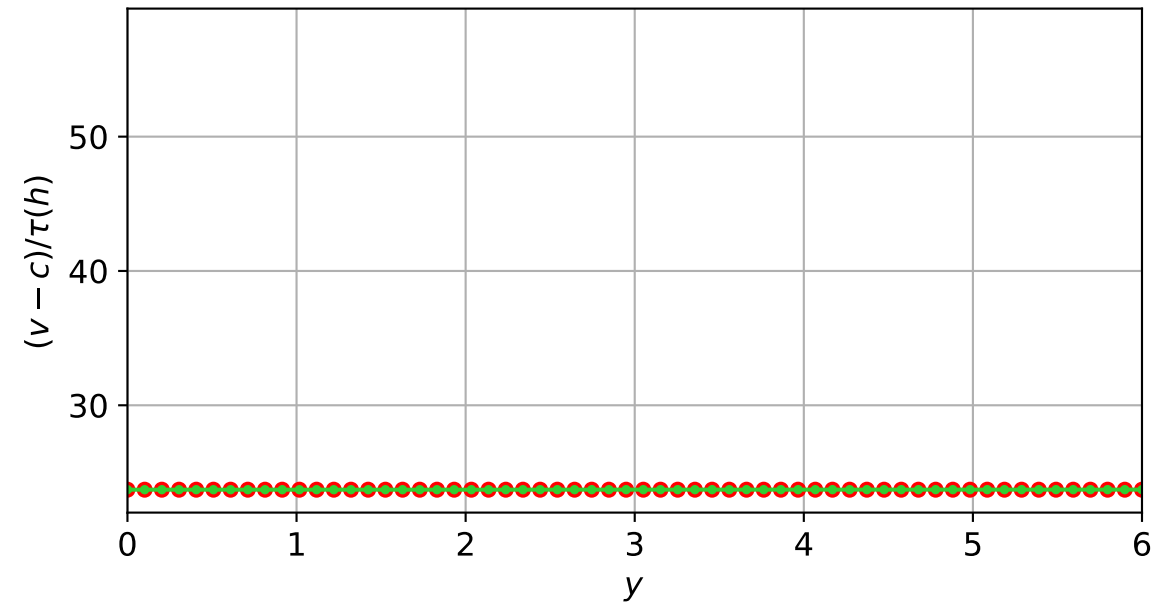
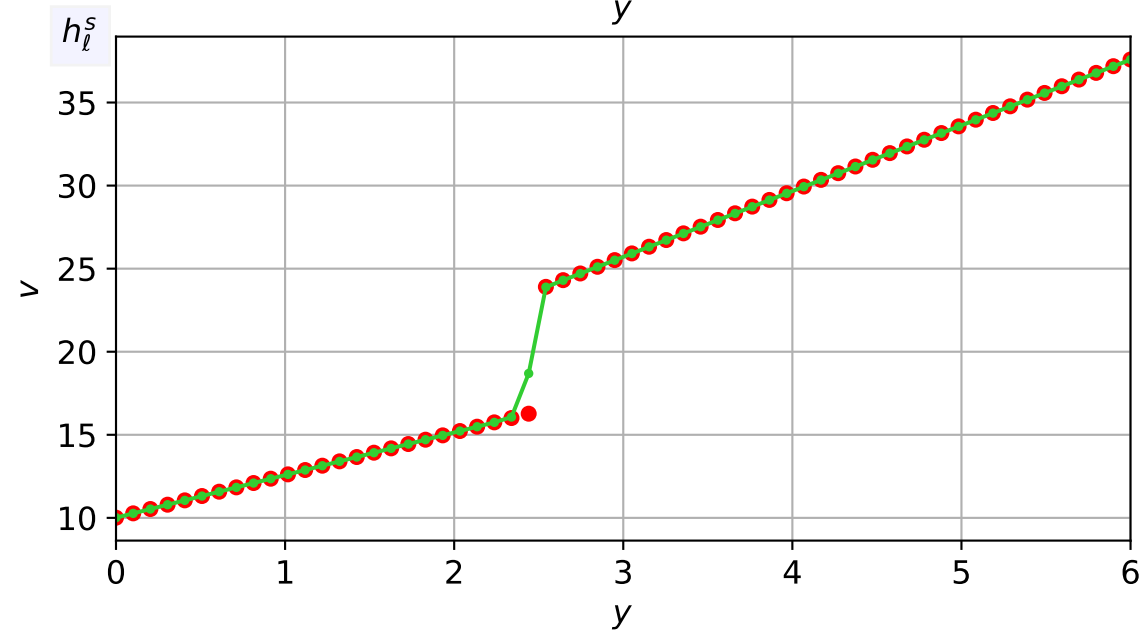
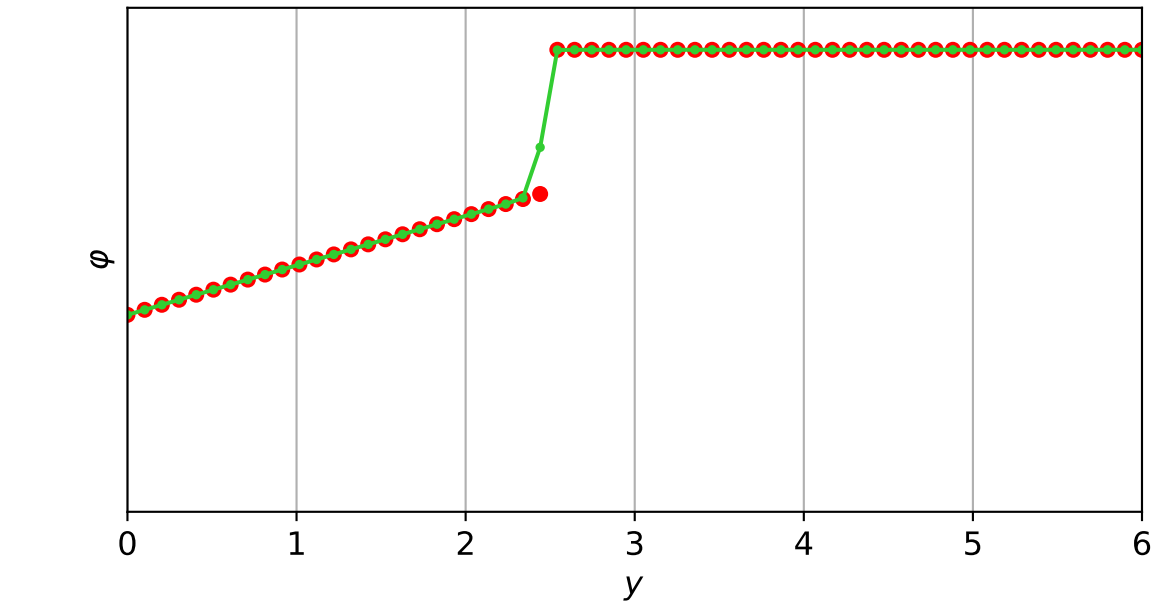
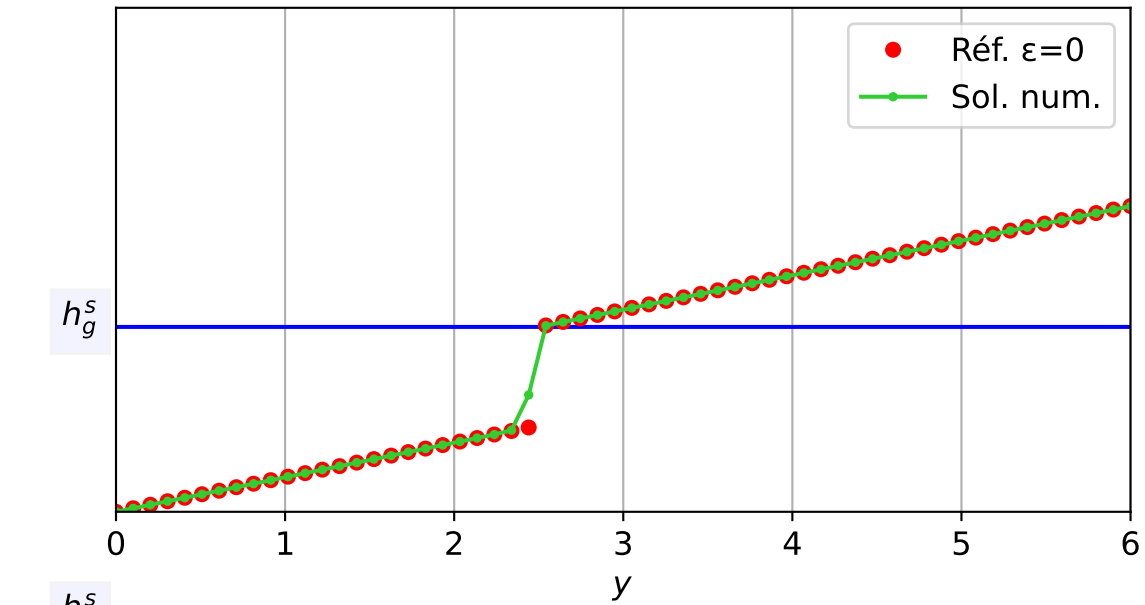
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 1.74e-03$  —  $t = 2.85005/12$  (7856 iter) — résidu eq. =  $9.09e-06$



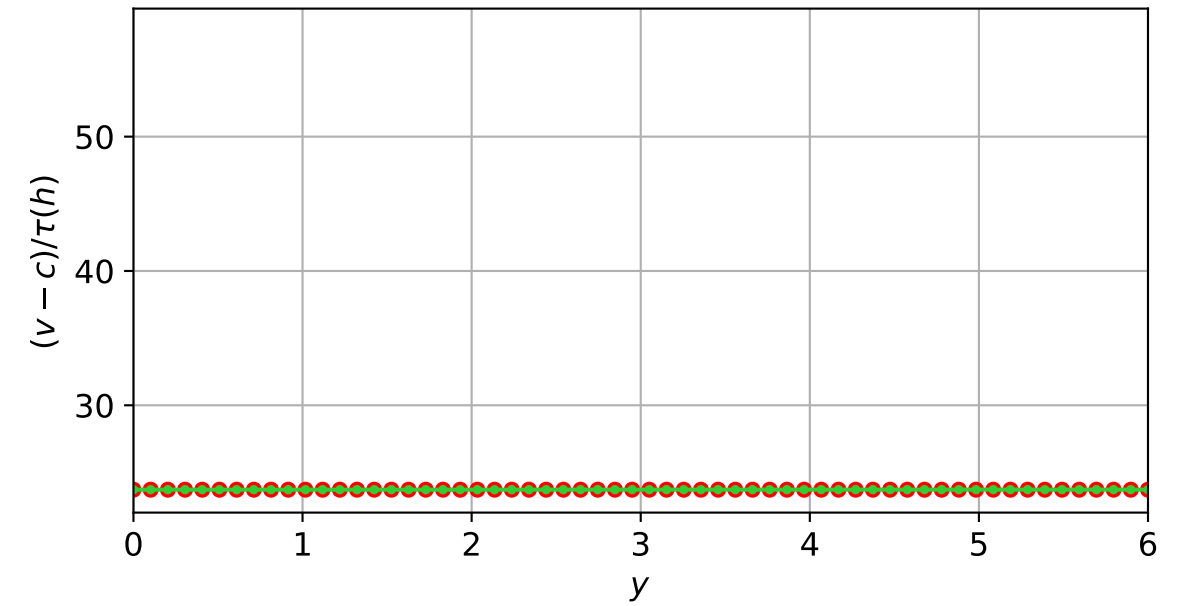
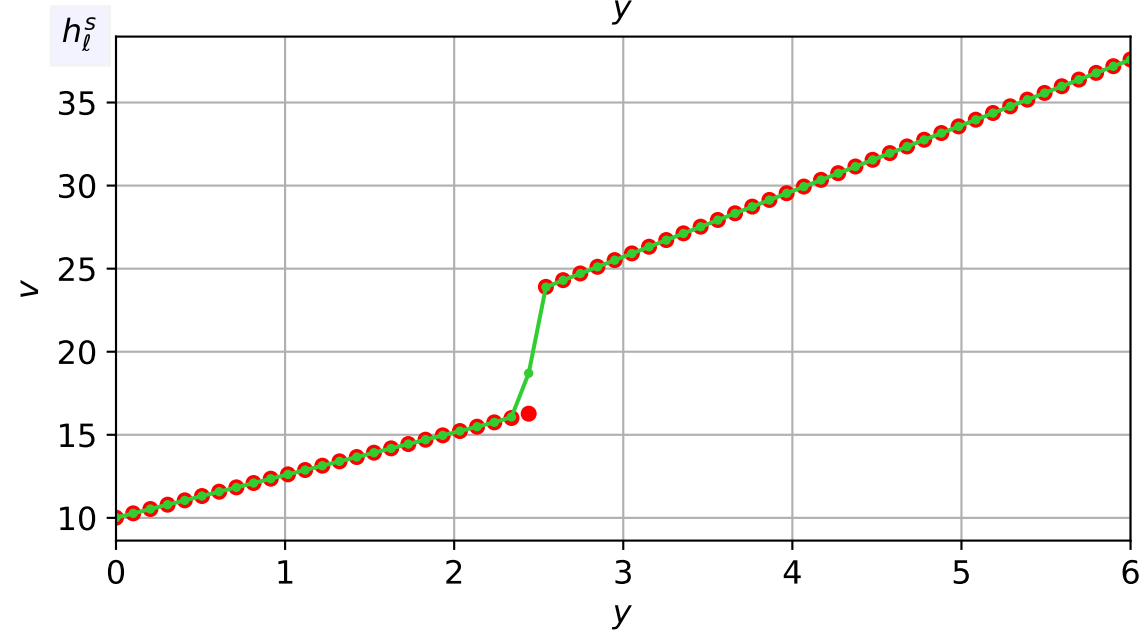
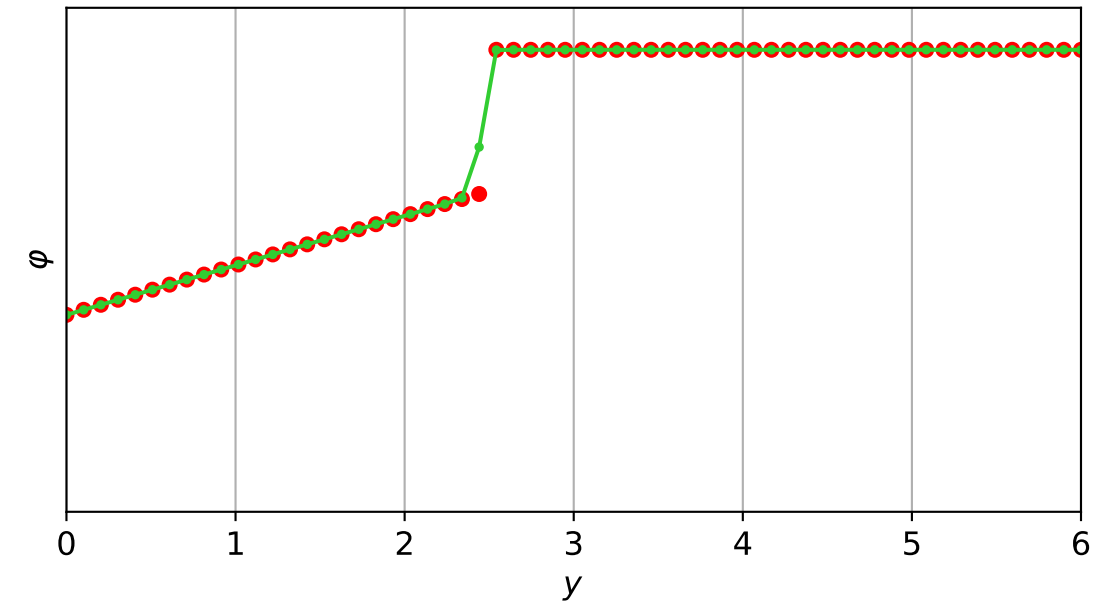
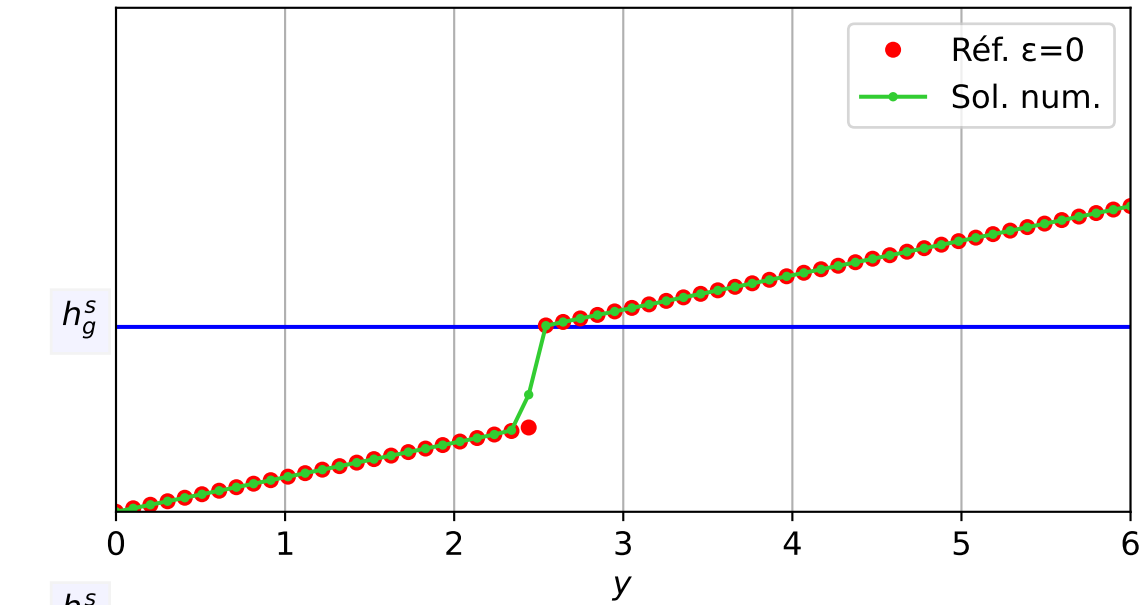
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 1.74e-03$  —  $t = 2.90036/12$  (7885 iter) — résidu eq. =  $8.07e-06$



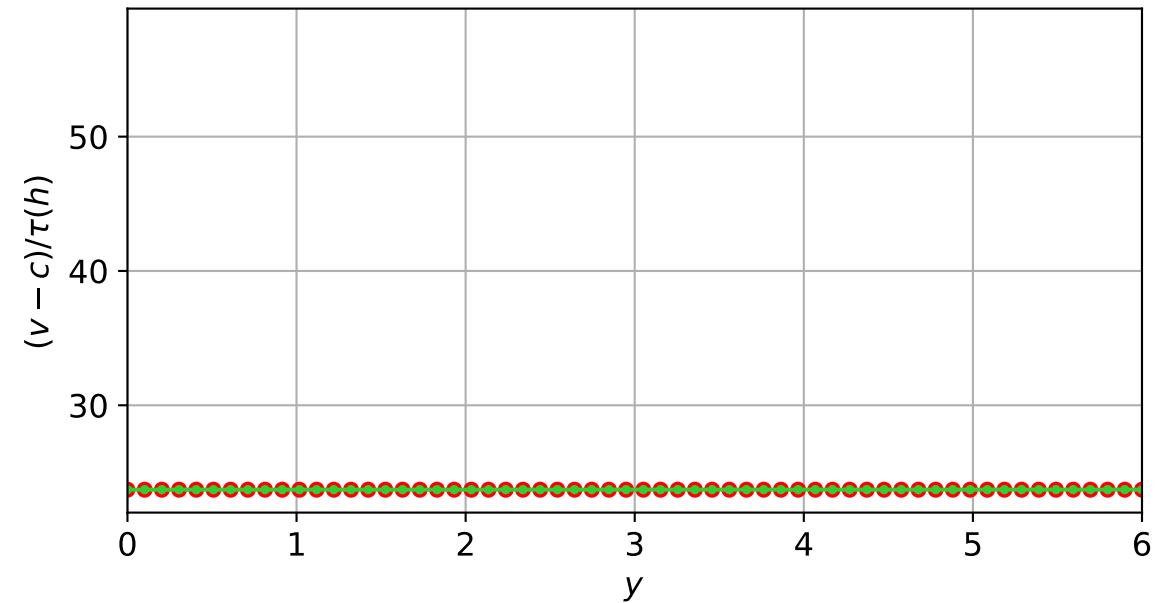
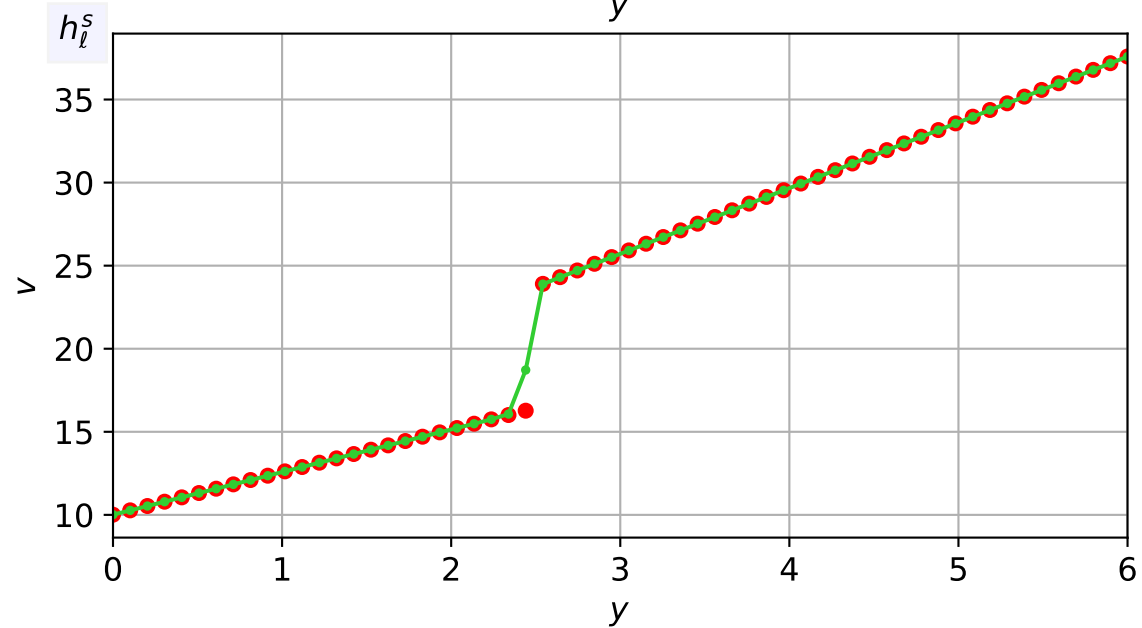
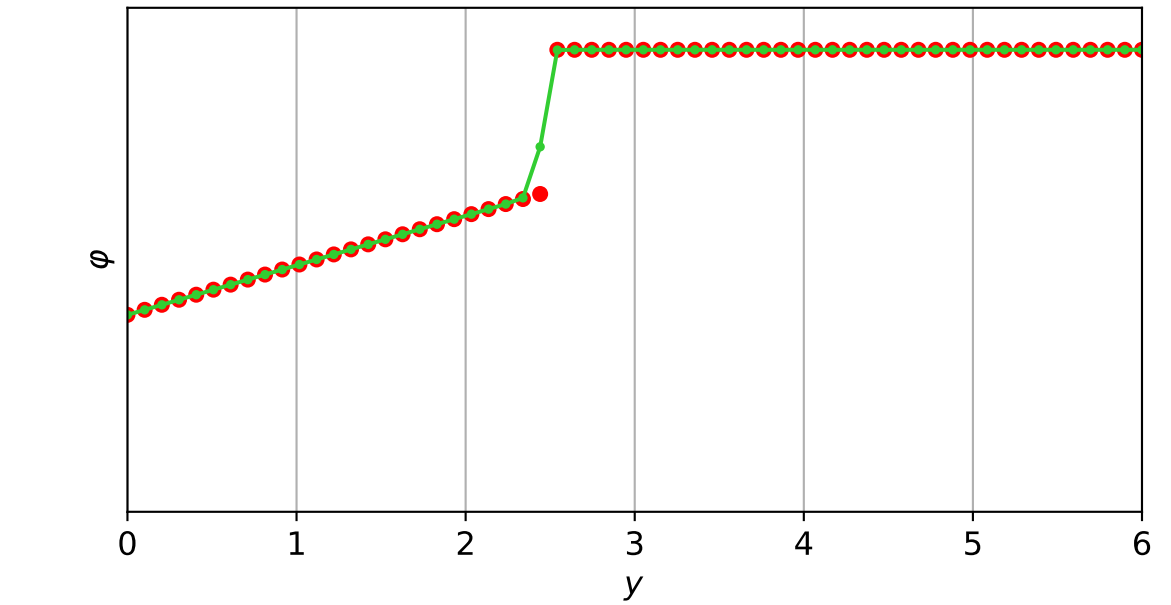
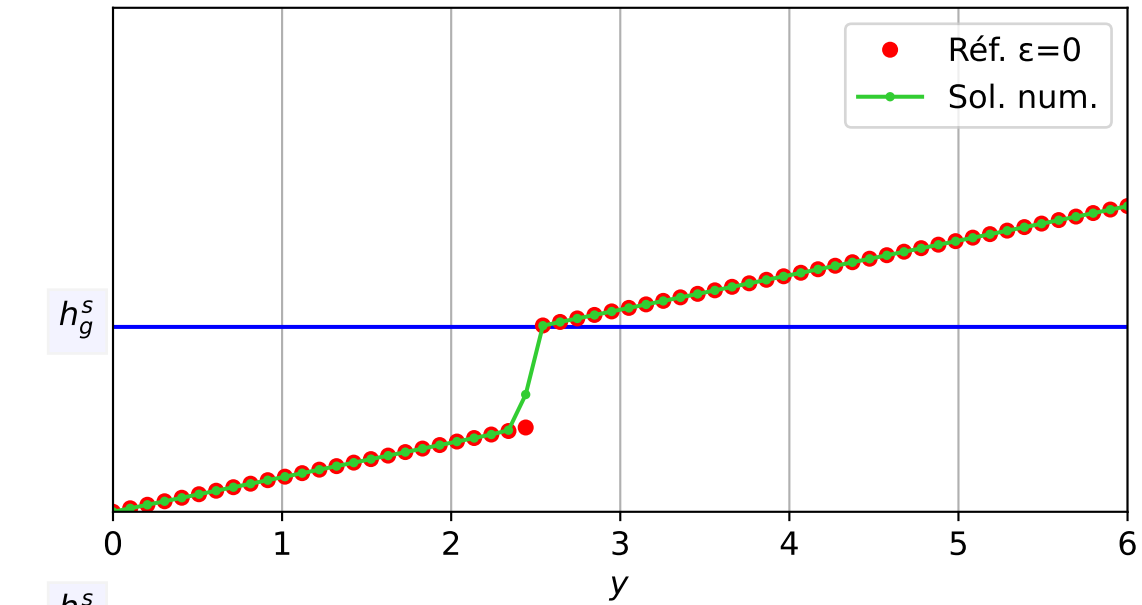
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.08e-03$  —  $t = 2.95103/12$  (7911 iter) — résidu eq. =  $9.28e-06$



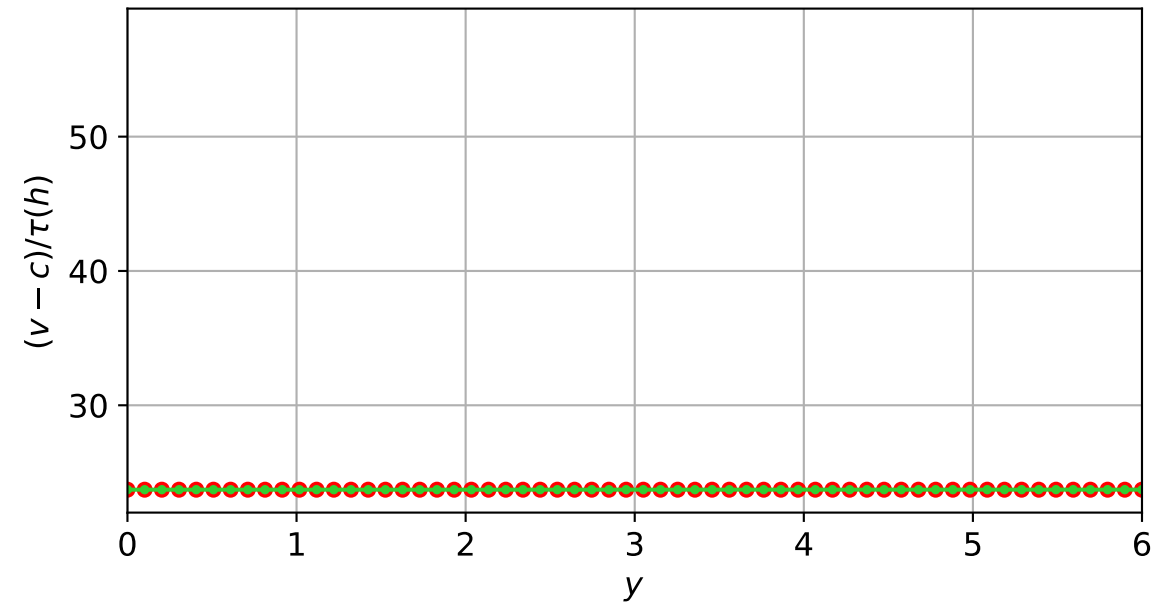
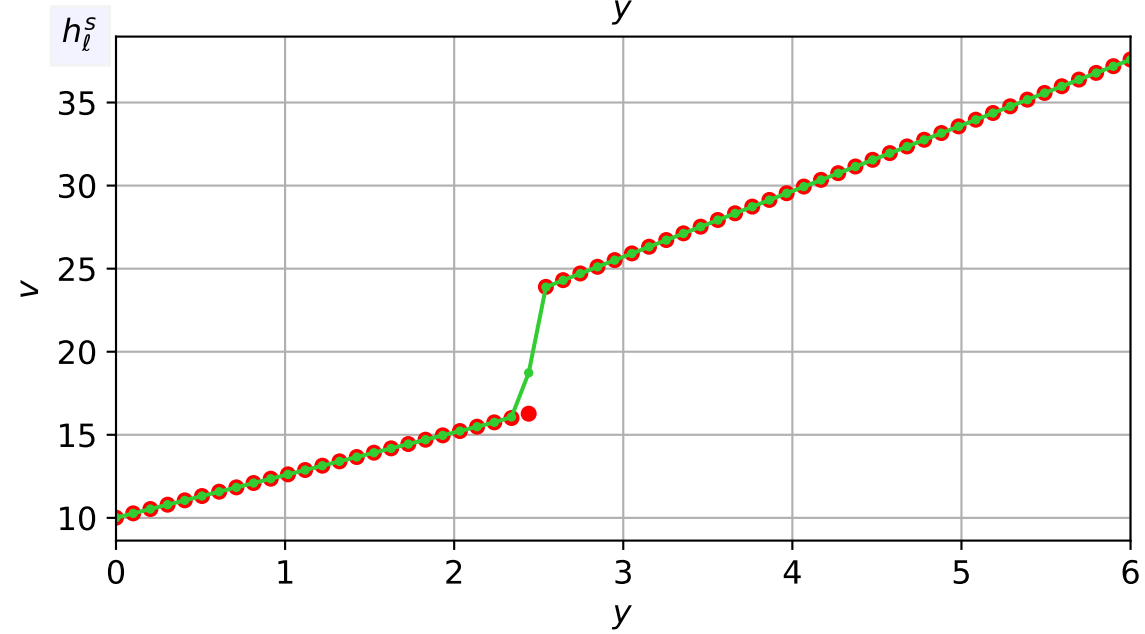
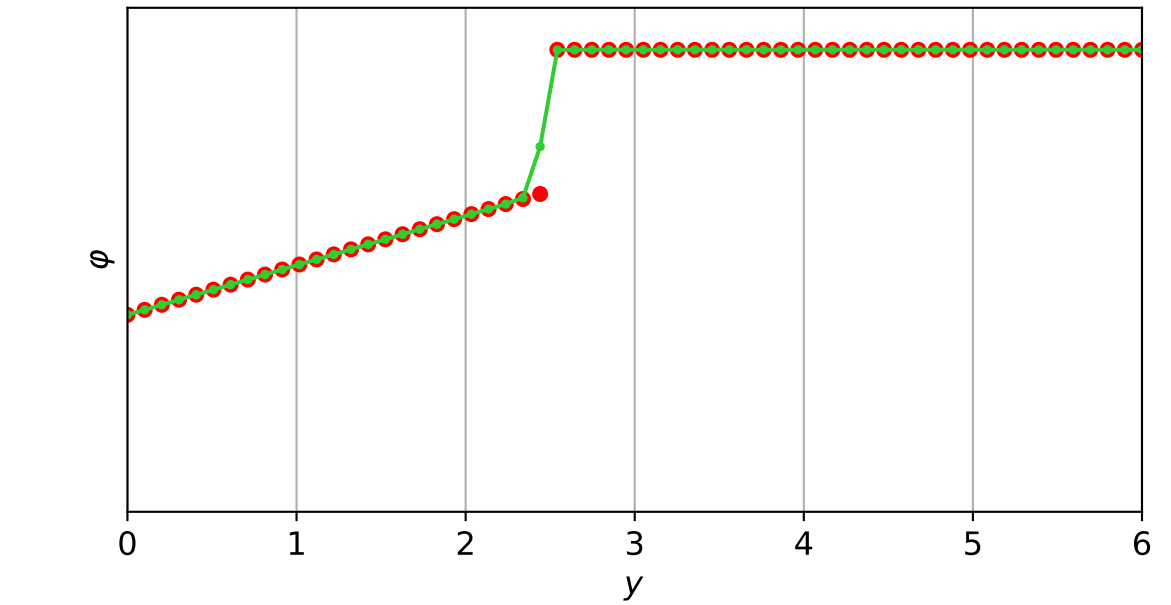
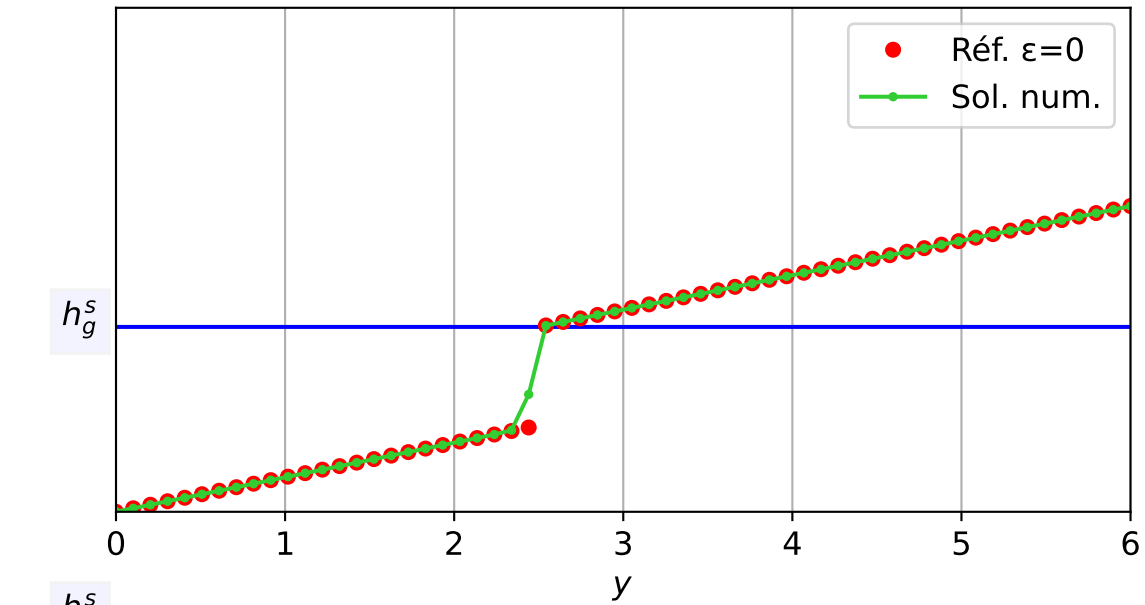
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.08e-03$  —  $t = 3.001/12$  (7935 iter) — résidu eq. =  $8.25e-06$



TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.0514/12$  (7957 iter) — résidu eq. =  $9.50e-06$

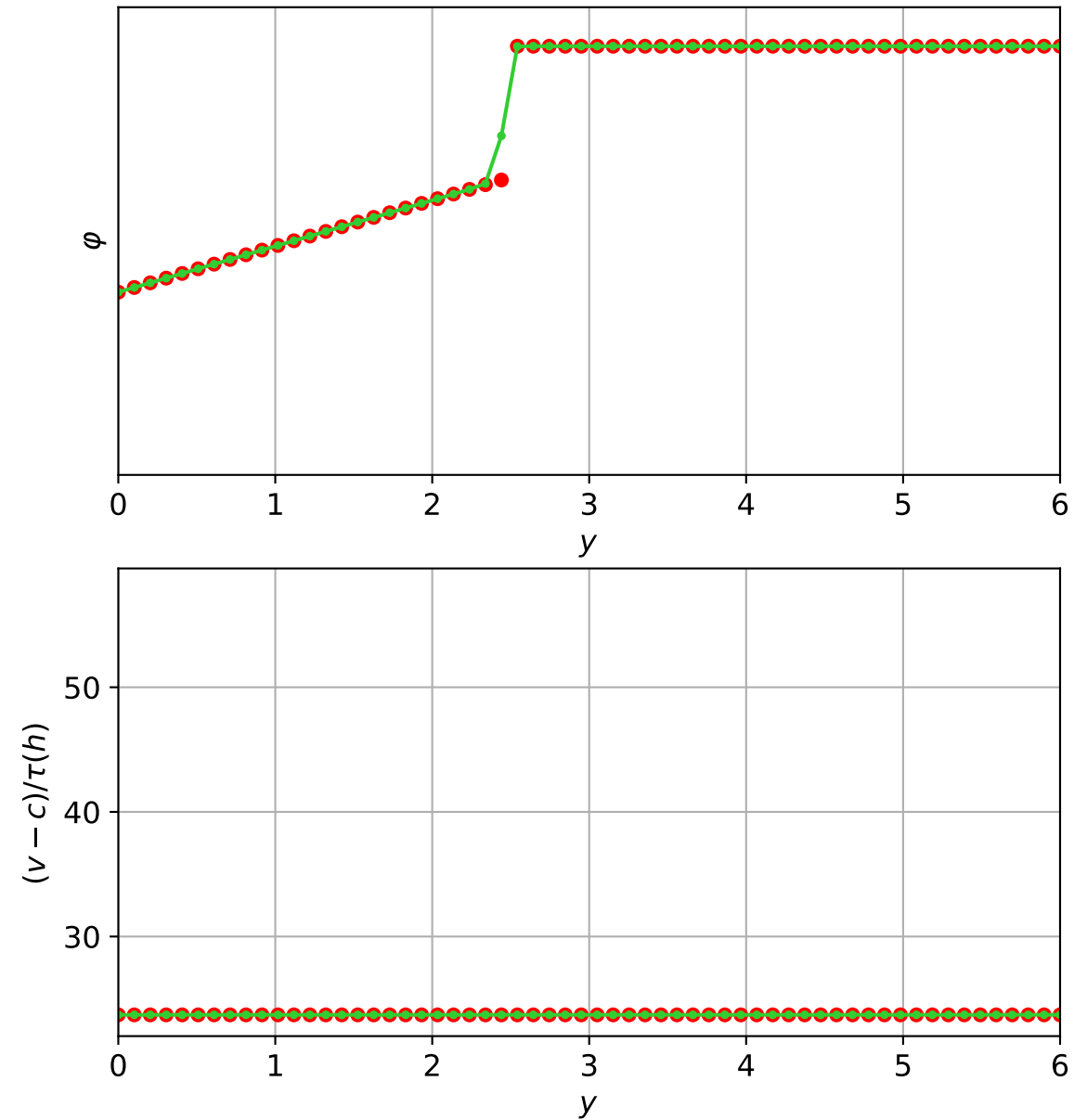
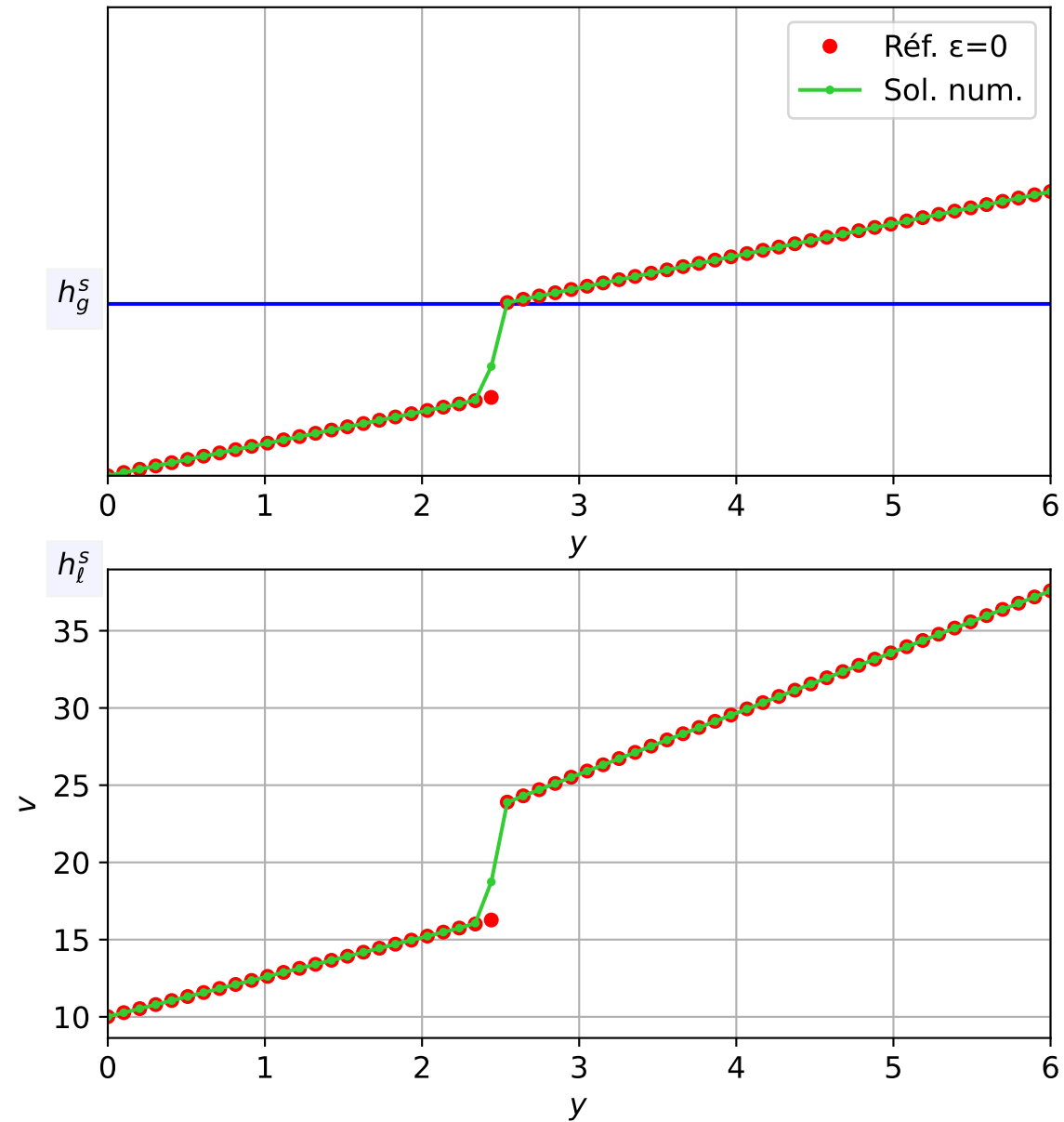


TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.1001/12$  (7977 iter) — résidu eq. =  $8.62e-06$

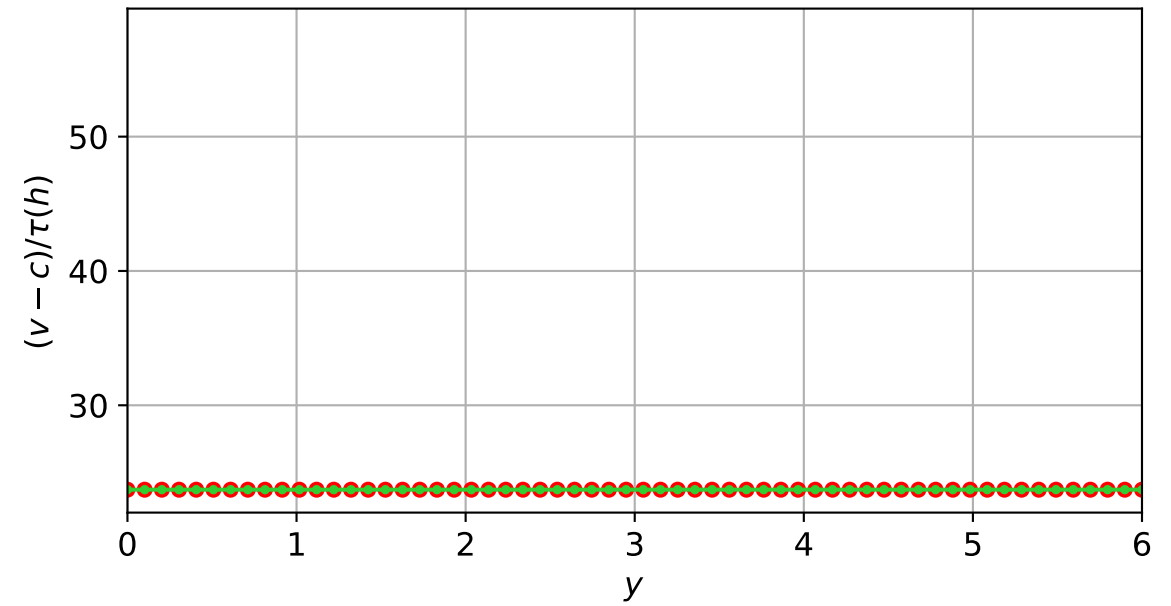
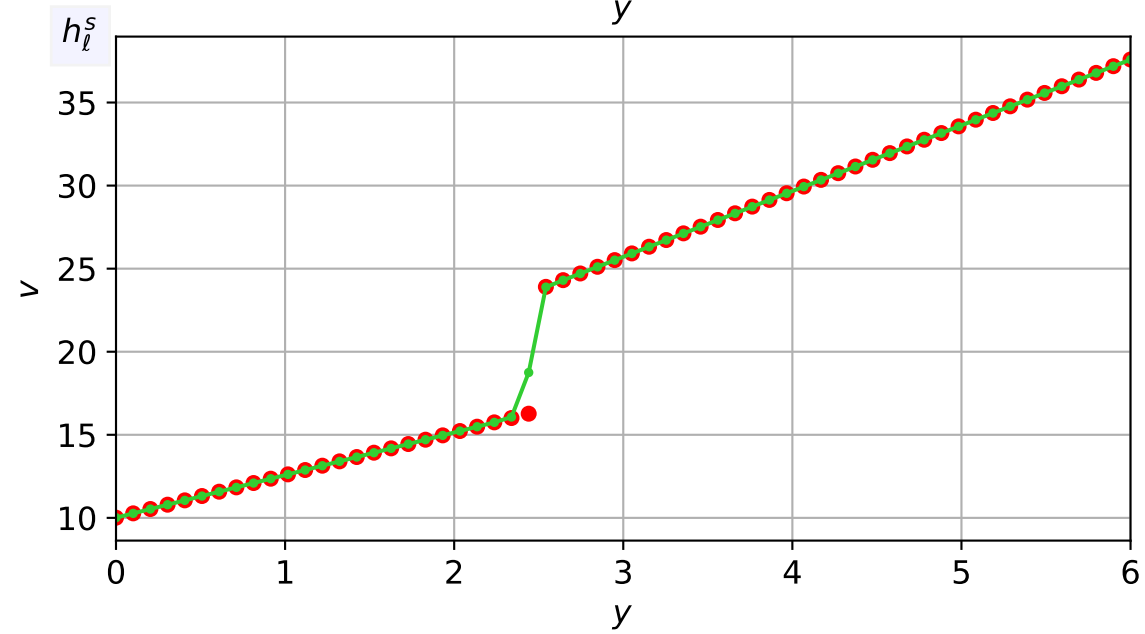
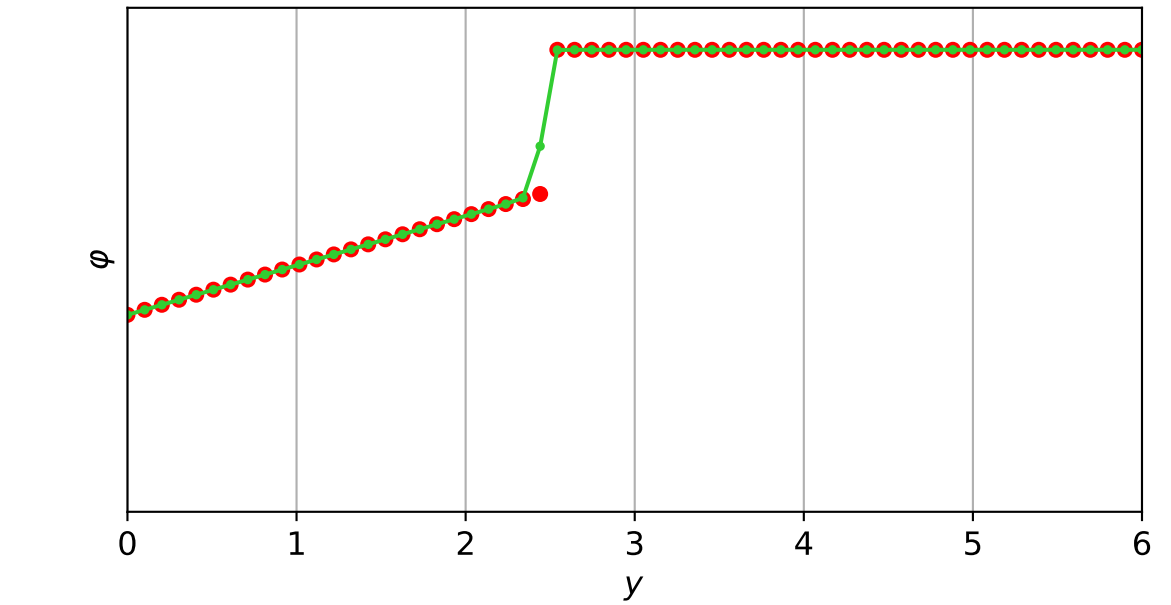
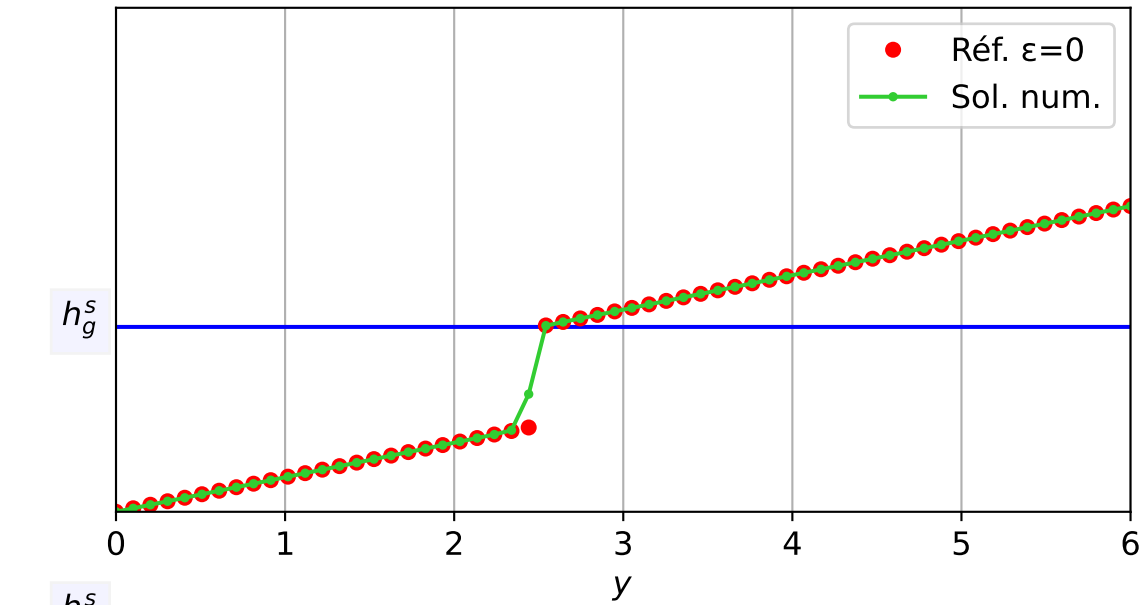




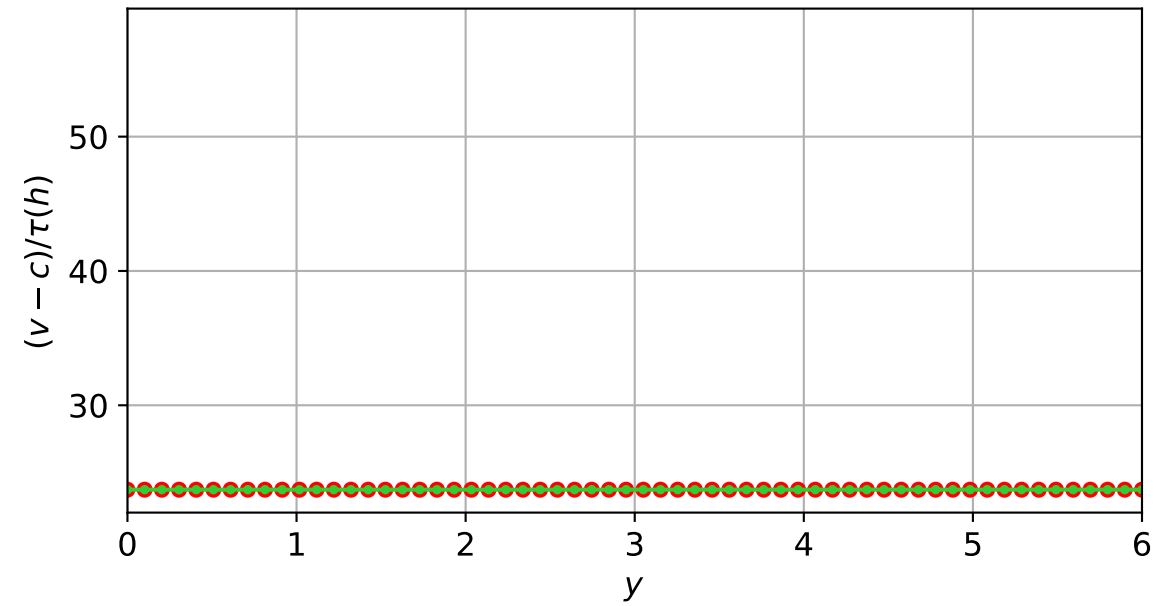
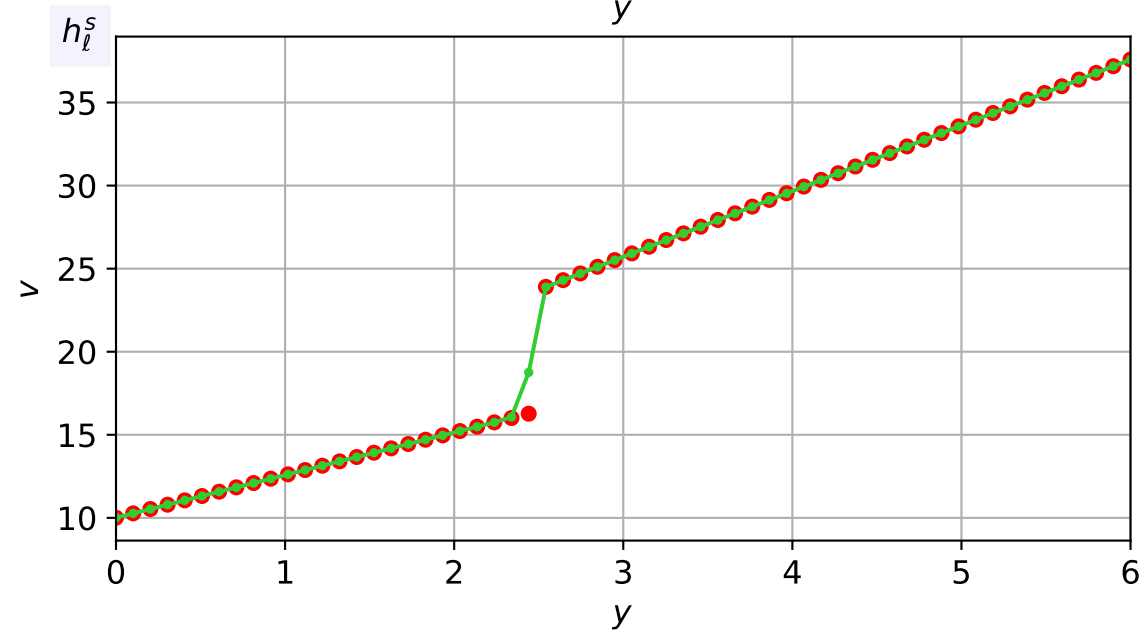
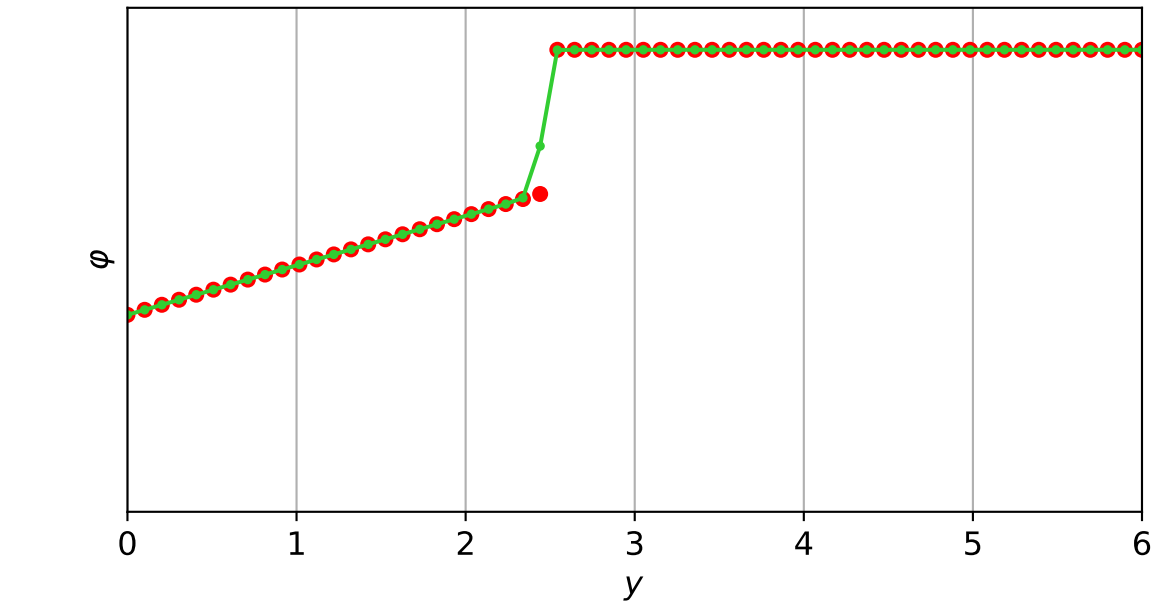
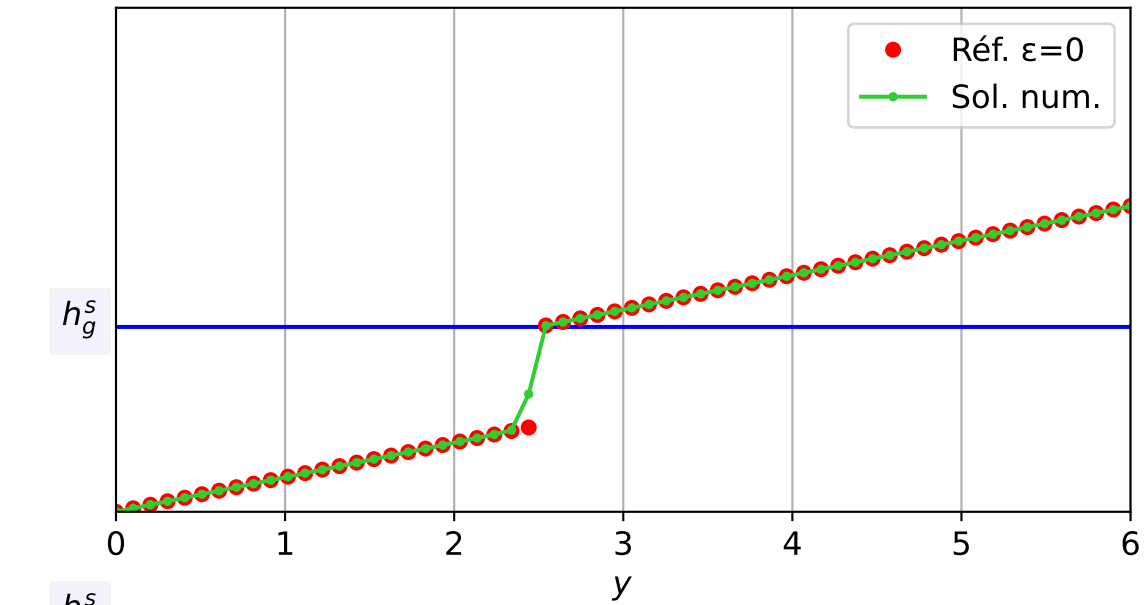
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.15125/12$  (7998 iter) — résidu eq. =  $7.77e-06$



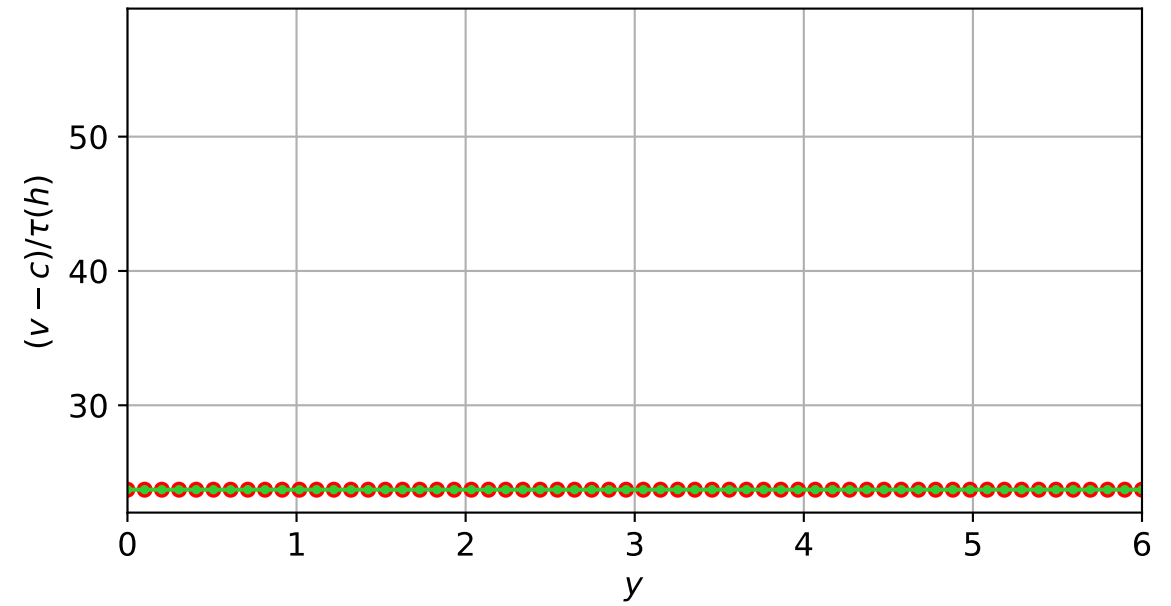
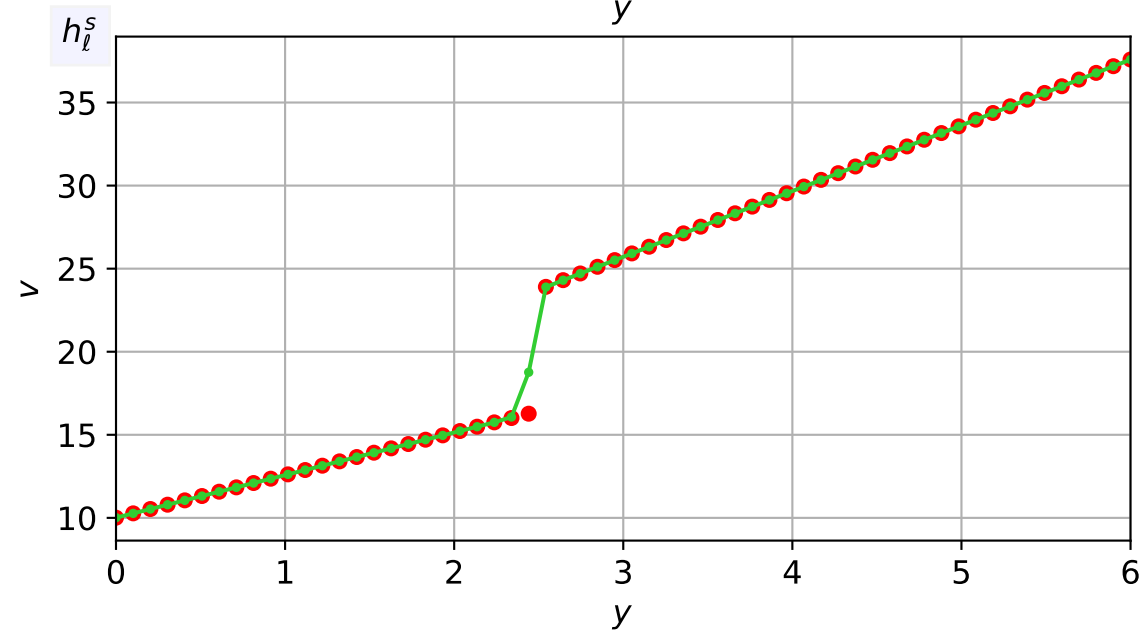
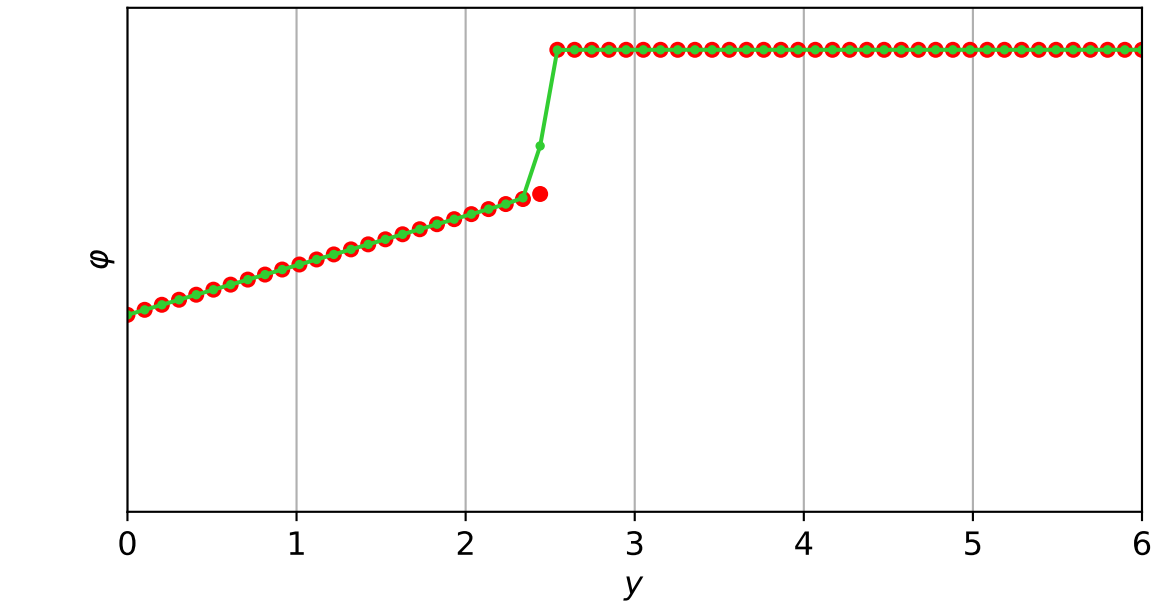
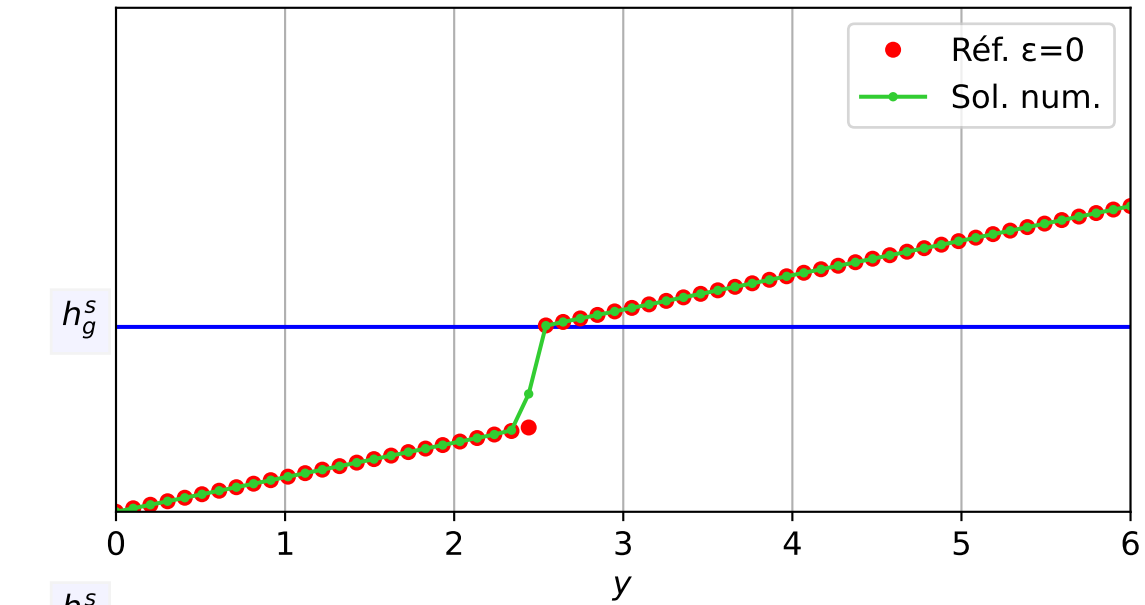
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.20239/12$  (8019 iter) — résidu eq. =  $9.03e-06$



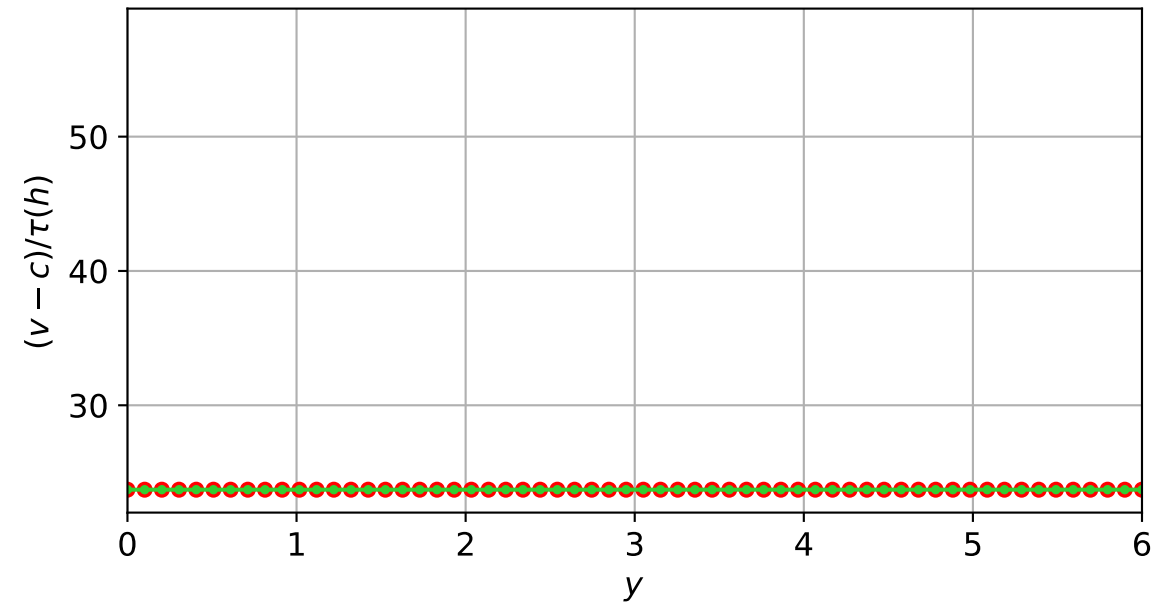
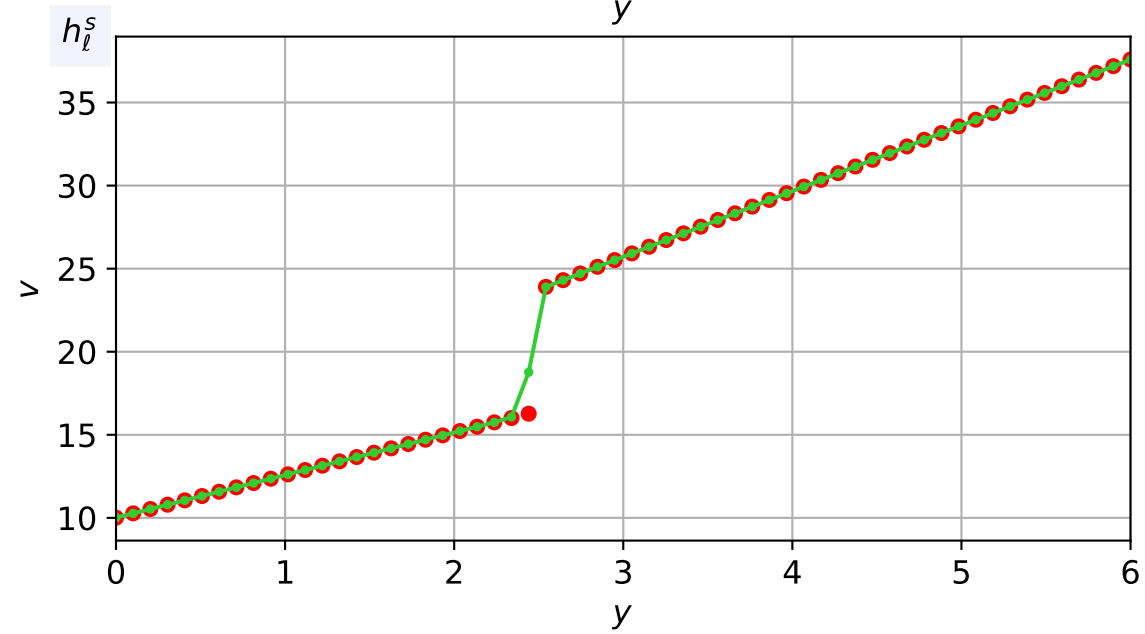
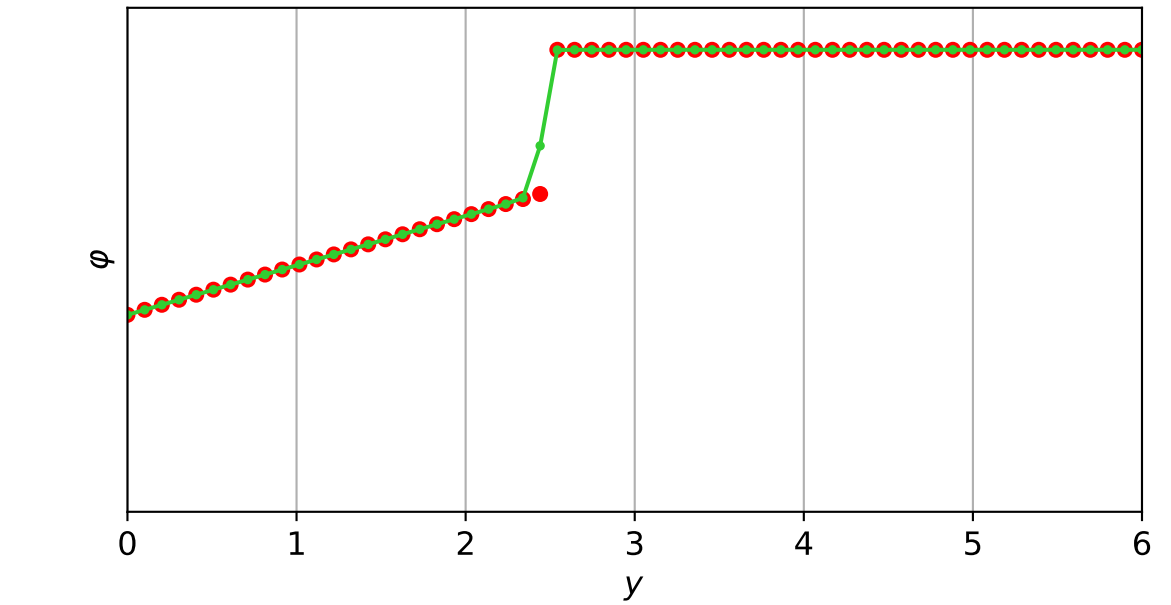
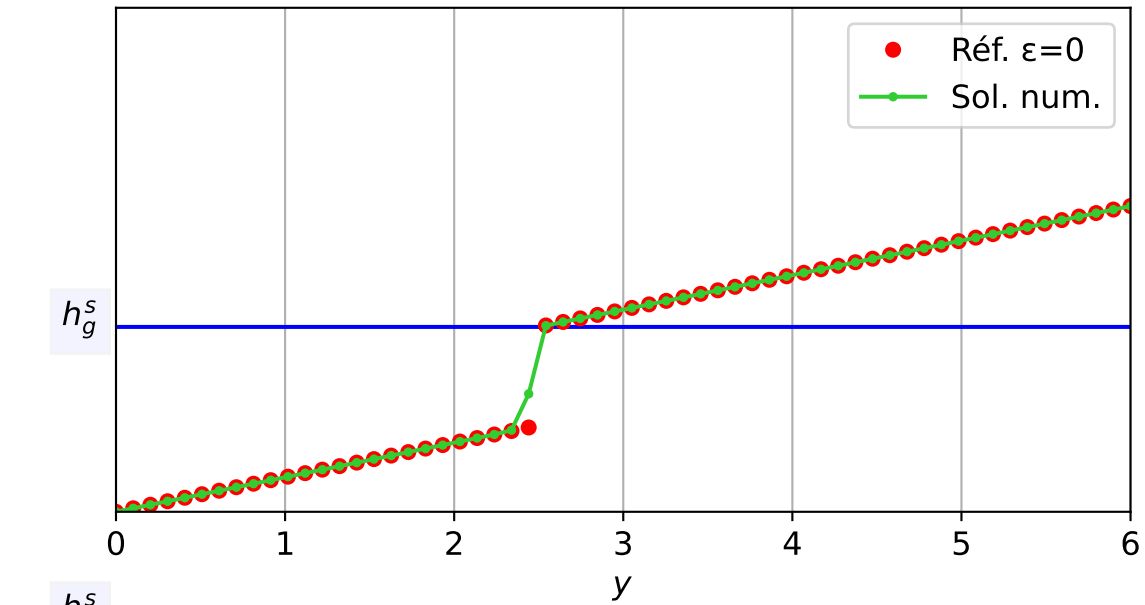
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.25109/12$  (8039 iter) — résidu eq. =  $8.13e-06$



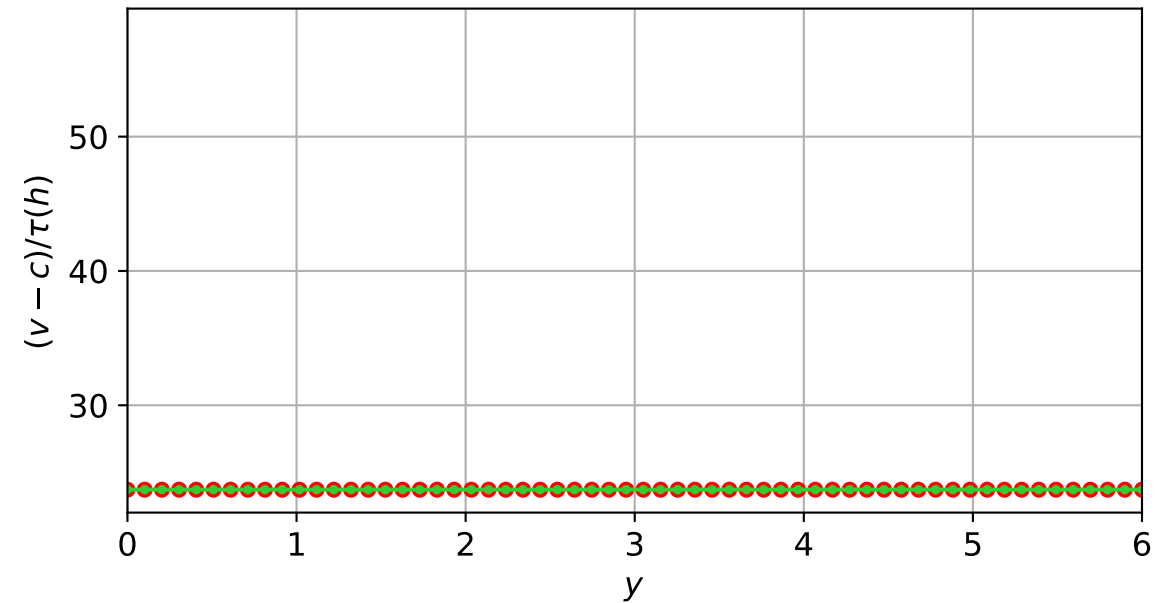
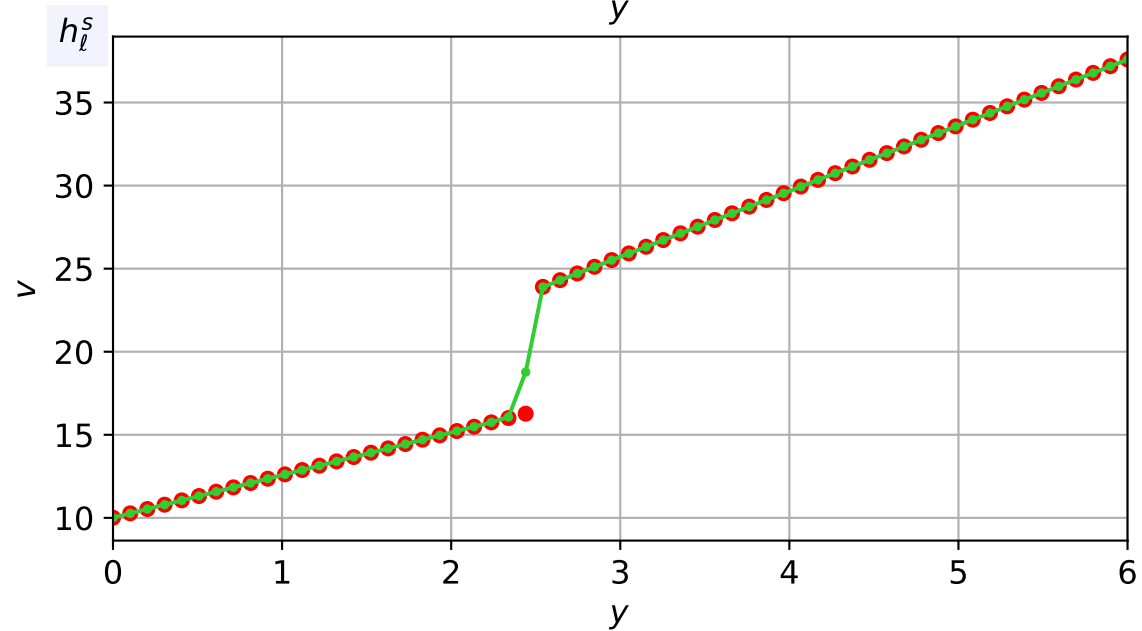
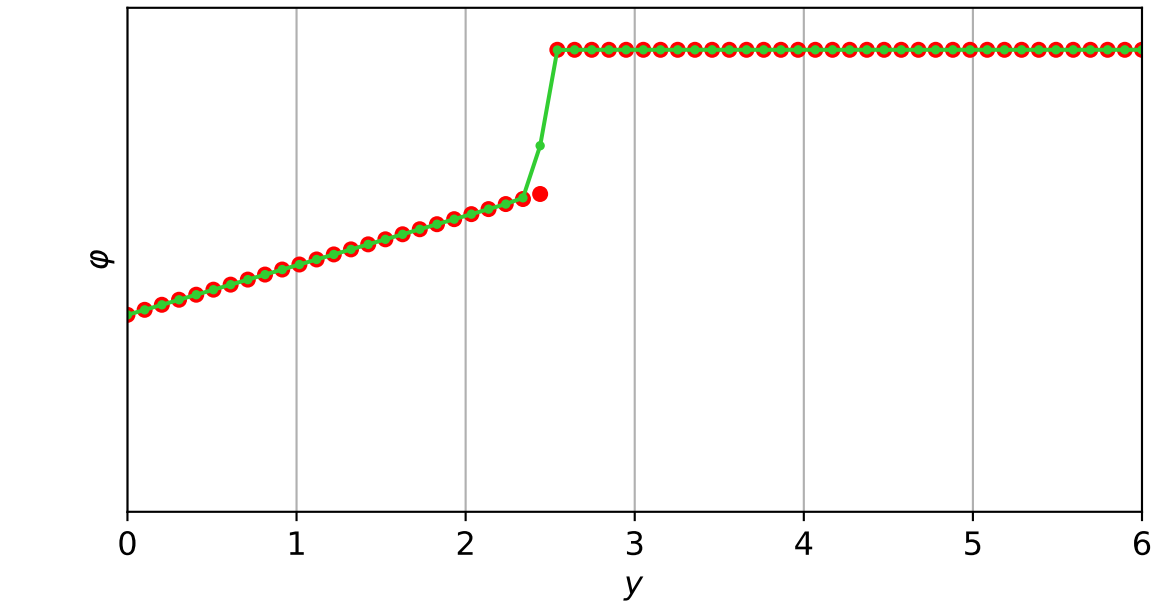
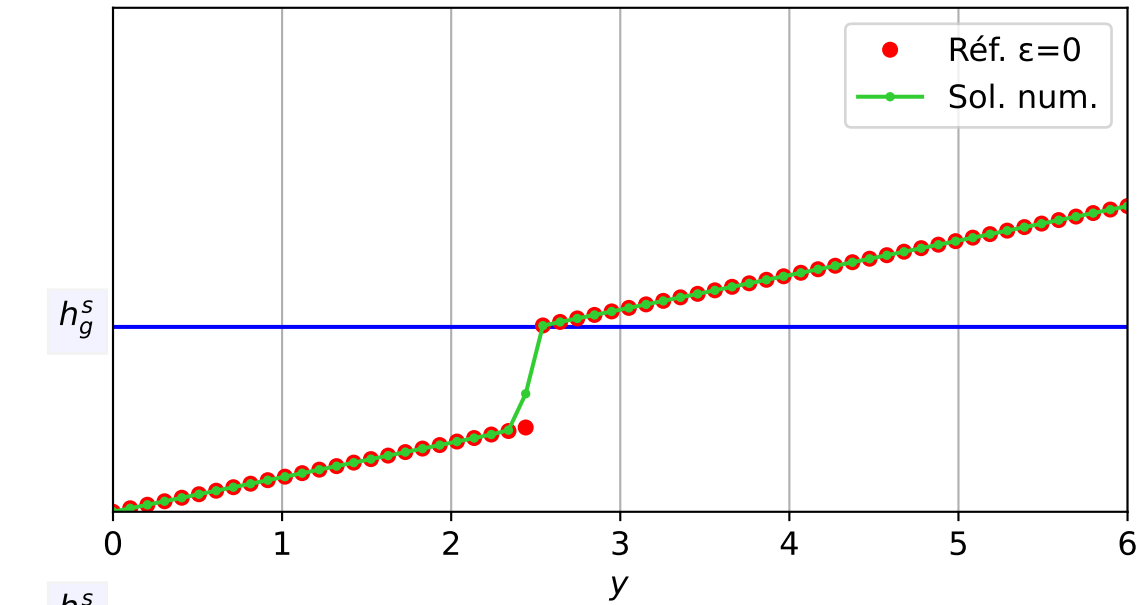
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.30223/12$  (8060 iter) — résidu eq. =  $7.27e-06$



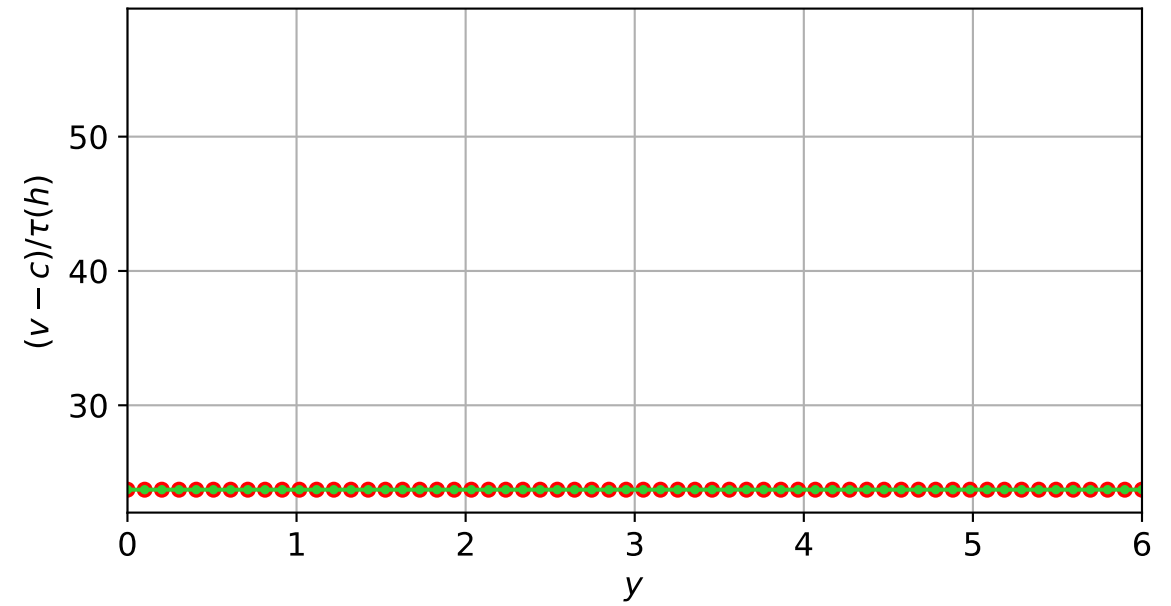
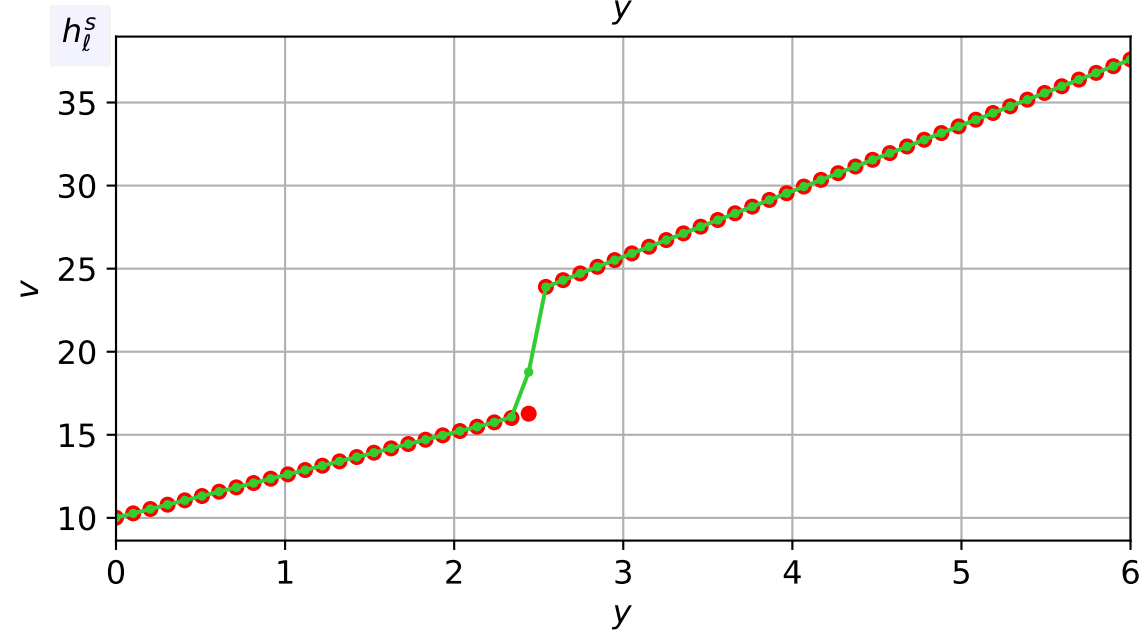
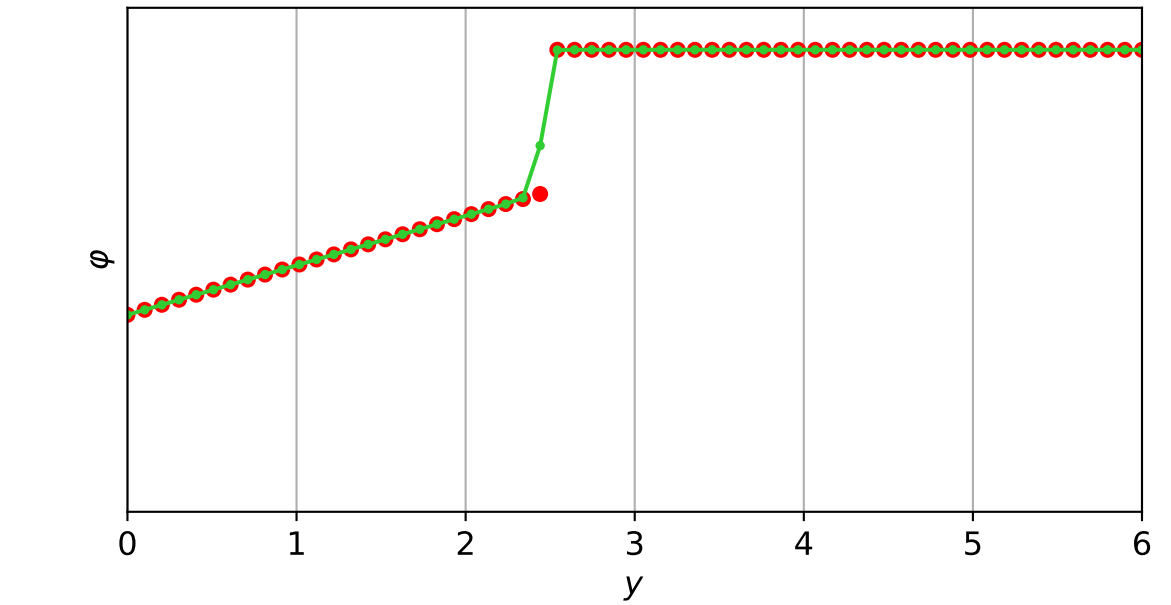
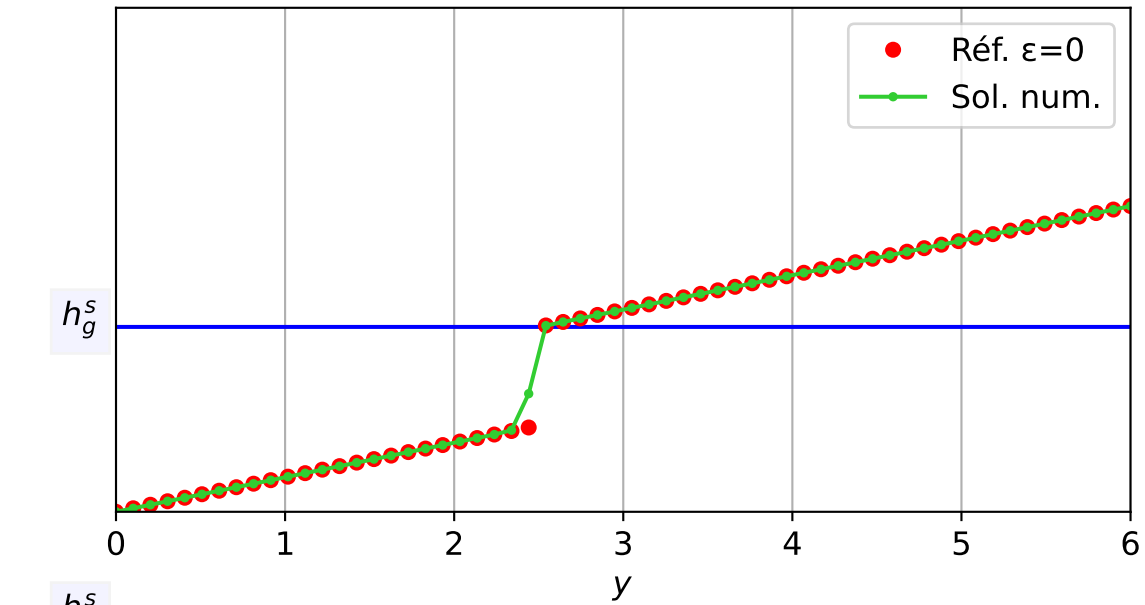
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.35093/12$  (8080 iter) — résidu eq. =  $6.52e-06$



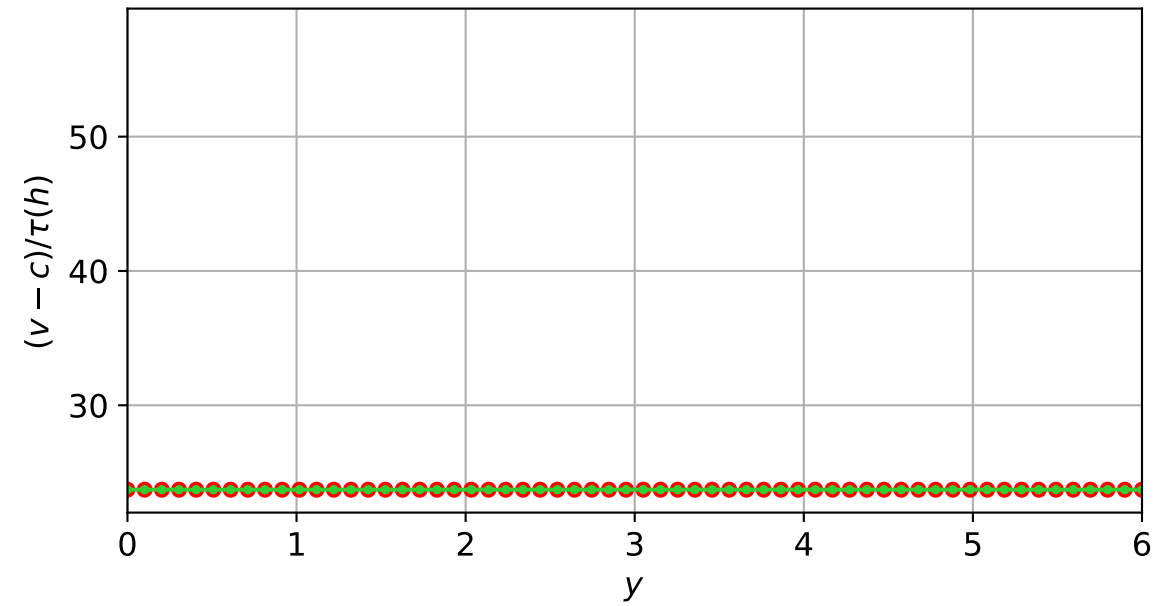
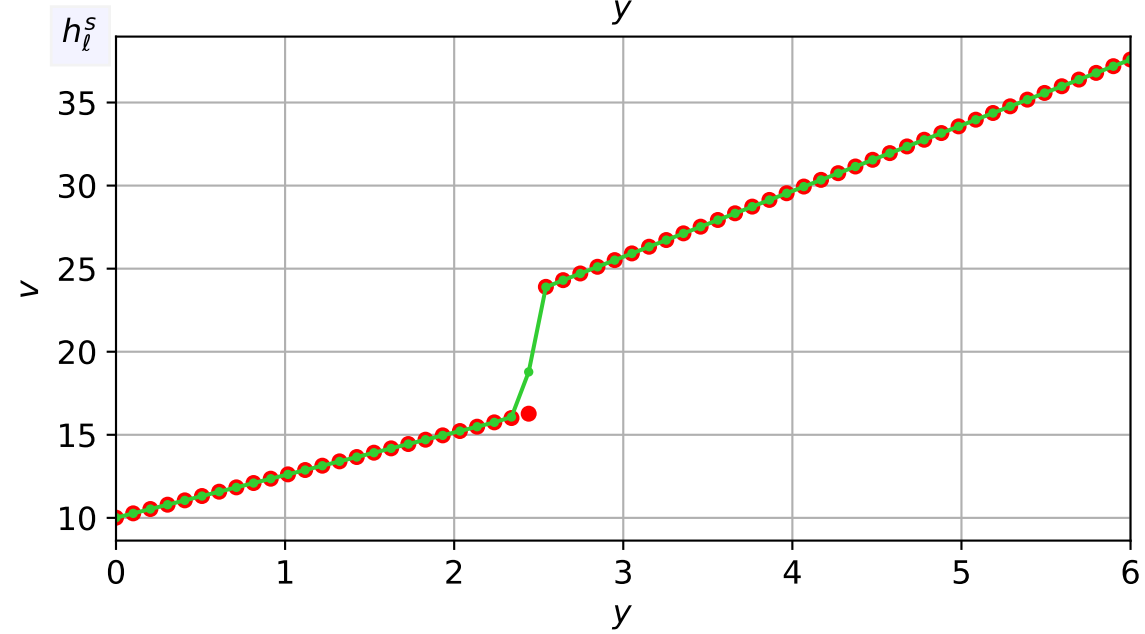
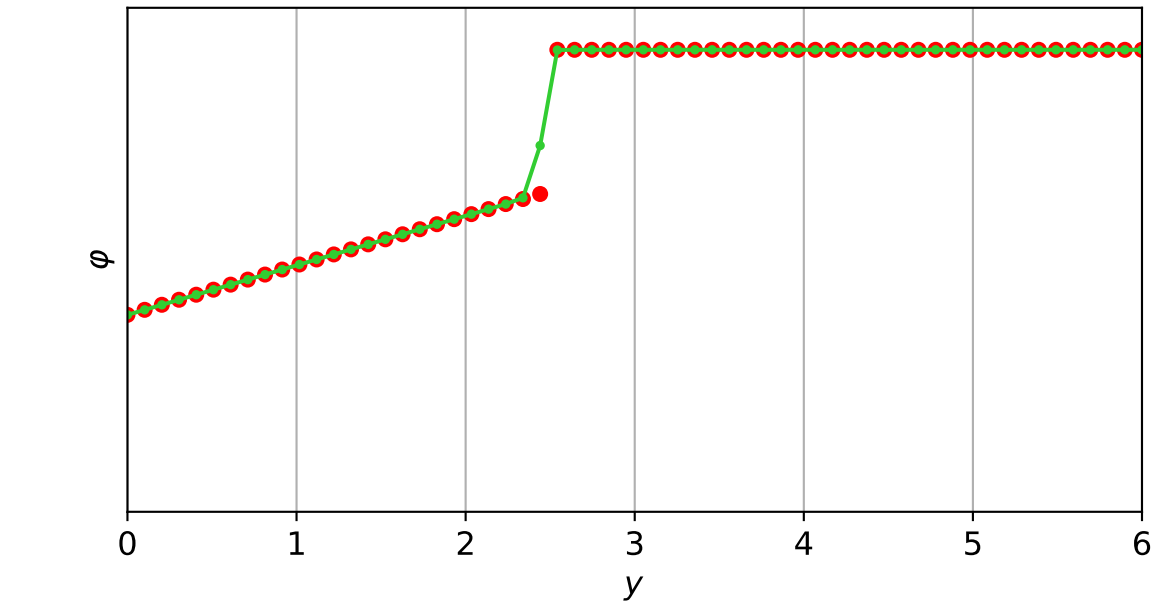
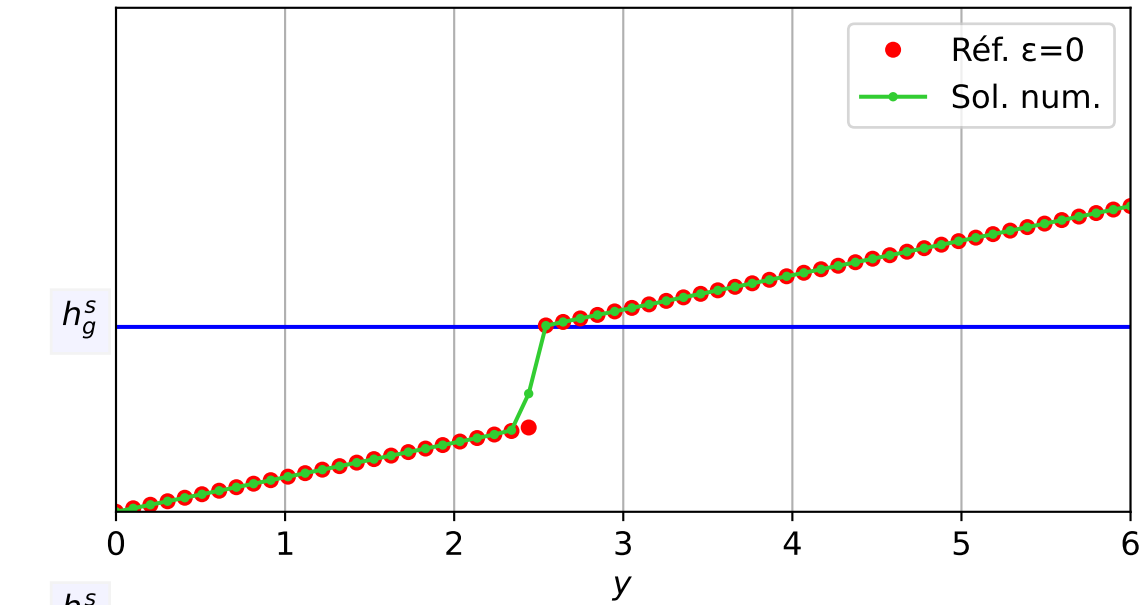
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.40207/12$  (8101 iter) — résidu eq. =  $5.82e-06$



TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.45077/12$  (8121 iter) — résidu eq. =  $5.21e-06$

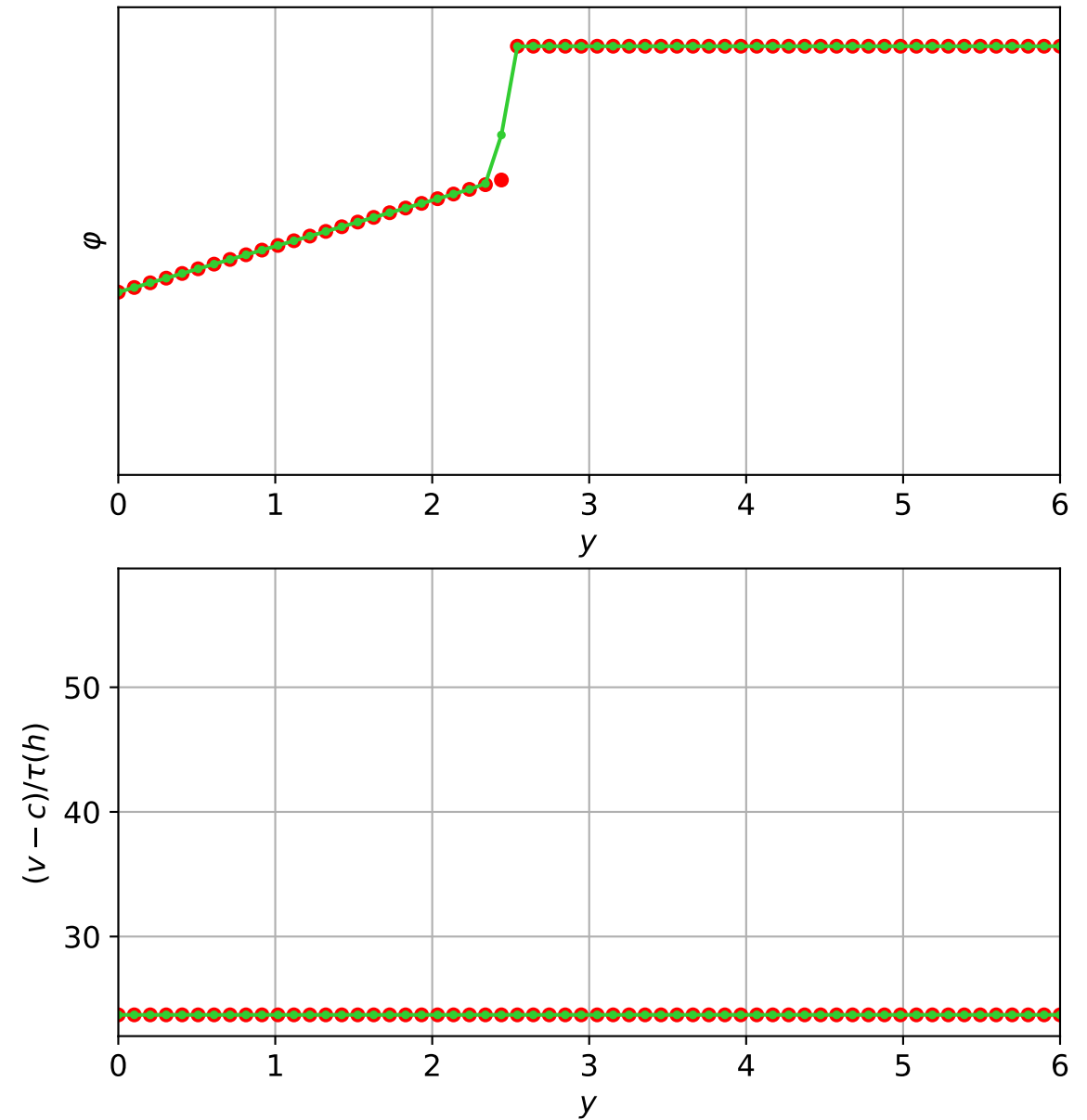
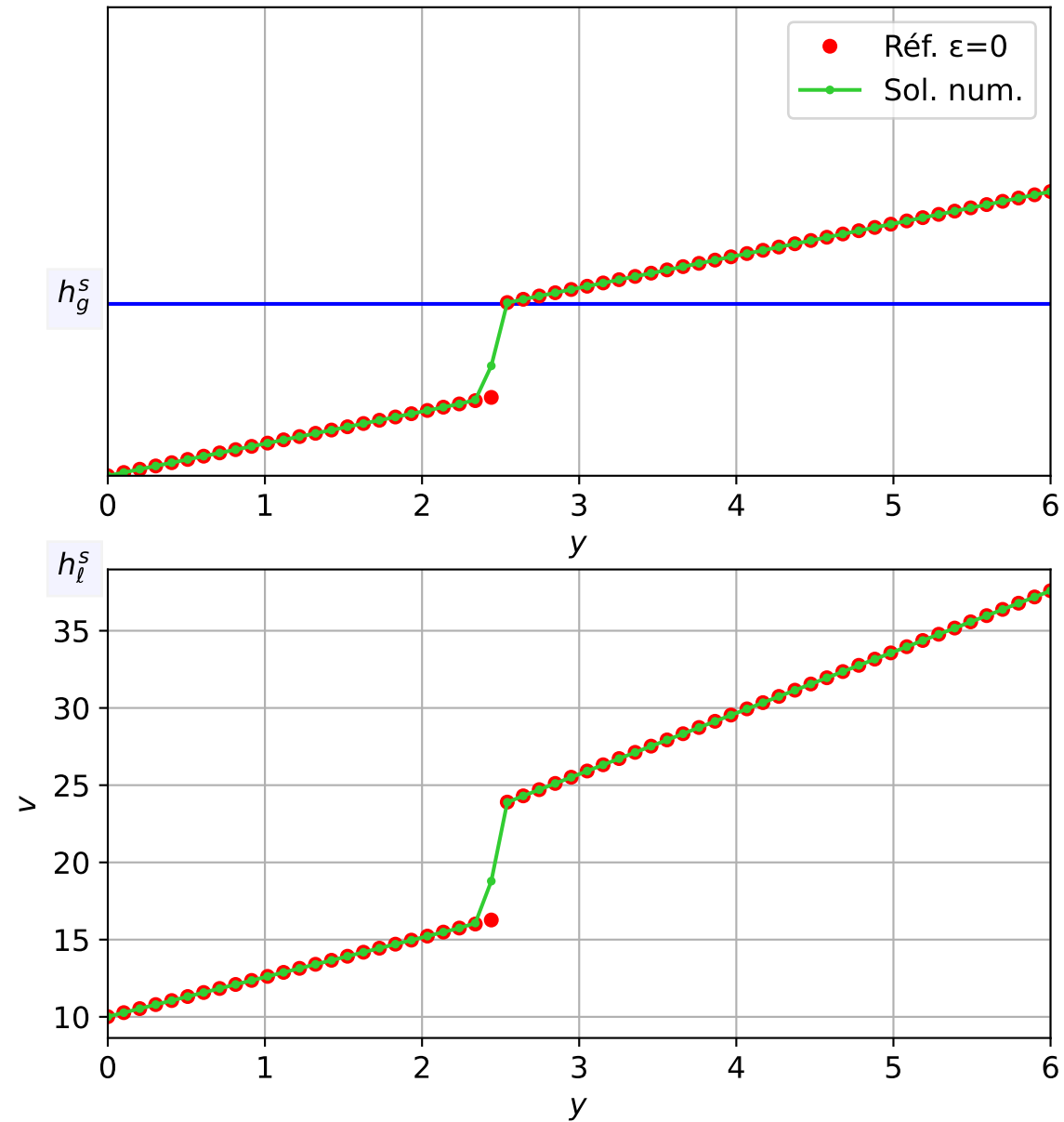


TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.50191/12$  (8142 iter) — résidu eq. =  $4.63e-06$

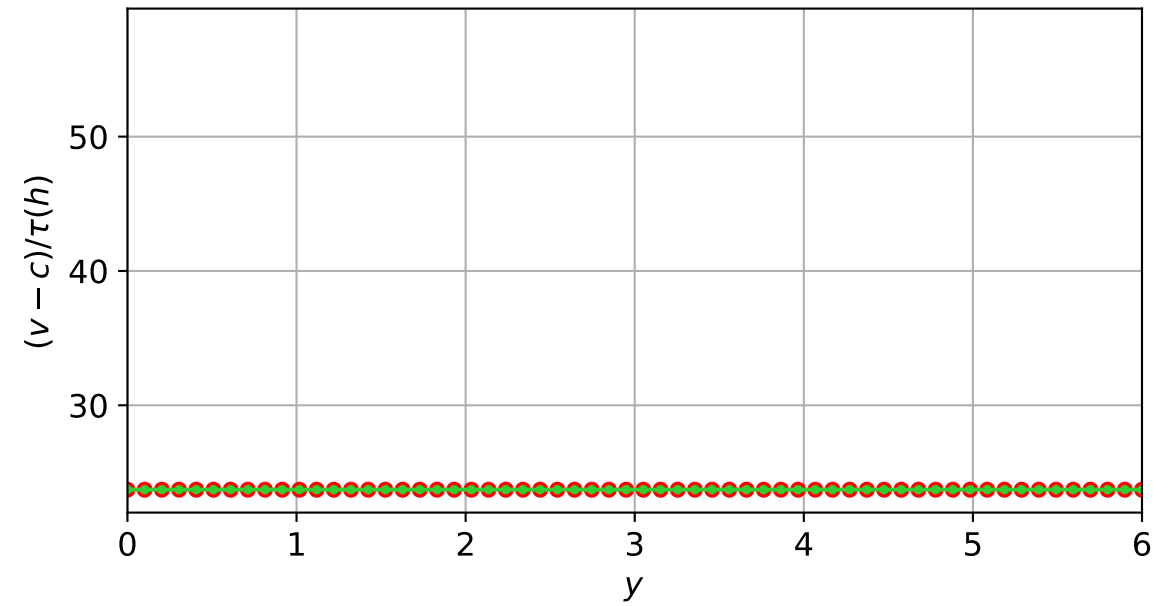
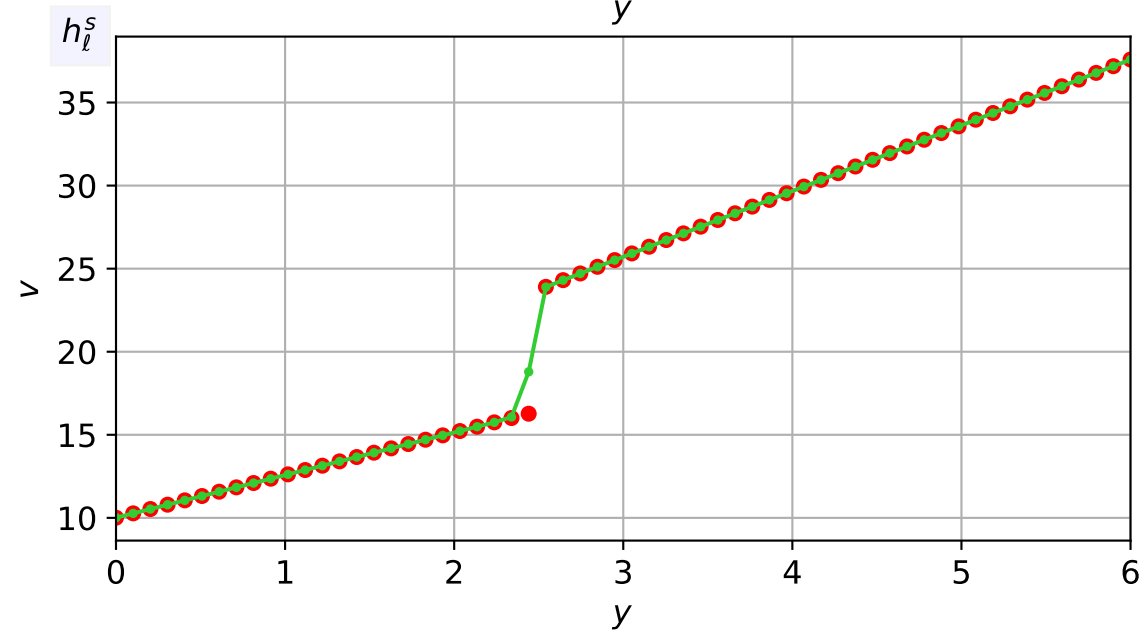
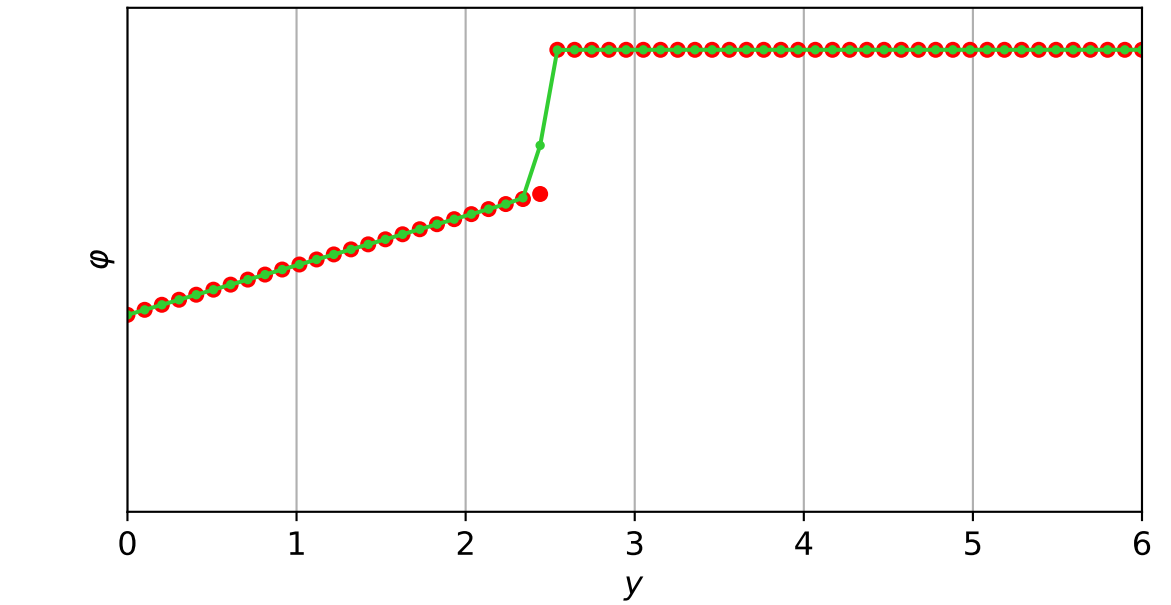
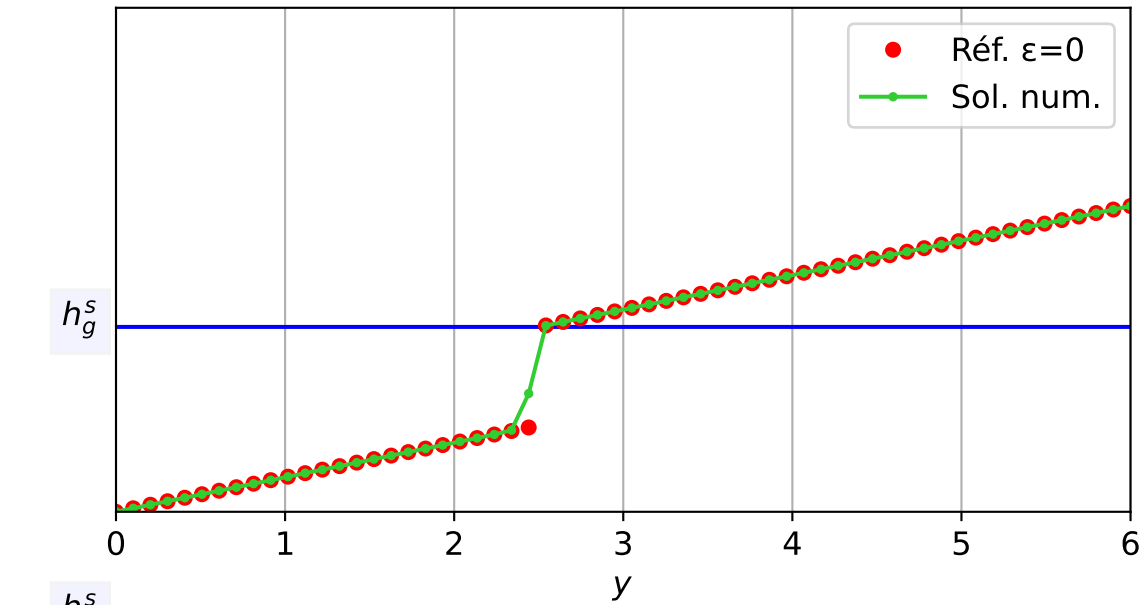




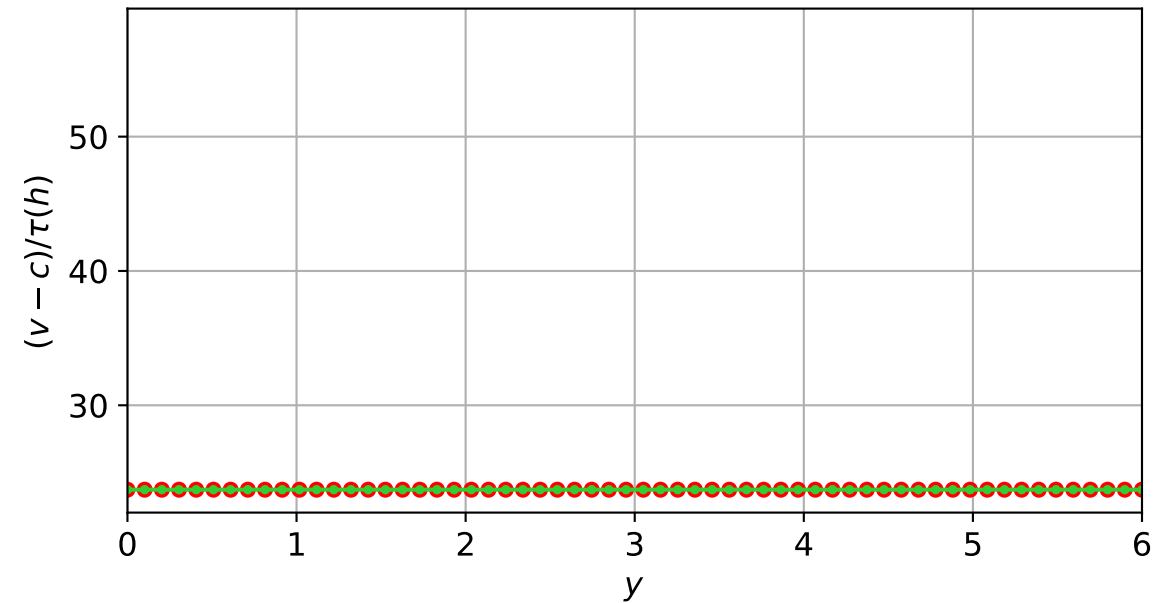
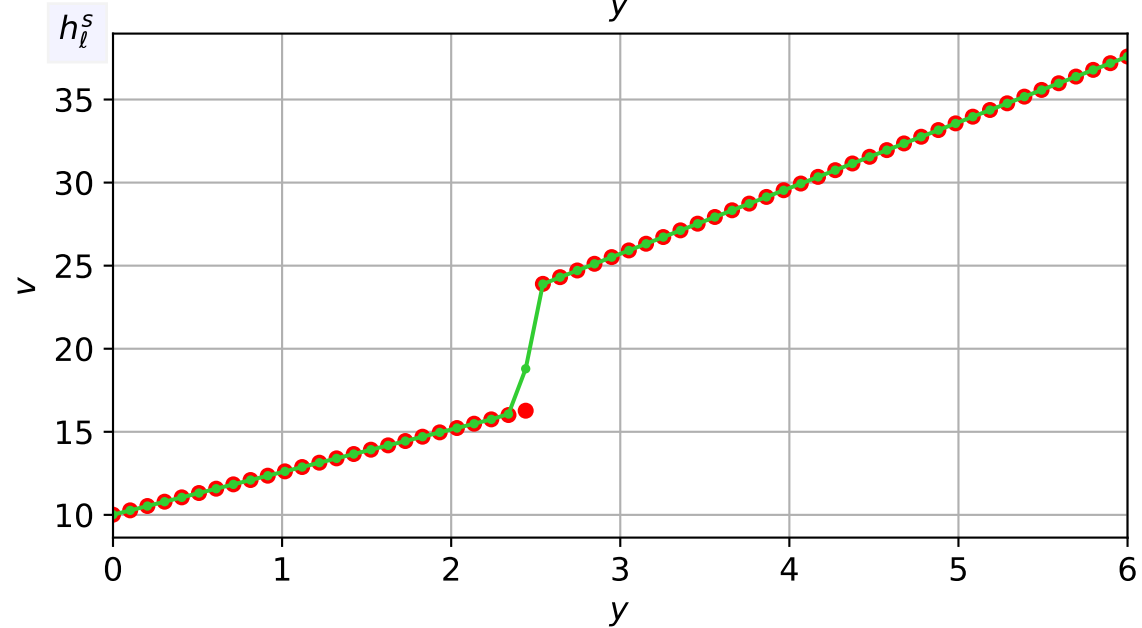
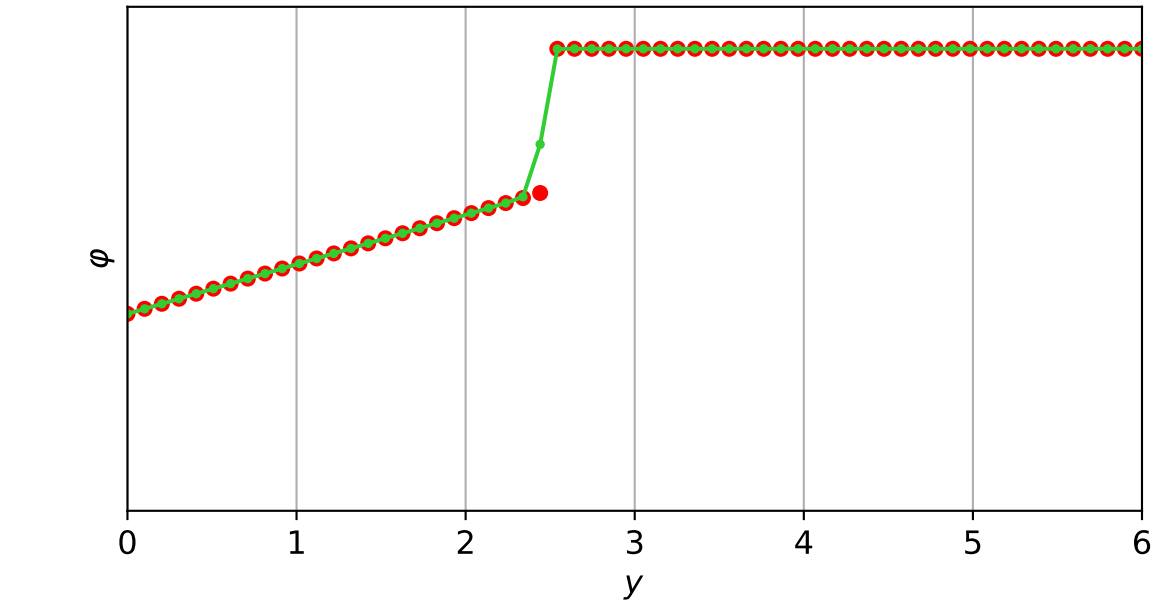
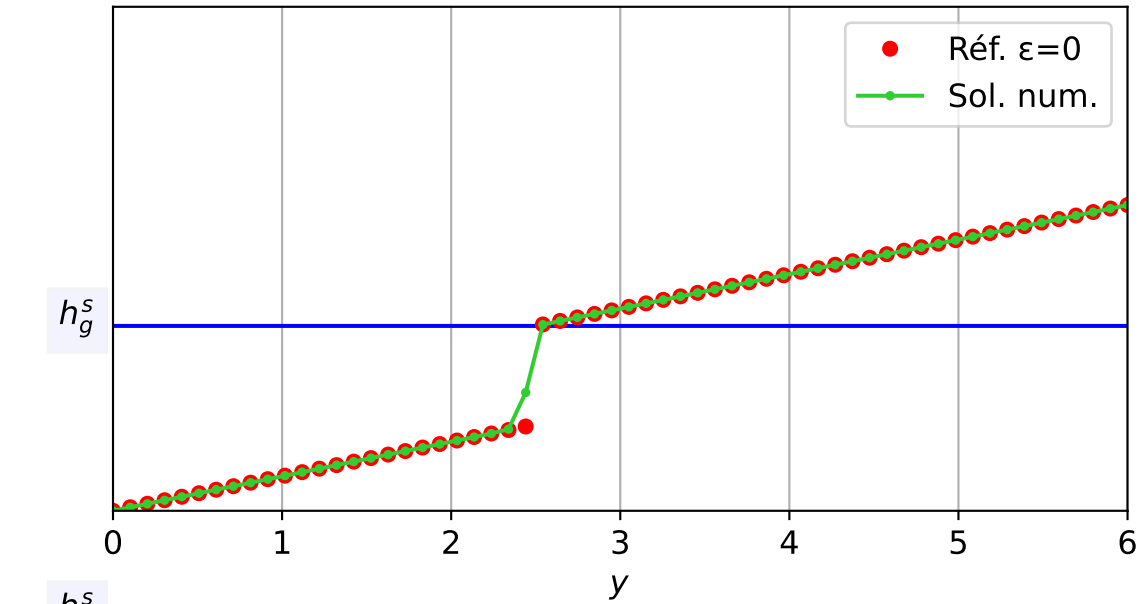
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.55061/12$  (8162 iter) — résidu eq. =  $4.14e-06$



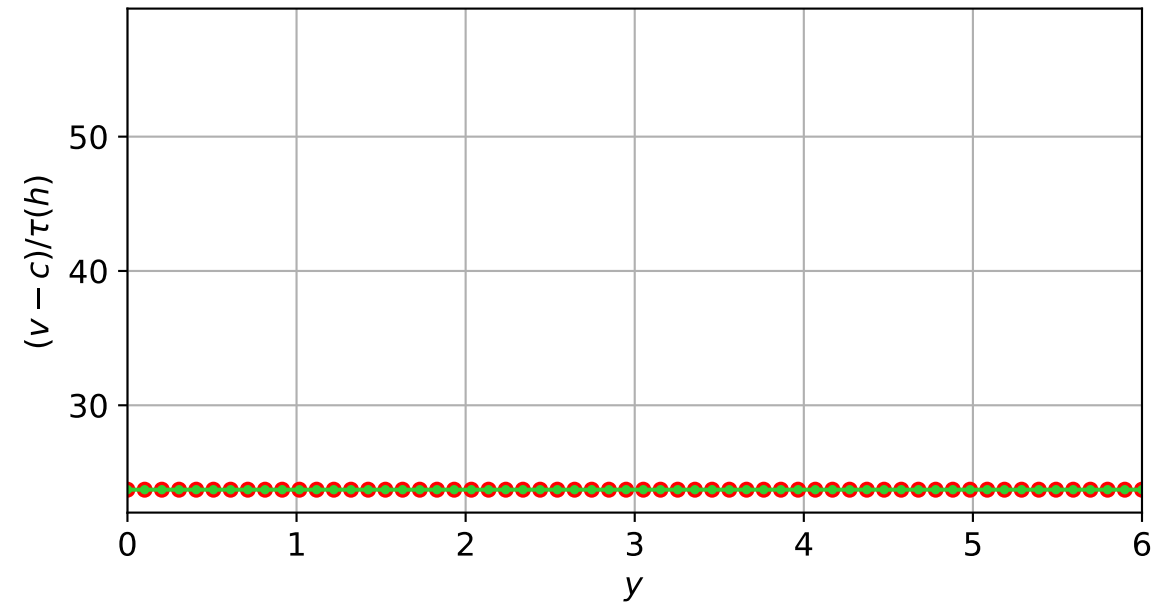
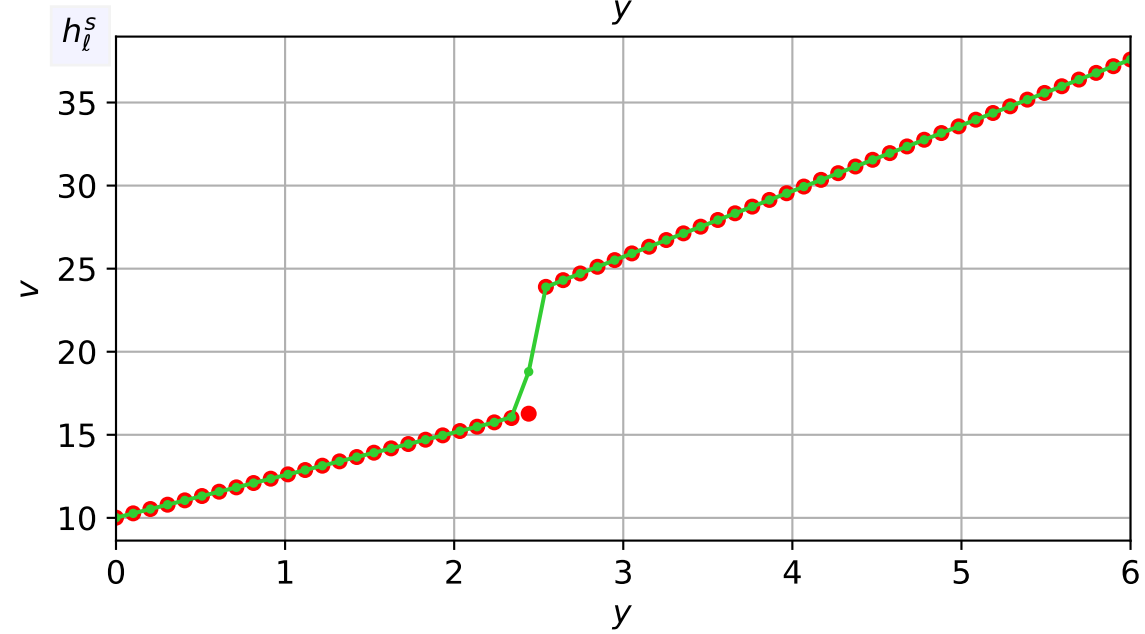
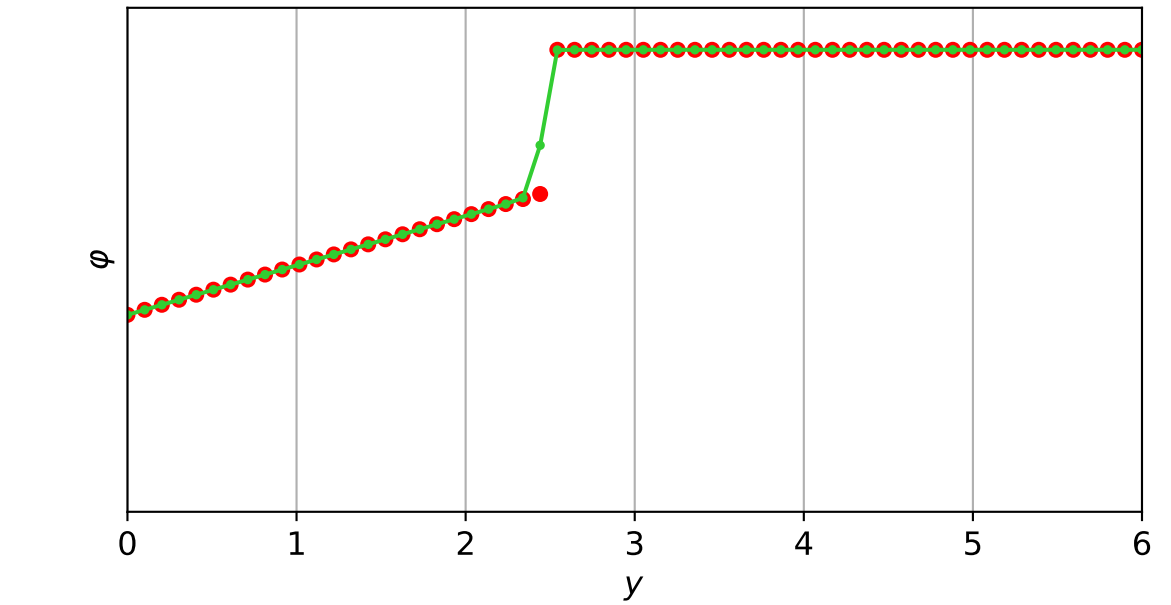
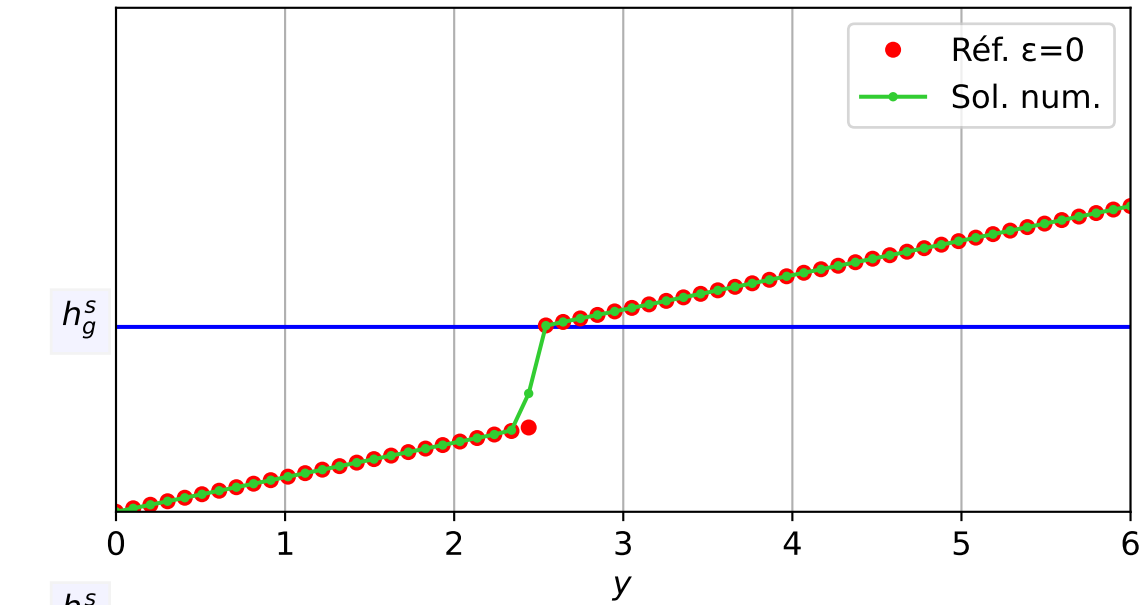
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.60175/12$  (8183 iter) — résidu eq. =  $3.68e-06$



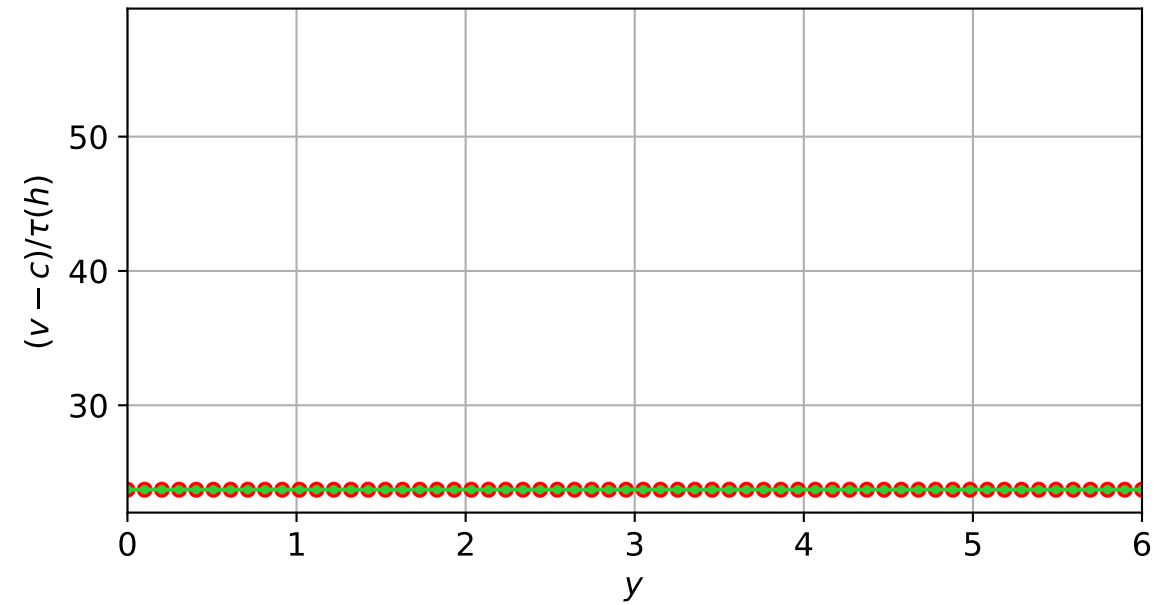
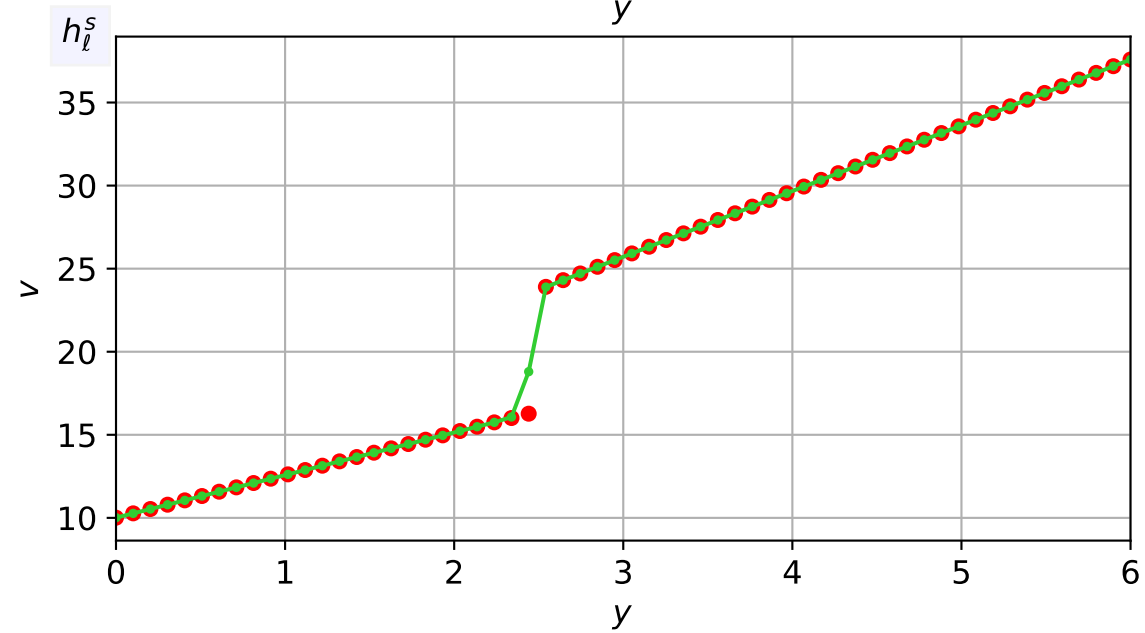
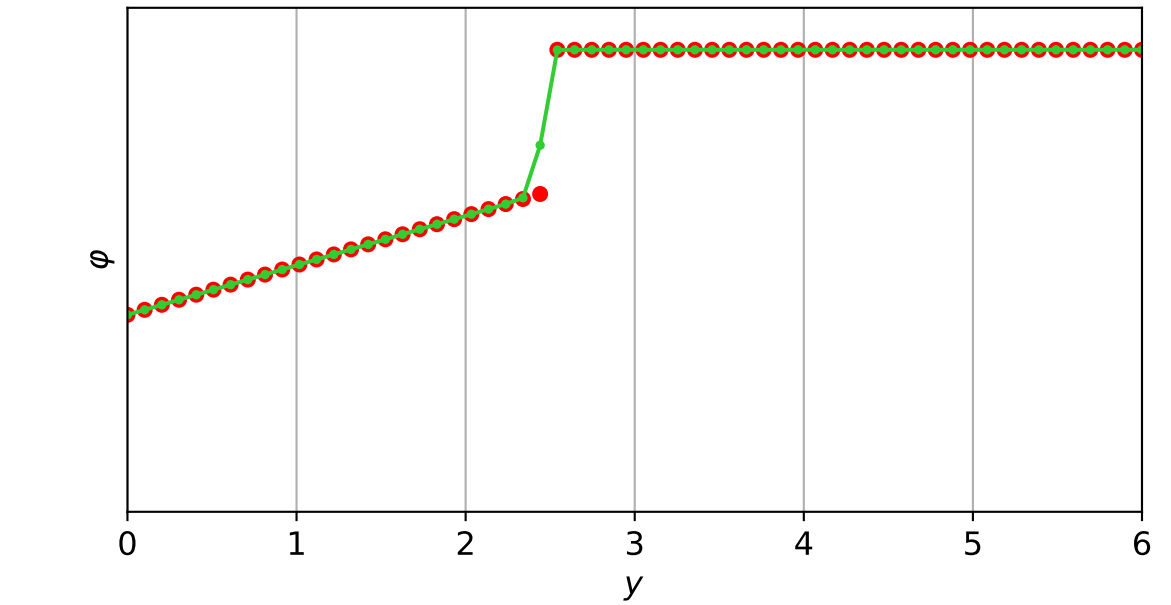
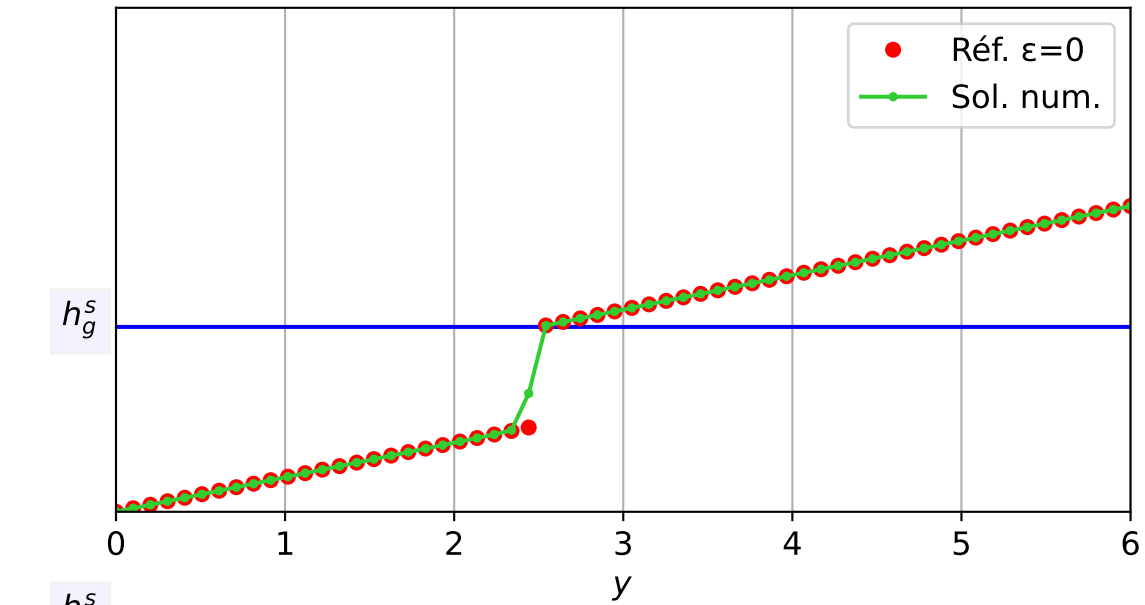
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.65045/12$  (8203 iter) — résidu eq. =  $3.28e-06$



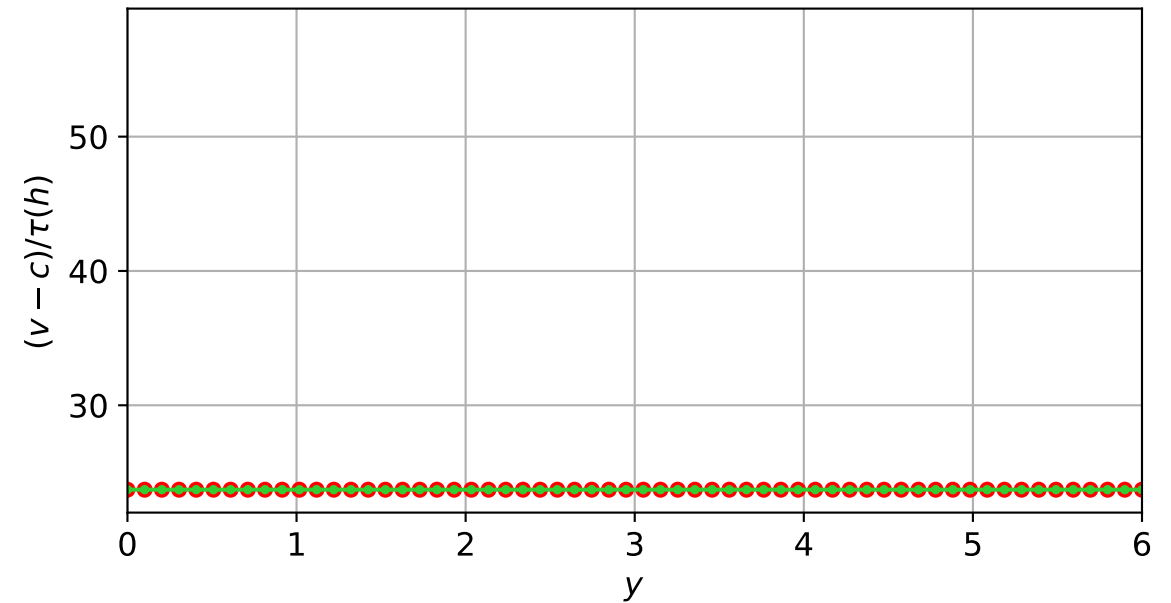
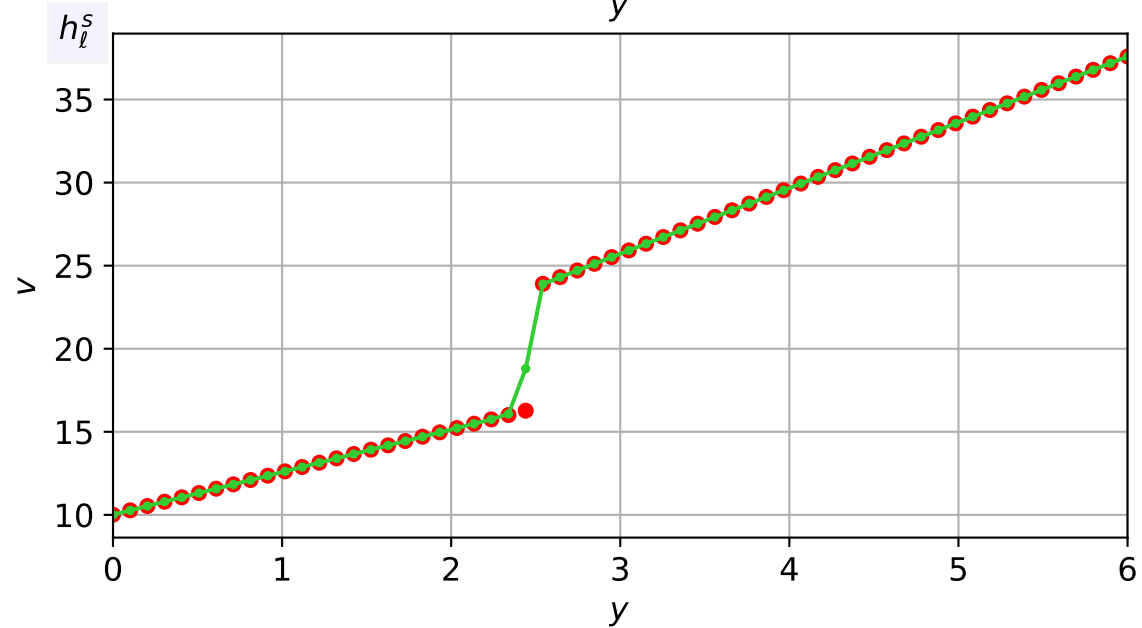
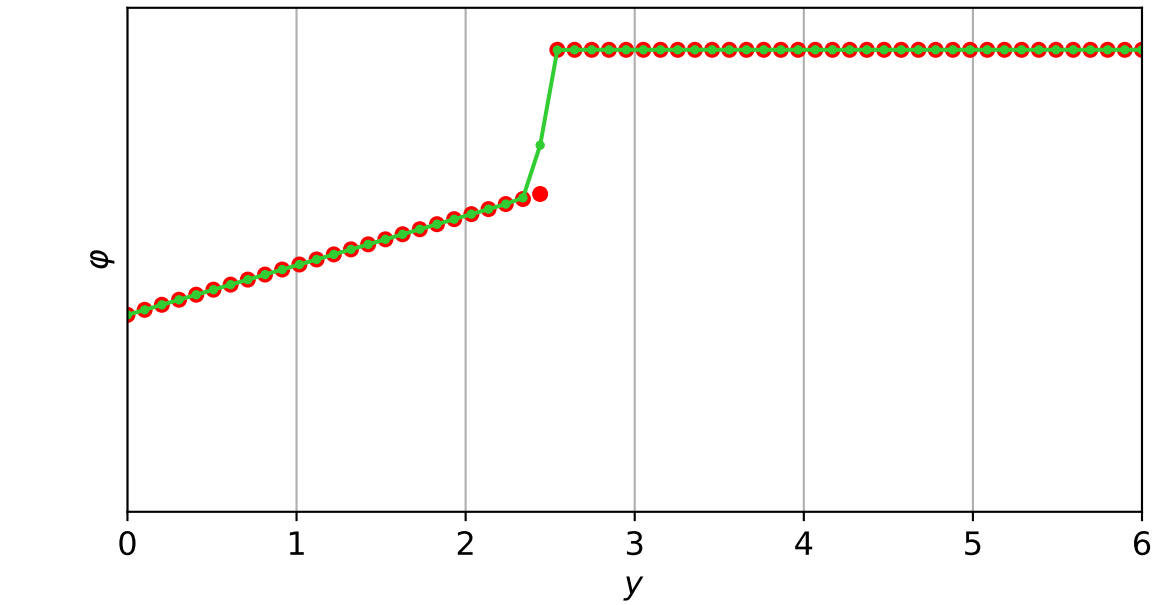
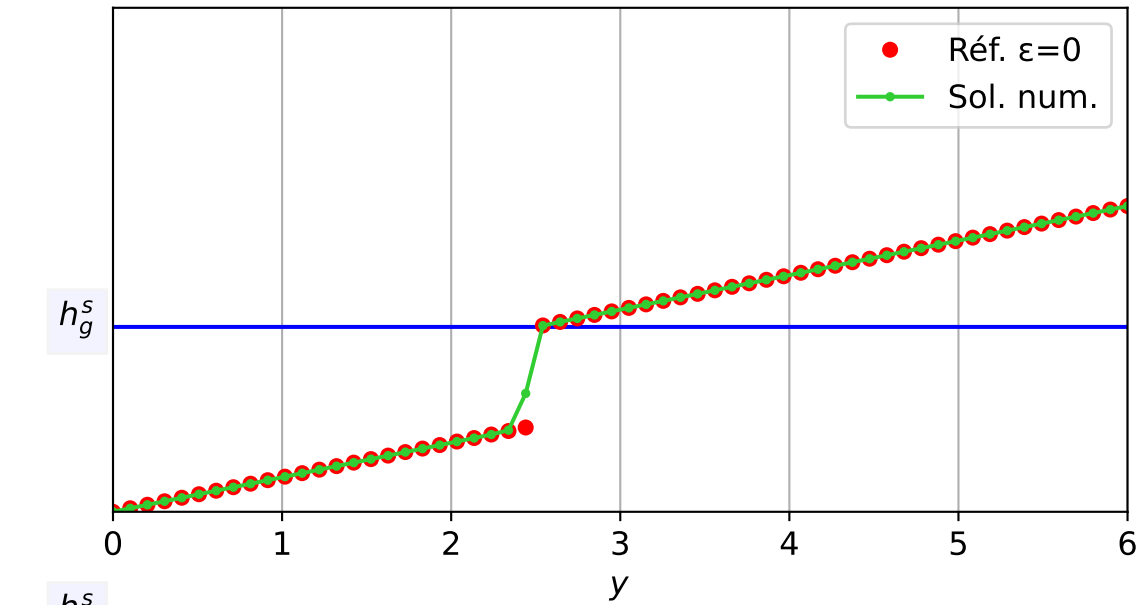
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.70159/12$  (8224 iter) — résidu eq. =  $2.91e-06$



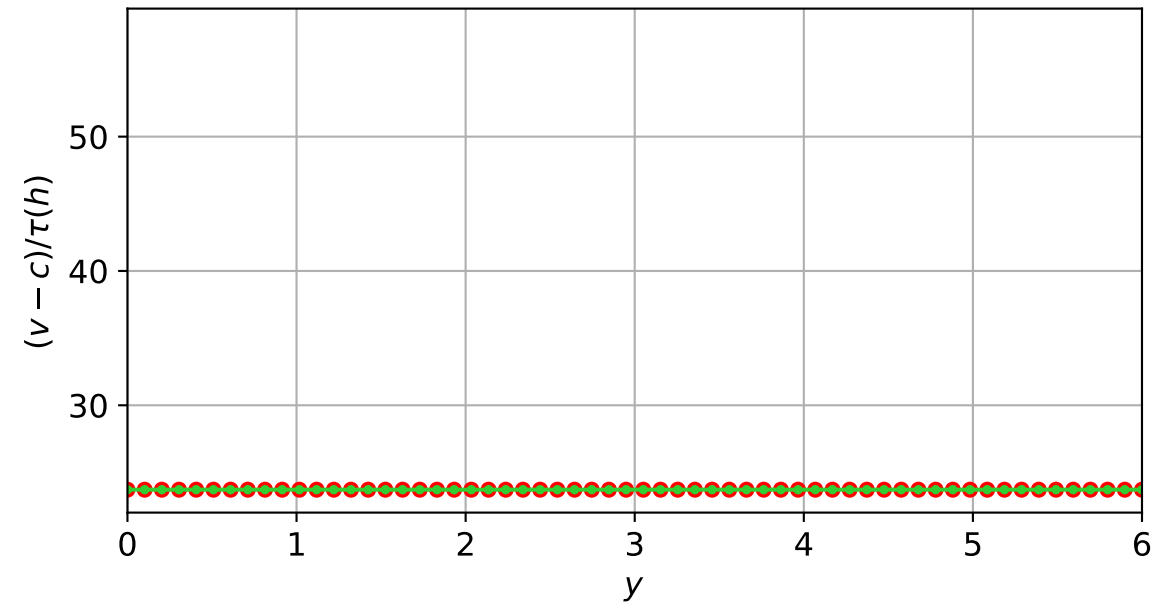
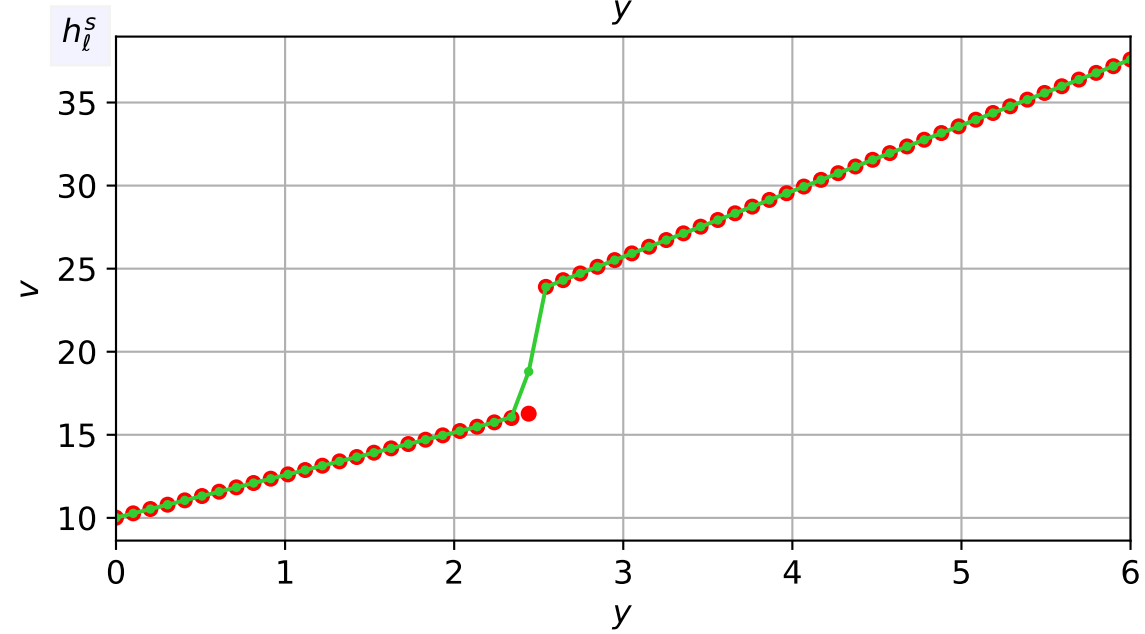
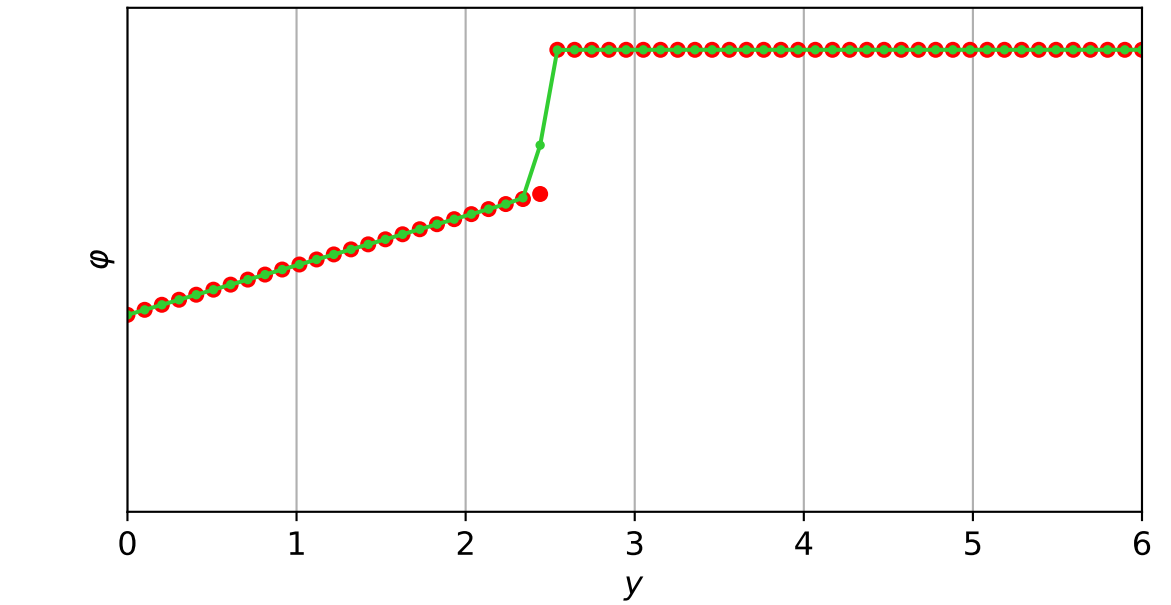
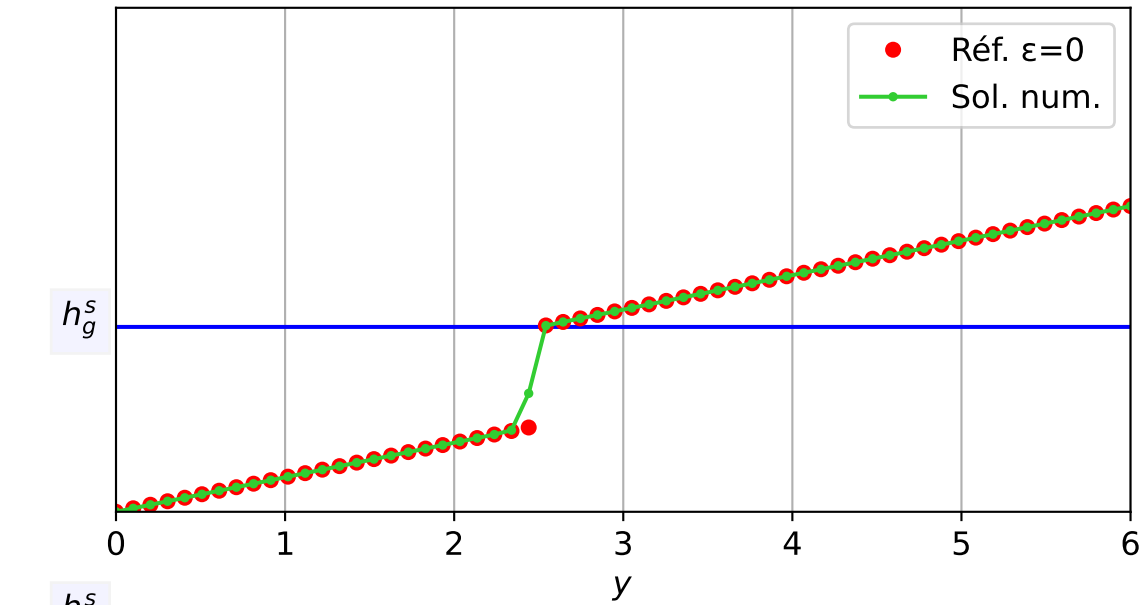
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.75029/12$  (8244 iter) — résidu eq. =  $2.60e-06$



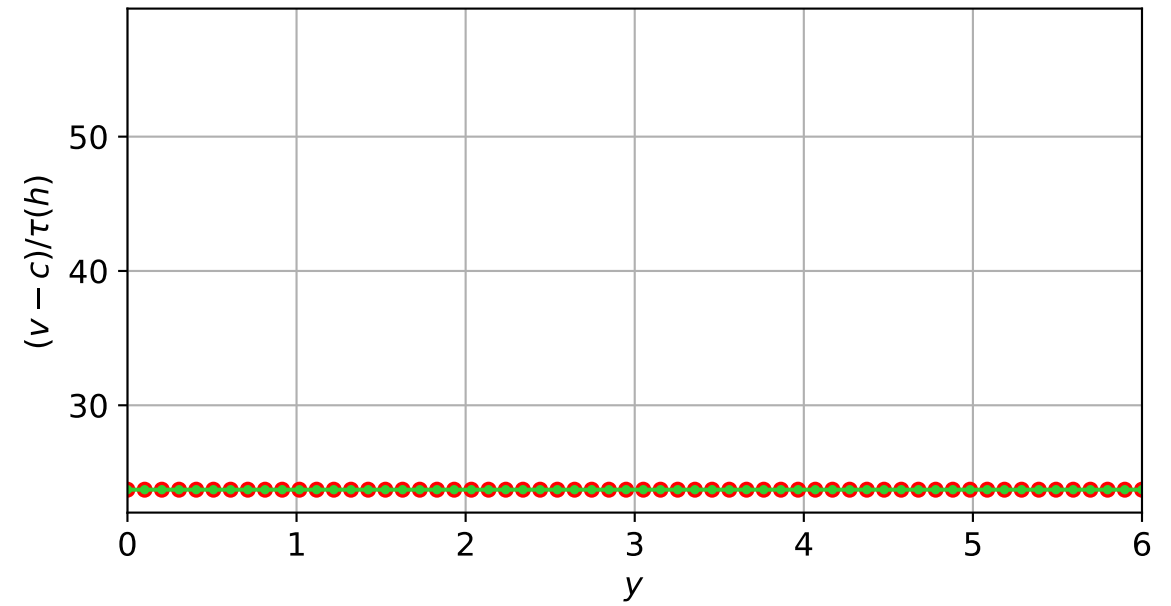
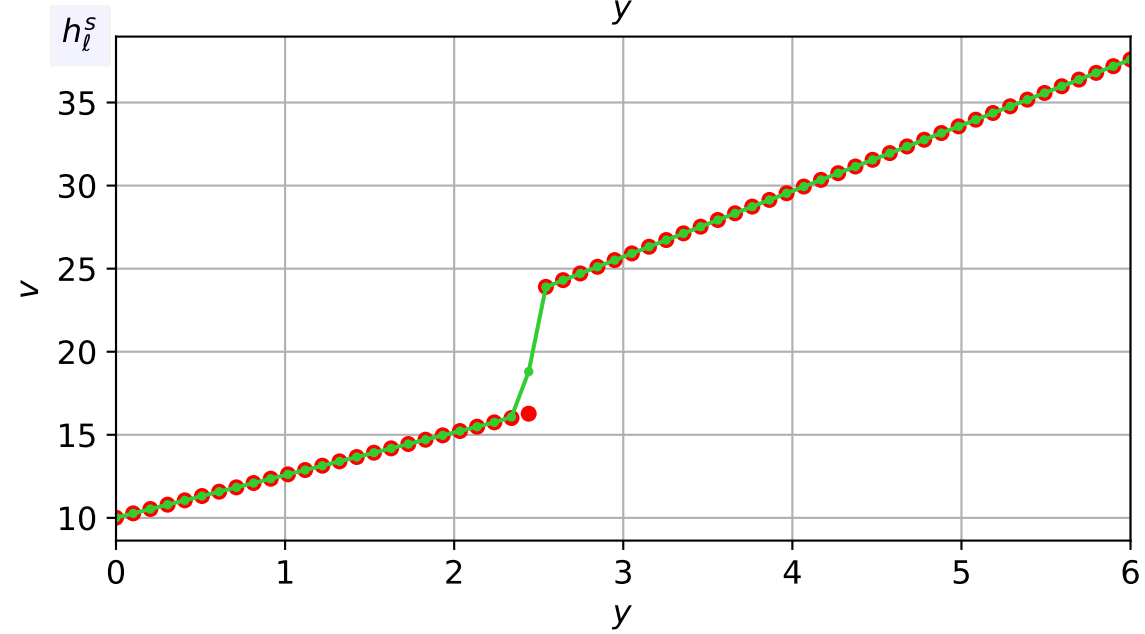
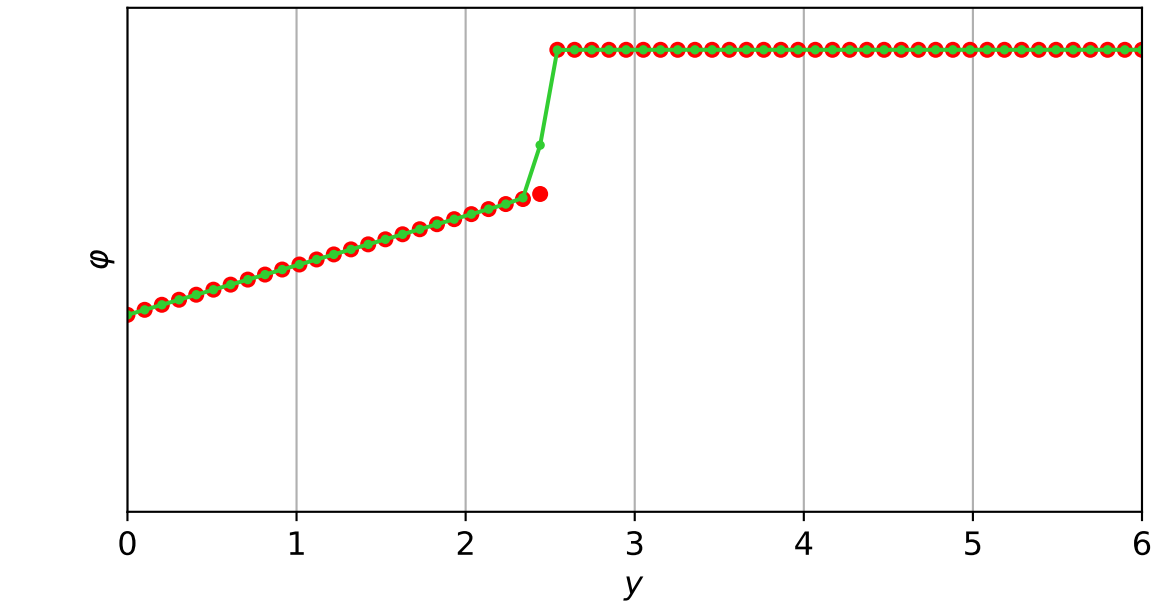
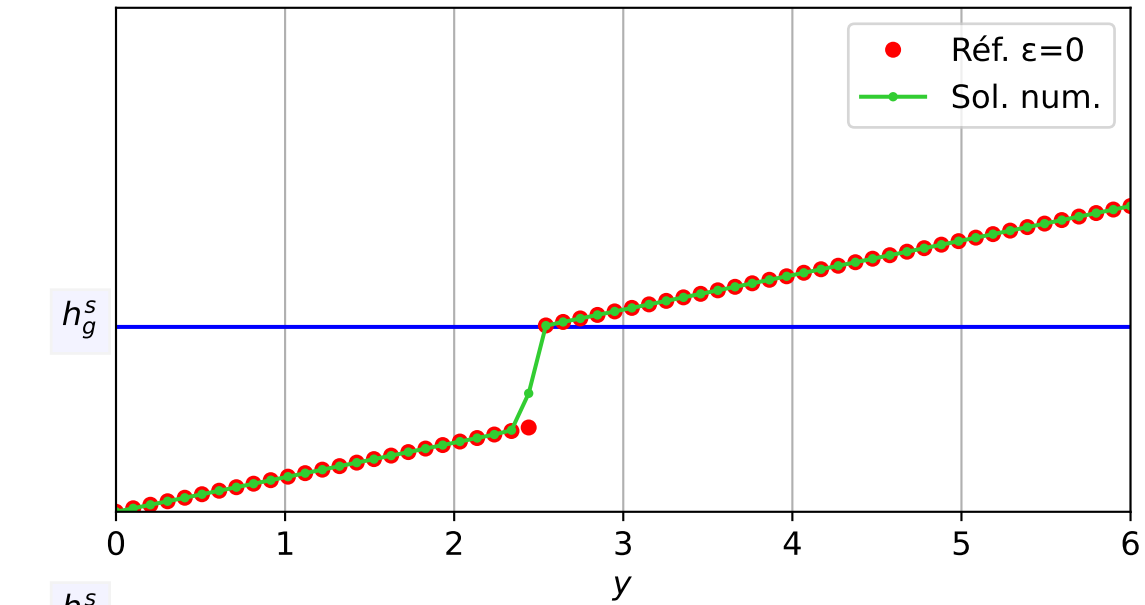
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.80143/12$  (8265 iter) — résidu eq. =  $2.30e-06$



TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.85013/12$  (8285 iter) — résidu eq. =  $2.05e-06$

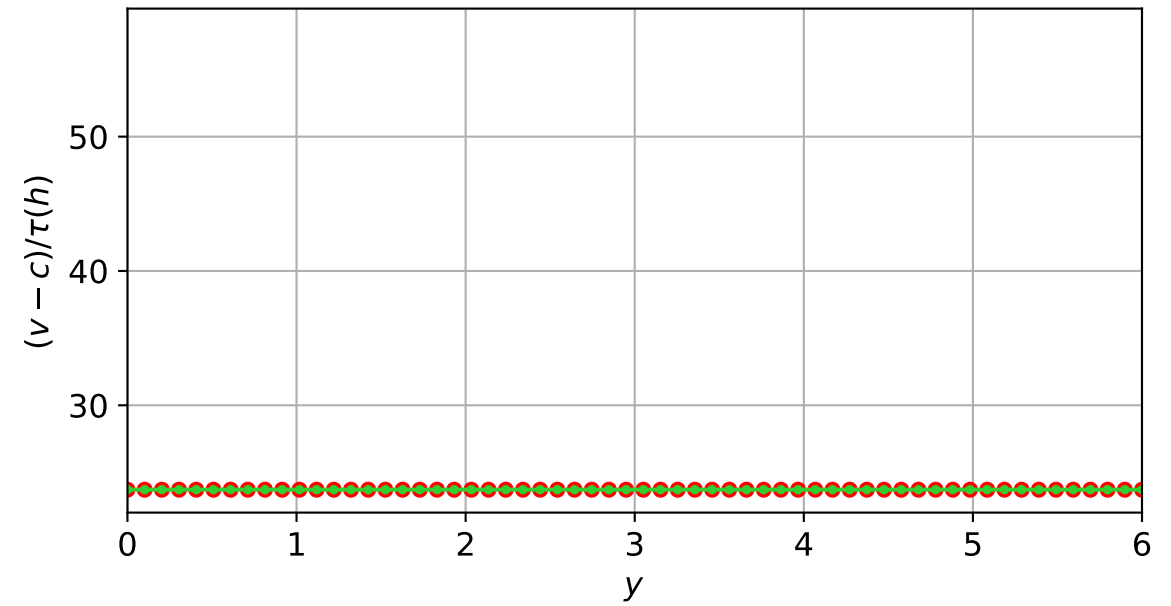
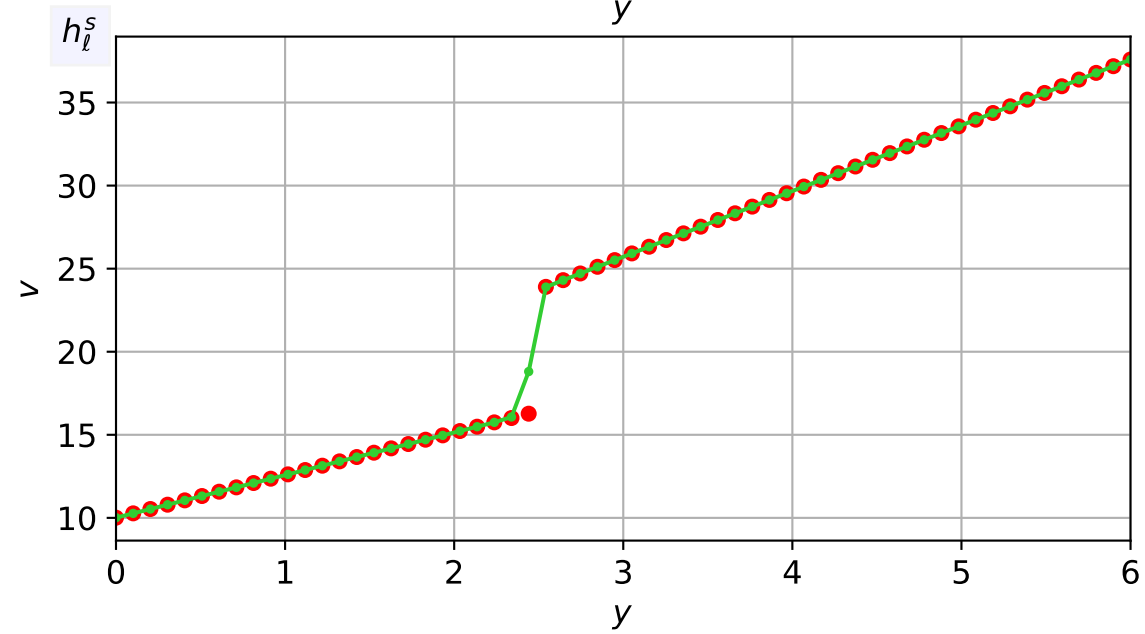
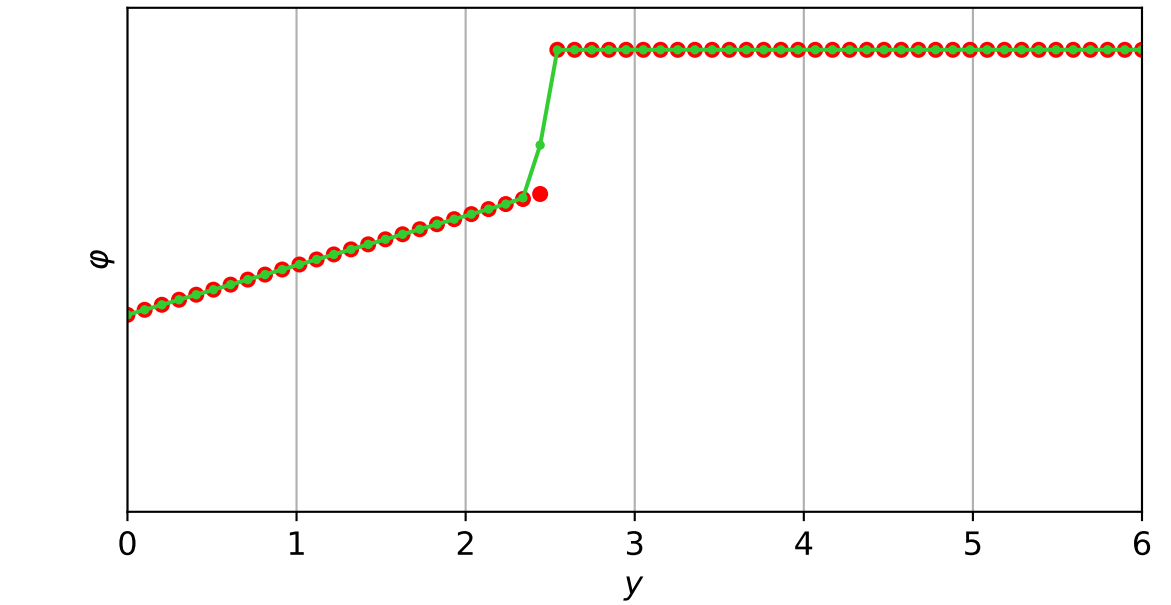
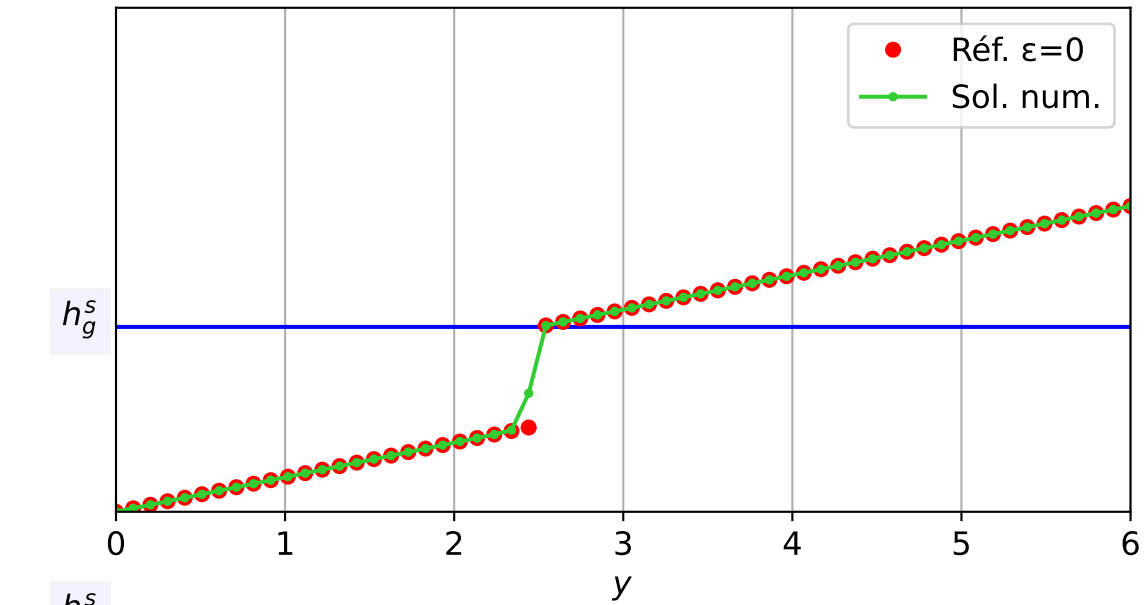


TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.90126/12$  (8306 iter) — résidu eq. =  $1.82e-06$

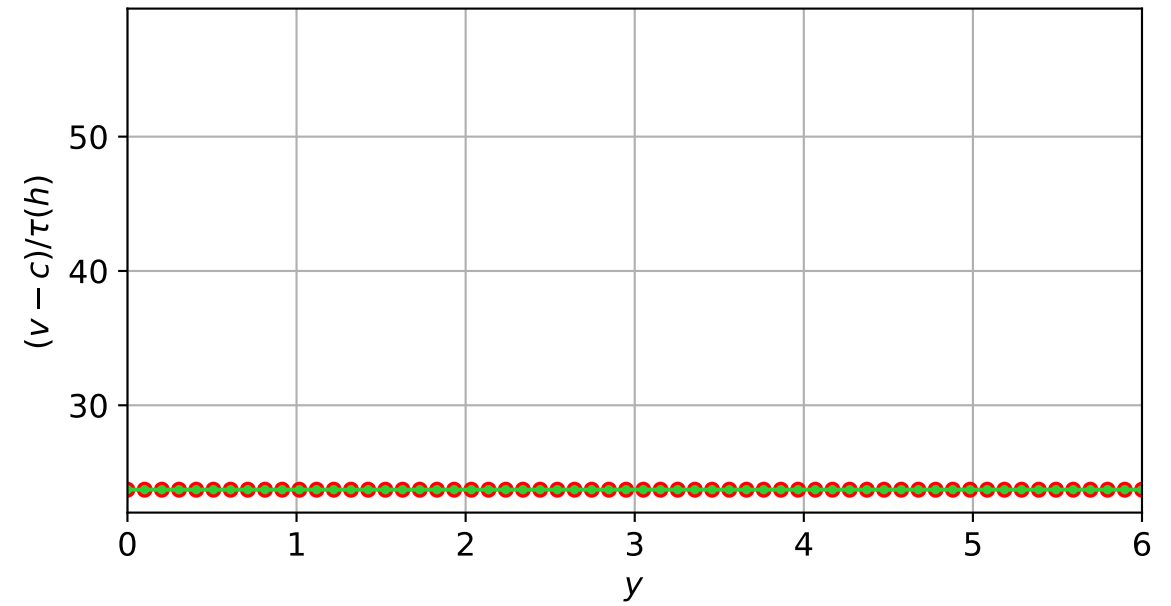
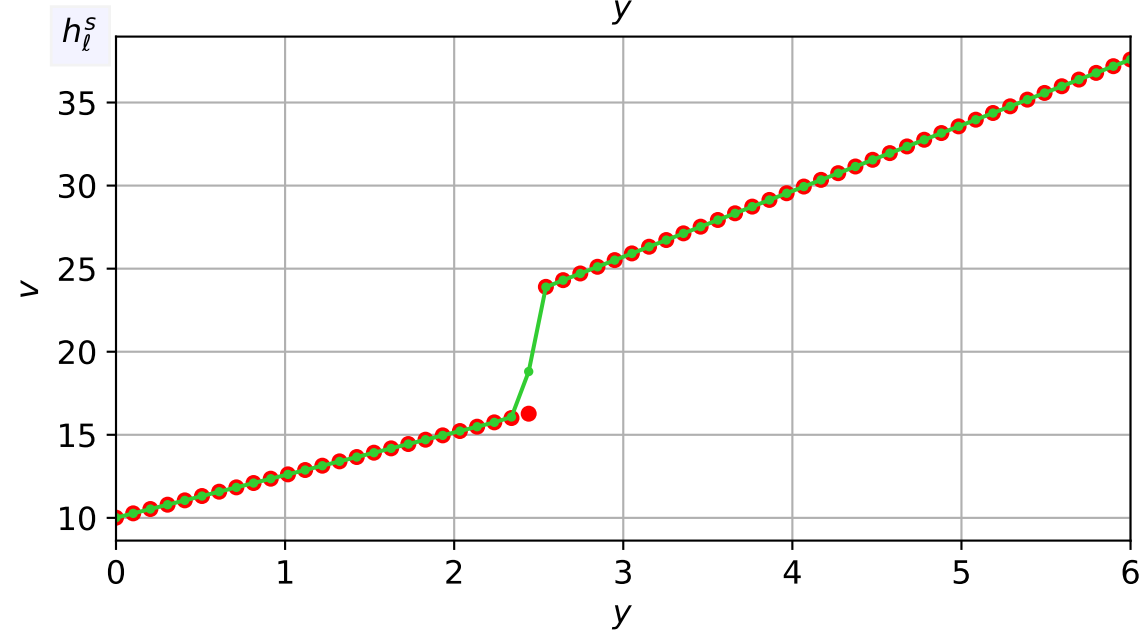
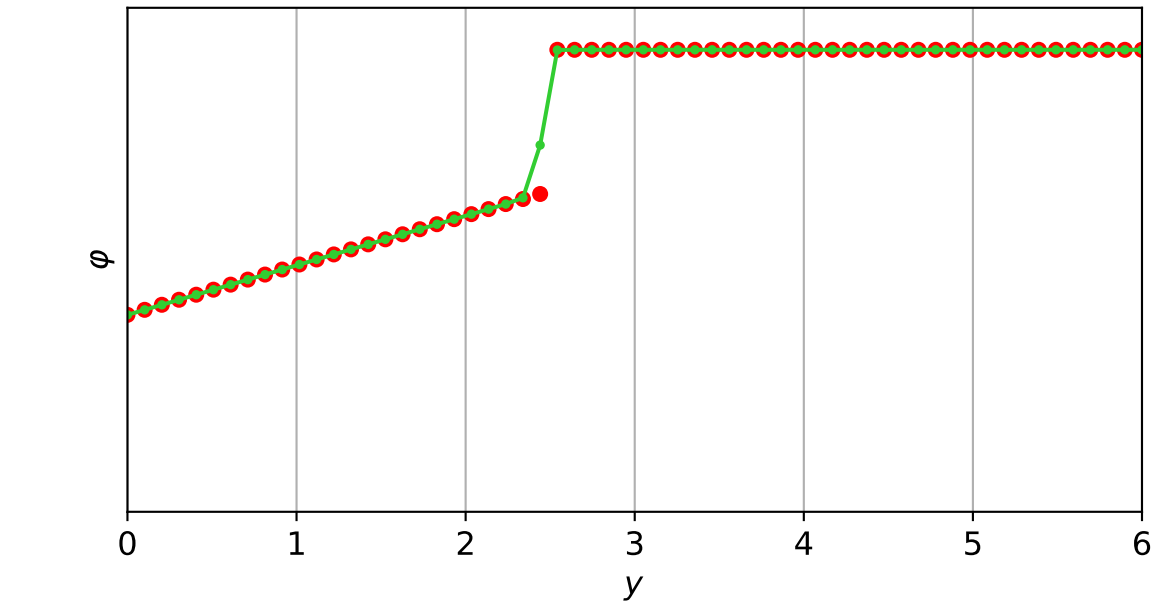
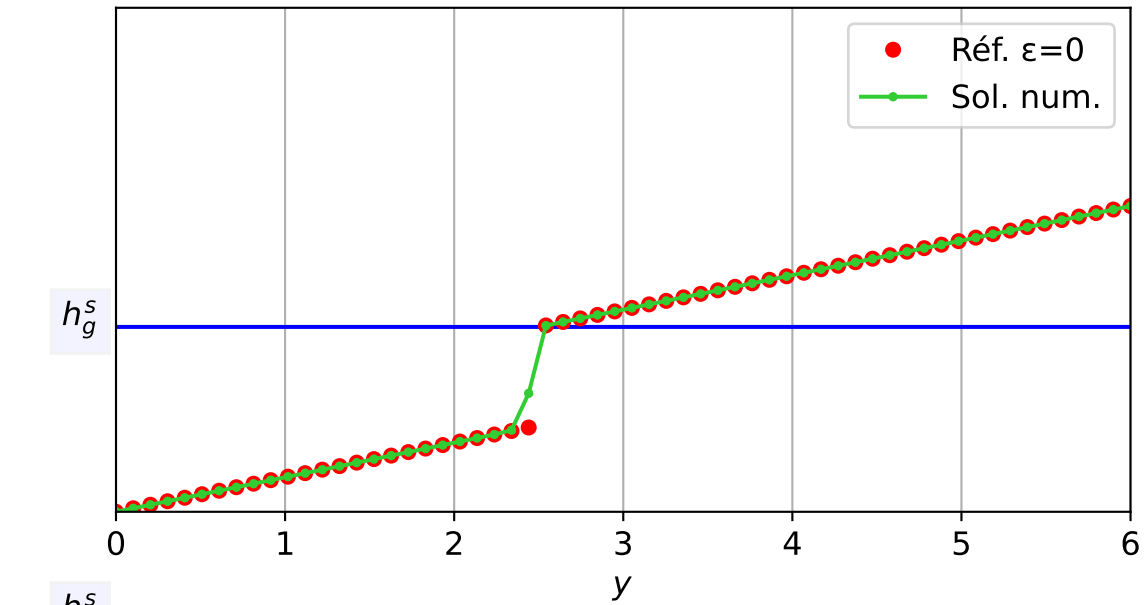




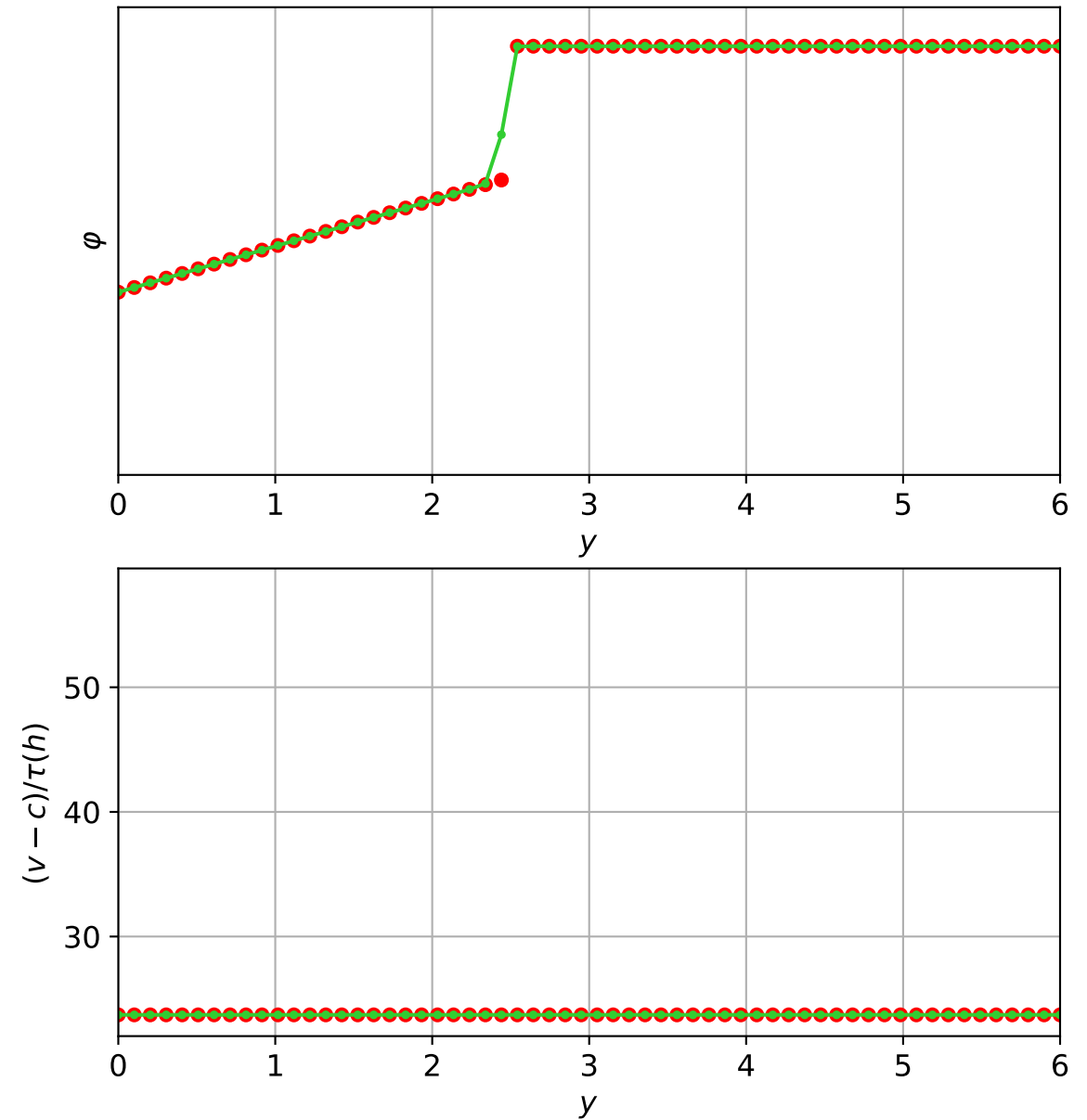
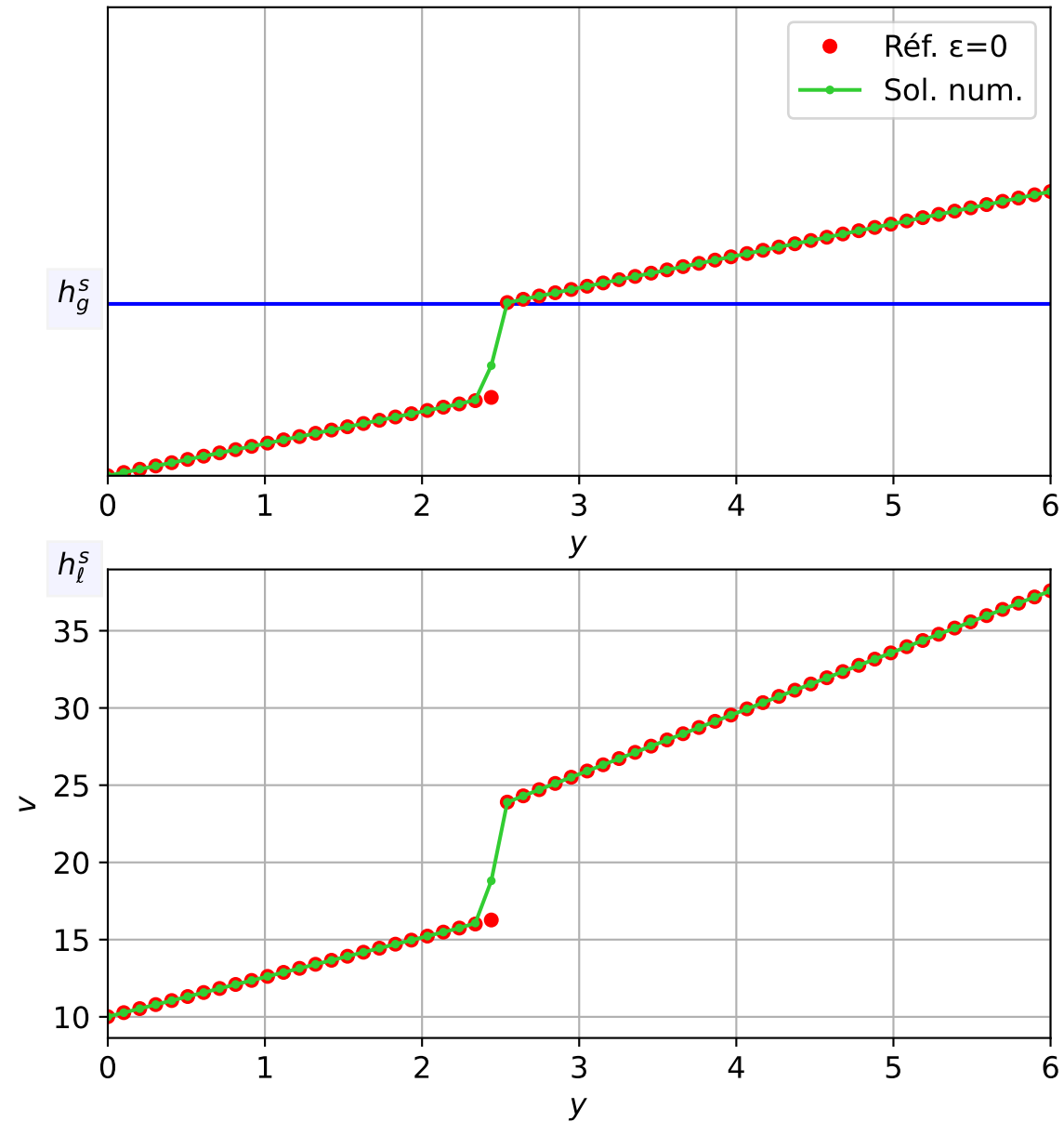
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 3.9524/12$  (8327 iter) — résidu eq. =  $1.61e-06$



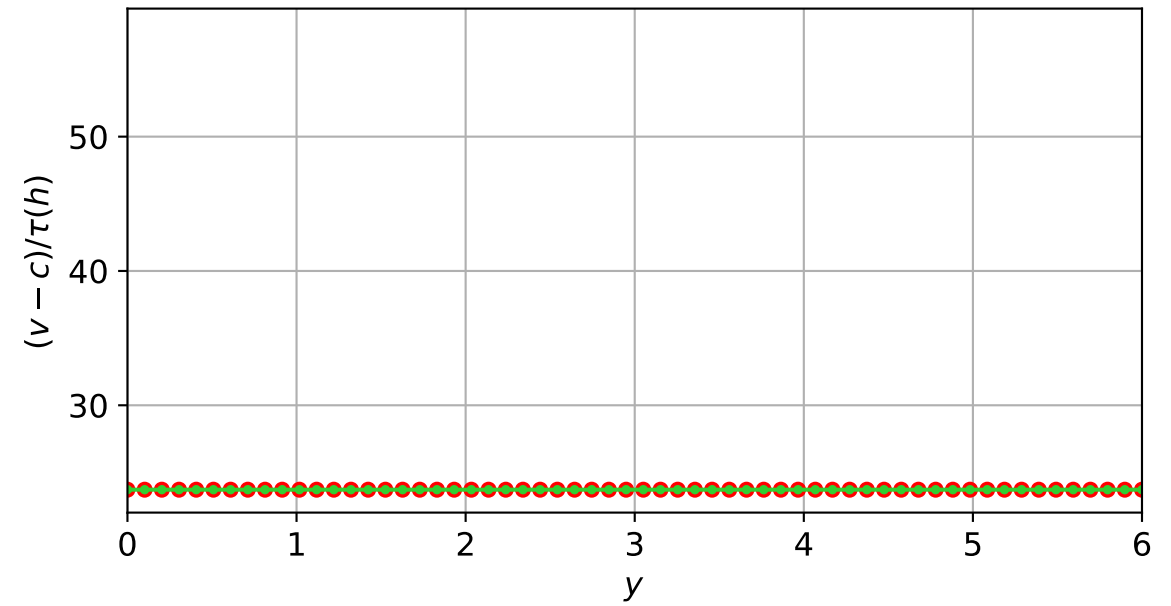
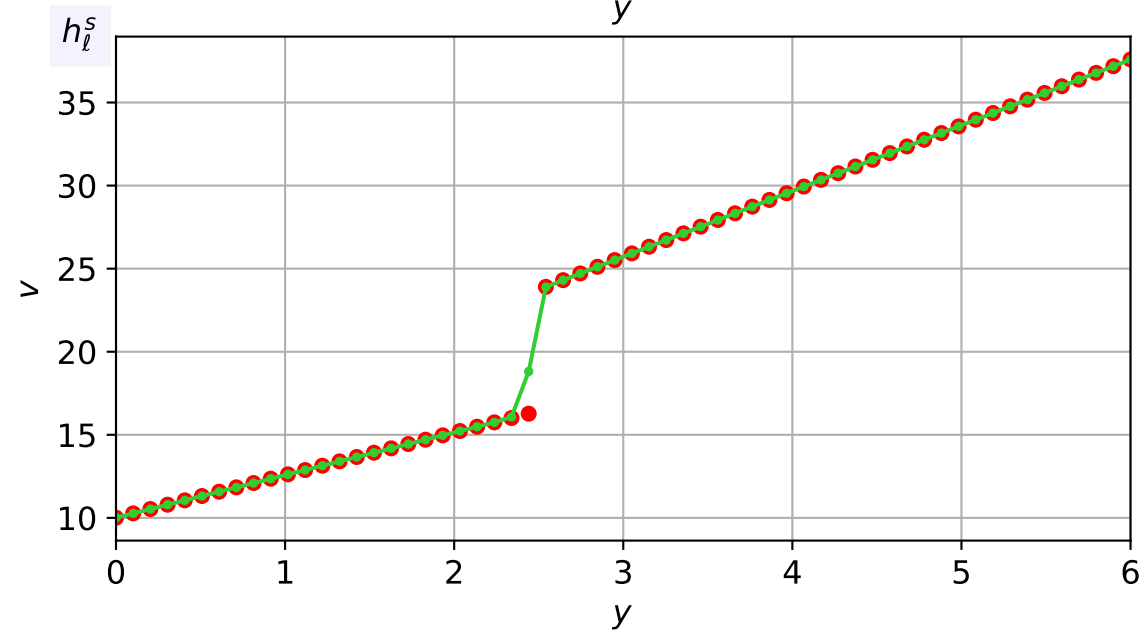
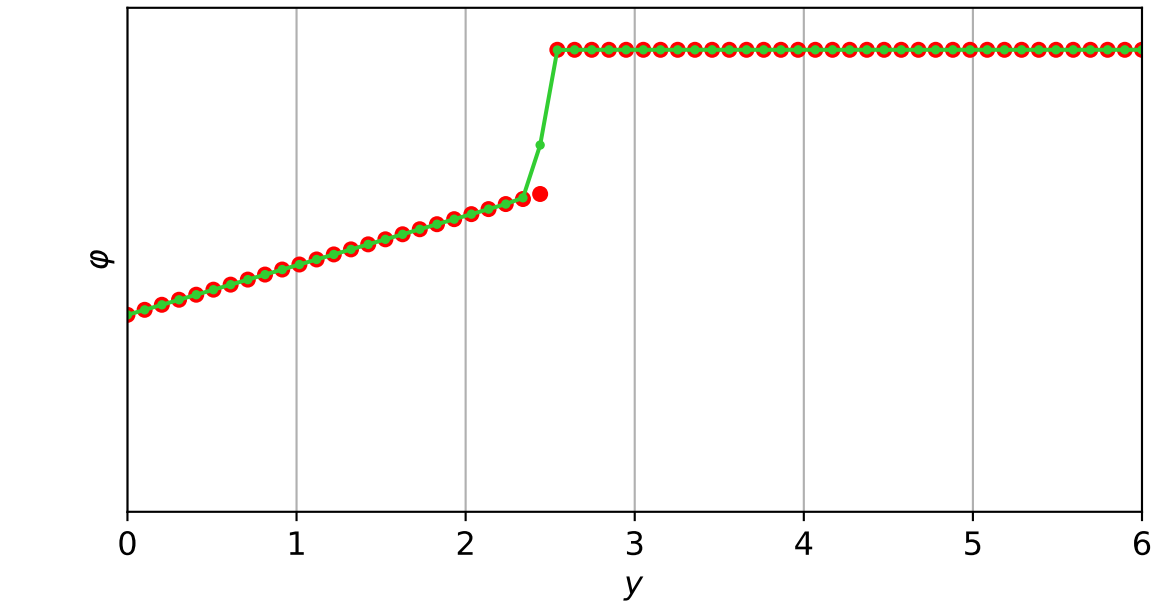
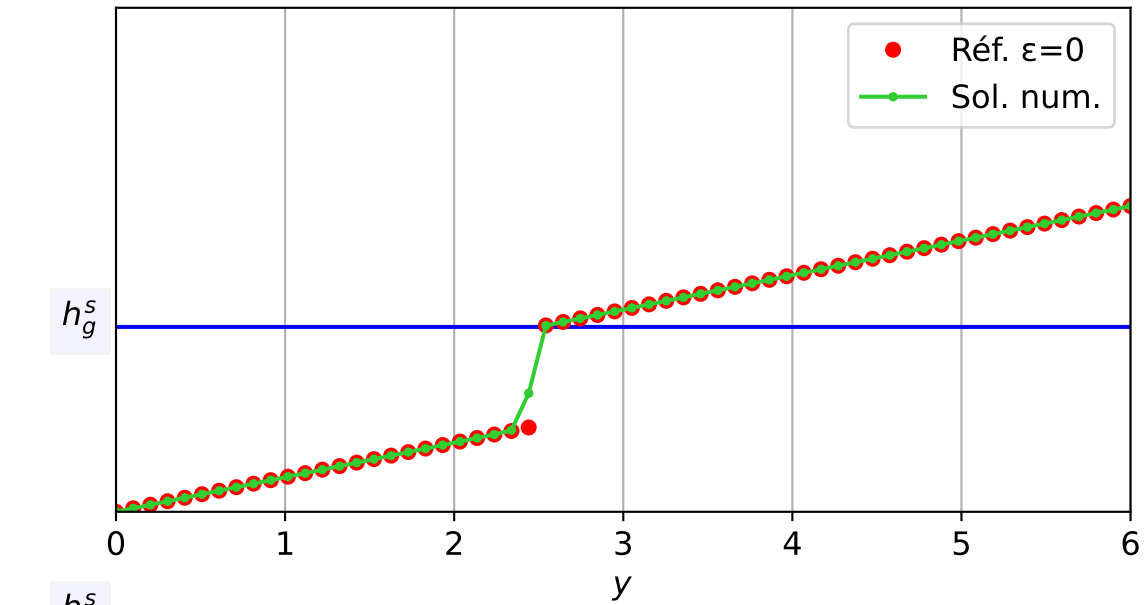
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.0011/12$  (8347 iter) — résidu eq. =  $1.43e-06$



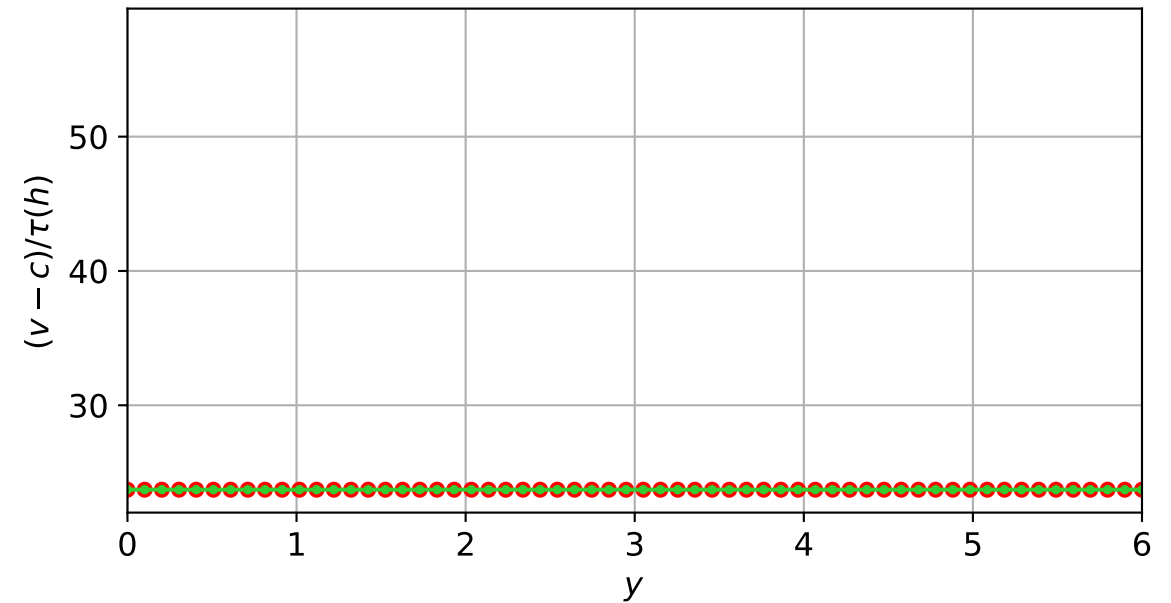
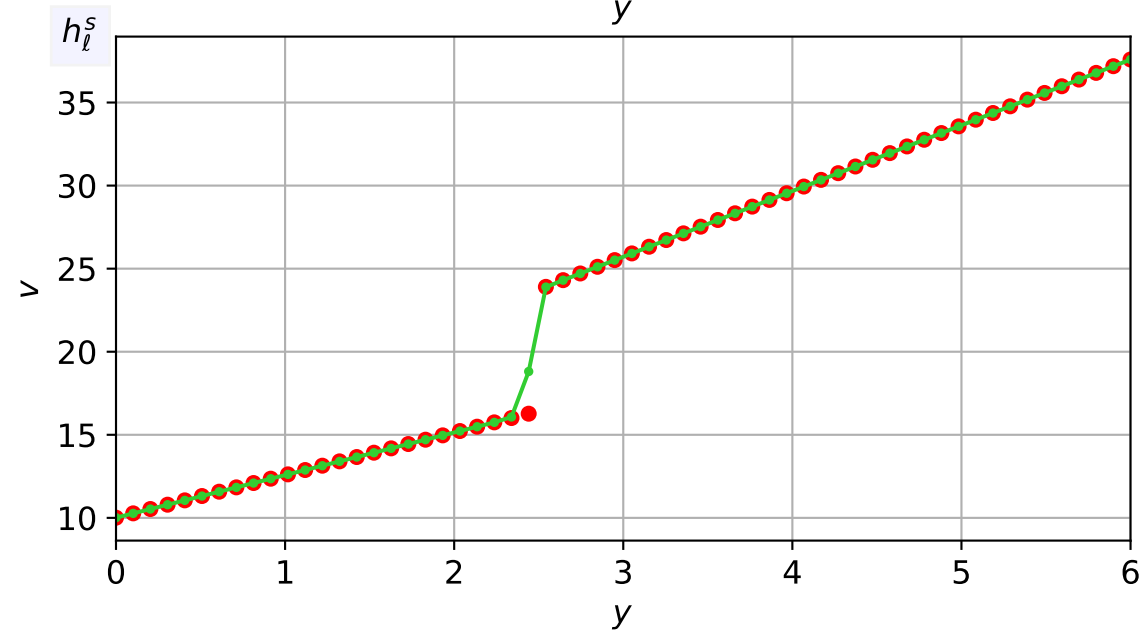
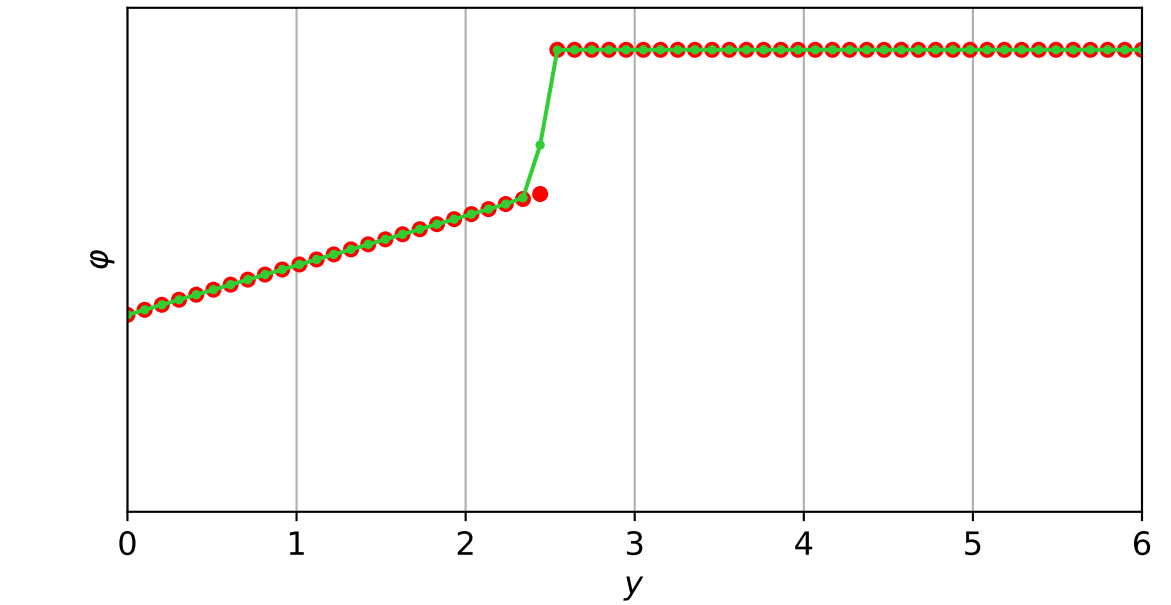
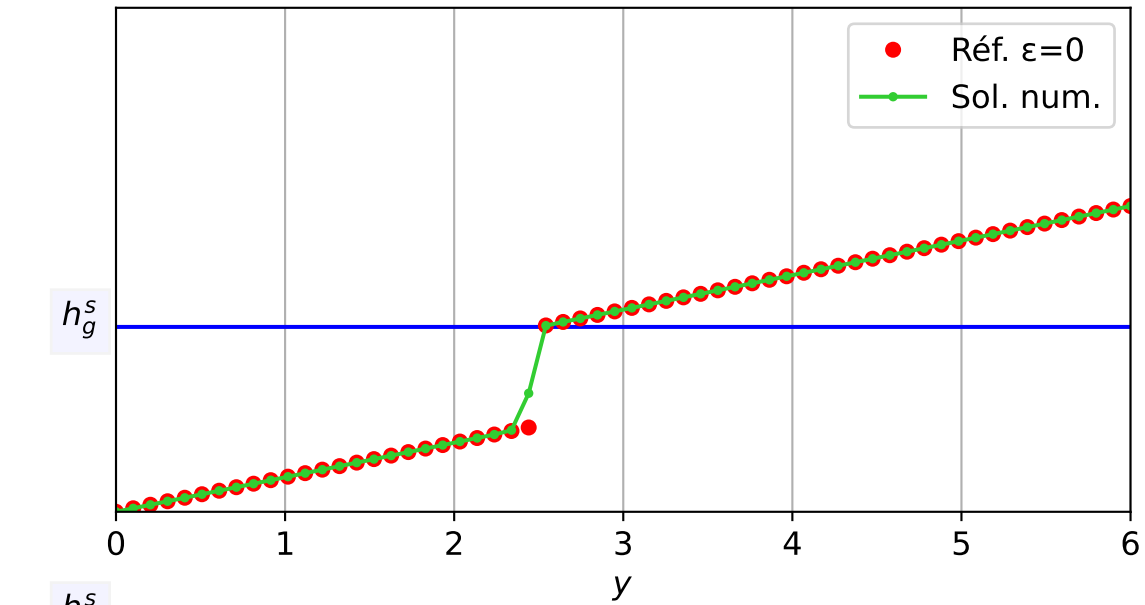
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.05224/12$  (8368 iter) — résidu eq. =  $1.26e-06$



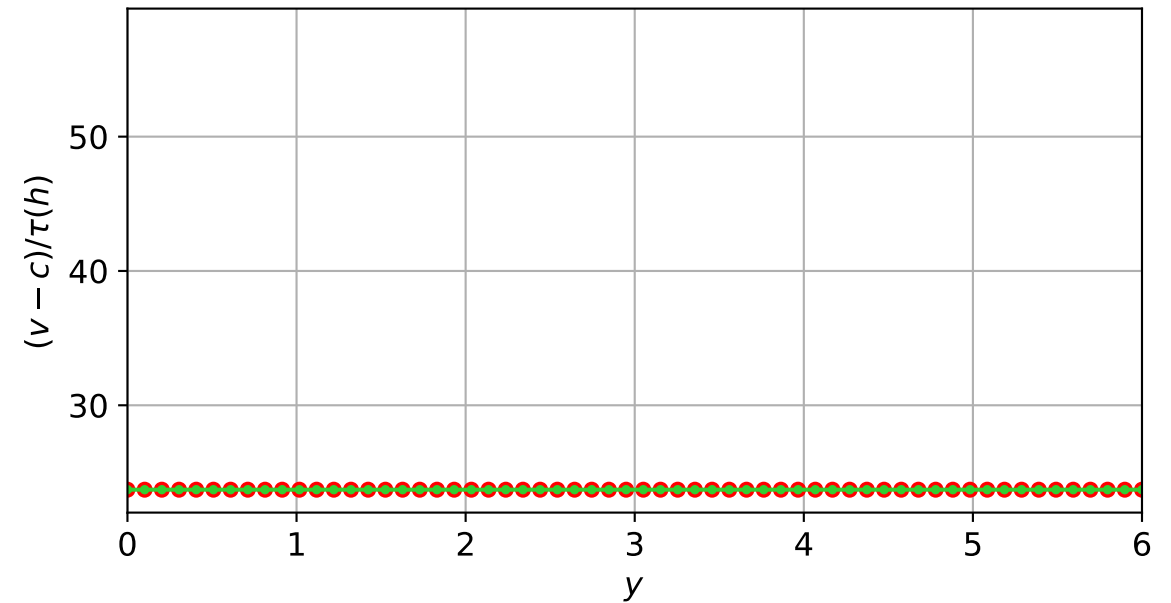
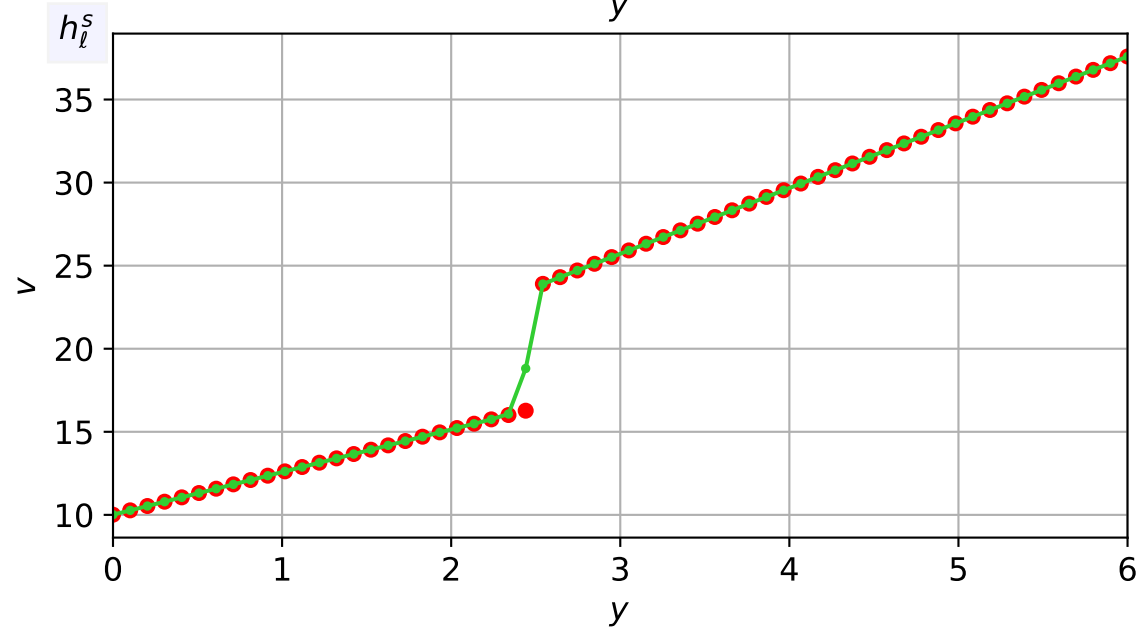
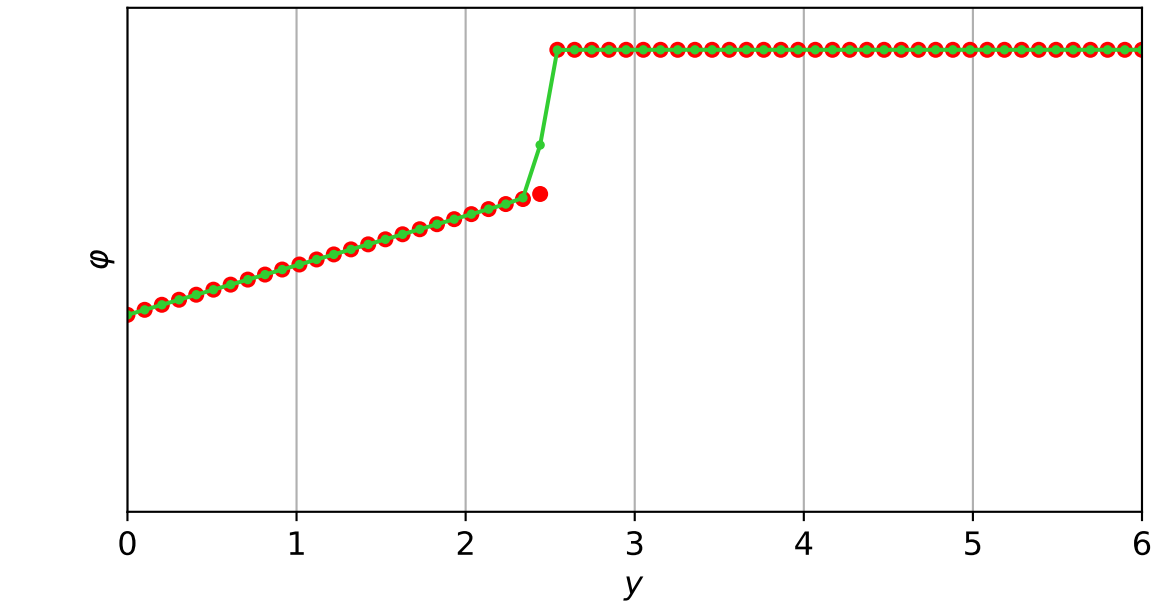
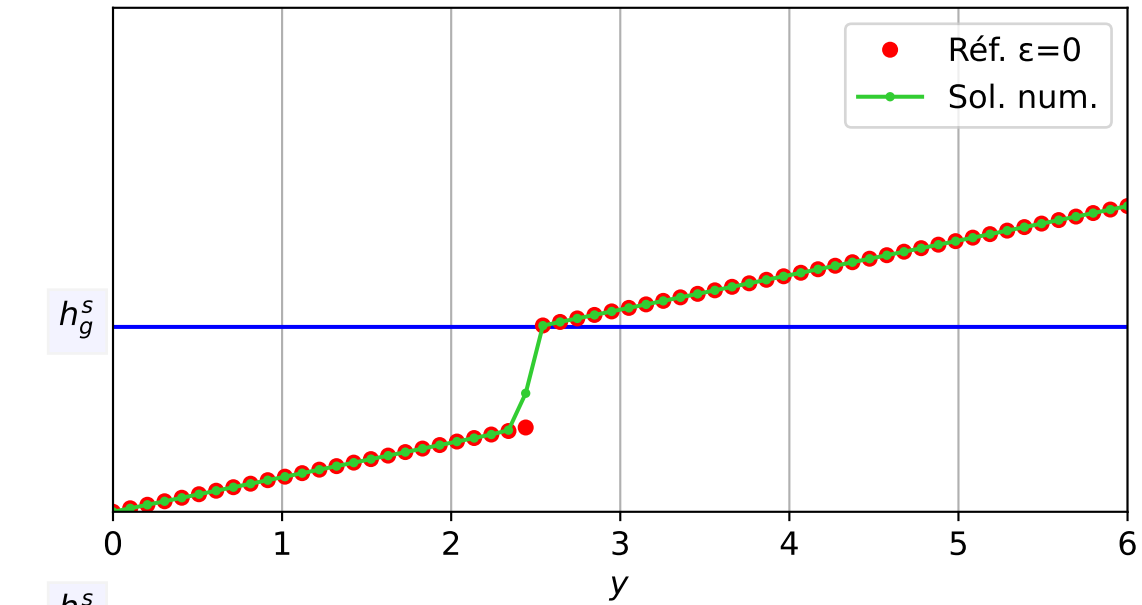
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.10094/12$  (8388 iter) — résidu eq. =  $1.13e-06$



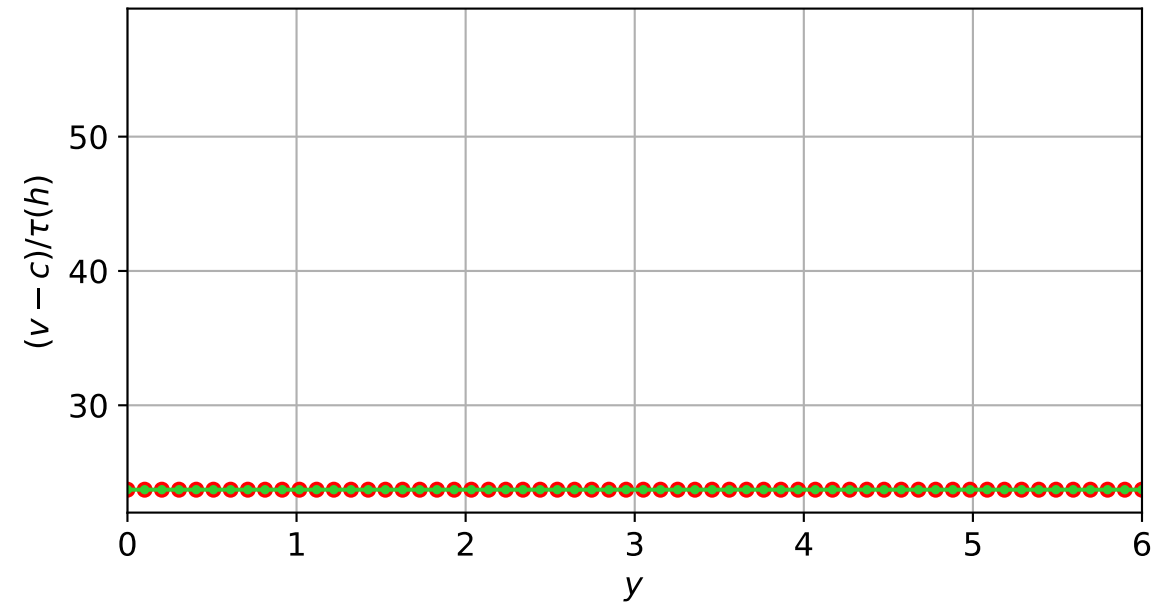
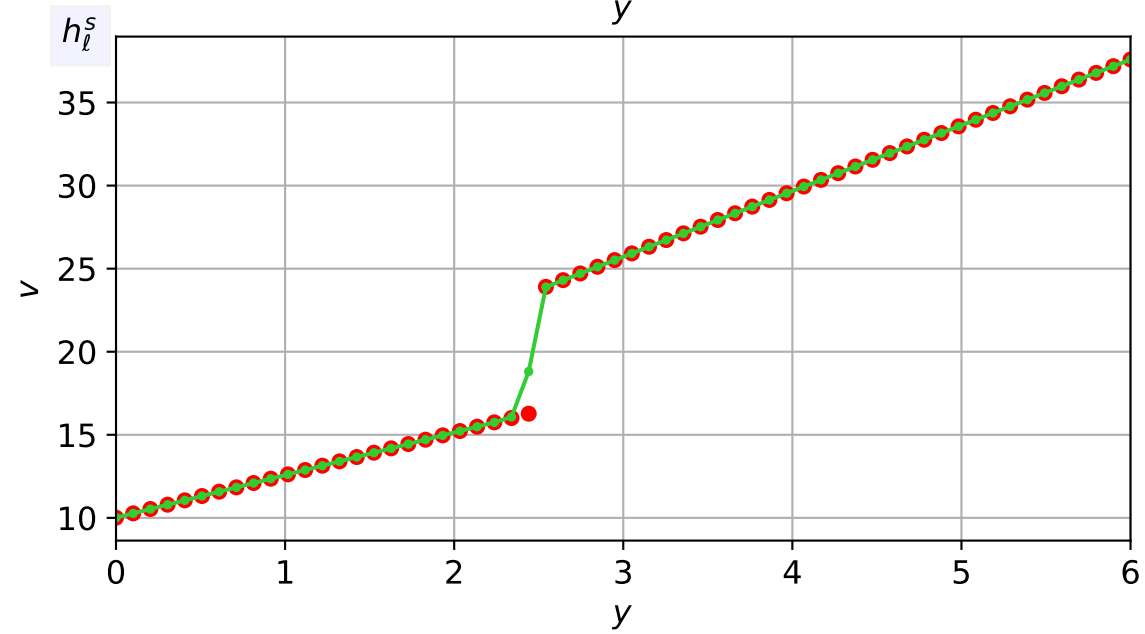
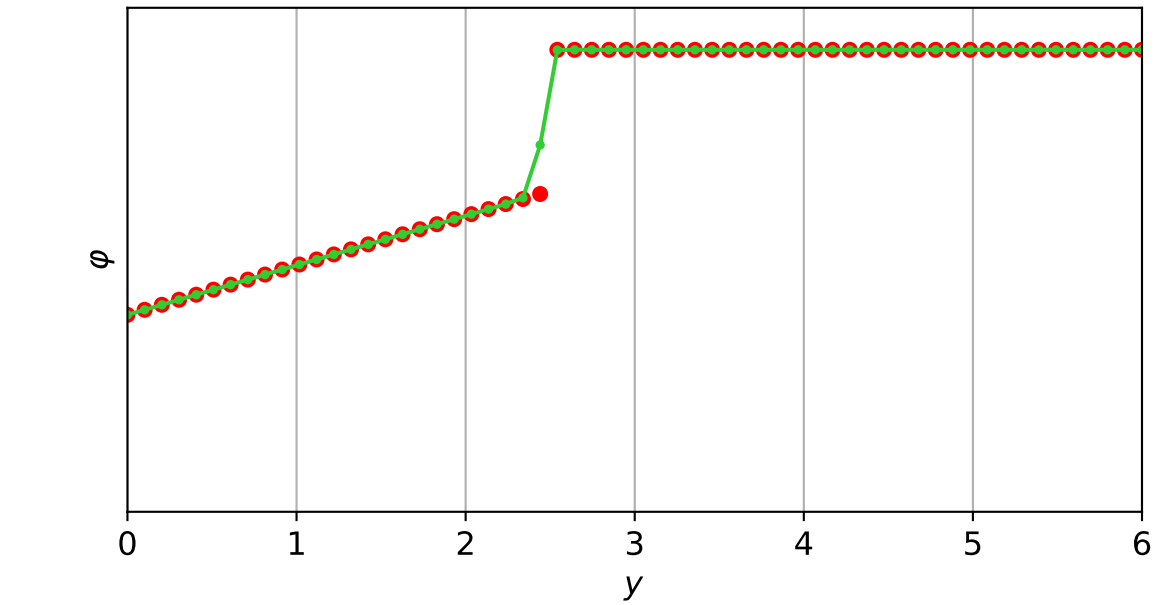
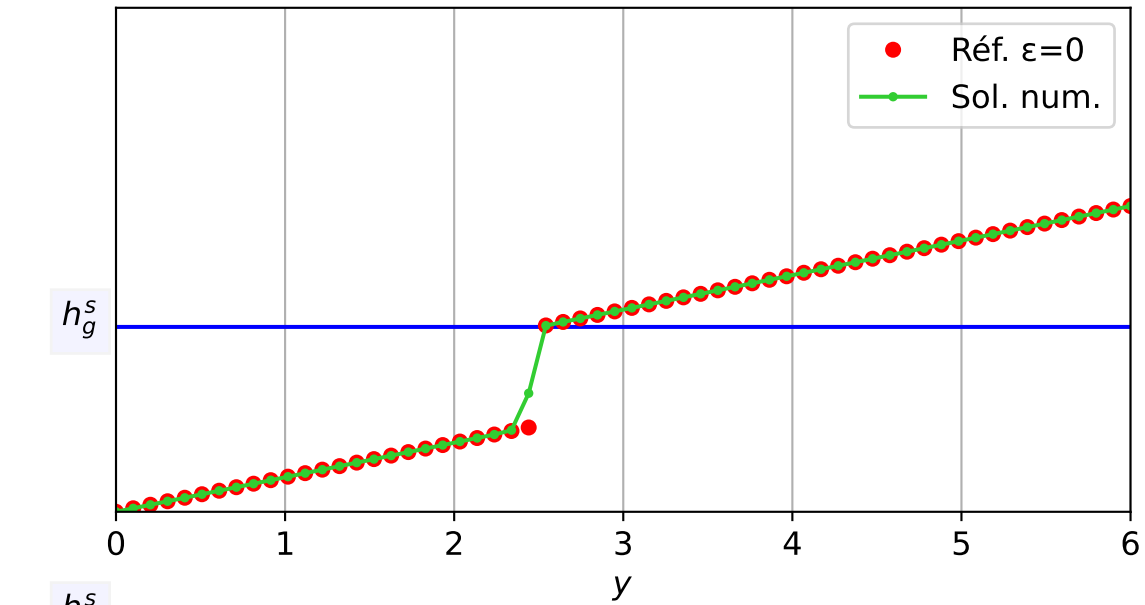
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.15207/12$  (8409 iter) — résidu eq. =  $9.95e-07$



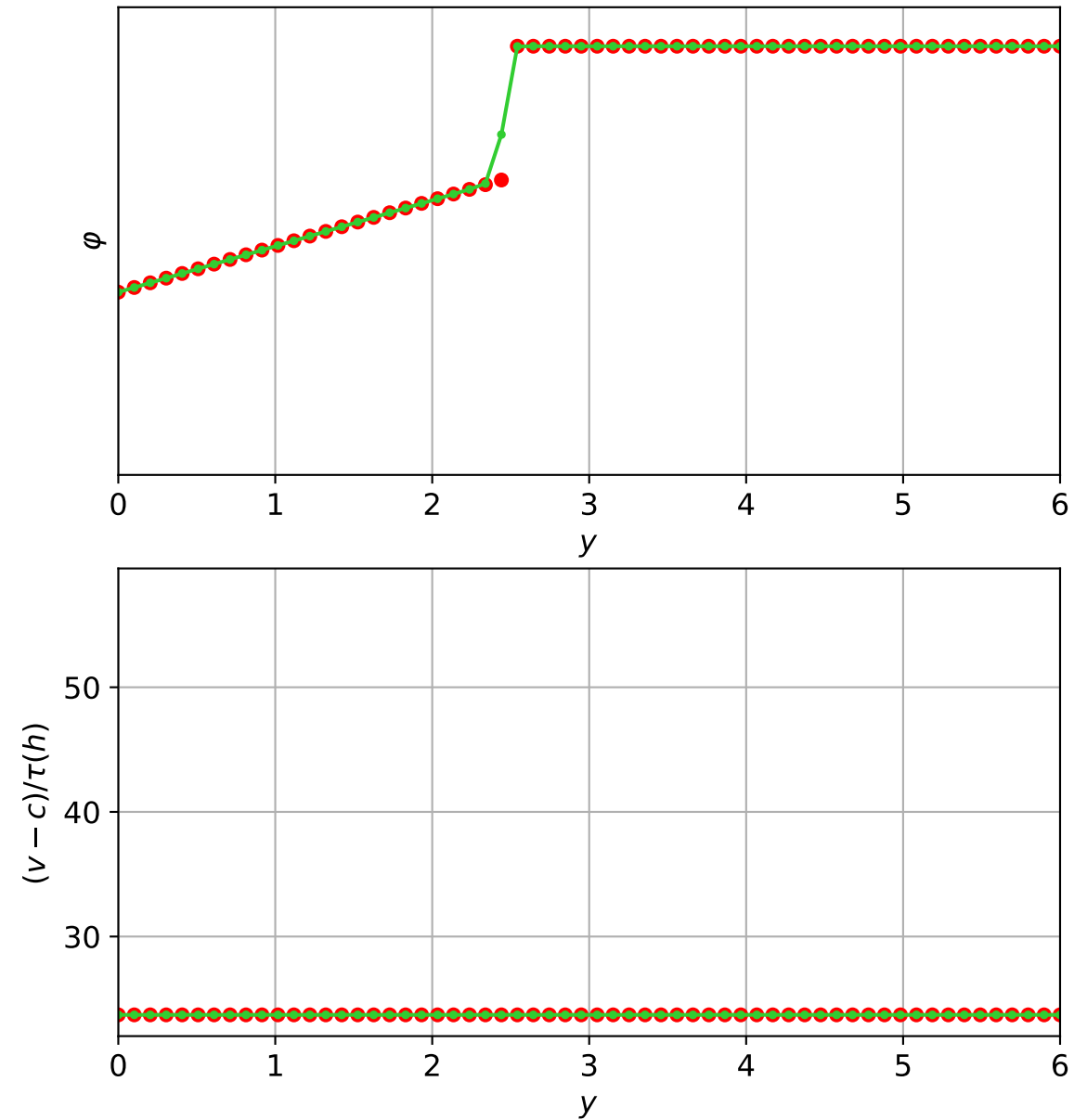
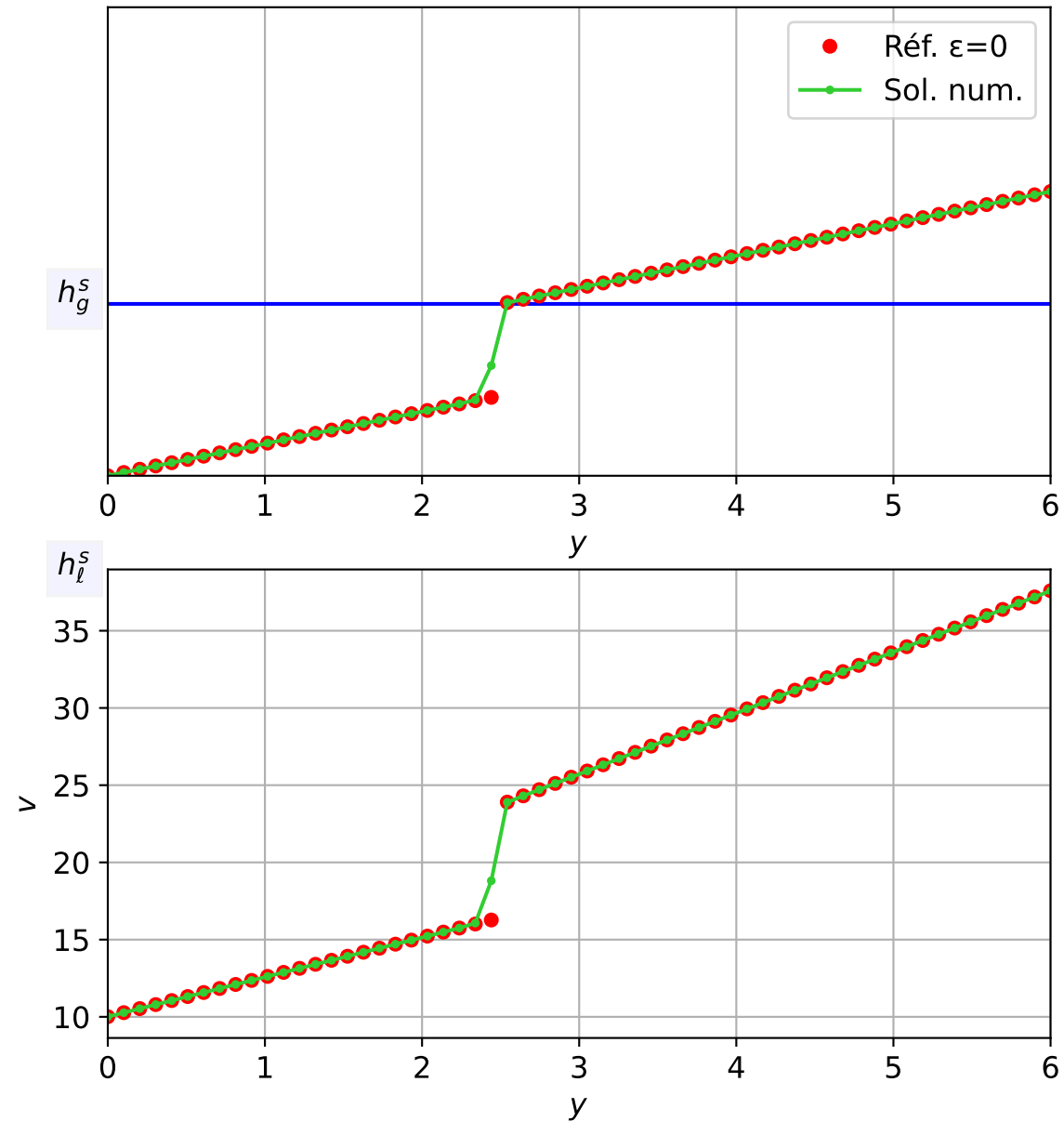
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.20077/12$  (8429 iter) — résidu eq. =  $9.02e-06$



TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.25191/12$  (8450 iter) — résidu eq. =  $8.84e-06$

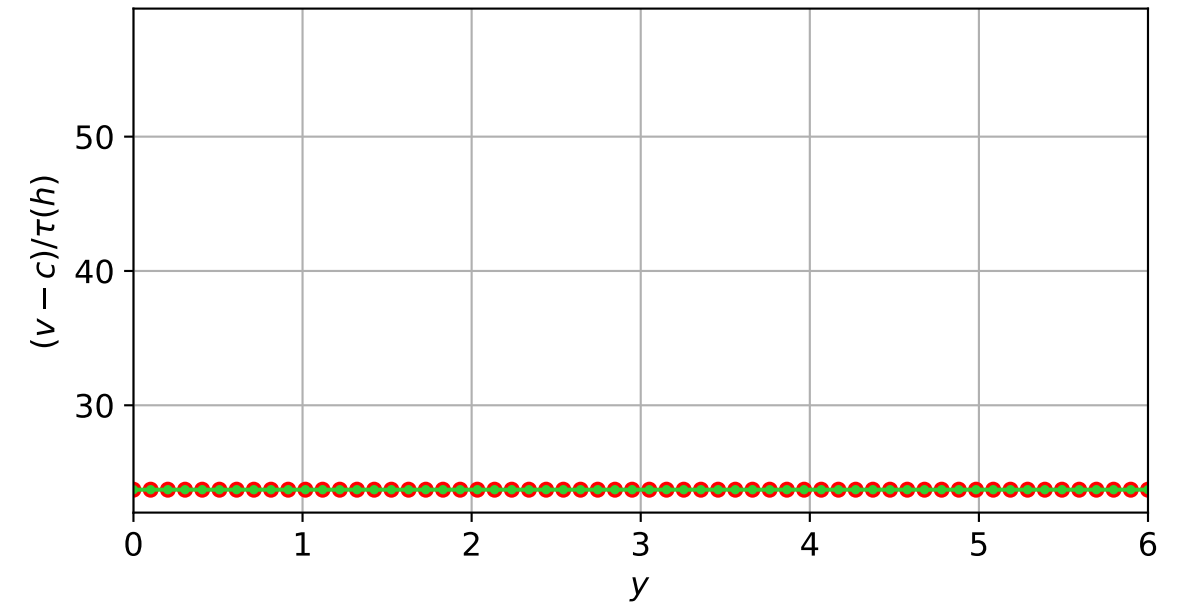
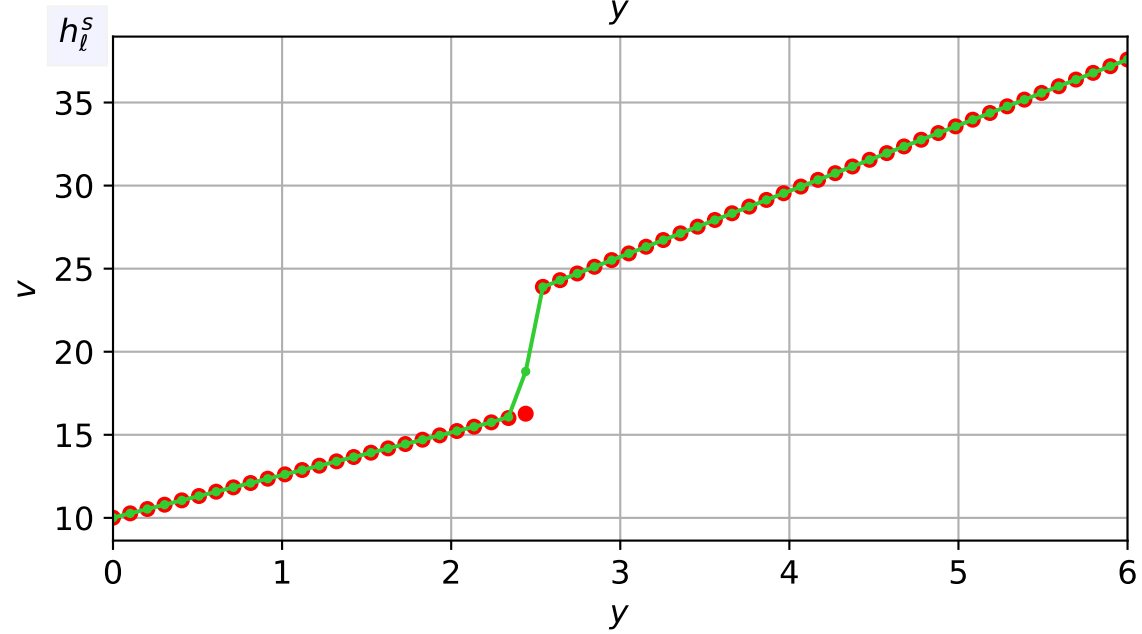
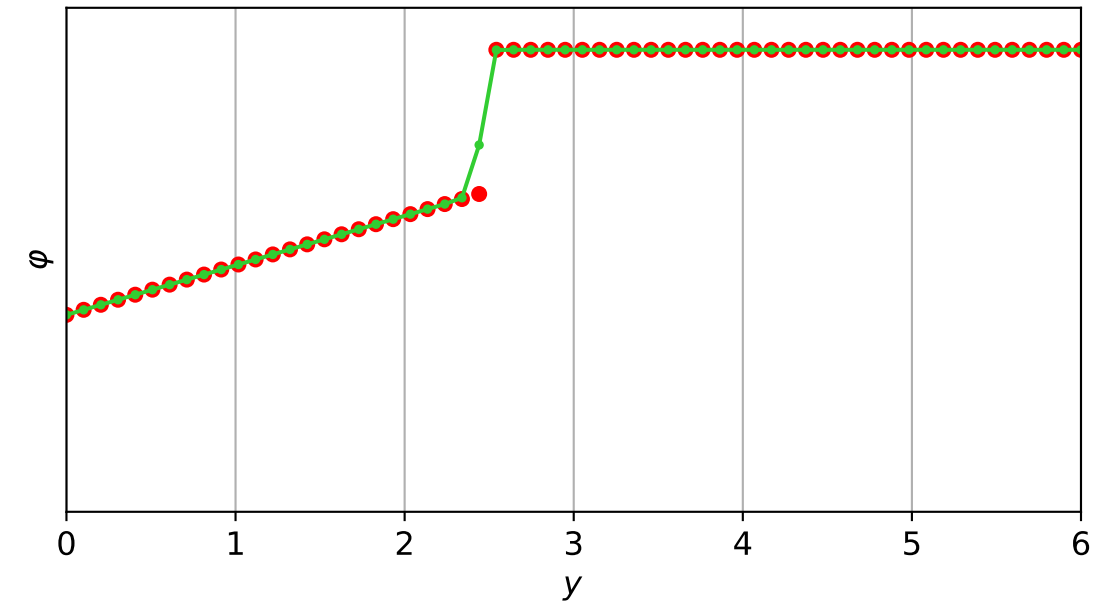
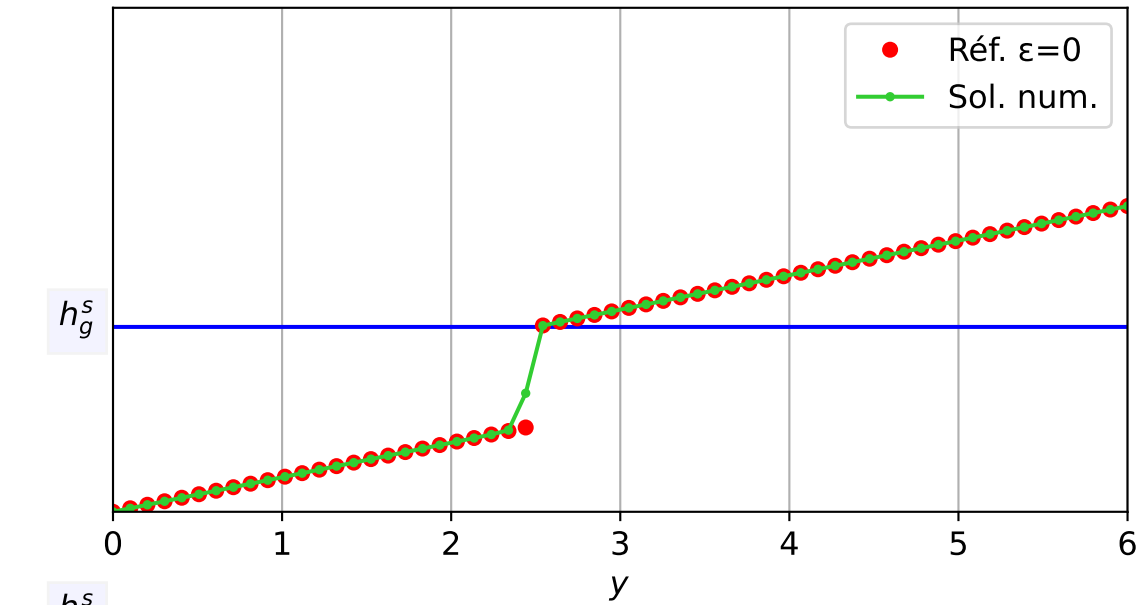


TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.30061/12$  (8470 iter) — résidu eq. =  $7.86e-06$

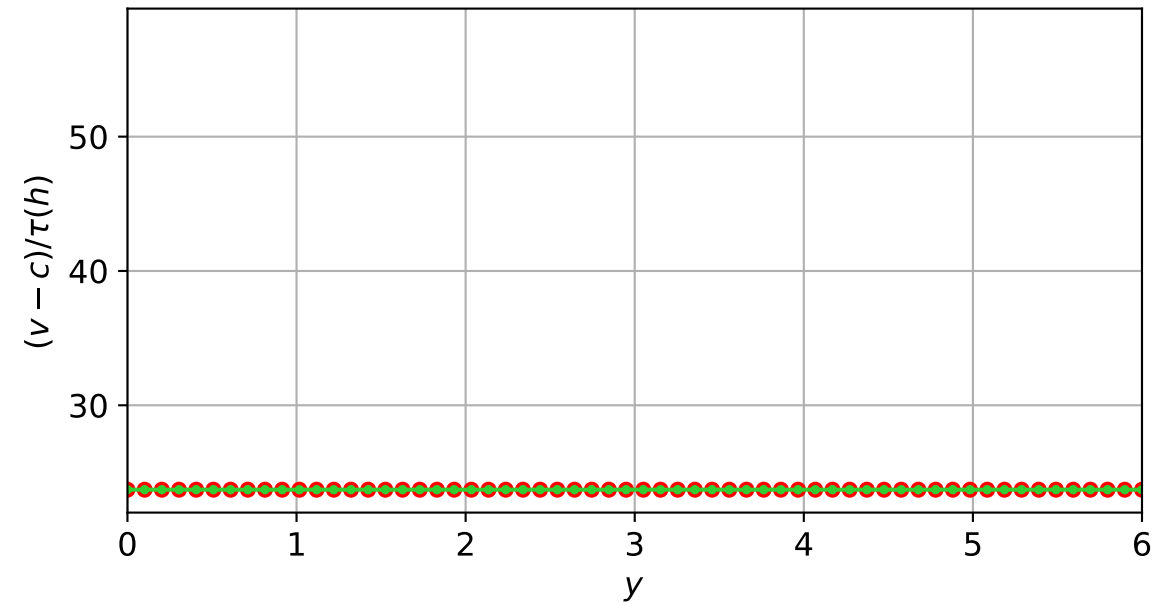
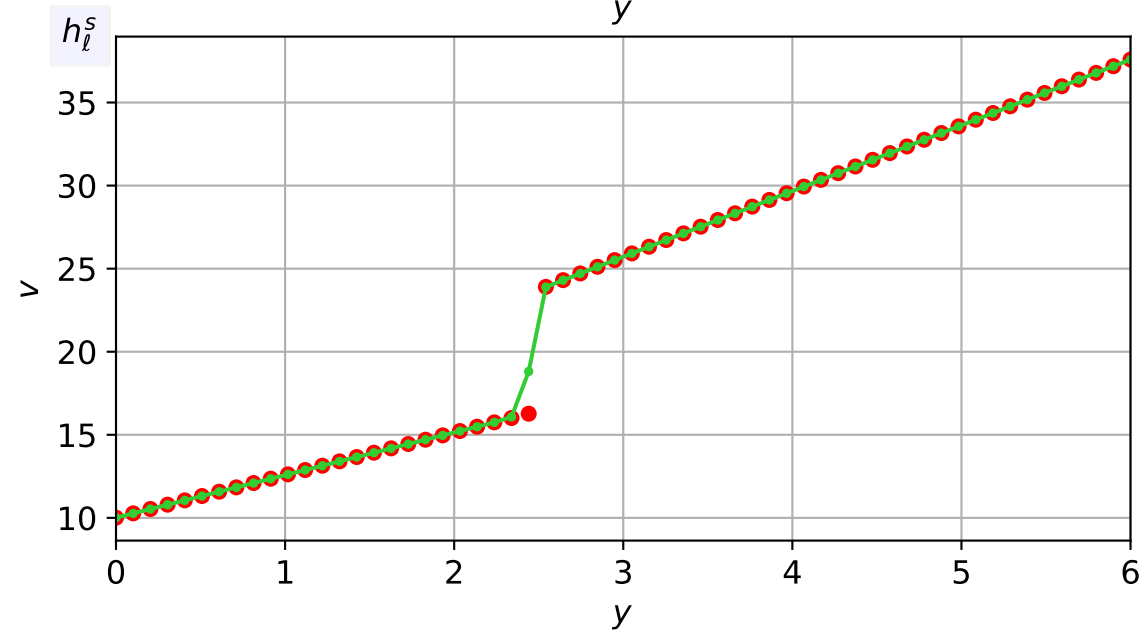
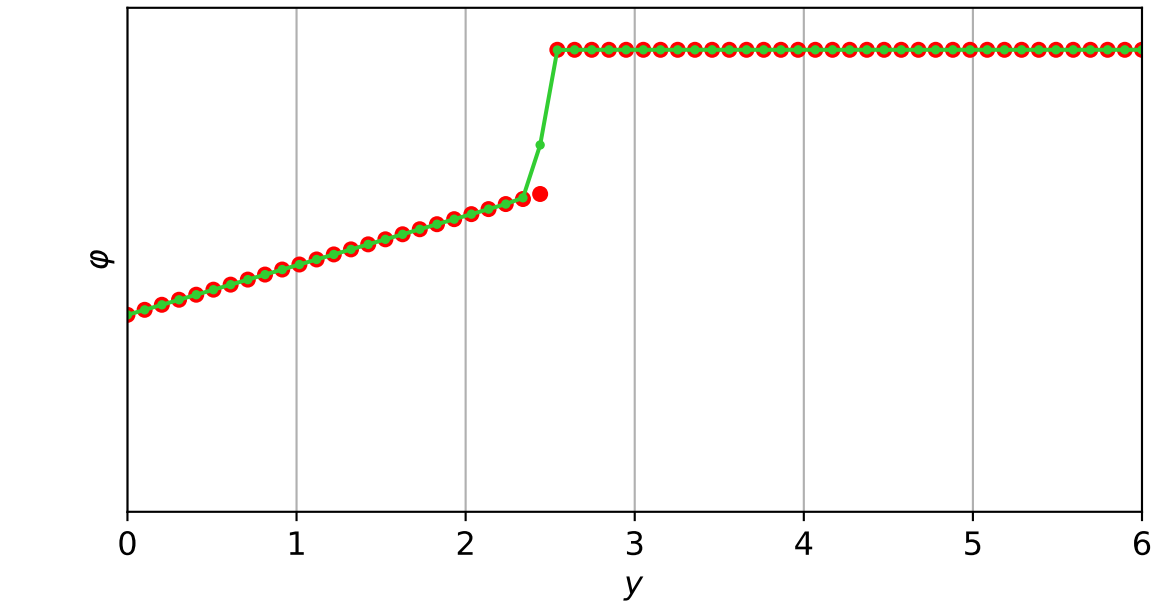
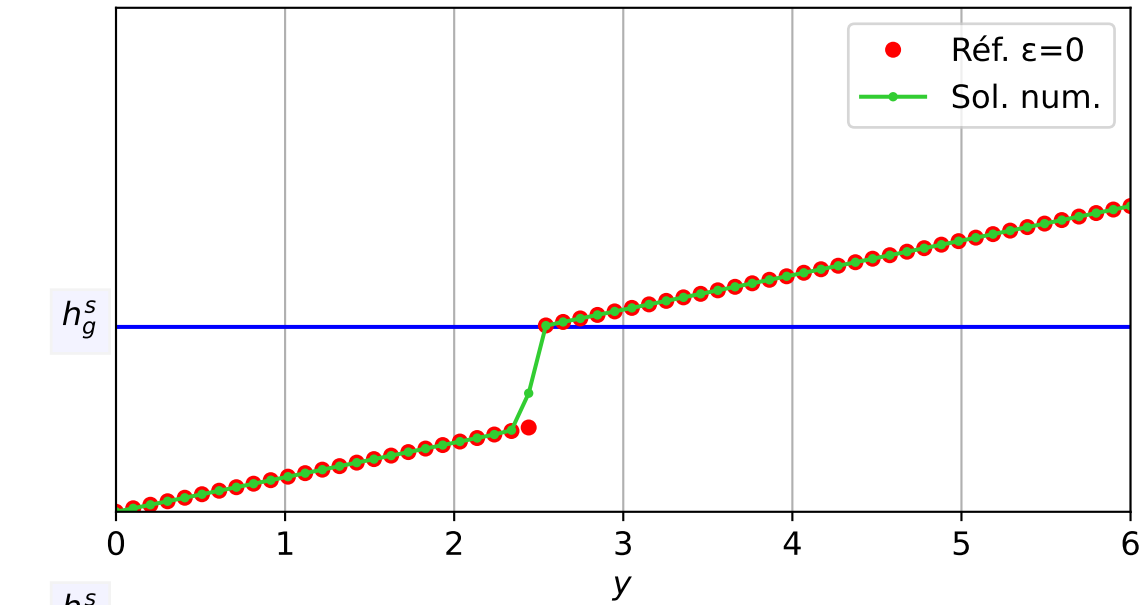




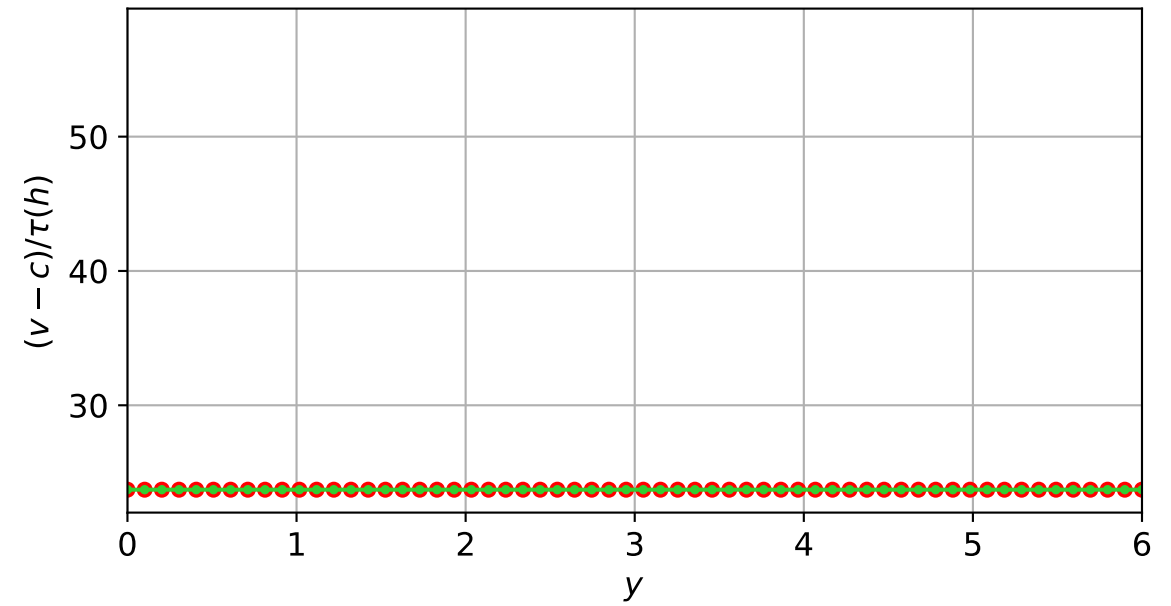
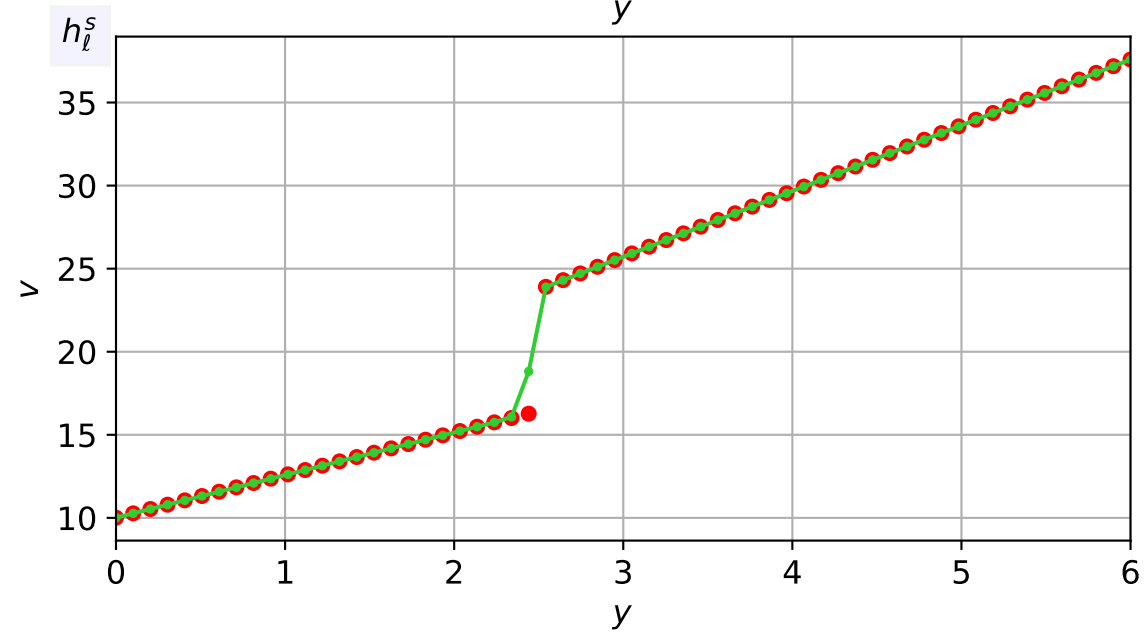
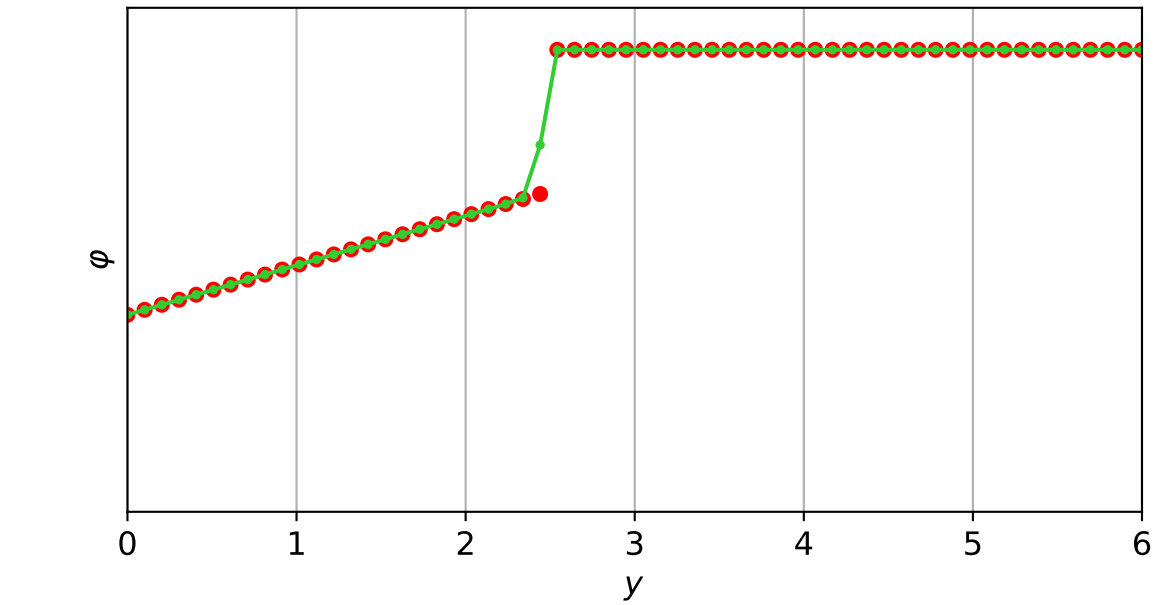
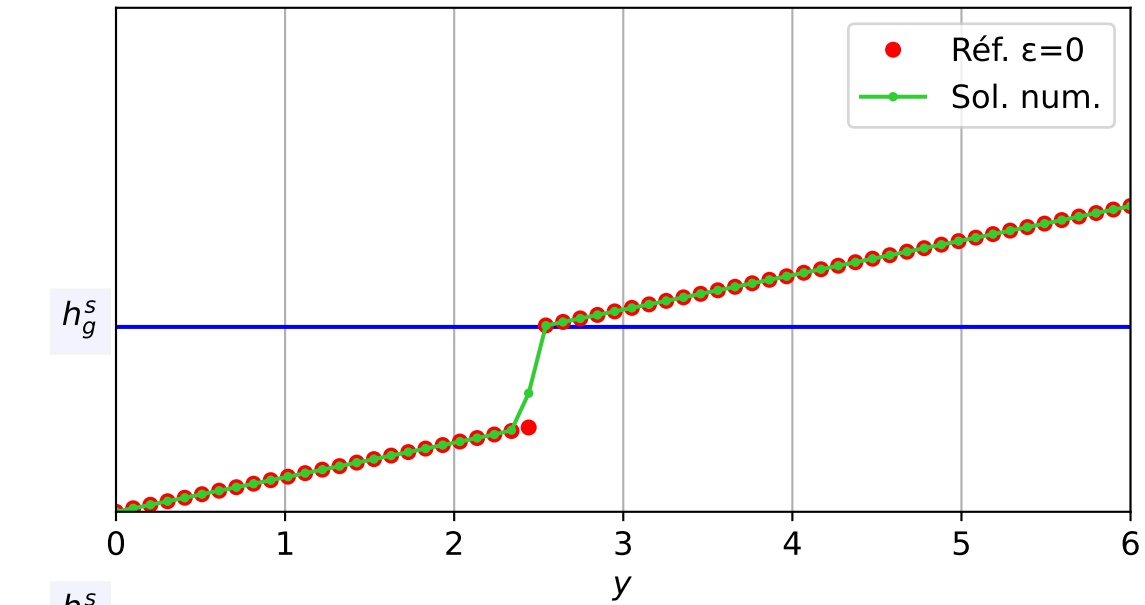
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.35174/12$  (8491 iter) — résidu eq. =  $6.94e-06$



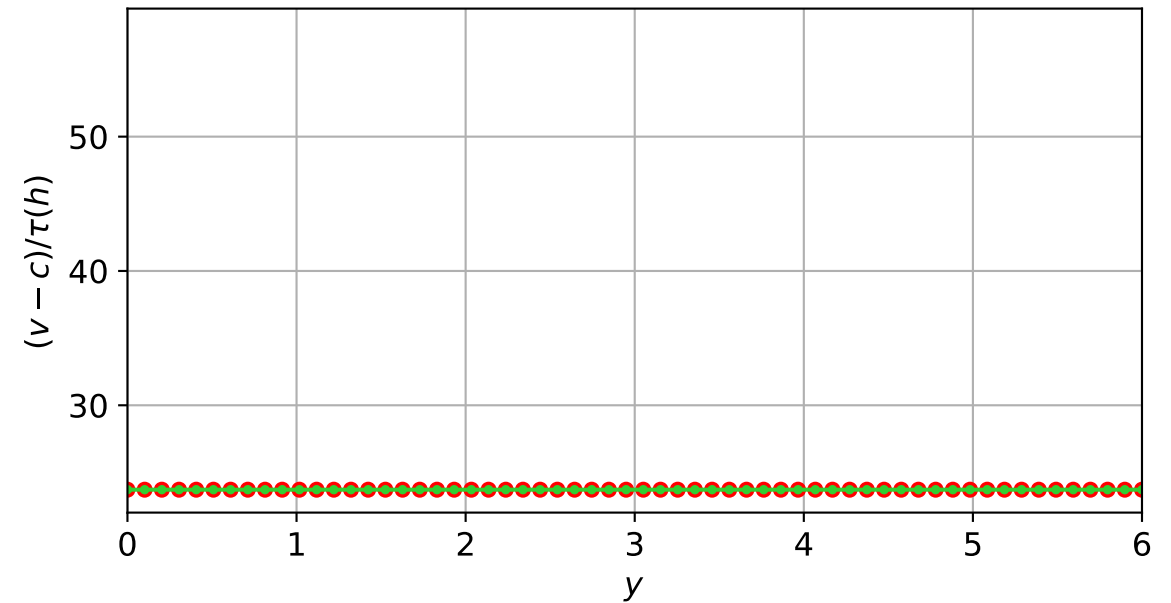
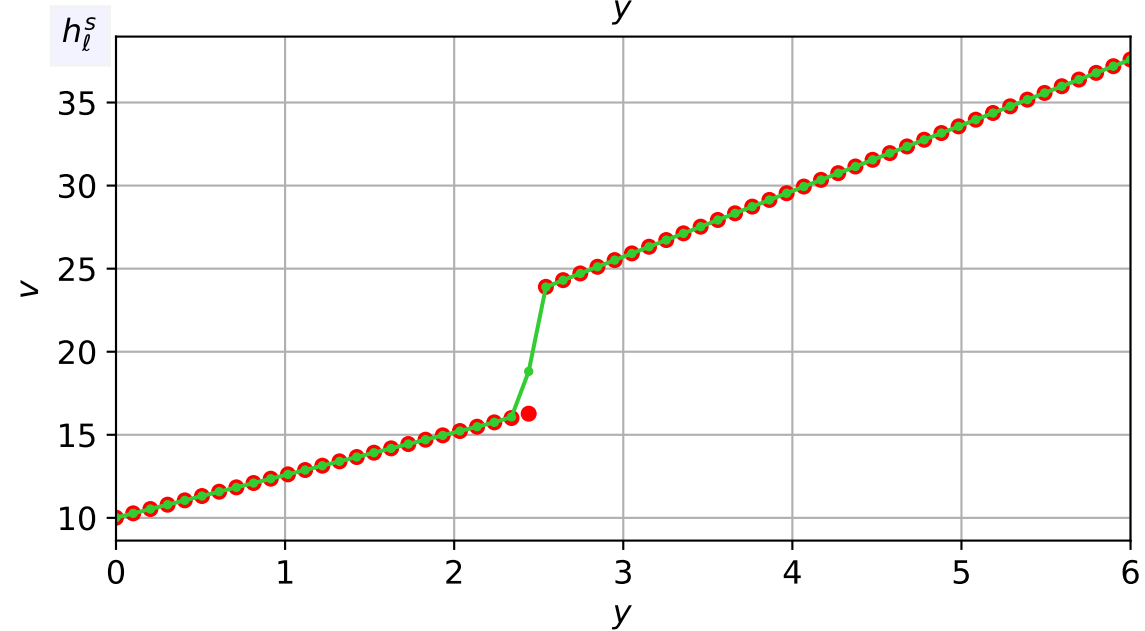
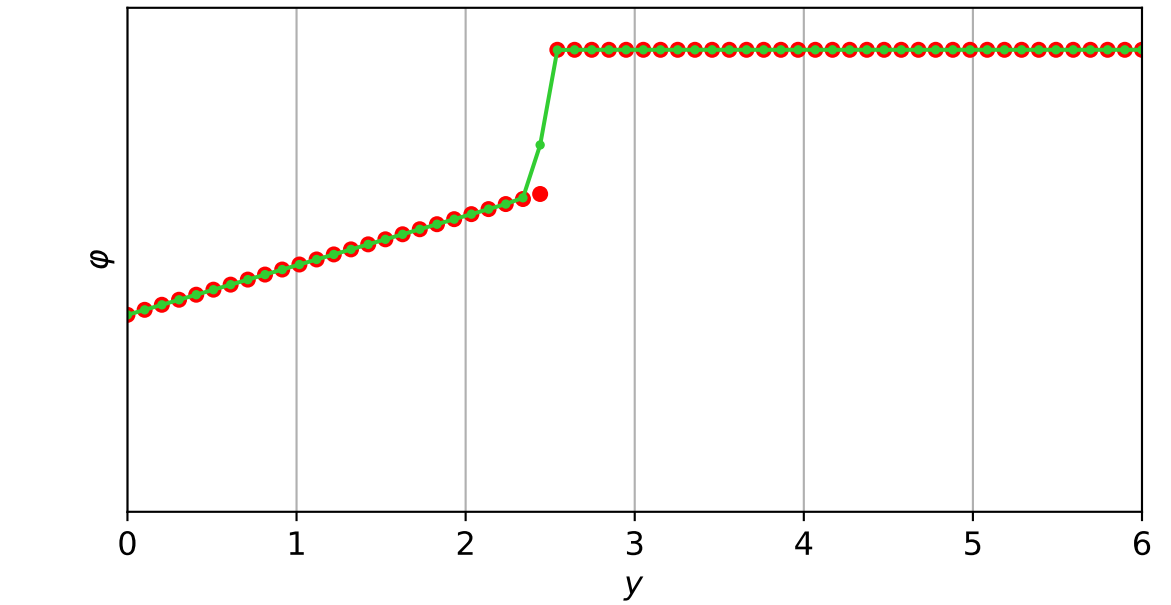
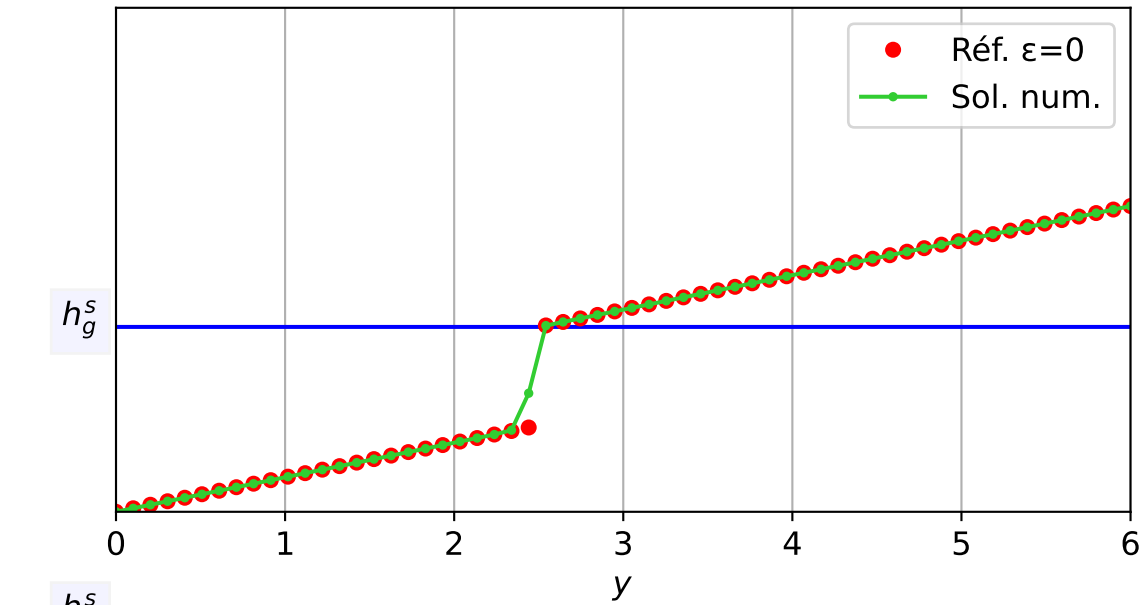
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.40045/12$  (8511 iter) — résidu eq. =  $6.17e-06$



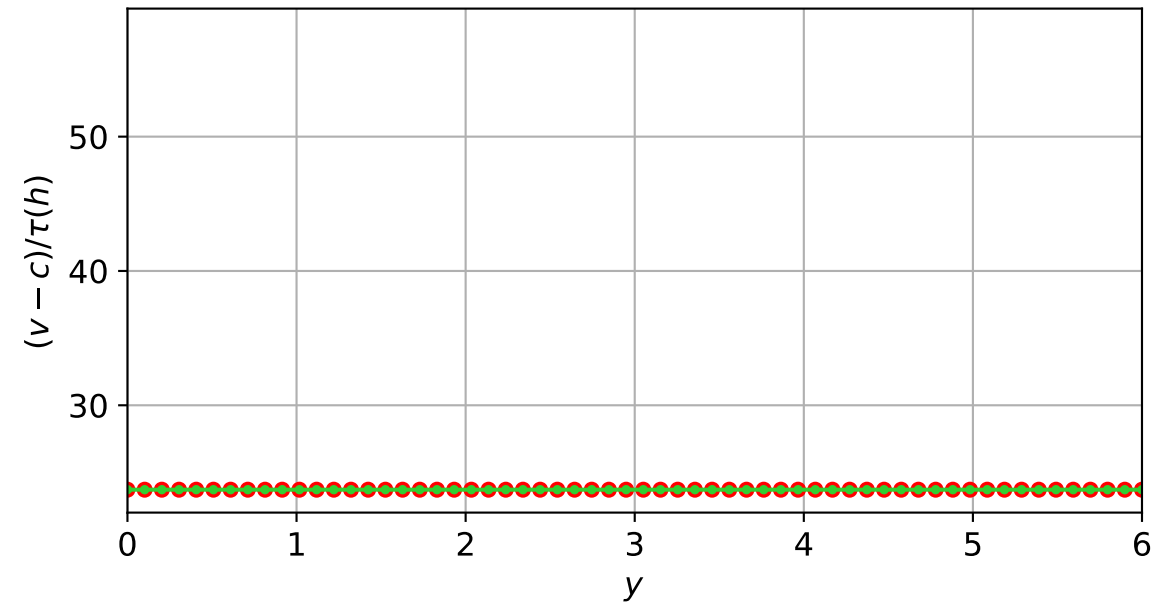
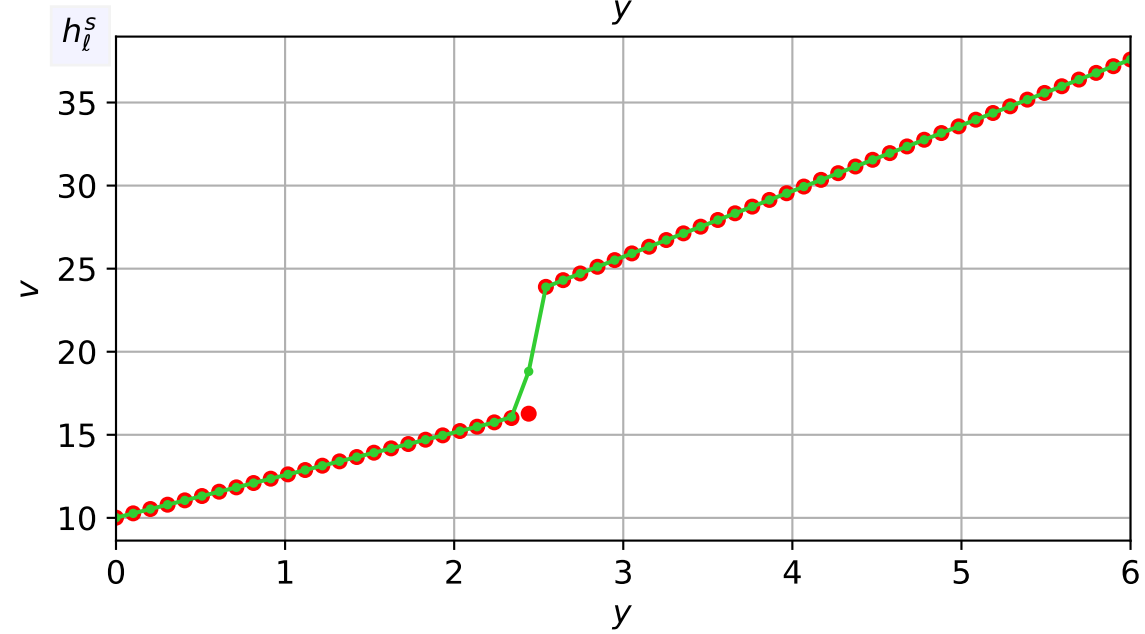
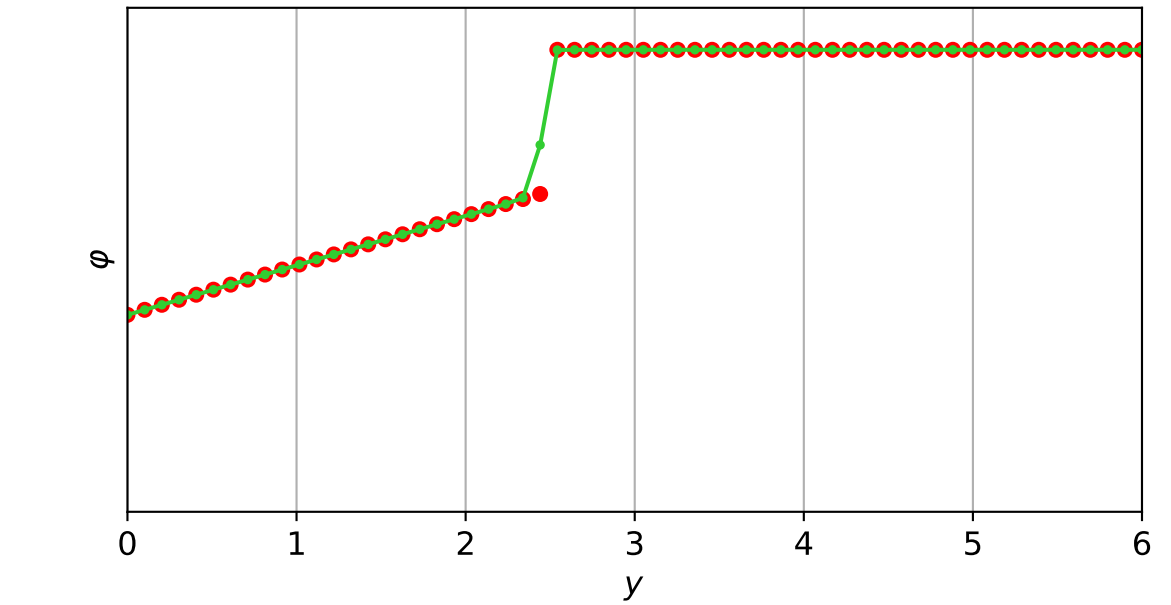
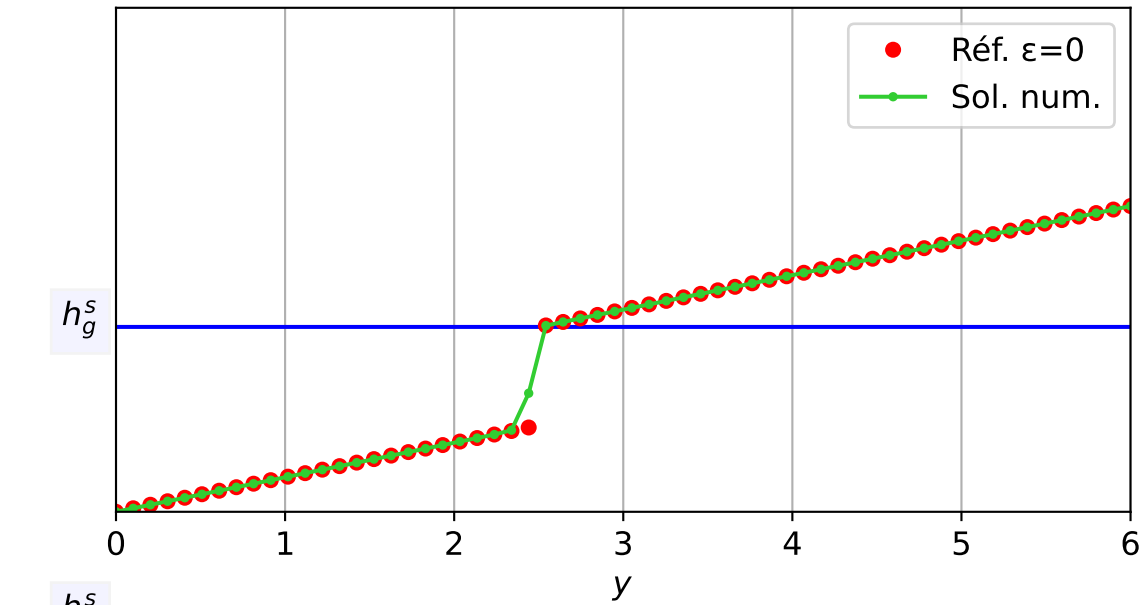
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.45158/12$  (8532 iter) — résidu eq. =  $5.45e-06$



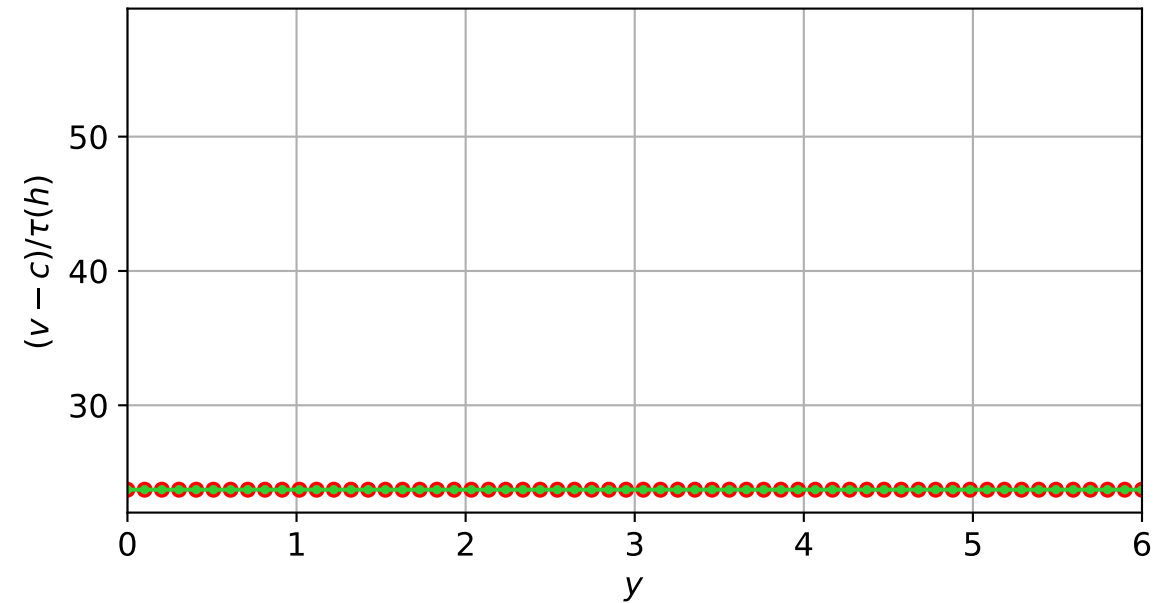
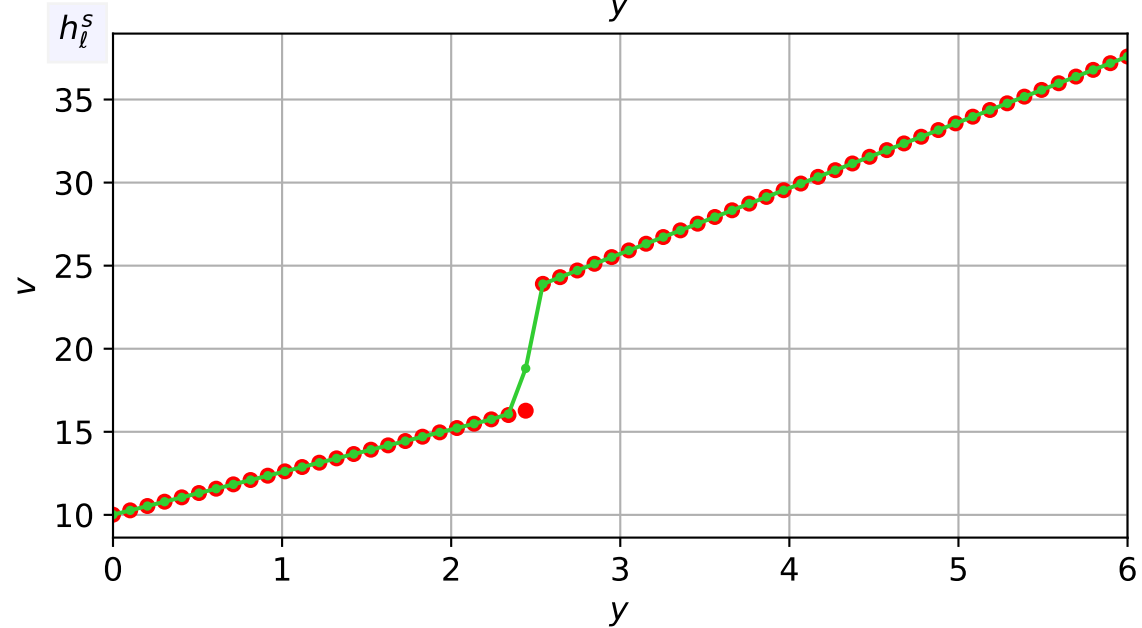
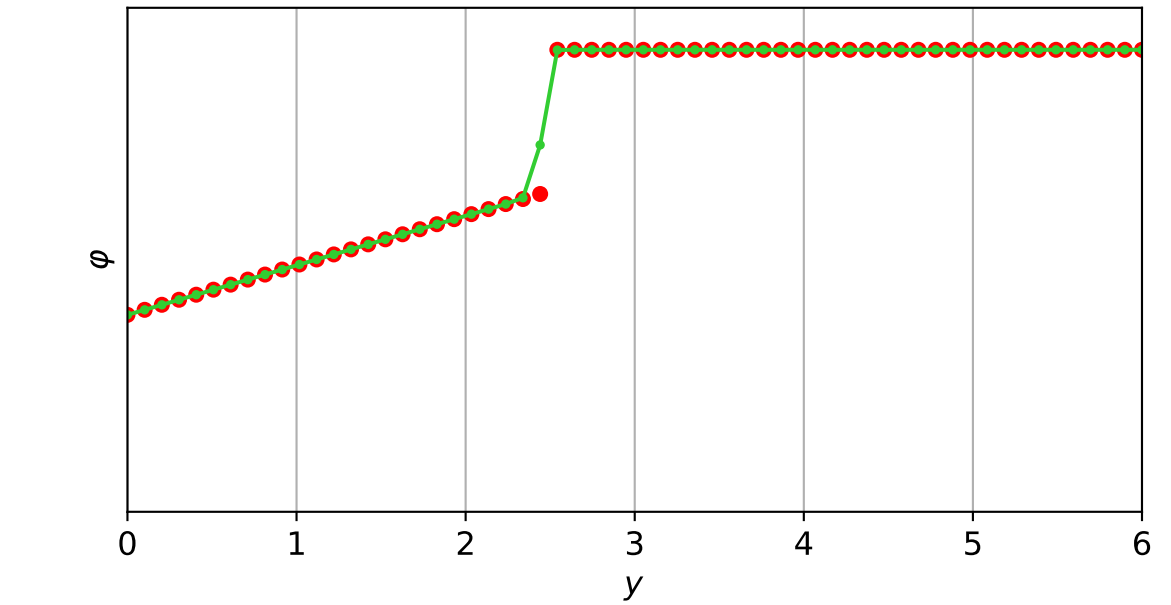
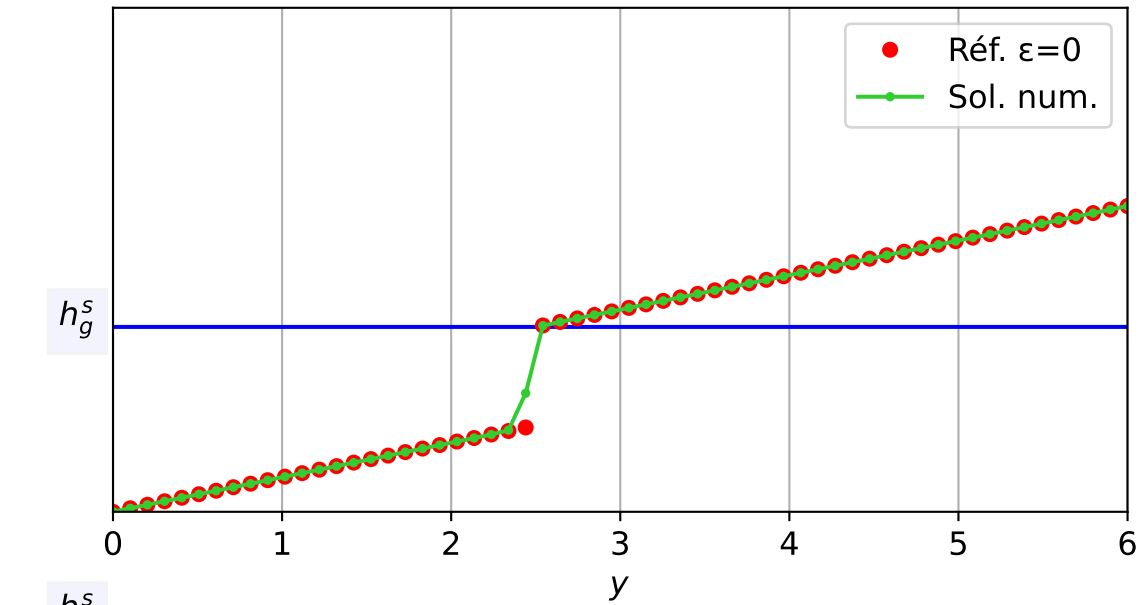
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.50028/12$  (8552 iter) — résidu eq. =  $4.85e-06$



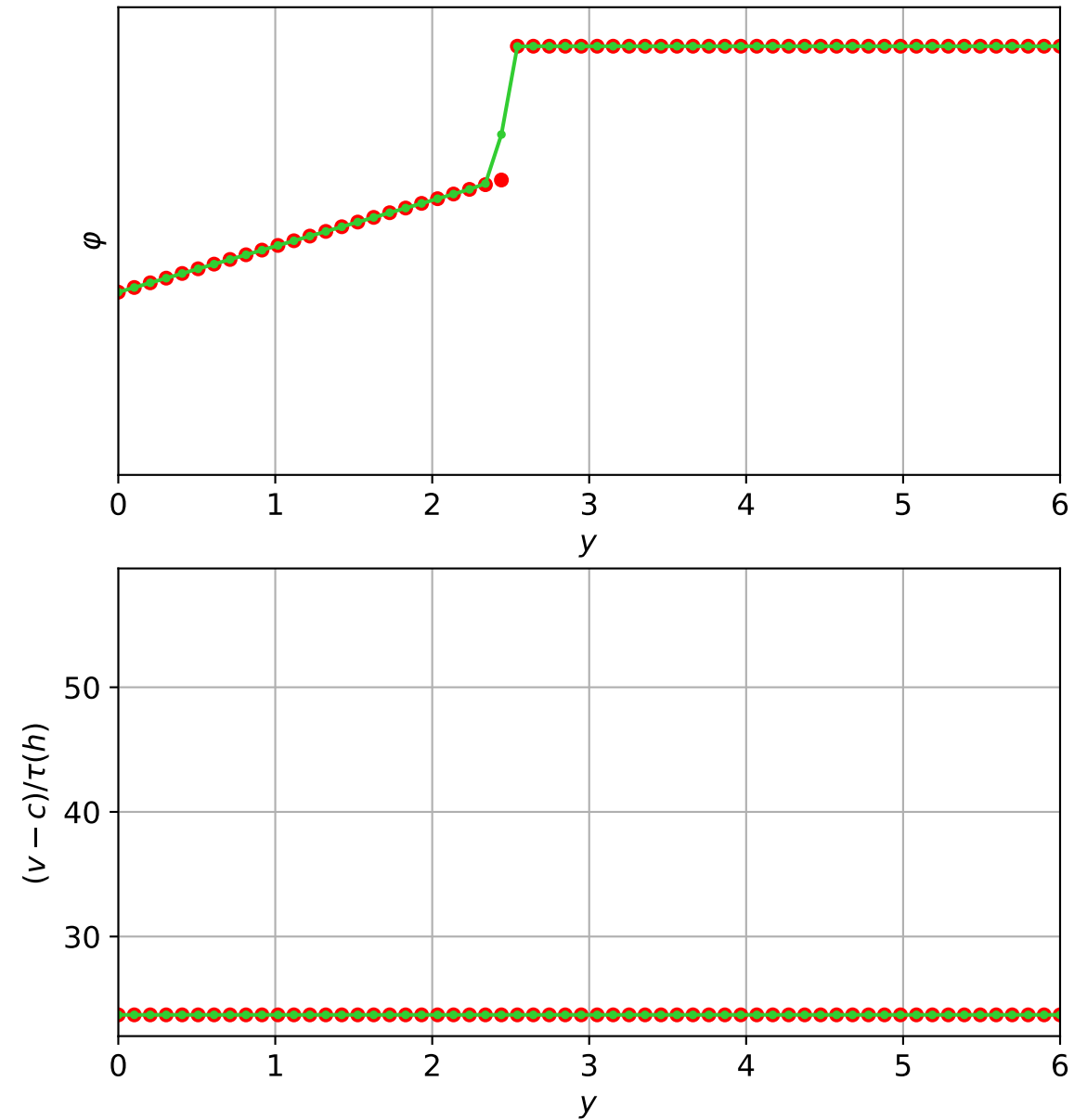
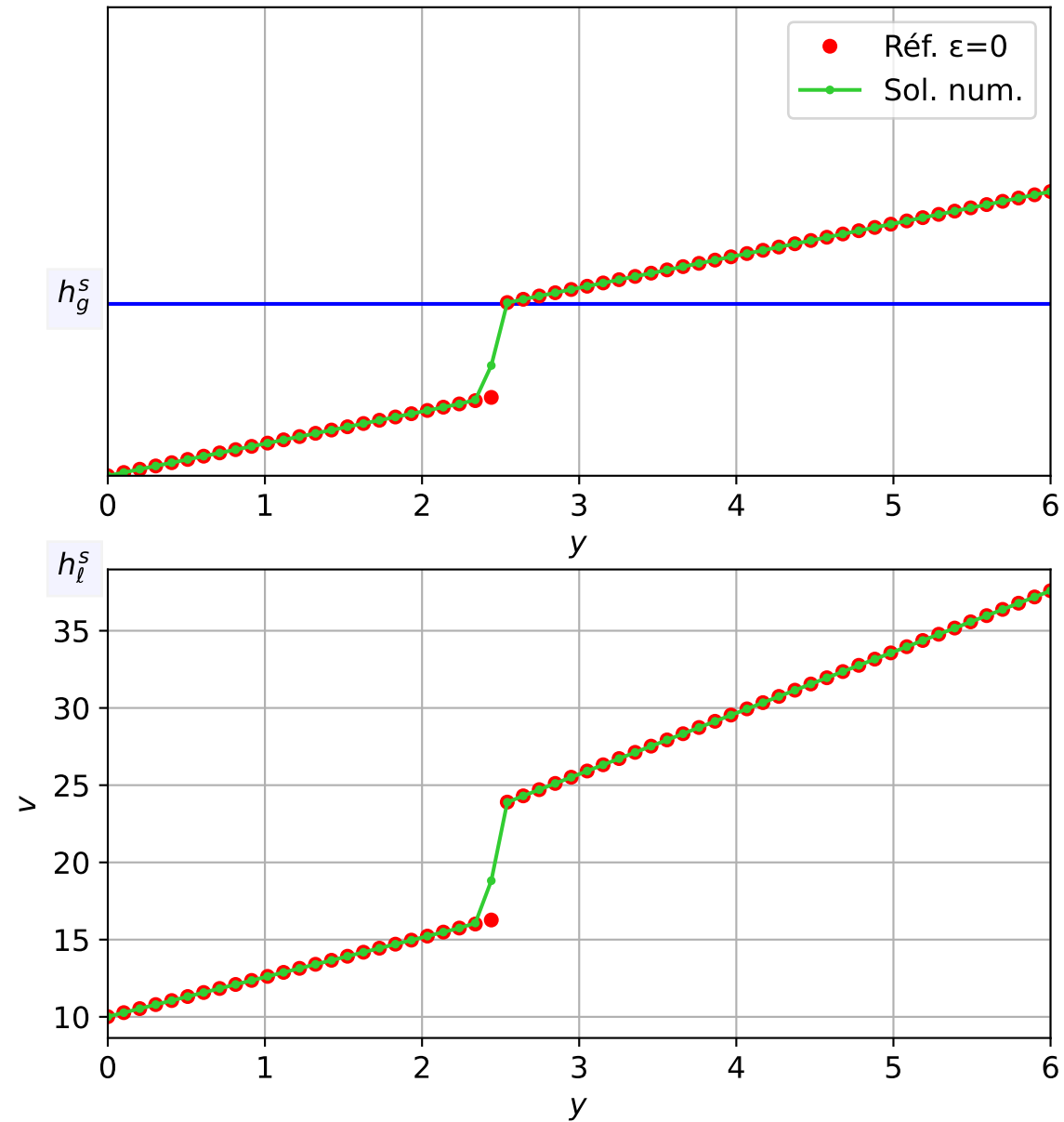
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.55142/12$  (8573 iter) — résidu eq. =  $4.28e-06$



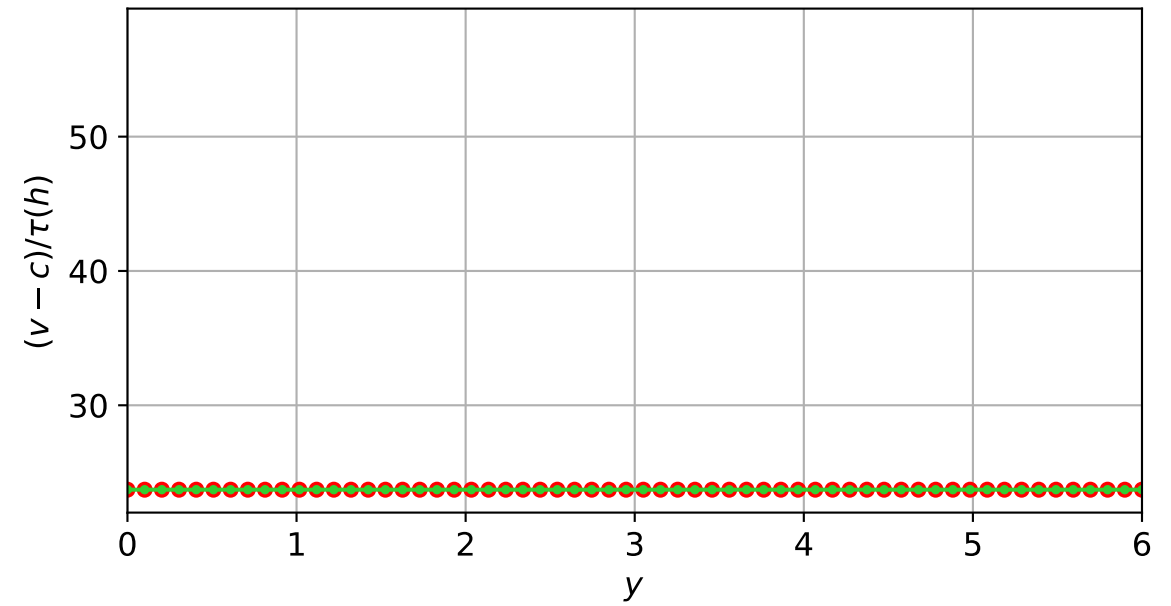
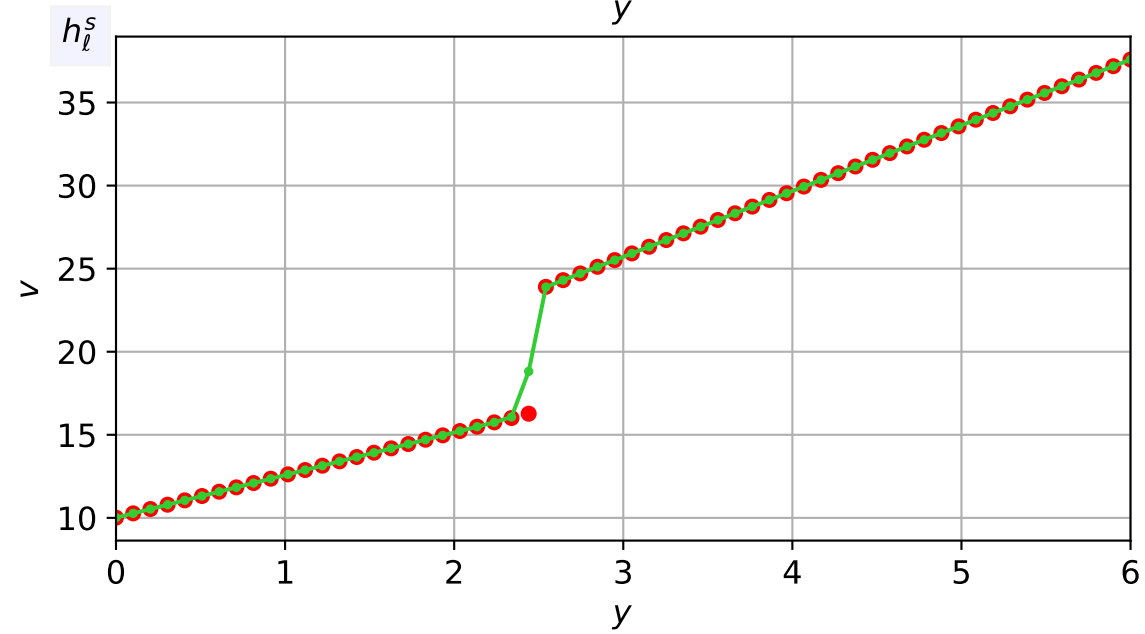
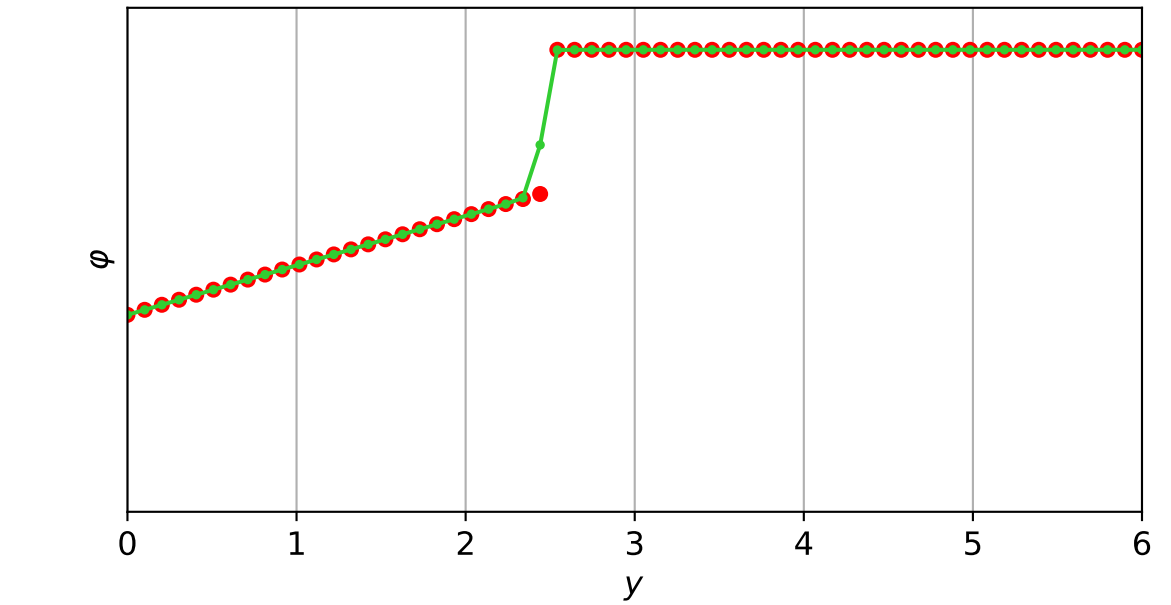
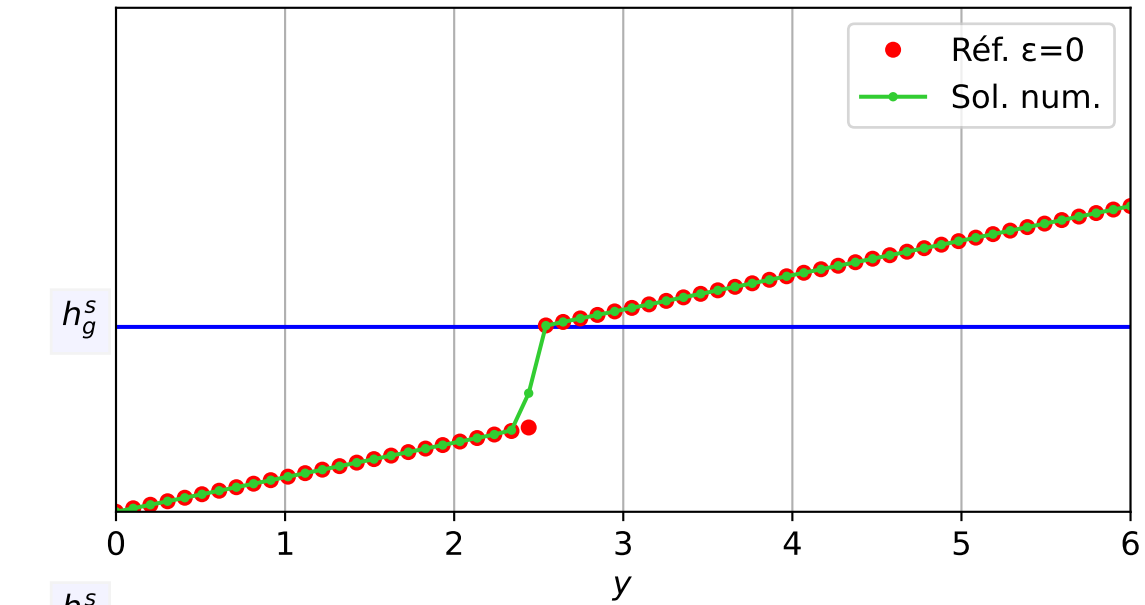
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.60012/12$  (8593 iter) — résidu eq. =  $3.81e-06$



TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.65125/12$  (8614 iter) — résidu eq. =  $3.36e-06$

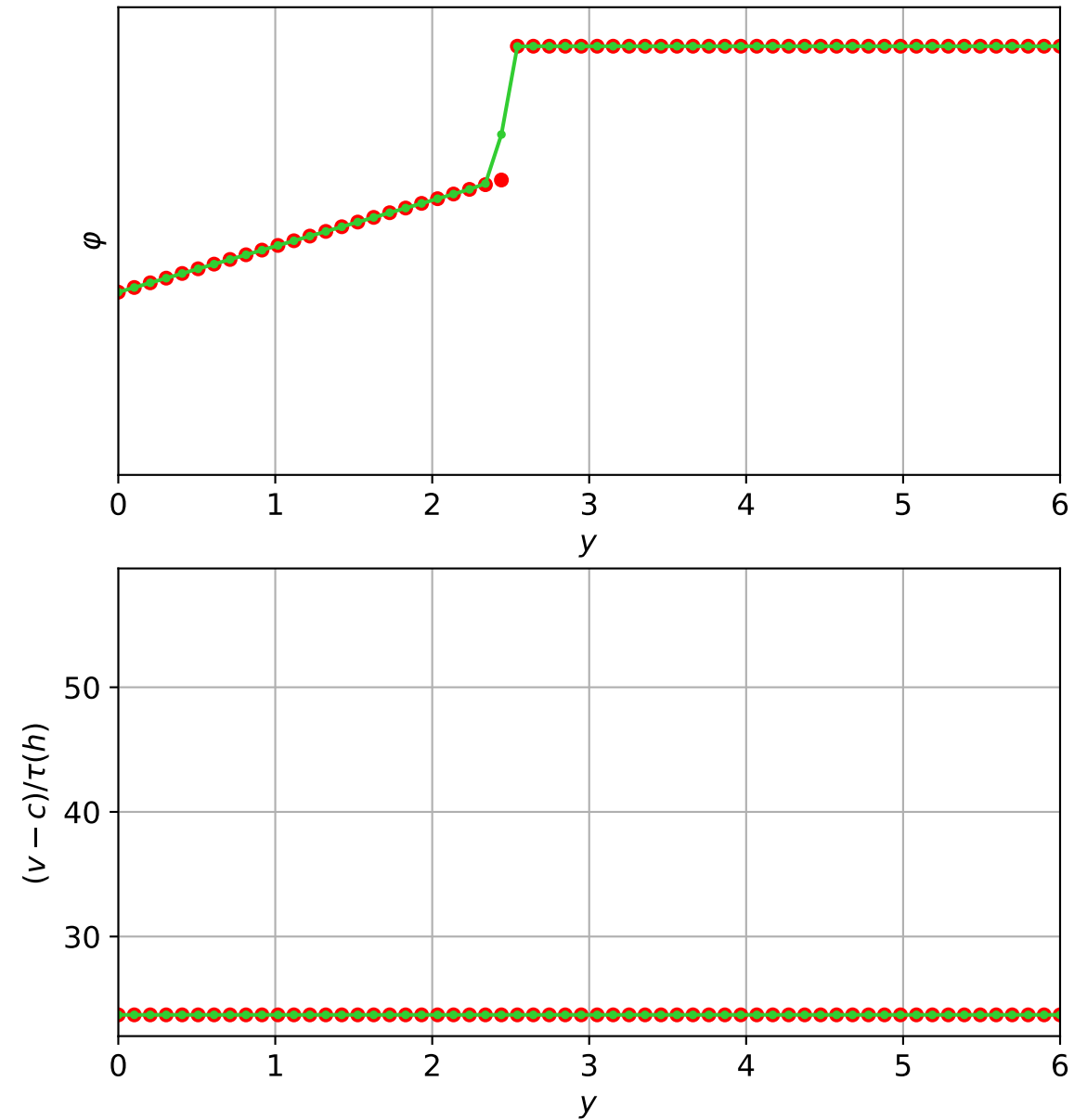
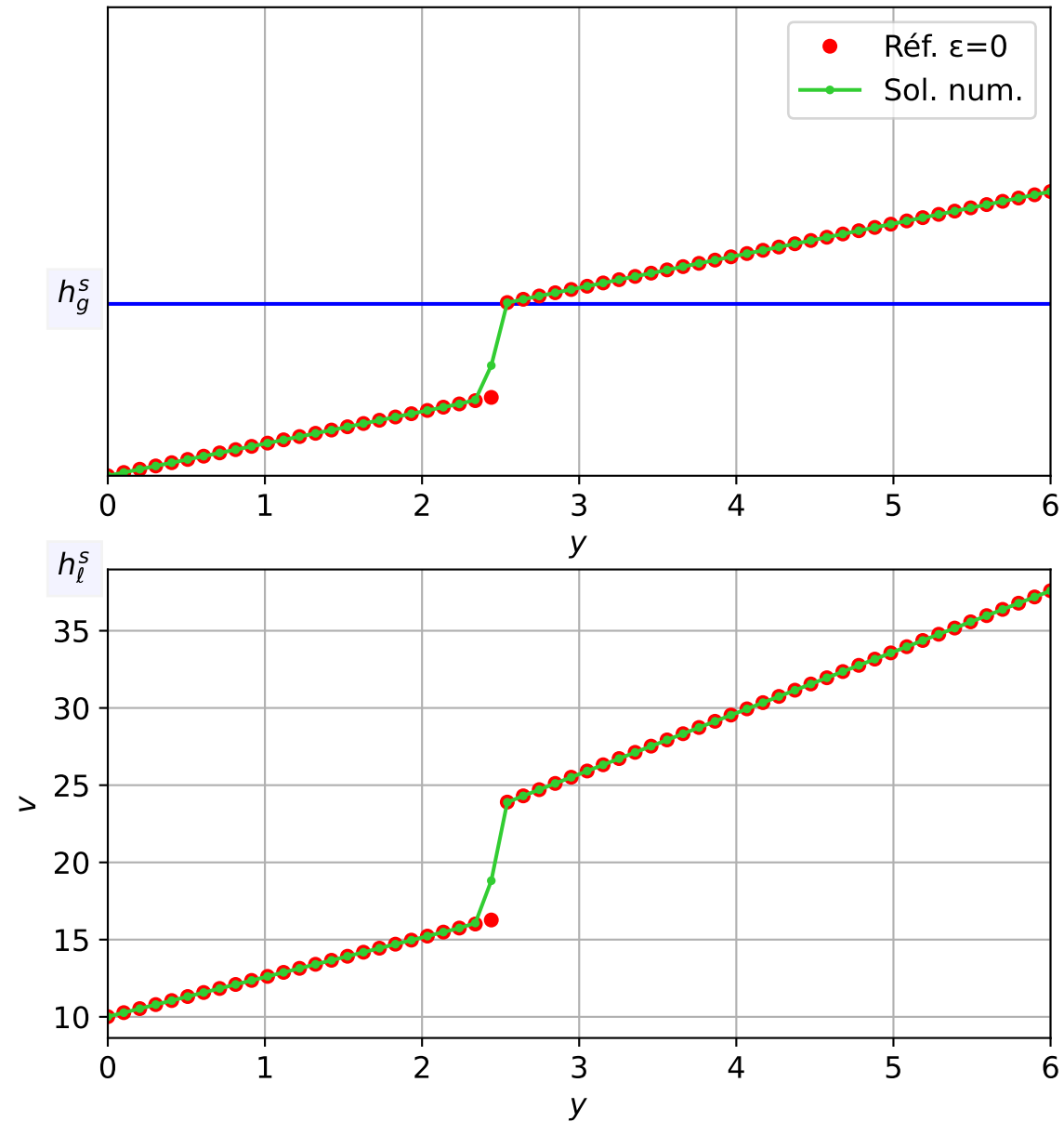


TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.70239/12$  (8635 iter) — résidu eq. =  $2.97e-06$

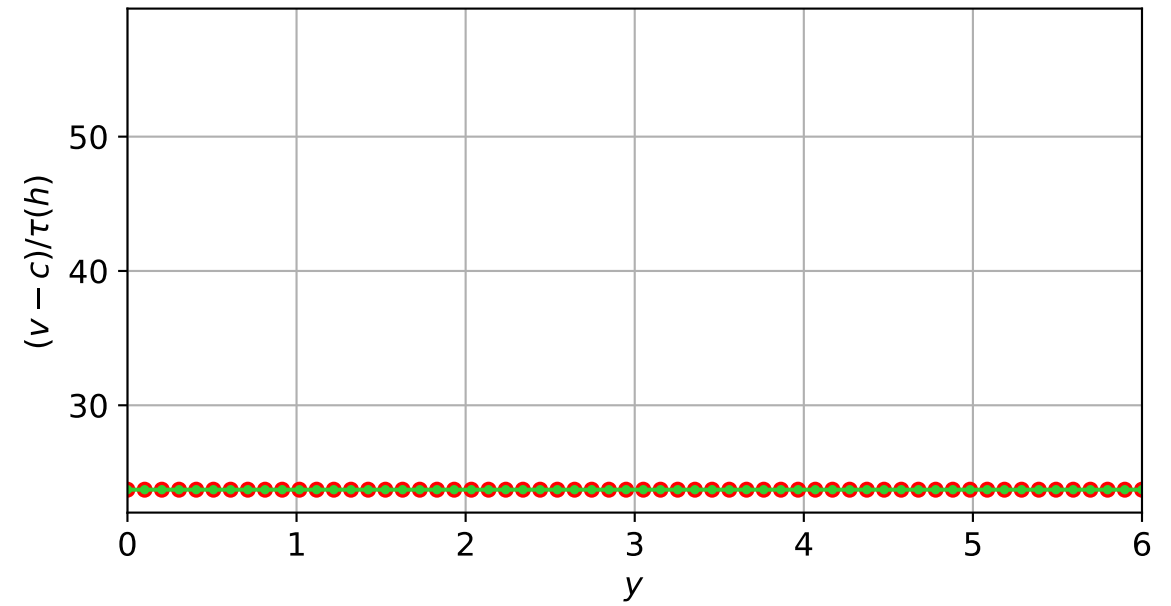
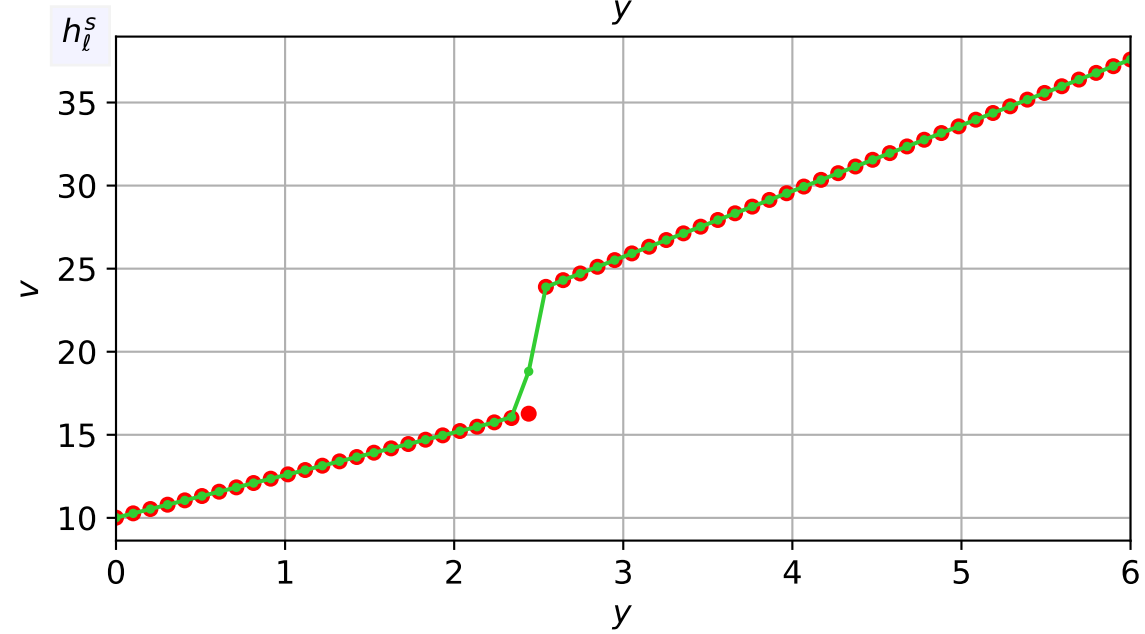
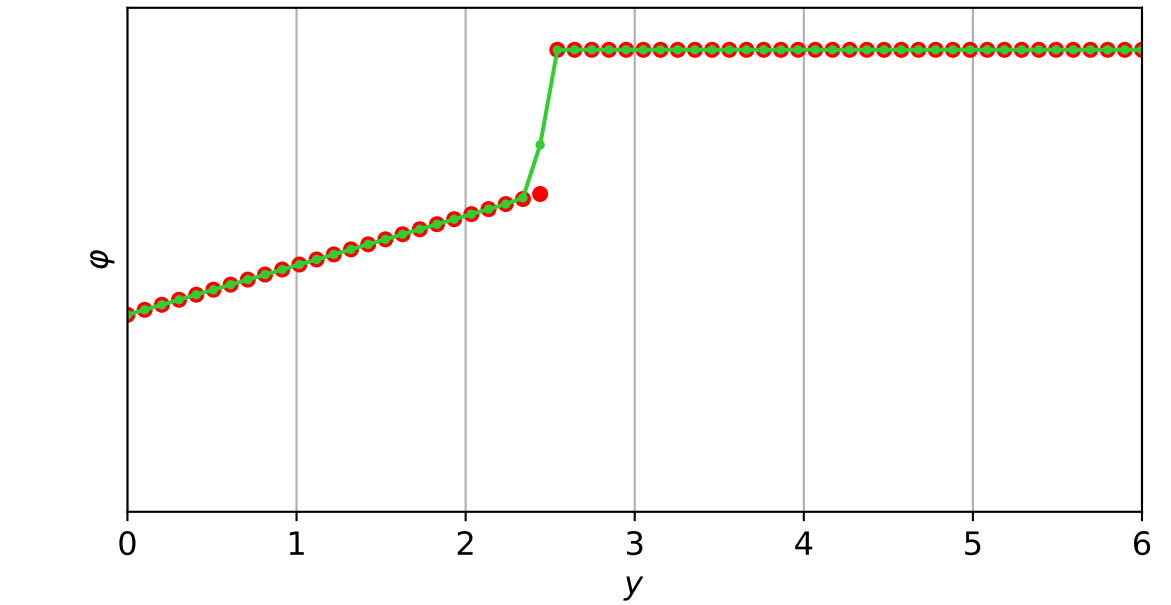
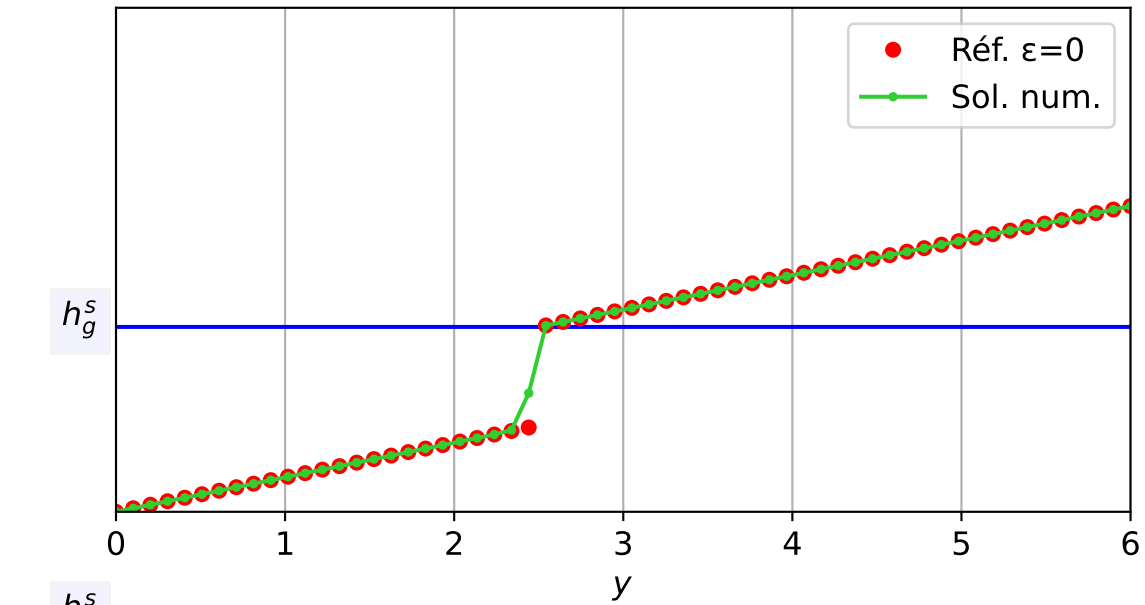




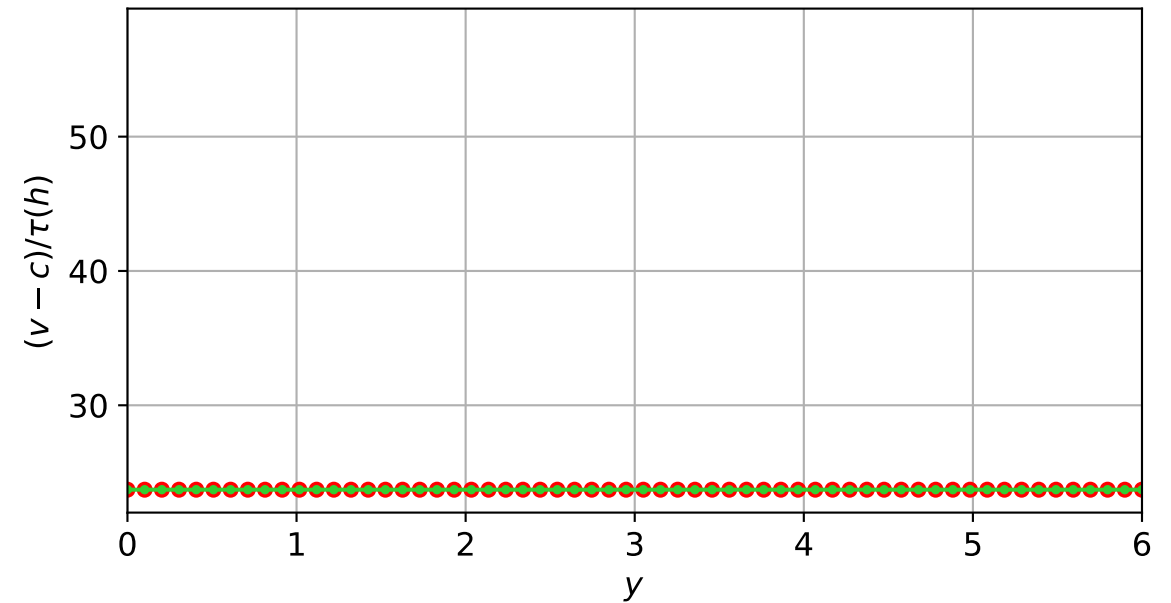
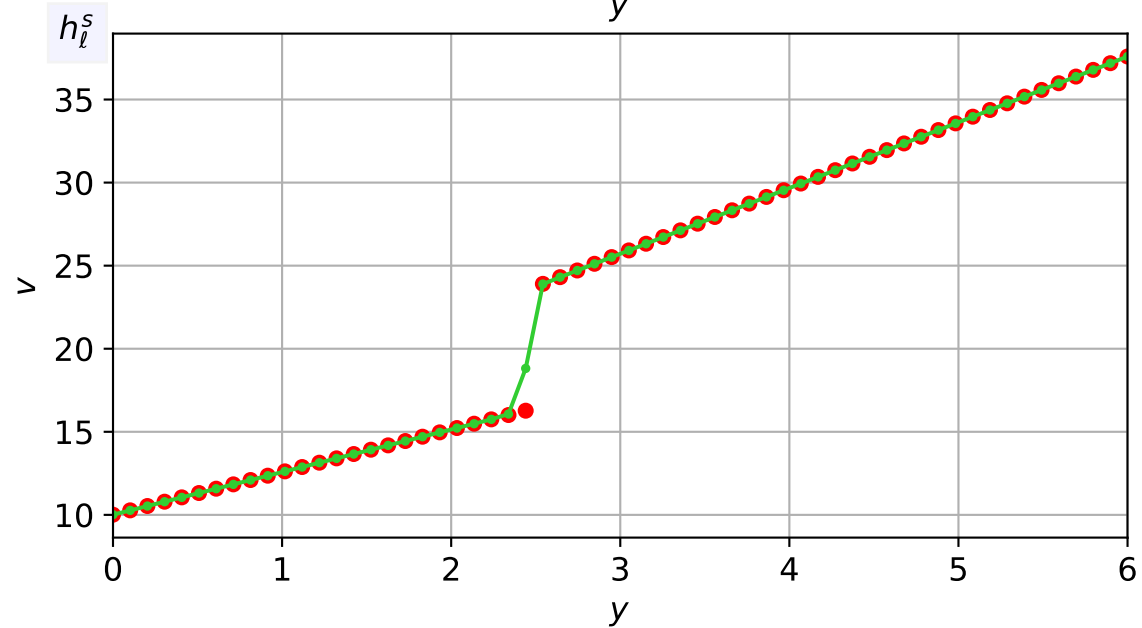
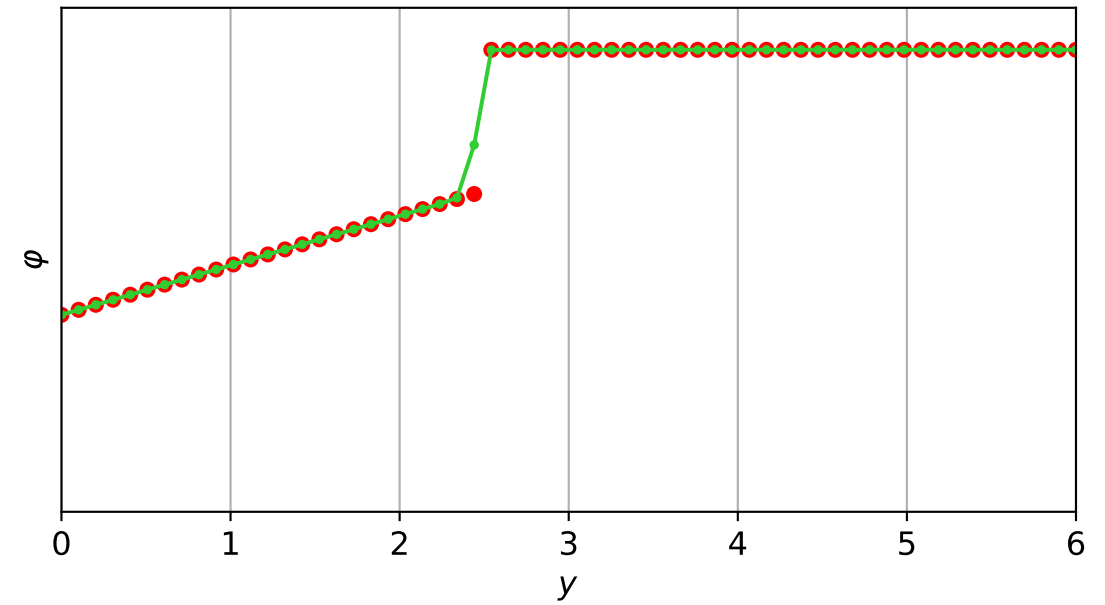
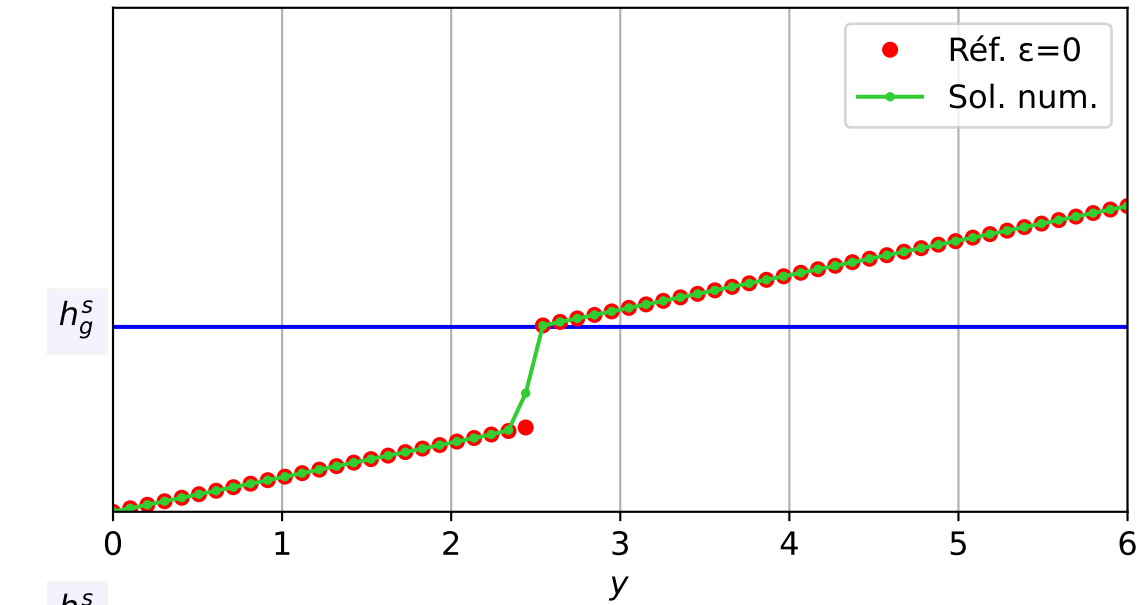
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.75109/12$  (8655 iter) — résidu eq. =  $2.64e-06$



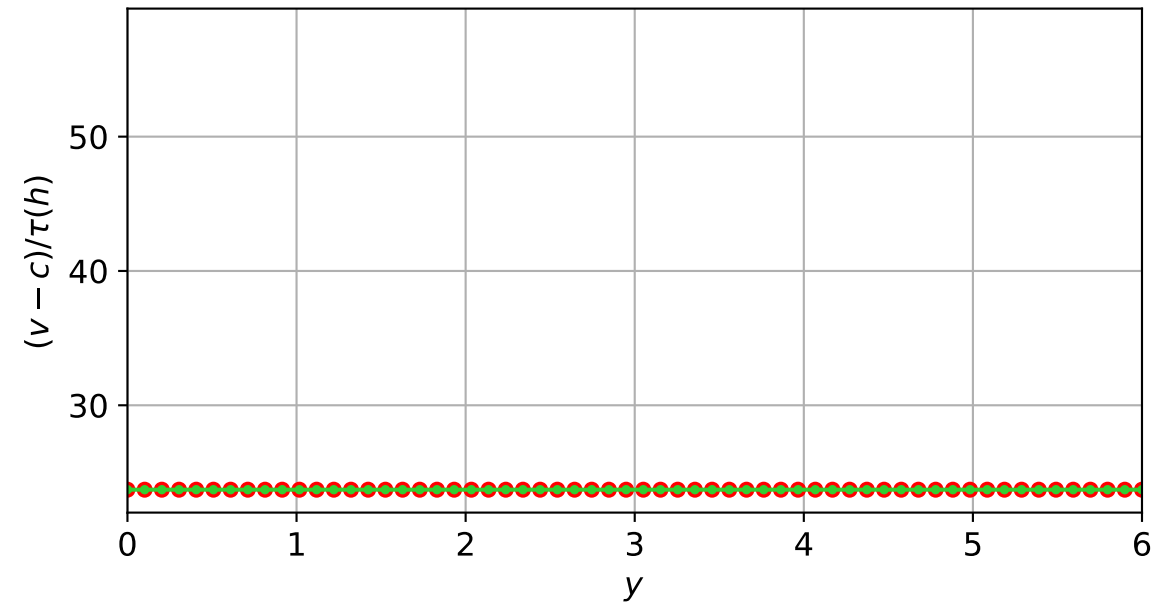
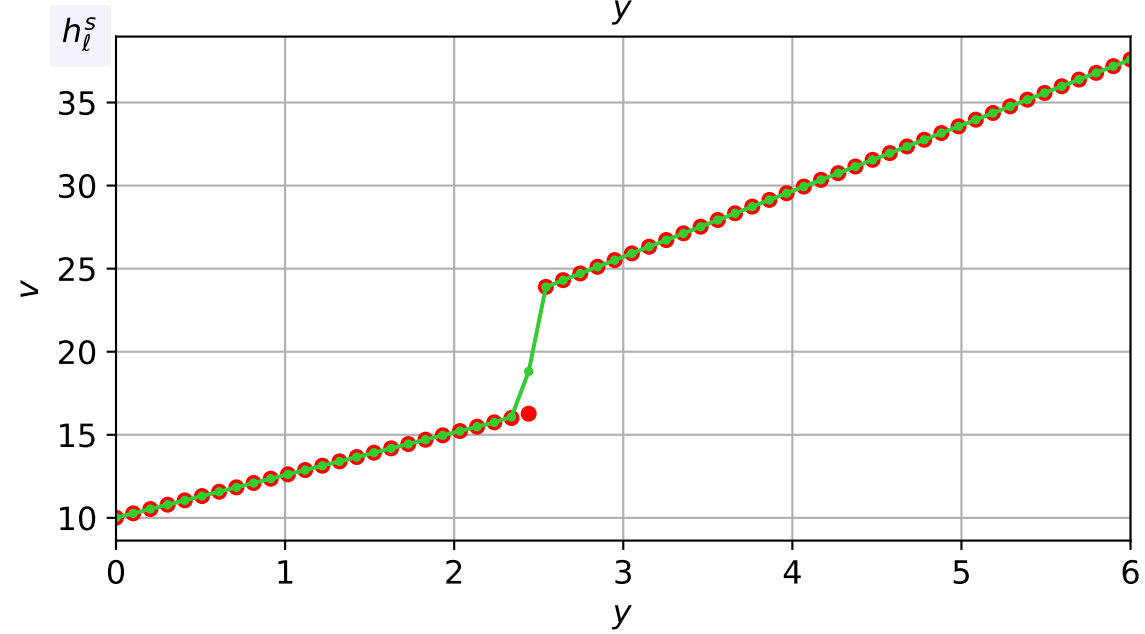
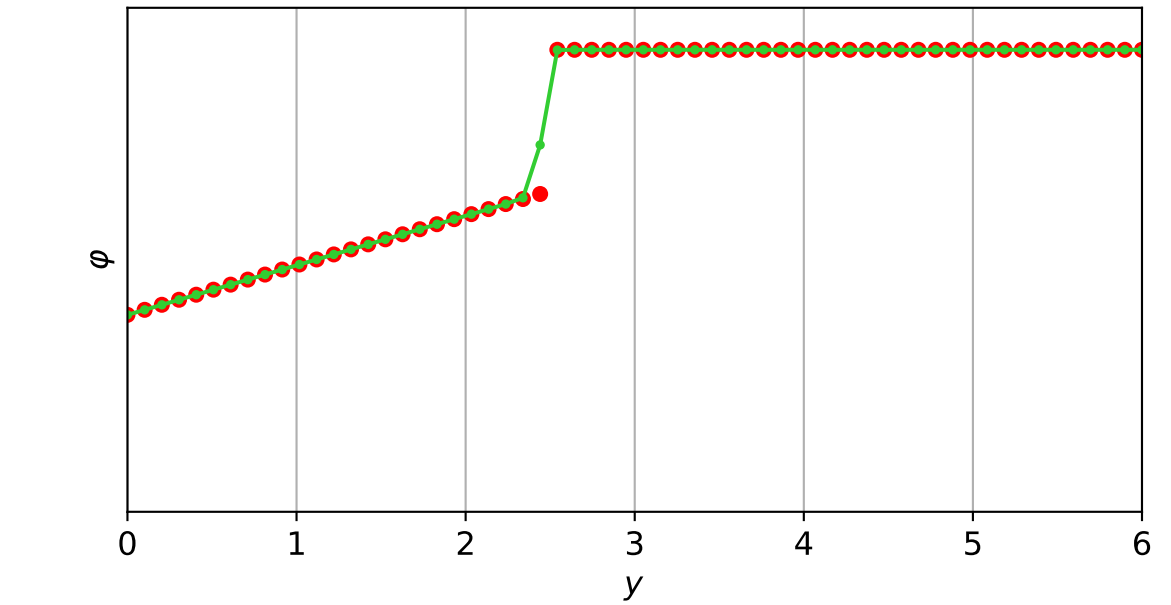
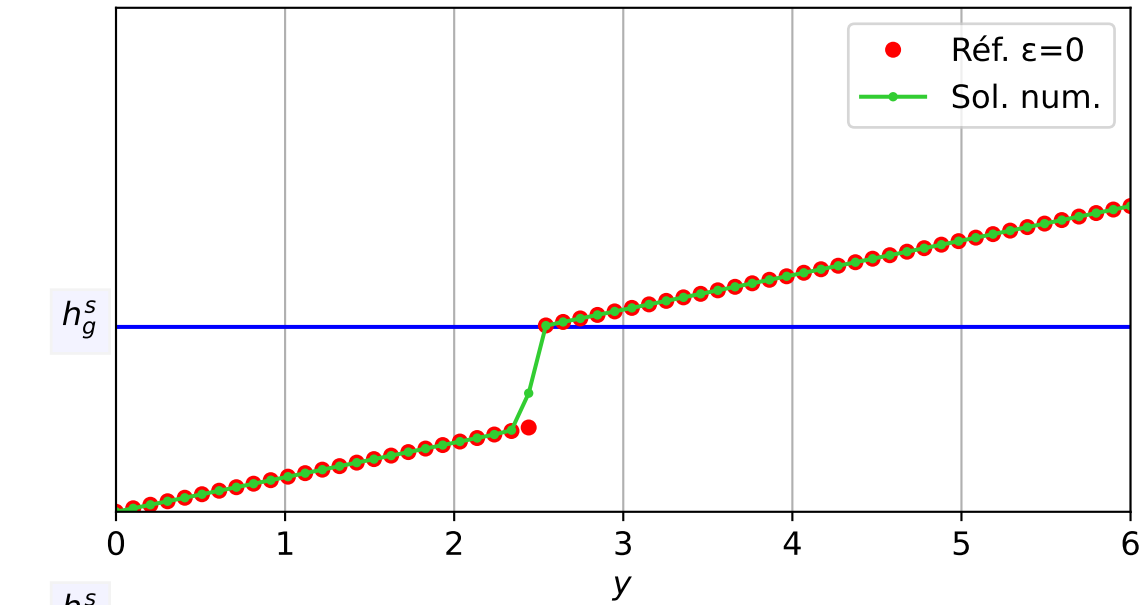
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.80222/12$  (8676 iter) — résidu eq. =  $2.33e-06$



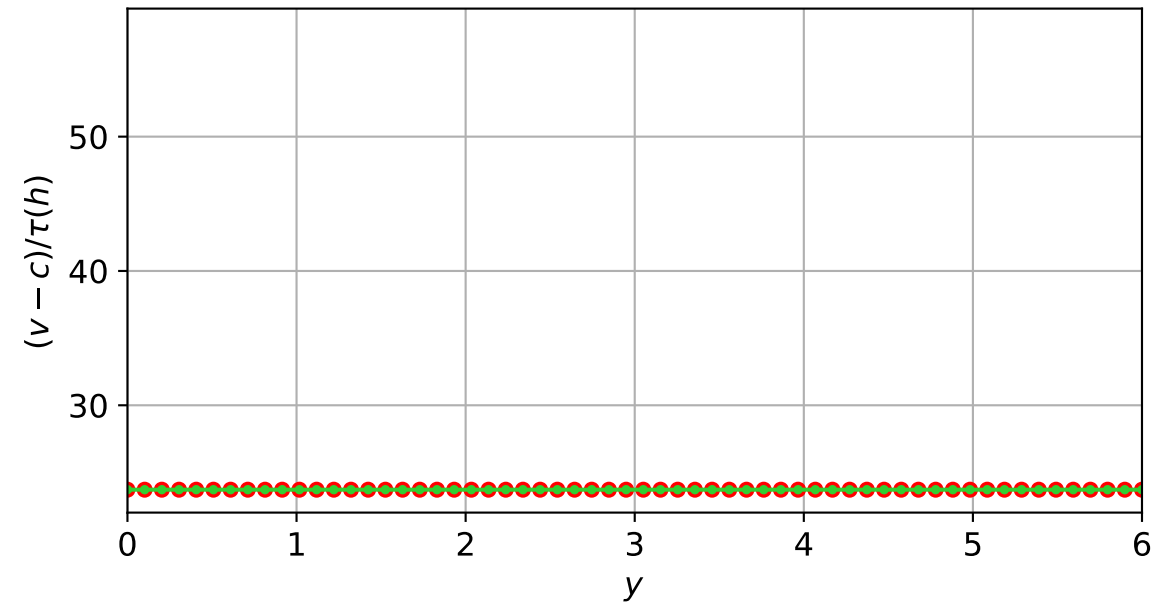
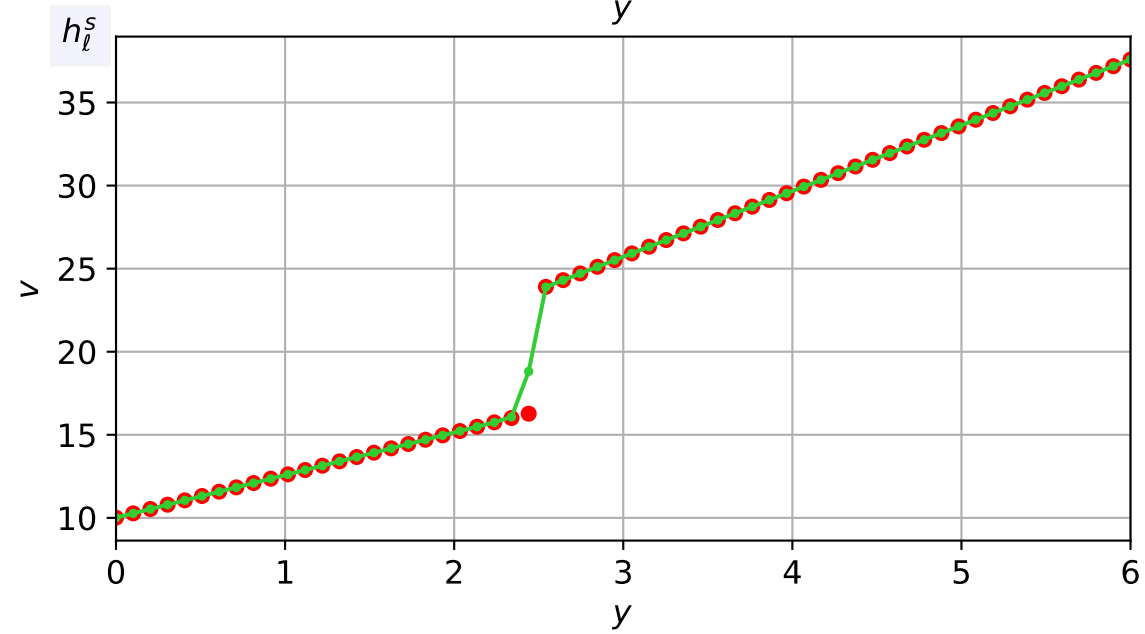
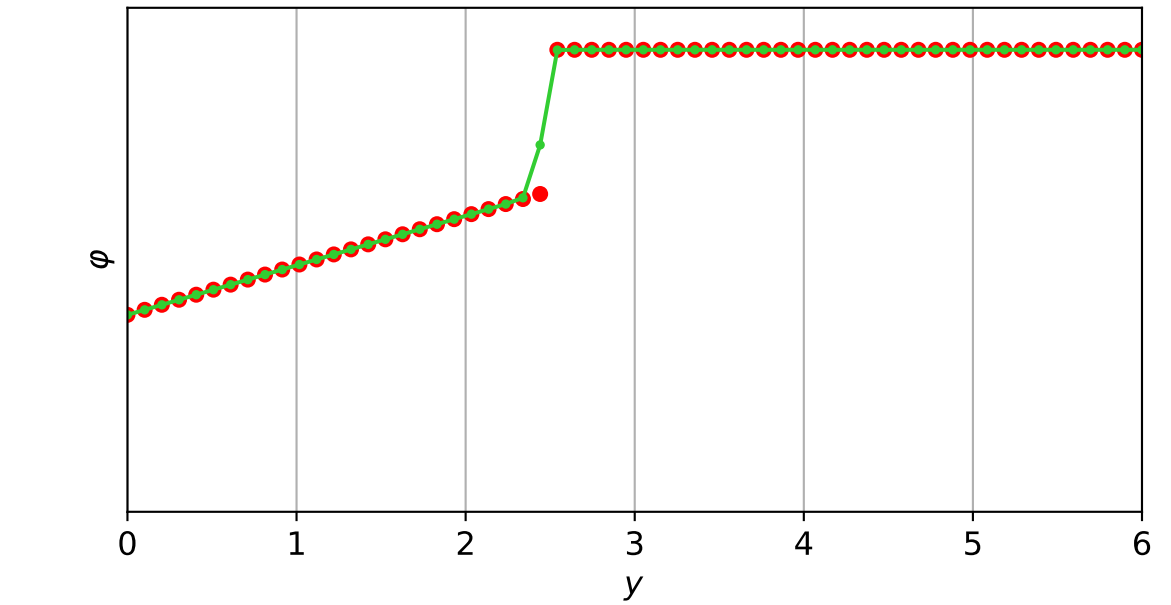
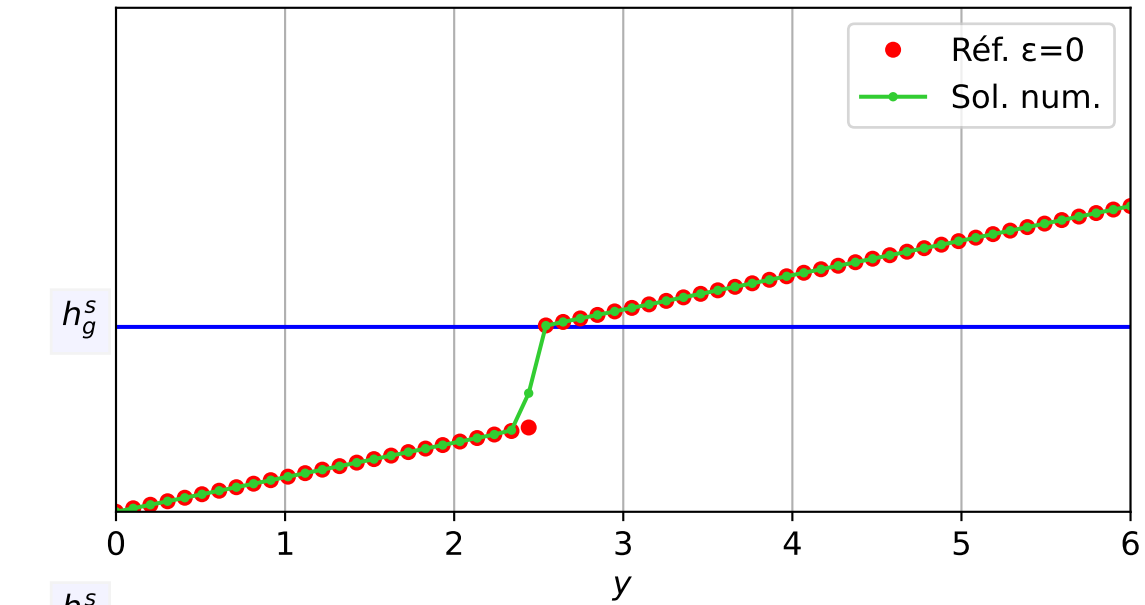
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.85092/12$  (8696 iter) — résidu eq. =  $2.07e-06$



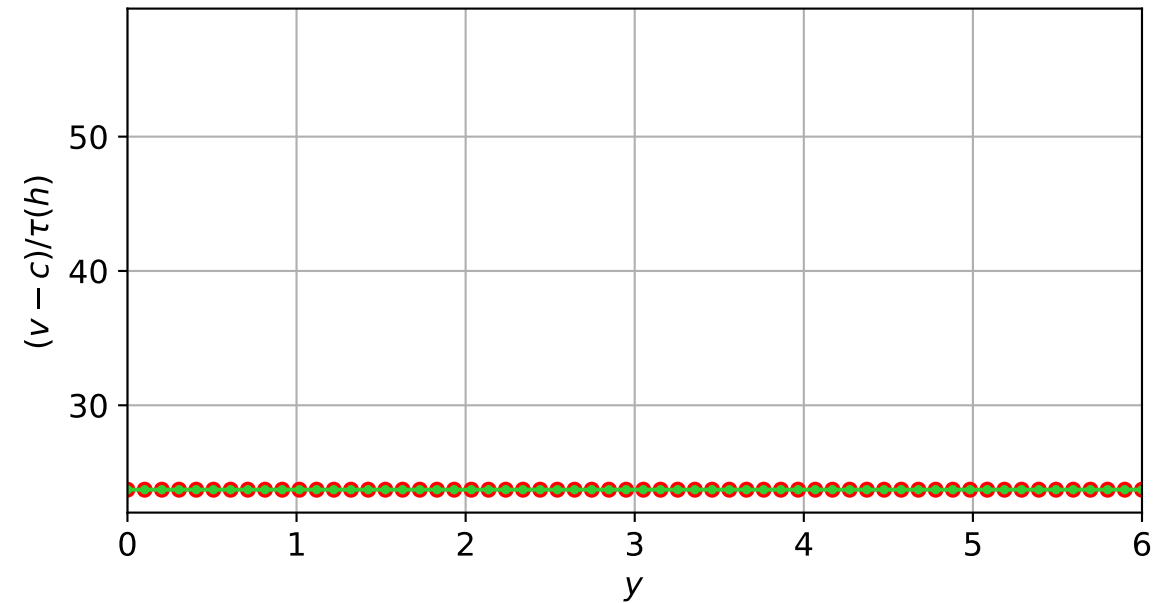
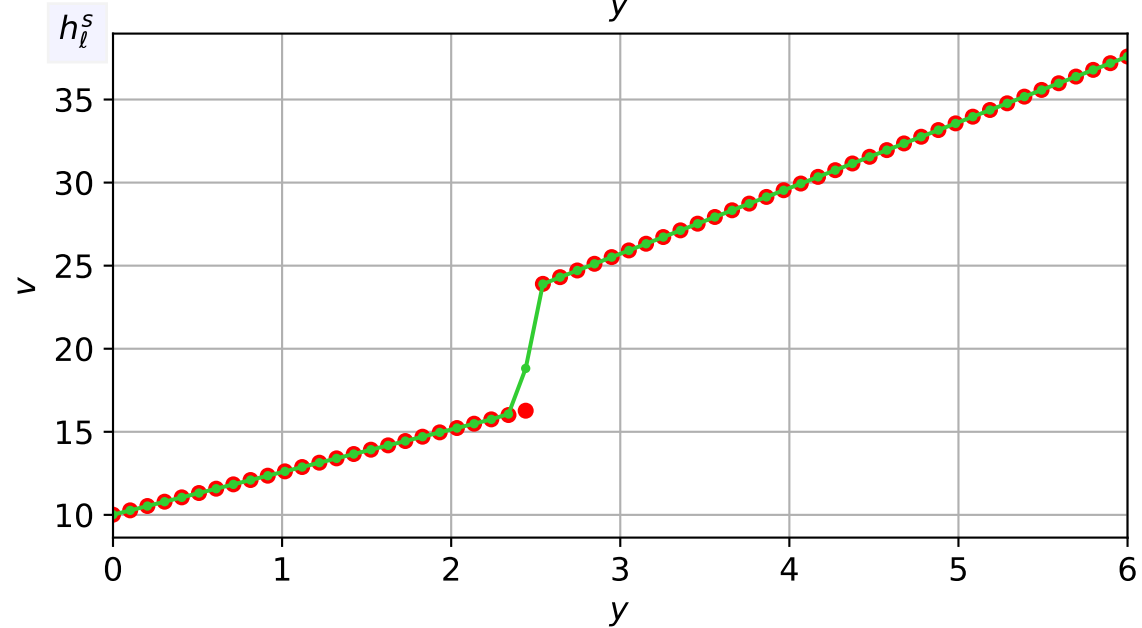
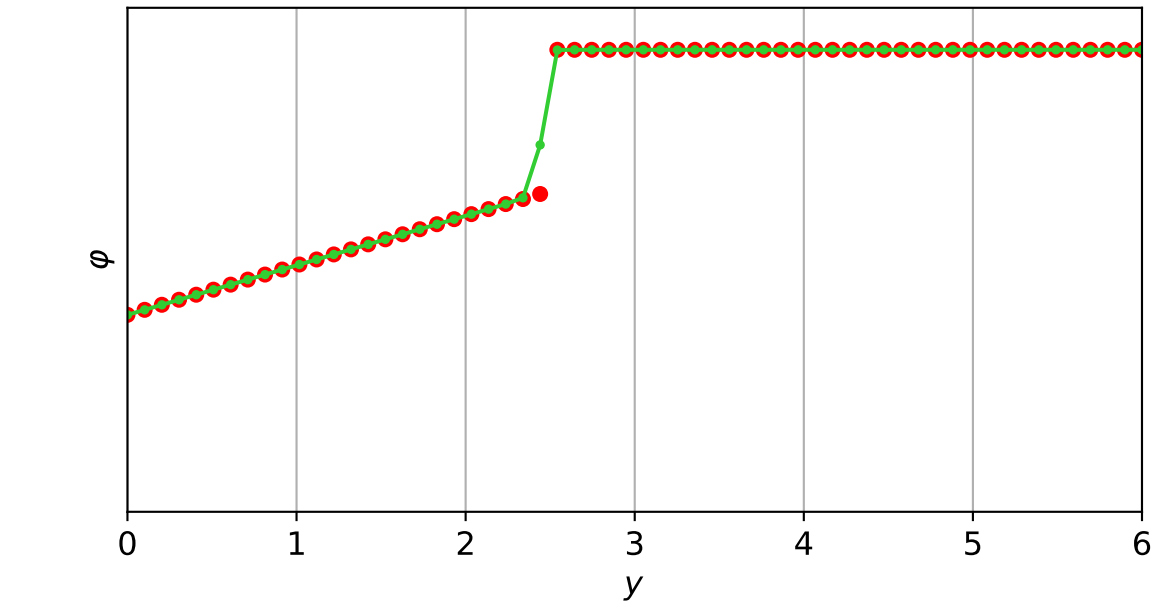
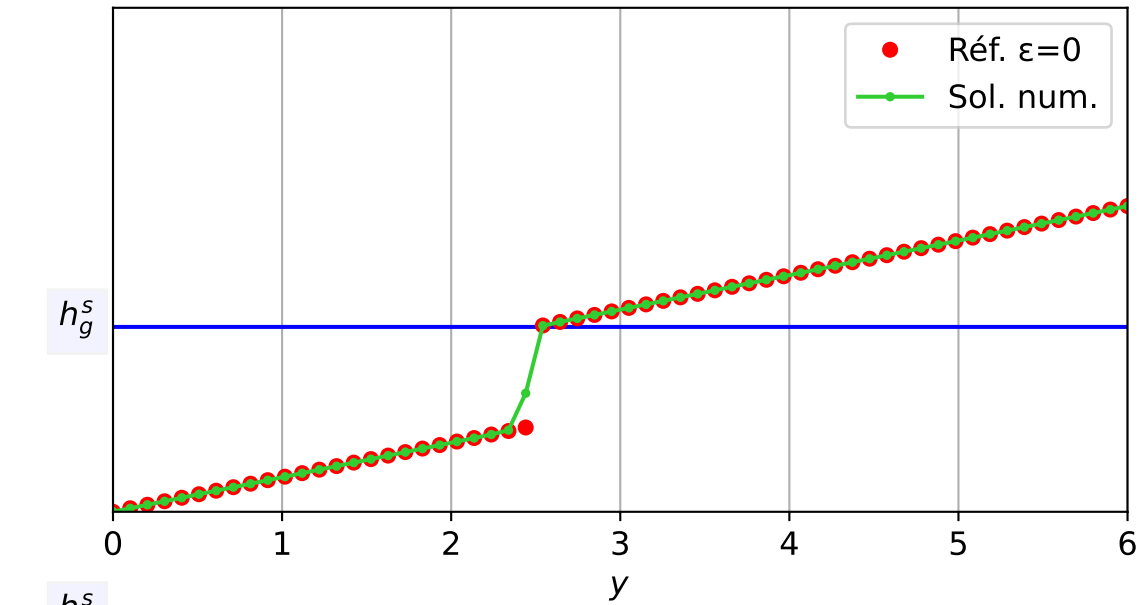
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.90206/12$  (8717 iter) — résidu eq. =  $1.83e-06$



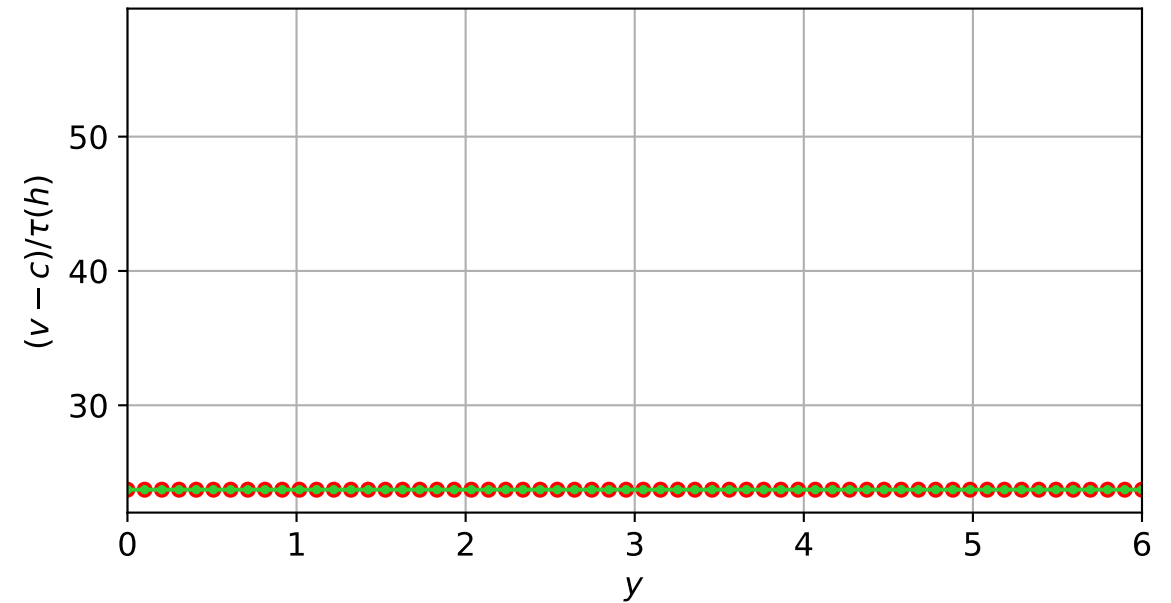
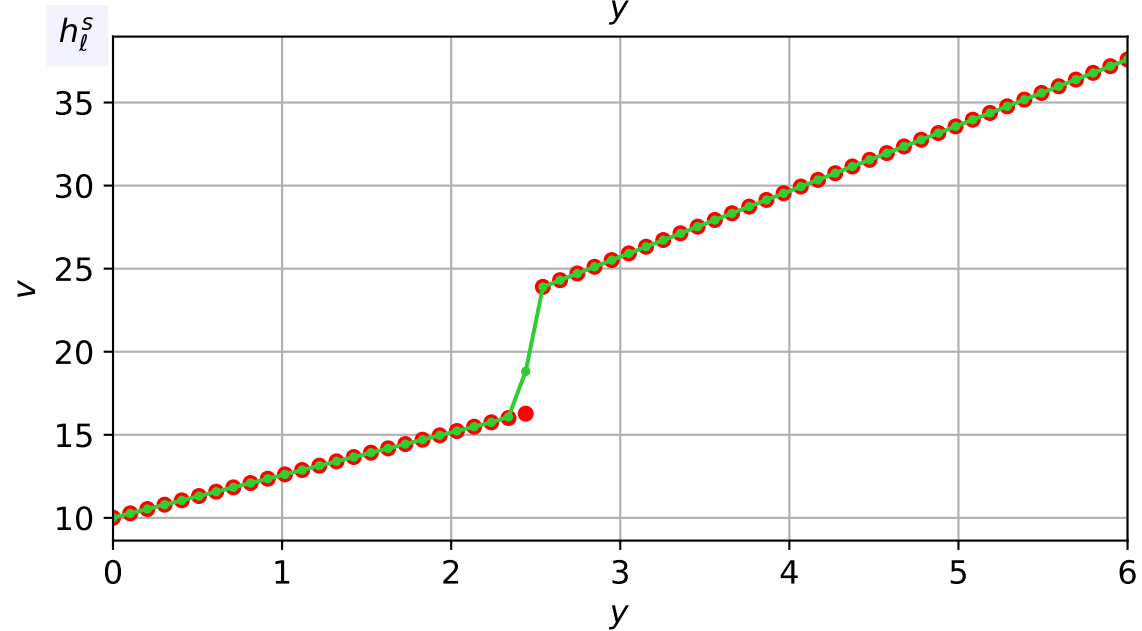
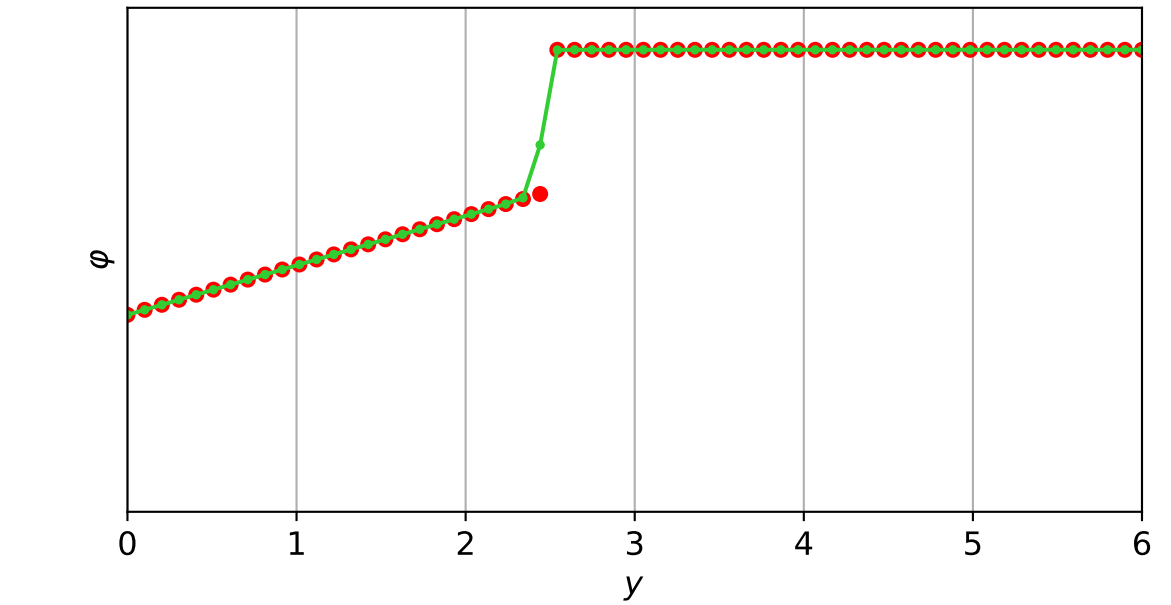
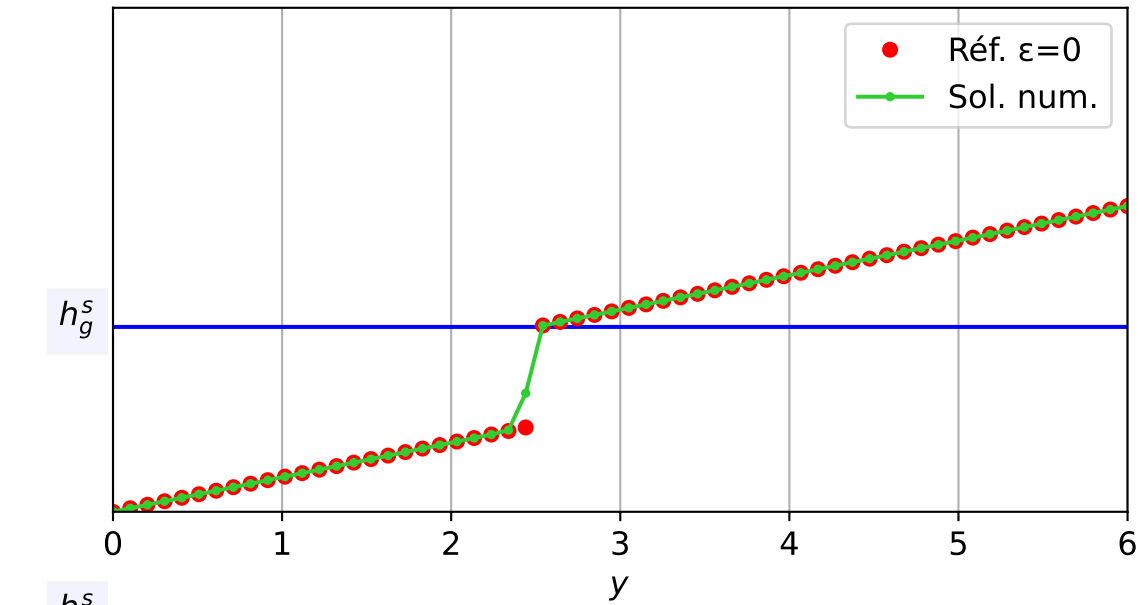
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 4.95076/12$  (8737 iter) — résidu eq. =  $1.63e-06$



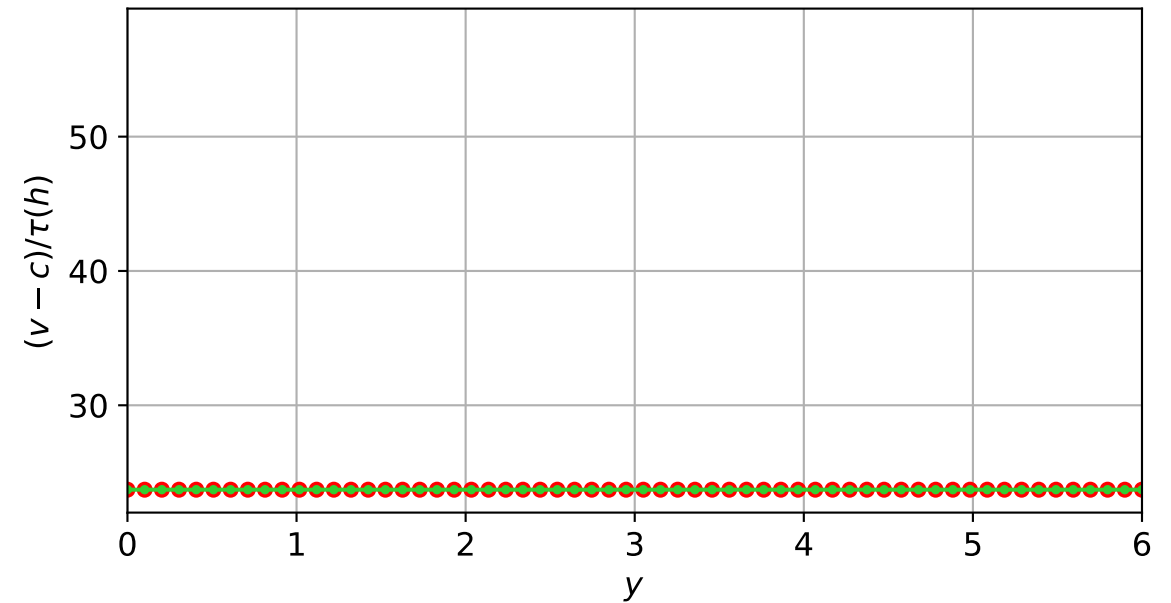
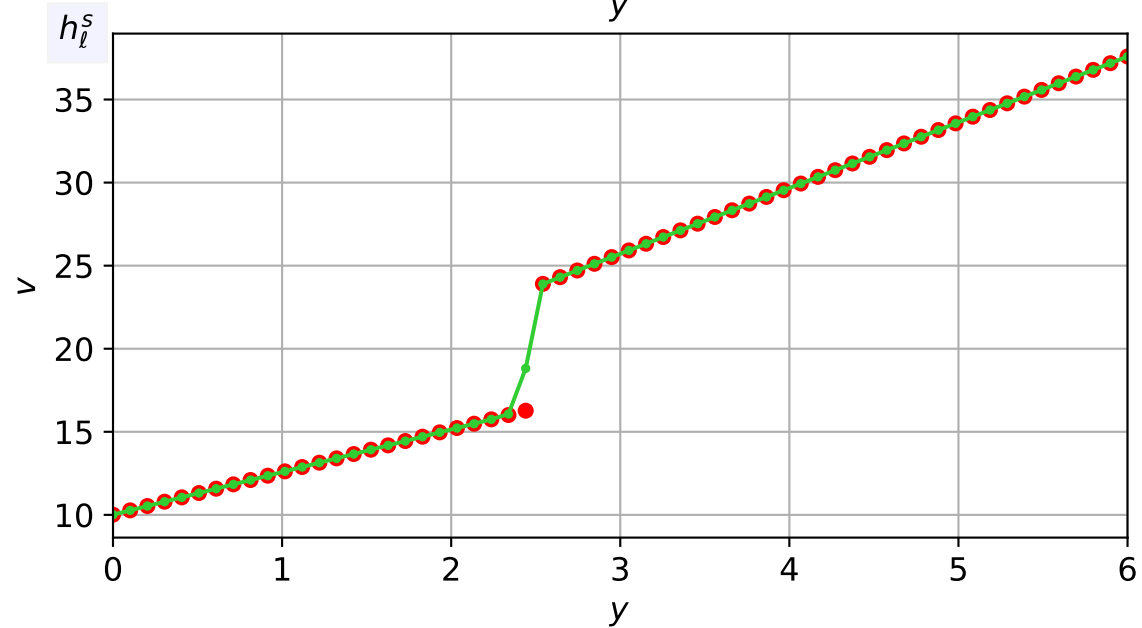
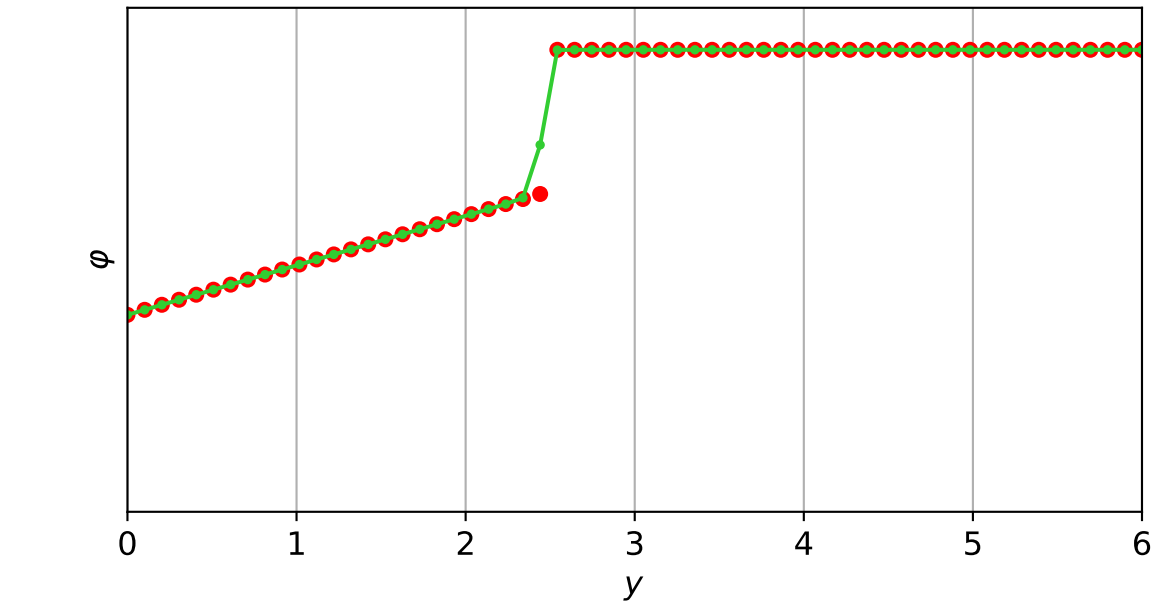
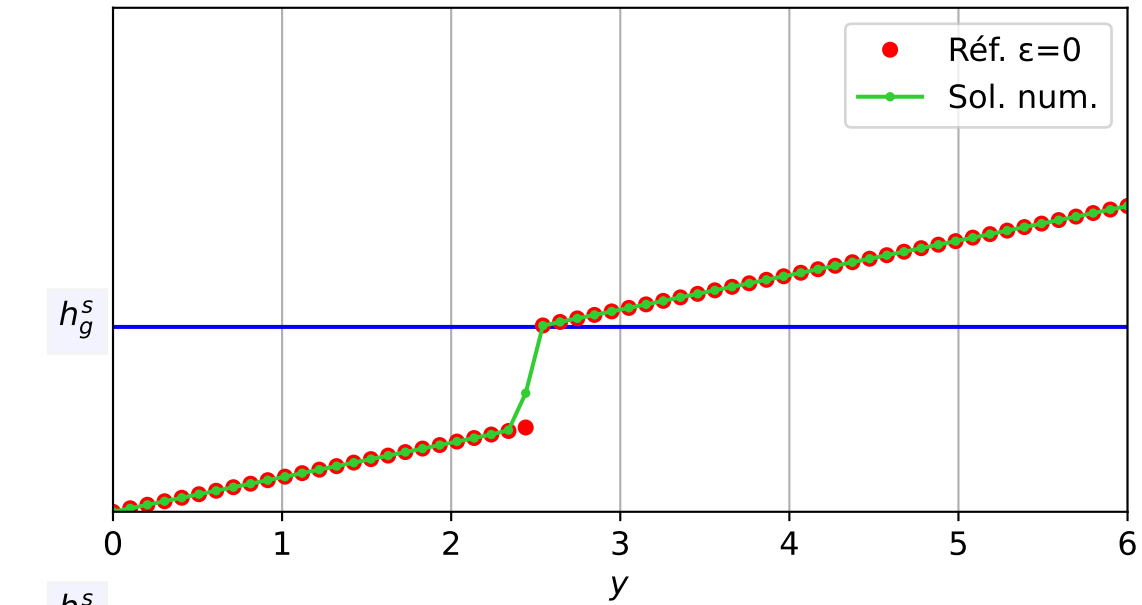
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.0019/12$  (8758 iter) — résidu eq. =  $1.44e-06$



TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.0506/12$  (8778 iter) — résidu eq. =  $1.28e-06$

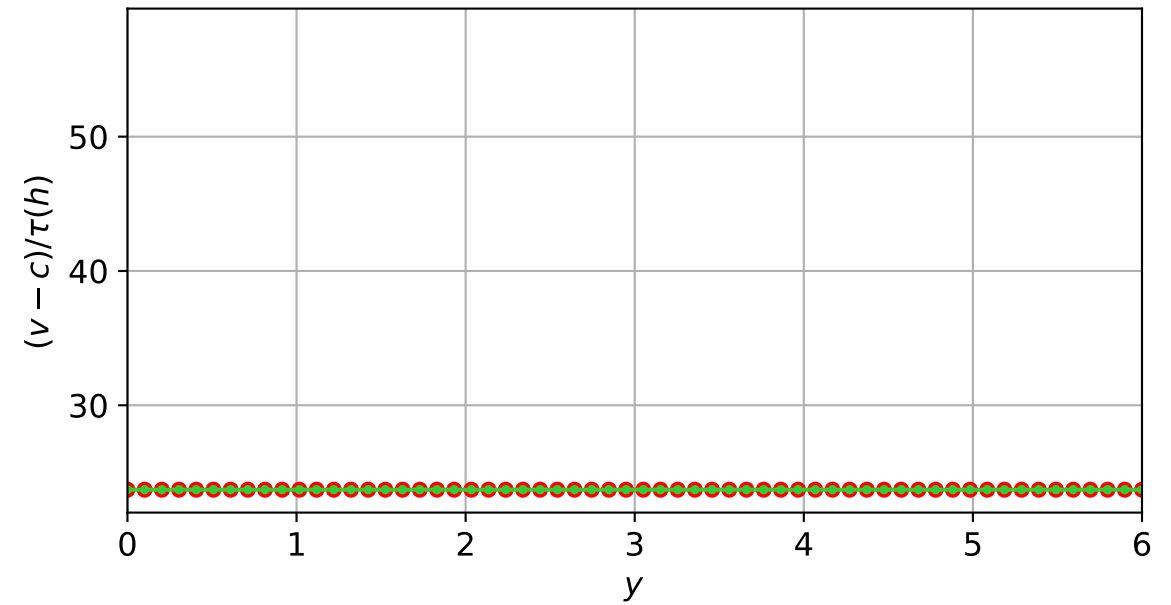
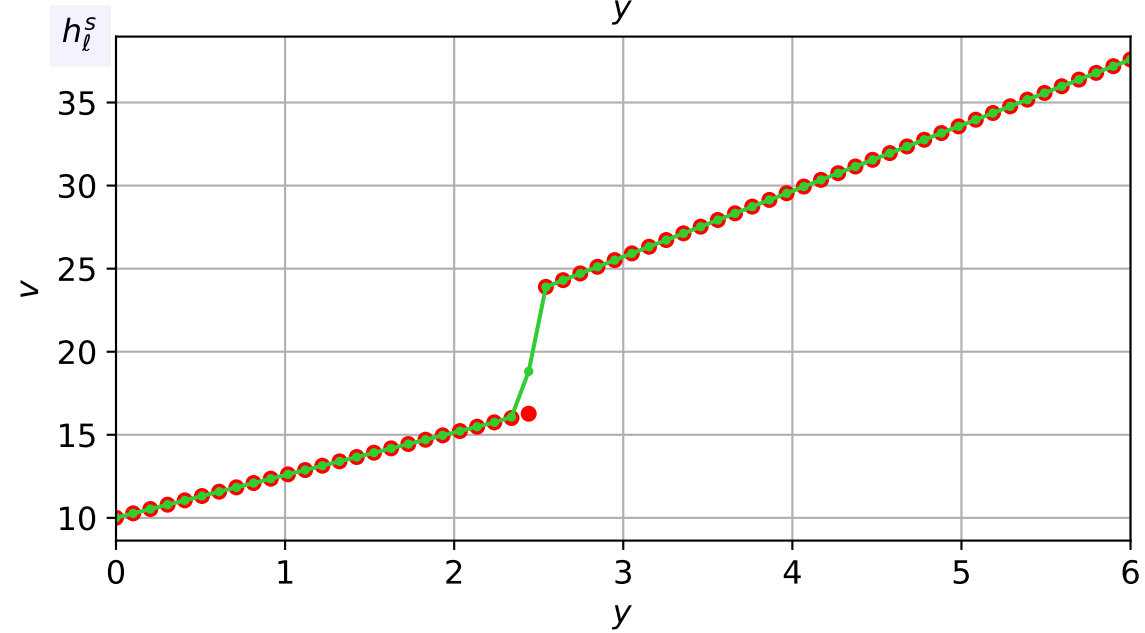
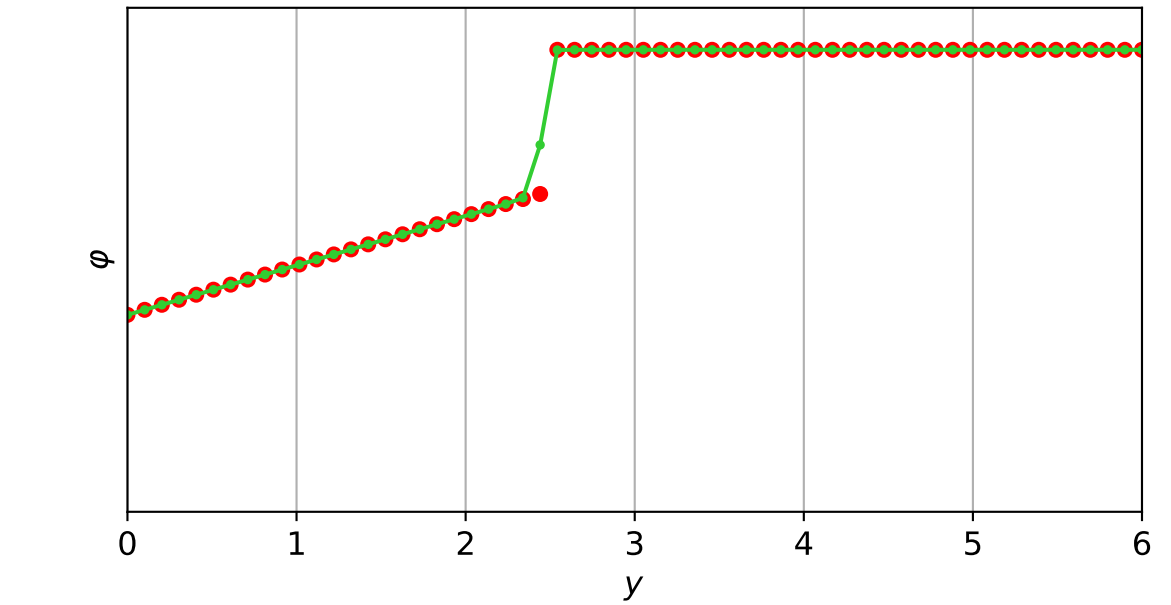
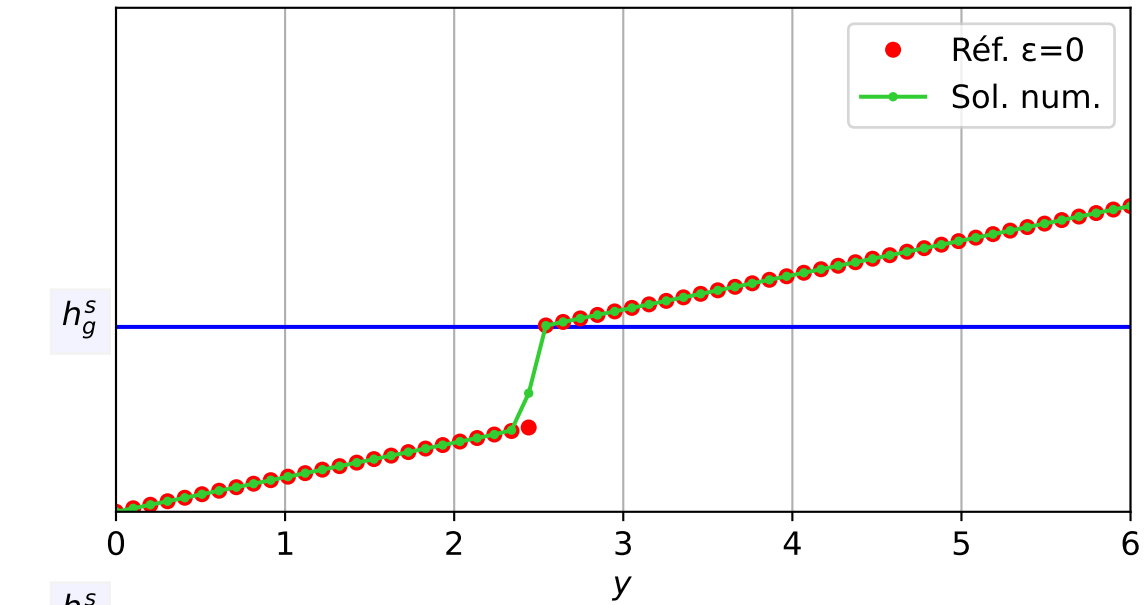


TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.10173/12$  (8799 iter) — résidu eq. =  $1.13e-06$

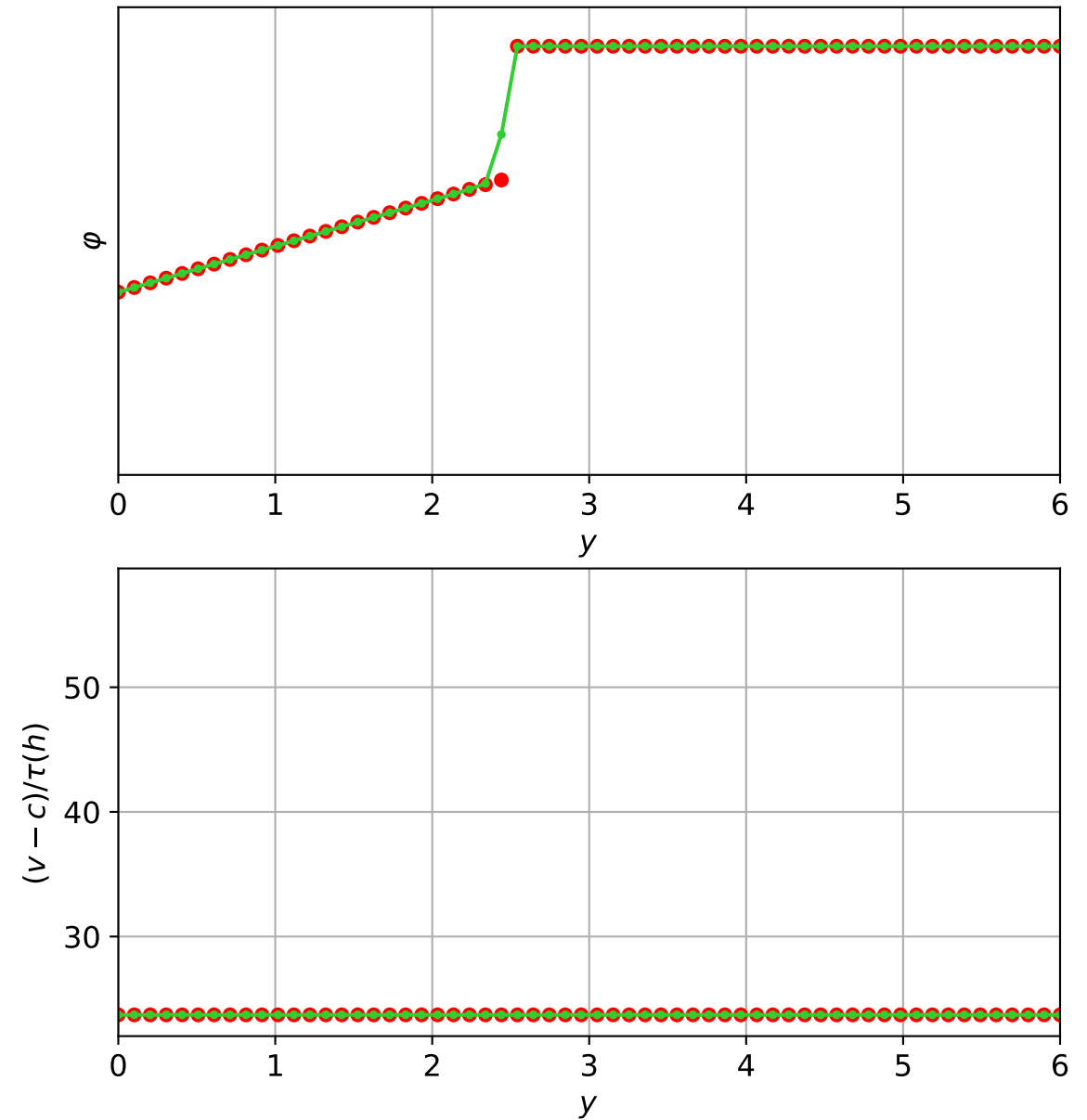
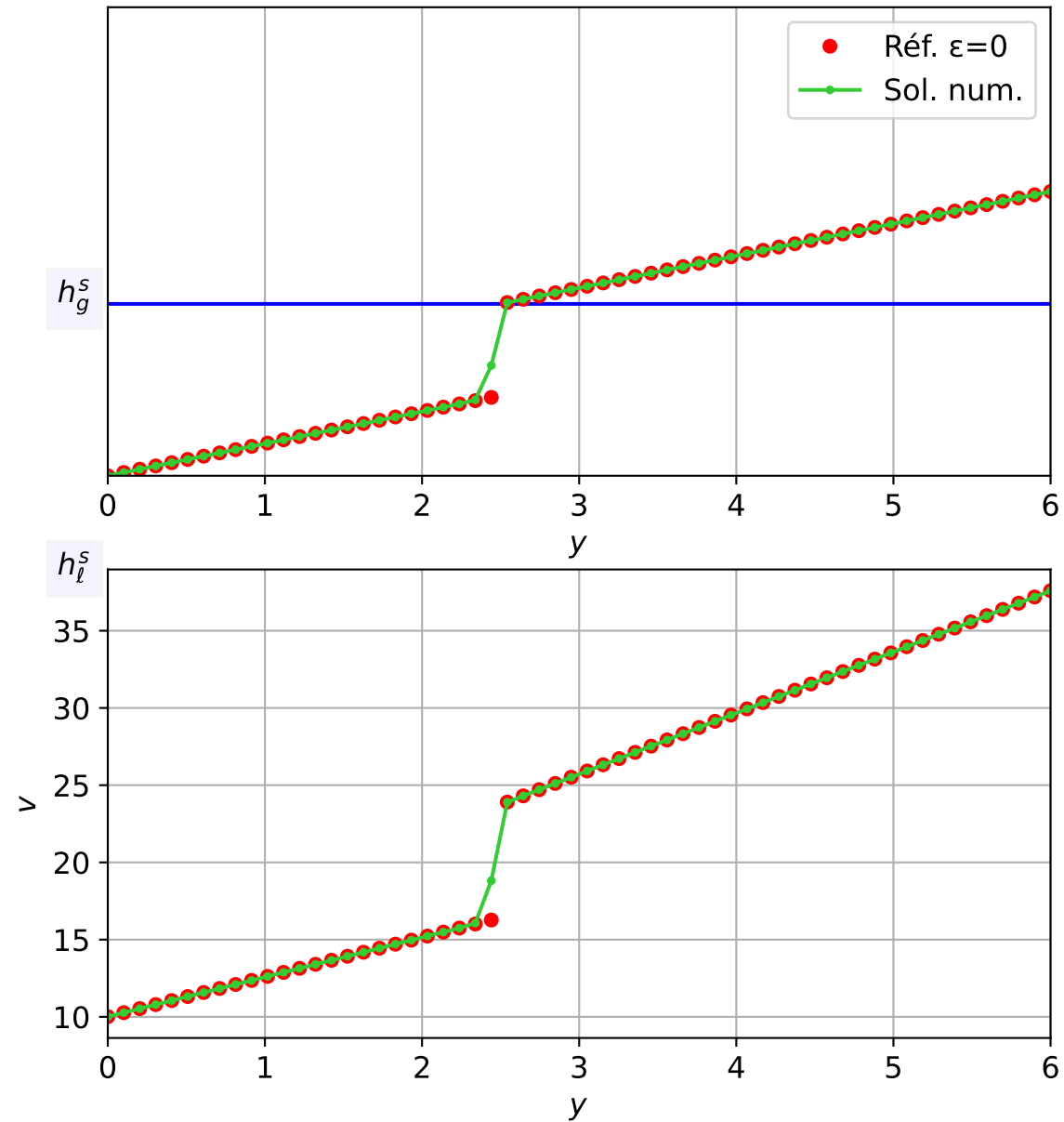




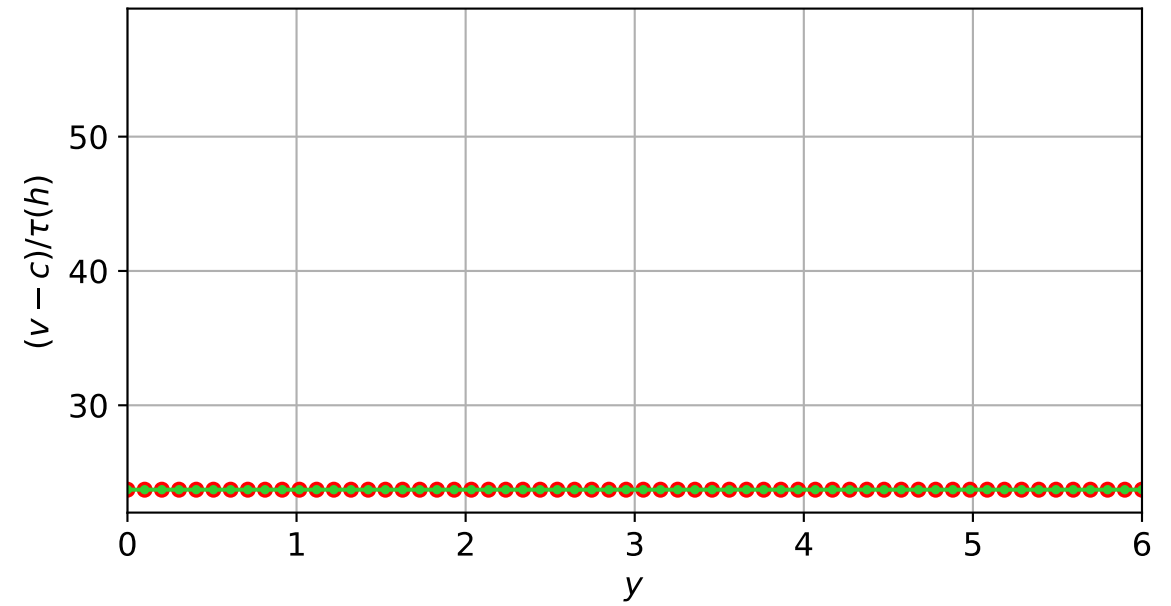
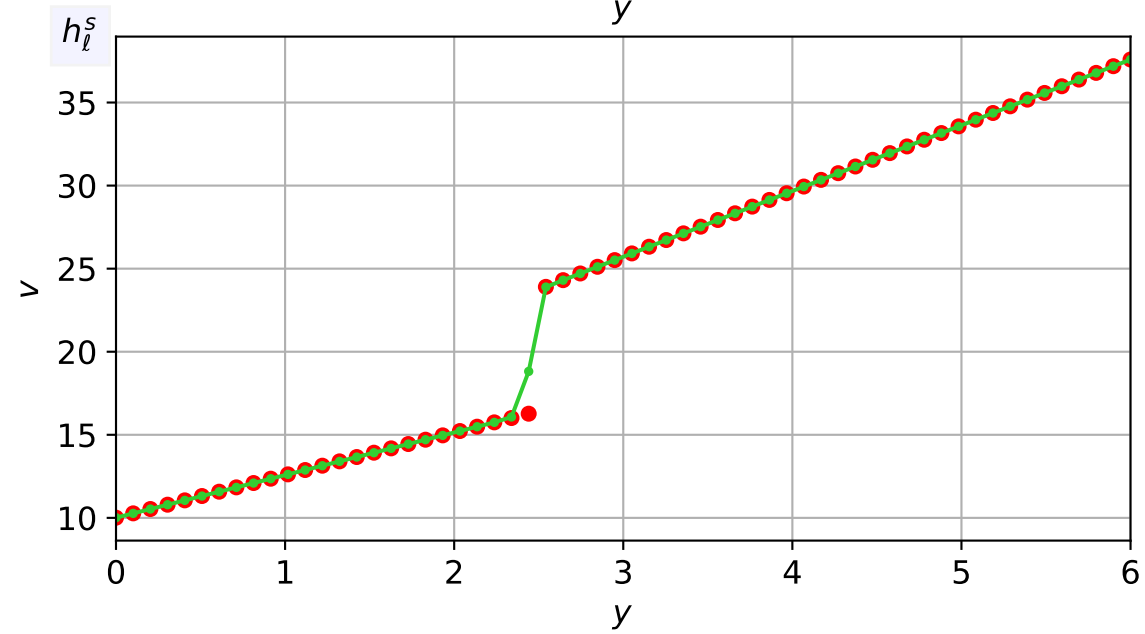
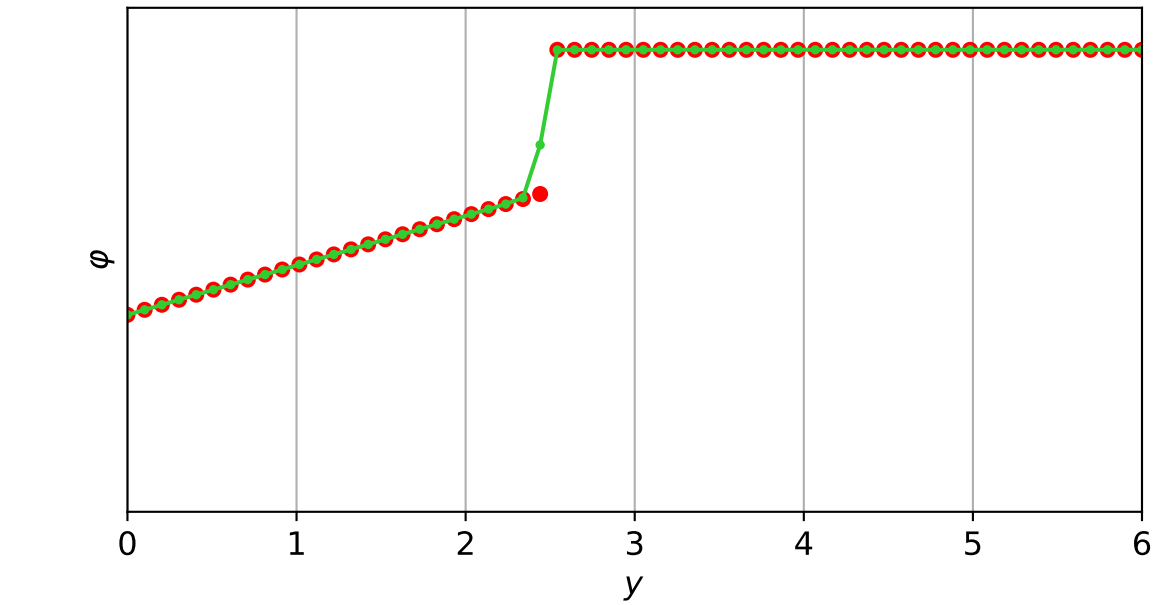
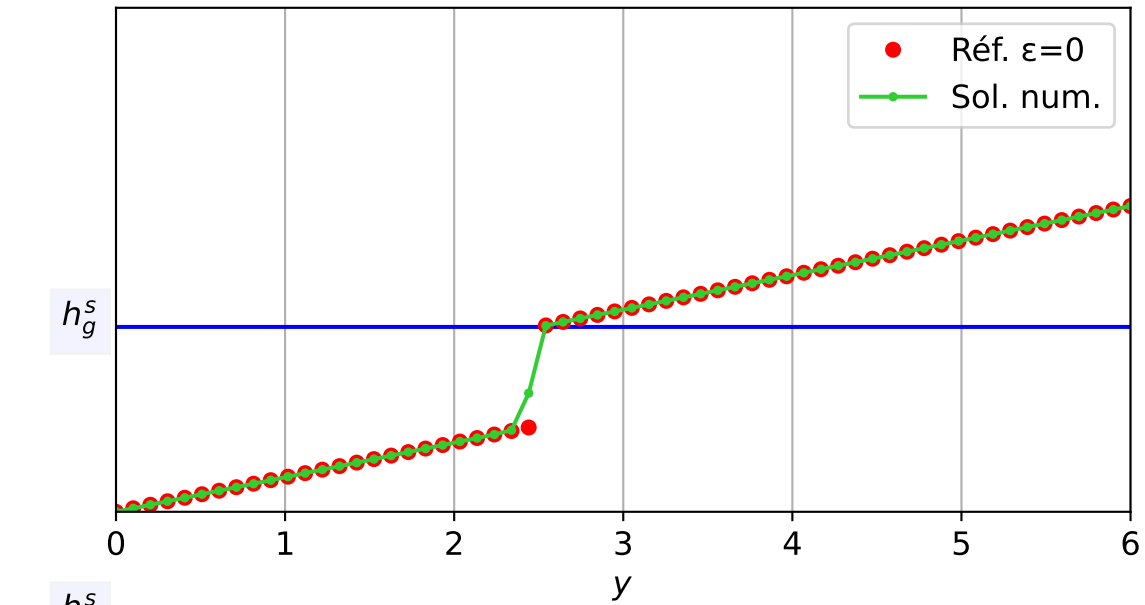
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.15043/12$  (8819 iter) — résidu eq. =  $1.00e-06$



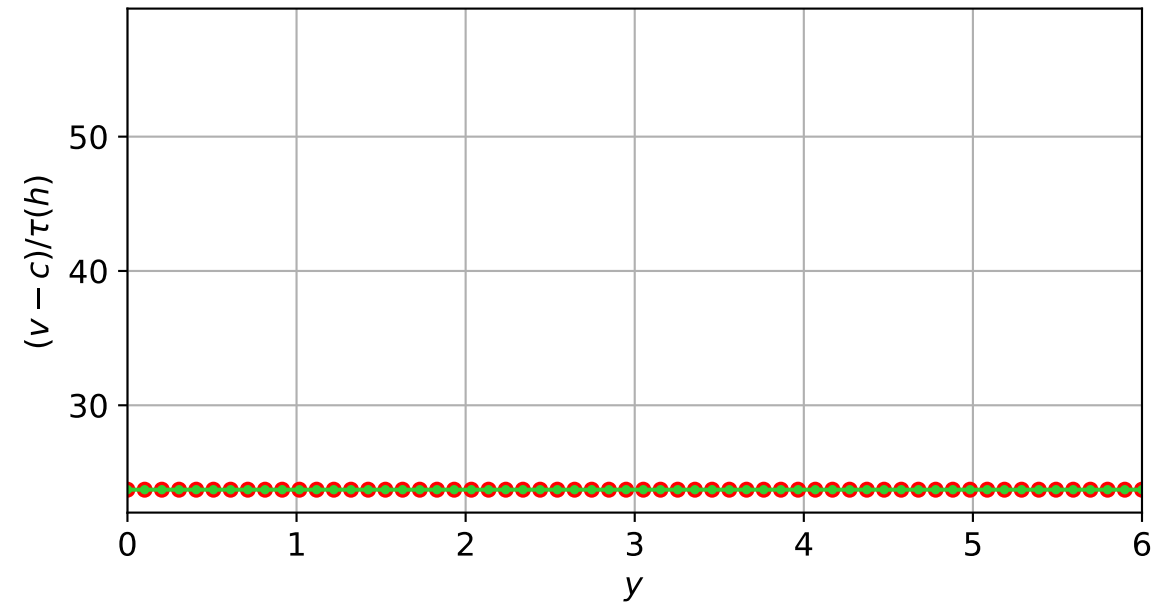
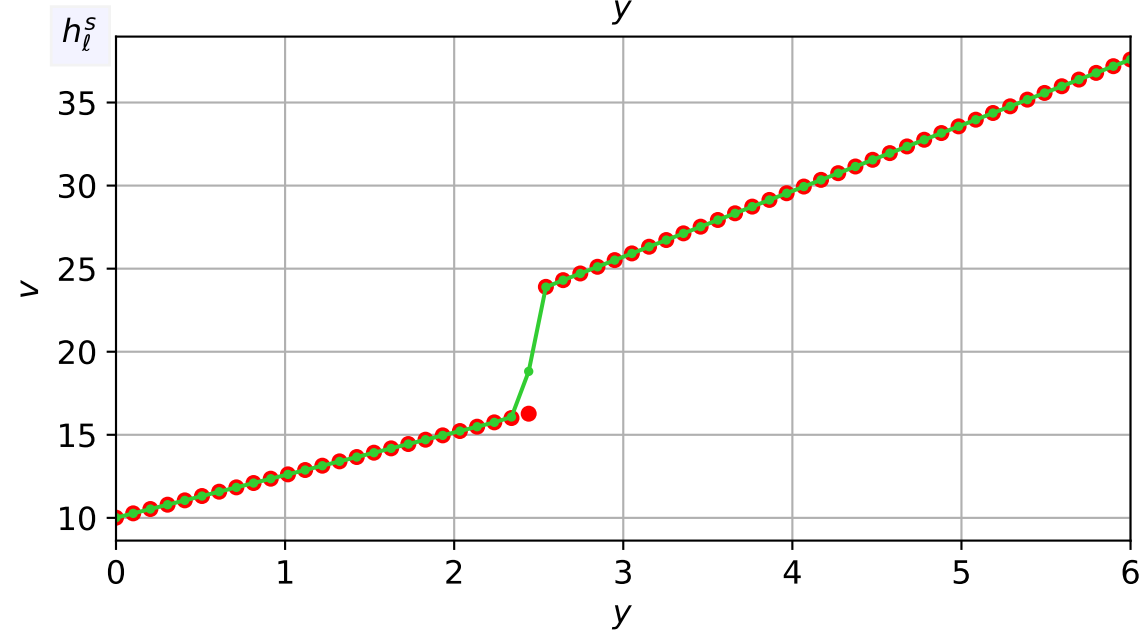
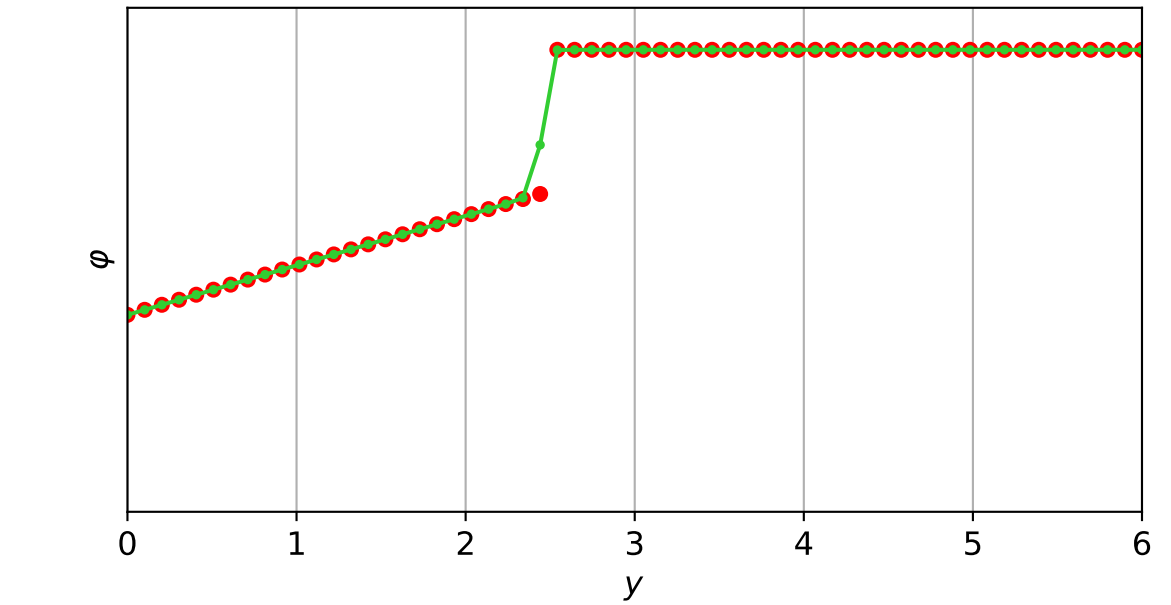
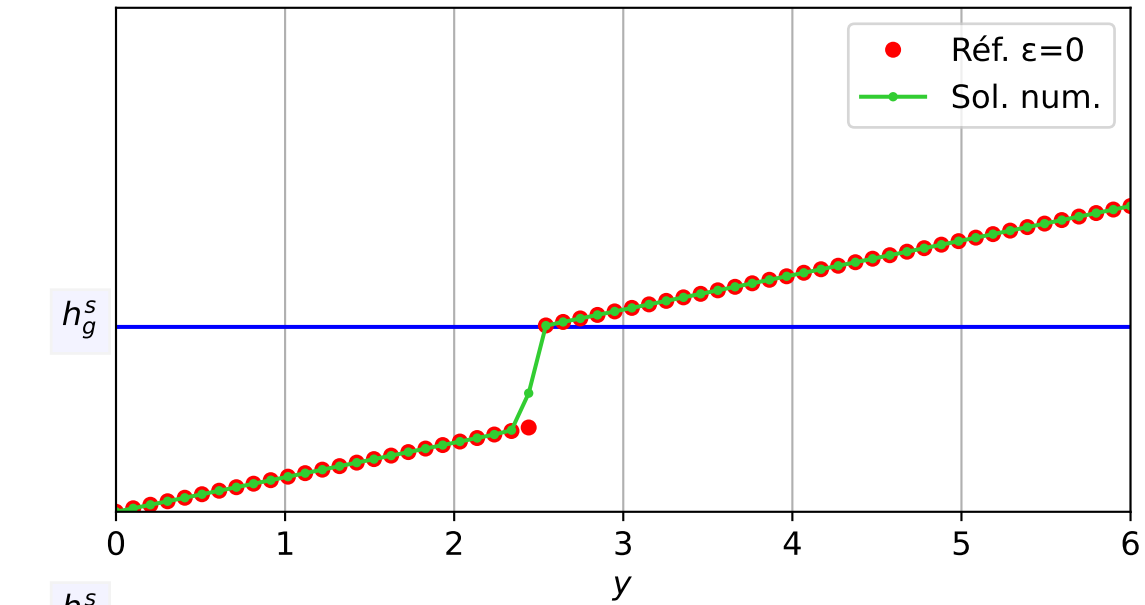
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.20157/12$  (8840 iter) — résidu eq. =  $9.44e-06$



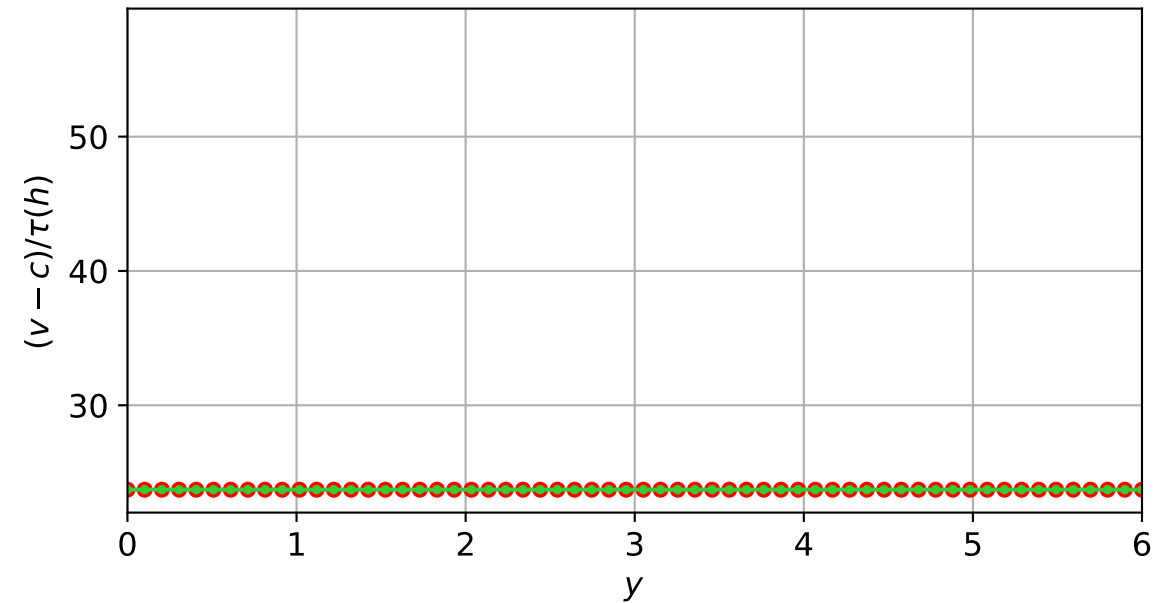
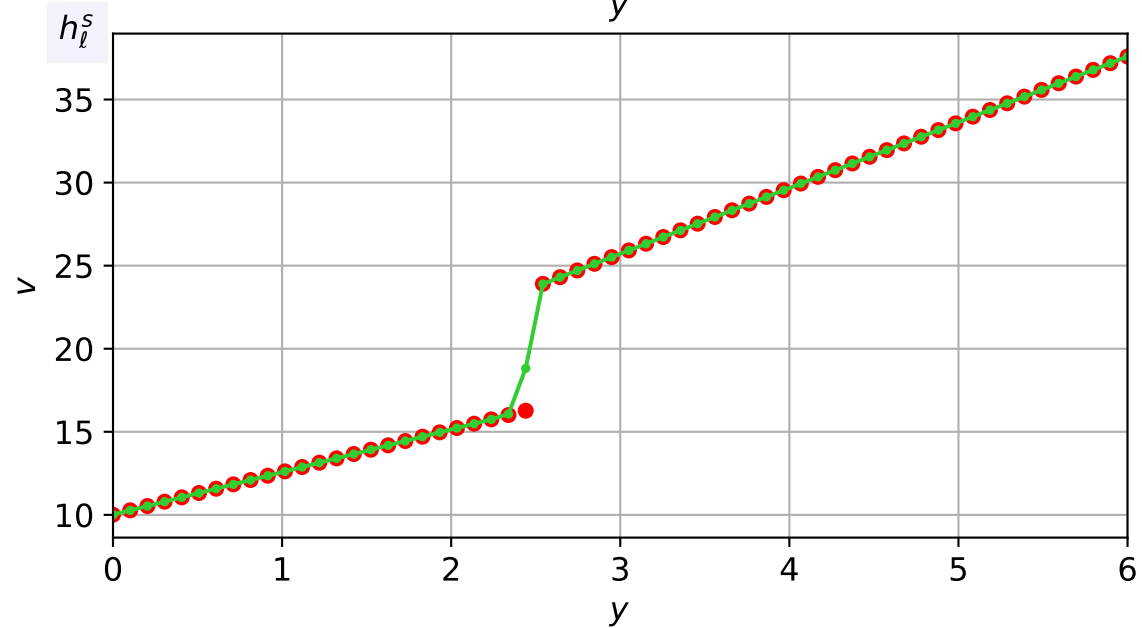
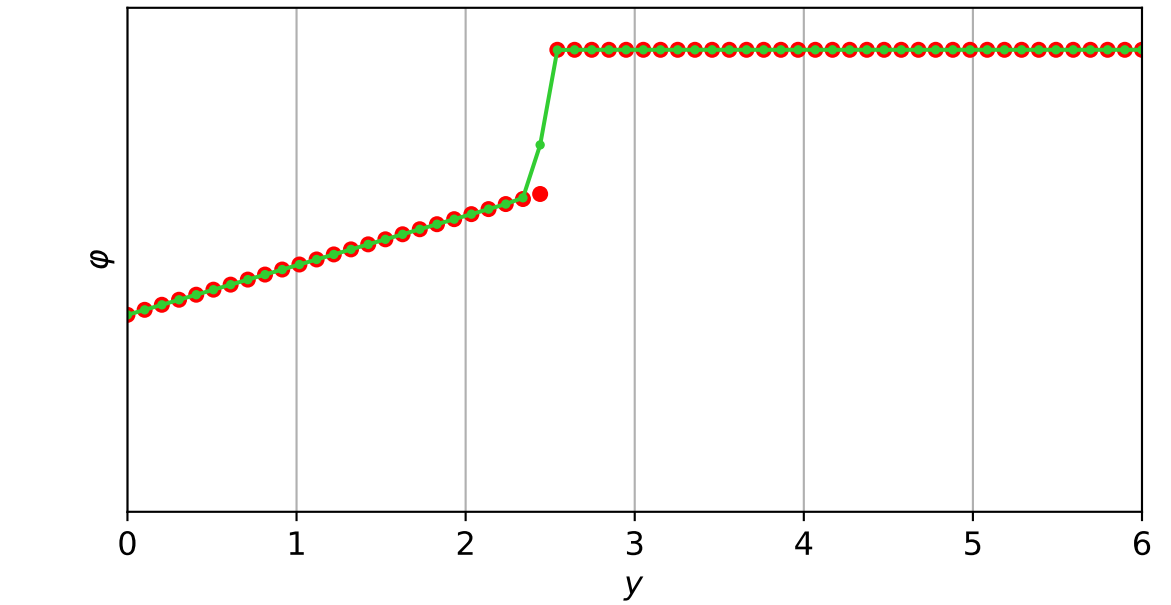
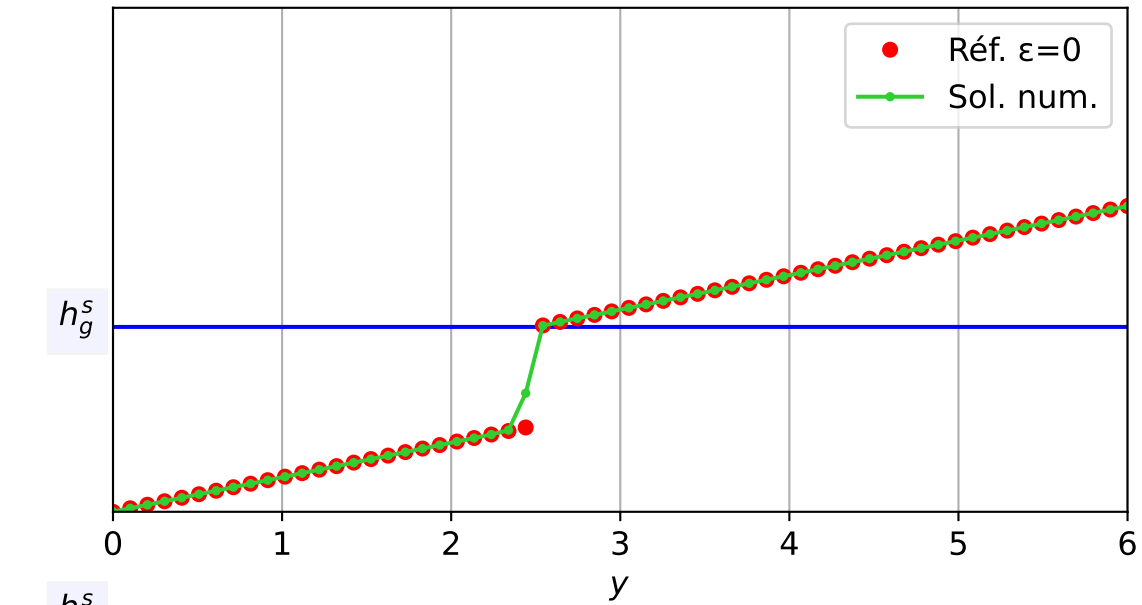
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.25027/12$  (8860 iter) — résidu eq. =  $8.35e-06$



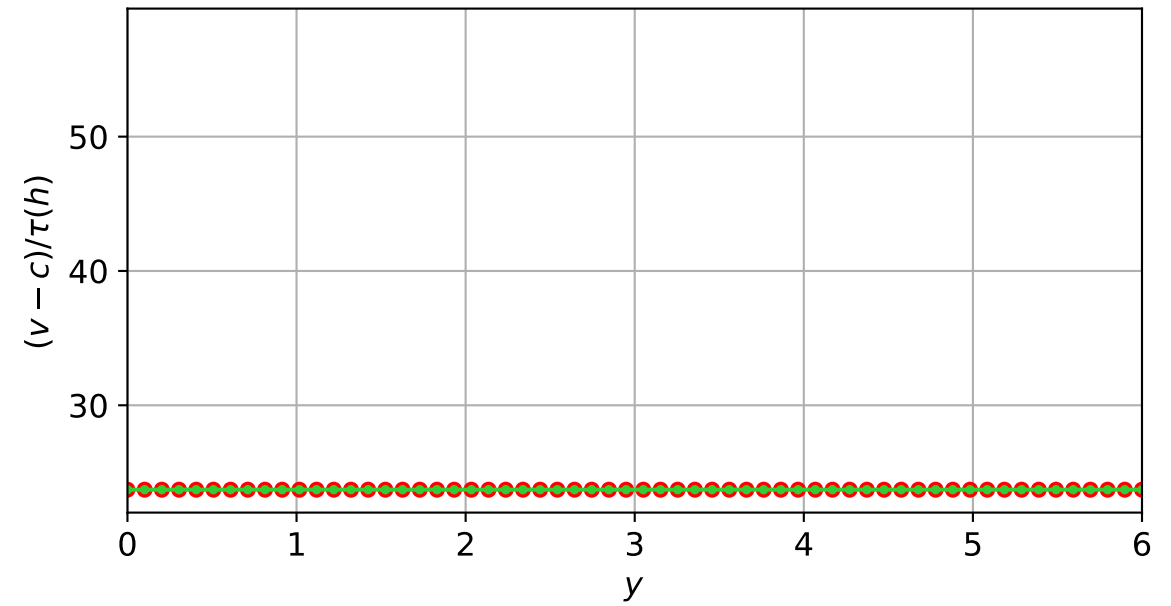
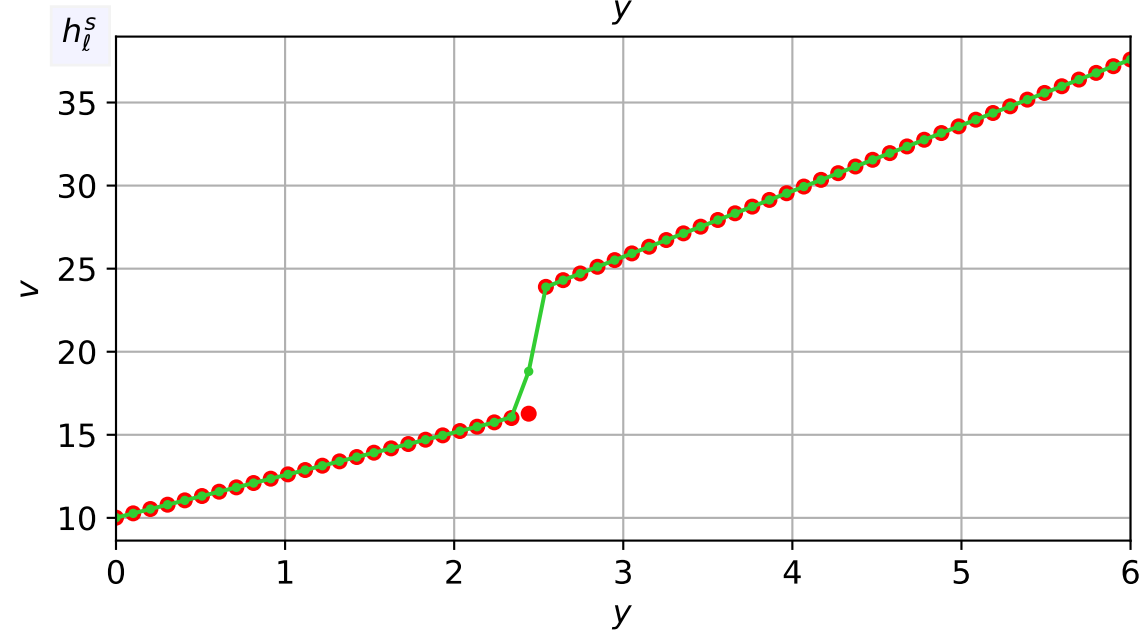
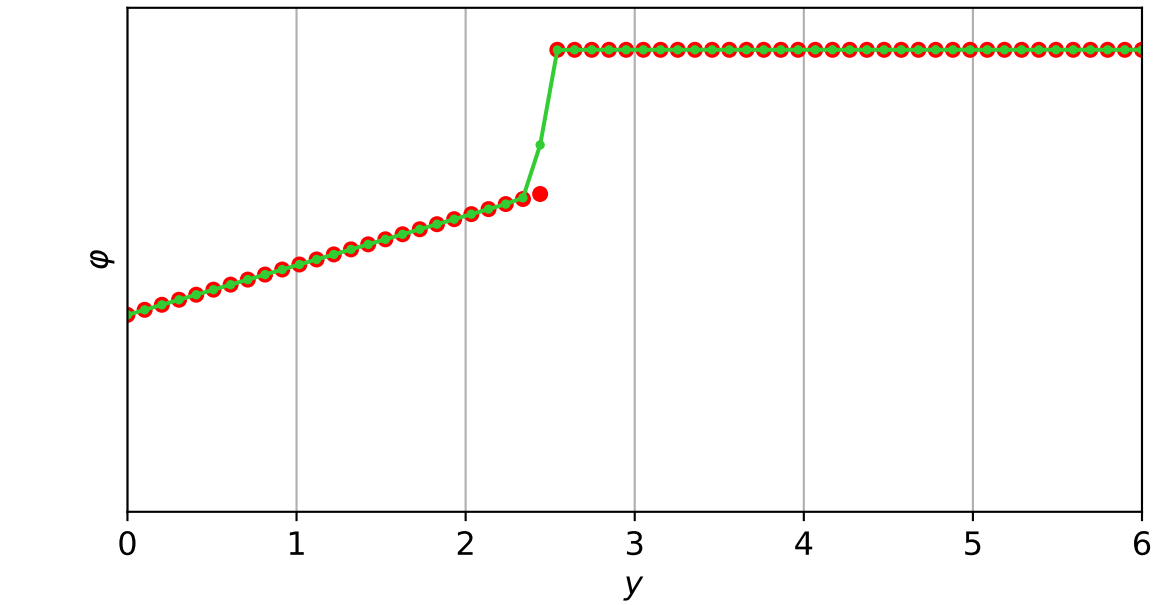
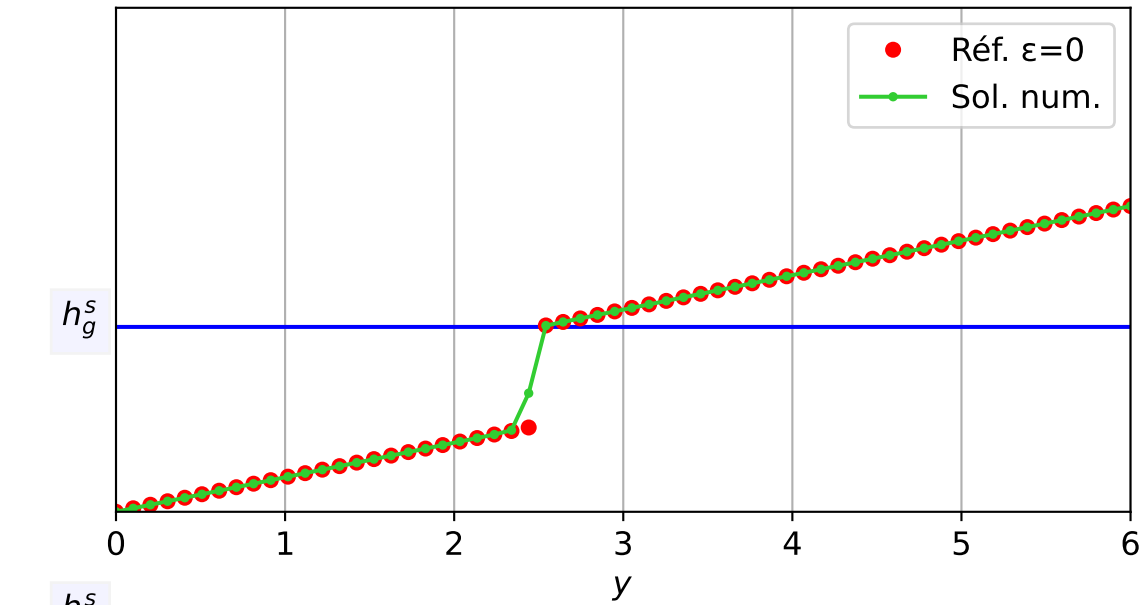
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.3014/12$  (8881 iter) — résidu eq. =  $7.36e-06$



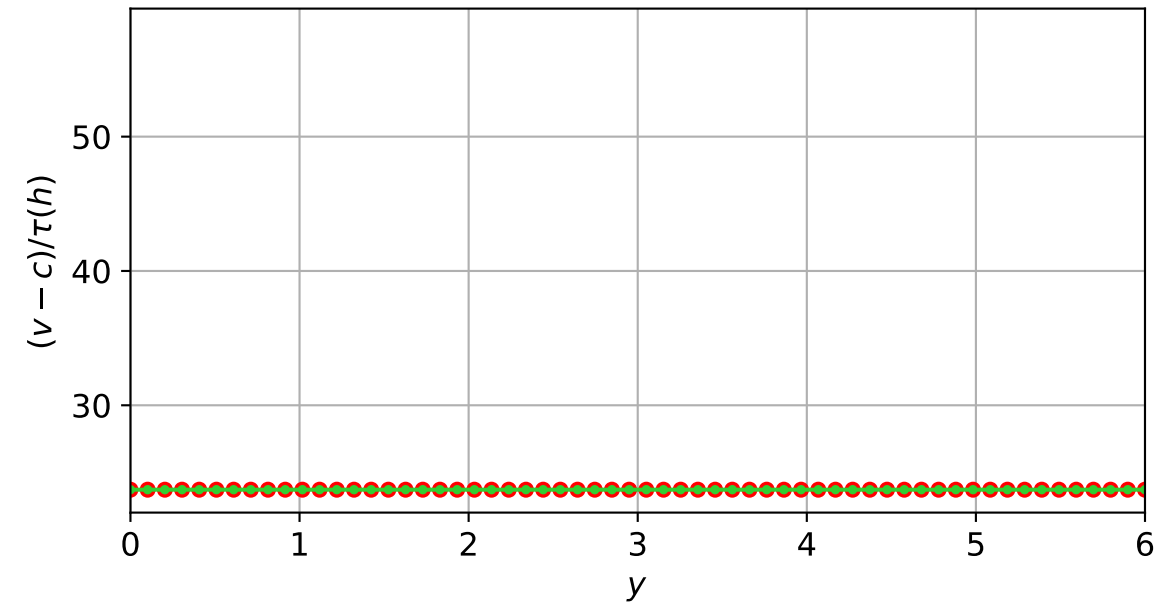
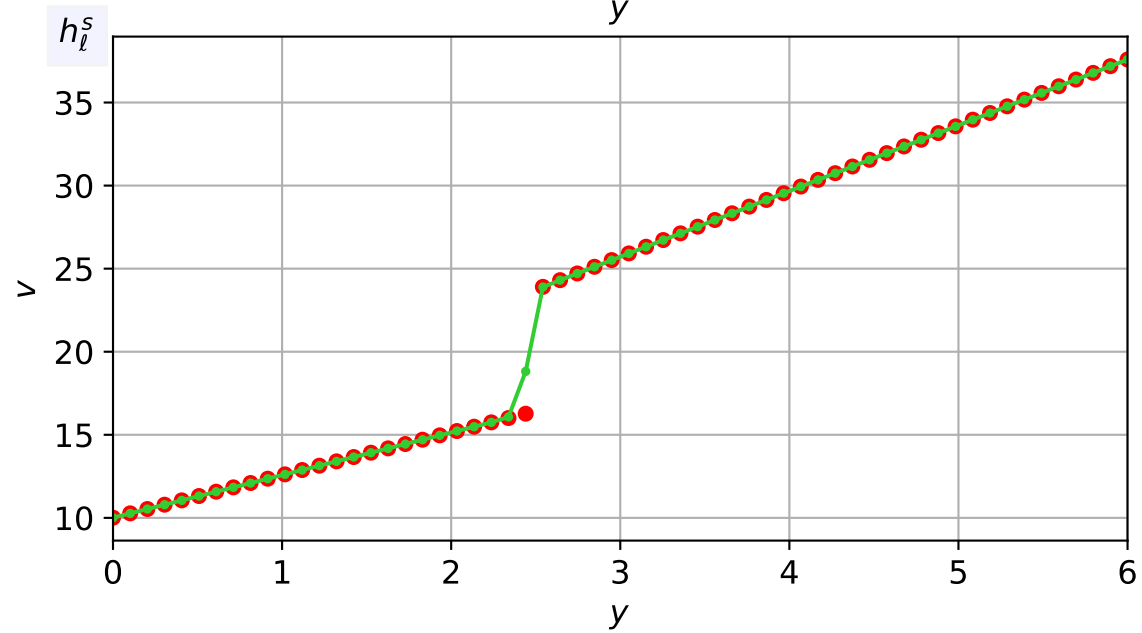
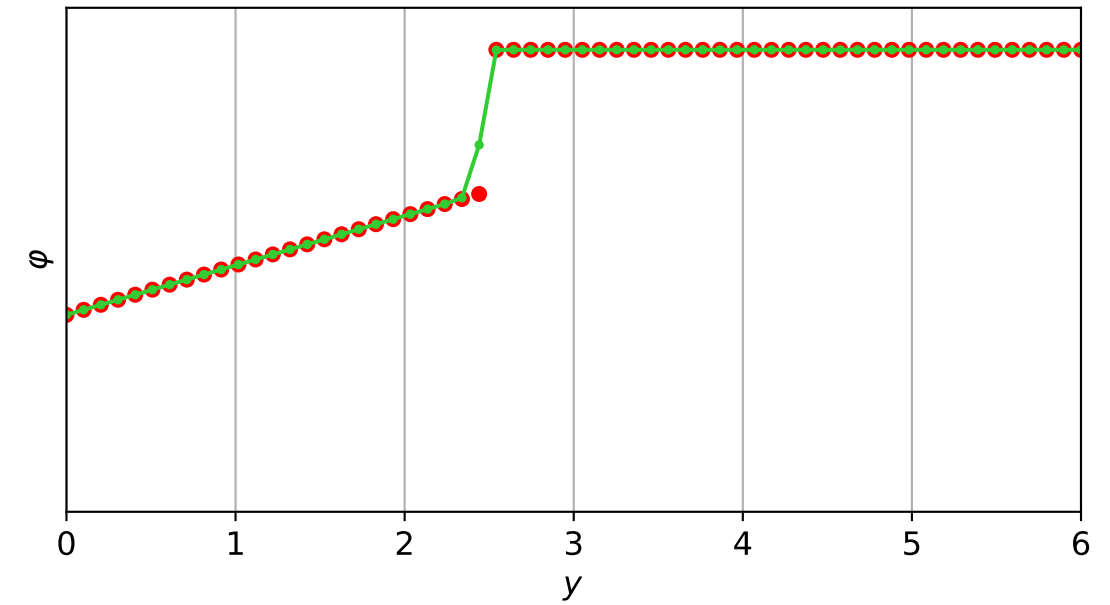
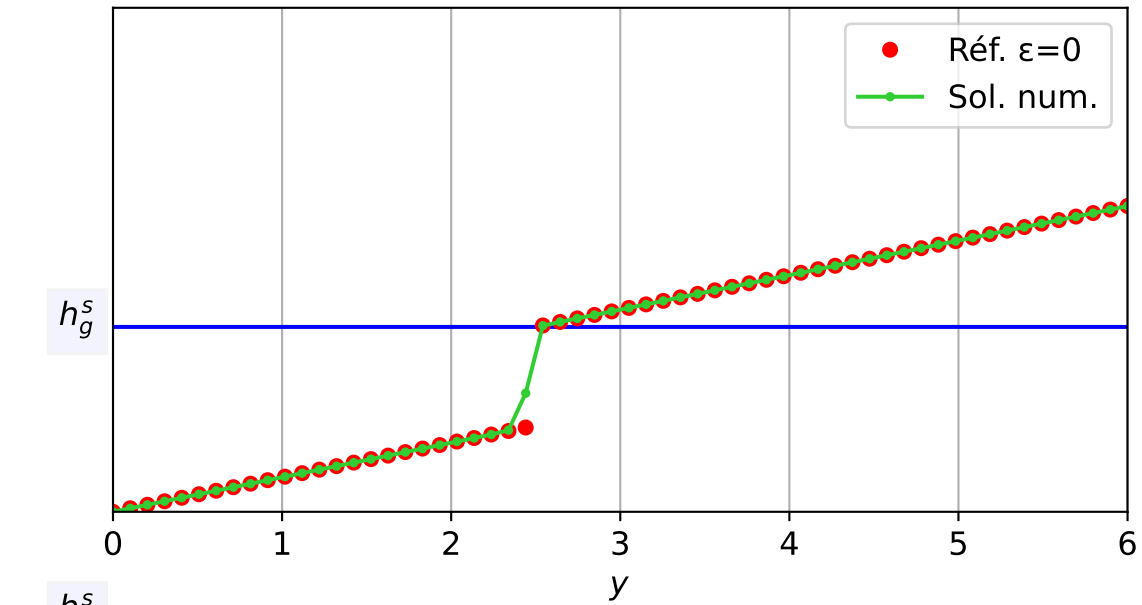
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.3501/12$  (8901 iter) — résidu eq. =  $6.53e-06$



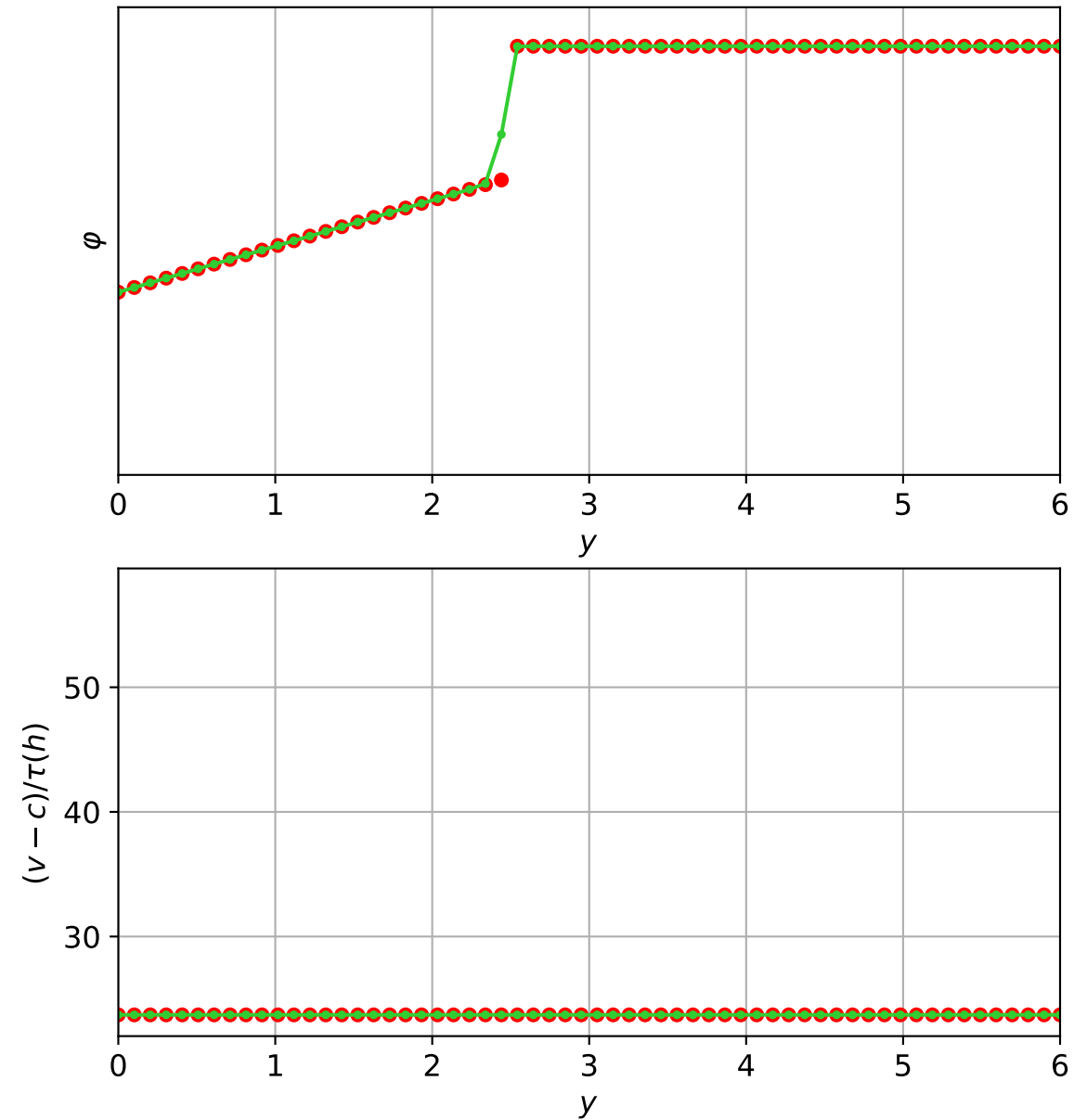
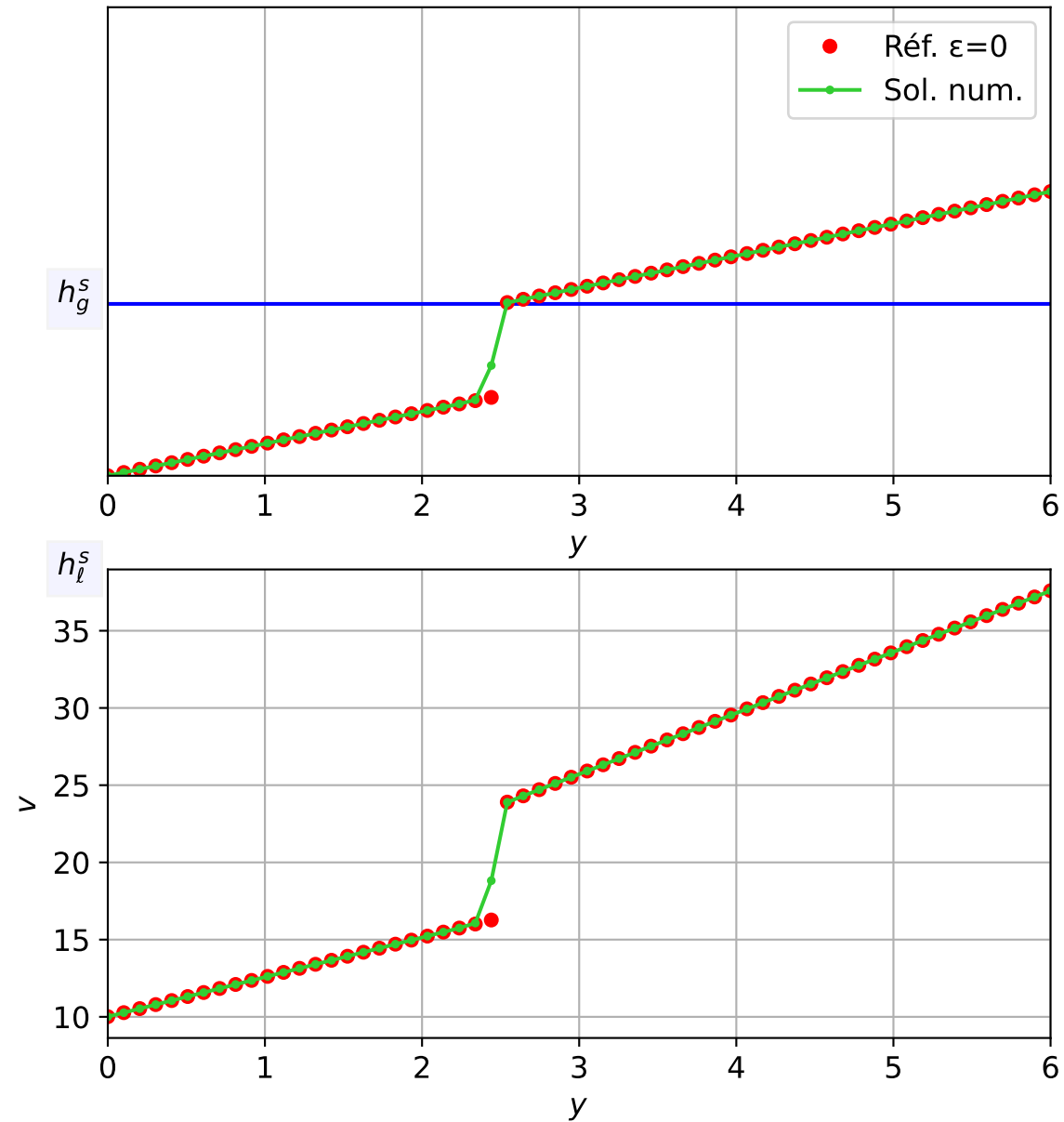
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.40124/12$  (8922 iter) — résidu eq. =  $5.77e-06$



TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.45237/12$  (8943 iter) — résidu eq. =  $9.65e-06$

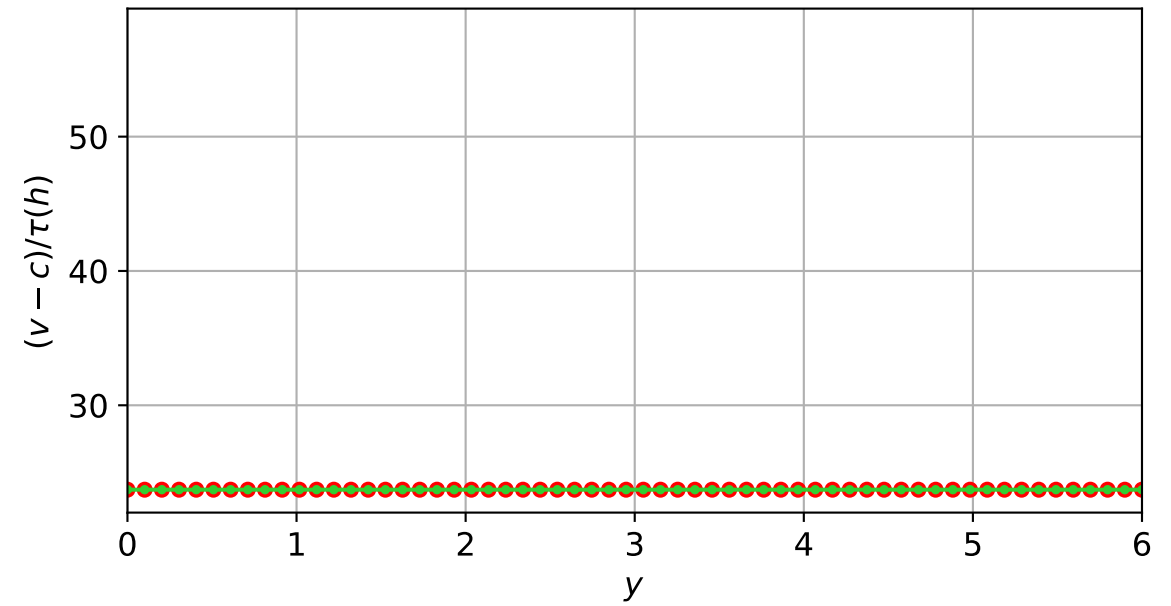
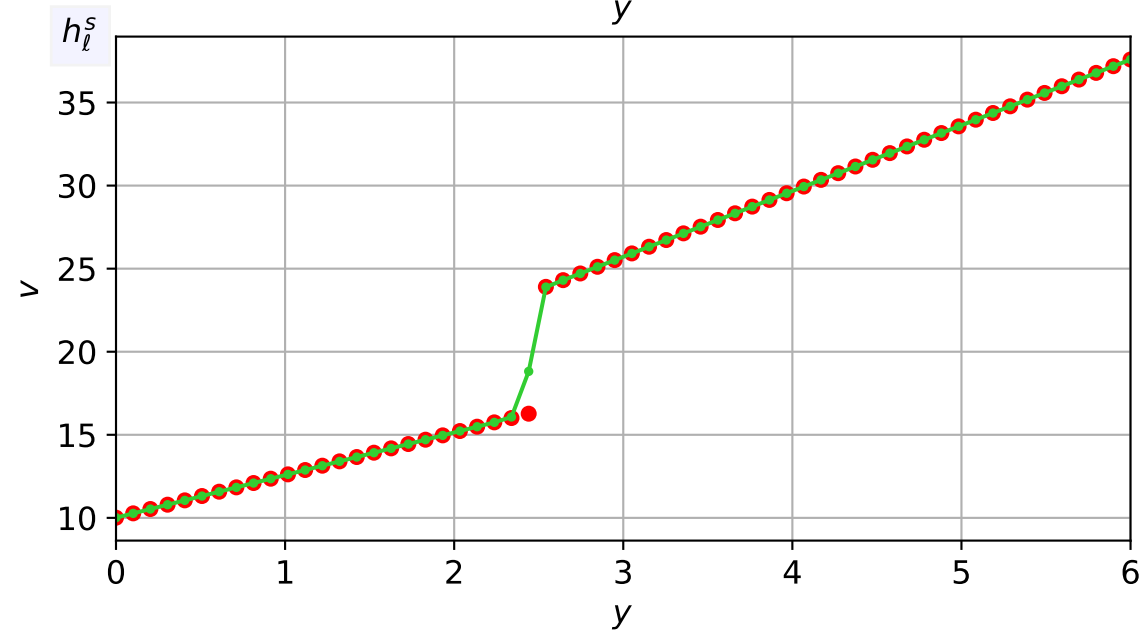
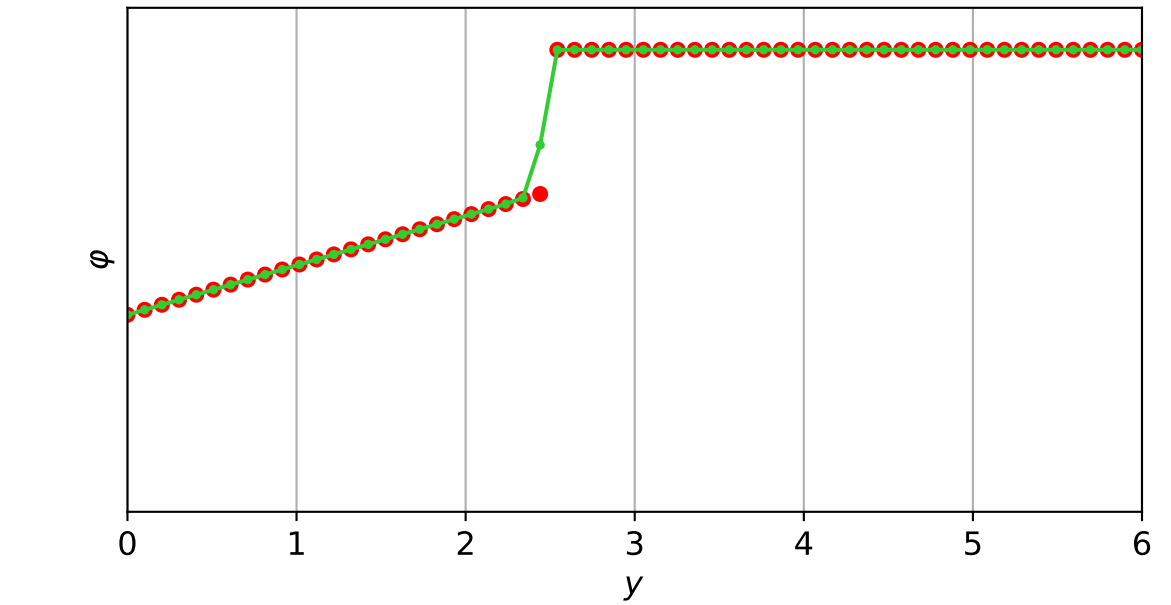
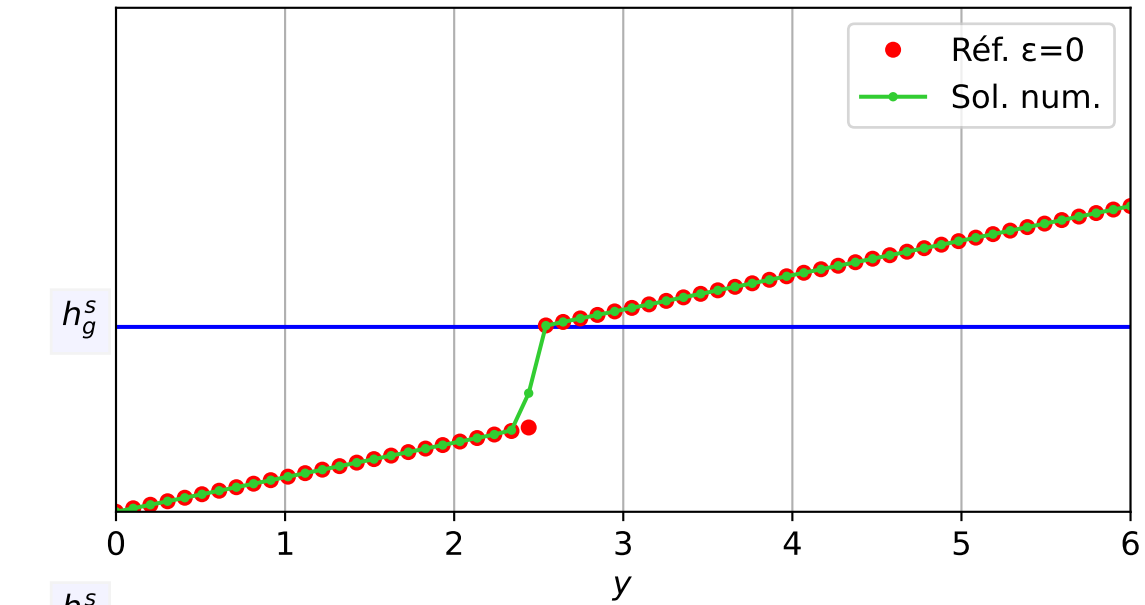


TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.50107/12$  (8963 iter) — résidu eq. =  $8.57e-06$

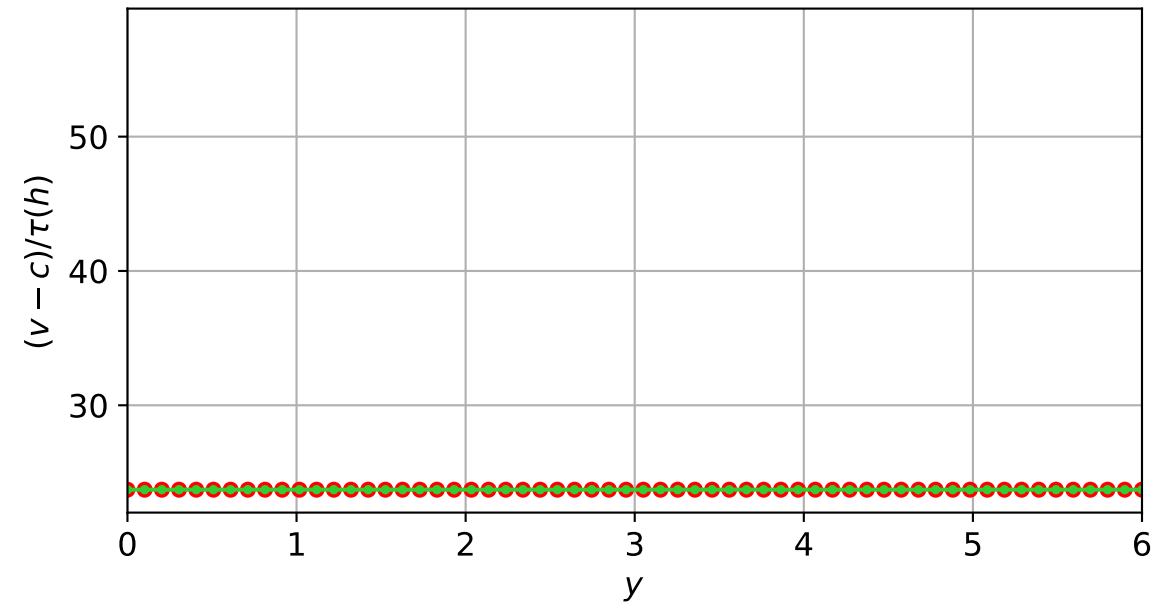
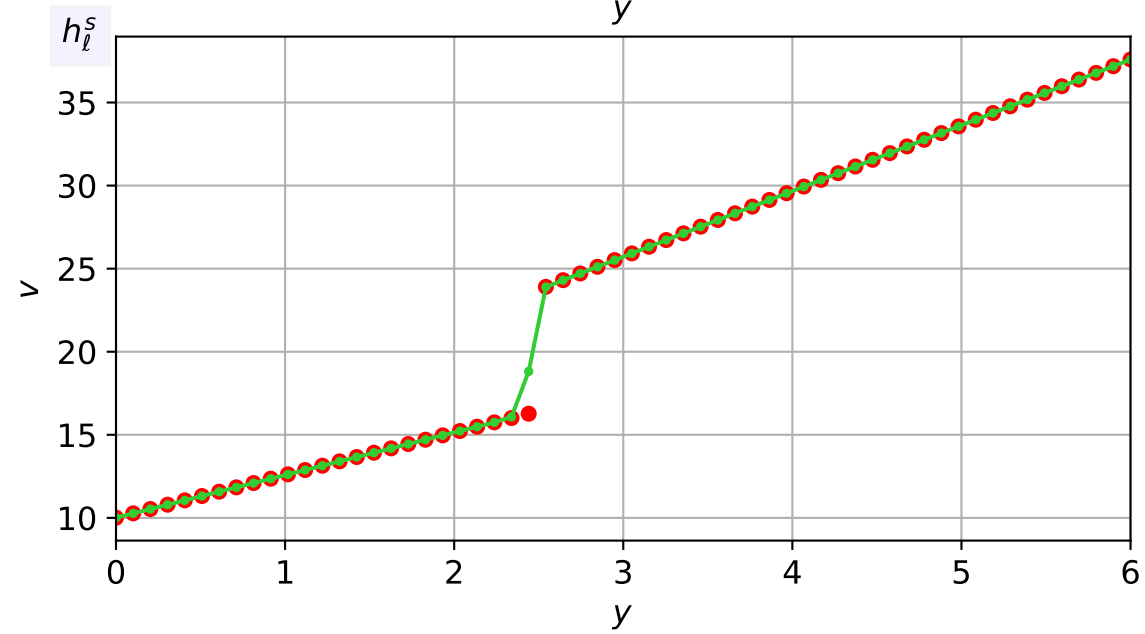
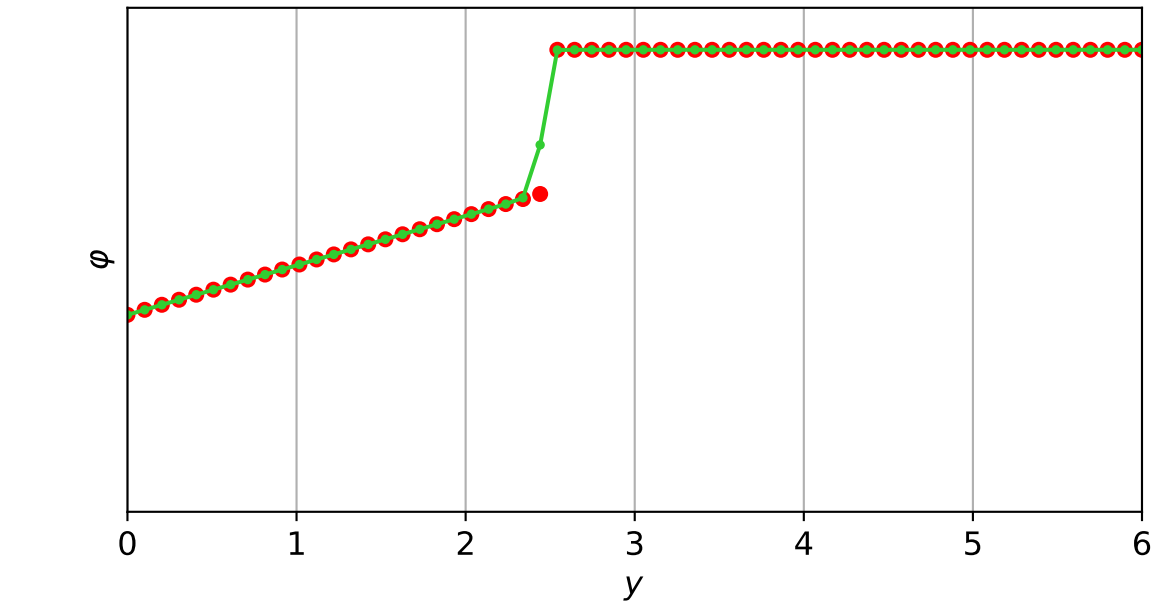
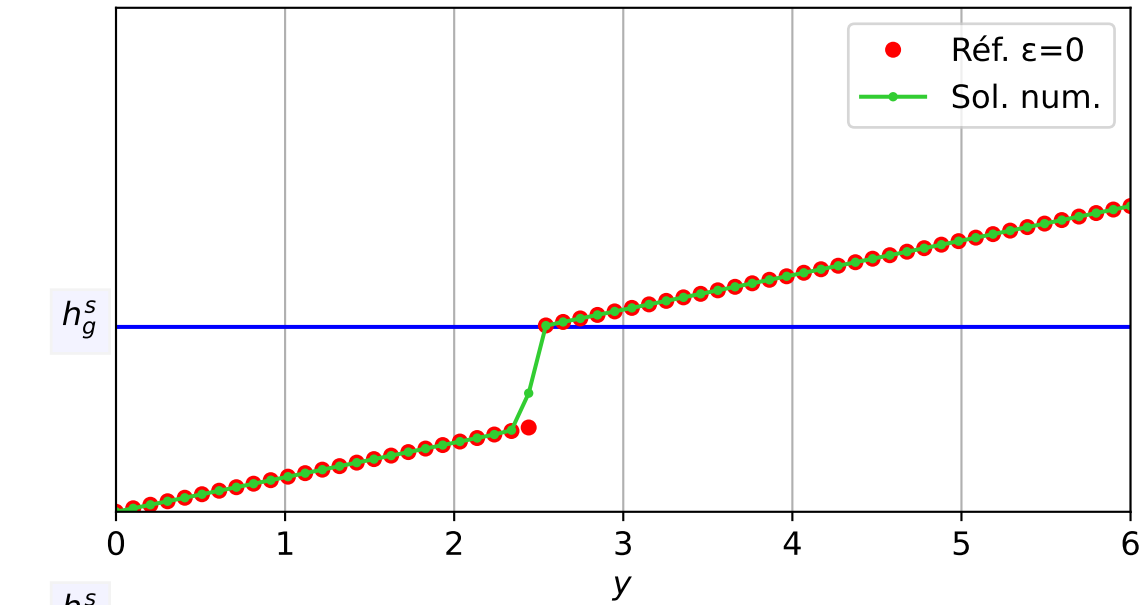




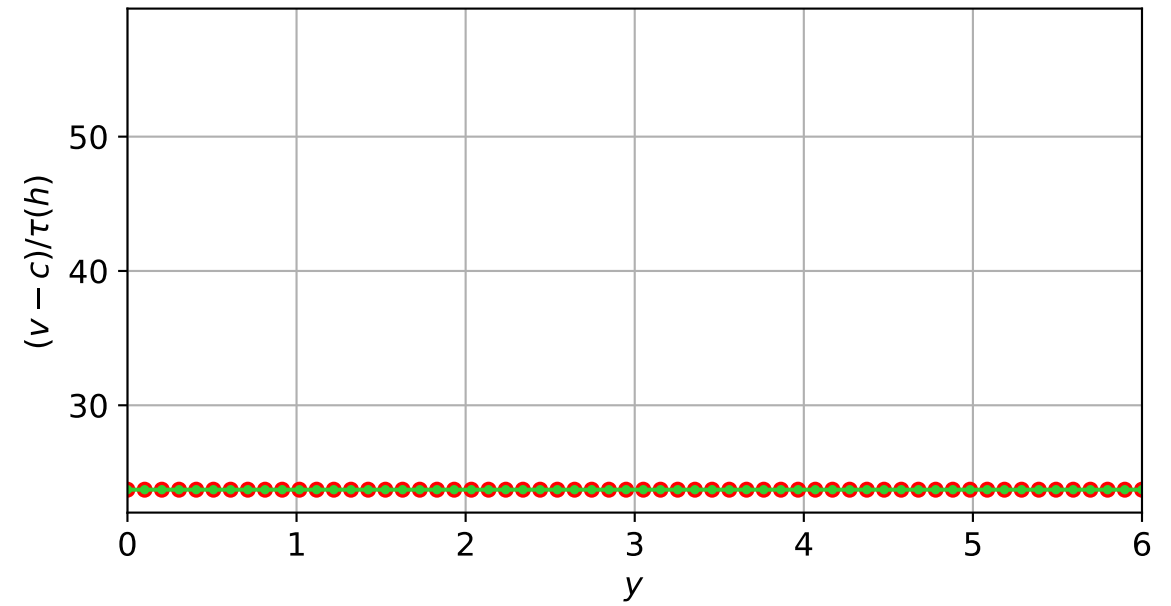
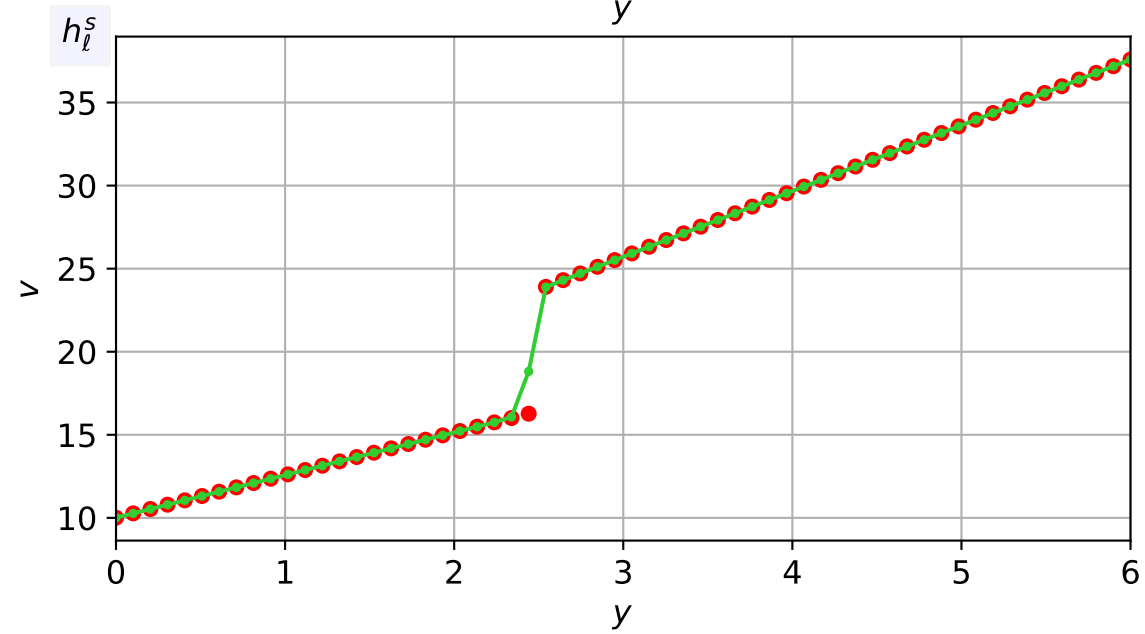
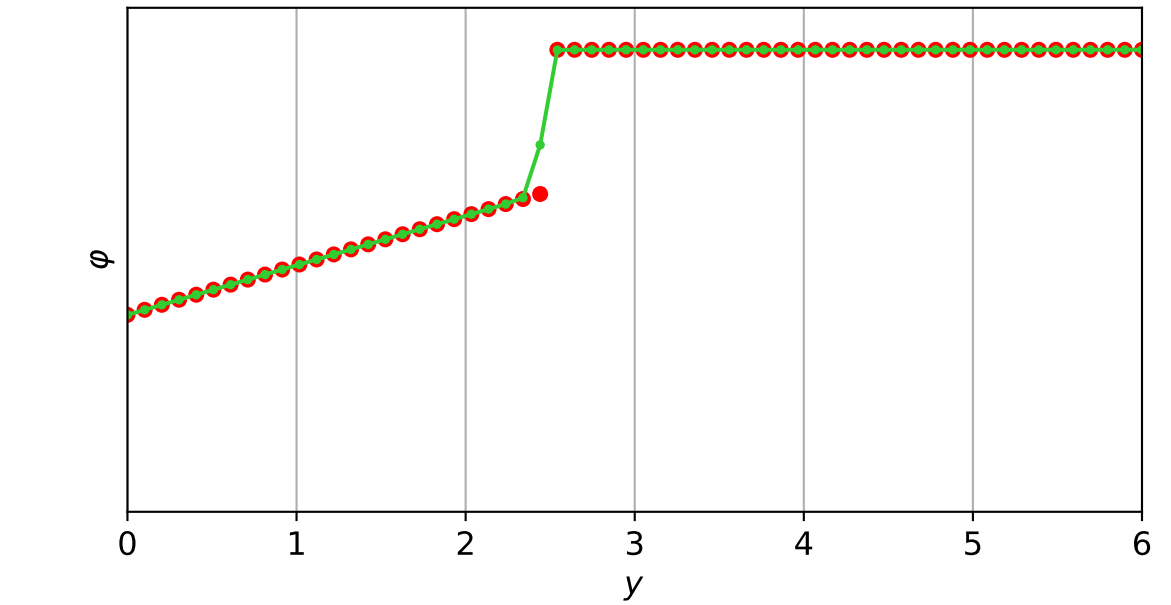
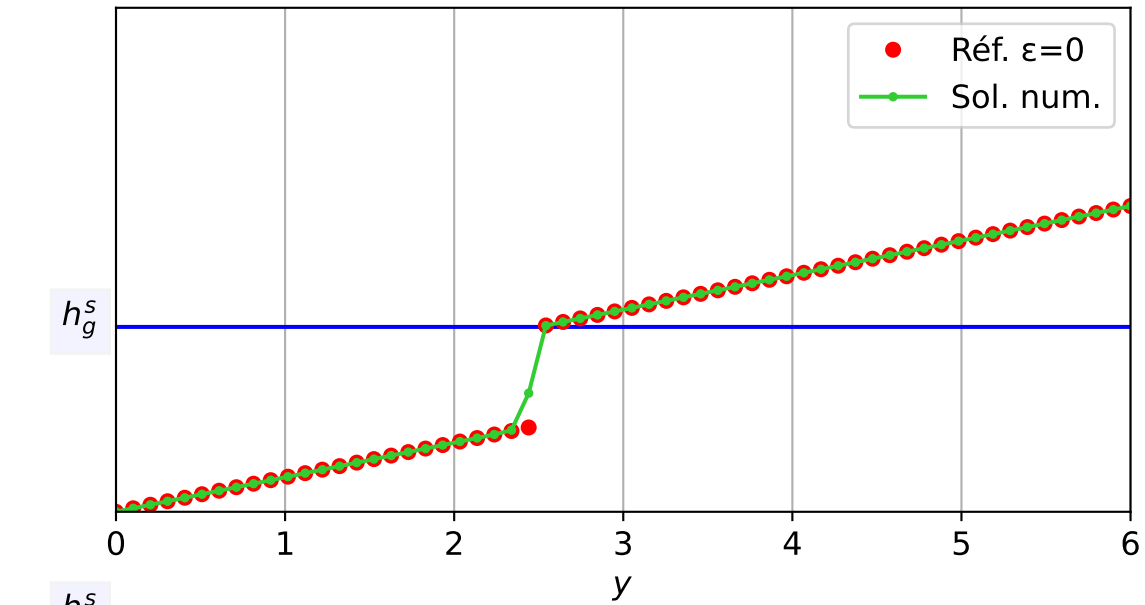
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.55221/12$  (8984 iter) — résidu eq. =  $7.57e-06$



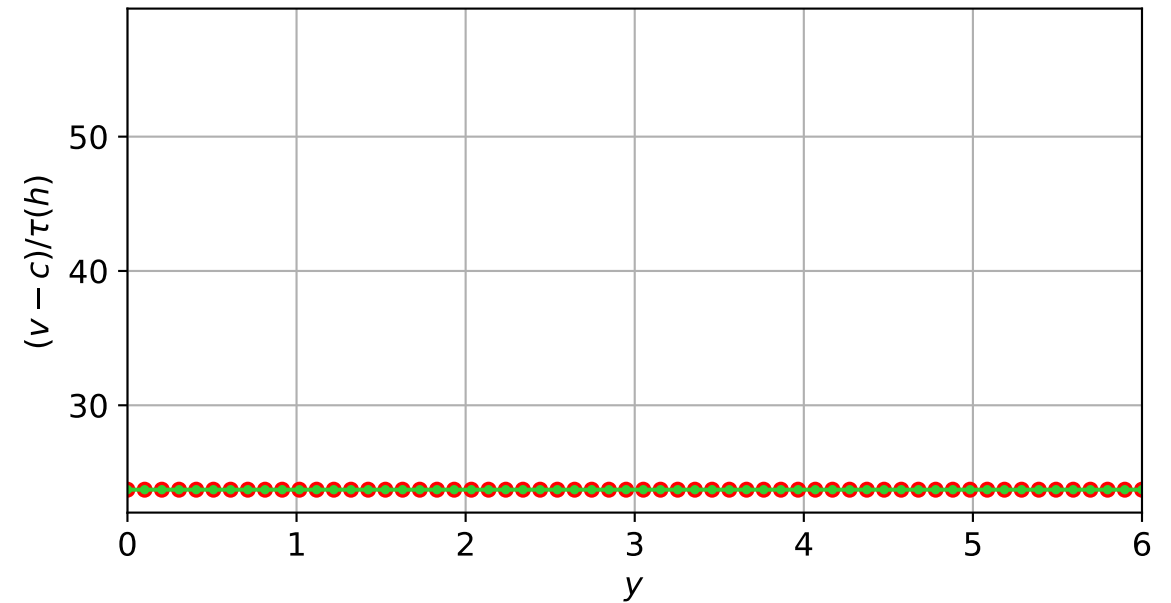
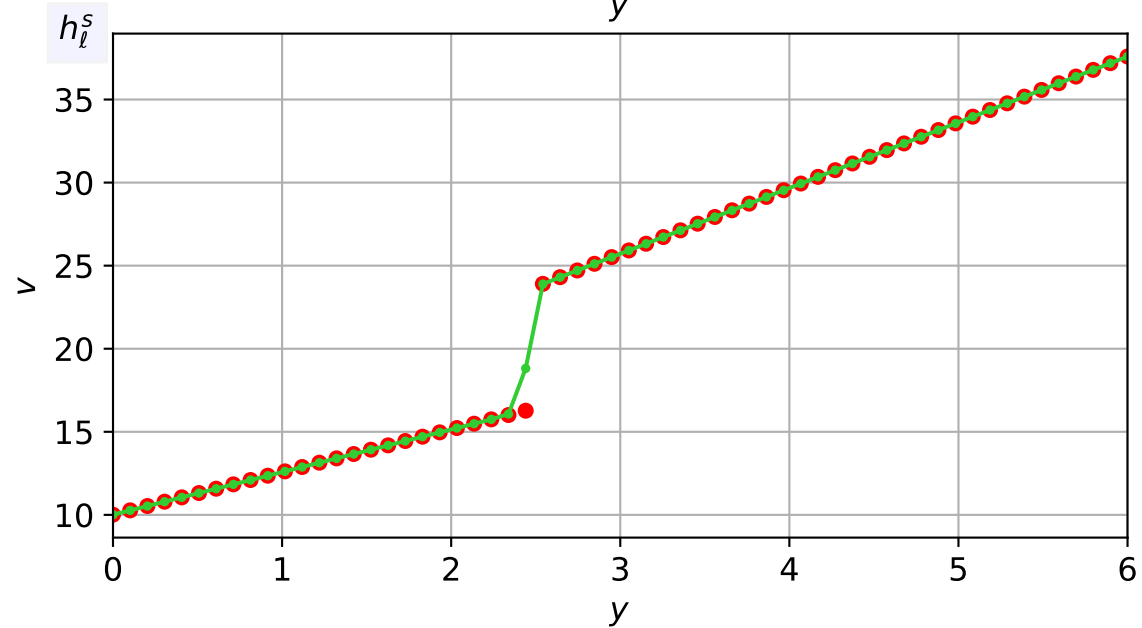
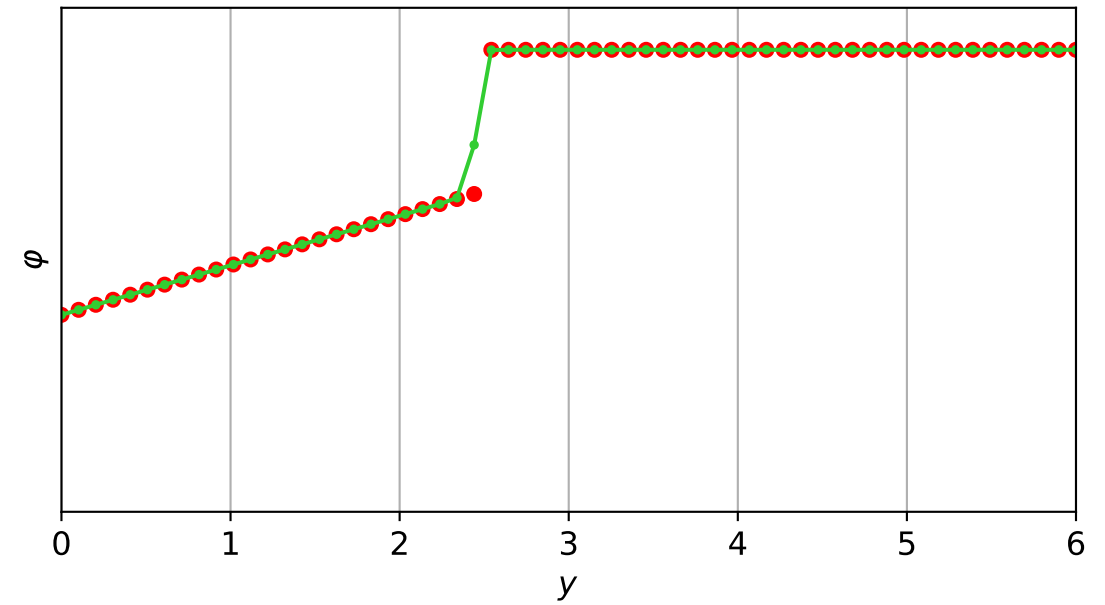
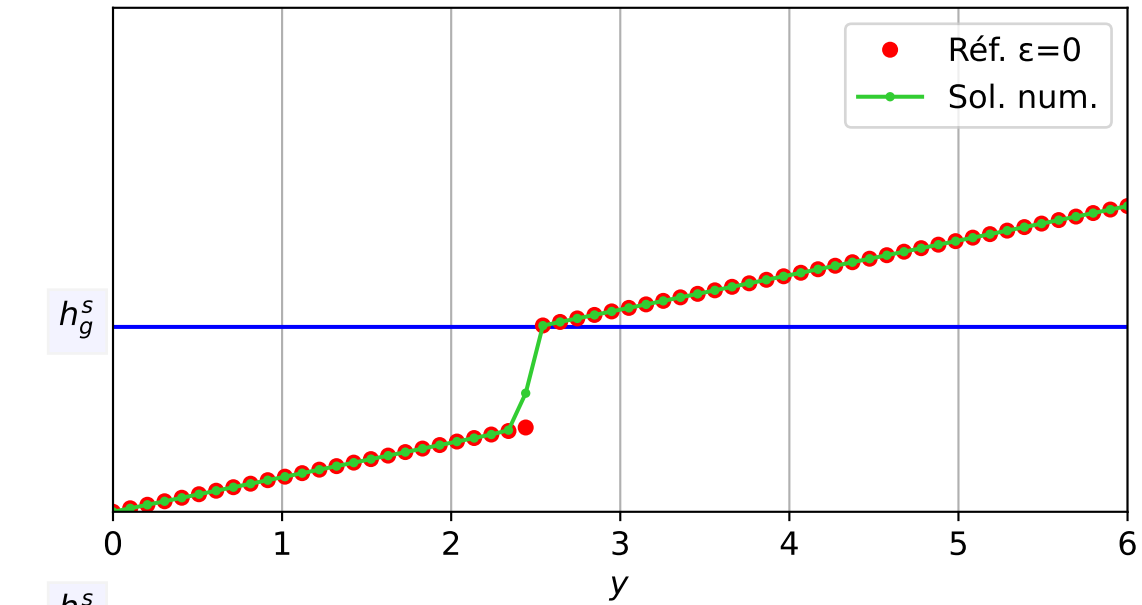
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.60091/12$  (9004 iter) — résidu eq. =  $6.73e-06$



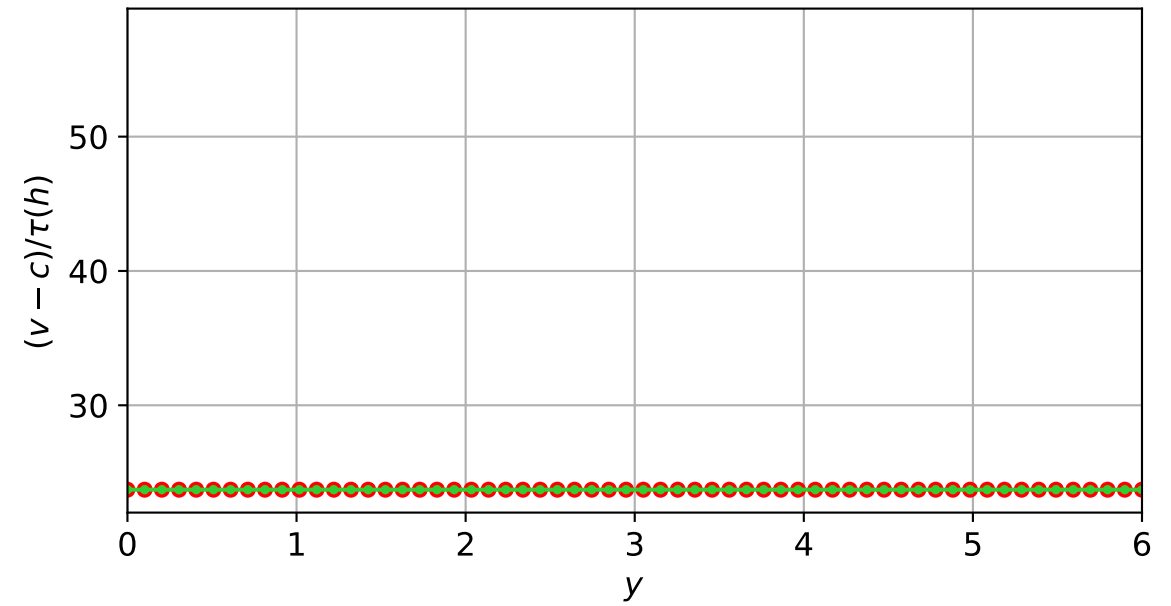
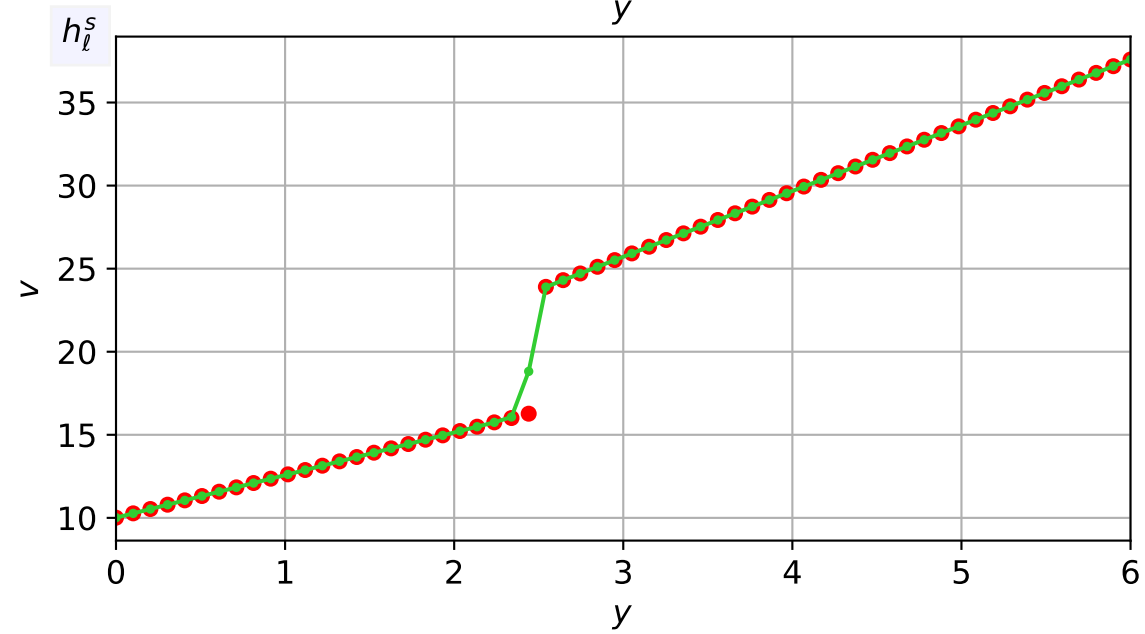
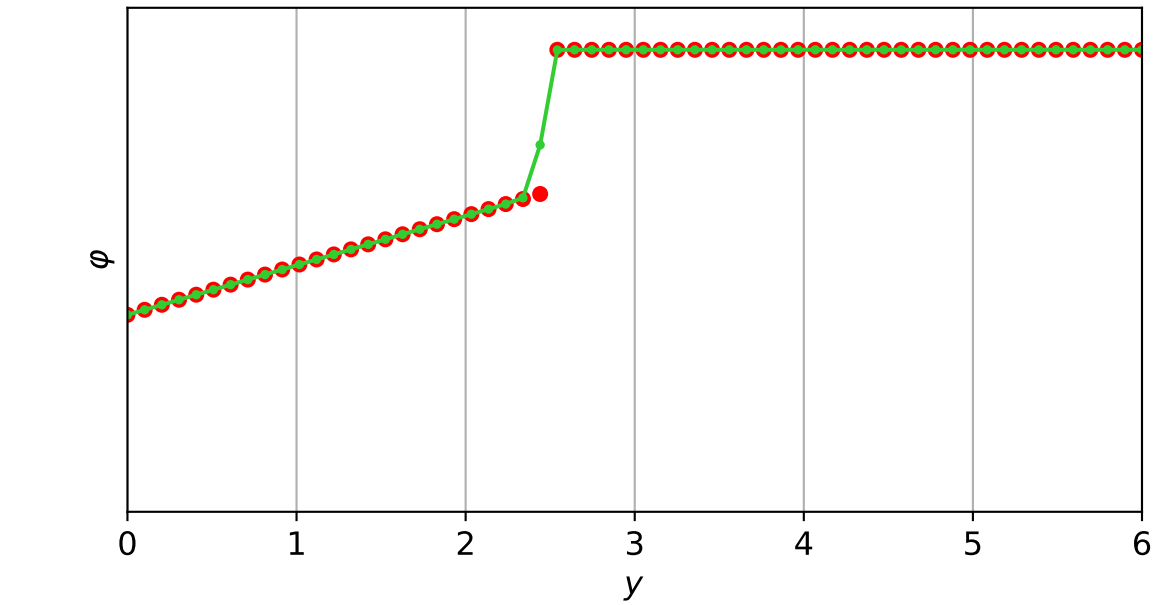
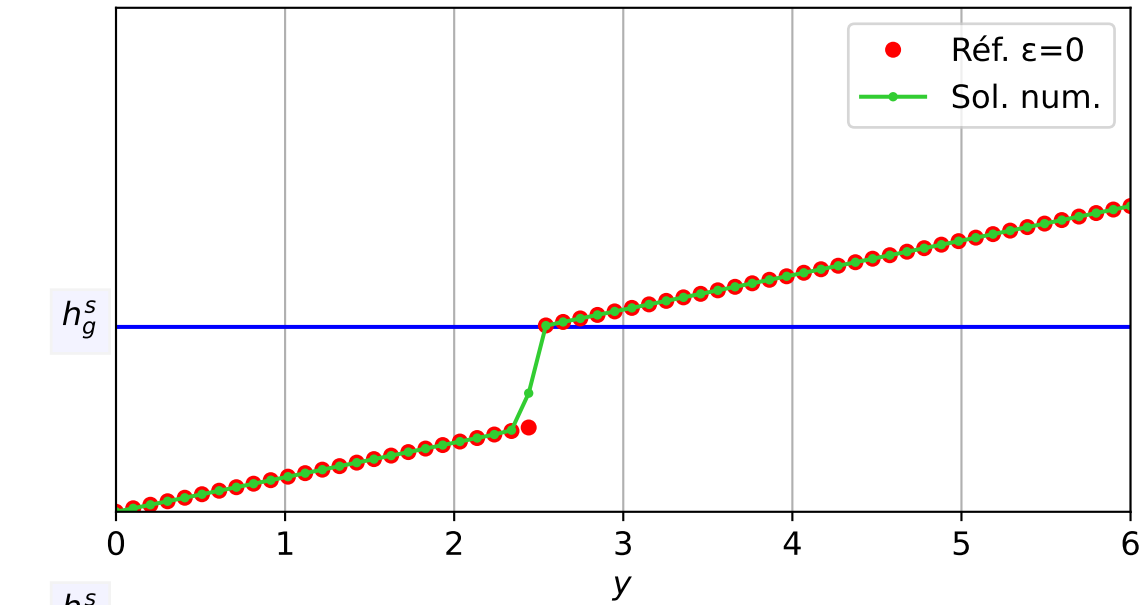
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.65204/12$  (9025 iter) — résidu eq. =  $5.95e-06$



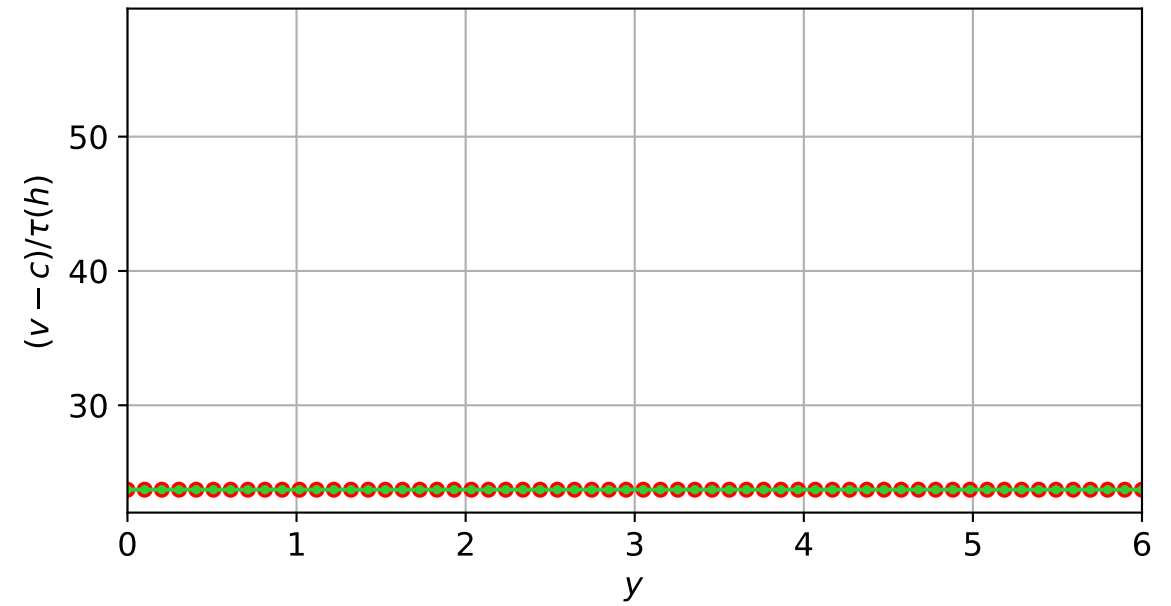
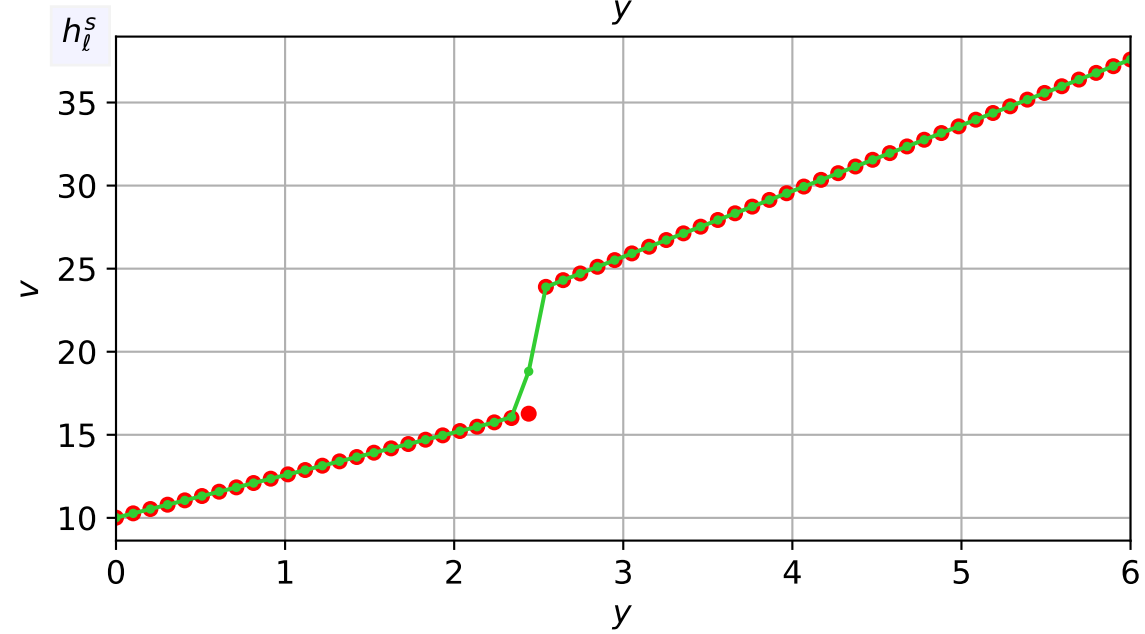
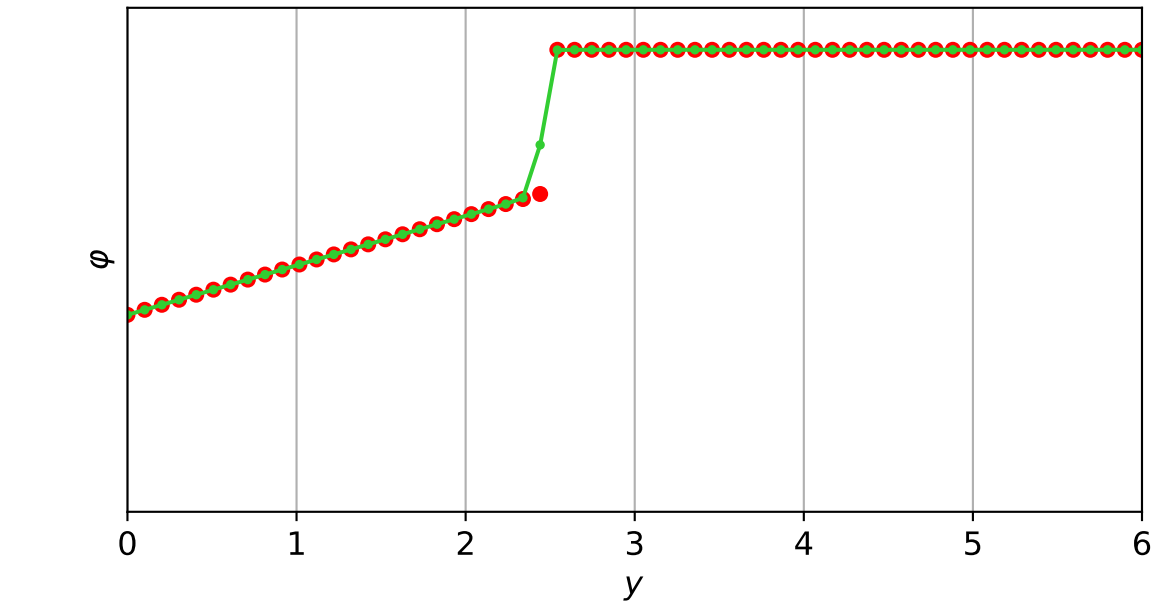
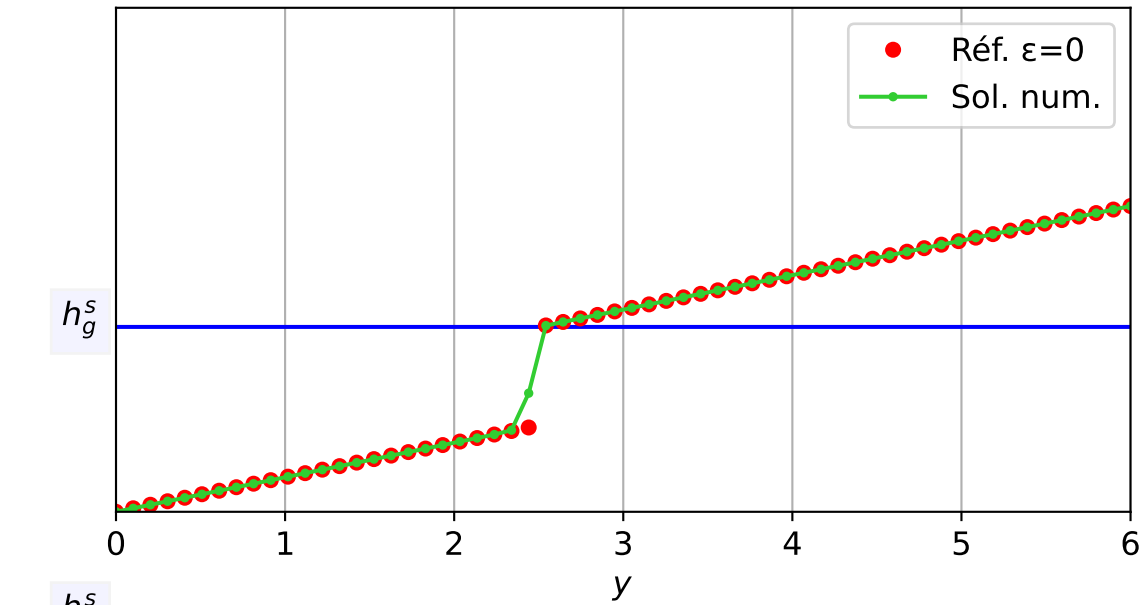
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.70074/12$  (9045 iter) — résidu eq. =  $5.29e-06$



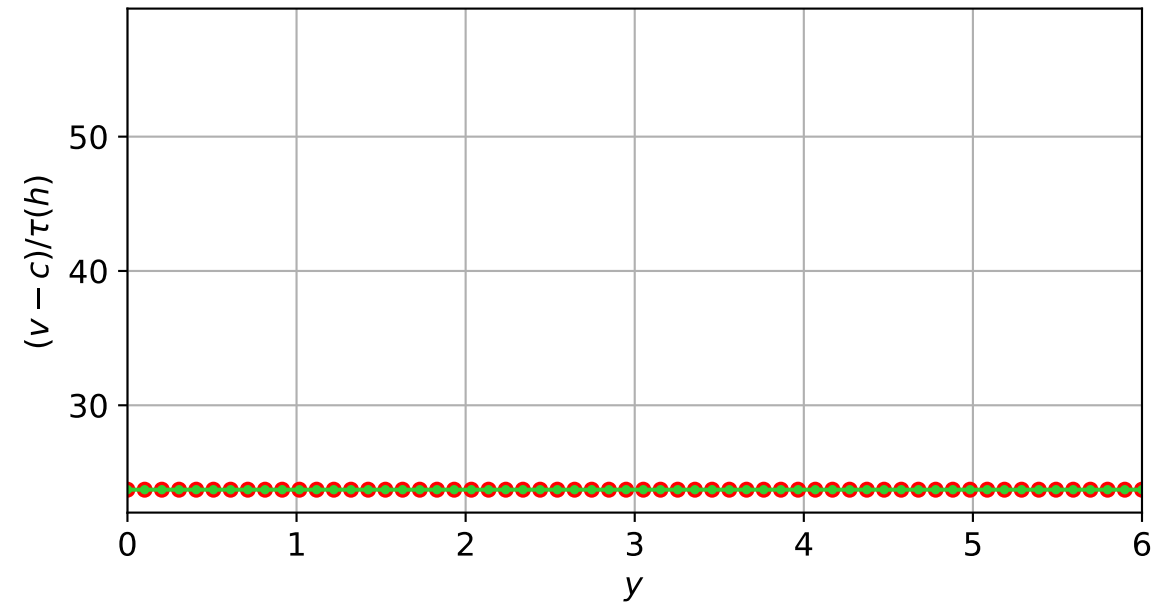
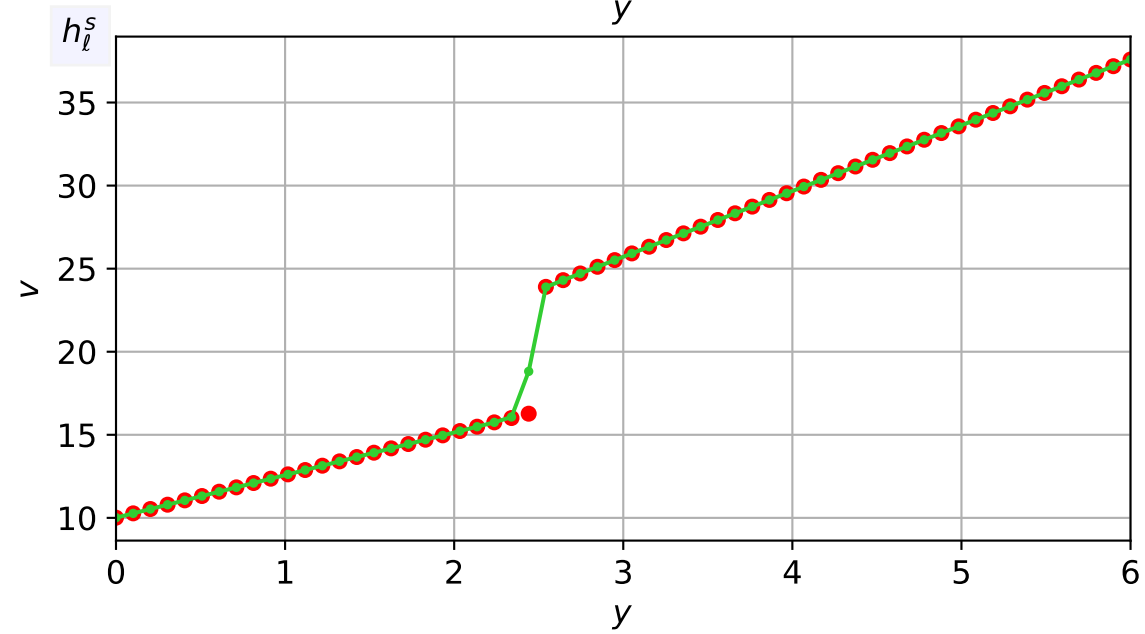
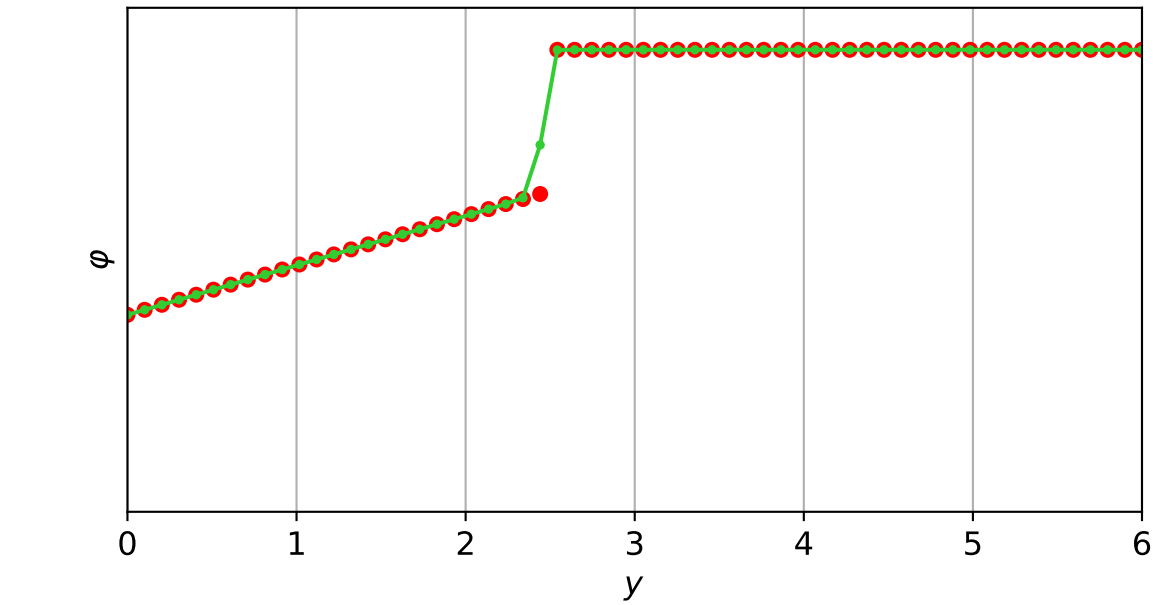
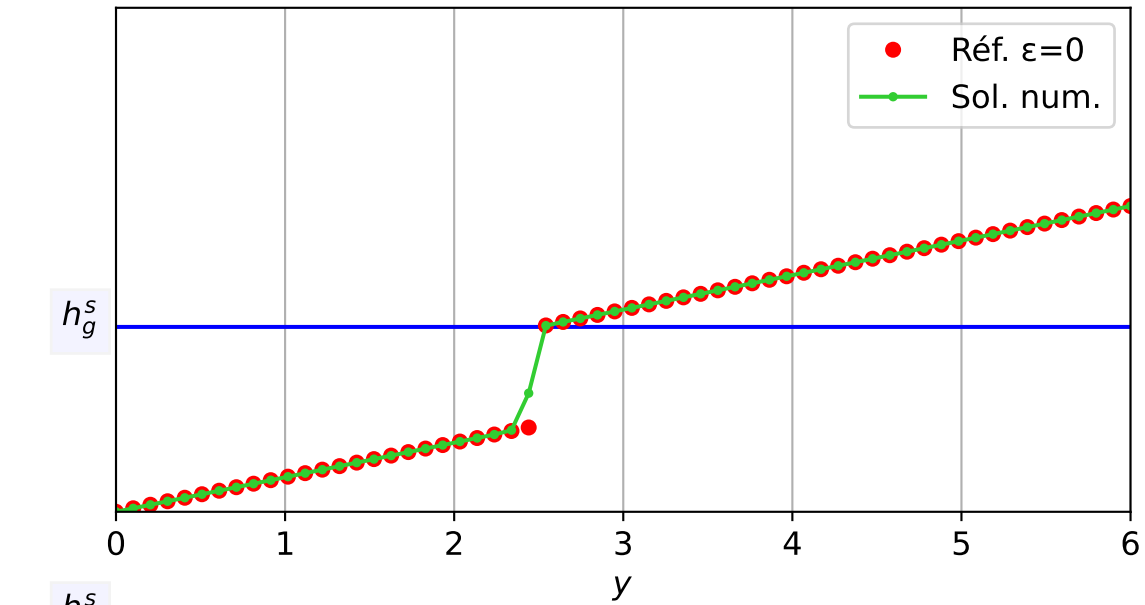
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.75188/12$  (9066 iter) — résidu eq. =  $4.67e-06$



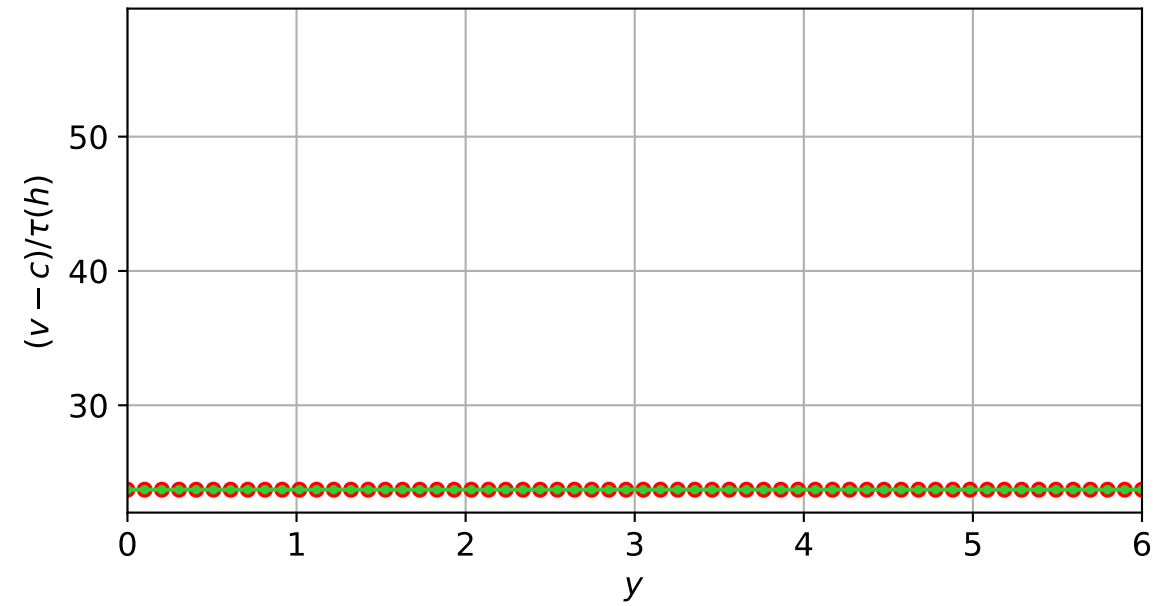
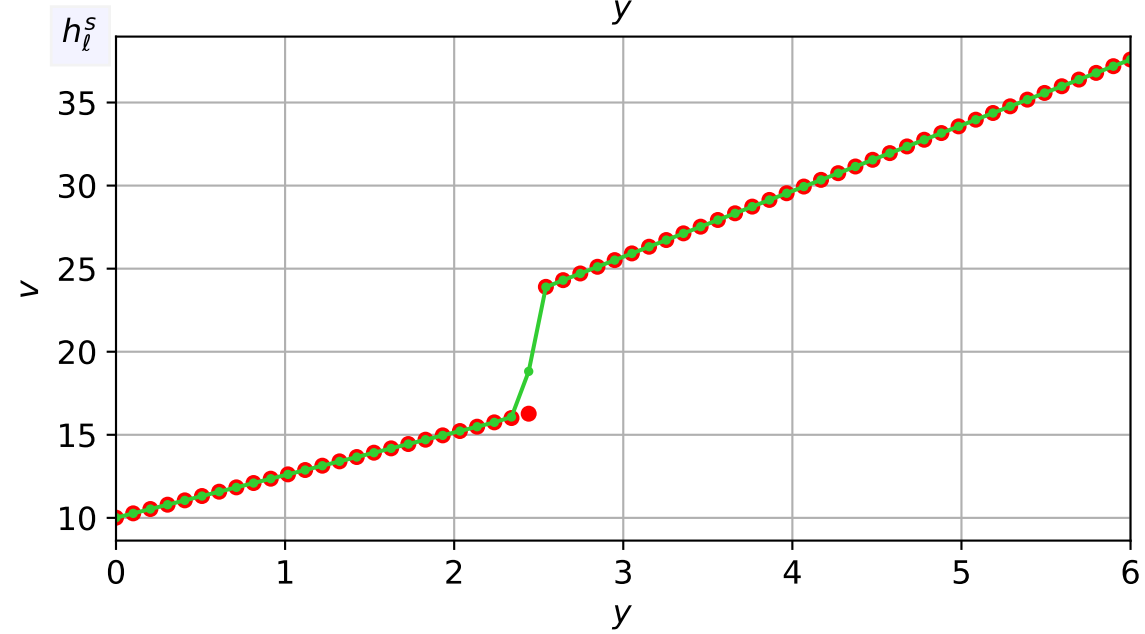
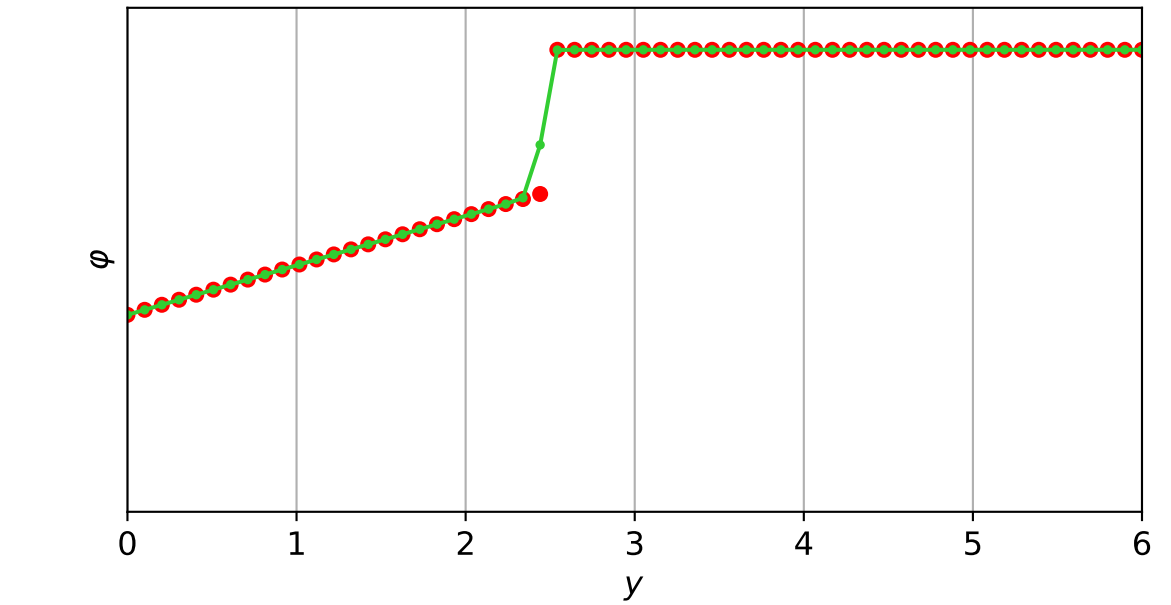
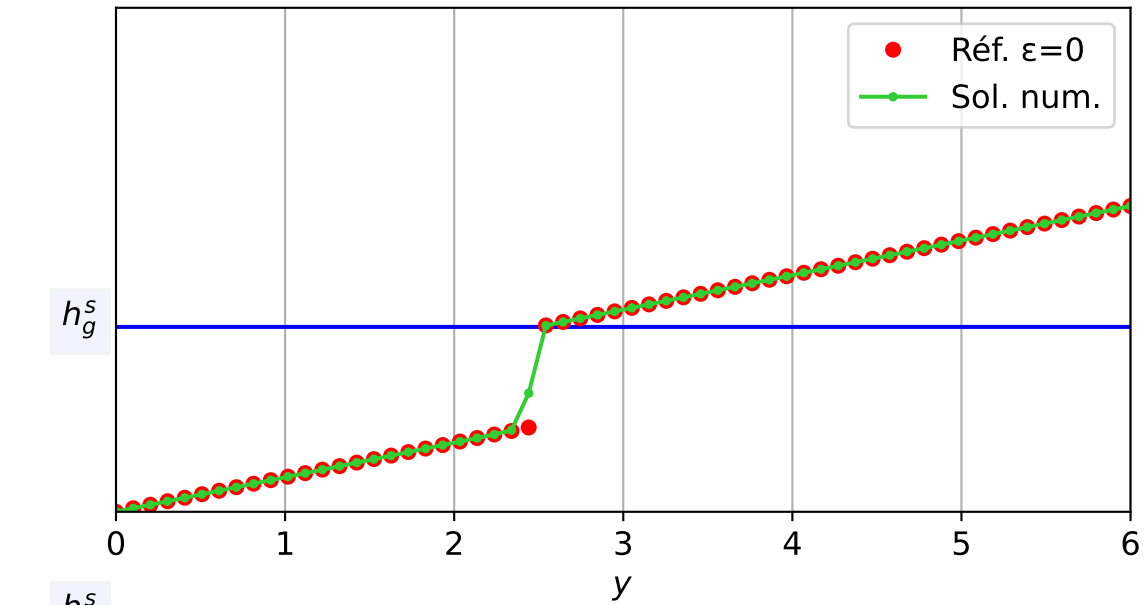
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.80058/12$  (9086 iter) — résidu eq. =  $4.16e-06$



TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.85172/12$  (9107 iter) — résidu eq. =  $3.68e-06$

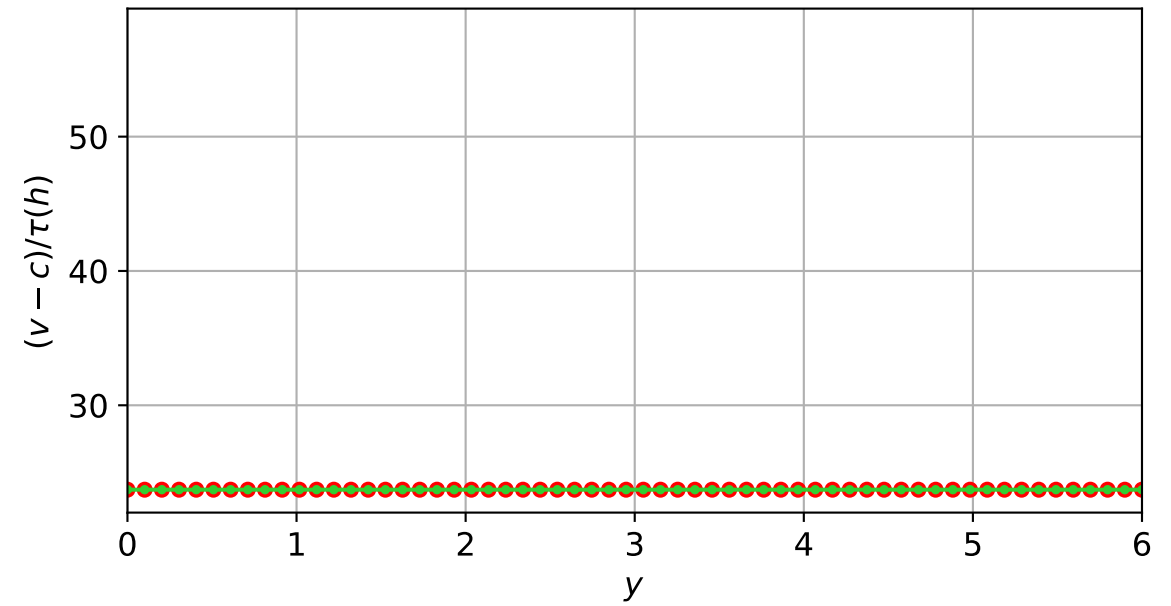
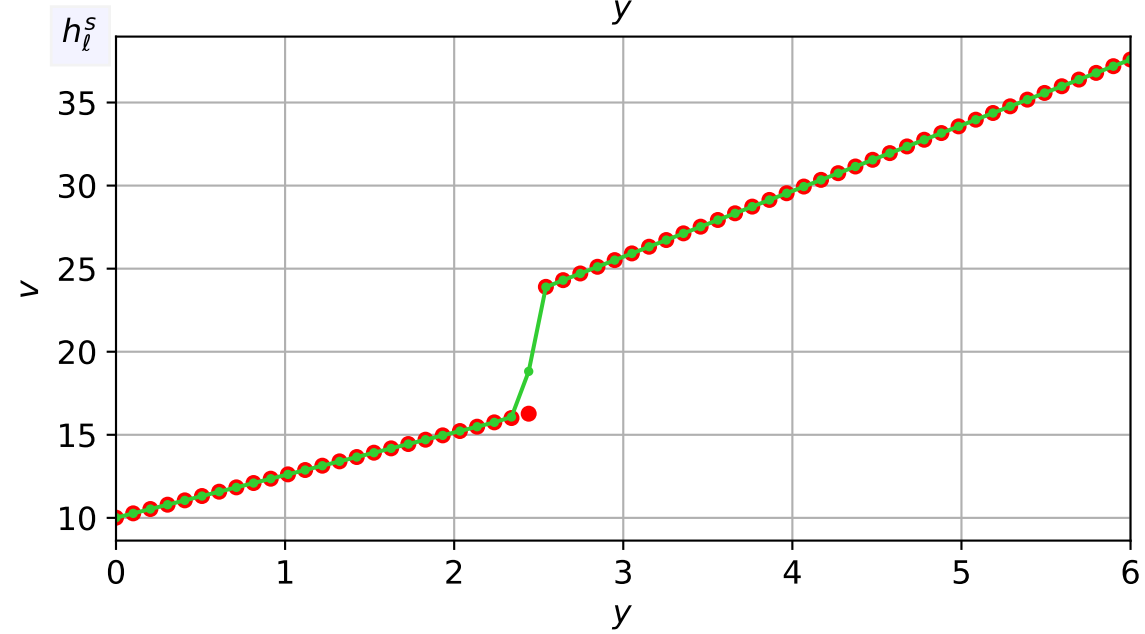
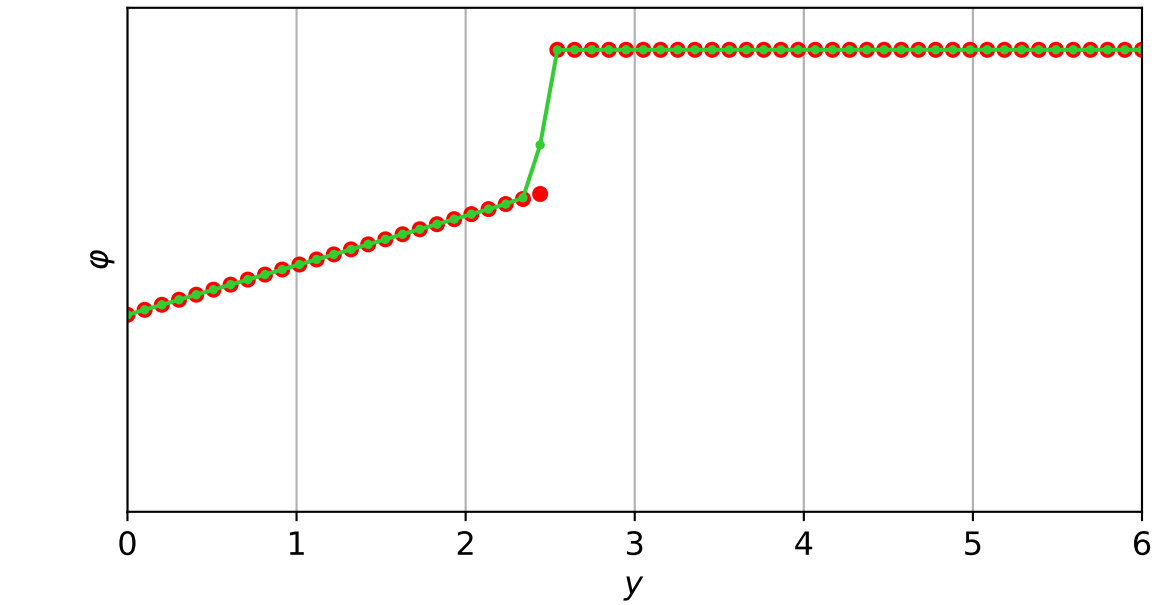
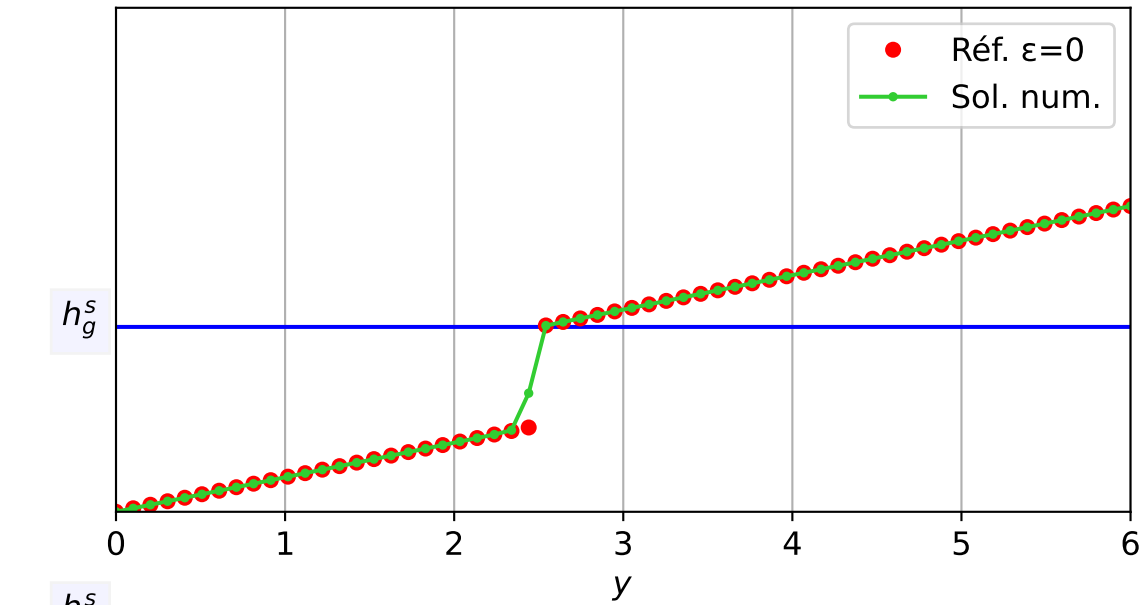


TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.90042/12$  (9127 iter) — résidu eq. =  $3.27e-06$

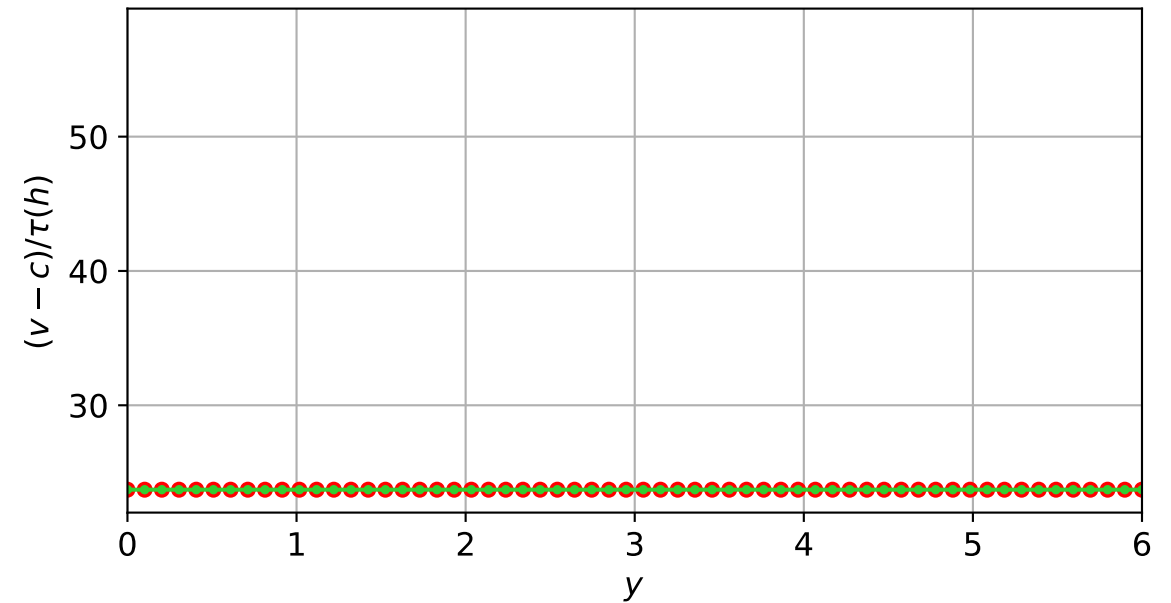
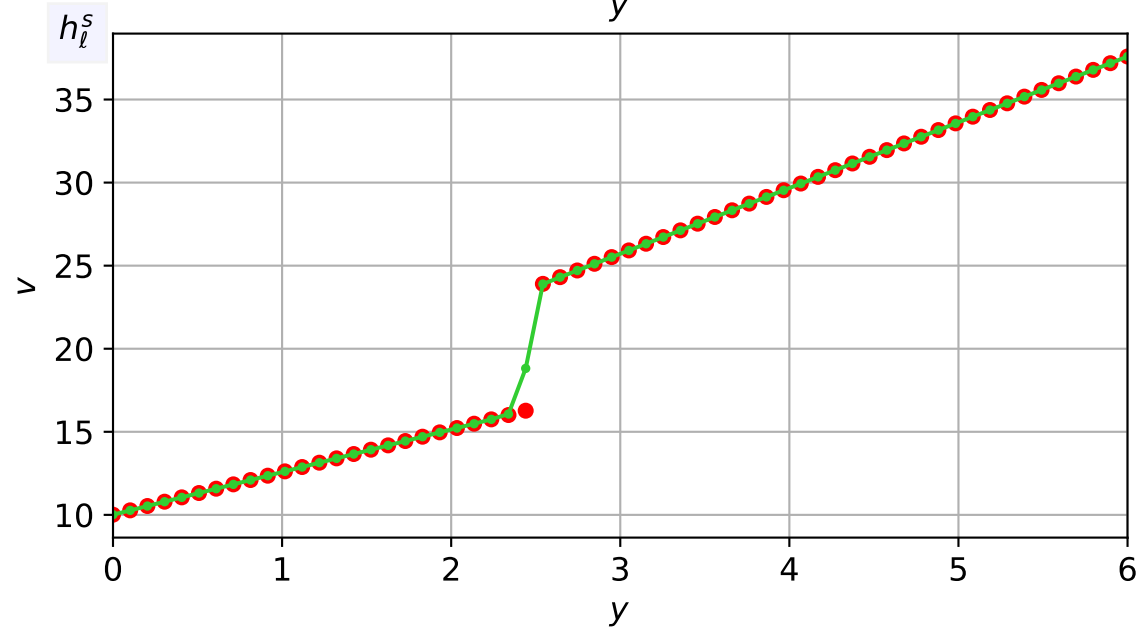
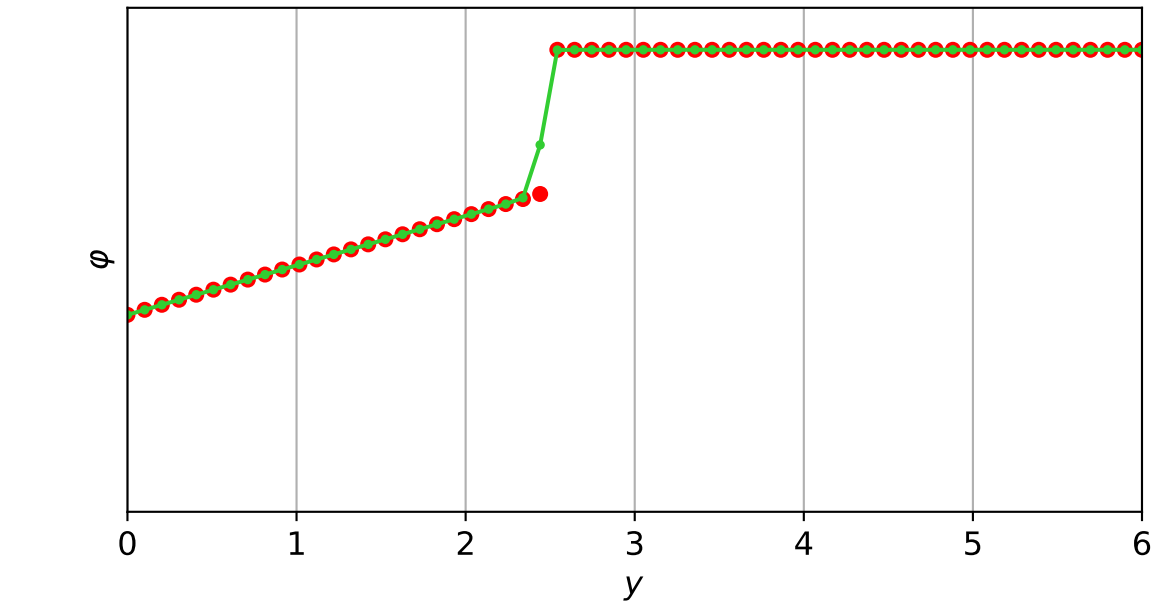
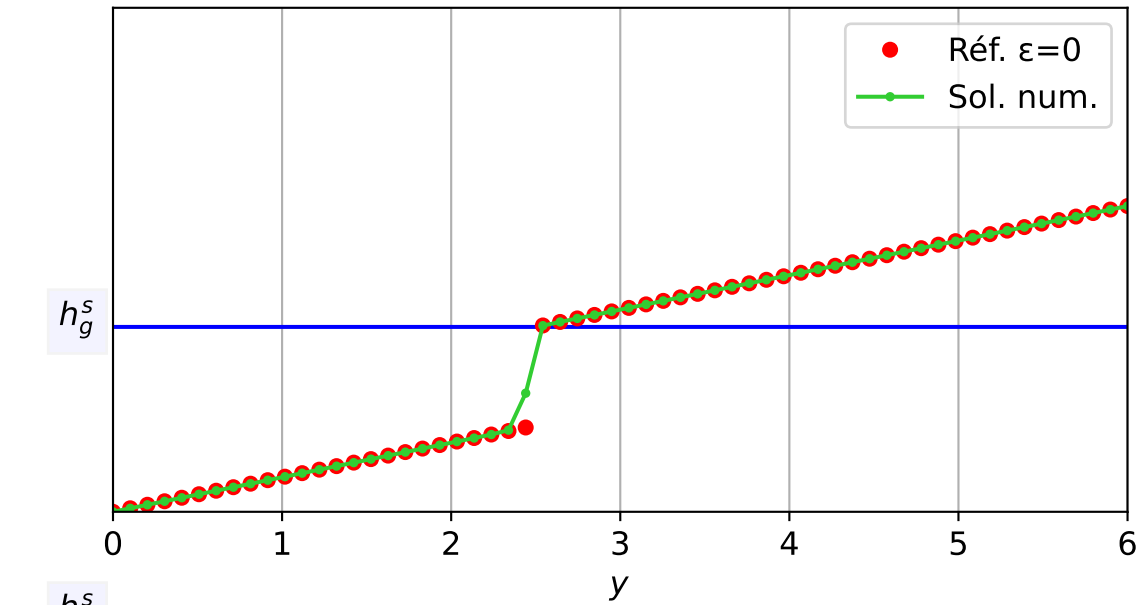




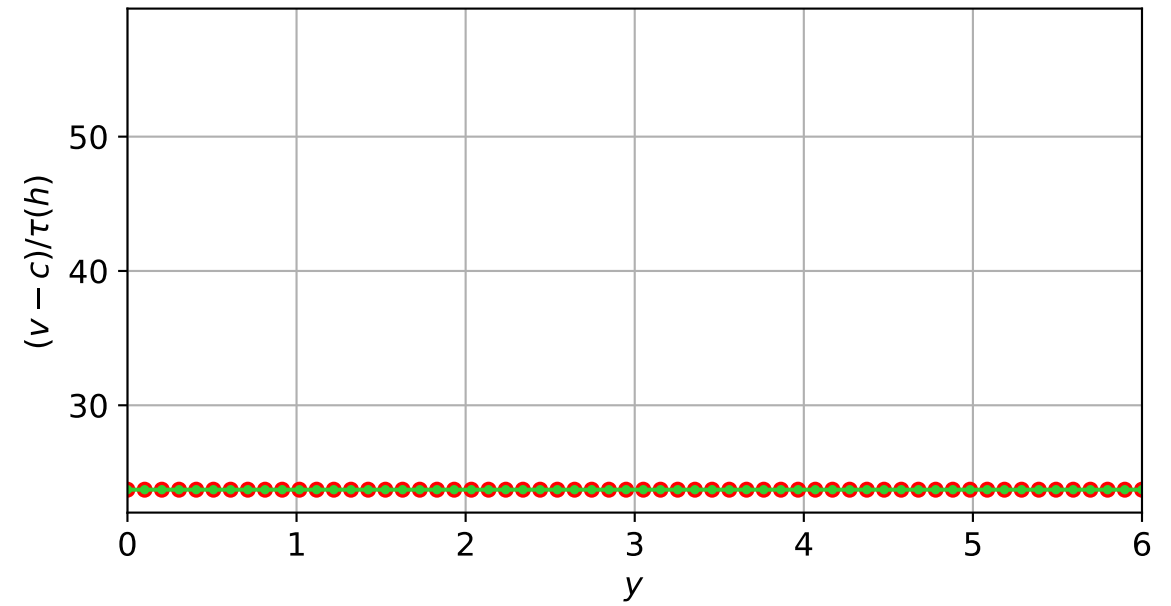
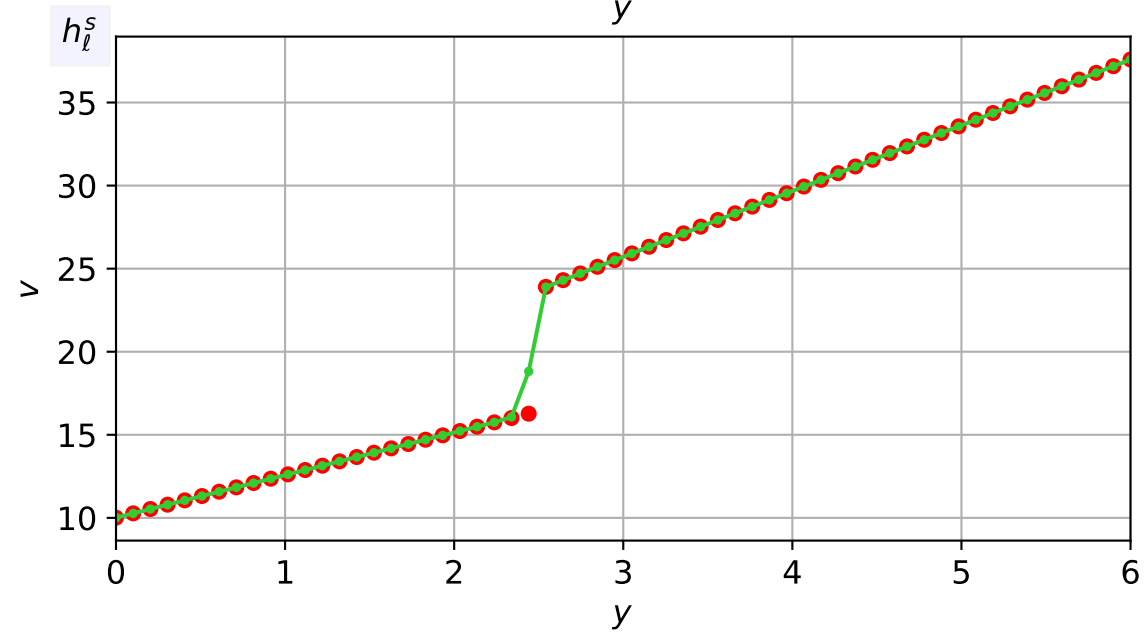
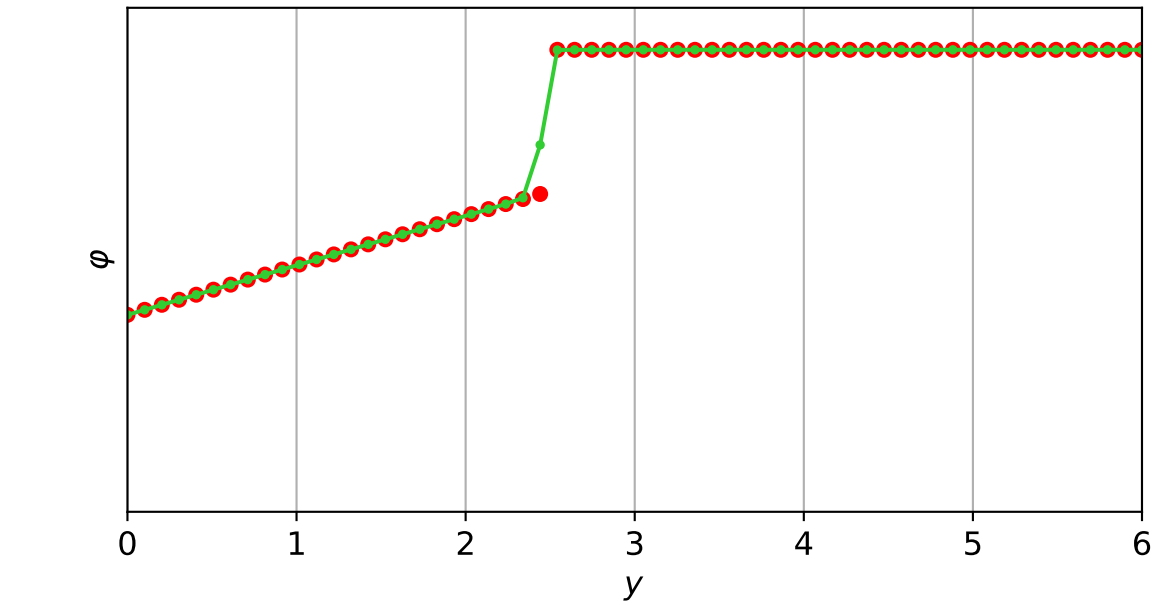
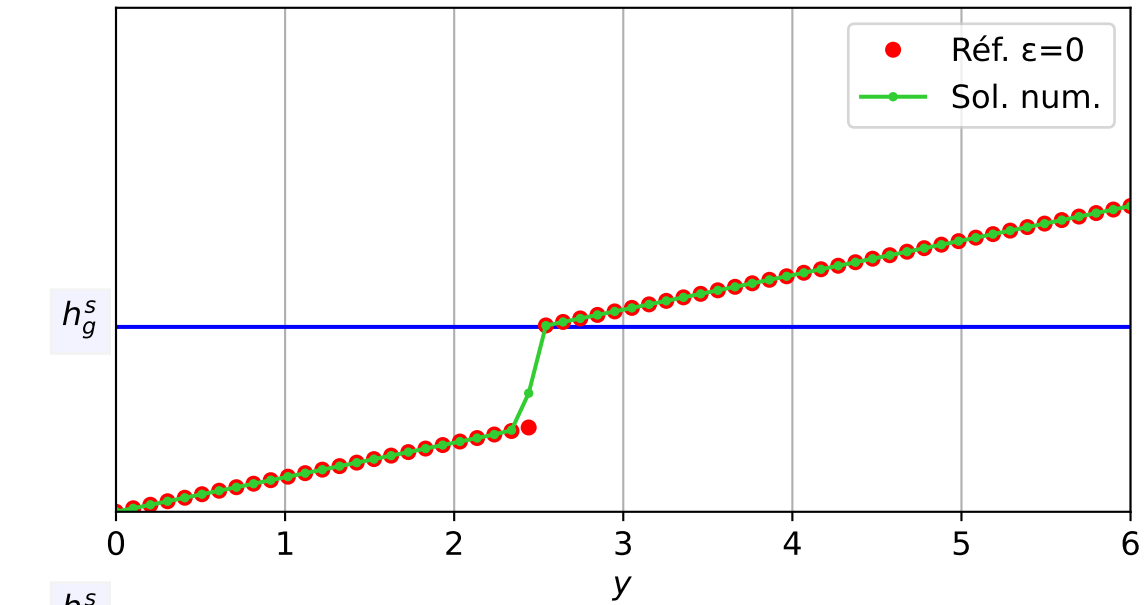
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 5.95155/12$  (9148 iter) — résidu eq. =  $2.89e-06$



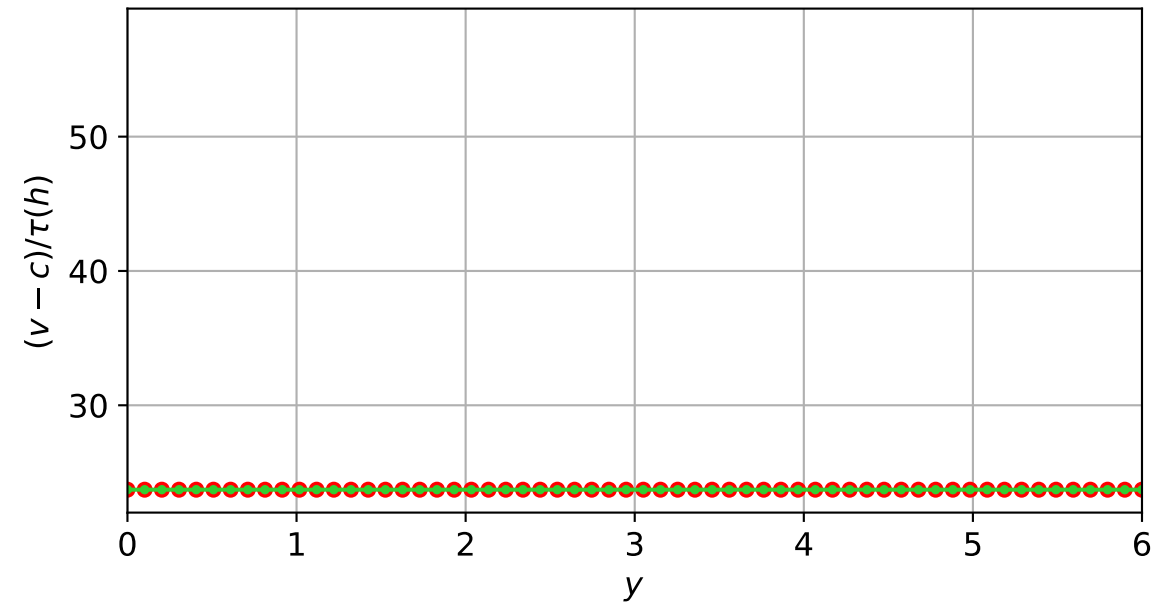
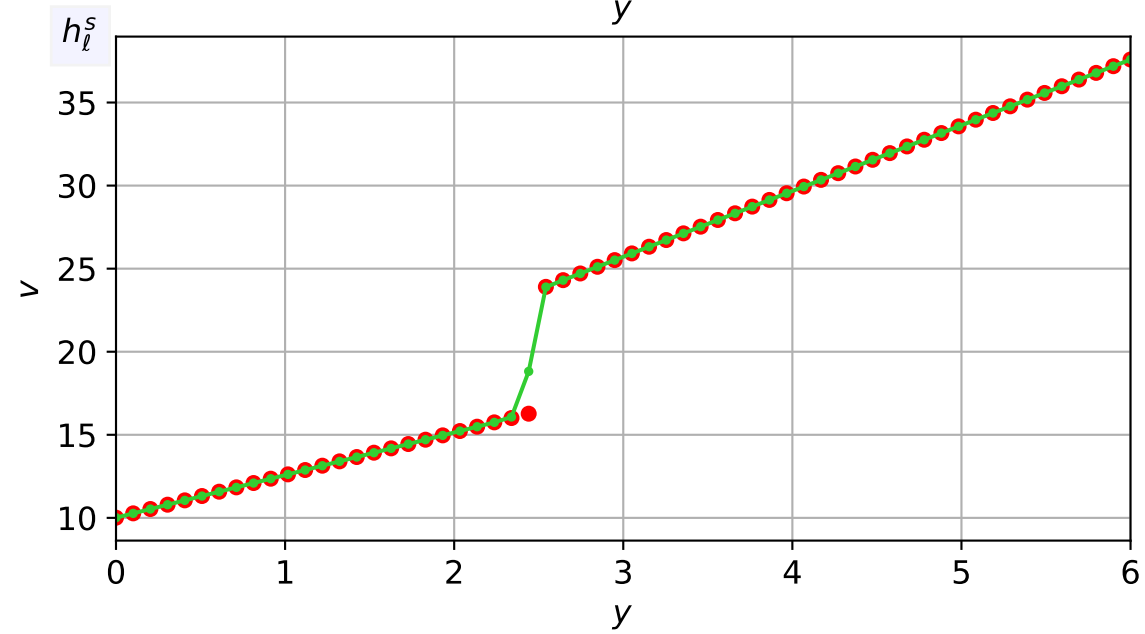
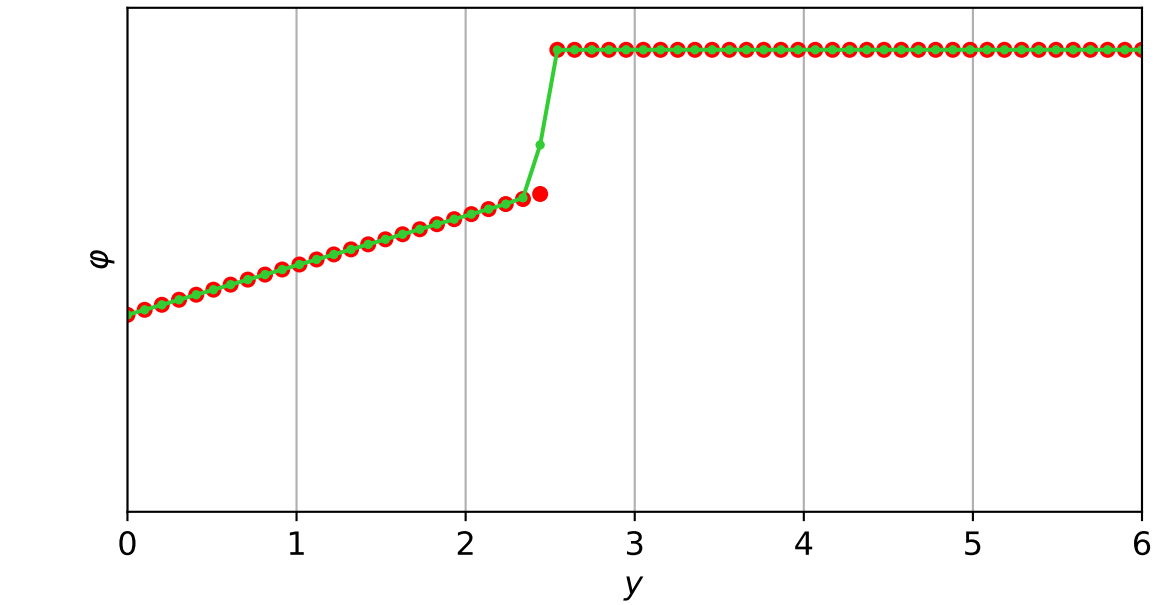
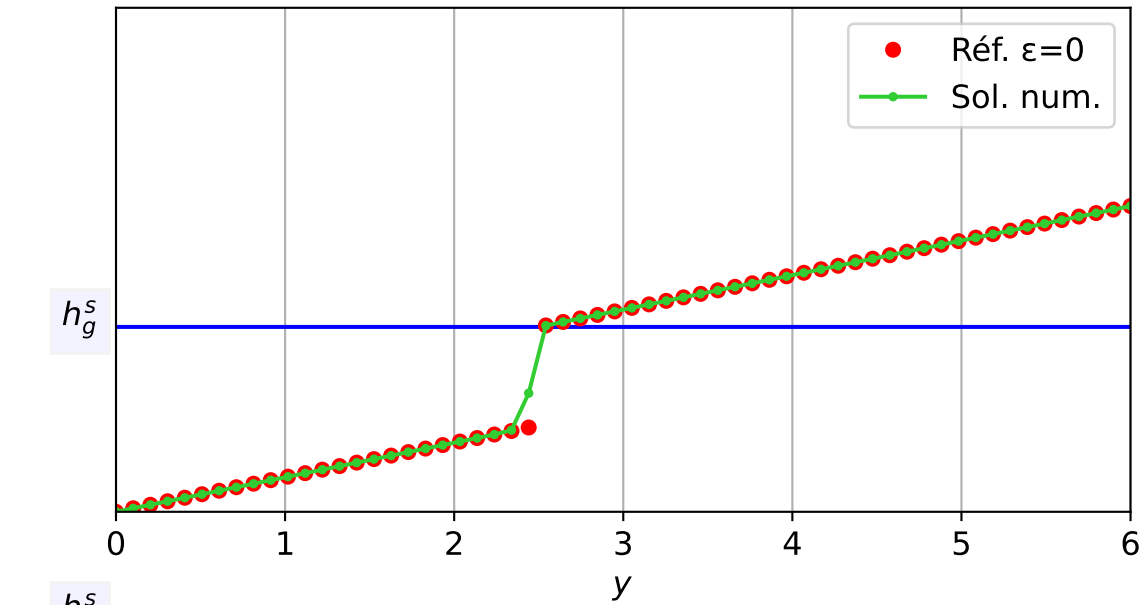
TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 6.00025/12$  (9168 iter) — résidu eq. =  $2.57e-06$



TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 6.05139/12$  (9189 iter) — résidu eq. =  $2.28e-06$



TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 6.10009/12$  (9209 iter) — résidu eq. =  $2.02e-06$



TEST-5\_MG\_AS : Validation AS | Asymptotique |  $\epsilon=1e-05$   
 $\bar{\epsilon} = 1e-05$ , schéma implicite,  $N = 60$ ,  $dx = 0.1$   
 $dt = 2.44e-03$  —  $t = 6.14879/12$  (9229 iter) — résidu eq. =  $9.64e-06$

